

AEwin™ Software

USER'S MANUAL

Rev 5

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Associated with: AEwin Software for Windows Part No. for:

DiSP 6320-7030

Express-8 7050-7001

PCI2-6301-7001

SAMOS 7030-7005

Sensor Highway Smart Monitor 9380-7003

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Added documentation for features new to this version:

The major changes include:

1. Most AEwin images updated to better reflect the current look of the UI.
2. Updated Waveform Streaming Section.
3. Updated Alarms sections.
4. Added section for Email Notifications.
5. Added section for I/O Board Setup.
6. Updated Display Mode section for 'Max Points' fields.
7. Consolidated sections for Line Display features and usage.
8. Updated sections for Preferences, many items were missing.
9. Added sections for Time Mark User Comment and Parametric Time Mark Comments.
10. Updated AST Setup for Scheduled ASTs and other features that had been added/missing (major update to this section).
11. Added section for Acoustic Property Matrix Generator.
12. Reformatting of Graph Setup section for clarity.
13. Updated Graph Color Selection section.
14. Updated Tables Menu section for description of Overview Pages.
15. Added section for Windowed Alarms.
16. Updated Location Options Tab description to include newly added options.
17. Updated Virtual Parametrics Section to include MODBUS channel.
18. Added sections for Intensity Analysis, C-Factor, TAFI, Material Property Calculation and Event Classification.
19. Added description of new AE features for modal AE (Pretrigger and Background Energies).

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1. INTRODUCTION

AEwin is a 32-bit WINDOWS, Data Acquisition and Replay program capable of running PAC's SAMOS, PCI-2, Sensor Highway Smart Monitor, Sensor Highway III, Express-8, USB, or Wireless based products. AEwin uses full WINDOWS resources, including Setting of any WINDOWS available screen resolutions, Printing, Networking, Multi-Tasking, Multi-Threading, etc. It is capable of operating in WINDOWS XP/7/8/10 operating systems. AEwin is fully compatible with PAC's standard (DTA) data files. This allows you to replay and analyze all your previously collected AE files. AEwin is easy to learn, operate and use. The software has all the acquisition, graphing, and analysis capabilities that you have come to expect in your AE system, plus many more new and enhanced features to ease your data analysis and visualization tasks. Multiple copies of AEwin can be run at once, one for acquisition with full display and analysis capabilities and one or more as Replay for additional analysis purposes.

Some of the general AE display and analysis capabilities found in AEwin include the following: AEwin provides a framework for easily adding graphs and graph screens to the main system WINDOW. AEwin has a very flexible line listing (dump) capability, which can be called at any time, on any screen, which can be scrolled up or down. AEwin has various user-selectable tool bars, including; Setup icon toolbar, acquisition control, line dump, status toolbar, and statistics tool bar. AEwin has many built-in, enhanced features, including Data Selection and Analysis, Graph zooming and panning, more flexible graph setup, more graphs, and built-in filtering functions (graphical filters and Post Filtering). Printing capabilities include Print a graph, a screen, or multiple screens to a printer, to the clipboard, or to a JPEG file.

Some of AEwin graphing capabilities include:

- Exceptional 2D and 3D graphing capabilities. Multiple graphs can be displayed on a screen, limited only by the resolution of the screen itself.
- Graphs are individually sizable on a screen, making for a very flexible arrangement. The user can set up one (or more) large graphs for visualization with multiple supporting small graphs alongside or around the main ones.
- Multiple Graphs are arranged on a screen; however, the user desires. Multiple screens can be set up, each accessible by selecting a user labeled tab. Therefore, a user can set themes for screen layout (e.g., Location & Cluster analysis, waveform analysis, AE activity analysis, AE Feature correlation analysis, Alarm Analysis, themes, to name a few)
- Many different types of graphs can be set up, including; histograms, point plots, 3-D graphs, waveforms, FFT's, multiple plots on a single graph with coloring options, etc.
- All graphs can easily be enlarged to full screen (maximized) by the touch of a button.
- All graphs have full Cursor Readout capability, either continuous or point-by-point movement.
- All graphs (2-D and 3-D included) can be infinitely zoomed and panned for close-up analysis.

In terms of Location determination, the standard version of AEwin has zonal and linear location modes, while the optional "Full Location" software adds 2D Planar Location capability to the list. In addition, there is also a 3D Location option available and a Spherical 3D location option. Key features regarding location include the following:

- Multiple (8) location groups standard, (32 optional), each capable of using all the AE channels in the AE system. Mouse-oriented sensor setup includes; Set sensors manually (click and drop), automated sensor setup for triangular setup, and easy sensor editing drag and drop, or table entry based sensor position.
- Multiple 1-dimensional, 2-dimensional, and 3-dimensional location modes.
- Flexible Clustering, cluster reporting, and Cluster statistics are available on any point plot graph screen (not just limited to location).
- User-selectable location structure visualization to ease setup. Options include; Plate, vertical vessel, horizontal vessel, sphere, and Free, each with specific setup menus geared for easy setup of that structure, including automated and manual placement of sensors, weld lines, nozzles, etc.
- Visualization overlay planes to ease setup and viewing of location setup and results. Overlay planes include; Sensors, grid, attenuation map, welds, and nozzles.
- Advanced Location setup capabilities provide improved event detection classification and accurate source location techniques.

1.1 System Requirements

AEwin software requires a PAC Micro-SHM, DaisAE, PCI-2, SAMOS, Express-8, Sensor Highway Smart Monitor, Sensor Highway III, USB or Wireless System with a Core 2 Duo processor (or equivalent). It is recommended that at least a 2 GHz processor, 2 Gigabytes of RAM (preferably 4 GB or more), 100-gigabyte hard disk, and 17" display monitor with 1280 x 1024 resolution be utilized. These days this type of computer power is easily available at a low cost. Please consult the factory if your computer is below these standards.

If AEwin is being installed explicitly for replay and analysis, it is also recommended to use a similar computer base as that described above.

1.2 Software Installation and Licensing

The AEwin program is generally pre-installed on all AE system computers from the factory. In special instances, such as software upgrades, however, you will need to perform the installation yourself from the supplied CD. Likewise, if your AE system was built at our factory, it will come pre-licensed. You will generally need to perform licensing yourself only when installing AEwin on a new computer.

If you are installing AEwin yourself, you will need to follow the simple installation instructions below. Software installation and licensing is a two-step process that will be discussed in the next two sections, for an example of "AEwin for Express-8". Other products such as PCI-2 and SAMOS will be very similar.

1.2.1 Software Installation

To fully install AEwin, you must perform the following steps.

- a) Exit all running applications.
- b) It is recommended that you un-install any previous versions of this product before installing the new one. This will NOT delete any of your data files or any layout files you have made. It will delete the example data and layout files, however, so if you have modified Layout.lay (or any other example layout or datafile) and wish to retain it, make a copy of it before un-installing.
- c) If you have a MISTRAS License Key already plugged into your system, please remove it now. This is a green or blue USB dongle that is used in licensing. Keeping it plugged in during installation could damage it.
- d) Run SETUP.EXE from the root folder of the software installation CD. You can find this through Windows Explorer.
- e) Upon executing SETUP.EXE, you will follow a typical Windows installation process. Select all the default folders (highly recommended but not mandatory).
- f) Install any hardware drivers if you are doing an acquisition. See your software manual for details.
- g) Install the Audio driver if using an audio board. See your software manual for details.
- h) If you have a MISTRAS License Key, plug it into a USB slot now. Windows may need to install the driver for the Key at this point. Simply follow the prompts and accept the default values.

1.2.2 Software Licensing

1.2.2.1 Licensing Basics

MISTRAS systems come pre-configured with licenses (either internal-file or USB Key-based). If you received such a system, then you should not have to do anything beyond plugging in the License Key (if you were issued one). If you install the software yourself on a new computer, then you will need to license it. The section "Licensing Instructions" below gives instructions on how to do so.

The software can be licensed using either of two methods.

- **MISTRAS License Keys** – These are green or blue USB dongles that can be purchased either with your system or at a future time. They can be used to run the licensed program on ANY computer they are plugged into. This makes them very portable. Note that these are not the same as the black or purple “Hardware Security Keys” used by older software versions.
- **MISTRAS License Files** – This is the software’s default licensing system. If you have not purchased a Key, you can acquire a License File from Customer Service by following the steps in the section titled “Licensing Instructions” below. This will license a single copy of the software to the computer of your choice.

1.2.2.2 Licensing Instructions

If you have a MISTRAS Licensing Key (Green or Blue USB Dongle).

- Plug the Key into a USB port. If this is the first time you have done this on this particular USB port, then Windows may want to install the driver. Let it do so now. Follow the prompts and accept the defaults. It should only take a few moments.
- Run AEwin.

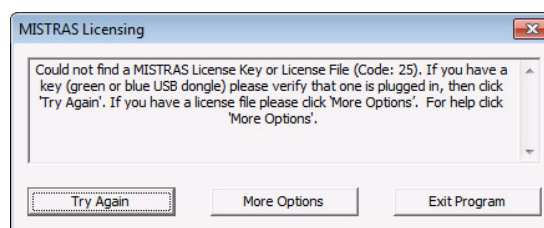


Figure 1-1. Initial Licensing Screen

If you don't have a MISTRAS Licensing Key.

- Run AEwin. You will most likely see a dialog box like this one.
- Click “More Options.” This will bring up the “MISTRAS Licensing” dialog box shown here.
- The center section labeled “MISTRAS License File” guides you through the process of acquiring and installing a license file. It is a two-step process as follows.
 - Contact Customer Service** – You will need to contact Customer Service to acquire a license file. The easiest ways are via email or by phone. In the sunken area of the menu is a block of information that Customer Service will need to create your license. You can copy/paste this into an email if you are using this method (be sure to enter your name, company, and sales order#), or simply have the dialog box open while you call so the information will be at hand. Once Customer Service has this information, they will create a license and email it to you. You can exit the software at this point if you wish, while your license is being made for you. If so, you can return to this dialog box later by following the same instructions above.
 - Install your License File** – Once you have received your license file from Customer Service, click the “Install License File” button and browse to the file. You should receive a message indicating success, at which point you can click the “Start AEwin” button at the bottom of the screen (which previously was “Exit AEwin”).
- AEwin is now licensed.

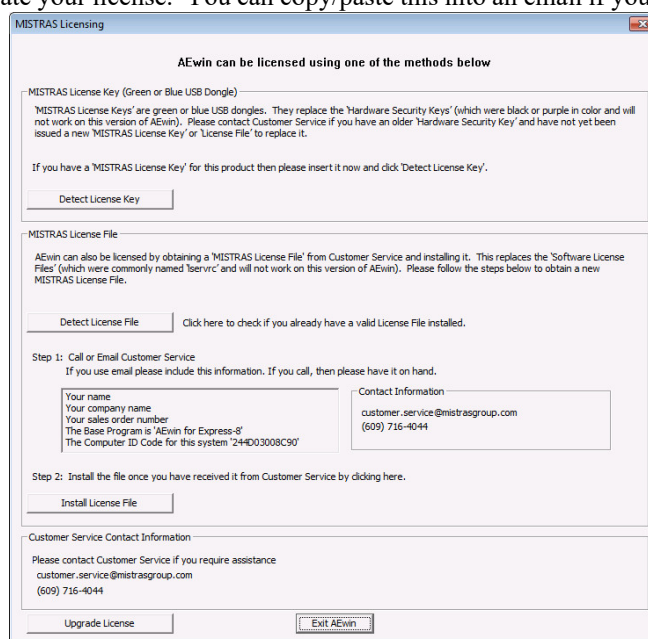


Figure 1-2. MISTRAS Licensing Screen

1.2.2.3 Upgrading an AEwin License

If at any time you decide you would like to purchase additional software options such as “3D Location”, “Post,” or “LeakTEC/Online Crack Alarms” (there are many more options to choose from), you can purchase them by contacting Customer Service. You will not need to re-install the software or return your License Key (if you use one) to the factory. Customer Service can upgrade your license via email.

- If you use a License Key, then Customer Service will send you a special file called a “Patch” file.
- If you use a License File, then Customer Service will send you a new one to replace it.

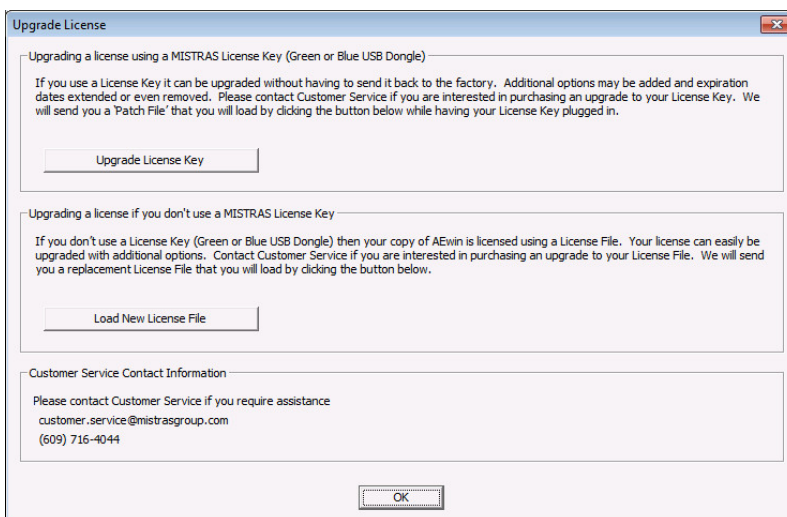


Figure 1-3. Upgrade License

In either case, once you have received the file, you will need to open the ‘Upgrade License’ dialog box. This can be accessed from one of two places.

- If the software displays a licensing error message on startup, such as the one in **Figure 1-3**, then click “More Options” to open the “MISTRAS Licensing” dialog box. From here, click the “Upgrade License” button found at the bottom left portion of the screen.
- If you can access the software normally, then open the “Options” dialog box from the “Help” menu. Click the “Upgrade License” button.

This will open the “Upgrade License” dialog box shown above.

- If you are upgrading a License Key, then make sure it is inserted into a USB slot. If you have more than one License Key, make sure that only one is inserted at any one time. Click “Upgrade License Key” and select the “Patch File” emailed to you by Customer Service.
- If you are upgrading/replacing your License File, then click “Load New License File” and select the License File emailed to you by Customer Service.

That’s it. The software may need to close. If it does not, or when you re-start it, the software will use your newly upgraded license.

1.2.2.4 Licensing FAQ and Troubleshooting

Q: My license had stopped working when I upgraded to a new version of the software. What do I do?

A: Starting in mid-2009, AEwin was upgraded to a new licensing system that is more flexible and user-friendly. The old “Hardware Security Keys” (black or purple USB dongles) have been replaced by “MISTRAS License Keys” (green or blue USB dongles). Likewise, the old “lservrc” license files have been replaced as well. If you have an old “Hardware Security Key” or “lservrc” license file, then this will need to be replaced with a new version. Please contact Customer Service for assistance.

Q: The software won’t install and gives the following error message: “There is a problem with this Windows Installer package. A program required for this install to complete could not be run. Contact your support personnel or package vendor.”

A: Make sure you run “Setup.exe” to install the software, not “AEwinSetup.msi.”

Q: After plugging in a MISTRAS License Key, Windows gives an error message like one of these:

- “You must be a member of the Administrators group on this computer to install this hardware: USB Device.”
- “You do not have sufficient security privileges to install hardware on this computer. Please contact your site administrator, or log out and log in again as an administrator and try again.”

A: The first time you plug a green Key into a particular USB port, Windows will attempt to install the driver for it. If you do not have sufficient privileges to do so, you will get this warning. In this case, you will need to log in as an administrator or contact your network or IT administrator for assistance. To avoid this problem in the future, be sure to plug the Key into all USB ports one by one (when you have admin privileges). This will install the driver for all ports. Note that the newer blue keys (introduced in early 2013) generally do not have this problem as they do not require a driver.

Q: Can I use my MISTRAS License Key on multiple computers?

A: Yes. License Keys are portable. Simply exit the software on your first computer, remove the Key, plug it into your other computer, and start the software there. You will have the same options as before.

Q: Can I install my MISTRAS License File on multiple computers?

A: No. License Files are created for a single computer. You can purchase additional license Files or Keys for replay purposes from Customer Service.

Q: Can I have multiple MISTRAS License Keys plugged in at the same time?

A: No. Only one License Key should be plugged in at the same time. If you have multiple licensed programs (e.g., AEwin for SAMOS, AEwin for PCI2) installed on your machine, then you should swap keys when you switch programs. Another option is to obtain License Files for one or more of the programs.

Q: What is the difference between green and blue MISTRAS License Keys? Are they the same?

A: The original MISTRAS License Key is green in color and requires a driver (installed automatically with AEwin). This key will be phased out in favor of a new key that is blue in color. This key works in exactly the same way but does not require a driver. The blue key will be supported by all products released after 2/2013. The green key will continue to be supported by all products.

1.2.3 Installing the Data Acquisition Board driver if you are doing acquisition

The various PAC Data Acquisition Boards (PCI-2, PCI-8, Express-8, etc.) and Audio Monitor Cards (ISA and PCI) require a driver to function properly in Windows.

Normally, when Windows is first booted after installing the card, it will prompt for a driver for it. Point Windows to the appropriate driver folder on the CD and follow the instructions.

The driver folders are located on the CD at the following locations:

- **AE Boards** – Drivers\[Name of the board], where the name of the board is something like PCI2, PCi8, Express-8...

If, for some reason, you need to install a driver manually, then read the document (drivers.rtf) found in the Drivers folder. Use the procedures described there only if the above automatic driver installation does not work. It is not normally required to install the driver this way. But if problems arise, that is how to do it.

1.2.4 Installing the PAC Audio board driver if using the Audio Board

The Audio Monitor Board (ISA and PCI) requires a driver to function properly in Windows.

Normally, when Windows is first booted after installing the card, it will prompt for a driver for it. Point Windows to the appropriate driver folder on the CD and follow the instructions.

The driver folders are located on the CD at the following locations:

- **ISA Audio Board** – Drivers\IsaAudio.
- **PCI Audio Board** – Drivers\PCIAudio.

If, for some reason, you need to install a driver manually, then read the document (install.rtf or readme.txt) found in the appropriate driver folder. Use the procedures described there only if the above automatic driver installation does not work. Note that ISA audio boards are rarely required as the ISA bus has all but been replaced by the PCI standard. ISA Audio boards are not automatically detected by Windows, and the driver will need to be installed manually. For PCI, it is not normally required to install the driver this way. But if problems arise, that is how to do it.

1.2.5 Warning Notes

Please note the following warnings:

1. If the software is installed on a computer without AE hardware boards (e.g., Express-8 boards) installed, the program will run in replay-only mode.
2. If the software is installed on a computer with AE hardware (e.g., Express-8) board(s) installed, it will not run in acquisition mode unless the Windows drivers for those boards are installed.
3. If a PAC audio board is installed in the system, the software will not be able to control it unless its drivers are also installed.
4. If the entire system came from PAC, the drivers and software are pre-installed at the factory.

1.2.6 Installing a Shortcut Icon for AEwin on the Desktop

Once the AEwin software is installed as described in the previous sections and assuming that all the default installation options have been selected, the software can be started by the following mouse operated (activated by pressing the left mouse button) sequence:

Start → Programs → MISTRAS Group → AEwin for Express-8

Other forms of AEwin, such as SAMOS and PCI2, will have similar shortcuts.

An AEwin desktop Icon can easily be created by performing the same sequence above except pressing the right mouse button when pointing to AEwin. Upon doing this, a menu will appear. You can either select “create shortcut” (which will create a second copy of the AEwin program title) and drag it to the desktop or you can select “Send to,” then select the “Desktop (create shortcut)” menu button, which will place this extra copy directly on the desktop. Once created, the Icon title can be modified by left-clicking on the title. Now the program can be run directly by double-clicking the AEwin Icon on the desktop.

2. AEwin LAYOUT (.LAY) FILES AND DATA (.DTA) FILES

This section discusses the various file types associated with AEwin. The most important files you will be working with every day in AEwin are the Setup Files (called Layout files in AEwin) and the Data files. Your working copy of AEwin will have examples of both these layout files and test data files. You will be creating, using, and modifying these files in the course of your work.

The initialization or setup files, referred to as “Layout” files, are distinguished by the suffix .LAY, hold all AE test setup information for the AEwin software. A .LAY file stores the hardware setup, graph setup, acquisition setup, filters, alarms, and location setup. Other program settings, such as graph item colors and various preferences, are stored in the Windows registry.

A .LAY file can be loaded from disk by means of the “Open Layout” option in the “File” Menu of the AEwin program. Once loaded, all the initialization information from the layout file immediately becomes effective for system operation. A .LAY file is written to disk by means of the “Save Layout” or “Save Layout As” option in the File Setup menu. All the initialization information currently established in the system is written to the file. A .LAY file can have any name followed by the .LAY extension.

There is one special .LAY file known as the Autoload file. It is distinguished by the special name Layout.LAY. The Autoload file, when present, is automatically read from the disk when the AEwin program begins to run. If the Autoload file is not present, the system is set up with default values held in the AEwin program itself. The Autoload file is a convenient feature to speed your work. Optionally, there is an item in the Preferences dialog box that can override this behavior by loading whatever layout file was last loaded/saved by AEwin.

AE Test Data Files are identified by the filename extension .DTA. You may specify any name for a data file using standard Windows file naming conventions. This is done in the Start acquisition dialog box from the Acquire Menu. If a new name is not entered, the system will use the default data file name, e.g., TEST0000.DTA for the first data written after AEwin is started. The next file will be TEST0001.DTA, then TEST0002.DTA, and so on. It is convenient to use numbers at the end of the filenames (e.g., ABCD000.DTA) as the system will then increment the numbers to help keep the files in order and organized.

Layout and data files may be written to any disk drive and any specified subdirectory. A good practice is to save both .LAY files and .DTA files to the same drive and subdirectory, keeping related test files together. It is also a good practice to name the files similarly (e.g., test.LAY and test0000.DTA) as this will help keep the layout files organized with the data files.

There has been an intentional name, and file-extension change from the “Initialization” (.INI) file of the “Legacy” (DOS) based PAC programs to the “Layout” (.LAY) file for setup files utilized by AEwin. The file extension for a data (.DTA) file remains consistent. There is a good reason for this. Due to the extensive changes in migrating to Windows, and the many, many extra capabilities that have been added to the setting up of the system, it is impossible to keep consistency in the setup files between the previous legacy software and the AEwin software. Therefore, to avoid confusion, the extension file name has been changed. Now, whenever a user sees an INI file, they will know that this can only be used with the previous legacy software, while the .LAY layout file can only be used with the AEwin software.

However, with the DTA file, the compatibility remains between the data file structure and AE .DTA data files can be read and displayed by either the DOS or the AEwin software.

3. OPERATING AEwin

The purpose of this section is to provide a quick software operational run-through, to provide familiarization with operating and using AEwin. This is the most important part of this AEwin software user's manual as it will provide a view of the important screens and commands necessary to get AEwin set up and running in both Replay and Acquisition modes.

3.1 Starting AEwin

The first step in the process of using AEwin is starting up the program on your AE system. As the AEwin program is a standard Windows program, it is started in the same way as any other WINDOWS program. This is accomplished either by using the “Start” command or by selecting the AEwin Icon from the WINDOWS desktop (if one exists).

It is assumed that the AE computer is booted up and in its opening screen, which is the desktop layout showing all the available program icons to select. If the PAC AEwin icon (as shown on the right) is on the desktop, all you need to do is select this by double-clicking the left mouse button once the mouse is positioned over the icon. If the icon is not on the desktop, you can create the shortcut by referring to section 1.2.6.



Alternatively, you can start AEwin by selecting the following sequence starting with the Start menu at the bottom left side of the screen:

Start → Programs → MISTRAS Group → AEwin for [Product Name]

The boot-up sequence for AEwin starts by loading the executable program AEwin.exe. Once AEwin is loaded, the program searches the startup “AE Data” subdirectory for the default Layout (setup file) called “Layout.LAY”. Upon finding this file, AEwin loads it and becomes fully configured and ready to run in accordance with the setup information in that default Layout.LAY file. If no Layout.LAY is found, AEwin chooses a common default starting setup that is ready for use. In either case, an AEwin startup screen similar to **Figure 3-1** will appear.

3.2 AEwin Screen Layout Familiarization

Figure 3-1 shows a typical 4-screen layout for AEwin. Above and below the graph area are various informational, status, and setup toolbars. Each bar is labeled with a name. Depending on the preferences selected for your program, you may have other toolbars or be missing some. The “View” menu in the main menu bar controls which bars are on or off. The following is a short description of each area identified on the AEwin screen. This section will get you familiar with the layout and the overall function. Further chapters will provide detailed insight into the use of each of these functions.

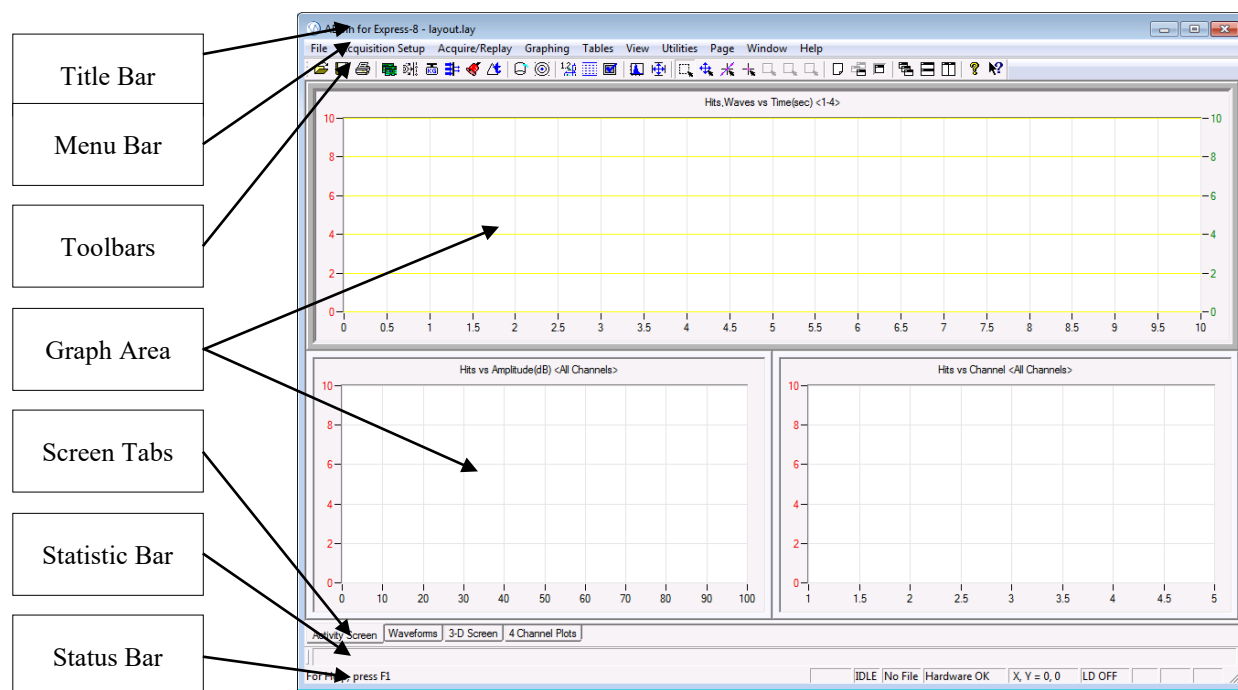


Figure 3-1. AEwin Opening Screen

Title Bar: This is a typical WINDOWS Title Bar with all the standard capabilities. At the leftmost position is the MISTRAS Group Logo icon from which you can manage the AEwin program with functions such as minimize, restore, maximize, move, size, or Close the program altogether. At the far right are the application buttons for Minimize, Maximize/Restore and Close. Back on the left-hand side next to the MISTRAS Group Logo icon is the program name (in this case, it is AEwin for Express-8), and next to that is the name of the current layout file being used by the program. If an asterisk (*) appears next to the file name, it indicates that one or more settings have been changed since the last time the layout file was loaded or saved.

Main Menu Bar: The main menu bar provides a series of text commands running across the line, including File, Acquisition Setup, Acquire/Replay, Graphing, Tables, View, Utilities, Page, Window, and Help. Most of these items contain pull-down sub-menus of their own, many of which in turn contains graphical dialog boxes and prompts of their own. You narrow down the range of choices until you arrive at the specific choice you want.

To display a pull-down sub-menu from the main menu entry, simply move the mouse pointer to the name of the menu item on the menu bar and click the left mouse button. This process is known as “clicking” (or “left-clicking”) on an item and will herein be referred to as such. To use the keyboard, simultaneously press the ALT key and the underlined letter of the menu item name at the same time. Once you have opened a pull-down menu item from the main menu, you can use the left and right arrow keys to move through each entry, allowing you to see the pull-down menu for each main menu entry. You can also use the up and down arrow keys to move through each selection of the pull-down menu. To select an item from a pull-down sub-menu, click on the desired action, or use the up and down arrow keys to select an action, and then press ENTER. Pressing the underlined letter of a menu item will also select that item. If an item in the pull-down menu appears “grayed out,” this action is not available in the current software or at the current time. Use the <ESC> key to remove the pull-down menu display.

Most of the selections in the pull-down sub-menus will display a dialog box requesting information necessary to carry out the command you requested. A dialog box contains areas where you enter information, select options or commands, or activate controls to execute or cancel a command. There are several types of boxes and buttons available within a dialog box. These boxes and buttons are described briefly below and will be described in further detail in subsequent sections of this manual. It is important that you be familiar with dialog boxes and be comfortable using them, for they are the means by which most functions are carried out in the AEwin software.

Icon Toolbars: Many of the menu items accessible from the Main pull-down menus can also be accessed by use of the Icon Toolbar and Acquisition Control Toolbar just below it. **Figure 3-2** shows the Icon toolbars.



Figure 3-2. Icon Toolbars

To determine the function of any of the icons, simply hover the mouse cursor over the top of the desired icon. Upon resting there for approximately 1 second, the Icon's function will appear. To select an icon as a shortcut command, simply click it with the left mouse button.

Graph Area: The graph area is the most important area of the screen. This is where the AE information in the form of graphs and line listings appears regarding the AE examination status. The graph area can be set up in many ways, with a minimum of 1 graph to a maximum only limited by the graph visibility. More about the graph setup, functionality, and flexibility will be discussed later in the Graph setup menu.

Screen Page Tabs: AEwin can show many different graph screens; these are called screen or graph "Pages." Accessibility to viewing any screen is provided via the Page tabs located just below the graph area. The Page tabs can be named to be synonymous with the graph theme as shown in **Figure 3-1** with the 4 Page tab names, Activity Screen, Waveform Screen, 3D Screen, and 24 channel plots. Using the right mouse button and clicking on a page will display a 'context' menu from which pages can be added, inserted, deleted, or renamed. The context menu is also a quick way to add new graphs and close existing graphs.

Statistics Bar: This toolbar area is reserved for the display of test statistics. Test statistics allow the user to view key Acoustic emission test indicators, including Cumulative AE counts, Total AE Hits, Total AE Events, Total # Waveforms, Cumulative Counts, Cumulative Energy, Time of Test, Disk Free Space, Parametrics, and others. These are very useful activity indicators during a test and very useful summary statistics after a test. The user can select any or all of these to customize his statistics display bar.

Status Bar: The Status bar provides useful information regarding AEwin and test status. On the left is a Text bar providing status and help information regarding the functions being carried out. Near the center is a status bar. In **Figure 3-1**, the word "Idle" appears in this bar. Other status information such as "Test Paused," "Replay," "Test Stopped," "Test Active," "Abort Test," etc., is displayed in this text field. Next to the Test status field is a filename field informing the user of the name of the file being replayed or the name under which an AE test is being saved. Next to that is a system diagnostics text field. In **Figure 3-1**, this status is "Hardware OK." "Hardware OK" is replaced with "Replay Only" if no hardware is detected or the program is in replay-only mode. Next to that is a cursor position, showing the current location of the Cursor on the graph. When selected, the X, Y positions will be read out in actual screen units.

3.3 Navigating the AEwin Menus

This section provides a detailed reference of AEwin's major menu items and dialog boxes. The first time through, we suggest you just look at the menus (to get familiar with them) and use the setups provided with your AEwin installation that has been predefined by PAC; later, you can make your own choices within the menus to control the system operation.

From the Opening Screen (**Figure 3-1**), pay attention to the Main Menu Bar near the top of the screen. This specific area is shown again below. Each main menu item will be described. This brief introduction to navigating the menus is provided in order to get ready for guiding the user in the sequence of starting and replaying a test.



Figure 3-3. Main Menu and Toolbars

3.3.1 File Menu

3.3.1.1 New Layout

The 'New Layout' menu item is used to remove all graphs and pages and restore the graph layout to a default setting of one page and one simple graph. This command does not alter any other settings.

3.3.1.2 Open Layout...

The 'Open Layout...' menu item is used to load a specified layout file's setup information into AEwin. Whenever a layout file is successfully opened, AEwin will save the name of the file and the folder that contained it in the Windows registry. The next time the Open Layout dialog box is shown, even if the program is shut down and restarted, AEwin will start it in that same folder.

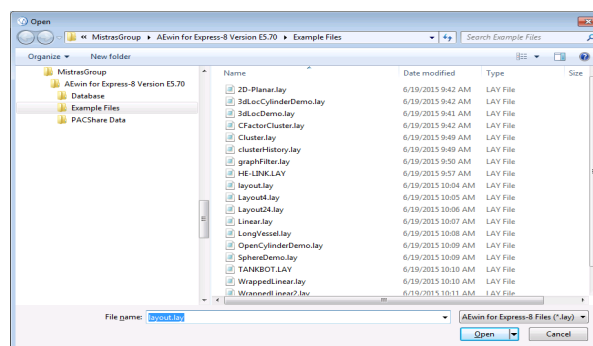


Figure 3-4. Open Layout...

3.3.1.3 Save Layout/Save Layout As...

The 'Save Layout...' and 'Save Layout As...' menu items are used to save the setup information in AEwin to a specified layout file. When the 'Save Layout As...' item is clicked, the 'Save As' dialog is displayed. Here you can enter a new layout filename to create or an existing one to overwrite. Once this is done, you will see the layout filename you entered on the upper left corner of the screen. You can use 'Save Layout...' from then on to update the file with any changes you have made since you have last saved it. As with the Open Layout function, AEwin saves the file name and folder in the registry, so they are available the next time the dialog box is opened.

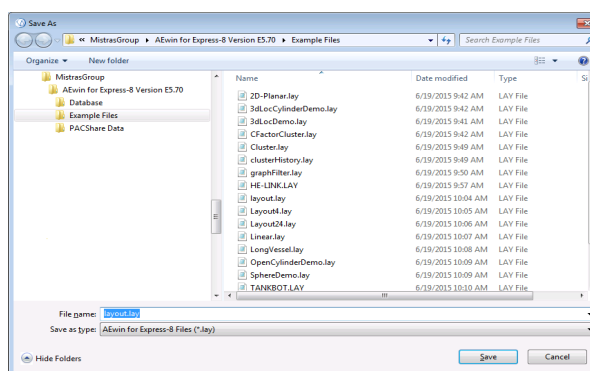


Figure 3-5. Save Layout...

3.3.1.4 Specify Data Folder...

The 'Specify Data Folder...' menu item opens a dialog box of the same name. It is used to select the default folders for the storage of Data and Layout files. These are the folders that are displayed by default in the various Open/Save Layout and Acquire/Replay dialog boxes. The 'Layout File Folder' is also where AEwin looks for the autoload 'Layout.lay' file on startup. By default, these are both set to your product folder found in C:\MISTRAS Group. Note that using the Open/Save Layout functions will overwrite the Layout File Folder field stored here, and saving or replaying a data file will overwrite the AE Data File Folder Field.

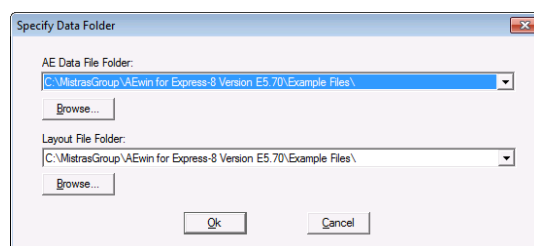


Figure 3-6. Specify Data Folder

3.3.1.5 Print Page(s)/Print Graph...

The 'Print Page(s)...' and 'Print Graph...' menu items are used to print a graph or multiple pages of graphs. 'Print Graph...' is used to send the currently selected graph to the printer, zoomed as much as possible while still retaining the aspect ratio. 'Print Page(s)...' is used to send a selection of pages to the printer one page at a time, each page zoomed as much as possible while still retaining the aspect ratio. You can select 'All pages' or a range of pages.

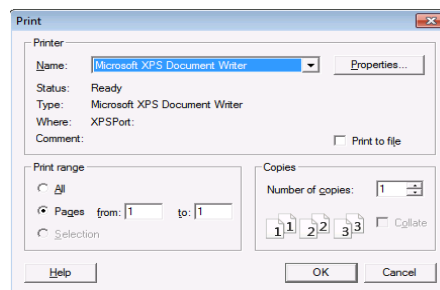


Figure 3-7. Print...

3.3.1.6 Export to JPG...

'Export to JPG...' is used to save pages as JPEG files in the same way that 'Print Page(s)...' sends pages to the printer. Here you enter a base filename and a destination folder. Each page will be saved as a separate JPEG file with its name formed from the combination of the base name and the page's title. e.g., If you had 2 pages named 'waveforms' and 'activity' and you set the base name to be 'TEST,' then the two files will be named 'TESTwaveform.JPG' and 'TESTactivity.JPG.'

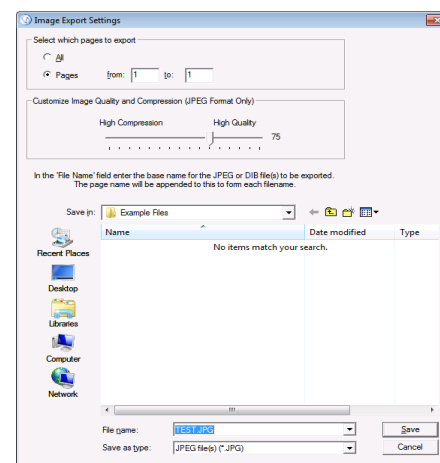


Figure 3-8. Export to JPG

3.3.1.7 Copy Screen to Clipboard

'Copy Screen to Clipboard' is used to save an image of the entire screen to the clipboard. This image can then be pasted in other applications such as word processors and image editors.

3.3.1.8 Exit

'Exit' is used to terminate the AWin program. If you are in the acquisition, you will be prompted to abort it; if you have made any unsaved changes to your current layout, AWin will prompt you to save your layout first, exit without saving or cancel.

3.3.2 Acquisition Setup Menu

3.3.2.1 Hardware...

This is the dialog box where all hardware settings are selected before data acquisition is started. There are a number of property pages that each deal with a different part of the hardware setup. Each is described below. Note that there are some differences in this dialog box between products. If a feature is described in this manual that does not appear in AWin, most likely, the hardware does not support it.

3.3.2.1.1 Standard Channel Setup

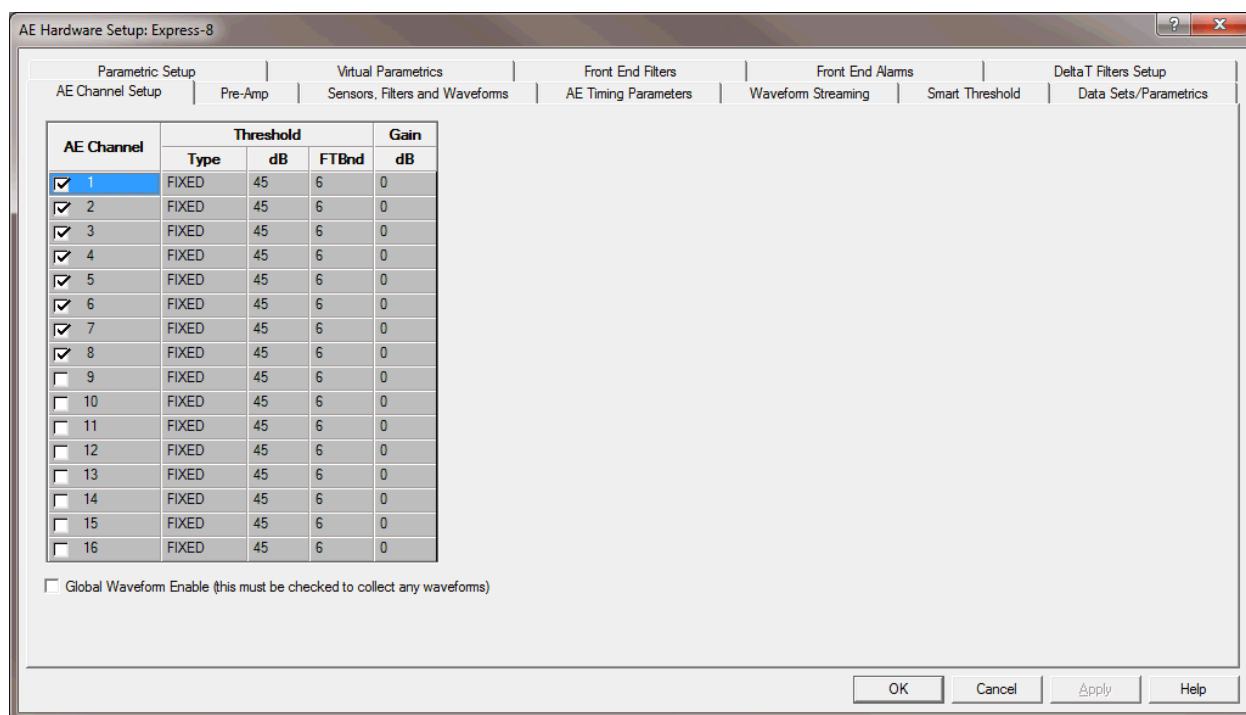


Figure 3-9. Standard Channel Setup

On the 'Standard Channel Setup' page, you can enable or disable each channel and edit their common settings.

AE Channel: You can enable or disable a channel for AE acquisition by clicking its checkbox. A check in the box indicates it is enabled.

Threshold, Type: Threshold is the prime variable that controls channel sensitivity. There are 3 possible choices for threshold type:

Fixed: Threshold remains at the level specified in the Threshold dB column throughout acquisition but can be manually changed during the acquisition by bringing up the hardware setup dialog box during acquisition.

Float: AEwin will automatically adjust the threshold level as a function of the ASL level throughout acquisition. It is useful under conditions of high, varying background noise. The value in the dB column represents the minimum threshold level. The value in the FTBnd column is the dB level above the current ASL level. Threshold adjustments are updated every 10 milliseconds. If AEwin does not show an FTBnd column, the band is fixed at 6dB and cannot be adjusted.

Example: If the threshold is set for 40dB and the FTBnd is set for 6 dB, a signal with an ASL of 50 would place the threshold at 56dB so only AE hits ≥ 56 dB would be captured. If the input ASL dropped to 25dB, the threshold would not be set to 31dB but would stay at 40dB, the value set in the threshold column.

Smart: Threshold level can change based on the current AE hit rate to maintain a specified rate of hits. Refer to the Smart Threshold Setup section below for more information. This threshold type is only available as an option to AEwin and is not standard. If it does not appear as a selection in the dialog box, it needs to be ordered (contact Mistras Customer Service).

Threshold, dB: The purpose varies based on the threshold type. Refer to the threshold type section above.

Threshold, FTBnd: Applicable only when the threshold type is 'Float.' Refer to the Float section above for more information. If the column does not appear, AEwin does not support this function for the acquisition system.

Gain dB: This is an additional gain located in the AE acquisition hardware. Not every product has on-board gain, and not all products support the same gain options. If the column does not appear, AEwin does not support this function for the acquisition system. 0dB is no gain; the other settings will lower the noise of the input signal at the expense of decreasing the dynamic range by the selected amount (typically 6 or 12 dB).

Global Waveform Enable:

If this box is left unchecked, AE waveforms will not be collected/displayed/stored during acquisition. While waveform filters selectively remove waveforms, this checkbox stops all waveforms. This function can be toggled from the tool bar at any time during acquisition. The current state of the waveform enable is shown by the state of the buttons on the toolbar. As shown below:



Figure 3-10. Waveforms Enabled



Figure 3-11. Waveforms Disabled

In Figure 3-10, the left button is depressed, indicating waveform collection is enabled. In this case, pressing the right button would disable the waveform collection. In Figure 3-11, the right button is depressed, indicating waveform collection is disabled. In this case, pressing the left button would enable waveform collection. Hover the mouse over the buttons to display text indicating the function of each button.

3.3.2.1.2 Channel Pre-Amp Setup

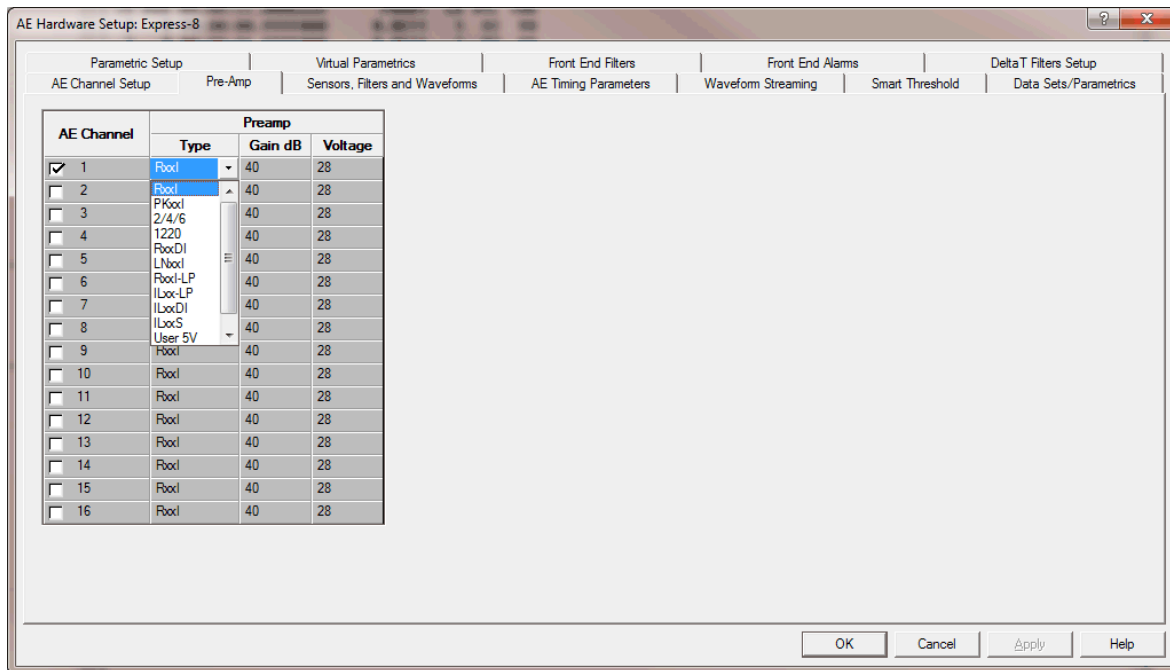


Figure 3-12. Channel Pre-Amp Setup

In order to produce correct AE measurements, AEwin must be configured with the correct pre-amp selection for each AE channel.

AE Channel: You can enable or disable a channel for AE acquisition by clicking its checkbox. A check in the box indicates it is enabled.

Preamp, Type: This drop-down list contains a selection of the most popular pre-amp types sold by Physical Acoustics Corp. The list includes both external pre-amps (such as the 2/4/6) and pre-amps that are integrated into sensors (such as the RxxI and PKxxI). In the event one of these preconfigured pre-amps is not used, User 5V or User 28V should be selected as appropriate. There are primarily two voltage levels of phantom power used to drive the pre-amps. Some of the AE data acquisition boards output 5V, some output 28V, and some can output either voltage based on a jumper or switch setting on the acquisition board itself. The table below summarizes the phantom power capabilities of each system:

AE System	Phantom Power Voltage	Selection Method
Express-8	5 or 28	Switch on board
PCI-2	5 or 28	Jumper on board
Sensor Higher Smart Monitor	5 or 28	Jumper on board
Sensor Highway III	5 or 28	Jumper on board
SAMOS (PCI-8)	28	Cannot change
DiSP (PCI-DSP4)	28	Cannot change
AE-USB	5	Cannot change
AE Wireless	5	Cannot change
AE Wireless -4	5	Cannot change

IMPORTANT: AE sensors can be irreparably damaged if they are exposed to a phantom power voltage higher than what they were designed for. For instance, connecting a 5V PK sensor to a 28V phantom power should be avoided. Always make sure the board is set to output the correct voltage. If it is inconvenient to visually check the switches or jumpers, a voltmeter can be used to verify the voltage before attaching a sensor (positive lead to the center pin of BNC, negative lead to the outside of the BNC, take care not to cross the voltmeter leads while they are attached to the BNC or the power supply could get damaged).

Preamp, Gain dB: User interaction with the selections in this column is dependent on the type of pre-amp selected. For some pre-amps, the value is fixed; for others, the value is selectable. If the value is selectable, it MUST be set to the value of the preamp being used. The selection does not change the actual gain of the pre-amp; it only tells AEwin what the gain value is. An incorrect setting will yield incorrect AE feature values.

Preamp, Voltage: This is a read-only field that shows the phantom power voltage expected for proper operation of the selected pre-amp type. This setting does not actually affect the voltage level output to the pre-amp. It is for informational purposes only.

3.3.2.1.3 Sensors, Filters, and Waveforms Setup

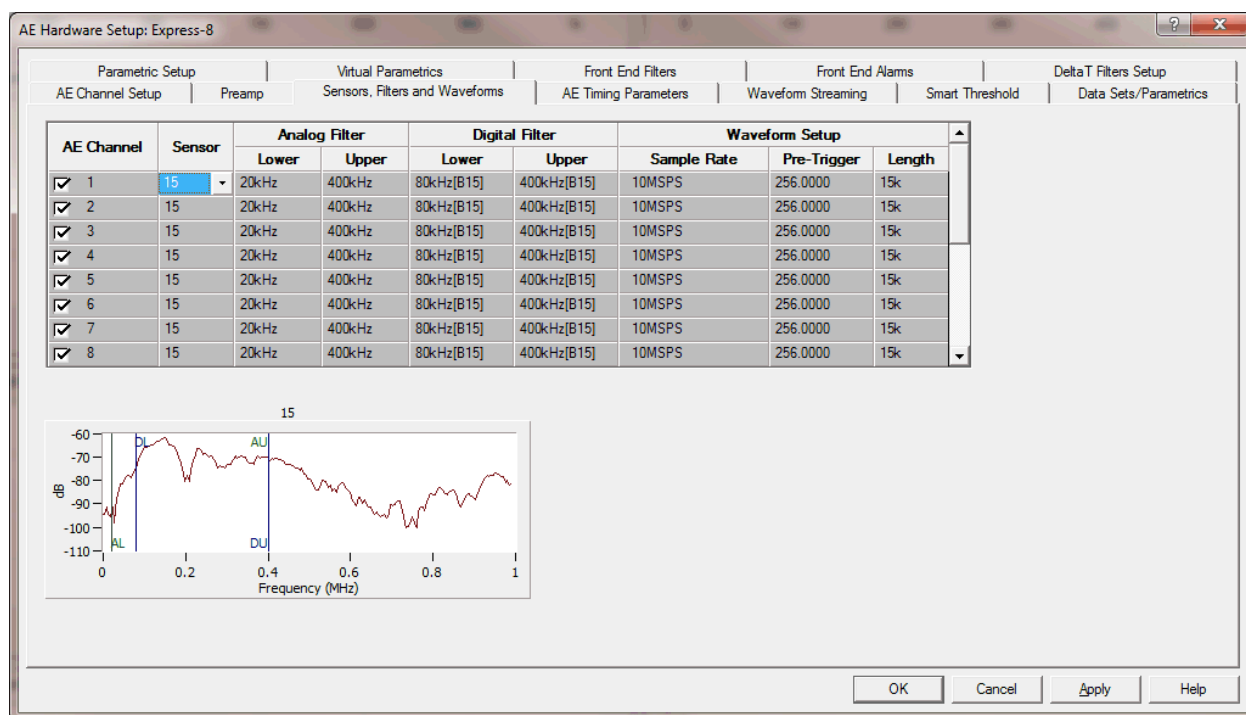


Figure 3-13. Sensors Filters and Waveforms Setup

When AEwin knows the type of sensor used, it can suggest optimal settings for signal filtering and waveform collection. These settings are suggested based on the resonant frequency of the sensor.

AE Channel: You can enable or disable a channel for AE acquisition by clicking its checkbox. A check in the box indicates it is enabled.

Sensor: This drop-down list contains a selection of the most popular sensor types sold by Physical Acoustics Corp. The sensor selection should match the sensor being used. The selection by itself doesn't actually impact data collection; it is primarily used to drive the recommended filter cutoff values that AEwin will suggest when prompted. After the selection is made, right-clicking on the sensor will display a pop-up menu where 'Set to suggested default for this sensor type' will appear. When this is clicked, AEwin will fill out the analog, digital, and sample rate fields with values that are optimal for the sensor.

Note that sensors will need to have appropriate settings in both this tab and the Pre-Amp tab. For example, if an R15I is used, the pre-amp should be RxxI, and the sensor should be 15. For a PK6I, the pre-amp should be PKxxI, and the sensor should be 6.

If your sensor does not appear in the list, either pick one with a similar frequency response (calibration curve) or pick anything and manually set the filter and sample rate values based on the sensor's calibration curve.

Note that the Mistras Group website (<http://mistrasgroup.com>) provides spec sheets for nearly all of the Physical Acoustics brand AE sensors.

Analog Filter: Selects the cutoff values for the analog filter. These should be set to limit the input signal to a range of frequencies of interest (within the sensors operating frequency range) and eliminate unwanted or uninteresting signals. The Upper value should be kept close to $1/10^{\text{th}}$ of the sample rate and should never be set to a value more than $1/2$ of the sample rate as that would introduce signal aliasing.

Digital Filter: Selects the cutoff values for the digital filter. If the column(s) does not appear, AEwin does not support this function for the acquisition system. These should be set to limit the input signal to a range of frequencies of interest (within the sensors operating frequency range) and eliminate unwanted or uninteresting signals. Digital filters can be high pass, low pass, band pass, or completely disabled. Not all products support all types of digital filters. A high pass filter will have 'None' in the Upper cutoff field, a low pass will have 'None' in the Lower cutoff field. A band pass will not have 'None' in any field, and [BX], where X is a number, will appear next to the cutoff frequency. When a band pass is selected in either column, the other column will automatically update to show the matching cutoff value of that filter. To disable a filter, select 'None' in both columns. For most products, the cutoff values of the filter scale with sample rate. So if a particular value cannot be selected at a particular sample rate, change the sample rate to see the new values that are available (this method does not work with PCI-2, which has the same filter cutoffs for each sample rate). Because the digital filters usually change with the sample rate, the digital filter values should be checked (and probably adjusted) whenever the sample rate is changed.

Waveform, Sample Rate: This is the rate at which the data acquisition board samples waveforms on a per-second basis. A sample rate of 1 MSPS (Mega Samples Per Second) means that one waveform sample is taken every μsec . A sample rate of 2 MSPS means that one waveform sample is taken every $\frac{1}{2} \mu\text{sec}$, etc. Refer to the analog and digital filter sections above for additional information about how sample rate and filters interact. In general, the higher the sample rate, the better, but it also depends on other factors, such as how much of the waveform is of interest. A 1024 point waveform at 10MSPS is only $102.4 \mu\text{sec}$ of the waveform, while at 1MSPS, it's 1.024 milliseconds (10x the time period). So a balance needs to be met between sample rate, waveform length, and the filter settings.

Waveform, Pre-Trigger: This value tells AEwin how long to record (specified in μsec .) before the trigger point (the point at which the threshold is exceeded). The minimum allowable pre-trigger value is zero. The maximum allowable pre-trigger value is calculated by dividing the hit length by the sample rate in MHz. If, for example, the hit length was 1 k (1 k = 1024) and the sample rate was 10 MHz, then the maximum allowable pre-trigger value would be $1024/10 = 102.4 \mu\text{sec}$. Hovering the mouse over the cell in the grid will produce a tool tip that shows how much of the collected waveform will be pre-trigger (as a percentage of the full waveform size).

Waveform, Length: Determines the maximum number of samples (as a multiple of 1024, i.e., 1k samples) collected for each waveform. The pre-trigger is included in this amount; it is not in addition to it. Hovering the mouse over the cell in the grid will produce a tool tip that shows how many microseconds of the waveform will be collected.

To the lower-left of the dialog box is a graph that shows a representative calibration curve for the selected sensor. Up to 4 vertical lines will be displayed; two green lines represent the lower and upper cutoff values of the selected analog filters, and two blue lines representing the lower and upper cutoff values of the selected digital filters. In addition to the color-coding, the letters AL and AU will be displayed to indicate which line is the analog lower and analog upper cutoff. The same applies to the digital filters (DL and DU). The graph will update when the sensor or filters are changed or when a different row in the grid is selected.

3.3.2.1.4 Timing Parameters

On the 'Timing Parameters' page, you can enable or disable each channel and edit Peak Definition Time (PDT), Hit Definition Time (HDT), Hit Lockout Time (HLT), and MAX Duration.

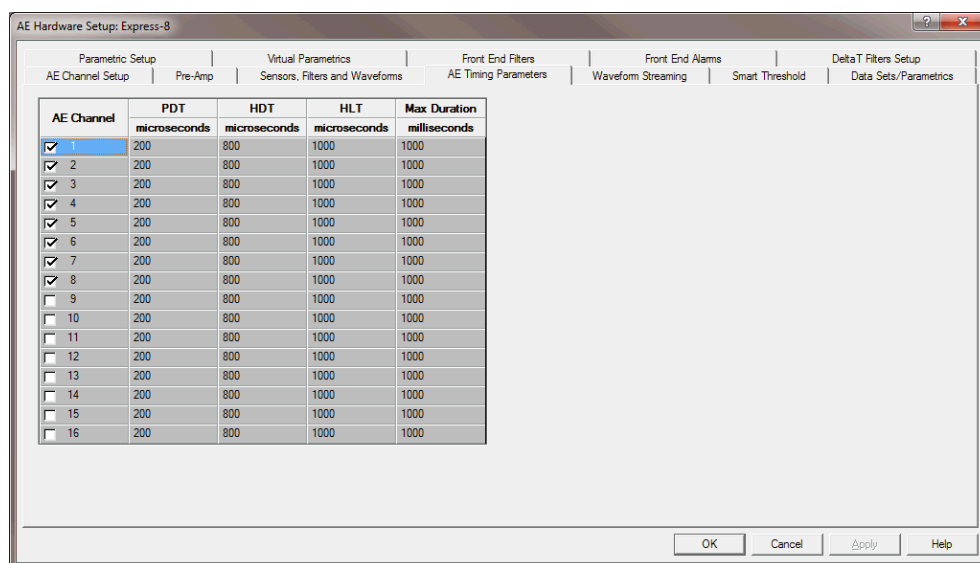


Figure 3-14. Timing Parameters Set up

PDT, HDT, HLT, and MAX Duration are timing parameters of the signal measurement process. In brief, a proper setting of the PDT ensures correct identification of the signal peak for rise time and peak amplitude measurements. Proper setting of the HDT ensures that each AE signal from the structure is reported as one and only one hit. With the proper setting of the HLT, spurious measurements during the signal decay are avoided, and data acquisition speed can be increased. Proper setting of Max Duration insures that there are no 'endless' hits under abnormal conditions, such as the threshold always being lower than the signal amplitude. A much more detailed explanation of these parameters and their proper application is described in Appendix I - The AE Timing Parameters.

3.3.2.1.5 Smart Threshold

The smart threshold algorithm automatically adjusts the detection threshold in order to obtain a target hit rate on a specified channel. This is not a standard feature of AEwin and is only available if the option is purchased.

This triggering logic should be used when the user wishes to limit data rates but still monitor the activity on each channel with maximum sensitivity. Smart threshold is especially useful for in-process acoustic emission monitoring, where the background noise level is high or changing, and the magnitude of acoustic activity varies significantly over the course of the test. It is also useful in applications involving waveform storage and advanced real-time processing. In these cases, it is desirable for the system to automatically adjust sensitivity to maintain a specified data collection rate. This avoids possible loss of data due to very high hit rates while providing continuous monitoring of channel data at a large range of amplitudes.

If Smart Threshold has been enabled in the AEwin license, a tab named 'Smart Threshold' will be available in the AE Hardware Setup dialog box (Acquisition Setup->AE Hardware..., or just press the F2 key).

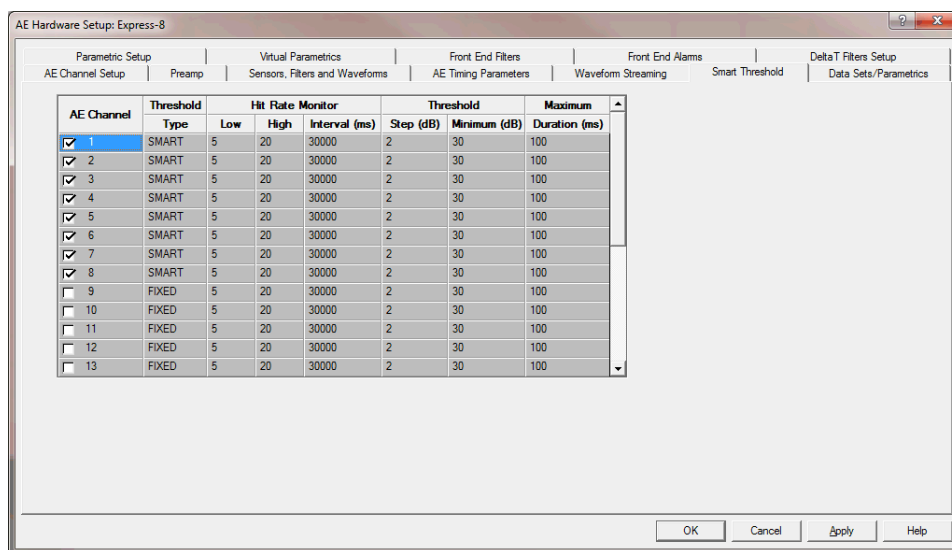


Figure 3-15. Smart Threshold Setup

AE Channel: You can enable or disable a channel for AE acquisition by clicking its checkbox. A check in the box indicates it is enabled. Note this is not an enable for smart threshold; selecting the threshold type as SMART enables smart threshold.

Threshold Type: Same as what is available in the AE Channel Setup tab. Available options are FIXED, FLOAT or SMART. The threshold type here (or in the AE Channel Setup tab) must be set to 'SMART' for smart threshold processing to occur.

Hits Low and High: Used along with the Interval setting to define an acceptable steady state hit rate. As long as the incoming hit rate is within this range, no threshold changes will occur. Values can range from 2 to 10,000 hits.

Interval (mSec): Used along with Hits Low and High settings to define an acceptable steady state hit rate. The value can be between 100 and 60,000 milliseconds.

Step (dB): Threshold change in dB steps taken when trying to reach an acceptable rate. The value can be between 1 and 10dB.

Minimum: Defines the lowest threshold value that can be reached even if the rate is too low. The value can be between 10 and 99dB.

STBnd: Normally should be kept at zero. A value other than zero initiates other processing that is described below. This processing was added for a specific application and is not necessary for most applications. It is not available in all products, so if this column is missing, the product does not support it.

Duration (ms): Detection of hits with durations longer than this number causes the threshold to be raised. Value can be between 1 and 1000 milliseconds. Note that the AE Timing Parameters tab of the AE Hardware setup dialog box contains a column for setting the maximum duration of a hit. If the timing parameter is less than the duration for Smart Threshold, the Smart threshold setting is ignored (because it can never reach the smart threshold duration).

If a channel is getting a data rate that is higher than that defined by hits high in an interval, the threshold is immediately raised, and a new interval is started. If the data rate is lower than that defined by hits low and the interval, the threshold is lowered only after the interval has expired. This allows the system to quickly reject noise if high amplitude activity begins.

A 2dB filter is implemented such that hits with amplitudes less than 2dB above the current threshold level are used in the rate calculations but are not sent to AEwin.

The system always keeps track of the current Smart Threshold level during a test so that the threshold for a subsequent test starts at the last level attained.

The Smart Threshold parameters can be set up at any time but are only active when the threshold type is set to SMART. The settings are only saved in the layout file if the threshold type is set to SMART.

Smart Threshold Processing:

Most products do not support STBnd, so it will have a value of 0 if the user cannot set a value.

Per Hit Processing (occurs each time an AE hit is collected):

The Hits Low is not evaluated for each hit; it only applies when the interval is expired, and the total number of hits in the interval is known.

If the current hit amplitude is $\geq$ the smart threshold evaluation level, increment the number of hits counted in the interval, else ignore the hit as it doesn't contribute to the smart threshold algorithm.

If the new count puts the count of hits in the interval over the Hits High value or if the duration of the hit $\geq$ the maximum duration, raise the hardware threshold by the step value and start a new interval.

If STBnd is < 0 , the new smart threshold evaluation level is set to the new hardware threshold - STBnd (subtracting a negative number yields an evaluation level $>$ the hardware threshold).

If STBnd is ≥ 0 , the new smart threshold evaluation level is set to the new hardware threshold.

If STBnd is > 0 , set a filter based on the hardware threshold + STBnd.

If STBnd is ≤ 0 , leave whatever filter was already in place.

Note that the filter is processed after the smart threshold processing. Hits that have contributed to evaluating the hit rate may eventually be filtered out by this smart threshold filter and therefore not sent to AEwin.

End of Interval Processing:

If the count of hits in the (now expired) interval is < Low Hits value, lower the hardware threshold by the step value (and do not go below the minimum threshold).

If STBnd is ≤ 0 , set a filter based on the previous hardware threshold value (the threshold before the step value was subtracted) plus 2dB.

If STBnd is > 0 , set a filter based on the new hardware threshold value (the hardware threshold after the step value was subtracted) plus the value of STBnd.

If STBnd is < 0 , set smart threshold evaluation level to the new hardware threshold value - STBnd.

If STBnd is > 0 , set smart threshold evaluation level to the new hardware threshold value.

3.3.2.1.6 Data Sets/Parametrics

AEwin features user-selectable data sets. This means that if you do not plan to use all the measurable AE parameters, you can leave some out and enjoy the compensating advantages of higher data throughput speed and more economical use of storage space. Selecting Data Sets/Parametrics from the Hardware Setup dialog box displays the Data Sets/Parametrics page that follows. Note that the above figure is specific to the DiSP product. Your product may have more/fewer options available.

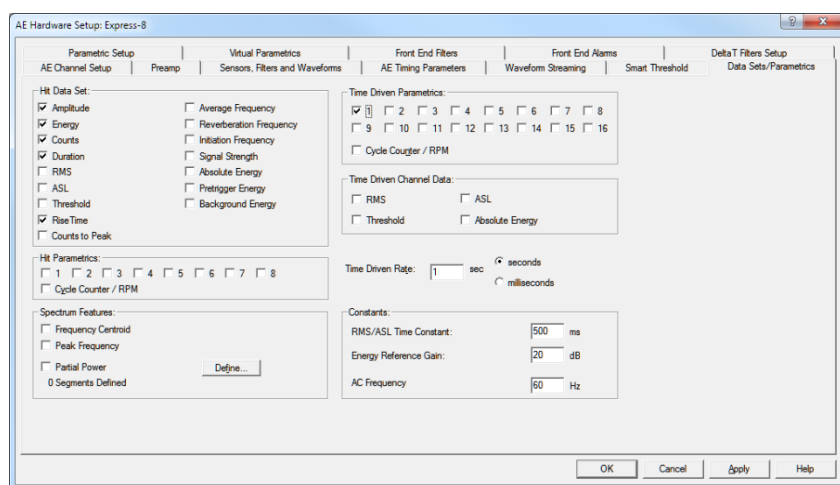


Figure 3-16. Data Sets/Parametrics

Hit Data Set

This box allows you to select which measured signal features are to be included in the reporting of each AE hit. The features to be included are so marked with a check in the checkbox next to the feature. Features can be included/excluded from the data set by selecting or deselecting the checkbox. "Time of test" is not shown because it is always included in the data set. A short description of each hit feature in the Hit Data Set is as follows:

- 1) **Amplitude:** Highest voltage in the AE waveform, expressed on the dBAE amplitude scale.
- 2) **Energy:** Time integral of the absolute signal voltage. The reported magnitude depends on the value selected for Energy Reference Gain. Proportional to Signal Strength.
- 3) **Counts:** Number of times the signal crosses the detection threshold.
- 4) **Duration:** Time from first to last threshold crossing (μ s).
- 5) **RMS:** Root mean square voltage during a period of time-based on a software programmable time constant, referred to the input to the signal processing board.
- 6) **ASL:** RMS, converted to the dBAE scale ($0\text{dBAE} = 1\mu\text{V}$ at the sensor, before any amplification).
- 7) **Threshold:** Detection threshold on the dBAE scale. Not particularly useful for Fixed Threshold because the value does not change during acquisition. It can be useful for Floating and Smart Threshold modes.
- 8) **Rise Time:** Time from first threshold crossing to highest voltage point on the waveform (μ s).

- 9) **Counts to Peak:** Number of threshold crossings from first to highest voltage point on the waveform.
- 10) **Average Frequency:** Counts divided by Duration, divided by 1000 (thus, kHz). Note that this is not a spectral-domain calculation but a calculation from time-domain features.
- 11) **Reverberation Frequency:** (Counts - Counts to Peak) divided by (Duration - Risetime).
- 12) **Initiation Frequency:** Counts to Peak divided by Risetime.
- 13) **Signal Strength:** Time integral of the absolute signal voltage, expressed in pVs (picovolt-seconds) referenced to the sensor before any amplification. Proportional to Energy.
- 14) **Absolute Energy:** Time integral of the square of the signal voltage at the sensor before any amplification, divided by a 10kΩ impedance and expressed in aJ (attojoules).
- 15) **Pre-Trigger Energy:** Time integral of the square of the signal voltage in Pre-Trigger portion of the signal at sensor before amplification, divided by a 10kΩ impedance and expressed in aJ (attojoules). **Note:** The pre-trigger energy (PTE) shows the background energy oscillation effect. (The energy in the pre-trigger portion of the waveform window is a good way to observe the fluctuations in the background energy.)
- 16) **Background Energy:** Minimum of the average energy over a preset length of time (100 sample points) in the waveform window.

Background energy (BE) is measured using a 100-sample rolling window (L in the image below) in the waveform window (A i in the image below). The rolling window length of time changes with varying sampling rate. Then, the minimum of the average energy over the preset length of time in the waveform length is measured and expressed as background energy. For example, a 100-μs preset length of time corresponds to a sampling rate of 1MSPS, which could be calculated by dividing 100 by sampling rate here 10^6 . The 100- μs window was stepped through each waveform window recorded during the test, and the energy was computed. Then, the minimum was found for each waveform.

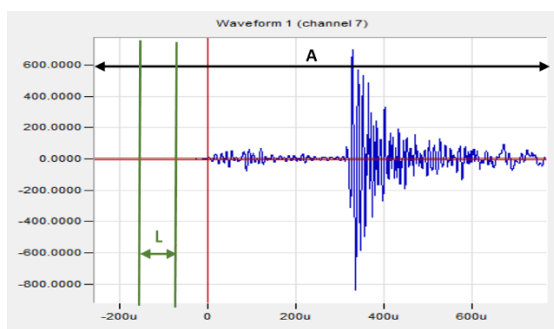


Figure 3-17. Pre-Trigger and Background Energy Measurement

Hit Parametrics

This box allows you to select which parametric inputs (e.g., external load signals) are to be included in the description of each hit. The number of hit parametrics available varies by product. Most products support 2, Express-8 supports 8. The Cycle Counter, which uses one of the parametrics as an input, can also be selected.

Spectrum Features

This box is available if you have the waveform option installed. Spectrum Features are derived from the power spectrum of the waveform associated with a hit:

- 1) Frequency centroid - the center of mass of the power spectrum graph.
- 2) Peak frequency - the point where the power spectrum is greatest.
- 3) Partial Powers - see 'Partial Power Features' below.

Partial Power – Partial Power is a feature that is derived from the power spectrum of the waveform associated with a hit. It is a percentage and is calculated by summing the power spectrum in a specified range of frequencies (up to four of these ranges can be specified), dividing it by the total power in another range of frequencies, and multiplying the result by 100.

To use partial power, you must

- 1) Check the partial power checkbox in the Spectrum Features group box.
- 2) Define at least one partial power (spectrum) segment. To do this, press 'Define...'. This brings up the Spectrum Segment Setup dialog box described below.

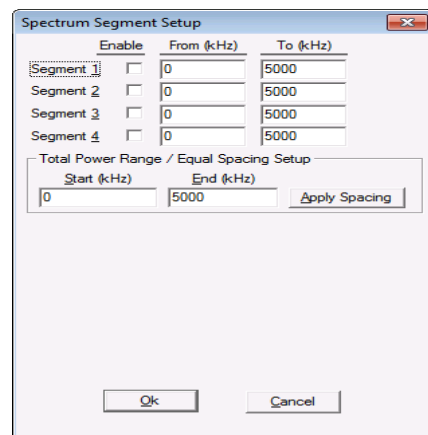


Figure 3-18. Spectrum Segment Set up

Spectrum Segment Setup is used to define the partial power features. For each of the 4 definable segments, the dialog box has a checkbox to enable the segment, an edit box for defining the start frequency, and an edit box for defining the ending frequency.

The group box titled 'Total Power Range/Equal Spacing Setup' contains two edit boxes. These edit boxes serve a dual purpose. The main reason for them is to define a range of frequencies over which the power spectrum will be summed.

The result of the summation is used in the calculation of total power for the partial powers calculation. Their other purpose is to allow the user to spread the frequency range evenly among the enabled segments. The spacing will occur when the user presses the 'Apply Spacing' button, which is located to the right of the edit boxes.

CAUTION: If the edit boxes are used to set up equal spacing, always make sure the edit boxes contain the Total Power range values when the dialog box 'OK' button is pressed. Otherwise, the partial powers will be based on the equal spacing range.

The Spectrum Segments are global for all active channels; if the sample rate and/or hit length are not the same for all active channels, a warning will be displayed to alert the user.

The following only applies to AEwin for DiSP:

The dialog box also displays information based on the sample rate and hit the length of the first active AE channel:

- 1) Effective Sample Rate – The sample rate of the decimated waveform (the FFT calculation is always for 1k of samples). For a 1k hit length, the effective sample rate is the same as the hardware sample rate. For a 2k hit length, it would be half the hardware sample rate. Etc.
- 2) Effective Maximum Frequency – This is one-half that of the Effective Sample Rate and is the maximum frequency that can be defined for any segment.
- 3) Effective Resolution - This is the Effective Maximum Frequency / 512 and is equivalent to the frequency range of a single FFT bin.

The system does not automatically set the input filters based on the sample rate to reduce aliasing, so the dialog box will remind the user to set the input filter to a value closest to the Effective Maximum Frequency.

Time-Driven Parametrics and Time-Driven Channel Data

Additional to the hit descriptions, AEwin allows the recording of RMS, ASL, Threshold, Absolute Energy, Cycle Counter, and Parametric inputs at regular time intervals independent of AE activity. These are called Time-Driven messages. This box allows you to select a set of features for them.

Time-Driven Rate – This controls the period of the readings and the data to be included in the records. Time-driven data is useful for leak monitoring and for recording changes in load and background noise levels. Also, time-driven data can be used for updating hit data graphs involving parametric values or time, even when no-hits are coming in. The interval you set must lie in the range 10 msec to 1,800 sec and must be a multiple of 10 msec. Typical values are in the range 1000-60,000 msec (1–60 seconds). For frequent parametric value updates, you may like to choose a short interval, but this leads to lengthy disk files and data listings, so it is not necessarily the best choice. When the time-driven data set is used to update graphs, it is a good idea to set the rate less than or equal to the ‘Seconds Between Updates’ field selected in the Display Mode dialog box.

Constants

RMS/ASL Time Constant– The time constant can be set from 10ms to 1000ms in steps of 10ms. This controls the response of the RMS/ASL value to a change in the input level to help smooth out the response to quick transients or bursts of activity. For example, if the AE input had a constant reading of 60dB and then immediately changed to 80dB, it would take the time specified by the time constant for the ASL to reach 80dB. Let’s say the time-driven interval was 100ms, and the time constant was 500ms. The 5 ASL readings would be something like 60, 65, 70, 75, 80 for the 500ms after the input changed to 80dB. Another example would be that a 100ms burst of 80dB occurred before returning to 60dB. In that case, the ASL reading would reach about 1/5 (~64dB) before returning to 60dB by the 5th (500ms) reading.

Energy Reference Gain – In some early analog PAC instruments, such as SPARTAN, Energy is a gain-dependent quantity. However, the newer digital instruments don’t have an equivalent gain setting, so the Energy in these systems was designed to give a reading as if 20dB of gain were applied. The Energy Reference Gain allows you to match the value of Energy reported by the system to that of a SPARTAN-2000 or other PAC instrument which has programmable gain. The value of the Energy Reference Gain has a valid range of 0 to 60 dB. For almost all situations, the value can be left at 20dB.

AC Frequency – Used with the TAFI option (only present if the option is enabled). The TAFI section of this manual has more information about this parameter.

3.3.2.1.7 Parametric Setup

The Parametric Setup page allows you to scale the voltages measured at the parametric input such that the system displays the corresponding load or pressure values. This page also allows you to tell the system what units (kN, psi) you want to be displayed on your graph axis and printouts, and it allows you to select from four available filters on gain.

The Parametric Scaling process used by AEwin is: $\text{Displayed Value} = (\text{Measured Voltage} \times \text{Multiplier}) + \text{Offset}$

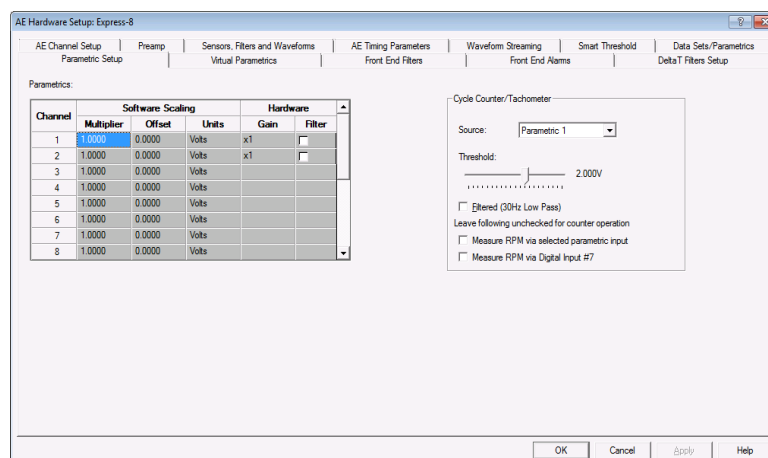


Figure 3-19. Parametric Set up

The default values are a multiplier of 1.00 and an offset of 0.00 with units of "Volts" so that the system will display the raw voltage delivered to the Parametric Input socket on the rear panel.

To scale the output of a linear load-measuring transducer, the easiest way to find the correct Multiplier and Offset is to go into a simulated test and turn on the Line Listing Display (F7). From the display, read the parametric voltages for two widely separated loads. Suppose these readings are V1 at load L1 and V2 at load L2. The correct values for Multiplier and Offset are then:

$$\text{Multiplier} = (L1 - L2) / (V1 - V2)$$

$$\text{Offset} = (V1 * L2 - V2 * L1) / (V1 - V2) = L1 - \text{Multiplier} * V1$$

Once you click OK to exit this dialog (or press Apply to commit the values without exiting the dialog), the scaled values of the parametric input will appear on all graph displays and data listings. Minimum and maximum values for graph axes should be specified in the scaled values, and the entered unit will appear on the graph (e.g., 0–1000 psi). The raw data on the disk will contain the original voltage readings, and the scaling will be done again each time the file is replayed.

One or more parametrics (depending on the hardware available) will have settings for gain and a low pass filter. Typical available gain settings are x1, x10, x100 and x1000. The low pass filter has a cutoff of 30Hz and should be enabled if the parametric input signal is noisy.

The **Cycle Counter/Tachometer** section allows you to control the source, threshold, and alternate function of the cycle counter; on some products^(*), the cycle counter can function as a tachometer. If the function is not available, the two RPM checkboxes will not be shown.

- The Source can be selected as parametric 1-2 (on most data acquisition boards).
- The Threshold slide control allows you to set the threshold for the cycle counter. The cycle counter counts every time a signal crosses the threshold. The threshold can be set within a -10.00 volts to +10.00 volts range. This voltage should be set based on the load signal coming into the system. It should be set low enough to assure constant, accurate triggers. Note that the left and right arrow keys on a keyboard can be used as a fine adjustment on the threshold level.
- The Filtered checkbox allows you to enable an independent 30 Hz digital low pass filter on the cycle counter input signal.

As mentioned previously, it may be possible to use the cycle counter input as a tachometer. Using the cycle counter circuitry in this manner precludes using it as a cycle counter. They are mutually exclusive functions. It may also be possible to use digital input #7 as the source instead of the analog parametric. If a digital input is used, none of the cycle counter controls apply to it. It is here as a matter of convenience to keep tachometer-related settings in one place. Refer to the board connector section of the manual to determine where digital input #7 is located.

The value of the RPM (from either hit or time-driven data or both) can be graphed by selecting RPM as an axis quantity. If the data acquisition boards do not support RPM, then RPM will not show up as a parameter for graphing.

^(*) PCI-2 systems require an FPGA of version 2.2 or higher on the board for any tachometer functions to be available. All data acquisition boards must have the same revision FPGA (2.2 or higher) in order for hit data set RPM measurements to be collected in multi-board systems. For time-driven RPM measurements, only the first board needs to have the RPM-enabled FPGA.

Express-8 systems also support tachometer readings.

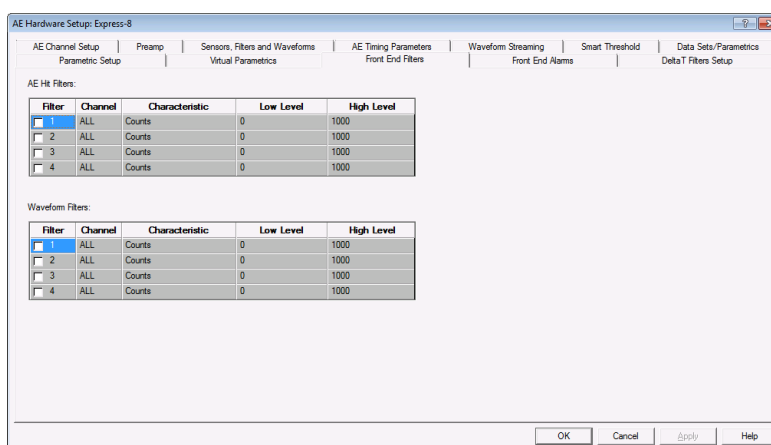


Figure 3-20. Front End Filters

3.3.2.1.8 Front End Filters

These filters apply to accept/reject criteria to the hit descriptions as the data is processed in the data acquisition board, so that rejected hits or waves do not even reach the main processor board, thereby improving overall system performance.

There are two kinds of front-end filters:

AE Hit Filters reject hits and any waveforms associated with them based on the criteria you set. **Waveform filters** work the same way but reject only the waveforms, allowing the hits to remain. Waveforms can take up a significant amount of storage space as compared to the hit data, so uninteresting filtering waveforms is recommended whenever possible.

The setup is identical for both filter types. There are four filters available at the present time. All four filters will always be displayed. To activate a particular filter, simply check the 'Filter' checkbox associated with the filter you want to use.

Use the Channel drop-down list box to select which channel (or all channels) to filter on.

Use the Characteristic drop-down list box to select an AE Characteristic to filter on. Each hit will be individually screened to see whether the measured value of the specified characteristic falls in the acceptance range from LOW LEVEL to HIGH LEVEL, inclusive. For the message(s) (hit and/or wave) to be accepted, the hit must fall in the acceptance range for all active filters affecting that channel. Change the filter LOW and HIGH levels by entering new values in the text boxes. If a hit is not within the acceptance range, the message(s) will be eliminated from the front-end signal processing.

3.3.2.1.9 Front End Alarms

There are four alarms defined on the alarms page. This page is almost identical to the front-end filter page, with the exception that the user may select the type of alarm to be Single Channel Single Hit or Single Channel Cumulative. When Single Channel Single Hit is selected, every hit that is received (on the selected channel) is tested against the alarm criteria. When Single Channel Cumulative is selected, the alarm criteria are applied to the current cumulative (summed) value of the selected parameter (counts, amplitude...) on the selected channel. An alarm is generated if one or more of the alarm criteria is exceeded. Upon alarm, the PC computer speaker will sound, and an alarm output on the I/O connector will indicate an alarm. In addition, an alarm popup box will appear on the screen indicating if the alarm is a warning or a trigger, which alarm was exceeded, and the value that triggered the alarm. To reset the alarm click the 'Clear All Alarms' button on the alarm popup box. See section 3.3.3.6 entitled 'Display Last Alarm' for more information.

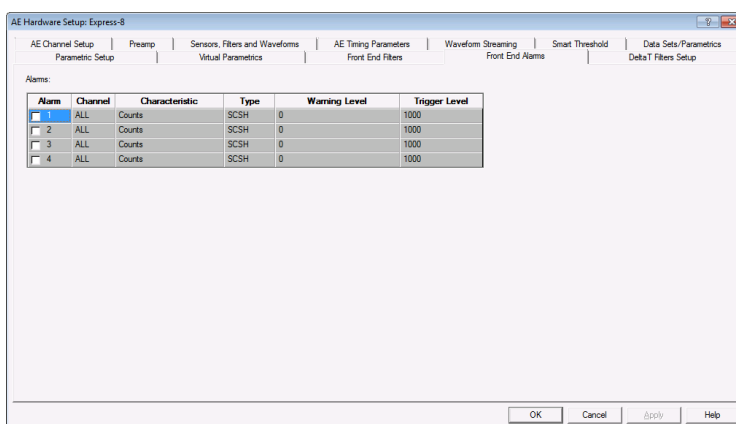


Figure 3-21. Front End Alarms

3.3.2.1.10 DeltaT Front End Filters (MISTRAS only)

MISTRAS Delta-T Front End Filters are used to eliminate “on board” AE hits that are part of the same event but outside the acceptance range selected in the DeltaT Filters Setup page.

This type of filter is very useful for cases where linear location is used. It will successfully filter hits that come from the outer areas of the sensors and beyond. Delta-T Front end Filters are very useful in 2 channel MISTRAS systems.

In cases where more than two channels are needed (such as planar location), a front-end filter cannot be used since the hit information is not available from other MISTRAS boards. For this situation, a “high-end” filter is needed, and Graph Filters can be used for this purpose. See the section on Graph Filters for more information.

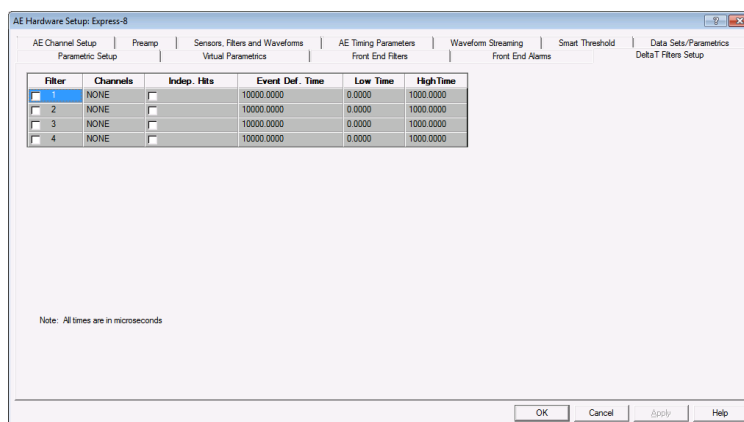


Figure 3-22. Delta-T Filters

The DeltaT Filters page is very similar to the Front-End Filters page discussed previously. Four different filters are available to allow the programming of each board differently. Alternatively, the first filter can be chosen as an “ALL,” which will have all boards filter using the same set-up parameters.

Each menu item and its function are discussed below.

- Filter:** This toggles between On and Off just like the Front End Filter page does.
- Channel:** This selects the AE channel numbers for the DeltaT filter. Because the DeltaT Filters operate on a per board basis, *pairs* of channels must be selected (e.g., 1-2, 3-4...), or ALL channels may be selected.
- Indep. Hits:** Allows independent hits (those hits not associated with an Event) to be accepted or rejected for transmission to the high-end, at the discretion of the user.
- Event Def. Time:** This is the maximum event time (in μsec) required in order for the AEDSP-32/16 to identify paired hits on the board as events. This should be set at the maximum time expected for an AE signal to propagate from the first to the second sensor plus 10% over range.
- Low Time:** This is the lower limit of acceptable Delta-T values in μsec . Below this is not an acceptable difference, and the hits will be discarded. Negative numbers can be used. For delta-T filters, a hit occurring closer to the even channel is assigned a negative delta-T.
- High Time:** This is the higher limit of acceptable Delta-T values in μsec . Above this is not an acceptable difference, and the hits will be discarded. As above, negative numbers indicate that the hit occurred closer to the even channel

The information setup in the DeltaT Filter page is downloaded into each specified AEDSP board with the rest of the set-up information. Upon acquisition, each AEDSP board has software event detection and filter routines implemented as follows:

As hits are received from both channels, the board has an event detection queue whereby it checks hit pairs (hits on opposite channels) against the Event Definition Time to determine if it is an event. To provide for asymmetric acceptance ranges between two sensors, the delta-T is always calculated from the lower numbered sensor to the higher numbered sensor. Hits whose delta-Ts are longer than the Event Definition time are considered independent hits. In this case, the first hit will pass to the high end as an independent hit.

In those cases where an event is detected (hits that have a delta-T within the Event Definition Time), its Delta-T is checked against the high and low levels for that board. If the Delta-T is outside of the range set in the “Low Level” and “High Level” fields (less than the Low Level and higher than the High Level), then both hits will be rejected, and neither hit nor waveform data will be passed to the high end. If the Delta-T is within the “Low Level” and “High Level” values, then the hits (and waveforms) are allowed to pass to the high end.

Delta-T filters are activated independently of any location processing. This allows the use of Delta-T filters in tests with extremely high data rates, whereby the data is stored and analyzed later with the slower location algorithm.

3.3.2.1.11 Waveform Streaming

Some AE boards or systems (PCI2, Express-8) are capable of streaming waveform data continuously into physical memory on the PC. The onboard DMA controller has used read samples from each channel's FIFO and transfer them into a circular buffer in the computer's memory. When a stream to disk event is detected (this being a threshold crossing, an external digital trigger, a manual trigger, or a set periodic interval), the waveform is pulled out of this buffer and written to the output data file. This data file (with a WFS extension) is created separately from the normal acquisition data file (with a DTA extension) used to store collected AE data. Once the output file is created, it can be viewed and analyzed with the WaveViewer program supplied on the installation disk or NOESIS™, an advanced data analysis pattern recognitions software package that is sold separately from AEwin.

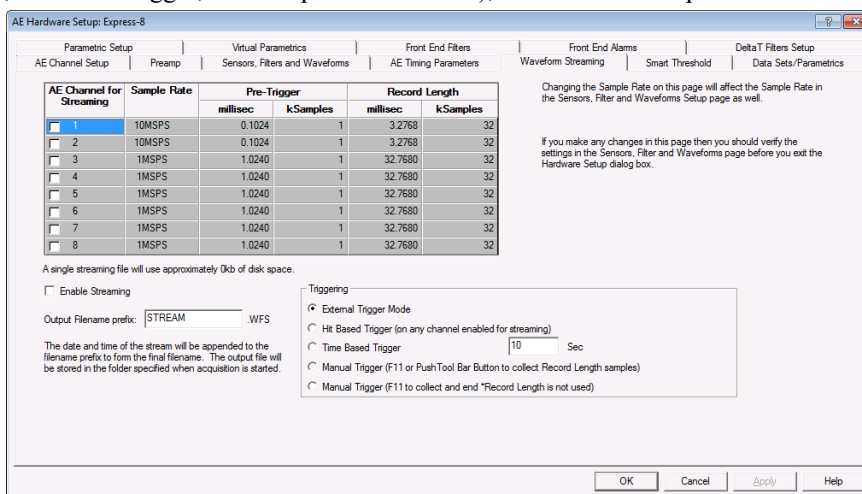


Figure 3-23. Waveform Streaming Set up

3.3.2.1.11.1 Per Channel Waveform Streaming Settings

AE Channel for Streaming:

A checked box indicates that a channel is enabled for streaming. When streaming is triggered, data from all selected channels will be stored in the file. Click the box to toggle this setting. A channel must also be enabled for AE data collection to be able to stream. If it is undesirable to have AE data from a channel that is streaming, go to the AE Channel Setup page and set the channel threshold type to 'Fixed' and raise the threshold level to its highest value (typically 99 dB).

Sample Rate:

Selects the sample rate of the streamed data. This setting is the same as the one on the Sensors, Filters, and Waveforms page. If it is changed here, it will automatically be changed in the Sensors, Filters, and Waveforms page (and vice versa) because AE and streamed waveforms are always collected at the same sample rate. Channels on the same board must have the same sampling rate, but different rates are allowed between boards.

The PCI bus-based systems (PCI2, SHSM) cannot transfer more than 10MSPS total. This means 1 channel can be enabled at 10MSPS or 2 channels at 5MSPS... PCIe bus-based systems (Express-8) can transfer much more data. Each PCIe8 board can support up to 50MSPS total among all the streaming channels on a board. Also, this upper bound is per board, so multiple boards can stream a total of 50MSPS each. However, at these rates, other system component throughput rates may limit the practical amount that can be streamed. Hard disk performance is one such factor. As a general rule, always try to keep the total MSPS for all enabled channels as low as is possible. Changing the sample rate also changes the time value for pre-trigger and record length.

Pre-Trigger:

This is the amount of data that occurred prior to the actual trigger of the streaming that should be saved to the file. It cannot exceed either 32k samples or 4 seconds, whichever is smaller. The pre-trigger amount cannot exceed the record length because the record length includes the amount of the pre-trigger. The value can be entered as either kilo-samples or milliseconds. When one value is entered into its field, the program will automatically calculate the other value. The program always stores the value as the number of kilo-samples; therefore, it may not be possible to obtain an exact value in milliseconds.

Record Length:

This is the total amount of data that will be streamed once the stream is triggered. The value can be entered as either kilo-samples or milliseconds. When one value is entered into its field, the program will automatically calculate the other value. The program always stores the value as the number of kilo-samples; therefore, it may not be possible to obtain an exact value in milliseconds.

3.3.2.1.11.2 Waveform Streaming Trigger Modes

Each time a valid trigger is detected, a new file is opened for streaming. Any trigger condition that is met while a stream is in the process is considered to be invalid and is ignored.

External Trigger:

In this mode, the streaming is triggered when an external pulse is applied to digital input #2 of the board that contains channel 1. The connector location of digital input #2 varies from system to system, so please refer to the section of the manual that describes connector pin functions to find the input's location and logic voltage levels. The pulse should come from a normally low input that goes to a high state for at least 1 microsecond before returning to low. Theoretically, it is possible to trigger this input with a pulse that is only 50 nanoseconds wide. Precautions should be taken to make sure that the input signal is clean or false triggers could occur. The signal must return to a low state for at least 10 microseconds before returning high to trigger another stream. If a stream is in the process when the trigger input goes from low to high, the transition will be ignored and will not affect the current operation. The input must go from a low state to a high state while streaming is idle in order to trigger any streaming.

Hit Based Trigger:

In this mode, the streaming is triggered whenever an AE hit is detected on any of the channels that are enabled for streaming. Additional hits that occur while streaming is in the process do not affect the triggering.

Time-Based Triggering:

In this mode, streaming can be set to occur on a regular periodic basis. Make sure that the record length is less than the specified interval. Otherwise, for example, if the stream is to be 1.5 seconds long and the interval is set to every 1 second, the interval from 1.5 to 2.0 seconds will not be stored in a file. The first interval begins when the acquisition is started.

Manual Trigger:

There are two types of manual triggers. Either will start when either the F11 key or the 'Trigger a Waveform Stream' acquisition toolbar button is pressed.

In the first case, the stream will terminate when the number of samples specified in the record length field is reached.

In the second case, either the F11 key or the 'Trigger a Waveform Stream' acquisition toolbar button needs to be pressed to terminate the stream.



Figure 3-24. Trigger a Waveform Stream Button

3.3.2.1.11.3 General Waveform Streaming Settings**Enable Streaming:**

This must be checked in order for streaming to be available during acquisition. This is a quick way to enable and disable the feature while still preserving all of the settings.

Output Filename Prefix:

This specifies the first part of the name of the output data file. It can be changed to any valid Windows filename characters. It is recommended that this be used to differentiate the data from different tests or experiments. The second part consists of the date and time that the stream was triggered.

Filename example: **STREAM20040804-090744.WFS**
PREFIXYYYYMMDD-HHMMSS.WFS

Here, STREAM is the default prefix. Following the prefix is the time and date of the stream. Four digits of the year (Y), two digits of month (M), two digits of the day (D), a hyphen, two digits hour (H, 0-23), two digits of minutes (M), and two digits of seconds (S). In the case of the example, the stream was started on Aug 4, 2004, at 9:07:44am.

3.3.2.1.11.4 Suggested Procedure for Configuring Waveform Streaming

- Select the channels for streaming from the first column by clicking on the check box.
- Select the sampling rate for each channel. Channels located on the same AE acquisition board must have the same sample rate. The program will enforce this.
- Select the pre-trigger. This can be specified in milliseconds or K samples (1K samples = 1024 samples). It is easier to specify milliseconds, but if the sampling rate is subsequently changed, the millisecond's value will be recalculated and displayed.
- Select the recording length in milliseconds or K samples (1K samples = 1024 samples). There is no set limit on recording length other than the available hard disk space. For file lengths greater than 1.5Gbytes, at least 256 Mbytes should be installed. As with pre-trigger, subsequent changes to the sampling rate will force the millisecond value to be recalculated and displayed.
- Select the Triggering Mode
- Specify the Output Filename Prefix
- Click the Enable Streaming Check Box. This box must be checked for streaming to be enabled.
- Press the 'Apply' button to verify the settings, fix any errors that show up.
- Go back to the Standards Channel Setup and verify that the waveform settings are acceptable.



Figure 3-25. Status bar indicating that streaming is enabled

After configuring all options, the system can be put into acquiring mode to begin data collection. If waveform streaming is enabled, the status bar at the bottom of the AEwin window will display “Waveform Streaming Enabled.”

When a waveform save is triggered, the text in the status bar will indicate it.

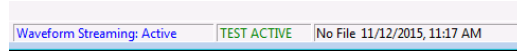


Figure 3-26. Status bar indicating that streaming is in progress

When the waveform save is complete, the text in the status bar indicates it.

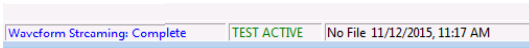


Figure 3-27. Status bar indicating that streaming is complete

This message will persist for 4 seconds before reverting to the “Waveform Streaming Enabled” text.

The file is placed in the last folder that was used to save a DTA file. If an acquisition is done and Autodump is enabled (producing a DTA file), the streaming data file will be created in the same folder. If Autodump is off, the streaming data will be placed in the last folder that was used for AE data. The streaming file is always created, and there is no way to disable it. The destination folder for the streamed data cannot be specified without also saving AE data.

3.3.2.1.11.5 Viewing a Streamed File

Once a WFS file is created, it can be viewed using the WaveViewer software package. This is included on the AEwin installation CD (if applicable to the product). Please see the WaveViewer User's Guide for complete instructions on using the software and viewing waveforms. The data can also be analyzed using [NOESIS™](#), an advanced data analysis pattern recognitions software package that is sold separately from AEwin.

3.3.2.1.11.6 Waveform Streaming Performance and Trouble Shooting

The rate at which waveforms can be saved to a data file is a function of many different variables. The PCI bus-based systems (PCI2, SHSM) is capable of streaming up to 20 Mbytes per second total over all enabled AE channels. This would correspond to 10M samples per second (10MSPS) to memory. The PCIe bus-based systems (Express-8) are much more capable and can stream as much as 50MSPS over all enabled channels per PCIe board. Unfortunately, the path from memory to the data file is dependent on hard disk performance and CPU utilization, so it is not possible for us to say what the highest achievable rate will be. The DMA controller requires physical non-paged memory in order to operate. AEwin reserves enough memory to hold 1 or 4 (product dependent) seconds of data per channel, so higher sampling rates require more memory. For example, if a channel is streaming at 10MSPS, that would require 20Mb of memory just for 1 second of data on that one channel. Memory requirements can add up quickly. There must be enough physical memory to map the buffer. The normal AE hit and waveform processing of the card also required physical memory as well. If AEwin locks too much physical memory, then overall OS performance will decrease. If there is insufficient physical memory available on your system to allocate the waveform DMA buffers, one of several failure messages will be displayed. If this happens, you can reduce the sampling rate or the number of streaming channels until there is enough physical memory for the buffers. Another option is to increase the physical memory installed in the system.

For long-duration waveform saves, the possibility exists that the CPU will not be able to write the waveform data to the output file at an equal or greater rate than the waveform is being read. If this happens, then a message indicating that the waveform has expired will appear. There are several solutions to this problem. The first is to reduce CPU utilization by turning off non-essential items. Disable the line listing display. Reduce the number of plots. Disable AE hit waveforms. Don't save AE data to disk. Other options include reducing the sampling rate of the channels or the number of channels being written to the data file. Upgrade your hard disk or add more physical memory.

3.3.2.2 Channel Group Setup...

The 'channel-group setup...' dialog box is used to define and edit channel groups. A 'channel group' is simply a named list of channels. You can select this group in the graph setup instead of selecting channels individually. The names of any channel groups selected for a graph are displayed on the graph's title for easy identification.

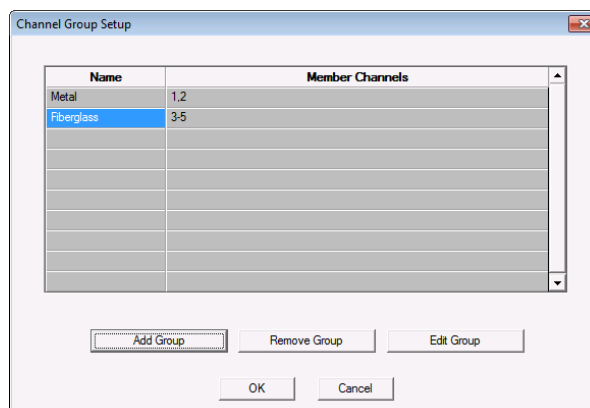


Figure 3-28. Channel Group Set up

To create a group, click the Add Group button. This brings up a dialog box shown in Figure 3-29. From here, you can name the new group and select its member channels.

To modify an existing group, highlight it and click the Edit Group button. This brings up the same dialog box. To delete an existing group, highlight it and click the Remove Group button. Note that when you remove a channel group, it is automatically removed from any graphs it was selected in.

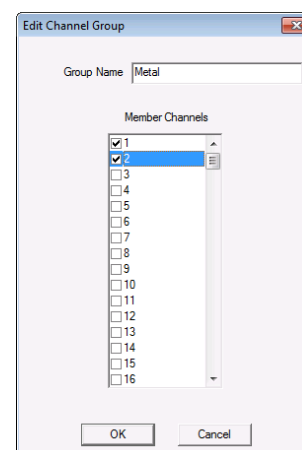


Figure 3-29. Add/Edit Group

3.3.2.3 Location...

The Location Setup dialog box is used to define and enable event groups (location groups). Use of Location involves more than just this dialog, however. For a full description of the Location dialog as well as other Location details see section 4.

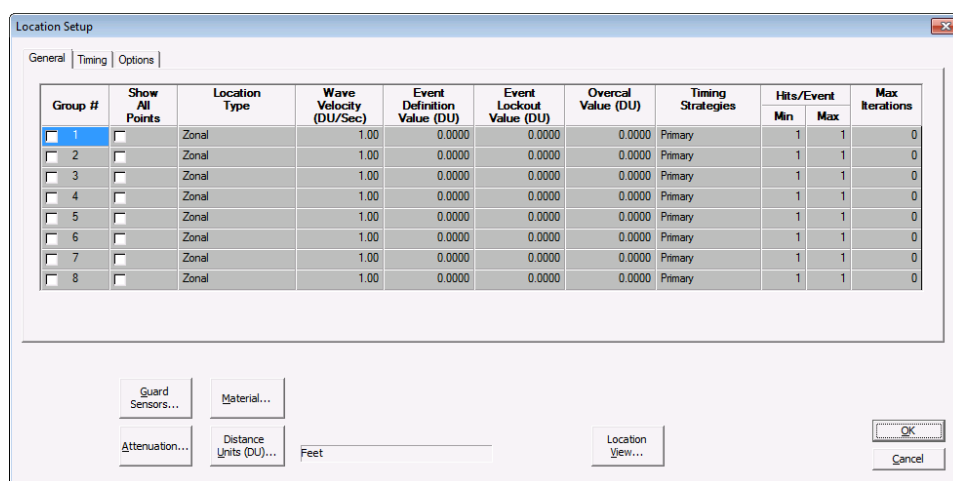


Figure 3-30. Location Set up

3.3.2.4 Title...

'Title...' is used to set the Test Title. It can be displayed on the statistics bar using the Statistics menu described below.

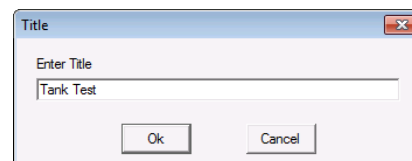


Figure 3-31. Title

3.3.2.5 Statistics...

'Statistics...' is used to toggle the display of various test statistics that can be shown on the Statistics Bar. The bar itself can be toggled from the View->Statistics Bar main menu item.

- **Test Title:** Displays the test title set in the 'Title' dialog box.
- **Time of Test:** Displays the elapsed time since the start of the test or replay.
- **Test Point Time:** Displays the elapsed time since the last test point (user-requested time mark).
- **Test Point:** Displays the total number of test points (user-requested time marks) recorded so far during the current test or replay.
- **AE Hits:** Displays the total number of AE hits recorded so far during the current test or replay.
- **Events:** Displays the total number of located events recorded so far during the current test or replay.
- **Waves:** Displays the total number of waveforms recorded so far during the current test or replay.
- **Cumulative Counts:** Displays the total number of counts recorded so far during the current test or replay.
- **Cumulative Energy:** Displays the total number of energy counts recorded so far during the current test or replay.
- **Free Space:** Displays the current amount of free space left on the disk on which the data file you are replaying or acquiring.
- **Parameters:** Displays the value of the specified parametric from the most recent hit or time data message recorded or replayed during the current test or replay.
- **Cycle Counter:** Displays the value of the cycle counter or RPM value from the most recent hit or time data message recorded or replayed during the current test or replay.

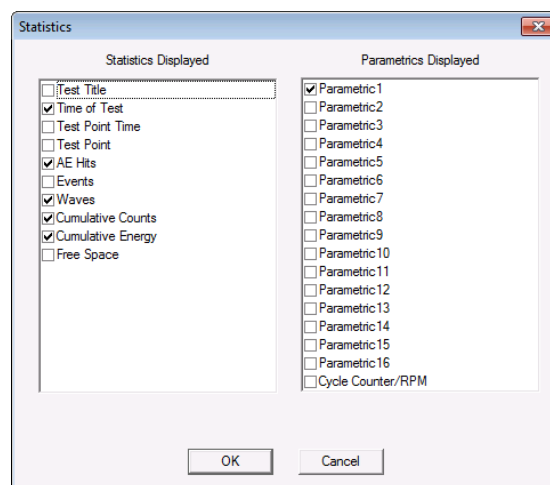


Figure 3-32. Statistics

3.3.2.6 Logistics Information...

The 'Logistics Information' dialog box is used to store information about a test. The Title column contains eight editable fields to be used as descriptions of the information that is to be entered by the tester into the field to the right of the title field (Field column). Examples of useful titles would be the date of the test, location of the test, tester name(s), the item being tested, identification or serial number, weather condition, etc.

The Field column also contains eight editable fields that may be filled in now or at test time, with information about the test to be performed. This information can then be preserved in the Layout and Data files.

In addition to these two columns of fields, there are several check boxes and buttons located at the bottom of the screen:

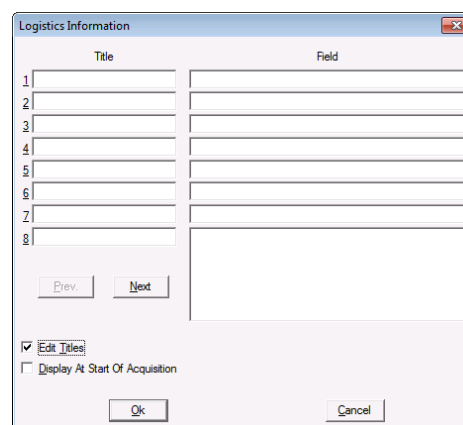


Figure 3-33. Logistics Information

Prev. and Next - Although there are only eight rows of the information shown, there are 16 rows available. These buttons are used to swap between each block of eight rows.

Edit Titles - When this item is checked, the information in the Title column fields can be accessed and edited. When this item is not checked, the information in the Title column fields will be for display purposes only; therefore, all text in the Title column will be grayed out and inaccessible.

Display at Start of Test - Checking this box will cause the logistics information dialog box to automatically be opened after initiating a new test. When this convenient option is checked, you will not have to remember to update the logistics information each time a test is begun because the program will automatically prompt you for logistics information each time a new test is begun.

3.3.2.7 Audio Monitor...

The audio monitor board (an optional purchase) contains the (audible) alarm and audio controls. The circuitry is designed to produce audible tones from high-frequency acoustic emission signals. This circuit can be selected to operate in either a homodyne or heterodyne mode. Operation in heterodyne mode allows the user to select the frequency of the AE signal to be monitored. Homodyne operation does not require frequency tuning and allows audio monitoring of AE signals at any frequency.

To activate the dialog box for setup of the PAC Audio Monitor board, select 'Acquisition Setup' then 'Audio Monitor...'.

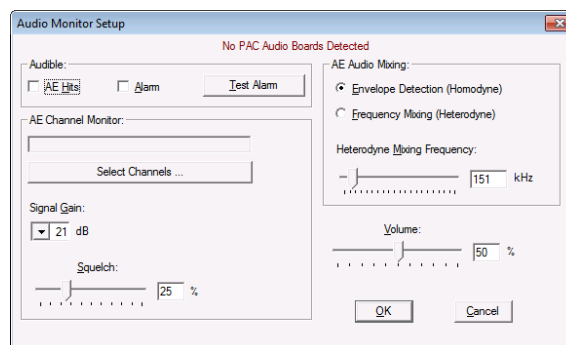


Figure 3-34. Audio Monitor Set up

In the group titled 'Audible,' there are two checkboxes. The first one, AE, enables the board to emit a sound when AE hits are detected. The second one, Alarm, enables the board to produce a tone whenever a front-end alarm warning or trip level is exceeded (these alarms are defined in the 'Acquisition Setup,' 'Alarms' dialog box). By clicking on the 'Test Alarm' button, you can hear how the alarm will sound. Click it a second time turn it back off.

The group titled 'AE Channel Monitor' contains controls for selecting monitored channels as well as setting gain and squelch. Click 'Select Channels' to choose which channels are to be used by the audio monitor. The 'Signal Gain' combo box contains a list of valid gain levels for the AE channels. This allows small-amplitude AE signals to be monitored. 'Squelch' is used to eliminate the sound of background noise. Valid entries are from 0 to 100%. This value should be set to a level so that under conditions of no AE signals, there is no audio output. Then as soon as a signal received is above the threshold as set by this control, the audio output for that signal will be heard. The keyboard left and right arrow keys can be used to move the slider +/-1%, and the page up/down keys will move it 10%. The mouse can be used to drag the slider as well; clicking the mouse on either side of the slider will move it 10% in that direction.

The group titled 'AE Audio Mixing' contains mutually exclusive checkboxes for selecting the method by which the AE tones are generated. When 'Envelope Detection (Homodyne)' mode is selected, each hit produces a click sound. When the 'Frequency Mixing (Heterodyne)' mode is selected, each hit is mixed with a signal whose frequency is entered in the edit box below the mode selection controls. This limits the audio to hits whose frequency is 10kHz above the selected frequency. For example, if the mixing frequency was set to 150kHz, then AE signals greater than 150kHz and less than 160kHz will be heard. To set the Mixing Frequency value using the slider, follow the same instructions as for the Squelch control.

The last control is for volume. It controls the volume of both the AE tones and the alarm tone. It ranges from 0 (off) to 100% (full volume). To set the value using the slider, follow the same instructions as for the Squelch control.

The changes made to the dialog box will go into effect immediately. Pressing Cancel will restore the audio settings to those in effect before the dialog box is displayed.

Note that the AE detection by the audio board is independent of the threshold levels of each channel, so it is possible to set the board up to hear AE, which is below the threshold of hits detected during a test.

The top line of the dialog box will display PAC Audio Board Version X.X if a PAC audio board is detected. If one is not, it will show No PAC Audio Boards Detected.

3.3.2.8 Auto File-Close...

In most tests, all data from test start to test end is saved to a single data file. In long tests, it is sometimes desirable to instead split the data into multiple files. When the Auto File-Close feature is active during acquisition, the program can automatically stop the test and start a new test under user-defined conditions. The default result is a numbered set of data files. Optionally it can close the file and begin saving data to another file without stopping the test. The result, in this case, is a numbered set of 'continued' data files that can be replayed together as if they were one file.

The program will name new files by incrementing the name of the first file. (i.e., if the name entered for the first file is test0001.dta, then subsequent files will be named test0002.dta, test0003.dta, etc.)

If checked, the program will also re-save the current layout every time a file is automatically closed. This saves the name of the new file in the layout as the default data file name for future tests.

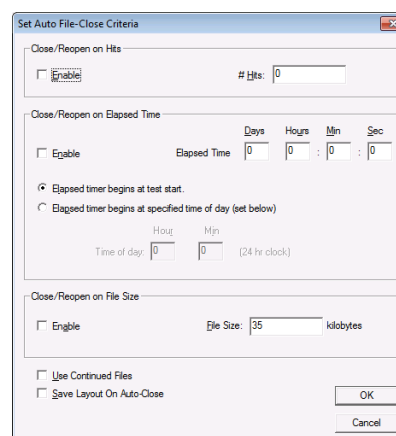


Figure 3-35. Auto File-Close

3.3.2.8.1 Enabling and Customizing Auto File-Close

You can enable the Auto File-Close feature and set the conditions under which it will work in the 'Set Auto File-Close Criteria' dialog box seen in **Figure 3-34**. There are four conditions under which an Auto File-Close operation will take place:

- #Hits:** Whenever the total number of hits in the test equals the #Hits value or a multiple of it, an Auto File-Close operation is performed. (i.e., If #Hits is 1000, then an Auto File-Close will happen at 1000 hits, 2000 hits, etc.)
- Time of Test:** Select the 'Elapsed timer begins at test start' option. Whenever the elapsed time equals the 'Time of Test' or a multiple of it, an Auto File-Close operation is performed. (i.e., If 'Time of Test' is 1hr 10 min, then an Auto File-Close will happen every 70 minutes.)
- Time of Day:** Select the 'Elapsed timer begins at the specified time of day' option and enter the time of day; in 24 hr. clock format, the file is to be closed. Also, enter the Elapsed time. Whenever the current time of day equals a multiple of the elapsed time, an Auto File-Close operation is performed. Some Examples:

Settings		Result	
Time of day	Elapsed time	Acquisition Start	Auto Closes Occur at
15:0 (3p)	0 3:0:0 (3hrs)	14:10:45 (2:10:45p)	3p,6p,9p,12a,3a,6a,9a,12p,3p...
15:0 (3p)	0 3:0:0 (3hrs)	08:34:15 (8:34:15a)	9a,12p,3p,6p,9p,12a,3a,6a,9a...
15:0 (3p)	0 3:0:0 (3hrs)	15:04:52 (3:04:52p)	6p,9p,12a,3a,6a,9a,12p,3p...

- File Size:** Whenever the current file size equals 'File Size,' then an Auto File-Close operation is performed. (i.e., If 'file size' is 64kb, then you will end up with a string of files, each one approximately 64,000 bytes in size.). Note that file sizes will vary, especially at high data rates.

You can enable any or all of these conditions. Whenever any condition is met, an Auto File-Close operation will occur.

3.3.2.8.2 Using Continued Data Files

Enabling (checking) 'Use Continued Files' will close the file in 'continued' form and open a new file without stopping data acquisition. The resulting set of files is referred to as a continued file set. They differ from normal data files as follows:

- Every file in the file set (except the last one) has an internal link to the next one. This means that when you replay one of these files, it will look for the next file(s) and replay them also. Note that replay can only find them if they are in the same folder, and the files should not be renamed (due to the internal reference to the name of the next file in the set).

3.3.2.8.3 Replaying Continued Data Files

- To replay the entire file set in order: Replay the 'first' file (the one with the name you gave it when you started the test) in the file set. The entire file set (if present in the same folder) will replay as if it were one file.
- To replay the file set from a certain file onward: Replay the file from which you want to begin replay. The file set from that point on will replay as if it were one file.
- To replay a single file only: Since the files are linked, the program will always try to find and replay all files that come after the selected one in order. To replay only one file, you must first copy it to another folder (or move/delete the others) then replay it as you would any other. Eventually, it will reach the end of the file and will warn you that it can't find the next file. If you want to eliminate this message from future replays of this file, you must remove its link using the Time Order/Repair utility (see the next section for details).

Note that if the program encounters a 'gap' in the file set (i.e., a missing file), it will display the "File Not Found" error described above. It can't replay beyond this point because the link is, in effect, broken. You can pick up where it left off by starting a new replay of the next existing file in the set.

3.3.2.8.4 Removing the Link from a Continued Data File

To remove the link from a continued data file (other than the last file, which has no link), run the Time Order/Repair utility on it. This will allow you to replay this file without the "File Not Found" warning being displayed. Doing this will prevent any files 'after' this one from being replayed as part of the complete file set, so be sure to make a copy first. Time Order/Repair will ask you if you want to replace the continue message with an abort message. Click 'Yes' in response. Time Order/Repair is accessed from the main menu Utilities item.

3.3.2.9 Disk Space Management

Free Space Alarm

The 'Free Space Alarm' is designed to warn the user in the event that the amount of free disk space remaining has run low during data acquisition. To enable this alarm, check the 'Enable Alarm' checkbox and set a minimum threshold (in Megabytes), below which the alarm will sound.

Data File Closing for Low Disk Space Condition

Windows™ systems do not perform well when there is zero disk space available. AEwin, by default, will stop writing data (DTA files) to the disk if it detects less than 1Mb of free space. It will continue acquisition, updating graphs with the data, but there will not be a permanent record of the collected data. AEwin will not automatically start saving data to disk once the low space condition clears. Not all output from AEwin checks this limit. STA (small files for internet monitoring) and WFS files do not check the space and will continue to be output (if enabled) even as the disk space dwindles toward zero. Therefore it is advised to set the alarm on low disk space that is comfortably above the Low Disk Space Threshold such that an alarm will occur with enough time for an operator to make more space available. The Low Disk Space Threshold should be adjusted as well based on how much data is expected to be written and the alarm response time so that no data is lost.

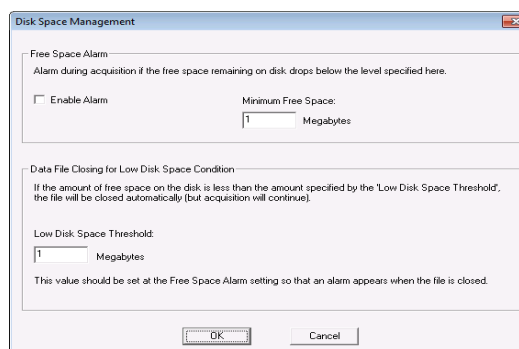


Figure 3-36. Disk Space Management

3.3.2.10 Alarm Preferences

The Alarm Preferences dialog box contains general alarm settings such as how or if an automatic alarm reset occurs and whether or not to generate a screen captures.

Periodic Auto Alarm-Reset: If the Periodic Reset feature is enabled, it will (during data acquisition) reset ALL active alarm Warn/Trip lines every time a period of time elapses equal to that set in this dialog.

Conditional: The Conditional Reset feature is used in conjunction with 'High End' alarms. High-End alarms include the 'Free Space' alarm (set in the Disk Space Management dialog box), LeakTEC, and On-Line Crack alarms (if this option is enabled). If this feature is enabled, a high-end alarm is automatically reset when the condition that caused its triggering no longer exists. i.e., If the 'Free Space' alarm is triggered because the amount of free space on the disk dropped below 5 Megabytes and the user then deleted enough files to raise the free space remaining above 5 Megabytes, the alarm would automatically reset.

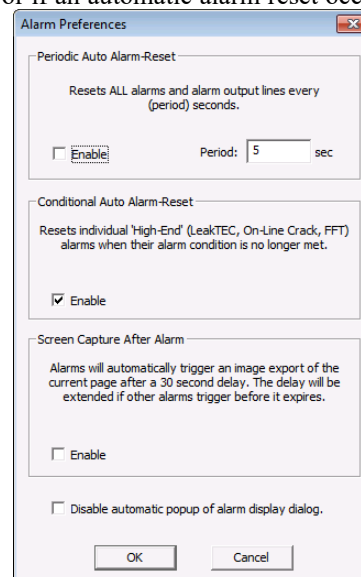


Figure 3-37. Alarm Preferences

3.3.2.11 Multivariate Alarm Setup

Multivariate Alarms are a highly flexible type of 'high-end' alarm that can be customized by the user for a variety of applications. The setup dialog box can be found on the Acquisition Setup menu.

A multivariate alarm definition is made up of a number of 'Alarm Criteria' arranged in a logical expression, as seen in Figure 1 below. A single 'criterion' is a single measurable condition that can be evaluated as true or false, such as 'amplitude of a single hit being within the range of 85 and 100 dB'. We will get back to what makes up a criterion shortly. For now, it is important to understand that a criterion can be evaluated as true or false.

The 'Enable Alarm' checkbox is used to enable or disable the entire alarm. The alarm definition also has a 'Name,' which is used in alarm reporting. You can set this in the 'Alarm Name' field.

Figure 3-38. Multivariate Alarm Set up

The criteria are arranged in a simple logical expression. They are grouped in pairs, the members of which are linked by an AND/OR operator. Each pair is likewise linked to the next one by an AND/OR operator.

The logical result of a pair is:

(Result of Criterion1) AND/OR (Result of Criterion2)

The expression then cascades to the next pair:

(Result of Pair1) AND/OR (Result of Pair2)

Which continues:

(Result so far) AND/OR (Result of the next Pair)

This continues until all criteria have been evaluated. Inactive criteria are ignored. More precisely, an inactive criterion evaluates to TRUE in an AND expression and evaluates to FALSE in an OR expression. If both criteria in a pair are inactive, then the entire pair behaves the same way with respect to other pairs (TRUE in an AND expression, FALSE in an OR expression). Thus it is possible to safely leave 'gaps' in the expression.

If the expression ever evaluates as true, then an alarm condition exists. If the multivariate alarm isn't already triggered, then it generates an alarm. How alarms are generated will be discussed later. For now, let's take a look at what makes up an alarm criterion.

Alarm Criteria

To enable a criterion, check its corresponding 'On' checkbox in the 'Multivariate Alarm Setup' dialog box, shown in Figure 1. To customize a criterion, click its corresponding 'Set' button. This will display the 'Alarm Criterion Setup' dialog box, shown in **Figure 3-38** below.

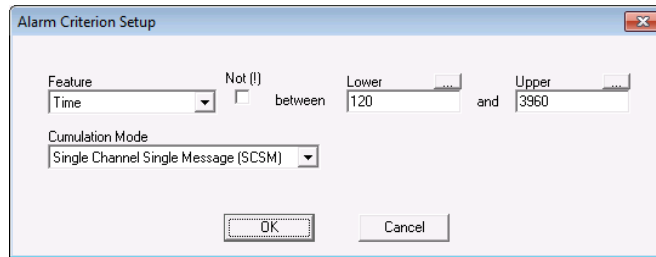


Figure 3-39. Alarm Criterion Set up

In this dialog, a hit-based 'feature' and a range of values that represent the alarm range (or the non-alarm range if 'Not' is used) are defined. Cumulation (summation) on a per-channel basis is also available.

Multivariate alarms are hit-based. Every hit that is acquired is evaluated by each enabled criterion. If the value of the hit's selected feature is within range (or outside it if 'Not' is used), then the criterion evaluates to TRUE; otherwise, it evaluates to FALSE. If cumulation (SCC) is used, then the cumulative value for that channel is used for evaluation purposes instead.

The controls here are as follows:

- **Feature** – This control is used to select a hit feature to be evaluated.
- **Not** – If this is unchecked, the 'Lower' and 'Upper' limits represent the alarm range. If this is checked, they represent the non-alarm range.
- **Lower** – This is the inclusive lower limit of the evaluation range.
- **Upper** – This is the inclusive upper limit of the evaluation range.
- **Cumulation Mode** – There are two cumulation modes:
 - **SCSM**: Single Channel Single Message – In this mode, each hit is evaluated separately. There is no accumulation.
 - **SCC**: Single Channel Cumulative – In this mode, the criterion accumulates the feature value of each hit (individually per channel) and uses that for evaluation purposes. These accumulators are cleared at the start of a new test. They are not cleared if the alarm itself is cleared during a test (either manually by the user or automatically by the 'Auto Alarm-Reset' feature).
- **[...]** – These two buttons are visible only when the Feature is 'Time.' When clicked, they popup the Time Converter dialog box seen in **Figure 3-39**. This dialog box makes it simple to enter or modify a time of test value in hours/minutes/seconds/etc. Without having to manually convert into seconds. Both the 'Lower' and 'Upper' limit controls have one of these buttons.

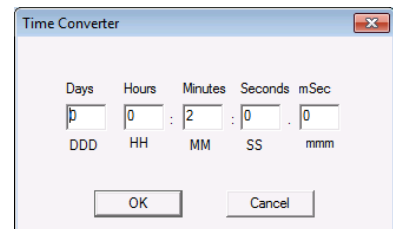


Figure 3-40. Time Converter

How an alarm is generated

The alarm definition is evaluated whenever an AE hit is processed during acquisition. If this results in the logical expression evaluating as TRUE, then an alarm may be triggered. It will not be triggered if the Multivariate Alarm has already been triggered and hasn't been cleared since.

Some Notes and Tips:

- 1) All criteria use the same set of channels, which are selected by clicking the 'Select Channels' button in the Multivariate Alarm Setup dialog box.
- 2) When using SCC (Single Channel Cumulative), a criterion accumulates values for each channel separately. What gets evaluated is the cumulative value appropriate to the incoming hit's channel.
- 3) An alarm cannot be triggered in the absence of data. In other words, be careful with criteria that specify things like "Cumulative Energy within the range of 0 to 4000". You might think that this would evaluate to TRUE immediately at the start of a test (because cumulative energy on all channels would then be zero). But since alarm criteria are evaluated only when a hit is processed, this will not happen. Take this into account when creating your alarm definitions.
- 4) During replay, any alarms are shown exactly as they appeared during acquisition. This preserves the data record and negates the need to use the same exact alarm definition during replay that was used during acquisition. For safety reasons, the Alarm Trip line (see the next section) is not set high when an alarm is replayed.
- 5) Alarms are automatically cleared at the start of a test.

What happens when an alarm is generated

- 1) An alarm message (16) is generated. If the auto dump is on, then this message is saved to the data file. If the line display is active, then it will be displayed there with the time it was triggered, the channel that triggered it, and the name of the alarm itself.
- 2) The Alarm Display dialog box is opened (or updated if already open). The details of the alarm and its name are shown. The alarm can be cleared (reset) by clicking the 'Clear Alarms' button. Note that doing this does not clear any accumulated values in criteria that use SCC.
- 3) The Alarm Trip line is set high.
- 4) An audible alarm is triggered if you have an audio board installed and enabled.

How to reset an alarm automatically

The 'Alarm Preferences' dialog box is used to automatically reset alarms based on a user-set criterion. This dialog box can be accessed either directly through the Acquisition Setup menu or from the 'Auto Alarm-Reset' button in the Alarm Status/Test dialog box. Refer to section 3.3.2.10 for more information about alarm preferences.

It is not recommended that the 'Auto Alarm Reset' capability be used with multivariate alarms unless your application requires it. In most cases, you will not want a multivariate alarm to be reset automatically. In some default layout files, the 'Conditional' alarm reset is on by default, so you may need to turn it off.

3.3.2.12 FFT Alarm...

FFT Alarms (part of the 'Waveform' Option) is used to monitor incoming FFT data during acquisition and alert the user if an FFT has an amplitude beyond certain limits within specified frequency ranges. Up to 10 'boxes' can be specified per channel. Each 'box' is defined by a frequency range and two levels (warning and trip). If the amplitude of the data in a particular box meets or exceeds one of these levels, then an alarm of the appropriate level is generated.

Box	Low Freq (Hz)	High Freq (Hz)	Warning Level	Trip Level
1	140000	160000	60	70
2	90000	110000	50	55
3	0.0000	0.0000	0.0000	0.0000
4	0.0000	0.0000	0.0000	0.0000
5	0.0000	0.0000	0.0000	0.0000
6	0.0000	0.0000	0.0000	0.0000
7	0.0000	0.0000	0.0000	0.0000
8	0.0000	0.0000	0.0000	0.0000
9	0.0000	0.0000	0.0000	0.0000
10	0.0000	0.0000	0.0000	0.0000

Figure 3-41. FFT Alarm Set up

3.3.2.12.1 FFT Alarm Setup

The 'Global Settings' section contains a setup that is not specific to any channel.

Enable FFT Alarm Functionality – Check/uncheck this to enable/disable FFT Alarm functionality.

Disable Conditional Auto-Reset of Trip Alarms – Check this option to prevent FFT Trip alarms from being automatically reset by the 'Conditional' type of 'Auto Alarm-Reset.' Refer to section 3.3.2.10 for more information about alarm preferences.

Disable Automatic Popup of Alarm Display Dialog – Normally, when an alarm is generated, a small dialog box is displayed describing the alarm. This can be disabled for FFT Alarms using this checkbox. The dialog box can still be accessed manually through 'Display Last Alarm' on the Acquire/Replay menu.

The large grid contains settings relating to a single channel. Use the 'View Prev Chan' and 'View Next Chan' buttons to view/modify other channels. The entire grid of settings for one channel can be saved to a buffer by clicking 'Copy These Settings to Buffer.' These settings can then be pasted to other channels by first changing to the new channel and clicking 'Paste Settings from Buffer.' Clicking 'Copy These Settings to All Chans' will copy the settings from the currently displayed channel to ALL other channels.

Enable Alarms on Channel X – Check/uncheck this to enable/disable FFT Alarm functionality on this channel. Note that if the global 'Enable FFT Alarm Functionality' setting is unchecked, this setting is ignored.

Box – Check/uncheck these to enable/disable up to 10 boxes on this channel.

Low Freq. [Hz] – Set the lower bound of the frequency range of this box here.

High Freq. [Hz] – Set the upper bound of the frequency range of this box here.

Warning Level – Set the Warning Level (in dBs) here.

Trip Level – Set the Trip Level (in dBs) here.

3.3.2.12.2 What happens when an alarm is triggered

- 1) An alarm message (16) is generated. If the auto dump is on, then this message is saved to the data file. If the line display is active, then it will be displayed there with the time it was triggered and the conditions that triggered it.
- 2) The Alarm Display dialog box is opened (or updated if already open). The details of the alarm and the conditions that triggered it are shown. The alarm can be cleared by clicking the 'Clear Alarms' button. Note that the display of this dialog box can be disabled. See the setup description for details.
- 3) If the alarm is an alarm 'warning', then the Alarm Warning Output Line is set high. If the alarm is an alarm 'trip,' then both the Alarm Warning and Alarm Trip lines are set high.
- 4) An audible alarm is triggered if you have an audio board installed and enabled.
- 5) The color of any Overview Page icons that show FFT Alarm status for this channel is updated to reflect the new alarm condition.

3.3.2.13 Alarm Status/Test

The Alarm Status/Test dialog box is used before and during the test for a variety of purposes:

- Testing of the I/O connections prior to a test.
- View the status of each alarm during the test.
- Manually reset one or more alarms during or after a test.
- Silencing the audible 'beeping' sound without resetting the alarms.
- Viewing the Alarm History.

A table of alarm types is displayed either on the left or at the top of the dialog box. Each cell represents one type of alarm (such as Free Space Low) or condition (such as 'Not in Acquisition'). During acquisition, these will become checked (and possibly unchecked) automatically to reflect current alarm conditions. If you check one manually, it will effectively trigger that alarm, even when not in acquisition. This is useful for testing the I/O board connections. If you uncheck one manually, it will reset that alarm.

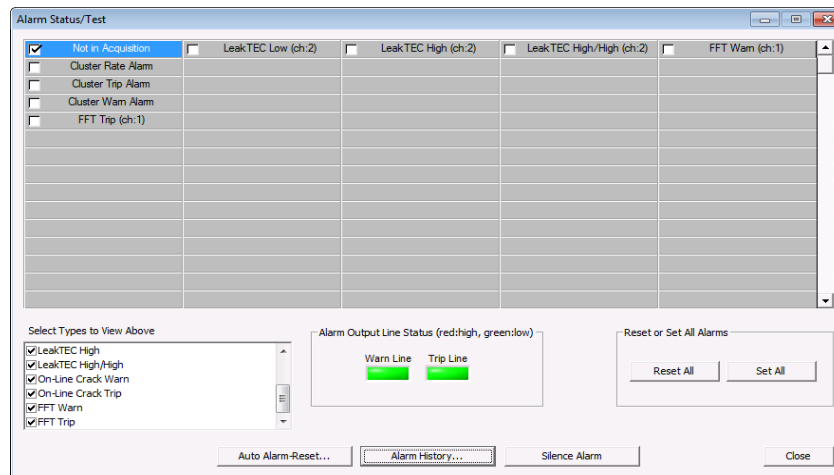


Figure 3-42. Alarm Status/Test

Nearly all possible alarms can be set/reset by clicking the appropriate button on the right side of the dialog box. In the center of the dialog box are colored indicators that display the status of the Alarm Output Lines (warn and trip, two of the digital output lines available on every system). They are colored green if they are reset (low) or red if they are set (high). You can silence the audible 'beeping' sound without resetting the alarm(s) by clicking 'Silence Alarm.' As a convenience, the Alarm Preferences dialog box can be displayed by pressing the 'Auto-Alarm Reset...' button. The 'Alarm History...' button displays alarms that occurred during the last acquisition (or the current one).

This is the same function available through the 'Display Last Alarm...' item in the Acquire/Replay main menu dropdown list.

Alarm	Software Option	AE Board Warn/Trip	I/O Board	Notes
System Status	N/A - Standard	N	Y	Low disk space alarm/Not in Acquisition
Cluster Rate	N/A - Standard	Y	Y	
Cluster Trip	N/A - Standard	Y	Y	
Cluster Warn	N/A - Standard	Y	Y	
Intensity Trip	Intensity (39)	Y	Y	
Intensity Warn	Intensity (39)	Y	Y	
LeakTEC Low	LeakTEC (18)	Y	Y	
LeakTEC High	LeakTEC (18)	Y	Y	Not available when Boat Inhibit option is purchased.
LeakTEC High/High	LeakTEC (18)	Y	Y	Not available when Boat Inhibit option is purchased.
Boat Inhibit LD	Boat Inhibit (43)	Y	Y	
Boat Inhibit	Boat Inhibit (43)	Y	Y	
On-Line Crack Warn	OL Crack (18)	Y	Y	
On-Line Crack Trip	OL Crack (18)	Y	Y	
FFT Warn	Waveform (3)	Y	Y	
FFT Trip	Waveform (3)	Y	Y	
Front End Warn	N/A - Standard	Y	N	Does not appear in Alarm/Status Test
Front End Trip	N/A - Standard	Y	N	Does not appear in Alarm/Status Test
Multivariate	N/A - Standard	Y	N	Does not appear in Alarm/Status Test
ASL	N/A - Standard	N	N	Does not appear in Alarm/Status Test
AST	N/A - Standard	Y	N	Trip only

3.3.2.14 ASL Alarm...

ASL Alarms are used in conjunction with Internet Monitoring. Unlike Front-End or Graphical Alarms, ASL Alarms do not sound an audible alert or display a dialog box when triggered. Instead, they save their alarm data in the STA files.

You can set alarm criteria on a channel-by-channel basis. Any channels not checked are not evaluated for alarms. The Low Level and High Level you set define the acceptable (inclusive) non-alarm range.

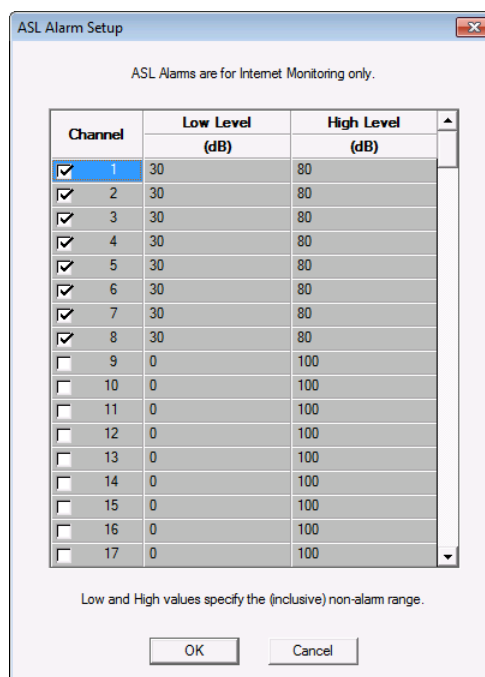


Figure 3-43. ASL Alarm Set up

3.3.2.15 Voltage Time Gate Setup...

Voltage Time gate is used for parametric-based AE data filtering. It is controlled by parametric 1 and allows the user to eliminate the collection of all AE data that occurs when the parametric falls outside a pre-selected range of values and other criteria.

AE data collection is either Enabled or Disabled. It specifies what to do when the voltage-time gate is generated.

Each threshold is specified with a level in Volts and $\geq$ or $\leq$, which indicates whether the signal has to be above or below the level in order to satisfy the threshold. Activation of each threshold is independent of the other; the user can activate either both or no threshold at all.

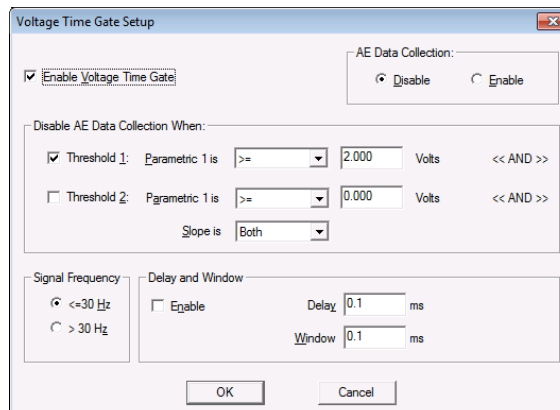


Figure 3-44. Voltage Time Gate Set up

The slope control can be POSITIVE, NEGATIVE, or BOTH. It is satisfied if the parametric is rising, falling, or at any time.

The slope and all active thresholds must be satisfied in order to generate a voltage-time gate. This allows the user to restrict the gate to crossings of thresholds with the parametric specifically changing a certain way (rising or falling).

Delay and Window specify an optional time window (the 'time' part of the voltage-time gate). When enabled, they cause the gate to be a time window of controlled duration, starting with some delay after the moment that the threshold and slope criteria are met.

Signal Frequency specifies a low-pass or high-pass filter that is applied before the parametric is fed into the voltage-time gate processing circuit. This provides the ability to respond to signals that change slowly (most cases) or quickly (fatigue cycles, etc.). There are two settings, one for parametric signals below 30 Hz and one for above 30 Hz.

3.3.2.16 Internet Monitoring Setup...

This dialog box allows you to set up the Internet Monitoring feature of AEwin. This is used in conjunction with other PAC 'Remote Monitoring' applications purchased separately. Please refer to their documentation for details.

From this dialog box, you can:

- Enable/disable internet monitoring.
- Set the time interval between file generations.
- Set the serial number for your unit.

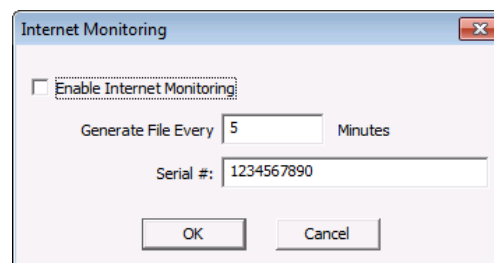


Figure 3-45. Internet Monitoring

3.3.2.17 Email Notifications

AEwin uses several methods to alert the user when an alarm triggers (line display, popup dialog, an audible tone, etc.). Quite often, though, the user isn't physically present to see these alerts. This is especially true on long-duration tests and unattended remote monitoring. For these and other circumstances, AEwin has an additional means to notify the user of an alarm. This is accomplished via email.

Through the use of the 'Email Notifications Setup' dialog box, you can configure AEwin to send emails to one or more recipients when an alarm or similar notification occurs. You can customize up to 4 independent setups, each one responding to different alarm types. These are labeled as "Address 1", "Address 2" ... "Address 4". Despite the name, each one can contain multiple email addresses.

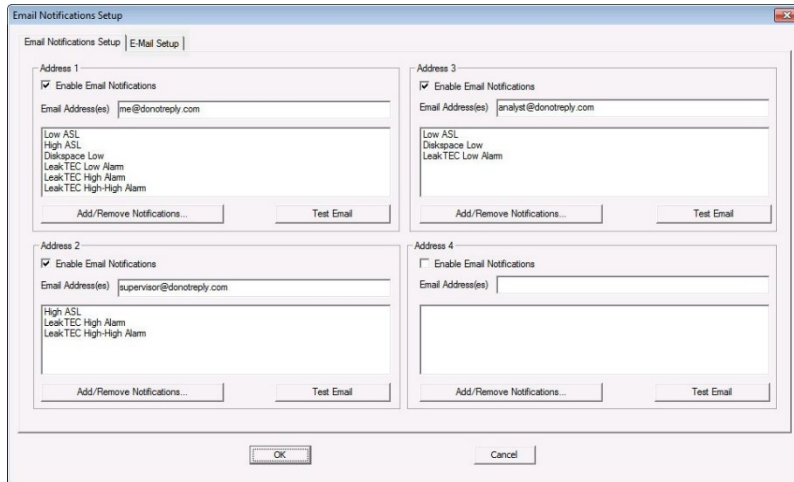


Figure 3-46. Email Notification Setup Dialog

The dialog box has two tabs: Email Notification Setup and E-Mail Setup.

3.3.2.17.1 Email Notification Setup:

There are up to four Address groups that can be defined and enabled. The edit field will accept multiple addresses if each address is separated by a semicolon. Multiple addresses are provided so notifications can be broken up into groups of different addressees for different notifications.

To setup an 'Address,' do the following:

- Check its "Enable" checkbox.
- Add one or more valid email addresses.
- Click the 'Add/Remove Notifications' button to designate alarm types.

Use the Add/Remove Notifications... button to display a dialog that contains a list of all available alarms or conditions that will send notifications whenever they go into an active state.

Available Alarms:

- Low ASL
- High ASL

ASL Alarm Levels are set from Acquisition Setup->ASL Alarm. They will only occur if Internet Monitoring is enabled (AEwin is outputting STA files for transfer to a database/web server usually maintained at Mistras Group) because the ASL level is only evaluated against the alarm criteria at the end of an internet monitoring period.

The email's Subject will contain the text entered in the AEwin System Name field of the SMTP Setup page followed by either "Asl Low" or "Asl High."

Example email Content:

```
System Name: My AEwin
Low Asl: Ch: 1, Average: 19 < Reference: 40
Local Date/Time: 1/26/2015, 3:55:18 PM
Test Started: 1/26/2015, 3:55:00 PM
Current Data File: No File
```

The email message contains the alarm condition (trigger value and reference level that was exceeded), the local date and time that the alarm occurred, the local date and time that the acquisition was started, and the name of the DTA file that was being filled at the time the alarm conditions were evaluated.

- Location Event

These emails are only generated for location groups and only under certain conditions. This requires the Event Classification option (41) to be enabled in the AEwin license. Not only does there need to be an active location group, but Event Classification in the Location Setup Options tab must be checked, and the Email Reporting Level in the Event Classification Options dialog (accessed from the Options... button to the right of the Event Classification check box) must be checked (after setting the appropriate Level for the application). Additionally, the text entered in the Event Labeling Source Type fields will be reported in any emails that are generated. Refer to the Event Classification documentation for more information about the reporting level and labels.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "Location Event" and the Event Label (from the Event Classification Options dialog box) associated with the event.

Example email Content:

```
System Name: My AEwin
Location Event: Event Label Text
ID DDD HH:MM:SS.mmmuuun      PARA1  CH  RISE  COUN  ENER  DURATION AMP
* Gp# 1[1,2] x = 5.875, y = 1 Event Label Text dT[ 0] Src Amplitude = 62.0
* 0 00:00:03.8154403 0.0000 1 208 25 35 400 62
* 0 00:00:03.8154405 0.0000 2 208 24 35 392 62

Local Date/Time: 1/30/2015, 8:46:08 AM
Test Started: 1/30/2015, 8:46:05 AM
Current Data File: No File
```

The email message contains the Event Label associated with the event (in the example, the text is "Event Label Text"), the line display listing for the event that generated the email, the local date and time that the event occurred, the local date and time that the acquisition was started and the name of the DTA file that contains the AE hits that formed the located event (if the auto dump was enabled when the acquisition was started).

- Graph Alarms

Many (but not all) of the graph types in AEwin support alarm generation. If the graph does support them, there will be a tab labeled 'Alarms' at the top of the Graph Setup dialog box. The graph alarms support independent enabling of warning and trip levels, and emails will be generated for any level that is enabled. Graph alarms are evaluated per graph, so any graph whose data crosses an alarm threshold will generate an email.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "Graph Alarm."

Example email Content:

```
System Name: My AEwin
Graph Alarm: 0 00:00:33.7342467 GRAPH ALARM Trip: Hits vs Time(sec) <All Channels> ::
Hits (1.0000) >= 1000.0000; (Time = 33.7342)
Local Date/Time: 1/30/2015, 8:20:22 AM
Test Started: 1/30/2015, 8:19:49 AM
Current Data File: No File
```

The email message contains the acquisition time, alarm type (warn/trip), the title of the graph that generated the alarm, the alarm condition (trigger value and level that was exceeded), Event Label associated with the event (in the example, the text was "Event Label Text"), the local date and time that the alarm occurred, the local date and time that the acquisition was started and the name of the DTA file that contains the AE hits that formed the located event (if the auto dump was enabled when the acquisition was started).

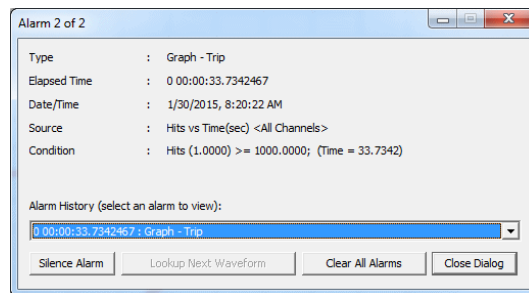


Figure 3-47. Example Graph Alarm Notification Dialog

- **Diskspace Low**

The Acquisition Setup->Disk Space Management... menu item will display a dialog box where low disk space criteria can be set. Enabling the Free Space Alarm will send an email message when the space on the where data files are saved gets below the level set as the Minimum Free Space.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "Diskspace Low."

Example email Content:

```
System Name: My AEwin
Diskspace Low: Drive E: Free Space Remaining (0 Mb) < 1 Mb
Local Date/Time: 1/30/2015, 10:09:19 AM
Test Started: 1/30/2015, 10:09:14 AM
Current Data File: TEST0060.DTA
```

The email message contains the affected drive letter and the current amount of free space (in Mb) on that drive, the local date and time that the alarm occurred, the local date and time that the acquisition was started, and the name of the DTA file that was being written to at the time the disk space went low.

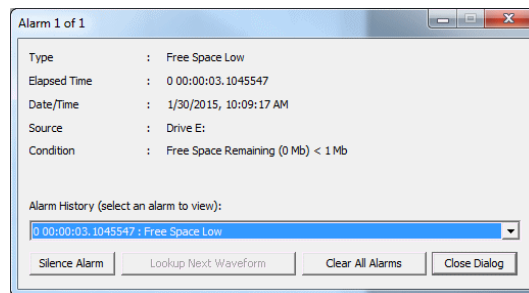


Figure 3-48. Example Low Disk Space Alarm Notification Dialog

- LeakTEC Low Alarm
- LeakTEC High Alarm
- LeakTEC High-High Alarm
- LeakTEC Sputtering Alarm

LeakTEC is a licensed option that provides enhanced ASL or RMS alarm condition testing. Any of the above alarms conditions can generate email notifications. Please refer to the LeakTEC documentation for more information about how the alarms are enabled.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "ALARM."

Example email Content:

```
System Name: My AEwin
ALARM: 0 00:00:10.9999000 LeakTEC ALARM Warning: LOW ASL(23) <= 40.0000 on chan 1 for
at least 10 seconds
Local Date/Time: 1/30/2015, 2:18:39 PM
Test Started: 1/30/2015, 2:18:29 PM
Current Data File: No File
```


The email message contains the acquisition time, alarm type (Low/High/High High/Sputtering), the alarm condition (trigger value and level that was exceeded), the AE channel number and the hold-off time, the local date and time that the alarm occurred, the local date and time that the acquisition was started and the name of the DTA file that was being written to at the time the alarm occurred.

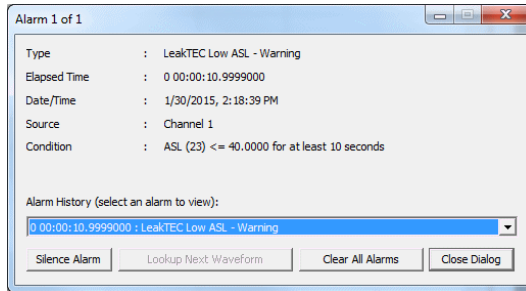


Figure 3-49. Example LeakTEC Alarm Notification Dialog

- Boat Inhibit – Long Duration Alarm
- Boat Inhibit

Refer to the section about Boat Inhibit for a description of the alarms. This is a licensed option.

- On-Line Crack Alarms

On-Line Crack Alarms is a licensed option that provides enhanced alarms based on a definable combination of hit rate, amplitude, and energy. Please refer to the Online Crack Alarm documentation for more information about how the alarms are enabled and what alarms can be triggered. When the email notification for the Online Crack Alarm is enabled, any crack alarm will generate a notification.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "ALARM."

Example email Content:

```
System Name: My AEwin
ALARM: 0 00:00:04.4541217 On-Line Crack ALARM Warning: HITS(15) >= 15 in <= 10 sec on
chan 1.
Local Date/Time: 2/3/2015, 11:16:33 AM
Test Started: 2/3/2015, 11:16:29 AM
Current Data File: No File
```

The email message contains the acquisition time, alarm type (Hits/Events, Warn/Trip/Energy Trip), the alarm condition (trigger value and level that was exceeded), the monitoring interval and the AE channel number, the local date and time that the alarm occurred, the local date and time that the acquisition was started and the name of the DTA file that was being written to at the time the alarm occurred.

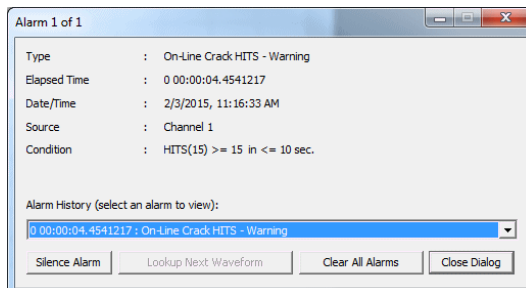


Figure 3-50. Example On-Line Crack Alarm Notification Dialog

- Cluster Alarms

Many (but not all) of the graph types in AEwin support cluster alarm generation. If the graph does support them, there will be a tab labeled 'Clustering' at the top of the Graph Setup dialog box. The cluster alarms support independent enabling of warning and trip levels based on cluster grades as well as a cluster rate alarm. Emails will be generated for any alarm that is enabled. Cluster alarms are evaluated per graph, so any graph whose data generates a cluster alarm will generate an email. Refer to the graphing section of the manual for more information about the clustering alarms.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "ALARM."

Example email Content:

```
System Name: My AEwin
ALARM: 0 00:00:02.4224520 CLUSTER GRADE LEVEL ALARM Trip:
Source: Duration(us) vs Amplitude(dB) <All Channels>
Location: Cluster-ID = 1, Grade = C, Points(>=50) = 50, Avg. X, Y, Z = 55.00, 376.00, 0.00.
Local Date/Time: 2/3/2015, 11:26:55 AM
Test Started: 2/3/2015, 11:26:53 AM
Current Data File: No File
```

The email message contains the acquisition time, alarm type (grade level/rate, warn/trip), the title of the graph that generated the alarm (as Source), the cluster location and alarm trigger information, the local date and time that the alarm occurred, the local date and time that the acquisition was started and the name of the DTA file that contains the AE hits that formed the located event (if the auto dump was enabled when the acquisition was started).

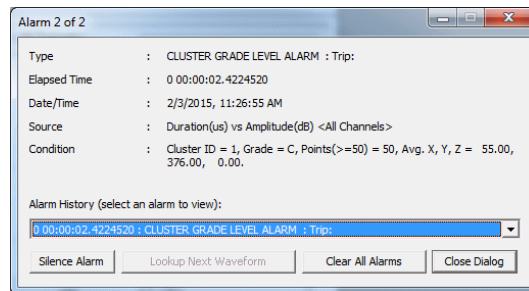


Figure 3-51. Example Cluster Alarm Notification Dialog

- **FFT Alarms**

FFT Alarms are applied to amplitude levels over a specific frequency range (or ranges) of the power spectrum derived from the AE waveforms. They are defined by using the Acquisition Setup->FFT Alarm... dialog box. Refer to the FFT alarm documentation for more information. Note that although the alarm definitions can be shown on the power spectrum graphs, the alarms do not require a graph to function.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "ALARM."

Example email Content:

```
System Name: My AEwin
ALARM: 0 00:00:03.6427660 FFT ALARM Trip on chan 1. Level: 45.3498, Freq: 60024.0000 Hz
Local Date/Time: 2/3/2015, 1:19:03 PM
Test Started: 2/3/2015, 1:19:00 PM
Current Data File: No File
```

The email message contains the acquisition time, alarm type (warn/trip), the channel number, the amplitude, and the frequency that caused the alarm. The local date and time that the alarm occurred, the local date and time that the acquisition was started, and the name of the DTA file that contains the AE hits that formed the located event (if the auto dump was enabled when the acquisition was started).

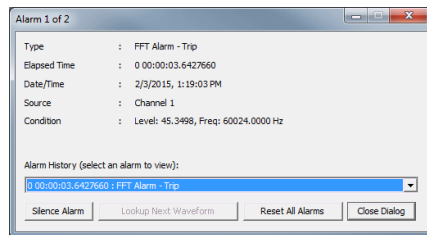


Figure 3-52. Example FFT Alarm Notification Dialog

- **Multivariate Alarm**

Multivariate Alarms are a highly flexible type of 'high-end' alarm that can be customized for a variety of applications. The setup dialog box can be found by selecting Acquisition Setup->Multivariate Alarm Setup... Refer to the Multivariate Alarm documentation for more information about its capabilities.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by "ALARM."

Example email Content:

```
System Name: My AEwin
ALARM: 0 00:00:02.5902185 Multivariate ALARM (Multivariate Test ALARM) Trip: Channel 1
Local Date/Time: 2/3/2015, 3:28:38 PM
Test Started: 2/3/2015, 3:28:36 PM
Current Data File: No File
```

The email message contains the acquisition time, the name given to the alarm in the multivariate alarm setup dialog box, alarm type (always trip), and the channel number. The local date and time that the alarm occurred, the local date and time that the acquisition was started, and the name of the DTA file that contains the AE hits that formed the located event (if the auto dump was enabled when the acquisition was started).

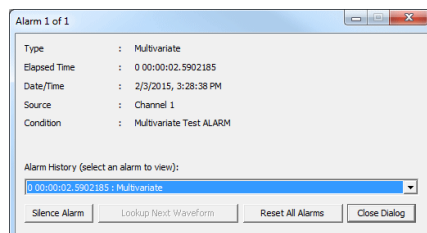


Figure 3-53. Example Multivariate Alarm Notification Dialog

- **Base Station/Networked Remote Messages**

This configuration has a version of AEwin running on a PC with no data acquisition boards communicating with one or more AEwin acquisitions systems over a network. It is not available without the 'Network Remote' option and is typically only used in applications where a large number of AE channels are required (more than would be available in a single AE system). This option is not available for all AE product lines. There are several conditions that will generate an email.

The email's Subject will contain the text entered in the AEwin System Name field of the E-Mail Setup page followed by one of the following condition descriptions:

- "Networked remote system dropped out."
This indicates the base station was not able to communicate with the AE system. Typically when this failure occurs, the base station will attempt to regain communication.
- "Networked remote system dropped in"
This indicates the base station was able to regain communication with an AE system that had previously dropped out.
- "HELP - Networked remote system needs Attention."
This indicates a potentially serious error with the AE system has occurred. If the AE system tries multiple times (typically 5) to enter data acquisition but is unable to do so, it will give up trying and notify the base station so that the base station can reconfigure acquisition to remove it from the list of available AE systems. The base station will no longer attempt to communicate with the affected system.
- "Basestation had an unexpected error."
If the base station is equipped with a watchdog timer and the timer expires, causing the base station to reboot, or if the base station just loses power, this message will be sent to indicate the disruption has occurred.

Example email Content:

```
System Name: My AEwin  
SH-22201-2 (ID=1) was removed from acquisition due to a system failure.
```

The email message contains the acquisition time, the name given to the alarm in the multivariate alarm setup dialog box, alarm type (always trip), and the channel number. The local date and time that the alarm occurred, the local date and time that the acquisition was started, and the name of the DTA file that contains the AE hits that formed the located event (if the auto dump was enabled when the acquisition was started).

- **Sensor Connection Status**

Some of the AE systems support sensor monitoring. If available, sensor monitoring will periodically check the voltage levels at the sensor and report if the measured readings are out of an acceptable range.

3.3.2.17.2 Email Filtering

AEwin can send email notifications to a group of addresses when something of interest occurs on the system, such as an alarm condition having been met or a location event having been detected. If a large number of location events are detected in a short period of time, a torrent of emails could be sent out. The email filtering provides a means to control the output of emails when certain conditions are met.

3.3.2.17.2.1 Notification Support

The following Notification types support email filtering:

- **Location Event:** This notification is only available if Event Classification (AEwin option 41) is enabled. Option 41 is typically available by a specific purchase. It will be listed in the Help → Options dialog box of AEwin if it is enabled.

3.3.2.17.2.2 Getting to the Email Filtering Setup Dialog Box

There are two ways to get to the settings.

1. From the Email Notification Setup dialog box (Main Menu: Acquisition Setup → Email Notification Setup...), if the notification is already active, double click the notification in the list box to display its filter setup.
2. From the Add/Remove Notifications dialog box (click the Add/Remove Notifications... button for the Address group in the Email Notification Setup dialog), highlight the notification, and a 'Filter...' button will appear below the 'Clear All' button. Press the 'Filter...' button to set the filter for the highlighted notification.

3.3.2.17.2.3 Filtering Setup

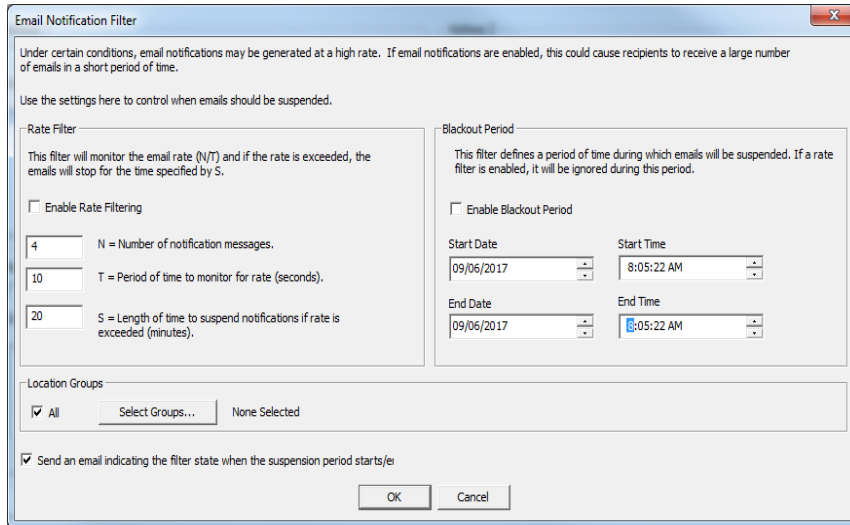


Figure 3-54. Email Notification Filter Dialog Box

Rate Filter

Once enabled, the rate filter will monitor how quickly emails are generated and suspend sending them for a defined period of time once the rate is met. There are two fields in the dialog box that define the rate. The first is N, the number of emails, and the other is T, the time to monitor. The third field, S, sets how long the emails will be suspended once the rate (N/T) is met. For example, if the emails are to be suspended for 30 minutes once 5 emails are generated in 10 seconds, the fields would be set as N=5, T=10, and S=30.

Blackout Period

For instance, when it is known that a large number of emails could be generated due to conditions that arise at a known date and time, a time span over which no emails will be sent can be defined. Two fields are provided for each the start and end times, one for the date and one for the time of day. The format of the fields is based on the localization settings in Windows (for instance, MM/DD/YY vs. DD/MM/YY or 02:00:00 vs. 14:00:00).

Location Groups

For the Location Event notification, the affected location group can be specified. By checking 'All,' any location event email can be blocked. If 'All' is unchecked, then specific groups must be selected. Only the emails originating from events by the selected groups will be blocked. Those groups left unselected will not be affected by the filter.

Notification of Active Suspension

By default, when an email suspension is starting and has ended, an email will be generated indicating so. The following is an example of the start and end suspension emails for a blackout period:

Example of a Start of Blackout Suspension:

Email Subject Line:

My AEwin: Email notifications are suspended due to the start of a blackout period

Email Body:

Do not reply to this message.

System Name: My AEwin

Email suspended: Friday, October 20, 2017, 13:38:00 Through Friday, October 20, 2017, 13:38:15

Blackout

Groups: ALL

Example of an End of Blackout Suspension:

Email Subject Line:

My AEwin: Email notifications suspended by a blackout period are now resuming

Email Body:

Do not reply to this message.

System Name: My AEwin

Email suspended: Friday, October 20, 2017, 13:38:00 Through Friday, October 20, 2017, 13:38:15

Blackout

Groups: ALL

Are now resuming. 4 messages were blocked.

Note that the end of suspension email includes the number of emails that were blocked during the period.

In addition to sending suspension notification emails, time marks will be generated and placed in the AE data file as a permanent record of when emails were suspended and resumed.

Example of a Start of Rate Suspension Time Mark:

211 0 00:00:08.2577123 Time Mark: Email suspended: 10/20/17 11:22:35 AM Through 10/20/17 11:23:35 AM, Rate:4/10, Groups:ALL

Example of an End of Rate Suspension:

211 0 00:01:08.2560050 Time Mark: Email suspended: 10/20/17 11:22:35 AM Through 10/20/17 11:23:35 AM, Rate:4/10, Groups:ALL, Resuming

Time marks do not indicate how many messages were blocked during the suspension.

Notes:

- If a location group is enabled in the filter but is not active for acquisition, it will simply be ignored. No error message will be emitted.
- The email rate is not monitored during a suspension period (be it triggered by an excess rate or a blackout). Monitoring only occurs when the emails are not already in a suspended state.
- If a rate filter is defined such that one starts just prior to a defined blackout and ends during the blackout, a notification will still be sent indicating emails have resumed; however, they will not actually resume until the blackout is over (at which time a blackout end notification will be sent as well).

3.3.2.17.3 Email Setup

Email requires an external SMTP server to host the outgoing mail. AEwin supports the use of servers that require authentication (user name and password). It is possible to use a public server, like Outlook or Gmail, but keep in mind that they will restrict the number of emails that can be sent. Other paid-for servers are available, such as SendGrid. Mistras does not recommend any particular provider. Note that to use Gmail, the security level of the account must be lowered as AEwin does not support OAuth 2. If a public server is used, it is recommended to avoid using a personal account, create one specifically for the system.

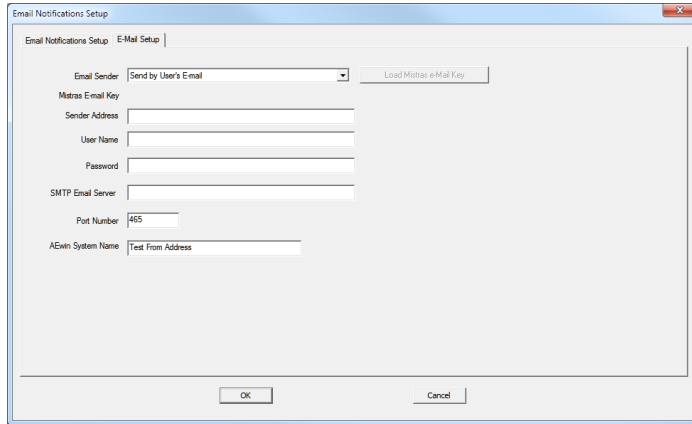


Figure 3-55. Email Server Setup Dialog

- Email Sender:** Options are User's Email and Mistras Email. Mistras Email is not supported and should not be used.
- Sender Address:** This is the email address of the account registered with the server.
- User Name:** Use the same value as the Sender's address.
- Password:** The password of the account registered with the server. AEwin will not display the value.
- SMTP Email Server:** The URL of the server. As an example, for an Outlook account, this would be smtp-mail.outlook.com. Check with the provider for the proper URL to use.
- Port Number:** The network port of the server. Again, as an example, for an Outlook account, this would be 587, and the provider should specify which port to use.
- AEwin System Name:** User-specified. To be used as a means to identify the AEwin system that sent the email. It will appear in both the email header and body.

3.3.2.18 I/O Board Setup...

The I/O Board is a 3rd party board used to relay state information (High/Low, On/Off, Triggered/Not Triggered, etc.) between the program and an external device through up to 96 lines. AEwin currently supports the following boards from Measurement Computing.

- PCI-DIO48H – 48 line board.
- PCI-DIO96H – 96 line board.

Before AEwin can interface with it, you must run the 'InstaCal' program from Measurement Computing to set up the board. Otherwise, AEwin won't see it. If you have more than one of these boards in your system, then AEwin will use the first one (you can use 'InstaCal' to designate the board order).

See the Measurement Computing device user's Guide for information on how to interface your device with this board. By default, all lines are pre-set as 'Output' lines. You can designate lines as either 'Input' (to program) or 'Output' (from the program) in 8-line blocks.

The I/O Board is not used during replay.

3.3.2.18.1 Output Lines

Output Lines are used to sending status or alarm state information (Triggered=HIGH, Not-Triggered=LOW) from the program to an external device. Typically a line is set to output the status of a single type of alarm on a single channel (e.g., LeakTEC Low Alarm on channel 5). However, a line can be set to output the status of multiple alarm types across any selection of channels and/or channel groups at the same time. In this case, if an alarm of any of the specified types is triggered on any specified channel, then the output line will be set HIGH.

The following alarm types are supported by Output Lines:

- LeakTEC Low Alarm
- LeakTEC High Alarm
- LeakTEC High/High Alarm
- Online-Crack Alarm Warning
- Online-Crack Alarm Trip
- FFT Alarm Warning
- FFT Alarm Trip
- Cluster Rate Alarm (*)
- Cluster Trip Alarm (*)
- Cluster Warn Alarm (*)
- Intensity Trip (*)
- Intensity Warn (*)
- Low Media Alarm (*)

Note: Types marked with an asterisk (*) are not channel-based but may be mixed with alarm types that are. A “Cluster Rate Alarm” output line will trigger if any graph triggers a cluster rate alarm. The same behavior holds for other cluster or intensity-based output line types. Many of these alarm types require certain options obtainable via Customer Service.

Additionally, the following non-alarm conditions are supported by Output Lines. These are not-channel based:

- Not in Acquisition
- In Acquisition
- Heartbeat Output

“Not in Acquisition” is triggered whenever the program *is not* currently in data acquisition. It is cleared when data acquisition begins (and also when the program closes). “In Acquisition,” as you might expect, is its inverse. It is triggered whenever the program *is* in data acquisition. These 2 line types are mutually exclusive. They may not be enabled on the same output line.

“Heartbeat Output” triggers and clears automatically at a regular pace (~1/sec), like a metronome, when in data acquisition AND not paused. This ceases if the test ends while it is paused or if AEwin closes for whatever reason (manually, due to power failure, OS fault, etc.). This is quite useful to applications that need to know that the AEwin test is operating properly.

3.3.2.18.2 Input Lines

Input Lines are used to receiving state information from an external device. They may represent one input type per line and represent any group or group of channels. Unlike Output Lines, Input Lines may only represent 'Channel Groups' rather than individual channels (see the section on Channel Groups for more information). This is necessary to prevent confusion when working with Overview Page icons. The following input types are supported by Input Lines.

- **Group On/Off** – This is used to designate a group of channels as ON or OFF. By default, an enabled channel is ON, unless a Group On/Off line is enabled for a group containing that channel, in which it uses the Input Line's state to determine this. When a channel is designated as OFF, the following occurs until it is designated as ON again:
 - The system stops acquiring data from all channels in the Channel Group. Note that there will still be channel-specific sections on Time-Driven messages (ID 2) for these channels, but the data (RMS, ASL, etc.) will be 0.
 - Any alarm capabilities enabled for these channels are disabled. This is especially important for LeakTEC Low Alarms, which would otherwise trigger in the absence of data.
 - The colors of any Overview Page icons that show Group On/Off status for this channel/group is updated to reflect the new status.
- **Alarm Reset (All)** – This is used to command the program to reset all alarms. Unlike other Input Line types, it has no channels or Channel Groups associated with it. When an Alarm Reset line is set HIGH, all alarms of all types are reset. The audible alarm (if in use) is silenced, and alarm Output Lines are reset. Alarms are not disabled by this action and may trigger again if conditions warrant. To reset the alarms again, set the line LOW, then HIGH again.

3.3.2.18.3 Setup Dialog

The I/O Board Setup dialog box is accessed from the Acquisition Setup menu. At the top of the dialog is a colored status message indicating the presence and health of the I/O board. If possible, this will indicate the type of I/O Board detected (e.g. 'PCI-DIO48H' or 'PCI-DIO96H'). Below is a list of the most common status messages.

Status Message	Condition
I/O Board Detected	I/O board found and properly set up by InstaCal.
No I/O Board Detected	This indicates that the I/O Board was either not physically installed or that InstaCal has not yet been run to set it up. If the board is present, then try running InstaCal.
Unexpected I/O Board Error	This could indicate a number of problems with the board. The most common problem is that the board was previously present (and set up by InstaCal) but was later removed. In this case, InstaCal is telling AEwin that the board <i>should</i> be there but can't be found.

There are a number of controls in this dialog box.

Enable I/O Board – This is a master enable the checkbox labeled 'Enable I/O Board.' All I/O Board functionality is disabled if this checkbox is not checked.

Enable MUX Lines – This is used only when a multiplexer PCI card is installed. Checking this box sets up 2 output and 3 input lines for use with this multiplexer. Keep this checkbox disabled unless you are using the multiplexer.

Port Setup – Clicking this button will open the Port Setup dialog box, which is used to designate which lines are to be used for input and which ones for output (see the section below for details).

The last control is a grid of settings, consisting of one line per I/O line. This is described below:

Port ID – This checkbox is used to enable/disable the individual lines. To the right of the checkbox is the name of the port and its pin on the board's main I/O connector. See the board's User's Guide for pinouts and other information related to this hardware.

Type – This is simply a label designating the line as either an Input or Output Line. This can be modified via the 'Port Setup' dialog.

Input or Output Settings – This is both a label indicating the specific input or Output Line type ('LeakTEC Low,' 'Group Speed,' etc.) and a button allowing the user to modify it. Clicking this field will open the I/O Line Setup dialog box. There are different ones for Input and Output Lines. This is where you set the line type and channels/groups. If the line is to be of the Group Speed type, then this is where the Low and High-speed settings are entered.

Channels/Channel Groups – This is a label describing the set of channels and/or Channel Groups selected in the I/O Line Setup dialog box. Clicking on this field will also open that dialog box.

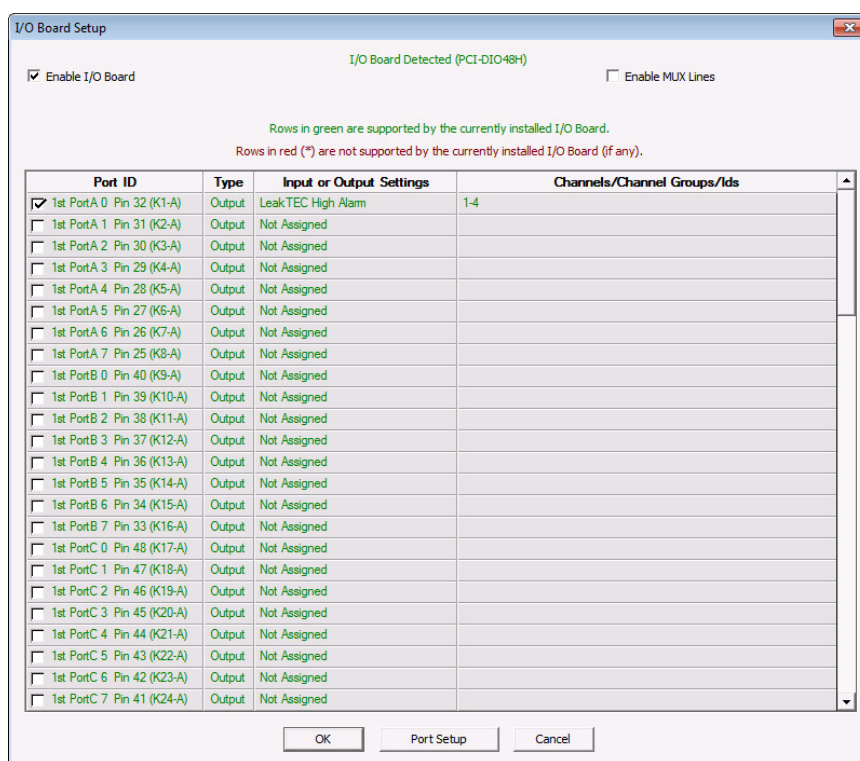


Figure 3-56. I/O Board Set up

3.3.2.18.4 Port Setup Dialog

The "I/O Board Port Setup" dialog box is used to designate which "Ports" are used for Input Lines and which ones are used for Output Lines. Each port is a block of 8 lines. The dialog box displays each port's name and pin range. To toggle a port's "Type" setting between 'Input' and 'Output,' simply click it. See the board's User's Guide for pinouts and other information.

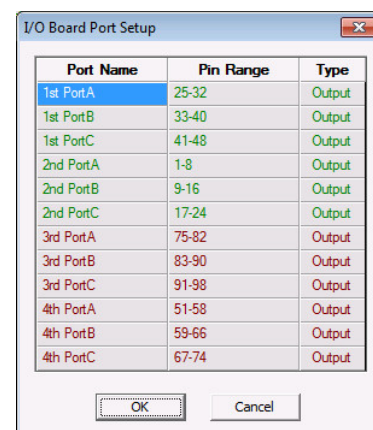


Figure 3-57. Port Set up

3.3.2.18.5 I/O Line Setup Dialogs

Clicking a cell in the “Input or Output Settings” column will display one of the following dialog boxes, depending on whether that line is designated as Input or Output. These are used to customize the selected I/O Line. A single Output Line can be used for multiple output types simultaneously. Use the checkboxes to select any that you wish to apply. Certain types are mutually exclusive or otherwise restricted. The dialog will alert you if these are selected. Many output line types are channel-based. Use the “Select” button to choose which ones are to be associated with this line.

For Input Lines, a single type may be selected from the drop-down list box. Some input line types are channel-based. Use the “Select” button to choose which channel groups are to be associated with this line.

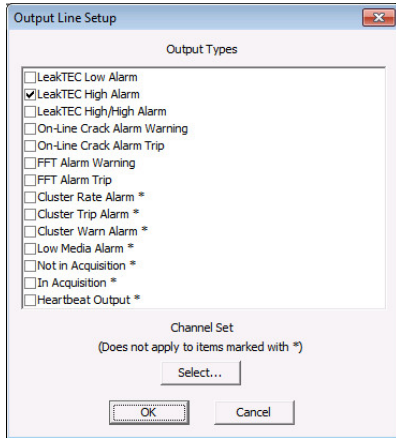


Figure 3-58. Output Line Set up

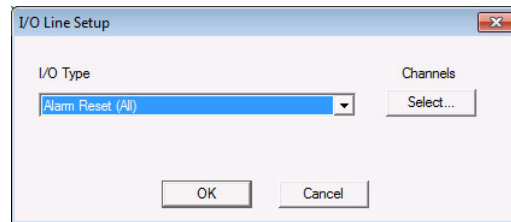


Figure 3-59. Input Line Set up

3.3.2.18.6 Tips and Suggestions

- I/O Board Input and Output Lines do not function during replay. I/O line state changes are stored in the datafile, and their effects (other than modifying Output Lines) are reproduced during replay. e.g., On/Off Icons change color when appropriate, and FFTs show Alarm level lines at the right frequencies based on speed.
- The use of Channel Groups is required when using Input Lines and highly recommended when using Output Lines. They help avoid confusion when coordinating with Icons and LeakTEC Alarm Graphs.
- The I/O Board Setup dialog box ensures that there is no channel overlap among Input Lines of the same type. This avoids the danger of having two on/off lines representing the same channel. Such a situation could create a problem if the two lines were in conflicting states. The dialog box will alert the user to any such problems and the opportunity to fix them when the Ok button is clicked. Channel overlap is not necessarily a problem for Output Lines and may be a requirement for some setups.

3.3.2.19 Display Mode...

Display mode allows you to select the form of display for the AE data, as well as the frequency of graph updates, via the Acquisition/Replay Modes dialog box.

There are two Acquisition/Replay modes available in AEwin:

- **Normal:** This is the standard mode for acquisition and replay. Statistics, graphs, tables, and the line display are all active in this mode.
- **High Speed:** This mode is used to achieve maximum data acquisition speed in tests where a very high emission rate is expected. Graphs, tables, and the line display are inactive in this mode. Only selected statistics will be displayed and updated; the priority task is transferring data to disk for analysis later.

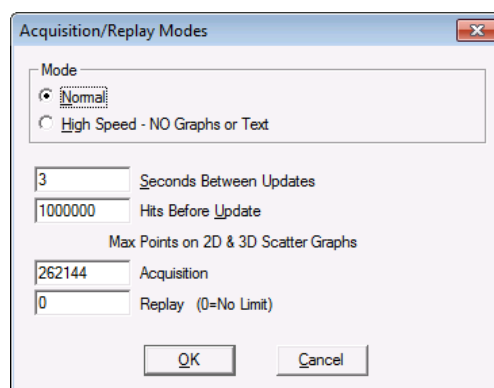


Figure 3-60. Display Mode

Seconds Between Updates, Hits Before Updates

During data acquisition and replay, graphs and tables are redrawn (updated) at intervals determined by these settings. They are updated whenever the specified time has elapsed, or the specified number of hits has occurred since the last update, whichever comes first. Depending on the anticipated data rate and test duration, you will want to select an update rate that is fast enough to keep you current without having the screen constantly redrawing. Type in the desired values in the text boxes provided. Recommended values for initial use are 3 seconds, 100 hits. Since screen updates affect the overall data acquisition rate, in high data rate situations, the user may want to use larger intervals for Hits Before Updates. For high data rate situations, it is advised to set the Hits Before Update to a very large value (e.g., 100000) and let the Seconds Between Updates determine the update rate.

Max Points on 2D & 3D Scatter Graphs

Many of the graph types available in AEwin support a limited number of points and have somewhat pre-defined resource (memory) requirements. Scatter graphs are an exception. Every point that goes into the graph is stored. As more and more data is collected, the memory requirements can grow without bound, having a severe effect on performance. AEwin supports setting separate values for acquisition and replay that will limit the number of points a scatter graph can hold. Setting a value of 0 will allow the graph to grow without limit and is the default for replay. The value can be as high as 1048500. There really aren't any hard and fast rules that can be applied as far as recommended settings. It depends on many factors such as how much system memory is available to AEwin, how many graphs are defined, how many plots are defined... The oldest data in the graph is replaced by the newer data. AEwin will display a message in the status bar when a graph reaches the limit.

3.3.2.20 Line Display...

The line listing mode allows the user to monitor any and all types of messages being processed by the AE instrumentation. The information is in a “line by line” text format listing, offering an additional monitoring capability over the graphing of AE data. With Line listing mode, the user can watch messages related to AE Hits, Events, Time-driven data (parametrics), comments, Auto Sensor Test (AST), waveforms, and administrative messages. Watching the line listing scroll by shows how the data are coming in is being processed, giving the user an idea as to the AE activity and severity.

The data is shown in numeric format. There is one row for each message, plus a title row every time it is needed. The title row shows the user which parameters are being displayed and where they are being displayed. This depends on which active parameters were selected for display in the Line Display Setup dialog box.

The line listing mode can be used in acquisition or replay and offers the same flexibility in both; however, it is recommended to use this mode sparingly in acquisition as its use might slow down the data acquisition speed by a significant amount. It is therefore recommended to use this when data acquisition rates are low, before a test as a prescreening method or after a test in the replay.

3.3.2.20.1 Enabling Line Dump Listing for Viewing

In normal use, AEwin is kept in graph mode, and the line listing is not displayed. In order to view the line listing, two steps need to be performed. First, one has to “Enable” the Line listing mode, and then the line listing mode needs to be selected for viewing. This is done as follows (whether in acquisition or replay).

To enable the line listing mode, simply click the mouse as follows;

Acquisition Setup → Line Display... → Select “Enable Line Display” Checkbox

To Display the “Line Display,” simply press the **F7** function key or click the mouse as follows;

View → Line Listing Display

As mentioned above, the function key **F7** provides a shortcut to toggle the line listing on and off once the Line Listing mode has been “Enabled” in the Test setup menu.

Note that whenever the line listing is displayed, it displaces the existing graphs based on their size and location. To restore the graphs, simply press one of the “Tile” functions (Tile horizontally or vertically).

3.3.2.20.2 Modifying the Line Display

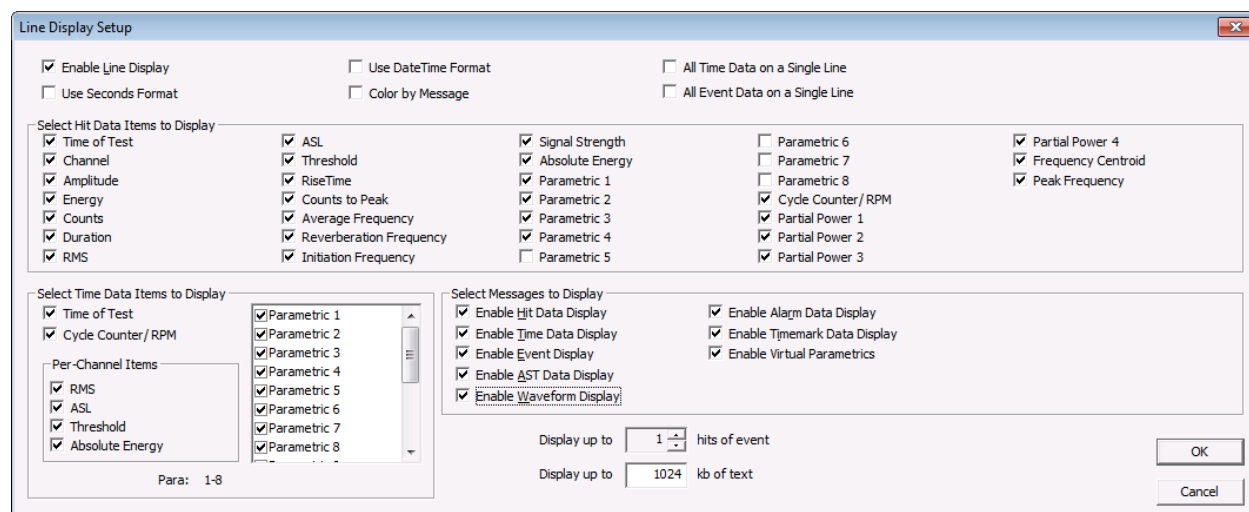


Figure 3-61. Line Display Set up

Here is a typical illustration for AE hit data:

ID DDD HH:MM:SS.mmmuuun PARA1 PARA2 CH RISE COUN ENER DURATION AMP

- The ID is the message ID. This will be 1 for a data message, 2 for a time-driven message, 173 for a waveform message, and 128, 129, or 130 for a test control message. A test control message is a RESUME, PAUSE, or STOP message.
- The TIME column is the time of hit read from the time of test counter. This time is displayed in days, hours, minutes, and seconds (or optionally as seconds) from the start of data acquisition, with a resolution of 250 nanoseconds.
- The PARA1 column is for parametric number one. The listed reading is scaled according to the values entered for Multiplier and Offset in the Parametric Scaling dialog box.
- The PARA2 column is for parametric number 2. Again the reading is scaled according to the values entered in the Parametric Scaling dialog box.
- The CH column would be for the channel number if the message was of type 1 or an AE data set.
- The RISE column is for RISE TIME. The range of this variable is from 0 to 65535 μ seconds.
- The COUN column is for threshold crossing COUNTS. The range of this variable is from 0 to 65535 counts.
- The ENER column is for ENERGY. The range of this variable is from 0 to 65535 energy counts.
- The DURA column is for signal DURATION. The range of this variable is from 0 to 16,777,762 μ seconds.
- The AMP column is for AE AMPLITUDE. The range of this variable is typically from 0 to 100 decibels.

In the Line Display Setup dialog box, you can:

Enable Line Display:

Note that the display can be visible (see the section on the View menu) but not enabled. Turning the display off can boost performance on very high data rate tests.

Use Seconds Format:

With this option, you can display TIME in seconds (total) rather than HH:MM:SS.mmmuuun format.

Use DateTime Format:

With this option, you can display TIME as a date based on when the acquisition was started in the form MM/DD/YY HH:MM:SS.mmmuuun format.

Color By Message:

With this option, you can have the various messages colored by message ID. This can make the display easier to read, though it is not recommended for very high data rate tests.

Line Listing Message Type	Line Listing Color
AE hit data	Blue
AE Event data	Violet
AE waveforms	Blue
Time-Driven Data (e.g., Parametrics, ASL, RMS, etc.)	Green
Administrative messages (Pause, Resume, Alarm, etc.)	Red

All-time Data on a Single Line:

With this option, Time Data messages (ID 2) are displayed using only a single line (each) rather than several shorter lines. The data on each row may be off-screen, but the columns of data will line up.

All Event Data on a Single Line:

With this option, Event, Data Messages are displayed using only a single line (each) rather than several shorter lines.

Select Hit Data Items to Display:

All possible hit features are listed here. Unchecking an item keeps it from being displayed on the line listing. It does not affect what features are collected, and having an item checked even though the feature is not in the data set is permitted. It affects any hit data listing, including those listed with events.

Select Time Data Items to Display:

All possible time-driven features are listed here. Unchecking an item keeps it from being displayed on the line listing. It does not affect what features are collected, and having an item checked even though the feature is not in the data set is permitted.

Select Messages to Display:

Items (message types) checked here will appear on the line display when they occur. Unchecking all but a single type of message allows the data to be read easier (especially when paired with 'Keep X Data on a single line) because the data of interest will line up in columns and not be broken up by other types of data.

Display up to X hits of event:

The event listing shows a lot of information about an event, including a list of the data for every hit that made the event. Sometimes it is desirable to limit the display of the hit information to one or no hits. Enter a number here to limit the number of hits displayed. No hits will be displayed if the value is zero (useful for 'event only' listing).

3.3.2.20.3 Sizing and Moving the Line listing Display

The line listing display, when selected to be viewed, pops up as full width, anchored text box, and anchors itself just below the AEwin toolbar. However, the Line listing display can be moved, floated, or re-sized simply by using standard Windows™ functions. To float, just "double click" on the border of the window. Once floated, the Line Listing Display window can be sized using typical Windows™sizing capabilities of moving the mouse to the edge of the window border until the double-sided arrow appears. Then the user can left-click the mouse and drag the window side to be larger or smaller as desired. The user can also move the floating Line Listing Display window anywhere on the screen simply by left-clicking the mouse when the mouse cursor is in the title bar (the blue title bar at the top of the Window) and can also anchor the Line Listing Display anywhere on the screen top, bottom, side, by moving one of the Line Listing Display Window edges to intersect with the side where it is to be anchored.

Holding the 'Ctrl' keyboard key while moving the line display will prevent it from docking.

Once the Line Listing Display has been moved, this position will be remembered.

3.3.2.21 Preferences:

The Preferences dialog box is used to modify settings that are not specific to any layout. These settings are remembered from session to session, so you generally only need to set them once. They are:

- **Only Show Installed Channels:** With this item enabled, Hardware Setup will only display entries for channels that represent hardware installed on the unit. If it is disabled, then Hardware Setup will display entries for up to the maximum number of licensed channels, including those that do not exist. This can be useful if you want to create layout files for use on other machines.
- **Load last used layout file on startup:** With this item enabled, the program will, on initial startup, load the last layout file that was loaded (or saved) the last time the program was run. It overrides the default action of loading layout.lay on startup.
- **Pre-enable Autodump:** With this item enabled, the 'Auto-Dump' checkbox in the Data Storage (Acquire) dialog box will be checked by default; otherwise, it is unchecked by default.
- **Use Wireframe on 3D Graph Rotation:** With this item enabled, 3D graphs will turn to a simpler to draw wireframe when grabbed with the mouse for pan/zoom/rotate. This only needs to be enabled if your graphics card is struggling with 3D graph manipulation.
- **Pre-enable Automatic File Naming:** When this item is enabled, the checkbox labeled 'Automatically Name File Using Current Time and Data of Test Start' in the Data Storage dialog box (the box that appears when the acquisition is manually started) will be checked (enabled) whenever the Data Storage dialog box is displayed.
- **Enable Image Export (All Pages) on File Close During Acquisition:** When this item is enabled, whenever AEwin closes a data file during acquisition, each page of graphs will be exported to a JPG file (one page per file). The file will be named with the name of the DTA file that is closing, followed by the name of the page as given by the page tab at the bottom of the page. E.g., TEST0000Page 0.JPG.
- **Enable Image Export (Current Page) on File Close During Acquisition:** The same operation as above, but instead of cycling through all pages, only the currently displayed page is exported.
- **Enable Image Print (All Pages) on File Close During Acquisition:** When this item is enabled, whenever AEwin closes a data file during acquisition, each page of graphs will be printed to the current default Windows printer.
- **Enable Image Print (Current Page) on File Close During Acquisition:** The same operation as above, but instead of cycling through all pages, only the currently displayed page is printed.
- **Enable Watchdog Timer:** This is only available on certain CPU boards. If available, the option will be selectable; otherwise, the checkbox is disabled. If available, checking the box will cause the computer to reboot and restart in the event that AEwin becomes unusable (or closes unexpectedly) once put into acquisition. Ending acquisition turns the watchdog off.
- **Replace 2D Scatter Graphs with Binned Scatter During Acquisition:** It is recommended that this item always be enabled. Binned scatter graphs require a known amount of resources as compared to a non-binned scatter, which can require more and more resources as acquisition continues. By using the binned scatter type, AEwin will behave much more consistently. The checkbox makes it easy to define graphs with the resolution of the standard point plot but has them automatically change to binned once acquisition starts. The setting does not affect replay. Point plots will stay as such during a replay.
- **Use Color Blind Point Styles on Colored Scatter Graph:** When this item is enabled, the point styles will be changed for each plot (rather than just relying on color to differentiate data). Note that it is also possible to change the colors assigned to each plot (Select Graphing from the main menu, then Colors and Plot Colors).
- **On Cylinders Draw Separate Grid On Caps & Body Instead of 1 Combined Grid:** When this item is enabled, extra grid lines will appear on the cylinder caps.
- **On Spheres, use +/- instead of N/S & E/W:** When enabled, the View Angle in the status bar will display '+' for North and East and '-' for South and West.
- **2D Graph Title/Axes Font Size:** Changes the size of the text in the 2D graph title line. The default size is 8. The size of the text in 3D graphs cannot be customized.
- **Draw Fixed Size Cluster instead of Observed Max/Min Clusters:** By default, clusters are drawn on the graph using the max/min values from data points that go into the cluster. Drawing fixed-size clusters only work with the 2D

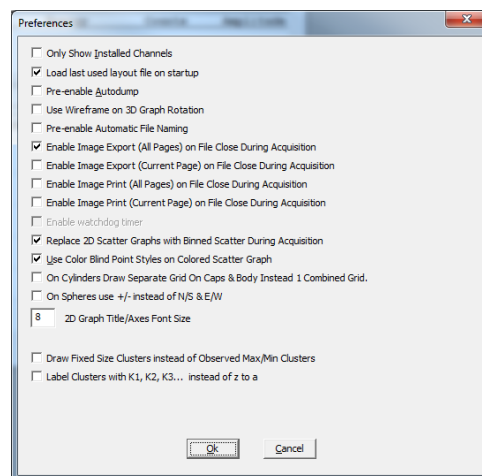


Figure 3-62. Preferences

Binned & 2D Colored Scatter graphs. For users who like to use 2D Colored Scatter graphs as a form of clustering by setting the color to be tied to the number of data points in the bin for the graph. Clustering has been enhanced to allow the user to use the actual bins as a real cluster so that now you can see the Energy, Count, and number of Points in the bin. When “Draw Fixed Size Clusters instead of Observed Max/Min Clusters” is on, the Clustering tab will hide the cluster size settings for the 2D Binned & 2D Colored Scatter graphs:

- Label Clusters with K1, K2, K3... instead of z to a: Controls how the clusters are labeled on the graph and in the cluster table. Clusters are typically shown with the labels ‘z’ thru ‘a’ to avoid confusing the cluster labels with grade labels (‘A’ to ‘E’) shown on some graphs like the Cluster Intensity chart. The Clustering tab of the Graph Setup dialog box has a field that controls how the clusters are sorted and displayed in the cluster table. The ‘z’ cluster is the cluster that is highest in the sort order, and the ‘a’ cluster is lowest in the sort order. This can be confusing when the table sort order is changed because it changes which cluster is the ‘z’ cluster. There is also a Cluster-ID shown in the table in that case. The cluster with ID 1 formed; first, ID 2 was second, etc. Now instead of ‘z’ to ‘a’ labels and a separate cluster-ID, the user can have the clusters labeled with K and the cluster-ID#, so that the cluster label does not change when the user changes the sort order in the cluster table. Those that prefer the traditional ‘z’ to ‘a’ labeling can simply leave this option unchecked.

3.3.3 Acquire/Replay Menu

3.3.3.1 Acquire...

An AE test is begun via the Acquire... menu item or by pressing F9. This will display a dialog box prompting you for information before the test begins.

This dialog box is very similar to the Save Layout dialog box discussed earlier, in which you are prompted for a filename, in this case, to store AE data. The exceptions here are the checkboxes in the upper left corner:

- Auto-Dump: If this box is checked, it means that Autodump is ON, and the test data will be recorded to the filename you specify in this dialog box. If you want a permanent record of your data, make sure that Auto-Dump is ON. If it is OFF, no data will be written to disk, and your record will be limited to the real-time graphic displays. Note that there is an option in the ‘Preferences’ dialog box (section 3.3.4.6) that will pre-enable the Auto-Dump checkbox.
- Enable Data Selection Features: If this box is checked, then support for Data Selection (including Cluster/Area History) will be enabled in acquisition. If it is not checked, then these features will not function for the duration of the test, but the system will exhibit better performance at high data rates. If you expect high data rates, then be sure to uncheck this and save the data to a file (Auto-Dump on). You can then use these features during a replay if needed.
- Automatically Name File Using Current Time and Date of Test Start: If this box is checked, then “File name” will not be used. Instead, the program will automatically name the file using the following convention: “YYMMDDHHMMSS_0.DTA” where:
 - “YYMMDD” corresponds to the year, month and day.
 - “HHMMSS” corresponds to the hour (24-hour clock), minute and second.
 - “_0” is appended to indicate that this is the first file in the acquisition test. If the data grows too large for a single file, then the program will open additional “continued” files using the same name as the first but will increment the “_0” each time to differentiate them. Refer to the Auto File-Close dialog box (section 3.3.2.8) for more information about ‘continued files.’

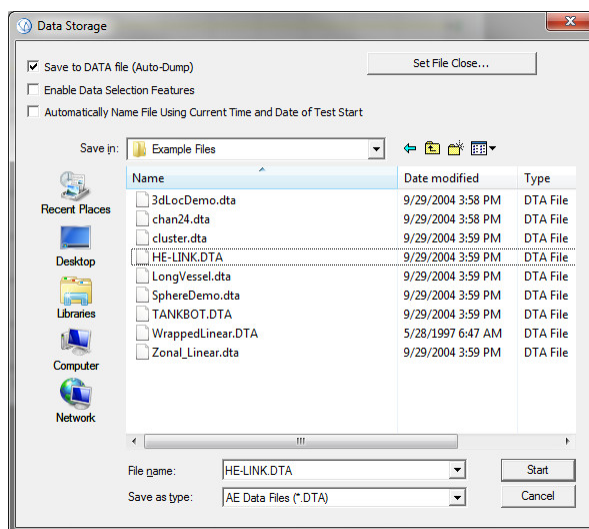


Figure 3-63. Data Storage

If Auto-Dump is ON, the program will save data to the filename specified. Assuming you don't have "Automatically Name File..." turned on, the program will then automatically increment the last letter or digit of the filename specified in the Test Storage dialog box each time you enter acquisition with AUTODUMP ON without specifying a different data filename.

The data file is automatically closed and reopened every second. This assures that the worst-case data loss is one second worth if power fails. A manual update to the disk can be accomplished by pressing pausing/resuming the test.

3.3.3.2 Replay...

This box is nearly identical to the Acquire dialog box discussed in the last section. Because Autodump is not needed here, it is not an option in this dialog box.

Once the filename has been selected, the controls are essentially the same as for Data Acquisition. During the replay, the system reads data from the disk, updating the graphs/tables at the rate specified in the Display Mode dialog box. The replay automatically ends when it finishes replaying the requested data file. If the file specified data file is a 'Continued file' (described in sections 3.3.2.8.2 and 3.3.2.8.3) then it replays the next file in its continued file set.

If the "Export Data to Text File" is selected, then the output from the line display will also be sent to a new file of the user's choice. Refer to section 3.3.4.5 for more information about configuring the line display.

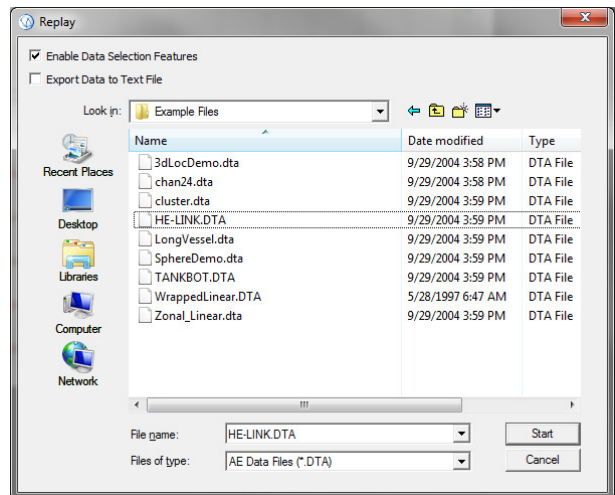


Figure 3-64. Replay

3.3.3.3 Resume/Pause/Stop/Abort

These four commands are used to control the test or replay.

- **Pause:** Click Pause to temporarily halt a test or replay. Note that buffered data will be processed, and the pause state will not be indicated until all data is processed.
- **Resume:** Click Resume to continue a paused test or replay.
- **Stop:** Click Stop to terminate a paused test or replay. This is the proper way to end a test. Any buffered data is saved (if auto-dump is on).
- **Abort:** Click Abort to terminate a non-paused test. This is a quick way to exit a test and can be used even if the hardware is not responding. It is recommended for use only in emergencies or when data isn't being saved to disk as any buffered data is lost.

3.3.3.4 Time Mark

Click this command (or press the ALT+T keyboard shortcut) to add a Time Mark (a.k.a. 'Test Point') to the data during the test. This message represents a point in time of significance to the tester (i.e., beginning of the loading cycle, important observation, etc.). It is saved with the data for future replays and is displayed on all-time-based graphs (graphs that have 'time' as their x-axis parameter) as vertical bars. It is shown on the Line Listing Display as an ID 211 along with the time of the test.

The statistic 'Test Point' displays how many Time Marks (Test Points) have been added to the data.

3.3.3.5 Time Mark User Comment

The time mark described in section 3.3.5.4 does not contain any additional information about why the user-generated a time mark, but it can be generated quickly (via the ALT+T keyboard shortcut). The Parametric Time Mark Comment can be used to add some descriptive text to the time mark that can be shown in the line display during acquisition or replay. This function is assigned the ALT+P shortcut and, when initiated, will display the Set Comment and Parametric dialog box. The dialog box has several pre-defined comments (HOLD, LOAD...) that can be chosen as well as a field to enter whatever text is appropriate to describe the time mark (Other). The time of the mark is taken at the time the OK button is pressed.

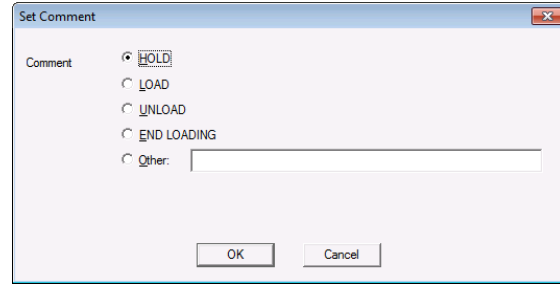


Figure 3-65. Time Mark User Comment

3.3.3.6 Parametric Time Mark Comment

This time mark has the same function as the Time Mark User Comment with the addition of being able to specify the value of parametric 1. This is useful in situations where a physical connection to parametric 1 is not available. The value entered for the parametric is not shown in the line display and does not affect the value of the parameters (hit or time-driven) stored in the dataset. The Parametric TM Management utility can use this value to modify the values in the dataset so that the parametric value can be used for graphing. Refer to the Parametric TM Management utility section for more information.

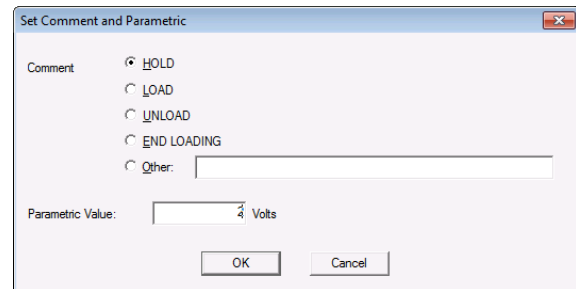


Figure 3-66. Parametric Time Mark Comment

3.3.3.7 Turn Waveforms ON/OFF

Click this during data acquisition to toggle waveform transmission on or off. These commands are only available if you have the Waveform option installed. Disabling the waveforms will not affect the collection of waveform-based features (like peak frequency, frequency centroid, and partial powers).

3.3.3.8 Display Last Alarm...

Click this to view the Alarm dialog box. This displays information on all alarms detected during the current test (or the previous test if no longer in test). Use the drop-down list box to select which alarm to view. All of the alarms can be 'Silence Alarm' – Mutes the audible alarm without clearing any alarm outputs.

'Lookup Next Waveform' – Only used with FT alarms. It will use the Selected Data Display to show the FFT that caused the alarm. The waveform can also be inspected. This requires that 'Enable Data Selection Feature' be checked in the 'Data Storage' dialog box (section 3.3.5.1) when the acquisition is started. This is also available in the replay.

'Reset All Alarms' - Resets all alarm outputs and turns off the audible alarm (if on).

'Close Dialog' – Closes the dialog box; does not clear any alarms or clear the alarm list.

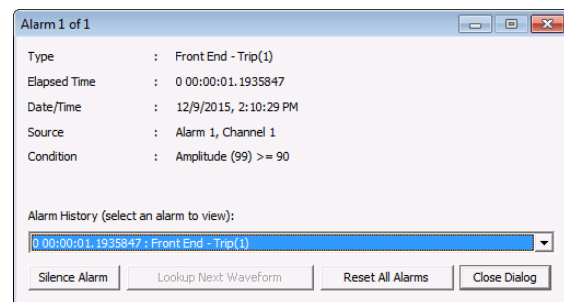


Figure 3-67. Display Last Alarm

This command is available only if there has been alarm activity during the current test (or the last one if ended). The alarm preferences dialog box (section 3.3.2.10) has a setting that will disable the automatic popup of the dialog. Note that most alarm types also have a setting in their configuration to disable the popup for that particular alarm type.

3.3.3.9 AST

The purpose of the Auto Sensor Test is to verify that nothing has gone awry with the sensor coupling before or after a test. The Auto Sensor test will run a short acquisition test while pulsing through to each sensor sequentially, using the current settings in the Test and Hardware Setup menus. An AST test should be run before and after an AE test and results compared to make sure there were no sensor sensitivity changes. Also, in order for AST to work, 1220A-AST preamps or Integral Sensors with the AST option (e.g., R15I-AST) are required. The AST Dialog Box is used to set up and begin the AST.

Before starting an AST, make sure that Amplitude, Counts, Energy, and Duration are selected in the Data Sets tab of the Hardware Setup dialog box (section 3.3.2.1.6) as these features are required by the AST.

AST should be run when the structure being monitored is in a quiet state. AST works by receiving signals generated as a result of pulsing sensors. AEwin has no way of differentiating signals that are from pulses from those that originate in the structure, so it assumes all hit data received while pulsing is occurring is a result of the pulsing. If the structure is emitting unexpected noise, it could interfere with the AST results.

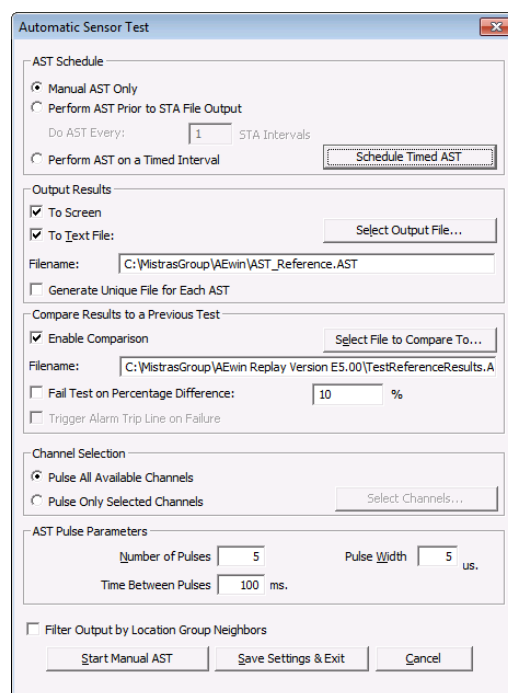


Figure 3-68. Automatic Sensor Test

AST Schedule:

Manual AST Only – Default behavior when not specifying to perform the AST automatically on a schedule.

Perform AST Prior to STA File Output – Used to perform the AST on a schedule and report any AST failures to a website. The STA file is a summary of data collection and messages from AEwin that is generated at a timed interval. The data in this file is eventually uploaded into an off-site database for use by a web server hosted by Mistras Group. There is a field to specify how many STA collection intervals to elapse before performing the AST. For example, if the STA interval in the Internet Monitoring Setup was set to every 15 minutes, setting the STA interval for AST to 4 would perform an AST every hour. If any failures occurred in the AST, they would be reported in the STA upload that would immediately follow the AST.

Perform AST Prior on a Timed Interval – Used to perform the AST on a schedule. Use the Schedule Timed AST button to display a dialog where the interval can be specified.

In this dialog, the date and time of the first AST of the day are specified as well as how often to repeat it. The start time should be set to the earliest time of the day. For example, if the current time is 2pm and the AST should occur at 15 minutes past the hour, every hour set the start time to 00:15:00, not 14:15:00. The schedule restarts at the start of every day, so setting it for 14:15:00 would mean there would be no AST from 12am to 13:15:00. The schedule can also be set to skip a number of days by entering a Repeat Daily Schedule Every N Days to a number greater than 1.

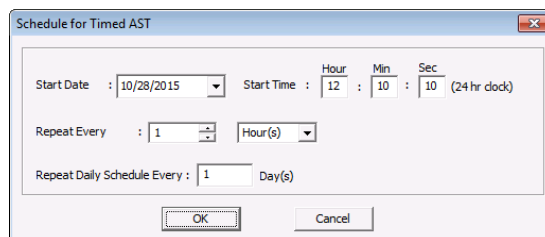


Figure 3-69. Schedule for Timed AST

Output Results:

To Screen – Check if the results should be displayed in the AST Results dialog box that will appear at the end of the AST.

To File – Check if the results should be saved to an ASCII format file. The name and location of the file to be created can be specified by clicking the Select Output File... button or by directly entering the path and file name in the Filename field.

Generate Unique File for Each AST – Check this to automatically create a filename based on the date and time the AST is started. The date and time are appended to the specified filename (as -YYMMDD-HHMMSS). This is especially useful when AST is run on a schedule so that files are not overwritten.

Compare Results to a Previous Test:

Enable Comparison – This must be checked to allow the AST to compare the current results to that of the specified reference file. The name and location of the file to be used as the reference is specified by clicking the Select File to Compare To... button or by directly entering the path and file name in the Filename field. The reference file should be the results of an earlier AST that is performed soon after all sensors are installed, i.e., when the sensors are known to have optimal coupling.

Fail Test on Percentage Difference – Check this and enter a value for the difference (1 to 100). If checked, AEwin will calculate the percent difference $((\text{actual}-\text{ref})/\text{ref} * 100)$ for the deltaT, Amplitude, and Energy between the values in the reference file and those just obtained. If any of the differences (displayed in the column labeled % CHNG, refer to) exceed, the % value entered to the right of the checkbox, a message box indicating the failure will be displayed at the end of the AST. The message box is not displayed if the AST is run on a schedule. Failures are also noted in the text file (if 'To File' is selected as an Output Results option) and in the STA file if internet monitoring is enabled.

Trigger Alarm Trip Line on Failure – When checked, if an AST failure occurs, the Trip digital output on the AE board that contains channel 1 will go high. This digital output can then be monitored by another piece of equipment, such as a PLC, or connected to a relay to allow for annunciation by, for example, a light or alarm. The output pin will go low again at the start of the next acquisition or if manually reset by the AEwin operator via the Alarm Status/Test dialog (section 3.3.4).

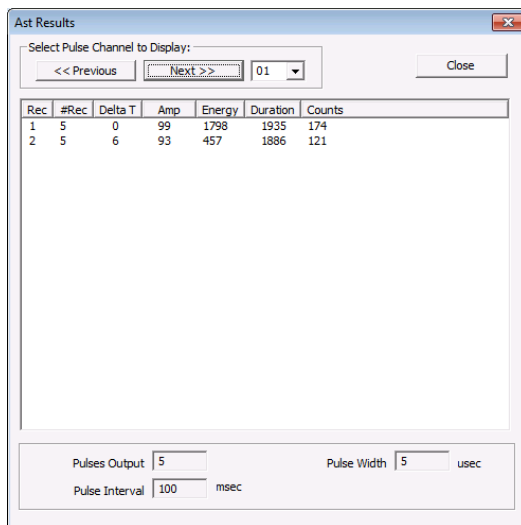


Figure 3-71. AST Result without Comparison

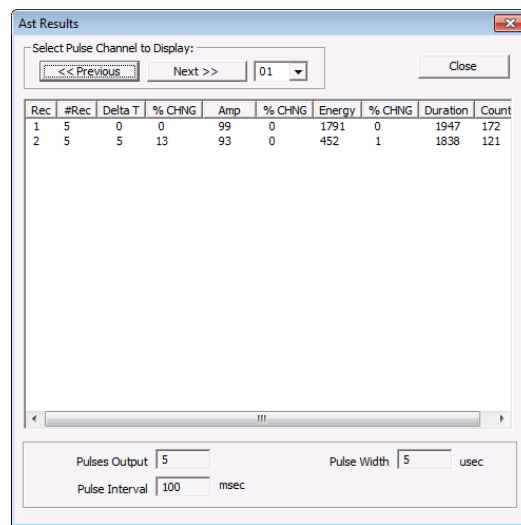


Figure 3-70. AST Result with Comparison

Channel Selection:

Pulse All Available Channels – When selected, all AE channels enabled for data collection will be included in the AST.

Pulse Only Selected Channels – When selected, a subset of the AE channels enabled for data collection can be selected by using the dialog box that is presented when the 'Select Channels...' button is pressed.

AST Pulse Parameters:

These settings control how the pulsing sensors produce the signals.

Number of Pulses: This is the number of times a single sensor will be pulsed. Each active sensor set in the Channel Selection will be pulsed this number of times. This number must be in the range of 1 to 100 (though a value of 5 or 10 is typically sufficient).

Pulse Width: This time, specified in μsec , determines the length of the pulse. It must be in the range of 1 to 20. The pulse needs to travel from the AE input to the pre-amp or sensor. If the cable length is very long (100's of feet), the capacitance of the cable can elongate and flatten the pulse to a point where it is not able to trigger the AST. Increasing the pulse width can help to reduce the effect and provide reliable triggers.

Time Between Pulses: This time, specified in milliseconds, determines the time span between pulses. One hundred to a few hundred milliseconds should be plenty of time for the pulse to be received by all sensors and processed by AEwin.

Filter Output by Location Group Neighbors – The AST output report is typically organized as a table of results for each receiver of a given pulser. When this option is checked, the receivers shown will be only those that belong in the same location group(s) as the pulsing sensor. This can be useful if not all sensors are on the same structure and one or more receiving sensors can't be reached by pulsing sensors.

Start Manual AST – Pressing this button will validate and save the current AST settings, then start an AST. If the system is not currently in the acquisition, starting an AST will first prompt for a filename to save the AE data (a standard DTA type file). Saving data is optional. If data is to be saved, make sure the 'Save to DATA file' checkbox is checked, and a valid file name is entered in the Data Storage dialog box that appears, and then presses Start. If Cancel is pressed, the AST will be aborted. If the system is in the acquisition, this dialog box will be skipped, and any AST messages and data will go to the current file (if one is being saved).

Save Settings & Exit – Pressing this will save the current settings in AEwin until it is closed. It does not force a save of the layout. To keep the settings when AEwin is started, make sure to save a layout file (File->Save Layout or File->Save Layout As... from the AEwin main menu).

Cancel – Any setting changes made from the time the dialog was shown will be discarded.

3.3.3.9.1 Reviewing the Report

If the 'To Screen' option is selected under Output Results, AEwin will display a dialog box that organizes the results by pulser (**Figure 3-55** and **Figure 3-56**). For each pulser, the pulser and all receivers are listed in a table. The pulser is listed because it too receives the pulses.

The columns are:

Rec..... The sensor id (number) of the receiving sensor.

#Rec..... The number of pulses that were received by the sensor.

Delta T The average of the difference of the pulser and receiver hit times for all pulses received.

Amp The average AE Amplitude for all pulses received.

Energy The average AE Energy for all pulses received.

Duration... The average AE Duration for all pulses received.

Counts..... The average AE Counts for all pulses received.

%CHNG .. The percent difference between the reference value and the value in the column to the left. It is used for failure determination.

Use the Previous and Next buttons to move sequentially from pulser to pulser, or use the drop-down list to the right of the buttons to select a specific pulser.

At the bottom of the result dialog box, the pulse parameters (Number Output, Pulse Width, and Interval) are displayed.

If the 'To Text File' option is selected under Output Results, AEwin will write the results to the specified text file. The file contains a header section with the date and time the AST ended, the name of the program that created the file, the original name of the file, and the AST pulse settings.

The header is followed by a listing of the same data found in the results dialog box.

Here is an example:

```
*****
****      Auto-Sensor Test      ****
**** Fri Dec 11 11:18:43 2015 ****
*****

PROGRAM      AWin for Express-8
THIS FILE     C:\MistrasGroup\AEwin\AST_Reference.AST

PULSES OUTPUT      5
PULSE WIDTH        5 usec
PULSE INTERVAL     100 msec

PULSER
1
REC #REC  DeltaT  AMP  ENERGY  DURATION  COUNTS
1   5      0     99   1897      1923     158
2   5      8     91   465       1774     107

PULSER
2
REC #REC  DeltaT  AMP  ENERGY  DURATION  COUNTS
1   5      9     88   275       1502     97
2   5      0     99  1103       2043     165
```

If a comparison to a previous run was enabled and any errors were found, the errors are shown in the text file with an explanation above any failures. The end of the file would contain an indication that there was at least one failure if any were found. The header will also contain the location, name, and collection time of the reference file used.

```
*****
****      Auto-Sensor Test      ****
**** Fri Dec 11 11:19:24 2015 ****
*****

PROGRAM      AWin for Express-8
BASE FILE     C:\MistrasGroup\AEwin\AST_Reference.AST Fri Dec 11 11:18:43 2015
THIS FILE     C:\MistrasGroup\AEwin\test-20151211-111924.AST

PULSES OUTPUT      5
PULSE WIDTH        5 usec
PULSE INTERVAL     100 msec

PULSER
1
REC #REC  DeltaT  CHNG  AMP  CHNG  ENERGY  CHNG  DURATION  COUNTS
1   5      0     0     99   0     1911    1     1923     159
AST Error, Delta Time Change 16 % > than Limit 5 % in Channel 2
2   5      9    16     90   1     453    3     1773     105

PULSER
2
REC #REC  DeltaT  CHNG  AMP  CHNG  ENERGY  CHNG  DURATION  COUNTS
1   5      9     0     88   0     271    1     1503     96
2   5      0     0     99   0    1094    1     2043     165

Test Results Failed the Reference File Comparison, (Percent Change > 5 % )
```

3.3.3.9.2 Messages, Line Display and Graphing

AEwin assigns message identifiers to each data type. For example, AE feature data is id 1, and time-driven data messages are id 2. As explained in the line display (section 3.3.4.5), the message-id is displayed in the line listing. To keep AEwin from processing AST hits (which are artificial in nature) as actual hits, AE data collected during the time AST is active is assigned an id of 10 instead of 1. The format of the message is identical to a real AE hit, only the id is different. The 'Input Data' section of the graph setup defines the source of the data to be displayed by the graph. One of the options is 'AST Hits.' When selected, only the data (Hit and Waveforms) during the AST will be displayed in that graph.

3.3.3.10 Acoustic Property Matrix Generator (APMG)

APMG is a standard feature of AEwin that utilizes auto sensor test (AST) functionality to perform automated round-robin testing of enabled channels. Each sensor is pulsed a designated number of times in a turn, with all sensors acting as receivers.

The APMG function is particularly useful for but not limited to the following applications:

1. **Automatic sensor testing to check sensor coupling status.** It can eliminate tedious manual work for sensor coupling checking with a pencil lead break.
2. **Automated distance and attenuation finding.** With all sensor coordinates being entered, the distance between any two sensors can be readily calculated and displayed. The amplitude attenuation from the pulse source to any other sensor can also be sorted out and displayed instantly.
3. **Automated velocity matrix generation.** With each channel as a pulse and all other channels as receivers, the velocity matrix or delta-T matrix from one sensor to another can be automatically generated. The velocity matrix associated with original waveforms can be used for rock tomography research.
4. **Refining location accuracy by using the ensemble average of the velocity matrix.** In many cases, the AE velocity used to do source location is either based on book value or onsite test with pencil lead break between a few sensors. Not only is the former not very accurate to a specific structure tested, but also the latter is tedious manual work with limited accuracy because only a few sensors are tested. With APMG, velocity matrix can be obtained automatically, and ensemble average velocity can be extracted so that to improve the accuracy of velocity for the specific structure tested.

The APMG function can be started from AEwin -> Acquisition/Replay -> Acoustic Property Matrix Generator, as shown in the figure below. Before start APMG, please make sure you have the waveform collection turned on. This is important if you want to export files later.

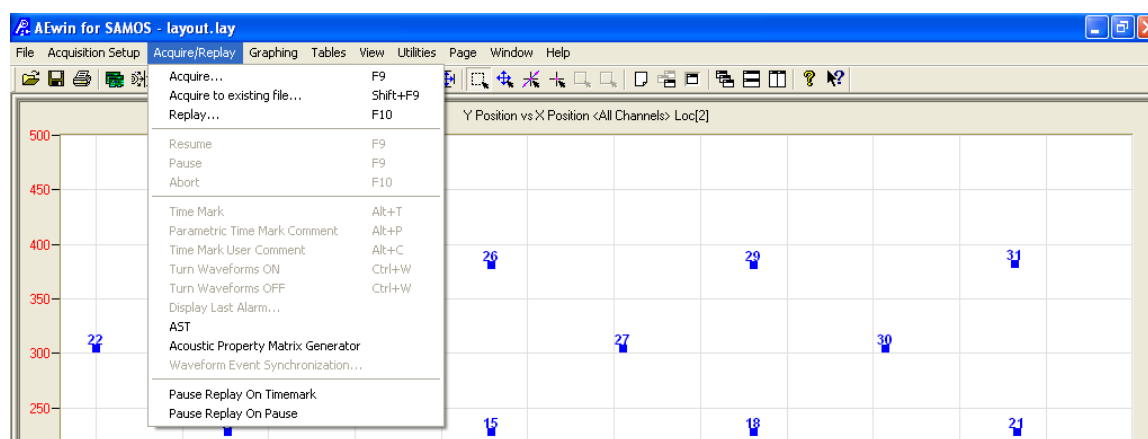


Figure 3-72. Acoustic Property Matrix Generator Start Menu

After entering the APMG, it has two pages or two tabs, as shown in the figure below. The first tab is the APMG matrix, where matrixes of different features, e.g., velocity, delta-T, linear distance from one sensor to another, amplitude, energy, duration, and count features, can be selectively displayed. The second tab is APMG features, where all AE features generated from a specifically selected pulse channel can be displayed and sorted according to feature values.

APMG MATRIX TAB

Unlike with AST, which is done with selected channels only but with no relation to the location group, APMG is uniquely associated with a selected location group because its coordinates are used to calculate distance and velocity automatically. So the first step is to select a location group from the drop-down box near the top left corner, as shown in the figure below. Doing so will display the member channels of that group and enable their checkboxes in the matrix below. The On/Off status of the location group is also displayed.

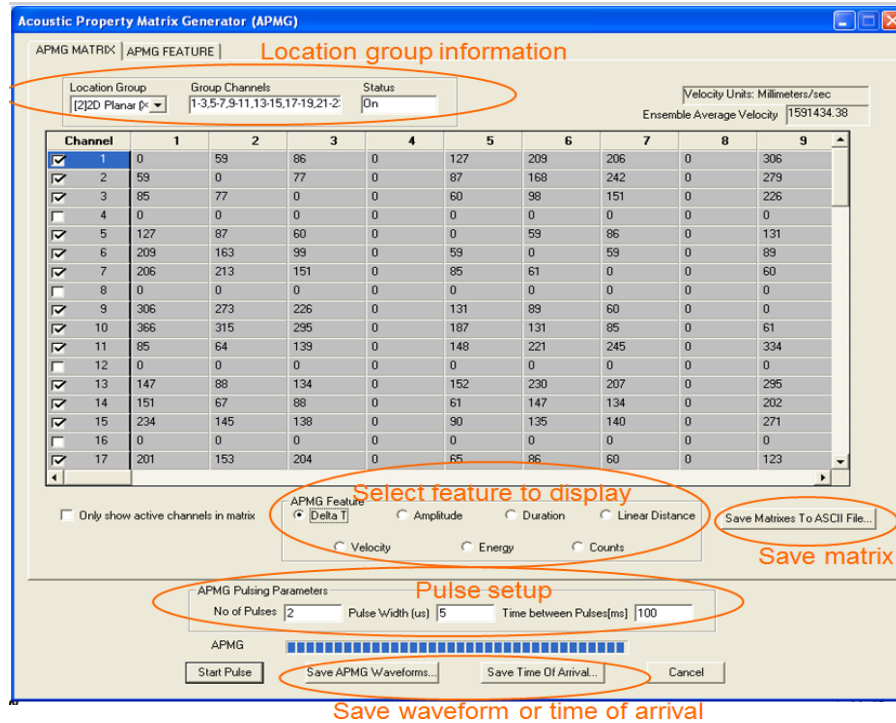


Figure 3-73. APMG Matrix Tab

In the APMG Matrix screen, the following information or actions are also displayed:

- **Ensemble Average Velocity.** This is the average velocity of the whole velocity matrix, excluding the channels that never receive the AST signal. This item is for information only and cannot be reset manually.
- **Selected features and their matrix display.** There are 7 features whose values can be displayed in the matrix, i.e.
 - Delta T (us)
 - Velocity (Unit / s)
 - Amplitude (dB)
 - Energy
 - Duration (us)
 - Counts
 - Linear Distance (Unit). The distance from the pulse sensor to other receiver sensors.

In the matrix display area, the first Column shows all pulse sensors in numerical order. The first row shows all receiver sensors in numerical order. Thus the data on each row shows the received features of a particular pulse sensor. At any time, any one of the above 7 features can be displayed in the matrix.

- **Number of Pulses.** This is the number of pulses to be fired in the APMG. If this value equals to 1, the Matrix table displays the exact features extracted. However, if this value is greater than 1, the features displayed are averaged from multiple pulses.
- **Pulse Width.** This is the width of the pulse in us. This rarely needs changing. You may wish to increase this above the default setting if your cables are very long. The smaller the Pulse Width, the higher the frequency bandwidth and the lower the pulse energy. If a tone burst preamplifier is used, the pulse is just used to trigger the tone burst signal in the preamplifier, and therefore, the pulse width is not important in such a case.
- **Time between Pulses.** This is the time interval between multiple pulses. It should be set wide enough to allow the furthest sensor to receive the APMG pulse.
- **Start Pulse.** This is a command button that triggers the APMG pulse and data acquisition.
- **Save Matrixes to ASCII File...** This is a command button. When it is clicked, the software saves the currently displayed matrix table into an ASCII file that can be called with other commercial programs, e.g., EXCEL Spread Sheet. File name shall be prompted to save the ASCII file. An example of such a saved matrix is shown in the figure below.

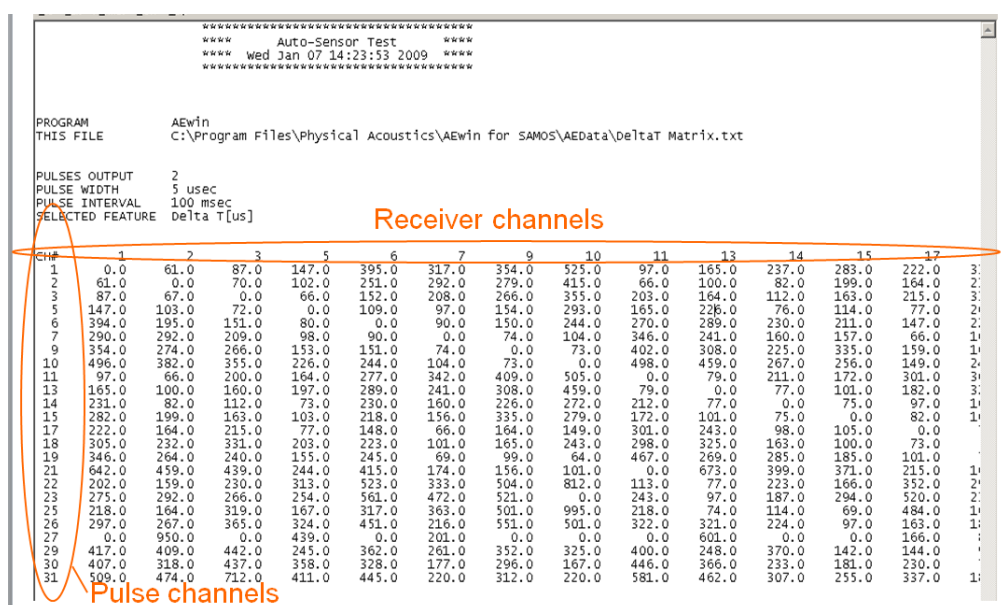


Figure 3-74. Example of saved Delta-T Matrix ASCII file

- **Save APMG Waveform...** This is also a command button. When it is clicked, it saves all waveforms to the ASCII file with suffix .TOW.
- **Save Time of Arrival...** This is also a command button. When it is clicked, it saves time of arrival data captured based on the threshold to ASCII file with suffix .TOA. This file provides a convenience to do tomography analysis without processing waveform. However, if a more accurate time of arrival data is required, the APMG waveform should be saved. A customized program can be used to extract the new time of arrival for tomography analysis from the APMG waveform.
- **Pulse Progress Bar.** This is for information only that shows the progress of the AST pulses.

In order to generate and see the APMG matrix, here are the three steps:

- 1) Select the location group.
- 2) Do pulse setup.
- 3) Click “Start Pulse”

After the “Start Pulse” button is clicked, the progress is indicated by a blue bar above this button. Once complete, the matrix will be populated.

The matrix is organized with rows designating pulser channels and columns designating receiver channels. If you wish, you can disable any that you don't want to use. Additionally, if you only wish to see enabled channels, then check the “Show active channels in the matrix only” button below the matrix on the left.

What the matrix actually displays can be selected from a bank of radio buttons below it. Values like ‘Delta T,’ ‘Amplitude’ and ‘Velocity’ can be viewed. These values are averaged among the number of pulses received. If you wish, these values can be exported to a text file by clicking the “Save Matrixes To ASCII File...” button.

APMG FEATURE TAB

The APMG Feature Tab (figure below) is an alternative display of the APMG data. The APMG feature data is displayed against each pulse sensor.

A particular pulse sensor can be selected through a pull-down dialog box on the top left corner. In addition to the Ensemble Average Velocity, the minimum velocity, maximum velocity, and average velocity in relation to this particular pulse sensor can also be displayed.

One important feature of this Feature Tab is that each column can be sorted upwards and downwards by clicking the interesting column. Then the related feature curve, e.g., the Attenuation curve for the Amplitude, can be plotted against the Linear Distance to the pulse sensor. All other features can also be plotted against the Linear Distance.

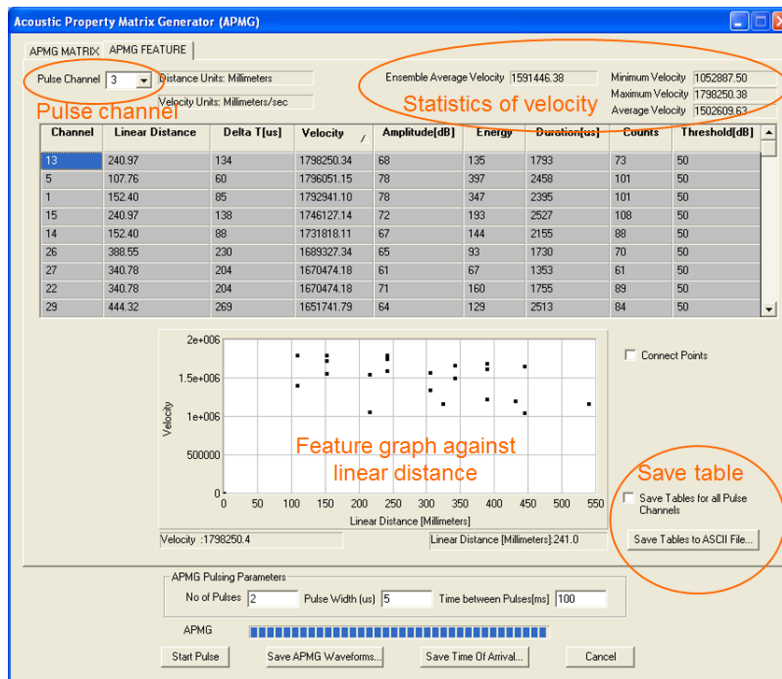


Figure 3-75. Example of an APMG Feature Tab

The pulse information and command buttons are about the same except that the Save Tables to ASCII File ... will save feature tables to ASCII for all pulse sensors in order. An example of saving ASCII format feature data is shown in the figure below.

2.TXT - Notepad

File Edit Format View Help

***** Auto-Sensor Test *****
 **** wed Jan 21 14:11:34 2009 ****

PROGRAM AEwin
 THIS FILE C:\Program Files\Physical Acoustics\AEwin for SAMOS\AEdat2\2.TXT

PULSES OUTPUT 2
 PULSE WIDTH 5 usec
 PULSE INTERVAL 100 msec

PULSER CHANNEL 1

REC	LIN_DIST	Delta_t	VELOCITY	AMP	ENERGY	DURATION	COUNTS	THRESHOLD
17	340.78	166	2052871.88	68	134	2674	128	40
26	444.32	226	1966011.22	62	59	2306	120	40
25	340.78	179	1903780.63	66	110	2624	122	40
7	388.55	207	1877030.38	69	182	2979	114	40
23	388.55	207	1877030.38	65	141	2870	144	40
9	457.20	244	1873770.38	66	135	2961	151	40
19	482.60	259	1863327.88	61	89	2449	111	40
5	240.97	131	1839431.63	72	313	3307	154	40
13	240.97	131	1839431.63	65	103	2935	122	40
2	107.76	59	1826492.75	76	371	3445	153	40
10	538.82	296	1820322.00	63	88	2487	134	40
3	152.40	86	1772093.00	73	190	3108	165	40
11	152.40	87	1751724.13	68	152	2433	132	40
6	304.80	177	1722033.88	67	146	3013	151	40
21	580.32	345	1682092.50	60	54	2097	97	40
31	655.18	394	1662888.00	55	50	2508	126	40
29	538.82	328	1642729.75	63	103	2642	140	40
22	304.80	191	1595811.50	67	121	2766	120	40
15	323.29	203	1592557.75	65	101	2653	148	40
27	431.05	278	1550547.75	56	44	2052	97	40
30	549.49	370	1485097.25	59	70	2562	115	40
14	215.53	146	1476206.38	60	80	2564	119	40
1	0.00	0	0.00	99	7704	6748	236	40
18	444.32	0	0.00	0	0	0	0	40

PULSER CHANNEL 2

REC	LIN_DIST	Delta_t	VELOCITY	AMP	ENERGY	DURATION	COUNTS	THRESHOLD
13	152.40	73	2087671.13	67	135	2916	155	40
19	388.96	189	2058011.88	65	121	3239	147	40
26	340.78	173	1969807.88	66	120	3393	150	40
22	240.97	124	1943270.50	71	343	3241	154	40
9	388.55	203	1914016.00	70	173	3168	163	40
23	304.80	160	1904999.88	64	106	2994	153	40

Figure 3-76. Example of Saved Feature ASCII File

3.3.3.11 Open Selected Dataset Display, Clear Selected Dataset, Waveform Event Synchronization

These items are covered in section 3.9, Data Selection and the Selected Dataset Display Dialog Box.

3.3.3.12 Pause Replay On Timemark/Pause

These menu items are used to automatically pause replay whenever a timemark or pause message is replayed. You can toggle these items on or off individually. They allow you to quickly jump to various pre-set points of interest in your data (time marks and pauses set during acquisition). You can resume replay in the normal fashion by pressing F9 or clicking Resume.

3.3.4 Graphing Menu

3.3.4.1 Graph Setup...

The 'Graph Setup...' dialog box is used to customize an existing graph (created using the 'New Graph' or 'Copy/Paste Graph' commands). You can enter this dialog box by selecting (clicking) a graph and selecting 'Graph Setup...' from the Graphing menu or by selecting the graph and pressing the F3 accelerator key, or by right-clicking the graph and selecting 'Graph Setup...' from the menu that pops up. The appearance of this dialog box will vary depending on the type of graph. The example at the right is representative of a simple activity graph.

Section 3.7 details the procedure for creating and customizing graphs. Below is a brief description of the various graph types available and the major pages of the graph setup dialog box.

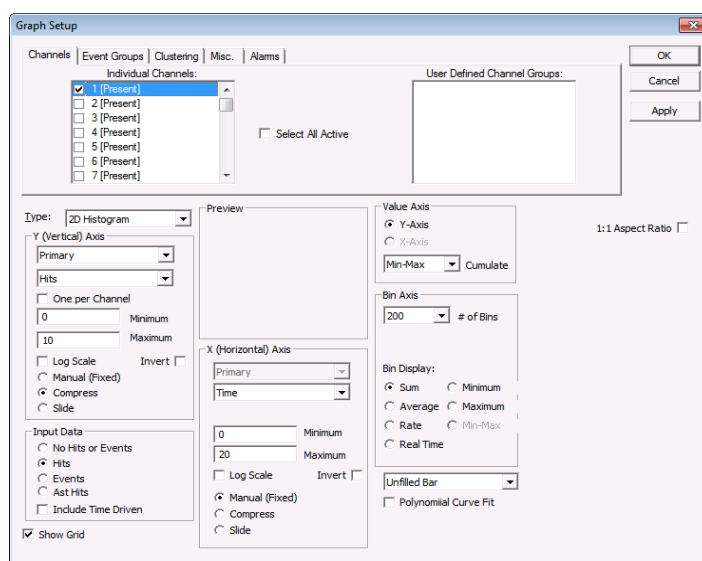


Figure 3-77. Graph Set up

Graph Types:

- Each graph type has a somewhat different set of controls specific to it. These are described below:

2D Histogram

This is the standard Bar (if 'Filled') or Line (if not 'Filled') graph. You can define multiple plots or use the 'One per Channel' checkbox to automatically create one plot for every enabled channel (up to a maximum of 6). The X-Axis is known as the 'Bin' Axis and is divided into a user-defined number of 'bins.' For example, if you set the X-Axis range as 0-100 seconds with 200 bins, then each bin would represent 0.5 seconds.

- Value Axis
 - Cumulate – Use this to select the bin cumulation mode. The default is 'None.' When set to 'Min-Max,' each bin displays not only its own data but that of its predecessors as well. When set to 'Max-Min,' each bin displays not only its own data but that of its successors as well.
- Bin Axis
 - # of Bins – Use this to select the number of bins to be used on this graph.
 - Bin Display – The 'Bin Display' determines how the bin's data is to be displayed. Statistics are kept in every bin. Pick one for them to display.
 - Sum – This will display the 'Sum' of all data in the bin, which is particularly useful when using counter-based features such as 'hits' or 'events.' Other features such as 'energy' or 'counts' lend themselves well to using a bin display of 'sum.'
 - Minimum/Maximum – This will display the 'minimum' or 'maximum' values in the bin.
 - Average – This will display the average value of all data in the bin.
 - Rate – This mode is most often used with an X-Axis of 'Time' or 'DateTime' to display a rate based on the width of the bin. The displayed value is equal to the bin's sum divided by the bin's width. For example, assume a graph with 200 bins and an X-Axis range of 100 seconds. Each bin is 0.5 seconds wide. A bin with a sum of 10 would display a 'rate' of 20.
 - Real-Time – Each bin will display the value of the most recent data it has received. If the Y-Axis feature is 'Amplitude,' then it will always display the amplitude of the most recent hit that applies to that bin.
- Bar Type – Found on the lower right corner, this control is used to choose the appearance of the bars.
 - Filled Bar – The bins are filled with a solid color.
 - Unfilled Bar – The bins are shown as hollow outlines but are otherwise the same shape as 'Filled Bar.'
 - Simple Line – This is the classic "line graph," which simply draws straight lines between the tops of bins.

- Polynomial Curve Fit – This checkbox applies polynomial curve fitting overlaid on the graph.

2D Scatter (Point Plot)

This is the basic point plot graph. Multiple plots can be defined. It is the simplest of the scatter-type graphs but quite flexible and precise.

- This is a good choice to use when precision is very important and when data loads are not high. It is an excellent choice to use in the replay, where presentation is more important than speed.
- High data loads can cause this graph to slow the speed of acquisition and replay. In this case, the '2D Binned Scatter' graph is a better choice for use in data acquisition.
- A safety feature can limit the number of points the graph can show at any one time. Once it reaches this limit, old points are replaced by new ones. This default limit is quite high and can be modified via the 'Acquisition/Replay Modes' dialog box (found on the 'Acquisition Setup' menu). This is often disabled for replay but is very important for data acquisition.
- Point Size – Found on the upper right side, this control allows you to control how large the points are drawn.

2D Binned Scatter

This variant of '2D Scatter' uses a different strategy for data display. It divides the graph area into a grid measuring a selectable number of bins high by the same number of bins wide. Thus, every dot represents not a discrete point but a small range of values.

- The advantage of this over '2D Scatter' graphs is that it can display large amounts of data quickly. 2D Scatter graphs can slow down when burdened by more than 100,000 points or so and will eventually run out of memory around 650,000 points (depending on the setup). This can be a problem during extremely long tests and replays. 2D Binned Scatter graphs rarely suffer from this 'data overload.' The disadvantage is a slight loss of resolution. Due to the binned structure, there are a finite number of values across an axis for a finite number of possible distinct points. To calculate the resolution of a graph axis divides its full scale by the number of bins. 200 bins is a good setting to maximize speed with only a slight loss of resolution.
- Multiple plots can be defined.
- Point Size – Found on the upper right side, this control allows you to control how large the points are drawn.
- Resolution – Found on the upper right side, this control allows you to set the resolution of the grid. This is analogous to the '# of Bins' control on histograms.

2D Colored Scatter

This variant of '2D Scatter' is nearly identical in operation to '2D Binned Scatter' but uses a single plot that is colored by value.

- On the right-hand side panel, you choose the 'Color Feature' (counts, duration, etc.), define the various color level maximums (or choose 'compress'), and set the display mode. The color levels define what color a point on the graph is based on its value. For instance, assume you set the color feature to be energy and set level 1 (yellow by default) to be 1000 and level 2 (purple) to 2000. If you get a single hit with energy 600, the point displayed will be yellow in color. If you got a single hit with energy 1001, then this point would be purple.
- Sum, Average, Minimum, and Maximum modes display (the color of) the sum, average, minimum, or maximum values respectively of all points that occupy the same grid location. 'Count' mode ignores the color feature and simply displays a count of the number of points that occupy the same grid location.

Windowed Histogram

This variant of '2D Histogram' works similar to a normal 2D histogram but only shows data within a specified "window" of time.

- As time passes, old data is discarded from the graph, showing only recent data. If you are interested in seeing the "most recent data" or "the current rate," then this is the graph for you. The user can specify the duration of the window and how often the graph is updated during that duration. It is during these graph updates that new data is added, old data removed, and "Windowed Alarms" (if enabled) are evaluated.
- Selecting this graph type adds a special "Windowed Histogram" page at the top of the screen. This must be set up carefully to make full use of this graph type. See the section below on the 'Windowed Histogram Page' for more info on how to customize this graph type.

3D Histogram

This is the standard 3D bar graph. It operates in much the same way that 2D Histograms do. Both the X and Z axes are 'Bin' axes.

- **Z (Depth) Axis**
 - This section is used to set up the depth axis. It is very similar to the 'Y-Axis' and 'X-Axis' sections.
 - **Feature** – Though not labeled, this list box is used to select the feature to be used on the Z-Axis.
 - **Minimum** – Enter the initial minimum axis limit here.
 - **Maximum** – Enter the initial maximum axis limit here.
 - **Log Scale** – Check this box to employ a logarithmic scale to the Z-Axis.
 - **Invert** – Checking this box will reverse the Z-axis.
 - **Manual, Compress** – Select the axis scaling method.
 - **Cumulate** – Use this to turn on (Min-Max) cumulation.
 - **# of Bins** – Use this to select the number of bins to be used on the Z-Axis.
- **Bin Display** – The 'Bin Display' determines how the bin's data is to be displayed. Statistics are kept in every bin. Pick one for them to display.
 - **Sum** – This will display the 'Sum' of all data in the bin.
 - **Average** – This will display the average value of all data in the bin.
- **# of X-Axis Bins** – Use this to select the number of bins to be used on the X-Axis.
- **Flat Traces** – Selecting this checkbox will change the bins to be pyramidal in shape. This can sometimes result in faster graph update rates during high data loads.
- **View Direction (Theta/Phi)** – 3D graphs can be angled using the mouse. These controls allow you to change the default viewing angle of the graph.

3D Scatter (Point Plot)

This is the standard 3D point plot graph. It operates in much the same way that 2D Scatter graphs do except with a depth (Z) axis.

- **Z (Depth) Axis**
 - This section is used to set up the depth axis. It is very similar to the 'Y-Axis' and 'X-Axis' sections.
 - **Feature** – Though not labeled, this list box is used to select the feature to be used on the Z-Axis.
 - **Minimum** – Enter the initial minimum axis limit here.
 - **Maximum** – Enter the initial maximum axis limit here.
 - **Log Scale** – Check this box to employ a logarithmic scale to the Z-Axis.
 - **Invert** – Checking this box will reverse the Z-axis.
 - **Manual, Compress** – Select the axis scaling method.
- **Point Size** – This allows you to control how large the points are drawn.
- **View Direction (Theta/Phi)** – These controls allow you to change the default viewing angle of the graph.
- **Structure Transparency** – This controls how transparent or opaque the structure (if any) is.

Waveform

Standard single plot Waveform graph.

- **Y-Axis**
 - **Auto** – Auto scales the Y-Axis.
 - **Auto-Symmetrical** – Auto scales the Y-Axis in a symmetrical manner (0 is always the middle value).
 - **Manual** – Scales the Y-Axis using manual settings on the 'Y (Vertical) Axis' panel on the left side.
 - **Manual-Symmetrical** – Scales the Y-Axis using manual settings on the 'Y (Vertical) Axis' panel on the left side but ensures that it is symmetrical.
- **X-Axis**
 - **Auto** – Auto scales the X-Axis.
 - **Manual** – Scales the X-Axis using manual settings on the 'X (Horizontal) Axis' panel.

Power Spectrum

Standard single plot Power Spectrum graph. The settings are nearly identical to the Waveform graph type.

Spherical Location

This graph type is used exclusively for the display of spherical events. Many settings are pre-configured and cannot be changed. A number of settings specific to this graph type are available.

- **Spherical Grid** – Use this to customize (or remove) the grid spacing.
- **Cartesian Grid** – When checked, the graph will surround the graph with Cartesian X/Y/Z axes.
- **Legs** – This is used to display/hide the legs on a spherical structure (if any were enabled on the structure).
- **Legs Grid** – This draws a grid of lines around the legs (if present).
- **Point Size** – This allows you to control how large the points are drawn.
- **View Direction (Theta/Phi)** – These controls allow you to change the default viewing angle of the graph.
- **Structure Transparency** – This controls how transparent or opaque the structure (if any) is.

Cylinder Location

This graph type is used exclusively for the display of cylindrical events. Its controls are nearly identical to those for Spherical Location.

Conical Location

This graph type is used exclusively for the display of conical events. Its controls are nearly identical to those for Spherical Location.

Felicity

This graph is used to show the Felicity effect (AE quantity vs. Load).

Additionally, at the top of the dialog box is a row of tab pages. Most of these are available for all graph types, while others are visible only when certain graph types are selected.

The 'Channels' page (at right) allows you to select which channels and/or channel groups will be active on your graph. Select channels from the left panel and channel groups (if any are defined) from the right panel.

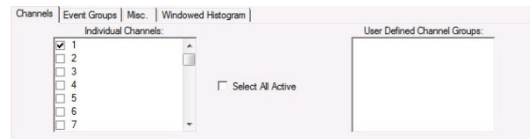


Figure 3-78. Channels

The 'Event Groups' page (at right) allows you to select which event groups will be active on your graph.

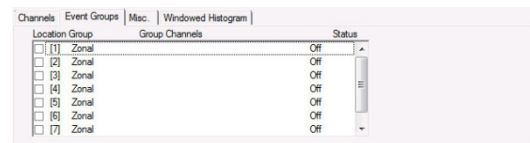


Figure 3-79. Event Groups

The 'Clustering' page (at right) allows you to customize your graph to display cluster information. See section 3.8 for details.

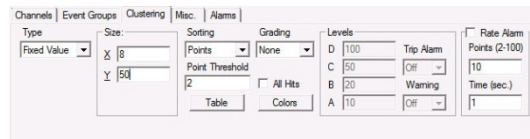


Figure 3-80. Clustering

The 'Misc' page (at right) allows you to set various graphing options that don't fall into any other category. From here, you can turn on the 'color legend.' This displays a small chart on the right of the graph describing what each color means. You can also designate your graph to be a 'history' graph. See section 3.10 for details on how to use the Cluster/Area History feature. 'Graph Name' is used to uniquely identify a graph. It will use this name whenever a Graph Alarm or Windowed Alarm is reported (in the line display or elsewhere).

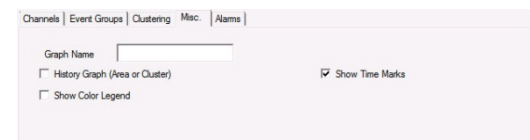


Figure 3-81. Misc.

The 'Alarms' page (at right) allows you to enable and customize Graph Alarms on your graph. See section 3.11 for details on Graph Alarms. Employment of the 'Graph Name' feature (in the 'Misc.' page) is highly recommended.

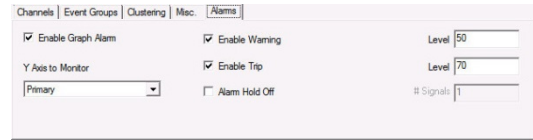


Figure 3-82. Alarms

The 'Windowed Histogram' page (at right) allows you to customize a 'Windowed Histogram' and add an optional 'Windowed Alarm' to it. This page is visible only when working with a graph of this type.

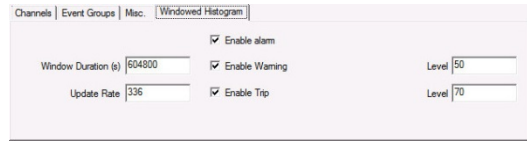


Figure 3-83. Windowed Histogram

- The 'Window Duration' represents the width of the window in seconds. For example, if you are interested in seeing only data from the last hour, then set this to 3600.
- The 'Update Rate' determines how many times during the window duration that the graph is updated and windowed alarms (if enabled) are evaluated. If you want the graph to update every 10 seconds during the 1-hour window duration, then set this to 360.
- **Performance Note:** This graph type can be very flexible, but certain combinations of settings can result in a slow graph, particularly when you have many of them. In general, the higher the 'Update Rate' and the higher the '# of Bins' (on the main section of the dialog), the slower the graph. This will vary based on the performance of your system, but if you have trouble, try lowering one or both of these settings.
- Windowed Alarms - These work like graph alarms except for the following:
 - Alarms will automatically clear themselves if the data goes below the threshold.
 - Alarms are triggered and cleared on a per-bin basis rather than on a per-graph basis. The alarm display dialog and the line display will show details on which bin was triggered and what range of values are represented by that bin.

See the section on Windowed Alarms for detailed information on how to set up and use this alarm type.

3.3.4.2 Graph Filters...

Graph Filters are used to restrict the data that gets plotted on a particular graph by specifying a graph parameter (such as amplitude, energy, duration, parametric) and a range of values to either plot (accept) or not plot (reject). The parameters can be the same as or different from the graph axis parameters of the graph to which they are applied. They only affect the data being plotted; they do not affect the data going to the disk. The Graph Filter dialog box is launched by either right-clicking on a graph and selecting 'Graph Filters' or by selecting Graph Filters from the Graphing pulldown of the main menu.

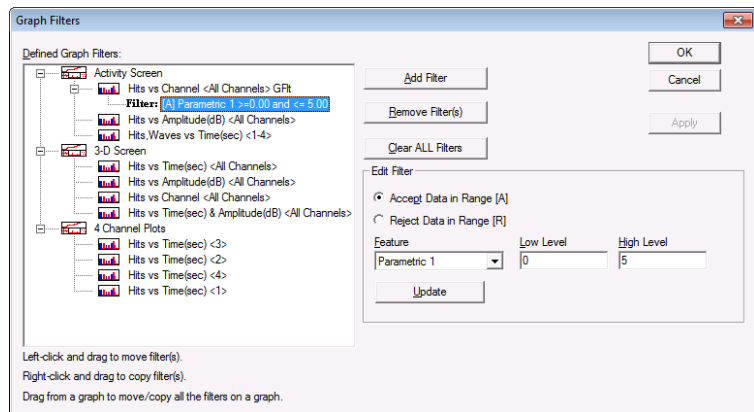


Figure 3-84. Graph Filters

The left side of the dialog box contains a tree control showing the 'Defined Graph Filters.' This control displays information in three levels. At the highest level are the currently defined pages of graphs. The next level shows all the graphs that belong to a page. The final level shows all the filters that are defined for the graph. If a given level has a '+' sign to its left, then it contains items that are hidden. Left-click with the mouse on the '+' to expand the level. Expanded levels have a '-' to the left of them. Left-click on the '-' to compress the level. The text for the screen is the same text that is used on the 'Page Bar' at the bottom of the page. The text for the graph is the same as the caption above the graph. When the dialog box is first displayed, it highlights the graph that is currently selected. So if the dialog box is displayed by right-clicking a graph and then selecting 'Graph Filters,' the graph that was clicked on will be the graph that is highlighted. Filters are displayed with the format [A/R] Feature >= Low-Level <=High

Level. [A/R] will be [A] for accept filters and [R] for reject filters. The buttons to the right of the tree control are enabled or disabled depending on which item is highlighted in control.

Add Filter: This will be enabled when a graph or filter is highlighted. Selecting it creates an empty filter under the highlighted item. The user must then use the controls in the 'Edit Filter' group to define the filters. The 'Update' button must be pressed to apply the current settings to the filter.

Remove Filter(s): This will be enabled as long as a filter is defined in any Screen or Graph that is highlighted or if a filter is highlighted. Pressing it will remove either the single filter that is highlighted or all the filters that belong to a highlighted graph, or all the filters in all the graphs of a highlighted screen. When the button is pressed, another dialog box will appear that will prompt verification. The filter removal operation can be canceled at this point.

Clear ALL Filters: This will be enabled as long as a filter is defined. Selecting it will remove all filters. When the button is pressed, another dialog box will appear the will prompt verification. The filter removal operation can be canceled at this point.

Accept/Reject: Determines if the data will be accepted (plotted) or rejected if it passes the filter test (i.e., falls in the range of the low and high levels).

Feature: Selects the feature used as a filter. The choices are the same as the graph parameters. Note that all graph parameters are listed here. The user must make sure that the selected parameter makes sense. Only hit data can be filtered (all-time-driven data will plot). If the graph is an event graph, then only the features from the first hit sensor will be used in the test, and the event will plot based on the result of the test on the first hit data.

Low/High Level: Determines the range of the filter. The range is inclusive of the limits (as denoted by the $\leq$ and $\geq$ in the graphical representation of the filter in the 'Defined Graph Filters' control). Any floating-point value can be entered, and the range is not validated.

Update: This button must be pressed to change the values in the 'Defined Graph Filters' control.

Apply: This button takes the filters as they are currently defined in the 'Defined Graph Filters' control and applies them to the system. The dialog box is not dismissed, but the settings are now 'locked-in.' Pressing the Cancel button after pressing Apply will not restore the filters to the state they were in before the dialog box was displayed.

Cancel: This button closes the dialog box and discards any changes made since either the dialog box was opened, or the Apply button was hit.

Ok: Applies the current settings and closes the dialog box. Note this will overwrite any settings previously 'locked in' with the Apply button.

Drag and Drop:

The text below the 'Defined Graph Filters' control explains how to move or copy a filter or group of filters.

If a filter is dragged with the left mouse button, it can be moved to another graph by dropping (release the mouse button) on a target graph.

If a filter is dragged with the right mouse button, it can be copied to another graph by dropping (release the mouse button) on a target graph.

If a graph is dragged, then all the filters in the graph will be moved/copied to the target graph.

If the target graph is off the display, hold the dragged item very close to the top or bottom of the control to initiate auto-scrolling.

If the target graph is in a collapsed page or graph ('-' to the left of the target), hold the dragged item over the collapsed page or graph, and it will automatically expand.

Tips:

- Reject is used over accept. If one filter on a graph is defined as [R] Amplitude $\geq 0 \leq 50$ and another filter on the same graph is [A] Amplitude $\geq 45 \leq 50$, then no-hit will be plotted. The [A] filter winds up filtering out all hits with Amplitudes > 50 and < 45 . The [R] filter removes all the hits in the [A] filter range because it includes the entire range of the [A] filter. Think carefully about what is to be accomplished and test the graph first whenever possible.
- No more than 20 filters per graph can be defined.
- The graphs indicate if they have any filters defined by placing 'Flt' on the caption at the top of the graph. To quickly make sure that there are no filters, bring up the dialog and check the Clear Button status. It will be disabled (grayed out) if there are no filters on any graphs.
- Don't forget to press the Update button whenever a filter is added or modified.

3.3.4.3 Colors

The 'Colors' submenu is used to set the various colors used in AEwin. Refer to the pictures below for examples of how some of the settings affect the graphs. Note that color settings are not stored in the layout files but are stored within Windows, which keeps the colors the same no matter what layout is used. The colors also remain the same each time AEwin exits and is restarted.

- 3D Histogram Colors: This is the same color selection as found in the Graph Setup dialog box for the y-axis of the 3D Histogram.
- 3D Structure Grid: Sets the color for the outline of the structure.
- Cluster Colors: Set the colors of the various cluster grades used on cluster graphs. This is the same selection as found by pressing the Colors button in the Clustering tab of the Graph Setup dialog box.
- Colored Scatter Graph: Set the colors of the various levels used by Colored Scatter Graphs. This is the same color selection as found in the Graph Setup dialog box for the 2D Color Scatter type.
- Graph Background Color: Set the background color of the graph frames.
- Legs Color: Set the color of the legs used on spherical location graphs.
- Nozzle Color: Set the color of the nozzles used on spherical location graphs.
- Plot Colors: Set the colors used by the plots of all other graph types. 'Plot 1' is used by all graph types other than Colored Scatter Graphs. 'Plot 2' through 'Plot 6' are used only by graphs that can use multiple plots (2D Histogram, 2D Scatter, 2D Binned Scatter).
- Sensor Color: Set the color of sensor icons used on event-based graphs.
- Structure Color: Set the color of the structure used on event-based graphs.
- Structure Outline Color: Set the color of the structure used on event-based graphs.
- Weld Color
- Import Color Scheme
- Export Color Scheme
- Recent Color File

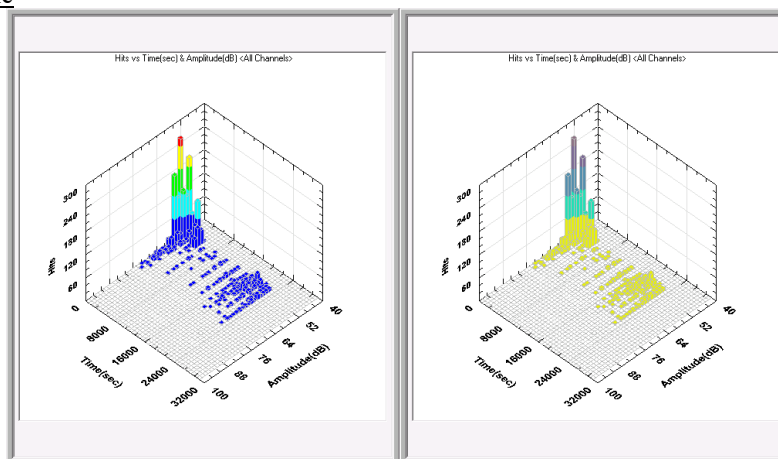


Figure 3-85. 3D Histogram Colors

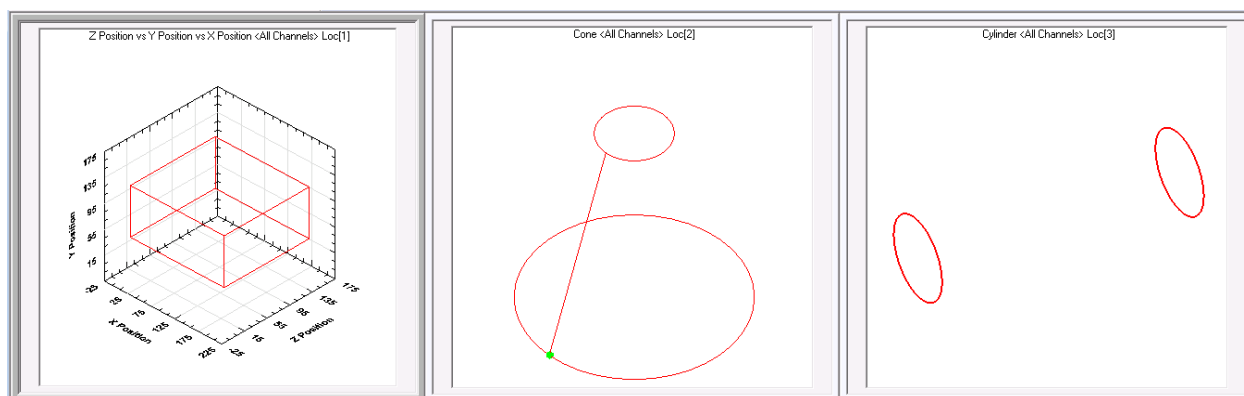


Figure 3-86. 3D Structure Grid Color Examples

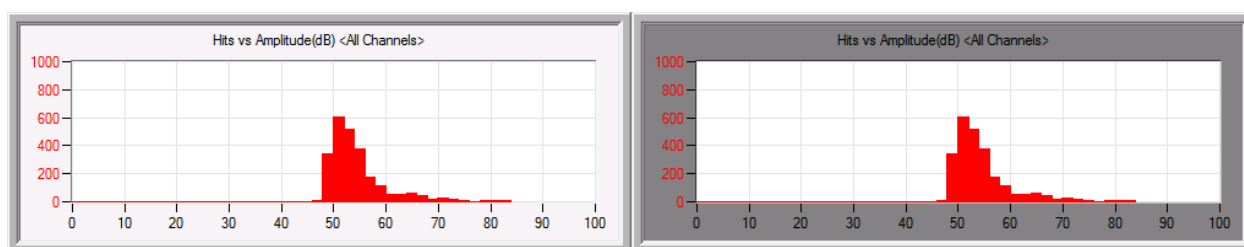


Figure 3-87. Background Color

3.3.4.4 New/Move/Close Graph or Table

- **New Graph:** Use this command to create a new graph. The new graph will be a copy of the currently selected graph. If no graph exists, then the new graph will be a simple 2D Histogram of hits vs. time. Use graph setup to customize it.
- **Move Graph or Table:** Use this command to move a graph or table using the mouse.
- **Close Graph or Table:** Use this command to close the selected graph or table.

3.3.4.5 Cut/Copy/Paste Graph or Table

- **Cut Graph or Table:** Use this command to cut a graph or table, presumably for later pasting on another page.
- **Copy Graph or Table:** Use this command to copy a graph or table, presumably for later pasting on the same or another page.
- **Paste Graph or Table:** Use this command to paste a cut or copied graph or table onto the current page.

3.3.4.6 Export Data to File...

This option on the right-click menu allows you to export the numerical values used in plotting a 2D Histogram onto a text file. These values are outputted to the corresponding x-y values found on the graph and are properly tabbed. This allows the user to import the values onto an Excel graph, for instance, for proper analysis. When the option has been selected on the right-click menu, the following dialog box will appear:



Figure 3-88. Select Output File

Here, the user can save the text file containing the numerical data onto any location of their choosing.

Note – This option is only enabled for 2D Histogram graphs. If the user attempts to right-click on any other type of graph, this option will be disabled.

3.3.4.7 Turn Waveform Threshold Lines On/Off

Waveform graphs can be displayed with horizontal lines denoting the threshold and a vertical line denoting the time of the first threshold crossing. Use these commands to turn the display of these threshold lines on or off on all waveform graphs. You can also turn threshold lines on or off on individual graphs by right-clicking the graph and checking (or unchecking) 'waveform thresholds.'

3.3.5 Tables Menu

With this menu, you can create specialized tables. A table is a window that displays mainly text-based information instead of graphical information. Like graphs, tables can be moved, resized, and closed from commands on the 'Graph' menu.

3.3.5.1 Overview Page

The 'Overview Page' is a window that displays a picture (bitmap) of the test article with various colored icons representing alarm states, virtual parametrics, or On/Off Input states of the specified channel(s). An icon is simply a colored square, usually with a number inside it placed by the user as desired on the Overview Page. They are often placed in such a way as to represent the physical location of a sensor or group of sensors on the test article.

Each icon is individually configured by the user. You assign a "Source," which is usually a channel or group of channels, and a "Status Input Type," which is an Alarm type or other reportable status. Lastly, you can designate a page of graphs. This allows the icon to act as a shortcut that, when double-clicked, will bring you to the designated page.

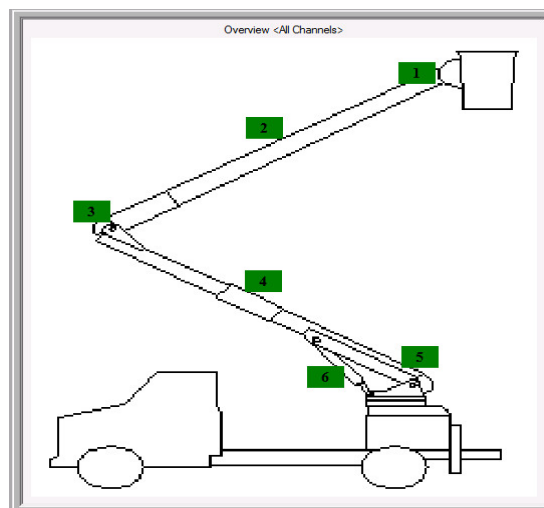


Figure 3-89. Overview Page

An Overview Page can be created by clicking 'Overview Page' from the Tables menu.

3.3.5.1.1 How Alarm and Virtual Parametric Type Icons Work

Most icons display the status of a particular alarm or virtual parametric. These use color to display the alarm state of the channel it represents (for virtual parametrics that have no associated alarms, green is used.). If it represents multiple channels, then it displays the **highest** alarm state of any of the channels it represents. These icons also display a number set by the user when they're first created. This number can be anything and has no effect on the icon's functionality. Icons associated with a virtual parametric also show their current value.

The states and their colors are shown below in order from 'lowest' to 'highest.'

Green	Ok. No alarm on this channel.
Orange	Low alarm on this channel.
Yellow	High or Warning alarm on this channel.
Red	High/High or Trip alarm on this channel.
Grey	The icon is not set up properly, or the alarm type selected does not have the channels it represents enabled. Check the appropriate alarm setup, the dialog box, and the Configure Icon dialog box.

3.3.5.1.2 How On/Off Type Icons Work

An On/Off icon uses color to display the On/Off state of the Channel Group it represents. This state is read from the I/O Board if installed. Unlike with Alarm/VP icons, this type uses Channel Groups exclusively. This helps to prevent potentially serious confusion with the I/O Board's On/Off Input lines. It is important that all channels represented by this icon have the same On/Off state. The best way to ensure this is to avoid channel overlap in your Channel Group Setup. Also, try to pair up I/O Board On/Off input lines and On/Off icons, with both using the same Channel Group.

On/Off icons display their state in color and in words, as shown below.

Green	The Channel Group is 'On.' It displays the word 'ON.'
Red	The Channel Group is 'Off.' It displays the word 'OFF.'
Grey	The state of the Channel Group is unknown. It displays '???'. This is not necessarily an error. Before a test or replay begins, the program does not know what the On/Off states are. If, however you start a test and continue to see an unknown state, it may mean that you haven't completed setting up the icon or that there is no I/O Board On/Off input line setup for the appropriate Channel Group.

How to Set the Overview Page Image

The background image shown on the Overview Page can be changed by right-clicking on display and selecting 'Set Image.' This opens a standard 'file-open' dialog box from which you can select any bitmap (*.BMP) file. Be sure to save your layout when you are done to retain any changes. Note: It is recommended that you save your image files in the same folder as your layouts. If the image file is deleted or moved, AEwin will not be able to find it when it loads the layout. If this happens, it will give you the option to locate it.

3.3.5.1.4 How to Add Icons to the Overview Page

- 1) Right-click the page to bring up the graph context menu.
- 2) Enter 'Add Icon Mode' by clicking the entry by that name on the context menu. You will see a checkmark next to it from now on until you uncheck it or check one of the other modes ("Move Icon," "Remove Icon").
- 3) Left-click on the part of the display where you would like an icon placed. This will open a dialog box requesting a number and a font size. The number is used as a label. The font-size determines the size of the icon.
- 4) Right-click on the new icon and select 'Configure this icon' from the graph context menu. This opens the 'Configure Icon' dialog box. At a minimum, you need to select a 'Status Input Type' and a 'Source.' The 'Source' is a set of Channels, Channel Groups, Virtual Parametrics, or Graphs as appropriate to the Status Input Type. You can also select a page to jump to when the icon is double-clicked.
- 5) Repeat steps 3 and 4 until you are finished. Then Right-click to bring up the context menu and click 'Add Icon Mode' again to exit this mode.

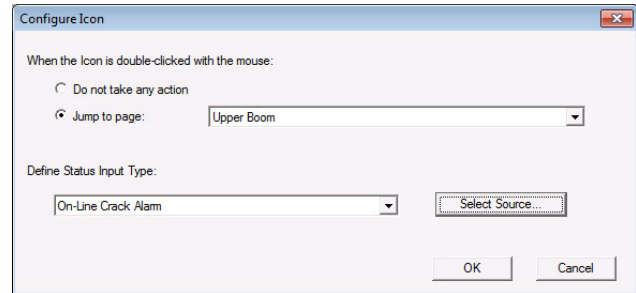


Figure 3-90. Configure Icon

3.3.5.1.5 How to Move Icons on the Overview Page

- 1) Right-click the page to bring up the graph context menu.
- 2) Enter 'Move Icon Mode' by clicking the entry by that name on the context menu. You will see a checkmark next to it from now on until you uncheck it or check one of the other modes ("Add Icon," "Remove Icon").
- 3) You can now drag and drop any number of icons. Left-click the icon you wish to move. While holding the left mouse button down, drag it to the desired location, then release the button.
- 4) Repeat step 3 on as many icons as desired. Then Right-click to bring up the context menu and click 'Move Icon Mode' again to exit this mode.

3.3.5.1.6 How to Remove Icons from the Overview Page

- 1) Right-click the page to bring up the graph context menu.
- 2) Enter 'Remove Icon Mode' by clicking the entry by that name on the context menu. You will see a checkmark next to it from now on until you uncheck it or check one of the other modes ("Add Icon," "Move Icon").
- 3) Left-click the icon you wish to remove. Click OK to confirm the deletion.
- 4) Repeat step 3 on as many icons as desired. Then Right-click to bring up the context menu and click 'Remove Icon Mode' again to exit this mode.

3.3.6 View Menu

With this menu, you can select which toolbars and information bars you would like to display on the screen. There are fixed positions for each of these toolbars defined on the AEwin screen, but some (Toolbar, Acq. Controls, Line Listing Display) can be moved according to the user's preference by grabbing (Left-clicking the mouse) the handle (vertical bar) at the left side of the toolbar and moving it to the desired location. The settings on this menu are saved internally (i.e., not in the layout file).

3.3.7 Utilities Menu

This menu contains a number of useful 'utility' programs for post-test analysis of data files. See "Appendix: Utilities Menu AEwin Software" at the end of Section 8 of this manual for details.

3.3.8 Page Menu

With this menu, you can create insert, delete and rename screen pages. This is an AEwin screen page management menu. You can also perform these functions with the icon keys and a right mouse click on the screen page tab.

3.3.9 Window Menu

This is a typical "Window" menu that you would find on any common WINDOWS program, and therefore the menu contents should be familiar. It allows you to arrange your graphs in cascade or tile patterns as well as edit their window settings. Select 'Maximize Graph' to enlarge the active graph and 'Restore Graph' to return it to its normal size.

3.3.10 Help Menu

With this menu, you can get some online help with the operation of certain features of the software. In addition, the "Options" selection informs the user which options have been installed in their AEwin software, and the "About AEwin" informs the user of the release version and date of AEwin.

3.4 Acquiring AE Data with AEwin for the First Time

This section is provided to help you understand the sequence of getting into and running an AE Test for the first time using the provided Layout (.LAY) files and also gives you insight into a more detailed sequence of carrying out an AE data acquisition test. Before starting in the acquisition, you should have your AE system already set up as per the "Getting Started" section of your AE system manual. This usually entails connecting sensors, preamplifiers, and cables to the AE system, connecting up your AE system, computer keyboard, mouse, and monitor, and booting up your system into AEwin as previously instructed. If you are not ready to connect up the AE system to conduct some type of AE test or at least any AE experimental test on a laboratory test, then skip this section and proceed to the next major section on Replaying an AE test with AEwin.

The key to carrying out an AE test is in setting up the hardware set-ups, graphics, location setups, and display modes to your particular requirements prior to entering the test. Although there are many set-up items, and they might seem confusing for first-time users, PAC supplies good default set-ups for most typical tests, and they are an excellent starting point. As you get more familiar with the system, you can modify these set-ups a little bit at a time and save your new layout (.LAY) setup files for your next AE test. This small step-by-step "Trial and Error" approach builds up your knowledge base until you are totally familiar with all the set-up options and what they mean. Until you get this knowledge, please depend on the PAC prepared.LAY files.

The following are the steps that should be taken when carrying out your first AE test using the PAC-supplied layout files. First, we will outline the steps, and next, we will carry them out.

1. Load the appropriate system Layout (.LAY) file into the system in order to set up the system to your desired operation and display conditions. We will be using the standard **Layout4.LAY** (or **Layout2.LAY** on Spartan and PCI-2 systems) layout file in this example, which sets up 4 (or 2) channels for acquisition.
2. Enter the Acquire menu.
3. Select the datafile name which you want to save to disk.
4. Start Acquisition.
5. Look at various screens, graphs, and data listings during acquisition.
6. Exit data acquisition.

The following is the step-by-step detail on starting and running your first AE test. We assume that you have sensors set up and will be exciting them to generate signals for AE generation or have them attached to a structure, bar, or plate with a pulser or some type of simulated AE source.

3.4.1 Loading a Layout File into AЕwin

From the Menu bar, use your mouse to select **Files**, then select **Open Layout...** A menu window called **Open** comes up. With your mouse, select the **Layout4.LAY** (or **Layout2.LAY**) file from the list of layout files so that this file name appears highlighted in the file box next to **File Name:**. This selects it as the layout file that will set up the AE hardware for a 4 (or 2) channel operation with a good, complete default set-up. To finish the sequence, move the mouse to the **Open** box area on the right side of the **Open** window box and click on it (left mouse key) or press <ENTER>. Upon executing this menu, the selected Layout file is read into the program. If you have made any changes to the current layout file, a message box will pop up asking you if you want to save those changes. If yes, select yes; if no, select no. AЕwin will then load and configure itself to the selected layout's setup. This completes the reading of the LAY (AЕwin setup) file.

3.4.2 Entering Acquisition

Entering acquisition involves steps 2, 3, and 4 of the above sequence. or the Mouse operated icon. To begin, left-click the mouse on the **Acquire/Replay** main menu item, then click on the **Acquire** selection. Alternately, you can enter the Acquire function by pressing the **F9** function key (on the keyboard), or you can select the Acquire icon in the Acquire/Replay toolbar. The Acquire icon is shaped like a traffic light with the green light showing indicating that AЕwin is ready to go into acquisition. This toolbar is usually to the right of the main icon toolbar or just below it. If you do not see this small toolbar (like that shown below), it may not be turned on from the **View** menu. To activate it, simply select the Tools menu and make sure the "Acq. Controls" entry is checked. If not, click on it, and it will be activated and will show up with the other toolbar icons.

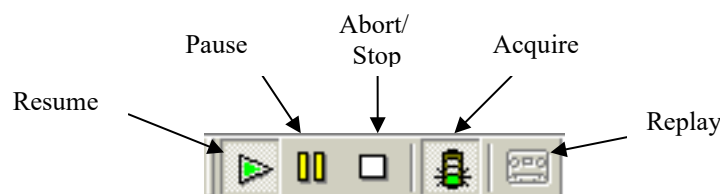


Figure 3-91. Acquire/Replay Icon Toolbar

At this point, the Acquire function, a **Data Storage** files menu box, will appear to signal you to enter a data file name and whether you want to save the file to disk. (An example of the Data Storage file menu box is shown in figure 60.)

First, make sure the "Save to DTA file (**Auto-Dump**)" box is checked if you want to save the data you will be collecting in this exercise. In AЕwin, this is always automatically checked, and you usually have to deselect (uncheck) it to prevent the generation of an AE data file.

Next, check the name shown in the **File Name:** box. If that name is acceptable, then you are ready to start the test by clicking on the **OK** button. If you want a different name, then enter the new name by typing it in when the **File Name** box is highlighted. Once you enter the name (you do not have to enter the .DTA extension), you can hit <ENTER> on the keyboard or click the **Start** button (located to the right of the Filename) with the left mouse button to begin acquisition.

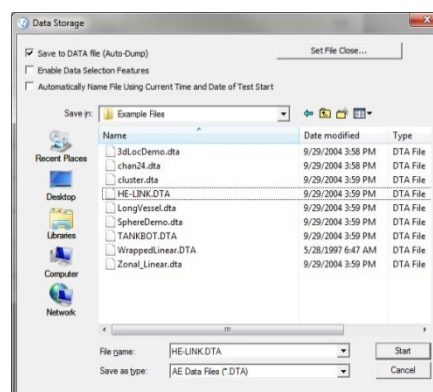


Figure 3-92. Data Storage

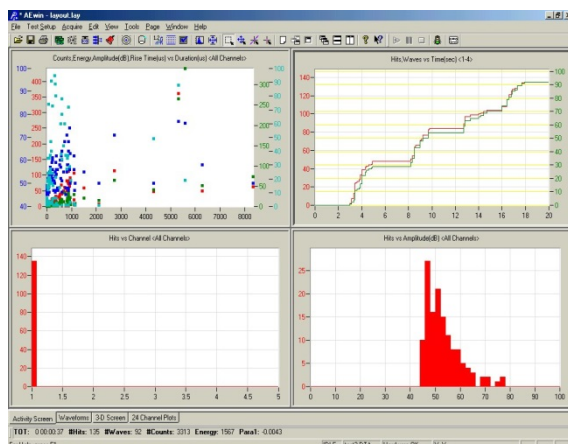


Figure 3-94. Example Activity Screen

At this point, AEwin enters the acquisition screen and waits for you to press the <ENTER> key to start the test. Press <ENTER> to start Acquisition.

The system is now in data acquisition and is collecting and displaying AE information and displaying it on the graphs. Tap the sensor against the table your instrument is set up on, or scratch it with your fingernail.

At first, you will see no change in your Data Acquisition Graph, but once the amount of time specified in **Acquisition Setup → Display Mode** menu parameter “Seconds between Update” time has elapsed (it is usually set for 1 – 2 seconds), you will begin to see a visual representation of your data. Continue tapping the sensors to see the changes to the graphs to verify that you are getting waveforms and graph updates. An example plot showing the first AE graph screen page (selecting the tab, Activity Screen) is shown in figure 61. You should be seeing data visualization like that shown, where graph 1 shows a group of 4 point plots including Counts, Energy, Amplitude and Rise Time (on the Y-axis) versus Duration (on the X-axis), graph #2 (to the right) is showing #AE hits and #waveforms collected versus time,

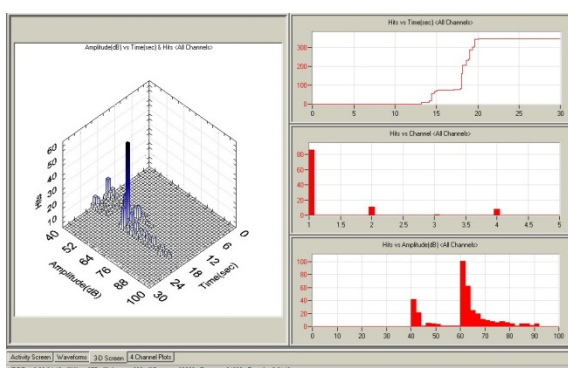


Figure 3-95. Example 3-D Screen

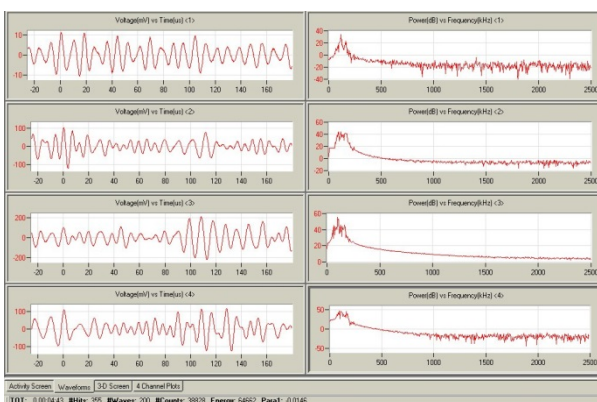


Figure 3-93. Example Waveforms Screen

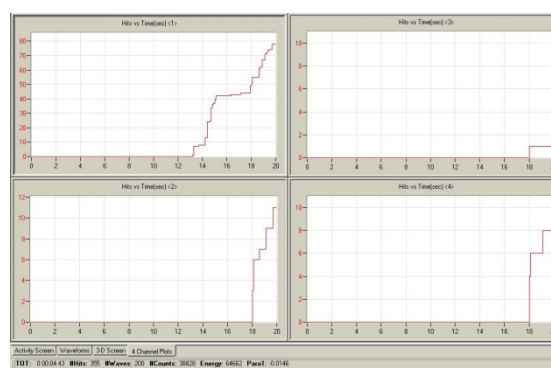


Figure 3-96. Example 4 channel screen

Graph #3 (below graph 1) shows Hits versus channel, and Graph #4 (lower right) shows the Hits versus Amplitude (or Amplitude Distribution).

3.4.3 Actions within Data Acquisition

There are several actions you can take in acquisition to view the graphs and other modes in the instrument. Try these items to see what happens.

3.4.3.1 Selecting and Viewing the Different Screen Pages

You can scroll between screens to view the activity between the different AE graph screen pages. This can be done simply by Left-clicking the mouse on any of the AE graph screen pages. Tabs are located just below the graphs on the left. In the example Layout file, we have defined four different graph screen pages. They are the Activity Screen (shown in figure 61), Waveforms (figure 62), 3-D Screen (figure 63), and the 4 channel plots (figure 64).

These graph screen pages give a good idea of the capabilities and the graph types that you can create with AEwin. On the Activity Screen page, you can see multiple point plots on one graph, multiple line graphs (AE Hits and Waves versus time) as well as bar charts. On the Waveforms Screen page (figure 59), you can see waveforms on each of 4 channels with the associated FFT located alongside the waveform for each of these channels. On the 3-D Graphs screen page (figure 60), you can see a nice 3-D plot with 3 associated graphs alongside. This graph shows that you can set up 3-D graphs (which can be rotated) with AEwin. It also shows that you can size your graphs differently. On the 4 channel plots (figure 61), you can set up graphs with multiple plots on them, one for each channel, so you can link AE with other test parameters. This can all be done in acquisition. As you observe and change graph screen pages, the test continues.

There are many other things that can be done within the acquisition. This includes maximizing a graph to full screen, zooming in on certain areas of a graph, viewing cursors and cursor readouts, printing graphs, saving graphs to a data file or the clipboard, just to name a few.

3.4.3.2 Zooming and Panning on a Graph Region

To zoom a graph, please note the graph-associated toolbar icons for zoom, pan, and cursor selection. This set of toolbars is near the center of the icon toolbar. This toolbar section is shown and defined in figure 62.

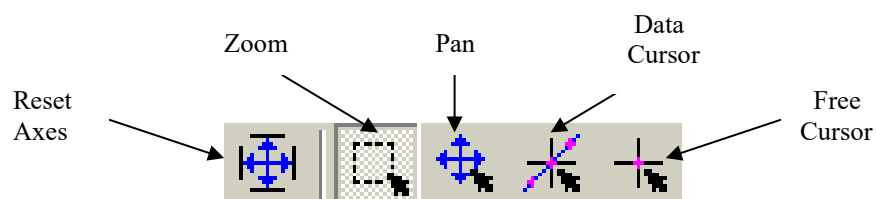


Figure 3-97. Zoom, Pan, Cursor Control Toolbar Icons

To use, select a graph by clicking on it. A good graph to demonstrate these functions on is the Activity Screen Point plots (see the upper left graph of figure 58). Select the graph by clicking on it. Next, click on the zoom icon. This is already selected. Next, with the mouse cursor, define a zoom rectangle by pointing to the starting point on the screen where you want one corner of the zoom rectangle to be. Next, press and hold the left mouse button and drag the mouse to the point where you want the zoom box to end. You will see the dotted box as you drag the mouse. As soon as you let go, the graph will zoom, and only the area that you selected in your zoom rectangle will be displayed in that graph. You will see that the axis scales have changed as well as the number of points in the point plot. You can now pan the plot (move around the zoomed area) by selecting the Pan icon button, moving your mouse pointer to the center of the graph screen that you have zoomed, click the left mouse button, and move the mouse around. You will see the graph pan as you move the mouse. You can continue panning in one direction by letting go of the left mouse button, repositioning the mouse and pressing down on the left mouse button again, and moving the mouse. You can even pan un-zoomed graphs (but there is not a lot of sense in doing that). Zooming and panning can be done on any 2-dimensional graph.

Once done and you want to return that graph back to the un-zoomed state, simply press the “Reset Axes” icon. This is the leftmost icon shown in figure 62. This will bring the graph and its axes back to the normal setting.

3.4.3.3 Zooming, Panning, and Rotation of 3-D Graphs

Zooming, Panning, and Rotation of 3-D graphs can be accomplished but not in the same way as that for 2-D graphs. For 3-D graphs, this requires a combination of the mouse and keyboard to accomplish. Before starting this exercise, click the Graph Screen Page tab “3-D Screen” and follow along to get familiar with the controls. You should see a graph like that shown in figure 60.

Rotating the graph is the easiest function. All you need to do is place your mouse pointer somewhere near the middle of the 3-D graph, press the left mouse button, and while holding the left mouse button down, move the mouse slightly around in a circle. You will see the rotation of the graph change. You can position it anywhere you want.

To zoom, simply place the mouse pointer somewhere near the center of the 3-D graph and when ready, press the **ALT** button on the keyboard, and while holding that down, press and hold the left mouse button down. Then slightly move the mouse down (while holding the ALT button and the left mouse button depressed). As you move the mouse down, the 3-D graph will zoom. As you move the mouse up, the graph will shrink.

To pan the 3-D graph, place the mouse pointer somewhere near the center of the 3-D graph, depress and hold the **SHIFT** button on the keyboard down and at the same time depress and hold down the mouse left button. Then slightly move the mouse around to see the graph pan.

Once done and you want to return that graph back to the un-zoomed state, simply press the “Reset Axes” icon. This is the leftmost icon shown in figure 62. This will bring the graph and its axes back to the normal setting, just like in the case of the 2-D graphs.

3.4.3.4 Using the Position Readout Cursor

Two cursor control icons are shown in figure 62. The cursor readout controls are active in both acquisition and replay, just as zoom and pan controls are. Each cursor provides a slightly different function. The “Data Cursor” identified above in figure 62 provides a discrete cursor movement only to legal data points on the screen. For example, for a waveform or a point plot, the Data cursor would only be able to identify actual existing data points. This, therefore, follows actual points on the screen. The other cursor, called the “Free Cursor,” identified as the rightmost cursor in figure 62, will be able to be moved to any location on the screen and provide a continuous position that is graph-based and not data-based.

The Cursor readout position is located on the status bar at the lower right side of the AEwin screen. The position readout is identified by the **X, Y =** readout box. The position readout will be in actual graph value readings such as Hits, Amplitude, etc. If the graph is a multiple plot graph, the data point read out will be for the exact graph point, so it will be correct regardless of how many plots are on a graph. If the graph is a Binned Scatter or Colored Scatter graph, then the readout will display the range of values represented by the data point.

The cursor readouts can be used on any graph screen. Simply select a graph screen by clicking on it. For this example, let's use the Point plot again on the Activity screen. First, click on it. Next, select one of the cursors. Once a cursor is selected, it usually is drawn with the center at the origin of the graph (the lower-left corner). Note the small hollow circle which identifies the center. Next, you need to position the mouse pointer over the center of the cursor and grab the cursor by pressing and holding the left mouse button down. Once done, you can move the cursor around the graph. If you have selected the Data cursor, you might have a little trouble moving it around the first few times, but you will get used to moving from point to point. The “Free Cursor” is a little easier to operate. Anyway, as you are moving the cursor around (with the left mouse button depressed), you can see the graph coordinates changing as you move. This is a very handy feature to have.

3.4.3.5 Maximizing a Graph to Full Screen

Sometimes you have many graphs on a screen, and it is a little difficult to see the axis values clearly. You can expand any graph to full screen by right-clicking the mouse in the desired graph area and selecting the menu item “Maximize Graph” from the graph menu. You can select the same function from the Main Menu bar, **Window → Maximize Graph** selection. To restore the graph to its normal size and place it on the screen, simply right-click the mouse in the graph area and select “Restore” or **Window → Restore Graph**.

3.4.3.6 Printing any or all Graphs

You can print out any graph, any graph screen, or all graph screens using the various Print commands located in the File menu. To print a given screen of multiple graphs or to print out all screens, simply select the “Print Page(s)” from the files menu. To print out a single graph, first click on the graph, then select the “Print Graph” menu item from the files menu. You can also export a graph or screen to a JPEG file by selecting the “Export to JPG” menu item on the files menu, or you can “Copy Screen to Clipboard.”

3.4.3.7 Pause, Resume, Abort the Test

The Pause, Resume and Abort functions are provided on the Acquisition/Replay Icon toolbar, as shown in figure 56. The icons are fashioned after similar visual controls used in the Audio tape recorder business. To pause your test, simply press the Acquisition/Replay icon labeled **Pause**. In the pause, you can see the test status box change to “Test Paused.” You can also see the Time of the Test clock stop. At this point, the system will stop collecting AE data, will flush all buffers to disk, but the test will not be exited, and the AE data file will remain open. Someone might want to Pause during a test to ignore some event that might add noise to the test (such as pressure or loading ramp-ups, etc.). To resume, just press the **Resume** icon button located next to the Pause button. This restarts acquisition and the Time of Test clock starts back up again, showing the new and current time, taking into account the pause time. To Abort the test, simply click the button labeled **Abort** (see figure 56). Selecting this function stops the test and saves only the data to the data file that already exists on the disk. The only difference between the Abort and the standard exiting of an AE test is that the buffer contents are ignored in the case of an abort while ALL data is saved and written to the AE data file in a normal test exit. This function was mainly implemented for SPARTAN products as it could take the ICC’s many seconds (when AE data rates were high) to empty their buffers. The abort function was convenient in those cases where the user wanted to quickly stop a trial test without waiting for data in the buffers to clear. The abort function is not needed in newer products as the transfer rates are very fast.

3.4.4 Exiting Data Acquisition

To exit data acquisition, first Pause the test by pressing the **Pause** icon button (see figure 56). Then stop the test by pressing the **Abort/Stop** key (the Abort/Stop key becomes a STOP function key whenever the system is in Pause). Upon receiving the Pause, all buffers are flushed to disk, and upon receiving the Stop, the AE data file is closed, and AEwin exits acquisition. The Status box (lower center of the screen) indicates “IDLE,” meaning that AEwin is ready to perform another test or a replay.

If you have made any setup changes before, during, or after acquisition and you want to update the LAY (layout) file (which you should), you may do so by going to the File menu and choosing **Save Layout...** or **Save Layout as...** menu item. “**Save Layout**” will overwrite your existing layout file with the new information, while “**Save Layout as...**” will allow you to rename the layout file as a new file. Then you can follow the guidelines to save the new .LAY file. Enter a name such as TEST1.LAY. It is not recommended to overwrite the existing Layout.LAY file since you may always want to keep this default layout file in case of problems in the future. As computer experience dictates, you should always save your work before carrying out a new task such as acquisition in the case for any reason the power or computer fails or that you simply forget after the test to save the LAY file. This is always a good practice.

3.5 Replaying AE Data

This section helps you understand the sequence of preparing for, recalling, and replaying your previously recorded AE test using the PAC-provided system Layout (.LAY) files and AE data (.DTA) files. It also gives you insight into a more detailed sequence of data analysis and manipulation of data during replay.

Replaying an AE test is almost identical to that acquiring data. As a matter of fact, many of the same steps that have been carried out to perform data acquisition are repeated. The following are the sequential steps that should be carried out when replaying an AE test using the PAC-supplied Layout and AE data files. First, we will outline the steps, and next, we will carry them out. For more details on the actual operation of software and menu items, please refer to the proceeding sections of this manual.

1. Read the appropriate system Layout (.LAY) file into the system in order to load the system to your desired set-up and display conditions. We will be using the **Layout4.LAY** (or **Layout2.LAY** on Spartan and PCI-2 systems) layout file in this example.
2. Enter the Replay function.
3. Select the data file (with .DTA extension) name which you want to replay. We will be calling the data file "Chan24.DTA" for our replay.
4. Start Replay.
5. Move between the various screens, graphs, and data listings during and after the replay.
6. Carry out some data manipulation functions during replay.
7. Exit Replay.

The following is the Step-by-step detail of replaying an AE data file using the PAC provided **Chan24.DTA** data file. For a replay test, you do not need to have any sensors connected to the instrument since you will be playing previously recorded AE tests.

3.5.1 Loading a Layout File into AEwin

This is identical to section 3.4.1 but is repeated here. From the Menu bar, use your mouse to select **F**iles, then select **O**pen Layout... A menu window called **O**pen comes up. With your mouse, select the **Layout4.LAY** (or **Layout2.LAY**) file from the list of layout files so that this file name appears highlighted in the file box next to **File Name:**. This selects it as the layout file that will set up the AE hardware for a 4 (or 2) channel operation with a good, complete default set-up. To finish the sequence, move the mouse to the **O**pen box area on the right side of the **O**pen window box and click on it (left mouse key) or press <ENTER>. Upon executing this menu, the Layout.LAY layout file is read into the program. If you have made any changes to the current layout file, a message box will pop up asking you if you want to save those changes. If yes, select yes; if no, select no. AEwin will then load and configure itself to the Layout.LAY setup. This completes the reading of the LAY file.

3.5.2 Entering Replay

Entering replay involves steps 2, 3, and 4 of the above sequence. To begin, click the mouse on the **Acquisition Setup** main menu item. On the submenu, click on **Display Mode**. Make sure the **Normal** mode is selected. Also, verify that the **Seconds Between Updates** is set for 1 or 2 seconds (this selects the graph update rate) and that the **Hits Before Update is 1000**. If not, set these values by clicking the mouse onto the particular box whose value you want to change and edit those numbers by typing new ones and deleting old ones with the delete or backspace keys on the keyboard. If you want, set in your desired values. When done, click the **OK** window on the bottom of the menu to execute the changes.

Now you are ready to enter the begin replay. This can be started by clicking that icon (that looks like a cassette tape, see figure 56), or pressing the <F10> Function key on the keyboard, or by selecting **Replay** from the **Acquire/Replay** menu. Once selected, a **Replay** window comes up to signal you to select a data file name for replay. Check the name highlighted in the **File Name:** box. If that name is acceptable, then you are ready to replay the test by clicking on the "Start" button. If you want a different file name to replay, then use the mouse to select it in the listed data file names. In this case, you want to select the name **Chan24.DTA**. Click the mouse on that entry. Then, to begin to replay, you can hit <ENTER> on the keyboard or click the "Start" button on the menu. At this point, AEwin begins replay and is displaying AE information as it appeared during data collection on the test.

3.5.3 Actions within Replay

All the actions carried out during acquisition can be carried out during replay. Please refer to section 3.4.3 for details relating to replay actions. There are several actions you can take in the replay to view the graphs and other modes in the instrument. Try these items to see what happens.

3.5.4 Exiting Replay

There is no need to exit replay; when the file is finished replaying, AEwin returns to the Idle status waiting for the user to continue manipulating the graphs in the replay, enter replay to analyze another file, make changes to the setup before continuing, or enter acquisition.

3.6 Using Line Listing Mode

Line (Dump) listing mode is a very familiar feature and function of PAC's AE software. PAC has provided a "line by line" listing capability with all its AE products for many years. This has also been referred to as "Line dump" in the past, but now we refer to it as "Line listing mode." The line listing mode allows the user to monitor any and all types of messages being processed by the AE instrumentation. The information is in a "line by line" text format listing, offering an additional monitoring capability over the graphing of AE data. With Line listing mode, the user can watch messages related to AE Hits, Events, Time-driven data (parametrics), comments, Auto Sensor Test (AST), waveforms, and administrative messages. Watching the line listing scroll by shows how the data were coming in is being processed, giving the user an idea as to the AE activity and severity. In AEwin, the Line listing has been improved with the following features over the capabilities found in our other legacy software programs:

- Ability to scroll the line listing up or down (used to only scroll down),
- Ability to select which data can be shown on the line listing
- Flexible display of the line listing with the AE graphs; the line listing can be adjusted to take up a small part of the screen, operate within a floating textbox, or take up the entire screen.

These features do add some extra utility to the line dump listing that makes it even more important and useful in AE testing. The rest of this section explains how to use these capabilities in AEwin.

The line listing mode can be used in acquisition or replay and offer the same flexibility in both; however, it is recommended to use this mode sparingly in acquisition as its use might slow down the data acquisition speed by a significant amount. It is therefore recommended to use this when data acquisition rates are low, before a test as a prescreening method or after a test in the replay.

The line listing mode color-codes the messages in order to make them easier to spot as they are scrolling by. Please note the following message types and colors as identified below.

Line Listing Message Type	Line listing Color
AE hit data	Blue
AE Event data	Violet
AE waveforms	Blue
Time-driven Data (e.g., Parametrics, ASL, RMS, etc.)	Green
Administrative messages (Pause, Resume, Alarm, etc.)	Red

3.6.1 Enabling Line Dump Listing for Viewing

In normal use, AEwin is kept in graph mode, and the line listing is not displayed (as shown in figure 56). In order to view the line listing, two steps need to be performed. First, one has to "Enable" the Line listing mode, and then the line listing mode needs to be selected for viewing. This is done as follows (whether in acquisition or replay).

To enable the line listing mode, simply click the mouse as follows;

Acquisition Setup → Line Display → Select “Enable Line Display” Checkbox

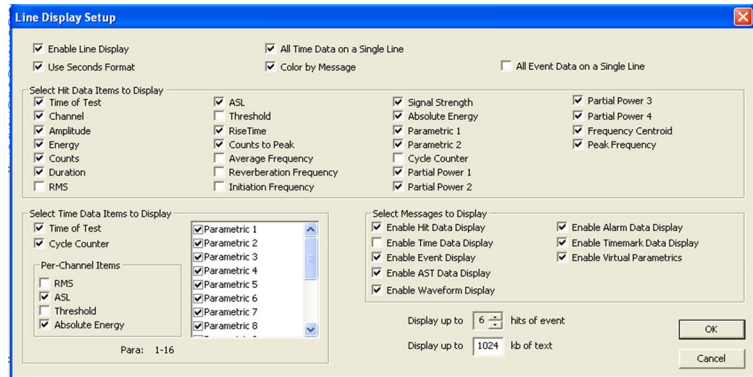
To Display the “Line Display,” simply press the **F7** function key or click the mouse as follows;

View → Line Listing Display

As mentioned above, the function key **F7** provides a shortcut to toggle the line listing on and off once the Line Listing mode has been “Enabled” in the Test setup menu.

Note that whenever the line listing is displayed, it displaces the existing graphs based on their size and location. To restore the graphs, simply press one of the “Tile” functions (Tile horizontally or vertically).

The line listing shown above shows various messages. The top line shows a message ID 2 or Time-driven message, while message ID 1 is an AE hit message, and ID 173 is a waveform message. As can be seen, all messages start with a time of test stamp, and all data is in time order. In addition, the Time-driven message shows some time-driven based parametric readings while the AE hit message shows various AE hit-based features.



3.6.2 Modifying the Line Display

Figure 3-98. Line Display Set up

Many times the user wants to view a limited set of messages in the line listing screen. For example, maybe it is only important to look at AE hit messages or even just a few features in an AE hit message, even though it is desired to collect much more comprehensive data than that to be viewed. The Line Display Setup menu (click the mouse on the Test Setup menu, then select the Line Display menu) allows the user to select exactly which data is to be viewed in the line listing display. The figure above shows the Line Display Setup menu. There are 4 sections to this menu. The Enable Line display at the top of the menu was discussed in the previous section; make sure the checkbox is clicked if you want to enable the line display mode. The next section is called the “Hit Data Items.” These are all the AE features that could be displayed on the line listing of all the features that were enabled in the **Acquisition Setup → AE Hardware → Data Sets/Parametrics** menu. Simply check those AE hit features you want to view in the line display, even though all the features selected in the Data Sets/Parametric menu will be processed to graphs and the data file.

The “Time Data Items” (shown on the bottom left of the Line Display Setup menu) allows the user to select which Time-driven data will be displayed with the Time-driven data message. The final “Messages” area (to the bottom right of the Line Display Setup menu) allows the user to select which messages will be displayed. In all cases, just click those checkboxes to enable their display in the Line Listing display.

This menu is accessible in acquisition or replay and can be changed at any time.

3.7 Clustering Setup and Operation

The purpose of clustering is to automatically identify and associate groups of hits or events which meet user-controlled “spacing” criteria. These clusters are drawn as rectangles around the data on a point plot screen to visually identify the active region(s). Cluster data can also be shown in a tabular form, allowing a quantitative assessment of the activity. Clustering offers the user computer-aided processing to identify regions of AE activity.

Cluster analysis can be activated on any 2D point plot (2D Scatter, Binned Scatter, or Colored Scatter), regardless of whether it is a Hit or Event-based graph. Clustering is also available on the Spherical, Conical & Cylinder Location

graphs types & 2D Histogram graphs allow 1-dimensional clustering along the X-Axis. Clustering on a location graph identifies active areas that require follow-up for defect growth, while clustering on a Hit based graph usually identifies hits & events with similar AE feature pattern's which can lead to hit classification of that data.

3.7.1 Clustering Settings

Clustering is controlled from the Clustering tab at the top of the Graph Setup dialog box shown below. There are a number of parameters related to clustering that can be adjusted, including the type, cluster size, point threshold, sorting & grading options. In addition, there are options for generating alarms based on clusters based on either the cluster grading (level alarms) or cluster growth (rate alarm). Each of these settings is described below.

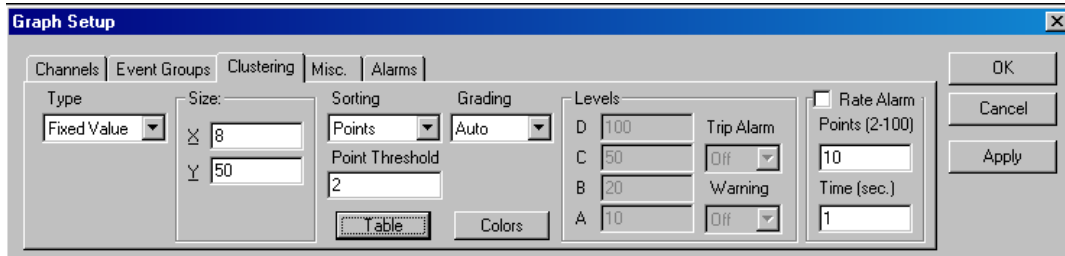


Figure 3-99. Clustering setup menu (at top of Graph Setup Menu)

Type & Size: Type controls how we want the Size parameter interpreted. Setting Type to None turns clustering off. Using Fixed Value allows you to enter the cluster dimensions you want directly into the Size X & Y field in graph axis units. In this case, this is the distance that 2 points must be within to form a cluster. Once the cluster is formed, the distance from the center of the cluster to any new points must be less than half this value. Axis % allows you to enter the cluster size you want as a percentage of the graph axis range. For example, a Size X entry of 10 means the cluster can be as large as 10% of the X-Axis. For example, if Size X were 10 & the X-Axis Maximum & Minimum are set to 10 & 50, then to form a cluster, 2 points must be within 10% of 40 from one another or within a distance of 4. For graphs using autoscaling & Axis %, the internal size used does not change as the graph scale changes. Instead, it is fixed at the value corresponding to the percentage of the initial axis limits.

Point Threshold: The point threshold is the minimum number of data points that must be in the cluster before it is reported. Internally the software maintains a list of up to 100 clusters on a graph, but it only reports clusters that are above the threshold up to a limit of 26 reported clusters. If your Input Data is set to Hits, the number of points in the cluster is the number of hits. If the Input Data is set to Events, then it corresponds to Events. The minimum number of points that can be in a cluster is 2, but clusters containing just 2 points are not usually of much interest. You can increase the Point Threshold beyond 2 (up to 100) to decrease the number of clusters reported in the Cluster Table.

Sorting: Sorting controls how clusters are reported in the cluster table. The available choices are Points, Energy, Counts, Counts to Peak. The cluster table displays clusters sorted from highest to lowest by one of these quantities. In order for Counts To Peak to be useful, you must have actually collected Counts To Peak as part of your data set. Otherwise, the software simply treats it the same as sorting on Counts. When a cluster table is displayed in the current graph window, you can use Ctrl-F6 to control the sort. Pressing Ctrl-F6 toggles the sorting through the list of available options.

Table: The Table button brings up the Table Setup dialog box shown at the right. The **Display** checkbox controls whether or not a cluster table is shown in the graph window. The F6 key is a keyboard shortcut for turning the cluster table on & off. The **Placement** setting controls where the table appears relative to the graph. The Shift-F6 key shifts the table through each of the 4 positions. The **Window %** setting controls how much of the graph window is allocated to the cluster table. If the text in the table grows too large for its window, scrollbars will appear to allow you to see all the text.

Colors: The Colors button brings up the Cluster Colors dialog box shown on the right. You may also access this dialog directly using the Graphing pulldown Colors submenu. Look for the Cluster Colors menu option. This dialog allows you to configure what colors are used in the cluster table & when drawing the cluster rectangles on the graph. These colors are used by all graphs doing clustering. Double-clicking one of the entries will cause a standard Windows color dialog to appear, allowing you to select a new color for the selected grade.

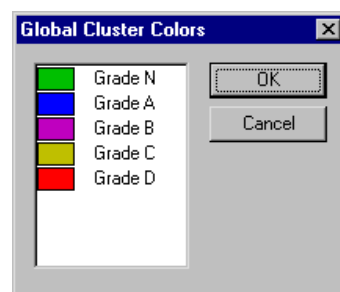


Figure 3-100. Global Cluster Colors

Grading: The Grading selection controls how clusters are assigned a color. After the clusters are sorted, a grade level is assigned to each cluster based on the sorted quantity. Choosing None causes all clusters to be classified as N (not significant) & they are drawn & reported in the Grade N color. Choosing Auto grades clusters by their percentage of the maximum observed in quantity used for sorting the clusters. It just divides the maximum observed value into 5 equal ranges (<20%, <40%, <60%, <80%, <=100%). For example, if the clusters are sorted on Points & the largest clusters had 250 points in them, then the grading levels used would be 50, 100, 150 & 200. If the Grading is set to Manual, then the Levels section will enable & you can enter your own grading levels. That can be convenient when you want to compare the cluster results from 2 different graphs as it allows you to enforce the same grading levels on both graphs.

Levels: The Levels section is used when the Grading parameter has been set to Manual. It disables when Grading is set to Auto or None. Clusters are drawn in up to 5 colors corresponding to 5 different grades. This section controls the boundary levels between the manual grades. If you are sorting on Points, have Grading set to Manual & entered levels of 10, 20, 50 & 100, clusters with less than 10 points are graded N & by default colored green (see Colors to change). Clusters with 10-19 points would be graded A, 20-49 points are B, 50-99 are grade C, and any with 100 or more points are considered grade D.

Warning & Trip Alarm: The Warning & Trip Alarm settings allow the user to generate alarm messages in the line display & set the Alarm Warning and/or Alarm Trip output lines high when a cluster reaches a user-determined grading level. The Warning & Trip Alarm settings require that Grading be set to Manual before they will enable in the dialog box & they only have an effect during acquisition. Both Warning & Trip Alarm can be set to Level A, B, C, or D or to Off. The Warning setting must always be lower than the Trip Alarm setting if the Trip Alarm setting is not set to Off. During acquisition, when a cluster level alarm is generated, an alarm message will be saved to the DTA file. If the line display is enabled, a text message containing the details will be displayed & the alarm dialog box appears with the details. The Alarm Warning & Alarm Trip lines will be set high as appropriate. The output voltage of the Alarm Warning & Alarm Trip lines is never altered in replay mode. When a DTA file is replayed, the same cluster level alarm messages will reappear that were observed when the file was recorded. These messages are not controlled by the current Warning & Trip Alarm settings during a replay. The appearance of these alarms is only controlled by the settings that were in use when the DTA file was recorded.

Rate Alarm: The Rate Alarm can be used to monitor a graph for concentrations of activity that occur at high rates. This type of cluster alarm works identically during acquisition & data replay & does not have to be enabled during data collection. The checkbox allows for simple on/off control. The Rate Alarm has 2 parameters, Points & Time. The Points setting specifies how many points must be observed within the Time period to set off the cluster rate alarm. The number in the Points field can range from 2-100, while the Time setting is specified in seconds & can be any floating-point number. For proper functioning, a Time value greater than 0 should be entered, or else the alarm will trigger anytime data is added to a cluster. Likewise, a large entry like 3.4E+38, the floating-point maximum, causes problems as well as it would be so long that the alarm would never trigger. If the software receives enough points in one cluster within the alarm time, a cluster rate alarm will be generated. For the settings above, that would be 10 points within 1 second of one another.

The cluster rate alarm is implemented to not generate extraneous messages. If the user says to check for 10 events in 1 second, the cluster has to collect at least 10 points before the rate is ever checked. Once the 10th point comes in, the software adds 1 sec. to the time of the first point & checks to see if the 10th occurred before or after that time. If it occurred before that time, then the rate is higher than the user setting & an alarm message is generated. When the cluster gets an 11th point, its time will be compared with that of the 2nd data point. If the new data maintains the rate above the user cluster rate & the cluster has already generated an alarm message, the cluster stays in the alarm state

and does NOT generate a new alarm message. If a cluster is in an alarm state & a new data point comes in after the alarm time, then the cluster alarm state will be reset to no alarm. If the cluster, later on, gets more data that is again above the user-specified cluster rate, it will generate another cluster alarm message. Basically, if the system is flooded with hits, one message will be generated & nothing else until the hit rate drops below the user threshold to reset the cluster & then the hit rate comes back up. When a cluster rate alarm is generated, the software will generate a status message in the line display reporting the cluster-ID & current cluster information.

3.7.2 Clustering Example

With so many settings, it may be useful to see an example of how to set up clustering. The software includes example layout and data files, **Cluster.LAY** and **Cluster.DTA**. The steps in setting up and running a clustering application are described below.

Step 1. If you wish, start by using New Layout under the File pulldown menu. Reconfigure the default Hts vs. Time 2D Histogram to meet your requirements. For example, you could set up a Y-Position versus an X-Position location graph setup as a point plot or set up an AE feature versus AE feature (e.g., Amplitude versus Duration, etc.) plot. When we plot an AE feature versus a feature, we usually call it a “Correlation” graph, referring to the way we are trying to “correlate” one AE feature against another to see how and if they are related. Our example layout **Cluster.LAY** and data file **Cluster.DTA** shows an example of a Duration versus Amplitude point plot. If you were starting with a new layout file & graph, you would set the graph type to 2D Scatter & set the X & Y Parameters to Amplitude & Duration. The example layout uses Manual scaling on both axes, with the Y-Axis limits set to 0 & 600, while the X-Axis limits are set to 0 & 120. The Input Data is set to Hits. Alternatively, you can just recall the example layout file & use the graph settings in it.

Step 2. Next, we must configure the Clustering function. We do this in the Graph Setup menu by selecting the tab labeled “Clustering” at the top of the Graph Setup menu (see figure 64). Start by picking Fixed Value from the Type dropdown Listbox.

Step 3. Next, set the Size X setting to 8 & the Size Y setting to 50.

Step 4. Leave the Sorting set to Points, but change the Point Threshold to 3 to eliminate the smallest clusters.

Step 5. Select Auto from the Grading dropdown list. Leave the Rate Alarm disabled & ignore the Table & Colors buttons for the moment. Clicking on the Graph Setup OK button will save the settings. If this is a new layout for you, you may wish to use Save Layout under the File pulldown menu before starting a replay.

Step 6. Now we are ready to run a test and view clustered data. Use the F10 key or select Replay under the Acquire/Replay pull-down menu to open a DTA file for replaying. Select the Cluster.dta data file and click on OK or press Enter to replay the data. If you use the setup as provided in the Cluster.lay layout file with the Cluster.dta data file, you will see results similar to those shown below.

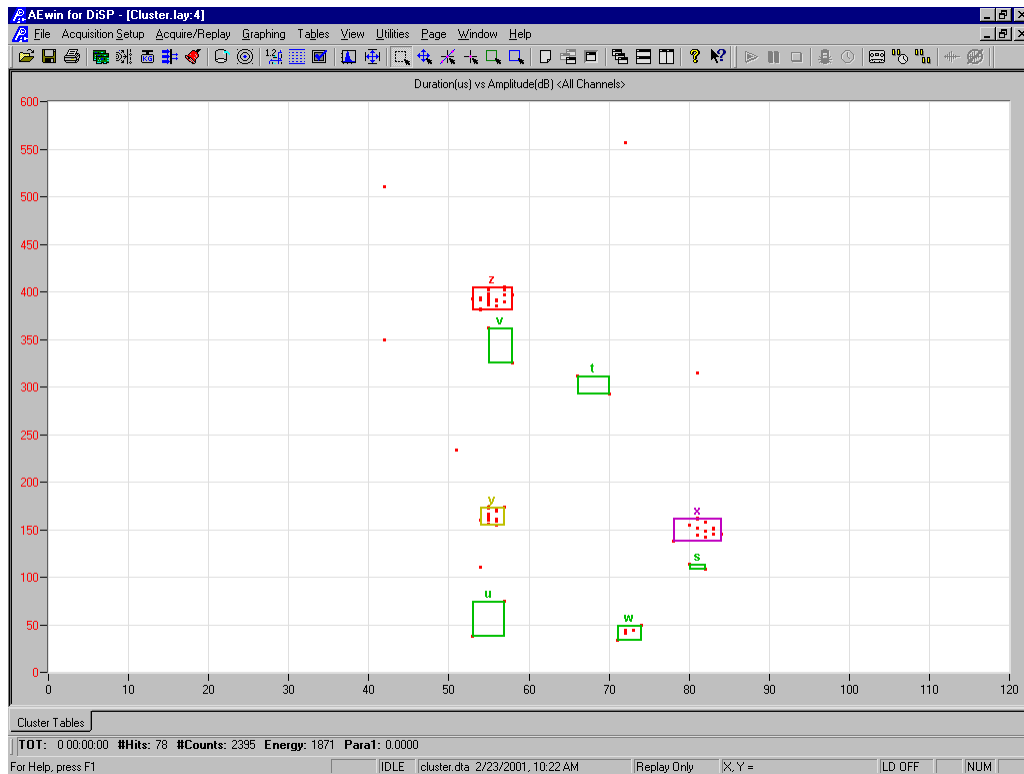


Figure 3-101. Cluster Graph

From the graph, we can see that we have identified 8 clusters based on the cluster configuration that we have set up. We can also see that there are cluster letters associated with each cluster and that each cluster is colored. Clusters are labeled with a letter starting with “z” as being the most intense and “a” being the least intense. Up to 26 clusters can be reported with the clustering capability built into the software.

Step 7. You can adjust the colors of the clusters by using the Colors submenu of the Graphing pulldown menu. Look for the Cluster Colors menu option. When you select it, the Global Cluster Colors dialog box will appear, as shown in Figure 66. You can also get to this dialog box using the Colors button on the Clustering tab of the Graph Setup dialog box. The Global Cluster Colors dialog box controls the color of each grade level. If you double-click one of the entries, a standard Windows Color dialog box will appear to allow you to select a new color for that particular grade level.

Step 8. This graphical representation of the clustered data is nice, but we may also want to see a more quantitative tabular listing. Press the F6 function key from the keyboard. By pressing the F6 key, you toggle the cluster table on/off. This cluster table shares the window with the cluster graph. When we use F6 to turn the cluster table display, we get the table as shown in figure 68.

The Cluster table has the following columns. The leftmost shows the cluster letter, starting with z as the highest. Next is the cluster-ID; this is the identification number assigned when the cluster was formed. As new points are added to the graph, the a-z cluster letter can change, but the cluster-ID always stays the same. The next column is the Points. This indicates how many points (hits or events) are in a given cluster. Total Energy is the accumulated energy in the cluster from all the hits or events. The Total Counts are also the accumulated number of AE Counts for the cluster. The Xrange & Yrange min & max values identify the boundary points of the cluster while the Avg. X and Y values identify the center of the cluster. Each line of the cluster table is colored by the grade level of the cluster, so it matches the color of the cluster box on the Cluster graph.

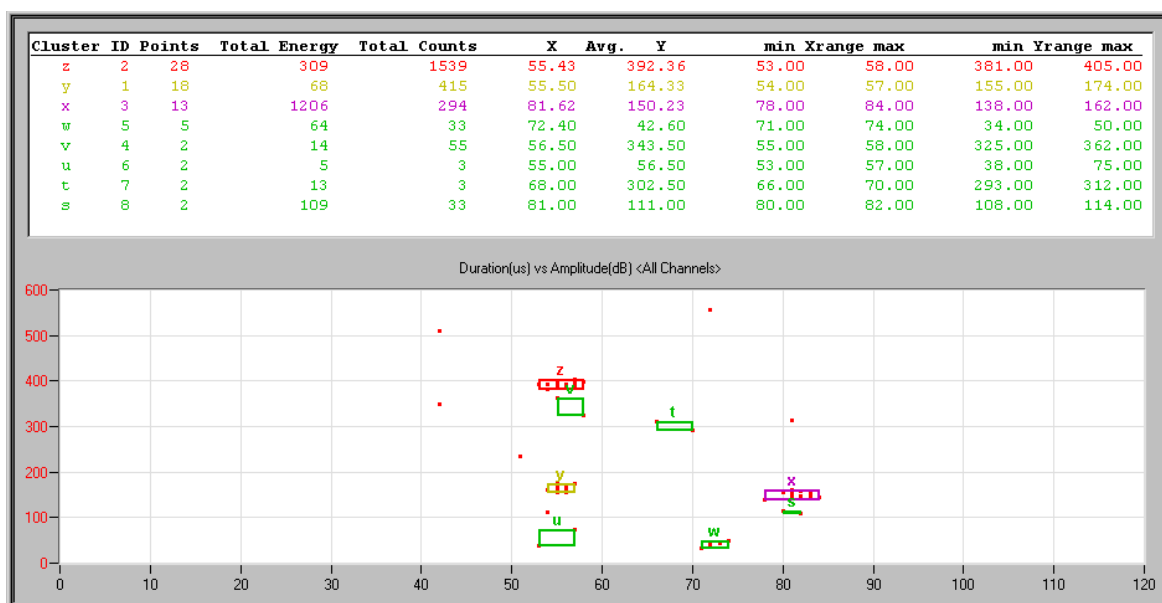


Figure 3-102. Cluster Table

Step 8. Try pressing shift-F6 while the table is displayed & the cluster graph is the active graph. Shift-F6 causes the cluster table to shift its position around the graph from top to bottom & then from left to right.

Step 9. Try pressing Ctrl-F6 to control the sort order in the cluster table. Each time you press Ctrl-F6, the table will switch between sorting based on Points, Energy & Counts & immediately update.

Step 10. The final step in the process of setting up and running the clustering function is the adjustment of the setting for optimal results. This is accomplished by changing some of the clustering configuration settings & replaying the test until you feel that you have the most logical results. If you have too many small clusters, make your Size X & Y settings bigger. Alternatively, you can adjust your Point Threshold higher to only display the largest clusters & ignore the smallest clusters. You might wish to try setting Grading to Manual & modifying the Level A thru D values to adjust the grading results.

In our setup, we set the cluster threshold at 2, which is a bit low. We can also see by figure 68 that there are 4 of the 8 clusters with only 2 hits within the cluster. Let's get rid of these by going back to the graph setup menu and changing the Point Threshold to be 3 instead of 2. After we click the OK button, we'll find the cluster table has been updated & now only hold 4 of the original 8 clusters. Note also that those points that are not part of the cluster seem to be randomly spaced, which tells us that it does not seem that these points or hits should have been identified as a cluster. Experiment further to get familiar with the clustering function.

3.8 Graphing in AEwin

The graphing capability in AEwin is very impressive, as it should be since data visualization is very important in AE testing. AEwin is able to provide multiple graphs, multiple plots within a graph, multiple graphs on a screen (or "Page" using Windows terminology), and multiple screens (Pages) full of graphs. The graph types can be multi-dimensional (2D and 3D) graphs, multi-colored graphs, and multiple graph types, including Point plots, Histograms, Waveforms, and Spectrum graphs. Graph axes can be set up to be fixed scaling (always stays at a fixed minimum and maximum value), automatic scaling (compression or sliding scale), logarithmic or linear scaling. The graphs can be set up to plot anything versus anything, limited only by the operator's imagination.

Before getting into setting up Graphs in AEwin, it is important to define the terminology used in graphing on AEwin. This will be used throughout the manual.

Plot: A plot is defined as a single graph within a graph (or graph window). Since a graph can be set up with multiple graphs inside, we need to distinguish the difference between the word graph as a graph window and the fact that there can be multiple graphs in a single graph window. We, therefore, call this a plot. So now we can say there are multiple plots within a graph or graph window.

Graph: A graph is a single Window on a Screen or Page. A graph is controlled by the Graph setup menu. The graph can have multiple plots of the same type within it, but these are all set up by a single specific graph setup menu.

Screen or Page: A screen or Page is the graph(s) that show up in the display area of the AEwin screen. This is the area between the top and bottom tool bars (see figure 4). A screen or page has a page tab located at the bottom of the screen to select between the screen pages.

Now that we have definitions, we can provide an example. Figures 58 –61 show examples of different screens. Four are shown; Figure 58 shows what we have labeled as an activity screen. The page tab at the bottom of the screen has been labeled “Activity Screen” to let the user know what types of graphs are set up on this screen page. Figure 59 shows a waveform and FFT screen page. Its tab also identifies it as “Waveforms.” Figure 60 shows an example of a 3D screen page with the appropriate Screen Page tab labeling. Figure 61 shows an example of some channel-by-channel line plots with an appropriate screen page tab.

In regards to graphs, figures 58 – 61 show multiple graphs on each screen. Figures 61, 63, and 64 show a 4 graph screen, while figure 59 shows an 8 graph screen. There is no limit to the number of graphs that can be placed on a screen; the only limitation is a physical one pertaining to the readability of the graphs. As far as multiple plots on a graph, the top two graphs of figure 58 show graphs with multiple plots. On the top left plot, there are 4 different point plots on one graph, while in the top right graph, there are two line plots.

The figures 58 – 61 are provided by the example **Layout4.LAY** and the data file **Chan24.DTA** gives a good idea as to the functionality of AEwin as far as graphing is concerned. Since all the major examples for graphs and graph setups can be found in this **Layout4.LAY** file provides an excellent reference for how to set up the different types of graphs. Please use this example as such. The following subsections will provide information on how to generally set up a graph and screen page.

3.8.1 Graph Setup Procedure

The following is the step-by-step process of setting up graphs in AEwin. This assumes that you are starting from a very simple single screen, single graph arrangement into multiple graphs, multiple screen setup.

1. Before starting, give a little thought to the number of screen pages your application will need, what the theme will be for each screen page (for tab labeling considerations) and how many and what type of graphs you would like on each screen. The better you plan this out, the easier and faster the graphing setup process will be. Use as an example the theme provided (and theme tabs) in the **Layout4.LAY** (or **Layout2.LAY** on Spartan or PCI-2 systems) layout file whose graphical results are shown in figures 58 – 61.
2. Find an existing Layout file that gets you partially to your goal. You can then edit that layout file to meet your exact needs. It is always easier to start with an existing layout. You can easily copy and paste graphs that have already been set up, saving a lot of time. If this is not possible, select **File → New Layout** which will start with 1 screen page (labeled Page 1) and one graph (a Hits versus Time graph).
3. Create the number of screen pages that you need by adding and deleting screen pages, then edit the Page tab titles to the desired name or description of what that page will contain.
4. Create the number of graphs you need on each graph screen by adding or deleting graphs.
5. Size your graphs as desired and tile them as desired so that they are laid out in the order that you wish.
6. Individually set up each graph in each graph screen to get the desired result. Enter the graph setup menu for each graph by right-clicking the mouse when it is inside the graph, then select the first pop-up menu called “Graph setup” to enter the graph mode.
7. Once you are done setting up your graph(s), test them out in acquisition or replay to make sure they act as expected. There will always be the need for some adjustment; you will rarely get things perfect the first time.

3.8.2 Screen Page and Tab Management

As with all Windows operations, there are various ways to add and edit Screen pages. The most convenient is by Right-clicking your mouse on an existing page Tab, adjacent to where you want to add the page. A Page/Graph setup menu will appear. To add a New Page after the selected page, simply select **the New Page** menu item. To insert a new page before the selected page, select the **Insert Page** menu item. In either case, a new page and page tab will appear. The page tab will be labeled as “Page X.”

There are other ways to add a page, including selecting the **New Page Icon** from the icon toolbar (it looks like a blank page), which will always place a new page at the end of the screen page tab list. You can also go to the main menu bar and select the **Page Menu** to make the **New Page** and **Insert Page** selections.

Once you have created a new page, it will be empty (no graphs). You will add the desired graphs and will label the page tab. You may add as many graphs as you wish to the screen page.

To label, the page tab simply right-clicks on that tab and select the **Rename Page** menu item. This will allow you to enter a page name. There are no major restrictions to naming a page tab. However, do not make the page tab title too long as it might take up too much space.

There is one other screen page editing function, which allows you to remove or delete a screen page. Simply Right-click on the screen page tab and select the **Delete page** function. The screen page and tab will immediately disappear. If you had graphs on the screen page that you deleted, they would be deleted as well.

3.8.3 Adding, Placing, and Sizing Graphs on a Screen Page

Once you have a Screen Page setup, you need to begin adding and placing graphs on it. If you know how many graphs you want and what types they will be, you can quickly add and edit the graphs. If all the graphs are to be the same style (e.g., waveform graphs, one for each channel), you should start by adding a graph, defining it, and then copying it as many times as you need. Then you can size the graphs and create the layout (position of the graphs) as desired, and then you can edit the graph setup to finalize the graph function. If each graph is to be different, you can either copy similar graphs and paste them into the new screen page, or you can simply add as many graphs as you want on the screen and edit the graph setup for each graph added.

To add a graph to the graph screen, simply right-click your mouse anywhere on the screen page and select the **New Graph** menu item. Alternatively, you can left-click the new graph icon from the icon toolbar menu. This icon shows 3 staggered graphs. Additionally, you can add a new graph by selecting the **Graphing → New Graph** main menu selection. Whichever way you select a new graph, a single graph will appear. To add multiple graphs on the screen, simply perform this function multiple times.

Once you have selected the number of graphs you want on the screen, you need to tile them. Tiling allows you to place them next to each other so that each part of every graph can be seen. There are two tiling options. You can use **Tile Horizontal** or **Tile Vertical**. The most convenient way to perform this function is to select one of these functions from the icon tool bar. The **Tile Horizontal** icon is shown as two graphs stacked on top of one another, while the **Tile Vertical** icon shows two graphs side by side. Alternatively, you can select these from the **Window** main menu. **Tile Horizontal** will favor a length-wise positioning of the graphs and will usually make the graphs longer than taller. **Tile Vertical** will give preference to a vertical direction. Choose one of these to place the graphs in a nice ordered manner.

Once the graphs are placed on the screen, we can alter their sizes. We do this using the standard Windows conventions of moving the mouse to the edge of a graph until we see a double arrow and then left-click the mouse and move the edge in or out to make the graph larger or smaller. We can also move graphs by right-clicking on the graph and selecting the **Move Graph** menu item. An example of sizing graphs can be seen in figure 63, where we have sized the 3D graph to be larger than the associated graphs.

The final function that we need to perform is that of editing or setting up the graphs to perform the desired display function.

3.8.4 Graph Setup Function

This section discusses the graph setup menu and how to set up a graph. First, we select the graph that we want to modify. We do this by right-clicking the mouse when it is inside the graph region of the graph that we want to work on. Once we do that, a menu appears. This menu gives us many choices, but we will select the top one called “Graph Setup.” Upon selecting this, a graph setup menu appears like that shown to the right.

This shows us how we have set up the Graph setup menu for a ‘Hits versus Time’ graph. It gives us a lot of insight into how the graph setup menu works and how to set it up. Let’s note first the layout of the menu. We see at the top of the menu a selection area for Channels, Event Groups, Clustering, Misc. and Alarms. These settings allow us to configure those items. When in the Channels Tab, we can select which channels and/or channel groups will be displayed on the graph that we are setting up. In this case, we are selecting all active channels for viewing in the graph. You can see that we have no groups set up in Event Groups if you select that tab. Clustering allows you to set up parameters that link closely spaced hits/events together for point plot graphs. Misc. allows you to set various options like turning on/off the legend control. Alarms allow you to create and customize graph alarms on this graph.

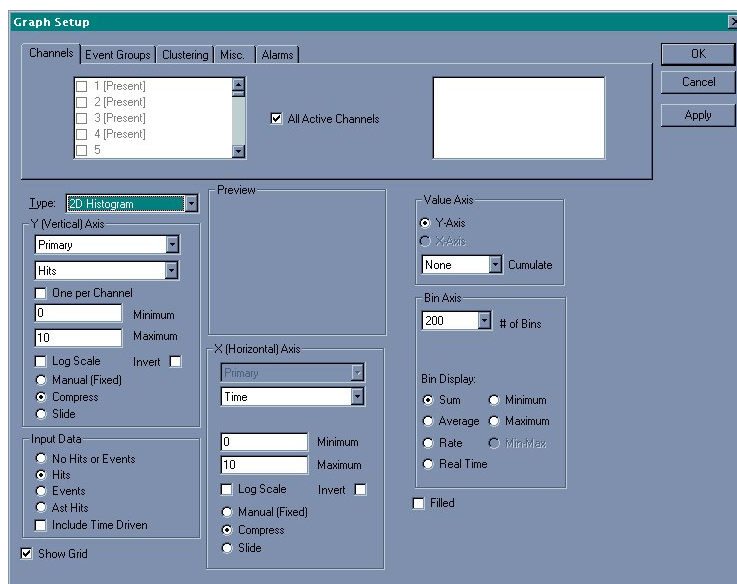


Figure 3-103. Graph Setup Menu for Hits versus Time Graph

The next menu area is the Graph Type selection box. This is located just below the Channel, Event Group, and Clustering selection area. Various graph types can be selected included 2D Histograms, 2D Scatter (Point Plots), 3D Histograms, waveforms, Power Spectrum plots, etc. With the selection of the graph type, the graph setup menu below this may change to offer the special configurations needed by that graph type. For the hits versus time application, we will use the 2D Histogram.

Below the Graph Type selection is the Y-Axis setup menu area. Going down one item at a time in this menu is the Plot # selection within the graph. This selection allows you to select more than one plot to be graphed. The one being selected for setup is the one shown. We have only selected one plot in this graph called the “Primary.”

Next is the Y-Axis parameter to be plotted on the graph. By clicking this, you can see many selections. However, we have selected Hits. This will display the number of hits (on all channels) detected on the Y-Axis.

The checkbox “One per Channel” allows us to select a unique per channel plotting color that will give us multiple colors by channel plot. In our case, for hits and all active channels, if we selected this checkbox, we will get a multiple plot graph, each plot a different color based on the hits from that one channel, up to 6 plots on one graph. If not checked, we will only get one color plot.

The next two edit boxes allow us to set the starting Y-axis settings (minimum and maximum).

The next checkbox, “Log Scale,” allows us to set a log or linear graph scale.

The next 3 radio buttons are related, allowing you to select one type of graph scaling. “Manual (Fixed)” will force the graph limits as set above to remain fixed. “Compress” will compress the graph so that the minimum graph value remains, but the maximum keeps changing as the graph limit is exceeded. And the “sliding” selection will allow both the minimum and maximum limits to change or slide as data is received that is beyond the current limits.

The next box allows us to select what type of Input data we are plotting. In our case, since we are plotting hit data, we want to choose the Hit data for plotting.

Below is a checkbox for displaying the grid on the graph.

The next column to the right shows the graph preview at the top and then the X-Axis settings.

The X-axis settings have similar setup boxes as were described for the Y-Axis and will not be described here. Just note how they have been set up to display the Time parameter and that we have set up the minimum and maximum graph values and a compression scaling so that we can see all data from the start of the test.

The last column on the right changes according to the graph type being set up. If a 3-D graph were being set up, this would be the column for the Z-axis setting. If a Point plot were being set up, there would be no extra column as the points do not need to be conditioned for display.

Since we have selected a graph type of 2-D Histogram, a “Value and Bin Axis” setting menu appears. The Value axis sets which axis the histogram or bars will grow. For our example, we are selecting the Y-Axis as the Hits will grow in the vertical direction. We set a cumulate mode if we want the data in the histogram to build upon the previous bin or bars.

The bin axis settings allow us to set the resolution and how the data that makes up a bin get counted in the histogram bar (or what we call a bin). In general, the larger the number of bins, the finer the bars. If your bin axis is time, you usually want to have a very high number. If your bin axis is a channel, you would like to match the number of bins and channels. In terms of the bin display, you can choose whether you want to sum up all the data in the bin or if you want to average the data in the bin, or if you want a rate to be displayed (e.g., Hits/second). To understand, this assumes that we have a test for 100 seconds and that we have 100 bins. Each bin will contain one second's worth of data. We can decide to “add it up” and display it as a bin sum, average the amount of data received in that time period or display a rate. In this case, since the bin size is 1 second, all the numbers would be the same for this setting, but for any other case, they would be different.

Learn more about these settings by making changes and seeing the effect.

At the bottom of the third column is a histogram setting that allows you to fill the histogram bars or keep them hollow.

Please note and try to make sense of all the selected menu settings. Although it might look complex at the start, if you have gone through these settings, one at a time, you will see that all is quite simple. Exit the graph menu by selecting the OK button at the top right of the Graph setup menu.

By looking at the top of the graph, we see its graph title of “Hits vs. Time (sec) <all channels>.” This summarizes what is being displayed. First is the graph type “Hits vs. Time (sec).” Next, we are displaying data from all channels.

Now that we are finished, we see that we have set up a Hits versus time graph and are ready to test it. Test it and make any corrections until you get the desired result. If we have other graphs, we set them up in a similar way, one at a time, until all the graphs are set up.

Now we are ready to proceed with a test or a replay. As a reminder, always save your Layout file before trying it out. It is not fun to have all your work disappear because the power went down or something happened to lose the layout.

3.9 Data Selection and the Selected Dataset Display Dialog Box

“Data Selection” is a process that allows you to retrieve waveform, hit, or event information from specified points on any type of scatter graph. You can use the mouse to designate points or groups of points to be added to a collection known as the “Selected Dataset” (sometimes abbreviated as “dataset”). These will be shown in the “Selected Dataset Display” dialog box, where you can individually inspect each item (waveform, hit, or event) in the dataset. You can easily add or remove items from the dataset and export them as a new data file. This can be used as a versatile graph-based filtering tool.

The Selected Dataset Display contains a mini-line display that shows the waveform, hit, or event messages in the dataset. The associated waveform(s) or power spectrum(s) (if any) are displayed above it. Furthermore, a simple X/Y location plot is available that displays the location of all events (if any) in the dataset. The user can hide or display this plot as desired, and some customization is available.

This process can be performed repeatedly, allowing you to progressively build your dataset. This is preserved unless you clear it manually or do something that requires it to be cleared (see below).

Data selection can be performed at any time during or after a test or replay. By default, this feature is enabled in Replay and disabled in Acquisition. You may change this when you start a test or replay via a checkbox in the Data Storage or Replay dialog boxes. Keeping data selection features disabled can improve acquisition performance during periods of high data rates.

How the Selected Dataset Is Organized:

The selected dataset is organized as a time-ordered collection of Waveforms or Hits or Events:

- Collection of Waveforms: When you perform data selection on a waveform-based scatter graph, then the resulting dataset will be a collection of waveforms ordered by time. The Selected Data Display dialog box will show you these.
- Collection of Hits: When you perform data selection on a hit-based scatter graph, then the resulting dataset will be a collection of hits ordered by time. The Selected Data Display dialog box will show you these, along with any waveforms associated with them.
- Collection of Events: When you perform data selection on an event-based scatter graph, then the resulting dataset will be a collection of waveforms ordered by time. The Selected Data Display dialog box will show you these, along with the member hits of each event and any waveforms associated with them.

You can perform data selection repeatedly without clearing the dataset in the process, so long as you use the same type of graph (waveform-based, hit-based, or event-based). If you use a different type of graph, then you are essentially trying to create a different type of dataset. In this case, the program will ask you if you want to clear the existing dataset or abort the new data selection.

Actions That Automatically Clear the Selected Dataset:

- Starting a new test or replay.
- Loading a new layout file.
- Changing location settings.
- Closing the program.

3.9.1 How to Add Data to the Selected Dataset

- 1) Start a test or begin a replay. Make sure that the "Enable Data Selection Features" box is checked in the Data Storage or Replay dialog box when you do so.
- 2) Select a graph to work with by left-clicking on it. This can be any of the following types: 2D Scatter, 2D Binned Scatter, 2D Colored Scatter, 3D Scatter, Spherical Location, Cylinder Location, and Conical Location. Its border will darken.

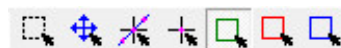


Figure 3-104. Selected Dataset Toolbar Icons – Add to Dataset

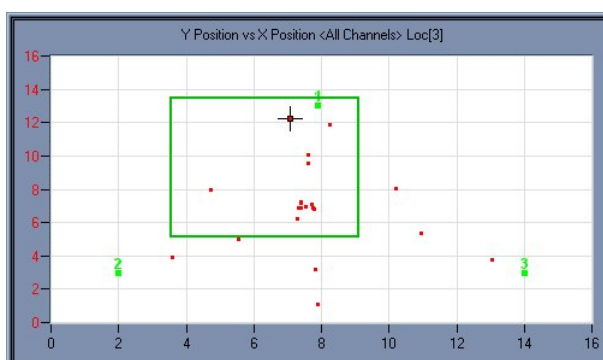


Figure 3-105. Area-Based Data Selection

- 3) Select the point(s) to work with. There are two ways to do this:
 - a) **Point-based:** You can select an individual point by using the data cursor. Activate the 'Data Cursor' from the toolbar. Position the cursor over the desired point. Then right-click the graph and select "Add This Point to the Selected Dataset" from the menu that pops up.
 - b) **Area-based:** You can designate a rectangular area on a 2D graph. Activate the toolbar button that looks like a green box. Hovering the mouse over it shows the text 'Use the mouse to add a group of points to the selected dataset.' Then use the mouse to select an area on the graph (as if you were zooming).
- This begins the data selection process. Depending on the graph and the size of the data file, this may take some time. Once finished, the Selected Dataset Display dialog box will be opened, and the cursors will be positioned on the appropriate graphs.
- 4) You can repeat steps 2 and 3 to add more to the dataset. There is no danger in selecting the same points more than once. The program will only include a point if it has not already done so. It is safe to brush with broad strokes.



Figure 3-106. Point-Based Data Selection

You can re-open the Selected Dataset Display dialog box at any time by selecting "Open Selected Dataset Display" from the Acquire/Replay menu. If this menu item is disabled, then it means that your selected dataset is empty, and there is nothing to show.

3.9.2 How to Remove Data From the Selected Dataset

There are 2 ways to remove items (waveforms/hits/events) from the selected dataset.

- A "de-selection" tool is available on the main toolbar. It looks like a red box. Hovering the mouse over it shows the text "Use the mouse to remove a group of points from the selected dataset." Employ this tool the same way you would the green-box area selection tool described in the above section. It is essentially an eraser. Any data you designate with it will be removed from the selected dataset. As with data selection, it is safe to erase with broad strokes. If you erase too much, you can add it back with data selection, as described above.
- In the Selected Dataset Display dialog box, you can use the "Remove this from the selected dataset" toolbar button (it looks like a blue X). Doing so will remove the currently displayed waveform, hit, or event.



Figure 3-107. Selected Dataset Toolbar Icons – Remove from Dataset

3.9.3 Selected Dataset Display Controls

The Selected Dataset Display dialog box has a number of controls, most of which are located on a toolbar at the top of the dialog. These are described below. As you move the mouse cursor over each toolbar button, a 'tooltip' will pop up to display the button's name. Some of these controls are only available when the dataset is organized as a collection of events. Other controls are only available if the Waveform Option is present.

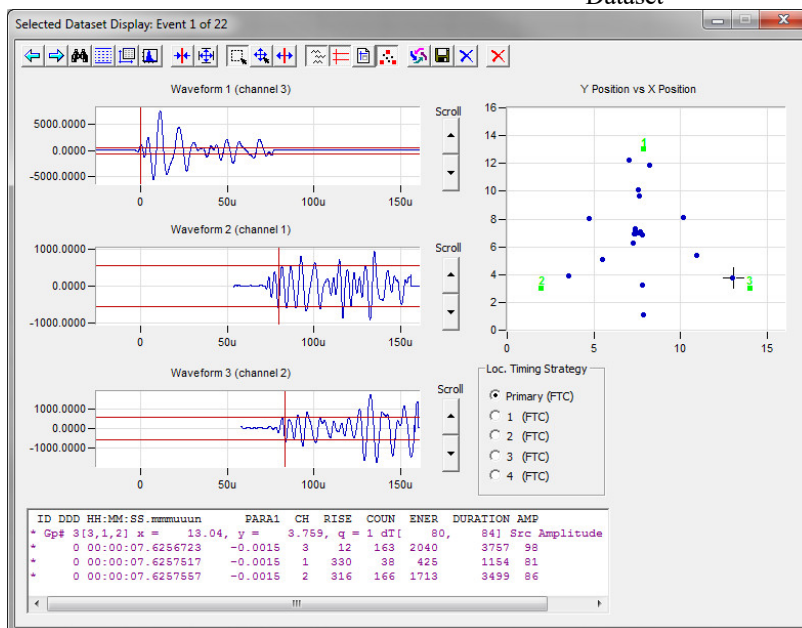


Figure 3-108. Selected Dataset Display

Prev/Next Buttons: If there are multiple waveforms, hits, or events associated with the selected point(s), then you will be able to view all of them. Use the prev/next buttons to step through them sequentially. The cursors (crosshairs) on any 2D point plot graphs will move appropriately.

Select Button: Instead of using the Prev/Next buttons to step through a large number of waveforms, hits, or events, the 'Select' button can be used to jump to the desired waveform, hit, or event by simply entering its number.

View All Button: You can view all selected waveforms, hits, or events together by clicking the 'View All' button. This will pop up a dialog box with a large line display showing all waveforms, hits, or events associated with the selected point or area. Note that this dialog shows the original unmodified waveforms and events (prior to any time-of-arrival changes you may have made, as described in section 3.9.5).

Set Power Spectrum Width Button: Clicking this button will open a dialog box allowing you to set the power spectrum width. You can set a specified width or accept the default.

Location Graph Setup Button: Clicking this button will open the 'Location Graph Setup' dialog box. From here, you can customize the location graph by selecting axis parameters, limits, and auto-scaling.

Reset User Changes Button: Clicking this button will reset any changes you have made by use of the time-of-arrival cursor (see section 3.9.5).

Reset Axes Button: Clicking this button will allow you to reset the axes of one or more graphs. A dialog box will be displayed for you to specify which graphs. You can also reset a waveform graph by right-clicking it and selecting 'Reset this graph's axes.'

Select Zoom Mode Button: Clicking this button will select the 'Zoom' mode for all graphs and clear. In Zoom mode, you can use the mouse to select an area of a graph to view in greater detail. Use the 'Reset Axes' button to view the original graph again.

Select Pan Mode Button: Clicking this button will select the 'Pan' mode for all graphs. In Pan mode, you can use the mouse to drag the viewable window of the graph to view data that is currently beyond the graph axes limits. Use the 'Reset Axes' button to view the original graph again.

Select Cursor Mode Button: Clicking this button will select the 'Select Cursor' mode for all graphs. In 'Select Cursor' mode, you can use the mouse to drag the red 'Time of Arrival' cursors on waveform graphs forward or backward to effectively change its time of arrival. While in this mode, the mini line display is replaced with a table showing the time of arrival for each graph. If you do this and then click the 'Force Recalculation of Event' button, the event location will change, and the new (modified) location will be displayed on the location graph as a black X. Use the 'Reset User Changes' button to undo these changes.

Synchronize Waveform Display Button: When this button is checked, the waveforms will be displayed in the same time reference. i.e., If the second waveform in an event has an arrival time of 20 us after the first waveform, then the entire waveform will be positioned 20 us to the right, so its time of arrival (red line) will be positioned at 20 us. If the button is not checked, then all waveforms are left-justified on the graphs (their time of arrivals will be positioned at 0 us). This does not modify the waveforms in any way. It is simply a display preference.

Display Threshold Lines Button: When this button is checked, red horizontal lines appear on the waveform graphs showing the threshold. When this button is unchecked, these lines are hidden.

Display Power Spectrum(s) Button: The power spectrums can be viewed by clicking the 'Display Power Spectrum(s)' toolbar button. This will replace the waveform graphs with corresponding power spectrum graphs. Click it again to return to viewing waveforms.

Show Location Graph Button: The location graph can be displayed by clicking the 'Show Location Graph' toolbar button. Clicking it again will hide the location graph. Note that this graph is only usable when displaying events and only when the event being viewed has more than one hit. You can customize this graph by clicking the 'Location Graph Setup' toolbar button.

Force Recalculation of Event Button: After changing the time of arrival of one or more waveforms in an event, click this button to recalculate the event location. If successful, the new event will be shown in the mini line display, and the location graph will be modified to show an X at the location of the modified event. If a new event location could not be created using the new time of arrivals, then the mini line display will show a message indicating the problem. Use the 'Reset User Changes' button to undo these changes. Note that certain actions can automatically force a recalculation of the event. These actions include: Changing the location timing strategy and using the Prev/Next/Select buttons to view other events.

Export Data File Button: Clicking this button allows you to create a new data file based on the selected dataset. See section 3.9.4 for details.

Remove Button: Clicking this button will remove the currently displayed waveform, hit, or event from the selected dataset.

Clear Selected Dataset and Close Button: Clicking this button will clear the entire selected dataset and close the dialog box.

Loc Timing Strategy Radio Buttons: The bandpass filters and timing strategies defined in the 'Timing' page of the 'Location Setup' dialog box can be applied to the waveforms/events. Use the radio buttons in the 'Loc. Timing Strategy' section to select a strategy. The waveform/power spectrum graph(s) and the mini-line display will be refreshed to show it after filtering. The default 'Primary' strategy does not apply a filter. Note that the selected strategy must have an enabled filter for a filter to be applied. See section 3.9.8 for information regarding the 'Custom Arrival' timing feature, which is particularly useful in this dialog box.

Scroll Buttons: The 'Scroll' buttons (next to each graph) are used to change which waveform is displayed in a particular graph. They allow you to 'scroll' through the list of waveforms up or down as desired. This is particularly useful when the selected event has more than 3 waveforms.

3.9.4 Exporting Data to a New Data File

One feature available in the Selected Dataset Display dialog box is the ability to create new data files based on the selected dataset. This can be used as a powerful and flexible way of filtering data. If you have made any changes to the time of arrival(s), then these changes are exported as well (see section 3.9.5). To start the process, simply click the 'Export Data File' button. This brings up a dialog box of the same name. Here is where you enter a filename for the new datafile and select from the following two options:

- Export selected data
- Export all data that is NOT selected (filter)

The first option is the most straightforward. This sends the entire data selection to the new file. The second file is the reverse of this. This sends all the data that is NOT part of the Selected Dataset to the new file. This type of filtering can be quite useful.

In addition to this, the new file will contain all of the following data messages from the original file:

- Time-driven Data
- Time-Marks
- Alarm Messages

You may choose to eliminate any or all of these from the new data file through checkboxes. Lastly, you may choose to exclude all waveforms as well.

When you are done choosing your preferences, click the “Start” button. This process may take some time for large datasets. The program will prompt you when it is complete.

3.9.5 Changing the Time of Arrival of a Waveform and Recalculating an Event

Another powerful feature of the Selected Dataset Display dialog box is the ability to ‘Modify’ a hit/waveform’s time of arrival and (if using events) recalculate the event location. To do this, simply use the ‘Select Cursor’ mode and use the mouse to drag the (red) time of arrival cursor on a waveform graph to the desired point in time. As you do this, you will see the new time of arrival updated on a table in the mini line display. If you are using events, then you can recalculate the event location by clicking the ‘Force Recalculation of Event’ button. This will recalculate the event using the new time of arrival. If your event has more than 3 hit/waves then you might have to use the scroll controls to access some of the waveforms to modify.

If you export a dataset that contains waveforms that have been modified in this manner, then the changes will be saved to the new datafile and accessible using the ‘Custom Arrival’ timing feature in location (see section 3.9.8).

3.9.6 Other Features

- 1) Many settings in the Selected Dataset Display dialog box are saved when the layout file is saved. This includes the various on/off preferences (‘display threshold lines,’ ‘show location graph,’ etc.), location graph setup, and custom power spectrum width.
- 2) If the Selected Dataset Display dialog box is open when a ‘cluster/area history’ operation is performed, then the dialog box will be updated with its data.

3.9.7 Notes

- 1) You can re-open the Selected Data Display dialog box from the Acquire/Replay menu. If this entry is disabled, then it means that there is nothing to display.
- 2) Try loading the ‘HE-LINK.LAY’ layout file and replaying ‘HE-LINK.DTA’; you can use this to try out Data Selection using 2D Planar location.
- 3) “2D Binned Scatter” and “2D Colored Scatter” graph types store their data in a ‘binned’ grid structure much like a Histogram (though in 2 axes). This means that each ‘point’ on these graphs represents an area rather than an absolute point. The cursor display (near the bottom right corner of the screen) shows the extent of the area represented by the point. Data Selection will locate all waveforms/hits/events that fall into that area (lower bound inclusive, upper bound exclusive).
- 4) Selected waveforms/hits/events are not highlighted with crosshairs on “3D Scatter” and “Spherical Location” graph types, though Data Selection can be initiated from them.
- 5) Data Selection functionality can be disabled by unchecking the “Enable Data Selection Features” checkbox in the Acquire/Replay dialog box (this setting is NOT saved in the layout). Doing this will grant a performance improvement in high data rate tests. It is recommended that you keep it disabled during an acquisition unless you know your test will not require a high data rate. Once disabled, it cannot be re-enabled until a new test or replay is performed.

3.9.8 Custom Arrival Timing Feature

When a user-modified waveform is exported, it is saved in its unmodified state to the new data file. The user-modified TOA (time of arrival) is also saved for future use. This can be accessed through the use of the 'Custom Arrival' timing feature.

Before starting the replay, enter the location setup. Set one of the timing strategies in the timing page of the location setup dialog box to the 'Custom Arrival' feature. If you want location plots to use the modified TOAs, then select this strategy in your location group's 'Timing Strategies.' Or, you can simply set the Primary strategy to 'Custom Arrival' so all location groups will use it. Since 'Custom Arrival' only works in replay, this will default to 'FTC' (First Threshold Crossing) during acquisition.

If you want to see the modified TOAs in Data Selection:

When the Selected Dataset Display dialog box is opened, it will plot the locations and waveforms based on the currently selected 'Loc Timing Strategy.' Note that the dialog will remember your previous setting and use it by default the next time it is loaded. If this strategy uses a 'Custom Arrival' timing feature, then all the locations will be plotted using the modified TOAs. If this strategy doesn't use a 'Custom Arrival' timing feature, then all locations will be plotted using the original TOAs.

Once the Selected Dataset Display dialog box is loaded, you can switch strategies through the use of the 'Loc Timing Strategy' radio buttons. Doing so will not re-plot all locations, but it will re-plot the current one AND reposition the TOA cursors on all current waveforms. You can switch back and forth between a timing strategy that uses 'Custom Arrival' and one that doesn't view the difference between the two.

3.10 Cluster/Area History

Cluster History and Area History provide the ability to view detailed graphical information about AE activity in a specified area on a 2D Scatter, Binned Scatter, or Colored Scatter graph (called the 'source' graph). This area can be defined by selecting a cluster (Cluster History) or by designating an area manually (Area History). This information is displayed on any number of graphs defined and designated by the user as 'History Graphs.' These graphs remain inactive until the user starts a Cluster or Area History process. When the user does so, the software 'replays' the data contained in the selected cluster or area onto all history graphs. The history graphs then remain active for the remainder of the test or replay, accepting/displaying any incoming data that would fall within the original bounds of the cluster or area at the time the process was started. This process can be repeated any number of times during a test or replay.

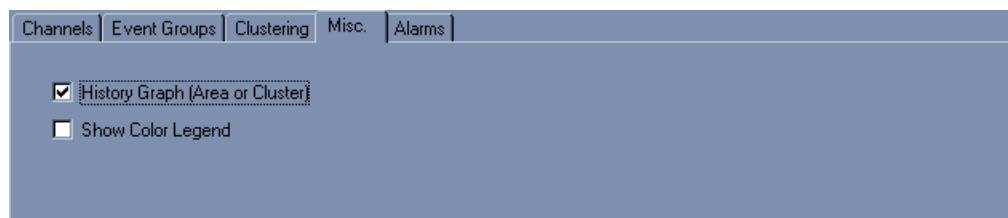


Figure 3-109. Area or Cluster History Enable

3.10.1 Setup

- 1) To use Cluster History, you need to create at least one cluster graph before starting the process, usually before starting the test or replay. Area History does not require this.
- 2) Create as many history graphs as desired. This can be done at any time before you start the history process, even after the test or replay has ended (step 3). Any graph can be designated as a 'History Graph' by checking the 'History Graph (Area or Cluster)' checkbox on the 'Misc.' page of the graph setup. The graph's title will then include the '(Hst)' tag.
- 3) Start an acquisition test or a replay.

3.10.2 Starting a Cluster History Process

Cluster History is started by right-clicking on a cluster graph and selecting 'cluster history' from the menu that pops up. If this option is grayed out, then it means one of the following:

- 1) There are no clusters on the specified graph.
- 2) You disabled 'Data Selection Features' from the Acquire/Replay dialog when you started the current test or replay.
- 3) You started a history or Data Selection process that hasn't been completed yet.

If there is more than one cluster available on the selected cluster graph, then the program will prompt the user to select one from a list.

3.10.3 Starting an Area History Process



Figure 3-110. Multivariate Alarm Set up

Area History involves manually designating a rectangular area on the graph. Activate 'Area History Mode' (blue square) from the toolbar.

If this option is grayed out, then it means one of the following:

- 1) The active graph is of an invalid type (i.e., histogram, waveform, 3D graph, etc.).
- 2) You disabled 'Data Selection Features' from the Acquire/Replay dialog when you started the current test or replay.
- 3) You started a history or Data Selection process that hasn't been completed yet.

If you are working with a 2D Scatter or 2D Binned Scatter graph that has multiple Y axes defined, it will prompt you to select which Y-Axis to work with. Then use the mouse to select an area on the graph (as if you were zooming).

3.10.4 Execution

The Cluster/Area History process is detailed below:

- 1) All history graphs are re-initialized. Any data on them (from a previous history process) is cleared.
- 2) The setup of all history graphs is modified to reflect the setup of the source graph.
 - Channel sets, location group sets, and the data source (hits/events) are made identical to the source graph.
 - The min/max limits of any graph axes which share the same graph quantity as on the source graph are changed to reflect the limits of the cluster or area.
- 3) Special graph filters are set on all history graphs to reject any future incoming data that does not fall within the (original) cluster or area from being displayed. These filters persist until changed by another history process, the user turns off the 'History Graph' checkbox on the graph or until the end of the test/replay.
- 4) All history graphs are enabled. This allows them to accept and display historical and future incoming data.
- 5) Historical data (the data within the cluster or area) is displayed on all history graphs.

3.10.5 Notes

- 1) A Cluster/Area History process can be started at any time during or after a test or replay, where 'Data Selection Features' was not disabled.
- 2) Try loading the 'ClusterHistory.lay' layout file and replaying 'HE-LINK.DTA'; you can use this to try out Area and Cluster History using 2D Planar location.
- 3) It is important to note that the bounds of a history graph reflect the bounds of the cluster or area at the time when the process was started. These bounds do not change if the cluster itself grows.

3.11 Graph Alarms

'Graph Alarms' is a feature that enables certain graph types to trigger an alarm whenever data is plotted on them above one or two pre-set thresholds ('warning' and 'trip'). They are available on 2D Histograms, 2D Scatter graphs, Binned Scatter graphs, and Colored Scatter graphs during data acquisition.

3.11.1 How to setup a graph alarm:

An 'Alarms' tab is available in graph setup (for graph types that can use them).

- 1) Check the 'Enable Graph Alarm' checkbox to turn on graph alarm functionality for this graph.
- 2) Check the 'Enable Warning' checkbox to enable the alarm 'warning' level. Check the 'Enable Trip' checkbox to enable the alarm 'trip' level. Either or both may be used at the same time.
- 3) Set the vertical axis thresholds for the alarm levels you have enabled. These thresholds are inclusive, so a threshold of 80 will be triggered by a signal of 80 or above.
- 4) Select which vertical axis controls the graph alarm (Primary, Secondary-1, etc.).
- 5) If desired, you may enable the 'Alarm Hold-Off' and specify its '#Signals.' See the section on 'Alarm Hold Off' below for information on this feature.

3.11.2 How an alarm is detected/triggered:

Graphs that have an enabled graph alarm will display horizontal lines at the levels of the thresholds set in the graph setup (Green for warning, Red for trip).

A graph alarm 'Warning' will trigger when all of the following occurs.

- The 'warning' level is enabled for that graph.
- New data is displayed on this graph at or above the green threshold line.
 - This needs to occur a number of times consecutively equal to the specified '#Signals' if used.
- No graph alarm (Warning or Trip) is currently active (on this graph).

An alarm 'Trip' will trigger when all of the following occurs.

- The 'trip' level is enabled for that graph.
- New data is displayed on this graph at or above the red threshold line.
 - This needs to occur a number of times consecutively equal to the specified '#Signals' if used.
- No graph alarm trip is currently active (on this graph).

'New' data is defined as any data that is plotted after the alarm was last reset. Note that the alarm is automatically reset at the start of the test.

3.11.3 What happens when an alarm is triggered:

When a graph alarm is triggered, several things happen:

- The audio monitor (if installed) sounds an audible beeping sound. This continues until the alarm is cleared or the test ends.
- The alarm output (WARNING or TRIP as appropriate for the level) is set high. This stays high until the alarm is cleared or the test ends.
- An alarm message (16) appears in the Line Display in red and is saved to the datafile if auto dump is on.

A small dialog box appears, displaying the conditions of the alarm. The alarm can be cleared by clicking the 'Clear Alarm' button. You can exit this dialog without clearing the alarm if you wish. It can be accessed later by selecting 'Display Last Alarm...' from the Acquire/Replay menu.

3.11.4 Alarm Hold Off

Normally (when the "Alarm Hold Off" is disabled), an alarm is triggered when a single data point ("signal") is plotted above the threshold.

By enabling 'Alarm Hold-off,' you can delay alarm triggering until a specified number of consecutive data points ("signals") are plotted above the threshold. This is useful when you wish to avoid having data spikes trigger alarms. It is important to remember that the string of data points must be consecutive for an alarm to be triggered. Any data points below the threshold will 'reset' the internal counters.

For example, if you set '# Signals' to 5, then an alarm would only be generated until 5 consecutive data points are plotted above the threshold. Let's say your threshold was 70. The following sequence:

- 70, 80, 75, 80, 70 would trigger an alarm.
- 70, 80, 60, 80, 70, 80 would NOT trigger an alarm because even though 5 signals passed the threshold, they were not consecutive.

Alarm 'Warning' and 'Trip' both maintain their own separate internal counters (though both use the same '#Signals'), so a particular sequence might trigger an alarm 'Warning' without triggering an alarm 'Trip.'

3.11.5 Tips:

- 1) Graph alarms are checked when the graphs are refreshed. If you want to increase the rate at which they are checked, then you should increase the rate at which the graphs are updated. See the settings in 'Display Mode...' under the 'Test Setup' menu.
- 2) Note that data does not have to be in the visible portion of the graph to trigger an alarm. Panning/Zooming of the graph will not hinder graph alarm operation.
- 3) Setting #Signals to 1 is identical to disabling 'Alarm Hold Off.' In fact, if you do so, the system will automatically disable 'Alarm Hold Off' for you.
- 4) Caution is recommended when using Compression on the horizontal axis of histograms, particularly when using "Sum" Bin Display mode. In this scenario, the bins will become taller as the graph compresses horizontally, an effect of the bins being combined as part of the compression process. Be aware that data is only checked against alarm criteria only when it is first plotted, not when the graph is refreshed during compression. A bin that suddenly grows above the threshold due to compression will not trigger an alarm.
- 5) Caution is recommended when using 'Alarm Hold Off' on graphs that have all the following settings selected:
 - Graph is plotting RMS, ASL, Threshold, or Absolute Energy.
 - The 'Include Time-driven' checkbox is checked in the 'Input Data' section of Graph Setup.
 - Graph is plotting more than one channel.

In this case, you are plotting channel-based 'time-driven data,' which can plot multiple points from a single time-driven data message. 'Alarm Hold Off' will advance its internal counter at most once per message, so this can be tricky if your graph is plotting more than one channel. If you want to use 'Alarm Hold Off' with Graph Alarms on the type of graph listed above, it is advised that you create multiple single-channel graphs instead.

- 6) It is recommended that 2D Scatter (not Binned/Colored) graphs that have a graph alarm enabled to use only Manual scaling on the Y (vertical) axis. Using compression or sliding scale won't hinder graph alarm operation, but it may slide the threshold line off the visible portion of the screen.

3.12 Windowed Alarms

'Windowed Alarms' are a variant of 'Graph Alarms' for use with the 'Windowed Histogram' graph type. They are setup and work in almost exactly the same fashion as 'Graph Alarms,' but differ in three important respects:

- 1) Windowed alarms clear themselves automatically if the data on the graph goes below the threshold. Remember that Windowed Histograms essentially display a rate. Old data is gradually removed over time, so the bins themselves can grow shorter. This can lower them below the thresholds.
- 2) Windowed alarms are triggered and cleared on a per-bin basis rather than on a per-graph basis. Careful setup of the graph axis limits and bin settings can allow you to create some very intricate and useful alarm setups. See the 'Setup and Examples' section below.
- 3) Windowed alarms have no 'Alarm Hold Off' setting.

Refer to the section on Graph Setup for more info on Windowed Histograms.

3.12.1 Setup and Examples

A 'Windowed Histogram' tab is available in graph setup (visible only for graphs of that type).

- 1) Check the 'Enable Alarm' checkbox to turn on graph alarm functionality for this graph.
- 2) Check the 'Enable Warning' checkbox to enable the alarm 'warning' level. Check the 'Enable Trip' checkbox to enable the alarm 'trip' level. Either or both may be used at the same time.
- 3) Set the vertical axis thresholds for the alarm level(s) you have enabled. These thresholds are inclusive, so a threshold of 80 will be triggered by a signal of 80 or above.
- 4) Optionally, you may give the graph a name in the 'Misc.' tab. If this is not blank, it will be used for alarm reporting purposes to uniquely identify the graph that triggered the alarm.

Example 1

You could create a graph of "Events vs. X-Position" with 10 bins. Each bin would represent 10% of the horizontal distance being tested with a linear location group. Such a setup would easily alert you to recurring trouble spots. The alarm message would tell you where (in X-Pos units) where the trouble-spot is.

Example 2

You could create a graph of "ASL vs. Channel" using the average bin display. This would alert you to, particularly noisy channels.

It is very important that you select your X-Axis and bin settings carefully. This can be tricky so let's continue the second example, where we were plotting ASL vs. Channel. We want to select a # of bins that most closely matches the number of channels we are working with. We want each bin to represent one channel with no overlap. If we were monitoring channels 1-18, the following settings would probably work best:

- # of Bins: 20
- Bin Display: Average
- X-Axis Min: 1
- X-Axis Max: 21

We chose 20 bins because this is the smallest value that we can select that is ≥ 18 . We chose 1 as our minimum because that is the smallest channel value that we want to plot. We chose 21 as our max because we want the X-Axis range to equal the number of bins. Otherwise, the bins wouldn't line up with the axis values properly. We chose 'Average' bin display because we are interested in trends. If we were looking for transient data, we might employ 'Maximum' bin display.

With a little experimentation, you will find a setup that works best for your needs. You could, in the above example, use 40 bins instead of 20 if you wanted narrower bins with gaps between each one.

3.12.2 How an alarm is triggered:

Graphs that have an enabled windowed alarm will display horizontal lines at the levels of the thresholds set in the graph setup (Green for warning, Red for trip).

Windowed histograms are checked for alarms when they are updated. How often this happens is based on the Window Duration and Update Rate. See the section on graphs for more details.

A windowed alarm 'Warning' will trigger during a graph update when all of the following occurs.

- The 'warning' level is enabled for that windowed histogram.
- A bin reaches or crosses the green threshold line.
- That particular bin isn't already in an alarm state (Warning or Trip).

A windowed alarm 'Trip' will trigger during a graph update when all of the following occurs.

- The 'trip' level is enabled for that windowed histogram.
- A bin reaches or crosses the red threshold line.
- That particular bin isn't already in a Trip alarm state.

It is important to remember that multiple alarms can be triggered by the same windowed histogram, one for each bin above the threshold. Alarms are cleared automatically in the same fashion, on a per-bin basis, when levels drop below the threshold.

3.12.3 What happens when an alarm is triggered:

When a windowed alarm is triggered, several things happen:

- The audio monitor (if installed) sounds an audible beeping sound. This continues until the alarm is cleared or the test ends.
- The alarm output (WARNING or TRIP as appropriate for the level) is set high. This stays high until the alarm is cleared or the test ends.
- An alarm message (16) appears in the Line Display in red and is saved to the datafile if the auto-dump is on.
- A small dialog box appears, displaying the conditions of the alarm. The alarm can be cleared by clicking the 'Clear Alarm' button. You can exit this dialog without clearing the alarm if you wish. It can be accessed later by selecting 'Display Last Alarm...' from the Acquire/Replay menu.

The alarm display dialog and the line display will show details on which bin was triggered and what range of values are represented by that bin.

3.13 List of Shortcut Keys

The following is a list and short description of all the shortcut keys available in the software. Many of the shortcut keys follow the tradition of PAC's DAQ and LOC software programs and should be familiar. Others follow standard WINDOWS conventions. Menu items usually do list the shortcut key, but the full list is provided below for convenience.

F1	Help Menu
Shift + F1	Context Sensitive Help
F2	Hardware Setup
F3	Graph Setup
F4	
Ctrl + F4	Close (Delete) Graph
Shift + F4	Delete Page (Entire Graph Screen)
F5	
F6	View Cluster Table (Toggle)
Ctrl + F6	Controls Cluster Table Sorting
Shift + F6	Shifts Cluster Table Position
F7	View Line Dump (Toggle)
F8	Location Setup
F9	Acquisition
Shift + F9	Acquire to Existing File
F10	Replay
F11	
F12	
Ctrl + C	Copy Graph
Ctrl + D	File New
Ctrl + H	Window Tile Horizontal
Ctrl + N	Add New Graph
Ctrl + O	Open Layout File
Ctrl + P	Print Page
Ctrl + S	Save Layout
Ctrl + T	Window Tile Vertical
Ctrl + V	Paste Graph
Ctrl + X	Cut Graph
Ctrl + W	Waveform Collection (Toggle)
Ctrl + Shift + C	Copy to Clipboard
Ctrl + Insert	Copy Graph (same as Ctrl + C)
Shift + Insert	Paste Graph (same as Ctrl + V)
Ctrl + Alt + O	Options Dialog

4. LOCATION

The purpose of this section is to familiarize you with setting up and using the software location modes. Location is the process of collecting incoming hits into events & analyzing the arrival times of the hits in an event to produce a source location. Individual location modes have several key differences, which are summarized in the table below.

Location Mode	Hits Used	Description	Analysis Type	Required Option
Zonal	1	Trivial solution to location. Identifies the sensor closest to the source using the relative arrival times of the hits. Returns coordinates of the first hit sensor as the source location.	Single	Standard
Linear	2	Designed for 1-dimensional location on a line. Performs a linear interpolation between 2 sensors' coordinates based on differences in the arrival times of the first 2 hits in the event.	Single	Standard
LinearXY	2	Linear without the assumption of sensors in a straight line & events allowed between any sensor pair. For example, sensors could be in a square with sources locating on the sides or diagonals.	Single	Standard
Tank Bottom	3	Designed for analyzing evenly spaced 2D circular arrays of sensors.	Single	Tank Bottom
2D Planar	3-8	Designed for an arbitrary array of sensors on a 2D plane.	Regression	Full
2D Planar (xy) 2D Planar (yz) 2D Planar (xz)	3-8	2D Planar designed to complement 3D Location. Locates using just 2 of the 3D coordinates & ignores the 3 rd . Useful for some events that fail normal 3D Location.	Regression	3D
2D Anisotropic	3-8	2D Planar allowing for differing velocities in the x & y directions.	Regression	Full
Cylinder	3-8	2D Planar analysis wrapped around a 3D cylindrical shell. The ends of the cylinder may be open or have flat, spherical, or elliptical endcaps. Assumes sound travel through cylinder shell rather than its interior.	Regression	Full
Anisotropic Cylinder	3-8	Cylinder allowing for differing velocities in the x & y directions. Only supports location on the cylinder body, not on the endcaps.	Regression	Full
Conical	3-8	2D Planar wrapped around a 3D conical shell.	Regression	Full
Spherical	3-8	2D Planar wrapped around a 3D spherical shell again, assuming sound travel via the shell only.	Regression	Spherical
3D Location	4-8	2D Planar extended to 3D coordinates assuming sound travel directly through space rather than via an external shell.	Regression	3D Location
3D T0 Location	5-8	3D Location with time zero input. Assumes all events are signaled via one channel that indicates the absolute start time of an event.	Regression	3D T0 Location

The most important difference in the location modes is the geometry they are designed for. The geometry, in turn, affects the number of hits used per event to calculate the location. The type of analysis also affects the number of hits used in an event. The single analysis modes use just enough hits to create one set of simultaneous equations, which are then solved to extract the source location coordinates. A more complex approach uses more hits & multiple regression analysis to produce the source location that best fits all the available data. By using data from extra sensors, the regression modes yield a more accurate source location with improved immunity to errors in the individual arrival times. For more details, see section 4.1, Theoretical Background.

Two location modes, zonal & linear, are standard features. Others are location enhancements that require additional licensing options. If you are not sure what locations options are enabled in your software, look for the Options dialog available on the Help pulldown menu. It will show a checkmark next to any purchased options. If you wish to purchase additional license options for other location modes, contact MISTRAS, Physical Acoustics Corp., at 1-609-716-4000.

4.1 Theoretical Background

The following section provides a brief overview of the math behind the location. It is definitely not required that the average user comprehend this material. Its purpose is to educate the inquisitive & allow the user additional insight into what location is doing behind the scenes.

The fundamental basis for the location calculation is just the simple time-distance relationship implied by the velocity of the sound wave. The absolute arrival time, t , of a hit in an event can combine with the velocity, v , of the sound wave to yield the distance, d , from the sensor to the source.

$$d = v * t \quad (1)$$

The distance between 2 points depends on the geometry of the problem. The majority of the location modes are a variation of 2-dimensional source location in a plane, although in many cases, the 2D plane will wrap around a 3-dimensional object. For 2 points in a flat plane, the distance equation is just the Pythagorean theorem expressed in Cartesian coordinates:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (2)$$

This calculation is complicated by the lack of knowledge of the exact time the event originated. To get around that problem, all the times are considered relative to the first hit in the event. Each arrival time difference implies a difference in distance to the sensor relative to the distance to the first hit sensor. For the second hit sensor relative to the first hit sensor, a differential equation can be written as:

$$t_2 - t_1 = (d_2 - d_1) / v \quad (3)$$

The distance equation (2) can be combined with the difference equation (3) to yield:

$$t_2 - t_1 = [\sqrt{(x_2 - x_s)^2 + (y_2 - y_s)^2} - \sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2}] / v \quad (4)$$

Where x_s & y_s are the unknown coordinates of the source. This equation contains 2 unknowns & cannot be solved by itself. To get a second equation with the same 2 unknowns, a 3rd hit is added to the event producing analogous equation:

$$t_3 - t_1 = [\sqrt{(x_3 - x_s)^2 + (y_3 - y_s)^2} - \sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2}] / v \quad (5)$$

These simultaneous equations can then be solved for x_s & y_s , although the algebra is fairly complex. Tank Bottom location uses this particular approach to produce a source location from 3 hits, as did the original version of 2D Planar location before it was extended to include multiple regression analysis.

If there are additional hits added to the event, there is a question of how to use the extra information. In the case of Tank Bottom location, the algorithm simply takes the extra hits & tries them 3 at a time to produce extra source locations. It iterates through all the 3 hit combinations of the hits in the event until it exhausts them or runs into a user-specified maximum. The problem with this approach is that it produces more than one source location per event & if there is any error in the timing values, the source location can be wildly incorrect. A better approach would be to average the data somehow to produce a single location.

Multiple regression analysis does just that, although it does not actually average the results of multiple 3 hit calculations directly. Rather it searches for the location that best fits all the available data. Each additional hit basically adds an extra equation to the set of simultaneous equations given above as (4) & (5). This can be generalized as:

$$t_i - t_1 = [\sqrt{(x_i - x_s)^2 + (y_i - y_s)^2} - \sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2}] / v \quad (6)$$

If one defines

$$\Delta t_i = t_i - t_1 \quad (7)$$

one can write:

$$\Delta t_i = [\sqrt{(x_i - x_s)^2 + (y_i - y_s)^2} - \sqrt{(x_1 - x_s)^2 + (y_1 - y_s)^2}] / v \quad (8)$$

Equations (7) & (8) give 2 ways of calculating the Δt of the i th sensor. Using the known arrival times of the hits, equation (7) calculates the observed time difference, $\Delta t_{i,obs}$. For a given set of source location coordinates, equation (8) defines the calculated time difference, $\Delta t_{i,calc}$. Multiple regression analysis is a general-purpose algorithm that minimizes the difference between 2 quantities, which in this case are the observed & calculated Δt values. To do that, a quantity called χ^2 (chi square) is calculated. In our case, it is defined as a sum across all sensors in the event:

$$\chi^2 = \sum (\Delta t_{i,obs} - \Delta t_{i,calc})^2 \quad (9)$$

This quantity is also called the fit value and is dependent on the x,y coordinates of the source location. The sum is recalculated for each potential source location. Assuming no error in the data, χ^2 will have a value of 0 at the source location. The location code searches for the values of x_s & y_s that minimize the value of χ^2 . The process is an iterative search because it is not possible to directly write down simple equations for the values of x_s & y_s that minimize χ^2 .

The search is performed using one of 2 algorithms, either a Simplex search (3D or 2D Planar modes) or Powell's Method (Spherical, Cylinder & Conical Location). Both search methods are based in general on code & algorithms described in "Numerical Recipes in C," 2nd edition (NRC) (A free copy can be downloaded from Google). The code has obviously been heavily adapted for the problem of acoustic emission source location, so there are significant variations from the NRC originals. The interested user may, however, find more information about Simplex & Powell fitting & searching in NRC in chapter 10 & more information on regression analysis in chapter 15.

The math gets more complicated when extended to 3 dimensions or involves wrapping around a 3D object, but the approach is still basically the same. An important point is that the regression approach is no longer over-determined in the case of the event containing the minimum number of hits, i.e., 3 hits for 2D Planar. In that case, however, regression analysis turns out to be mathematically equivalent to solving the complicated algebra of equations (4) & (5) & yields the same location coordinates.

4.2 Location Setup Dialog

When setting up location groups, you use the Location Setup dialog box (F8 shortcut key). It has 2 tabs labeled General and Timing. Source locations are produced by analyzing events. Events are just collections of hits & their associated time of arrival. The settings on the General page control how hits are combined into events & how events are analyzed by the location modes to produce source locations. The settings on the Timing page control how the time of arrival of each individual hit is determined and even whether or not a hit is considered significant or excluded from the location analysis.

There are several features to make setting up location groups easier. First, you should be aware of the F5 function key. It allows the user to quickly set all rows in a column of the grid to the value of the current cell. It works on both

the General & the Timing tabs. In the example setup shown below, pressing F5 would set all rows of the Overcal column to the current cell's value of 200. Alternatively, you can Right-click a cell & select Set Column from the context menu that appears. The context menu for the General tab also has Copy Group & Paste Group options. Right-click & select Copy Group on the source row, then Right-click the destination row & select Paste Group. All of the group settings & the associated sensor & structure settings will be copied to the new row. By default, Paste Group will use the first group as the source if you did not use the Copy Group option to specify a source row.

Group #	Show All Points	Location Type	Wave Velocity	Event Definition Value	Event Lockout Value	Overcal Value	Timing Strategies	Hits/Event Min Max	Max Iterations
1	☒	Zonal	1000000.00	1000.0000	0.0000	200.0000	Primary	1 1	0
2	☐	Linear	1000000.00	1000.0000	0.0000	10.0000	Primary	2 2	0
3	☐	2D Planar (XY)	1000000.00	1000.0000	0.0000	10.0000	Primary	3 8	256
4	☐	3D Location	1000000.00	1000.0000	0.0000	10.0000	Primary	4 8	256
5	☐	Spherical	1000000.00	1000.0000	0.0000	10.0000	(Primary).1,2	3 8	256
6	☐	Cylinder	1000000.00	1000.0000	0.0000	10.0000	Primary.1,2	3 8	256
7	☐	Tank Bottom	1000000.00	1000.0000	0.0000	10.0000	Primary	3 3	0
8	☐	2D Anisotropic	1000000.00	1000.0000	0.0000	10.0000	Primary	3 8	256

Figure 4-1. Location Setup – General Page

4.2.1 General Tab Settings:

The layout for the General tab is shown above. The grid on the General page of Location Setup contains a list of location groups & their type. If you click an entry in the Location Type column, an arrow appears, allowing you to change the type. As mentioned above, many location types (3D, Spherical, Cylinder) appear only if you have purchased the appropriate option.

Group #: This column acts as an On/Off setting. Unchecked groups are ignored when doing a replay.

Show All Points: This column allows the user to show all source results for events that this group can successfully locate. Independent of which groups have Show All Points checked, the software always allows all enabled groups to analyze a given event to get the best results for each group and then compares the results from all successful groups to come up with the best result. For example, linear location is always presumed to give a better result than zonal location if there are enough hits(2) to produce a location. If no groups have Show All Points checked, the software will report one source location per event & it is the best result that is shown on graphs & in the line display. There are times when a side-by-side comparison of the results of 2 locations groups is useful. In that case, you can mark the groups as Show All Points & then whatever source locations they can calculate will be reported. The software will appear to sometimes produce more than one source location per event & if you look at the line display, you will see extra event listings with the same first hit times. The extra location results will also show up as extra points on your graphs too, but they only count as 1 event in the Statistics bar. In all cases, you will see the software's best result as one of the points. There will be times when an event cannot be located by more than 1 group, so having Show All Points does not guarantee a result from a given location group. It is only a guarantee that you will see it if it can produce a useful result.

Location Type: The Location Type column lets you specify which location mode to use with this particular location group. If you click on a cell in this column, a down arrow appears on the right-hand side of the cell. If you click the down arrow, a drop-down list of the available location choices will appear. The available choices do depend on the

software licensing options as mentioned in the introduction, so you may not see all the location modes that are discussed in this manual.

Wave Velocity: The Wave Velocity is the expected rate of propagation of the acoustic emission signal you wish to monitor. It is entered in the user's Distance Units per sec. For example, if the Distance Units are in Inches, then the Wave Velocity takes units of Inches/sec. If the user's Distance Units are time-based, this column is ignored by the location algorithm. The Materials button will allow you to access a database of known velocities for various materials. See the section on the Material Properties dialog for more information. It is always recommended that you determine the appropriate value by measuring the velocity on your test structure using lead breaks, for example.

Event Definition Value: The Event Definition Value (EDV) is the length of an event in the user's Distance units. If the user's Distance Units are time-based, then the EDV is directly the maximum time allowed between first & last hits in an event. If the EDV is distance-based, then it is the physical length between the first and the last hit sensor of the event. The goal is to select an Event Definition value that will give enough hits to perform source location with the desired accuracy. For example, in the figure below, the 2D location requires a minimum of 3 hit sensors to locate; however, more sensors can provide a more accurate source location; therefore, the inner circle radius defines the Event Definition Value and in the case shown, it will detect channels 9, 10, 15, 16, 17, 21, and 22, so up to 7 channels (hits) will be able to form the event if the EDV value as shown is entered.

Since the location calculations use time information, any distance units will be later converted to time units using the velocity entered in the Wave Velocity column. The velocity units are the user's Distance Units per second. For example, if the Distance Units are set to meters, then the velocity units are meters/sec.

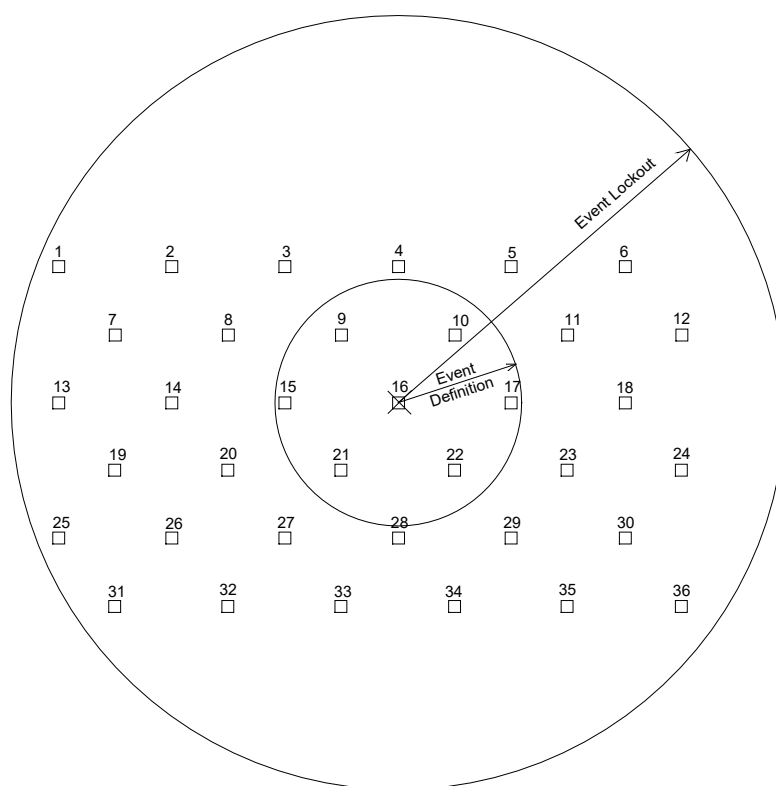


Figure 4-2. Relationship between Event Definition and Event Lockout.

Event Lockout Value: The Event Lockout Value is also a length of time or distance entered in user units, but it controls the interval between consecutive events. The Event Lockout of the last located event blocks hits in all location groups until it has expired. The Event Definition Value & the Event Lockout Value are both specified relative to the start of the event (time of test of the first hit). An Event Lockout Value of 0 allows overlapping events in 2 *different* location groups. Overlapping events are not permitted in the same location group, no matter what the Event Lockout Value is. An Event Lockout Value larger than the Event Definition Value allows hits to be ignored after an event, for example, if reflections could generate extraneous acoustic signals. In the example shown in the figure above, all AE hits between the Event Definition Value and the Event Lockout will be ignored from the point of view of being part of the event in progress or the next event.

Overcal Value: The Overcal Value or, more fully, the over-calibration value is also a length that is specified in the user's Distance Units. Before data acquisition or a replay begins, the software calculates a table of time differences between sensors using their known coordinates & the user enters wave velocity. The Overcal Value is also converted to a time value if needed. After the first hit in an event, these calibration times are compared to the time differences of later hits relative to the first hit sensor. In order for a hit to be included (or accepted) as part of the event, its time difference must be less than this calibration time, but the Overcal value provides an allowance for a margin of error. The time difference can be higher than the calibration time by the Overcal Value & still be included in the event.

Timing Strategies: The Timing Strategies column is used to specify which strategies listed on the Timing page will be applied to a particular location group. You set up the strategy on the Timing page, but you control whether it is used with a particular location group using the Timing Strategies column. See the section on the Timing page for information on the settings for strategies & use of the Timing Strategies column.

Hits/Event Min & Max: The Hits/Event (Hits per Event) columns are related to the regression modes. In theory, the multiple regression can sum across any number of sensors (hits) in a given event when calculating a fit, but there are limits in the code. First, the absolute minimum number is determined by the mathematical requirements specific to the location mode. For Spherical, Cylinder Location, and all 2D Planar regression modes, the event must contain at least 3 hits to produce a location. For the 3D location, a minimum of 4 hits is required to analyze an event & produce a location. All regression modes can make use of up to 8 hits in an event.

The Hits/Event Min & Max settings allow the user control over the analysis between these limits. For example, in 2D Planar, a minimum of 3 hits are required to produce a location result, but then this is not an over-determined result. If there is an error in the timing data of any one of the 3 hits, the location calculated will be in error. With 4 or more hits, the regression analysis produces the best fit location that averages out errors in the data. Thus, the user might want to adjust the Hits/Event Min setting to 4 for 2D Planar to eliminate analysis of 3-hit events. This will reduce the number of events observed & generally reduce scatter in the results. In some cases, the user might want to only look at events with a specific number of hits. Setting the Hits/Event Min & Max settings to 4 will limit the analysis to events with 4 or more hits using only the first 4 hits to form the event.

Max Iterations: The Max Iterations column is also related to the regression modes. In this case, the setting controls the maximum number of locations evaluated in the search for the best fit location. It is not strictly speaking the same as the number of times through a search loop for 2 reasons. First, some searches can test a variable number of locations during one pass through the loop. Second other searches use loops within loops. Due to the variable nature of the search algorithms, the actual number of locations tested can be slightly higher (0-3) than the Max Iterations setting. A lower Max Iterations setting will terminate long location analyses sooner, but the higher speed results in lower accuracy in the best fit location produced.

When the Location Type is not set to a regression mode, the Max Iteration column & the Hits/Event Min & Max columns will be disabled. In that case, the Max Iterations column displays 0 as iterations are not used for the other modes. The Hits/Event Min & Max settings will be updated with the number of hits required for the location mode. For example, the Linear location mode requires 2 hits to produce a location, and if Location Type is set to Linear, Hits/Event Min & Max will both display 2 & the user will not be able to adjust the setting.

4.2.2 Timing Tab Settings:

Timing strategies process hits and their accompanying waveform to produce a time of arrival. A list of arrival times a series of hits are grouped together to form an event. The location modes analyze the arrival time differences (delta t's) in the event to calculate the source location. For an enabled location mode, the software always analyzes the event at least once. The first analysis is performed using only the arrival times from the Primary Strategy. If you use additional strategies with one of the regression modes, the software performs a second event analysis to refine the source location by adding the arrival times produced by the secondary strategies (1-4) to the event.

The settings for the strategy are on the Timing page, but you control which location groups use them with the Timing Strategies column on the General page. When you click on a cell in the Timing Strategies column, the Select Strategies dialog appears as shown below.

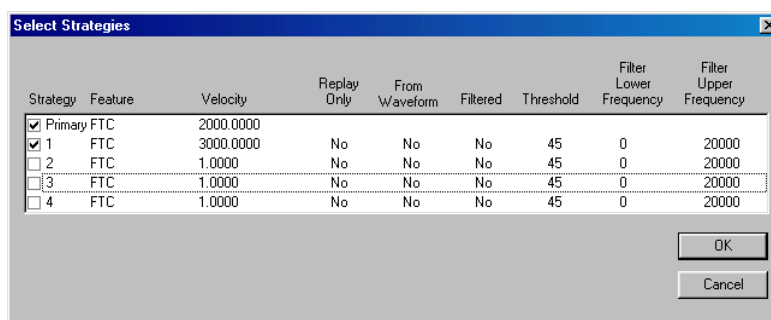


Figure 4-3. Select Strategies Dialog

Simply check or uncheck the strategies you want to use. The dialog summarizes the current strategy settings from the Timing page. After you click on OK, the cell in the Timing Strategies column will update to list the enabled strategies. A notable exception is the Primary Strategy which always appears in the list as either Primary or (Primary).

With additional strategies enabled, unchecking the Primary Strategy causes the arrival times from the Primary Strategy to be omitted from the second event analysis, but they are still used to perform the first analysis. If the location mode cannot produce a useful source location after the first analysis, the software will not attempt to refine the location with the secondary data. In that respect, the Primary Strategy acts as a filter for the second analysis even when the Primary results are not being used in the second analysis. To reflect this implied filtering, the Timing Strategies column displays (Primary) whenever the Primary Strategy is not to be included in the secondary analysis. In the figure showing the General settings page above, Location group 5 has Strategies 1 & 2 turned on with the Primary turned off and displays "(Primary),1,2". Unchecking all the strategies is equivalent to checking only the Primary Strategy. In both cases, the event is analyzed just once with the Primary Strategy, and the Timing Strategies column just displays "Primary."

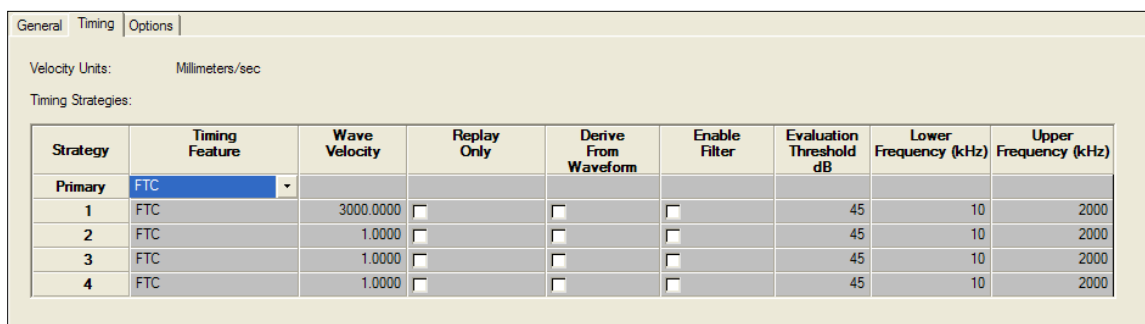


Figure 4-4. Location Setup – Timing Page

The current layout for the Timing page of the Location Setup dialog is shown above. For each strategy, there are a number of options available. Those to the right of Derive From Waveform are all related to waveform processing.

For the Primary Strategy, the only available option is the Timing Feature option. The other columns are not used because they are unneeded or not allowed for the Primary Strategy. For example, the Primary Strategy always uses the velocity associated with the location that is listed in the Wave Velocity column on the General page. The Primary Strategy is always applied during both acquisition & replay. The Primary Strategy relies on the results reported by your hardware in a single hit message, so waveform processing options are not available. The software later searches & pairs each hit message with an accompanying waveform message making the waveform processing options available for the secondary strategies.

Timing Feature: The Timing Feature column controls how the arrival times for a hit are determined. There are 2 basic choices: FTC (First Threshold Crossing) and Peak Timing (PT). All the arrival times are based on the time of test associated with the hit and a variable offset. During data acquisition, the hardware actually sets the time of test of a hit based on a threshold crossing where the Threshold level is controlled by the user in the Hardware Setup dialog. The FTC setting actually just defaults the arrival time to the time of test for the hit with no offset. The Peak Timing parameter adds the risetime (RT) as determined by the hardware to the hit time of the test. Risetime is defined as the delay between the first threshold crossing and the highest amplitude peak during the Peak Definition Time (PDT). The risetime feature must be enabled in the Data Sets tab of the Hardware Setup dialog for this option to work.

In addition to FTC and PT, there is a third timing feature called Custom Arrival. This is used when replaying a data file that has had the time-of-arrival of some or all of its waveforms modified through data selection. If this timing feature is used when replaying such a datafile, then the modified time-of-arrival's will be used instead of the original ones. For any other timing feature, the original time-of-arrivals will be used. Custom Arrival is designed for use in replay. If it is used during acquisition, it will default to FTC behavior.

Wave Velocity: The Wave Velocity column contains the velocity used during analysis by the location modes. The regression modes use a velocity to convert arrival time differences into differences in distances to the sensors. There is a velocity associated with each hit arrival time. For the Primary Strategy, that velocity always comes from the location group Wave Velocity setting. During the second analysis, the velocity from the secondary strategies is used when the arrival times are added to the list of hits in the event.

Replay Only: The Replay Only checkbox controls whether the strategy is applied during both data acquisition and replay or only in replay mode. Since secondary strategies cause events to be analyzed twice & the arrival times may be extracted from a digitally filtered waveform, secondary analysis can slow data acquisition rates. In cases where high data acquisition rates are important, you may prefer to postpone the secondary analysis until after data acquisition is over. The Replay Only flag just makes it easier to disable the strategy during acquisition without having to continually access the Location Setup menu to toggle the strategy on or off.

Derive from Waveform: The Derive From Waveform checkbox controls whether the strategy attempts to extract the arrival time from the waveform. If it is disabled (unchecked), the arrival time can still vary from that produced by the Primary Strategy if the Timing Feature for the secondary strategy varies from the Primary Timing Feature setting.

Enable Filter: The Enable Filter checkbox controls whether a digital filter is applied to the waveform before the arrival time is determined.

Evaluation Threshold: The Evaluation Threshold is similar to the Hardware Setup Threshold used during data acquisition. If the Timing Feature is set to FTC (First Threshold Crossing), the Threshold directly impacts the arrival time or even whether an arrival is detected. The Evaluation Threshold is applied to the waveform after the optional digital filter is applied. Note that the Hardware Setup Threshold affects what data is recorded during acquisition. The user can set the Evaluation Threshold to values higher than the Hardware Setup Threshold used during data acquisition & simulate the results that would have been recorded during acquisition with a higher Threshold. Beware, though; setting it to lower values cannot reveal data that was not recorded in the first place.

Upper & Lower Frequency: Lower Frequency & Upper Frequency columns control the bandpass cutoff frequencies. Turning on the digital filter & adjusting the filter cutoff frequencies can help isolate individual acoustic emission modes with differing wave velocities. Note that you should be aware of the sampling rate used during acquisition and avoid setting the Lower Frequency cutoff above $\frac{1}{2}$ the lowest sampling rate on any channel. You should also avoid targeting frequency ranges that your sensors do not respond to or that have been removed by your hardware's analog filters. Otherwise, you could be analyzing meaningless waveforms.

4.2.3 Options Tab Settings:

Figure 4-5. Location Options

There are a number of options available on the Options Page of Location Setup that control how location processing occurs. There are 3 different groups of options. Some apply to all location modes, some to only the regression location modes & some are specific to Tank Bottom Location. Here is a description of each group of options.

4.2.3.1 Options for All Location Modes

4.2.3.1.1 Allow Outside Events

The 'Allow Outside Events' checkbox controls what happens to events that produce source locations outside the structure. Usually, this is due to timing errors from an odd combination of hits, like the 2nd hit sensor being further away from the 1st hit sensor than the 3rd hit sensor. The default behavior is to reject them as failed events & not display the calculated source locations, but the user can optionally display them. This option does not apply to the Zonal or Linear location modes, but it does apply to Tank Bottom location & all the regression location modes. For example, if 2D Planar location is paired with the Plate structure, events that locate outside the Plate limits will be rejected. The option has no effect for 2D Planar or 3D Location when they are set to use the Free structure. The software does allow a margin of error at the structure limits so that events that locate very close to the limit still show. In general, that margin of error is an extra 2% of the structural dimension. For the Plate structure, that is 2% of the length and 2% of the height. For the Tank Bottom structure, it is 2% of the radius. When you have to Allow Outside Events enabled, you may wish to use manual axis limits on any 2D location graphs as outside events can sometimes have coordinated with very large values.

Figure 4-6. General Loc. Options

4.2.3.1.2 Free First Hit Only for Failed Events

The software assembles groups of hits into event candidates, which are then analyzed by whatever location groups the user has set up. When these potential events failed to produce a valid location for whatever reason, the software has always freed up all hits in the candidate event in the past. This meant it would not reuse those hits in other candidate events. A single stray hit due to noise can easily obscure a true event that occurs right after the noise hit. Now there is an option to only free the first hit of a failed event. The other hits will be allowed to continue to be considered for use in other candidate events. Checking *Free First Hit Only for Failed Events* will allow the software to find more potential events at the expense of more processing time, slowing down the replay of the user's data. This option is ignored during acquisition. During acquisition, all hits in a failed event are always freed to minimize the impact of location processing on the data acquisition speed.

4.2.3.1.3 Allow Overlapping Events

Normally, after an event is found, all the hits that made up the event candidates are freed up and not used for further event analysis, even if they are not in the found event. When this option is enabled, only the hits that made up the event that was found are freed. Any other hits in the group of candidates are put back for possible use in another grouping of hits for event testing.

4.2.3.2 Options for Regression Location Modes

There are 5 options specific to the regression based location modes. These are 2D Planar (& XY, YZ & XZ variants), 3D Location, Spherical Location, Cylinder Location & Conical Location, 2D Anisotropic, Anisotropic Cylinder.

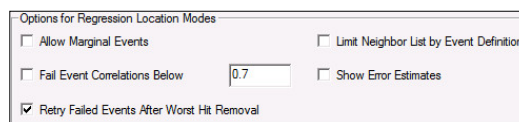


Figure 4-7. Regression Location Options

4.2.3.2.1 Allow Marginal Events

The Allow Marginal Events checkbox has the largest impact on 2D Planar & 3D Location. By default, this checkbox is not enabled. Marginal events are events that the software expects to have low accuracy. These are linear events for 2D Planar & planar events for 3D Location. See the Notes sections on those location modes for a more detailed discussion of what linear & planar events are & why they are problematic.

This option is useful for comparing with the results of PAC's older DOS packages that did not automatically filter out these kinds of events. If the user wishes, he can enable this checkbox & see the results of these events, but be aware that it is not recommended as the calculated results have a 50% or greater chance of being incorrect. The incorrect locations can sometimes be useful, though. For example, if a 2D linear event shows a result that is +5 from the line of sensors, then -5 is an equally valid answer according to the mathematics. This is also true for distance from the plane for 3D events. In both cases, the software has no way of deciding between the 2 locations, but you could always do a visual inspection at both locations. Note that marginal events often produce locations outside the structure, so you may want to enable Allow Outside Events if you are using Allow Marginal Events.

4.2.3.2.2 Fail Event Correlations Below

4.2.3.2.3 Retry Failed Events After Worst Hit Removal

All of the regression location modes now calculate a correlation coefficient. These values are reported as "q" (= quality) in the Line Display:

```
* Gp# 1[2,3,1,4,5,6] x = 11.45, y = 20, q = 0.9902 dT[ 68, 71, 302, 316, 323]
```

Correlation is also available for use on graph axes. For events with 4 or more hits (5 or more for 3D Location), the reported correlation values range from 0 to 1, with 1 being a very good event & 0 being a bad event. For events with 3 hits (4 for 3D Location), the correlation coefficient should be very close to 1.0. Note that for these events, a value of 1 does not guarantee a high-quality event as timing errors cannot be detected with only 3 hits. Other location modes like Linear & Zonal will always report correlation coefficients of 0 as they do not calculate them.

There are 3 location settings relevant to correlation coefficients. There is a Fail Events Correlations Below checkbox along with an edit field to set the failure level. The regression location modes can optionally reject events with low correlation coefficients, which are less likely to produce useful location information. The failure level can be adjusted based on the user's personal preference.

The third new setting is a checkbox labeled Retry Failed Events After Worst Hit Removal. This allows failed events to be reanalyzed after the worst hit of the event has been removed. Worst hit removal & reanalysis can be applied to events that fail due to low correlation values & can significantly improve the correlation coefficient, which implies improved location accuracy as well. Here are a few examples:

Worst Hit Removal & Reanalysis on Events with Low Correlations

Time of Test

* 0 00:00:23.3043055					
- Gp# 1[5,6,4,7,3,8,2,9,1,11]	x =	7.897	y =	-4.841	q = 0.483
+ Gp# 1[5,6,4,7,3,8,2,9,1]	x =	4.4,	y =	-3.246	q = 0.9967
* 0 00:02:11.5468993					
- Gp# 1[2,1,3,12,4,11,5,6,10,9,7,8]	x =	4.129	y =	7.312	q = 0.4556
+ Gp# 1[2,1,3,12,4,11,5,6,10,9,7]	x =	2.835	y =	5.095	q = 0.9822
* 0 00:02:29.9029040					
- Gp# 1[5,6,4,7,3,8,2,9,1,11]	x =	9.131	y =	-6.009	q = 0.413
+ Gp# 1[5,6,4,7,3,8,2,9,1]	x =	4.521	y =	-3.38	q = 0.9968
* 0 00:03:22.0759543					
- Gp# 1[8,9,7,10,6,11,5,12,4,1,3,2]	x =	-6.047	y =	-7.491	q = 0.3598
+ Gp# 1[8,9,7,10,6,11,5,12,4,1,3]	x =	-3.915	y =	-4.642	q = 0.987

- = No Correlation Filtering, + = Correlation Filtering + Worst Hit Removal + Reanalysis

Worst hit removal & reanalysis can also be applied to failed outside events when Allow Outside Events is not checked. Reanalysis after worst hit removal can shift the results of the event analysis from outside the structure to inside the structure, sometimes with the removal of just 1 hit. Here are some results for events that originally occurred way outside the user's structure:

Worst Hit Removal & Reanalysis on Outside Events

Time of Test

* 0 01:05:04.1100330					
- Gp# 1 [5,6,4,7,3,10,12]	x =	71490	y =	-50270	q = 0.9988
+ Gp# 1 [5,6,4,7,3,12]	x =	15.55	y =	-13.91	q = 0.8813
* 0 01:09:39.3894840					
- Gp# 1 [8,9,10,5,1]	x =	-31600	y =	-31380	q = 0.9899
+ Gp# 1 [8,9,5,1]	x =	-15.99	y =	-17.58	q = 0.978

- = Outside Events Allowed, + = No Outside Events + Worst Hit Removal + Reanalysis

Worst hit removal & reanalysis is never applied to marginal events. Events are considered marginal due to a problem with the geometry of the sensors included in the event. For example, for 2D Planar, an event where all the hit sensors are in a straight line is considered marginal. Removal of any hit, still leaves all the hits in a straight line, so the event is still marginal. For that reason, marginal events are never reanalyzed.

4.2.3.2.4 Limit Neighbor List by Event Definition

Enabling this option will restrict the list of neighboring sensors by comparing the sensor separation to less than two times the event definition time of the location group. This can help speed up location processing and reduce the number of events found.

4.2.3.2.5 Show Error Estimates

Enabling this will add additional information about the results of the event calculation to the line display, including the error estimates, fit, max deviation, and the channel that was the 'worst-hit.'

4.2.3.3 Options for Tank Bottom Location

There are 2 options specific to the Tank Bottom location. One allows the software to display more than 1 source location for a given event by varying which 3 hits it uses in the calculation. The Force Outside Events to Rim option works in conjunction with the Allow Outside Events to control where outside events are plotted. If Allow Outside Events is disabled, the Force Outside Events to Rim does nothing. If Force to Rim is disabled, the events are plotted at their calculated coordinates. If both Allow Outside Events & Force Outside Events to Rim are enabled, then all the outside events are plotted on the rim of the tank bottom structure. For examples of how the options affect your Tank Bottom location results, see the notes for the Tank Bottom location mode in section 4.4.9.

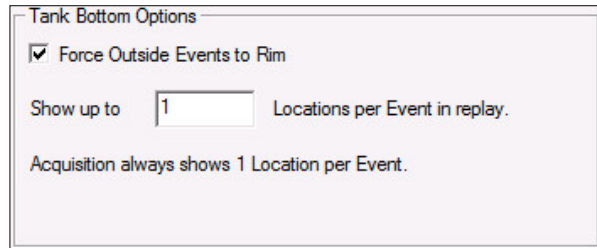


Figure 4-8. Tank Bottom Location Options
For examples of how the options affect your Tank Bottom location results, see the notes for the Tank Bottom location mode in section 4.4.9.

4.2.4 Distance Units Dialog

Pressing the Distance Units button opens the Distance Units dialog box. The user can select his desired units from the dropdown list. The user is given an option of converting old numerical values to the new units by simply checking the "Convert all numerical values based on new units." option. This will cause the event definition value, event lockout value, the overcal settings, the wave velocities & sensor positions to be updated as appropriate.

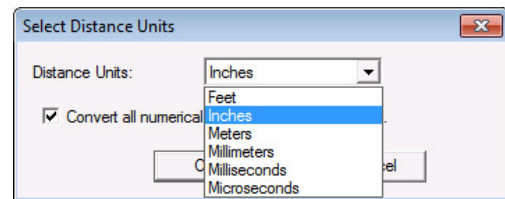


Figure 4-9. Distance Units

4.2.5 Material Properties Dialog

Pressing the Material... button opens the Material Properties dialog box. This dialog box is just a simple database of velocities for acoustic emission in various materials. When you find a useful velocity, simply highlight it & press Ctrl-C, Ctrl-Ins, or Right-click & select Copy. You can then insert it into any field in the Location Setup dialog with Shift-Ins or Ctrl-V.

If you wish to add some of your own numbers, you can simply type a new entry for the Material Name, fill in the velocities & press the Add button. The Delete button removes an entry from the database, while the Update button lets you replace bad values.

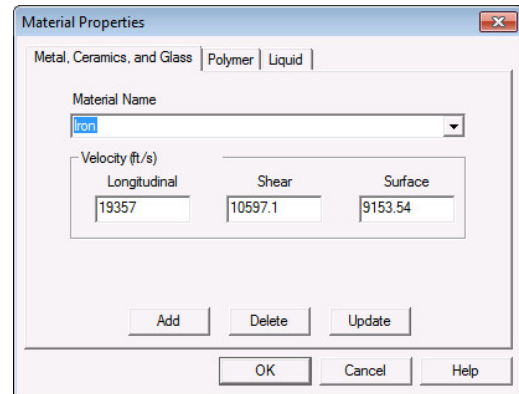


Figure 4-10. Material Properties

4.2.6 Attenuation Dialog

Pressing the Attenuation... button opens the Attenuation dialog box shown at right. The sole purpose of this dialog is to allow the software to account for the decay of the acoustic emission as a function of distance traveled & to estimate of the initial amplitude at the source. It uses the amplitude observed at the first hit sensor, the distance to the source for the event, and the attenuation table entered by the user in this dialog. The attenuation table documents amplitude as a function of distance. The units of distance are controlled by the Location Setup Distance Units setting. The Update button will cause the graph to display the current values in the table. The user may use the Save & Recall buttons to save or recall an attenuation table to/from the disk

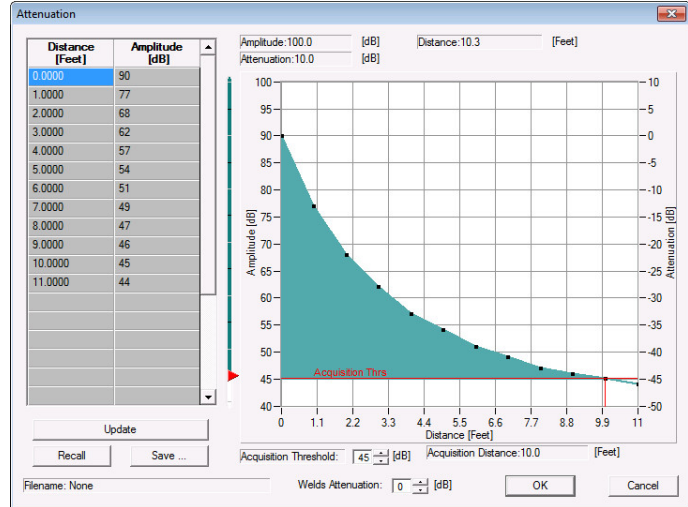


Figure 4-11. Attenuation Dialog

The basic procedure uses the table to calculate the difference between an amplitude at the source & at the observed distance from the source. If the source to first hit sensor distance falls within the range of values in the table, an amplitude at that distance is calculated using linear interpolation between the 2 nearest distances in the table. Using the table above as an example, if the source was determined to be 2.5 feet from the first hit sensor, the table has the 2 closest entries as 68 at 2.0 feet & 62 at 3.0. The amplitude predicted at 2.5 feet would be 65 ($= 68 - (68-62)*0.5$). The correction applied to the amplitude at the first hit sensor would be 25 dB ($= 90 - 65$). If the first hit amplitude was 60, the line display would report a Source Amplitude of 85 for this example. The amplitudes are not extrapolated beyond the maximum distance. If the distance is beyond the maximum distance in the table, only the maximum correction of 46 ($= 90 - 44$) is applied. If the user never accesses this dialog to fill in the table, no correction is applied & the source amplitude reported is identical to the amplitude of the first hit sensor.

4.2.7 Guard Sensor Dialog

Pressing this button opens the 'Guard Sensors' dialog box. Guard Sensors can be used to lockout out events on selected location groups whenever a hit is first detected on a guard channel. It is designed for situations in which one wishes to ignore events that originate outside of the location group.

In the example setup shown, sensors 1, 3 & 5 are protecting Location Group 1 from outside events. If there is a hit on sensors 1, 3, or 5, then Location Group 1 will not process any events for the next 1000 us. If the Event Only flag were checked, only the next event on Location Group 1 would be locked out instead of being locked out for the next 1000us.

The Guard Data Filter Setup controls what hits get saved into the DTA file during an acquisition if guard sensors are enabled. The default setting is to just save all hits, but you can restrict the data saved to just those hits that make up events & the guard sensor hits. This can be useful in situations where external noise creates a large number of hits that end up being useless for the purpose of location. Not only do the extra hits take up space in the data file, but they also slow down replays of the data file too.

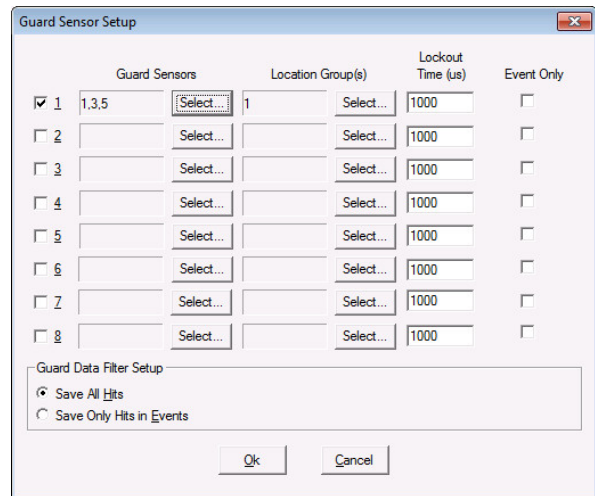


Figure 4-12. Guard Sensor Set up

4.3 Location View & Sensor Placement Dialog

Before discussing the Sensor Placement Dialog, there is one important point about the physical placement of your sensors that you should understand, and it applies to all of the regression location modes. *You can place sensors anywhere, but you will get better results with the regression location modes if your sensor layout is made up of an evenly spaced triangular array of sensors. The optimal arrangement would be a*

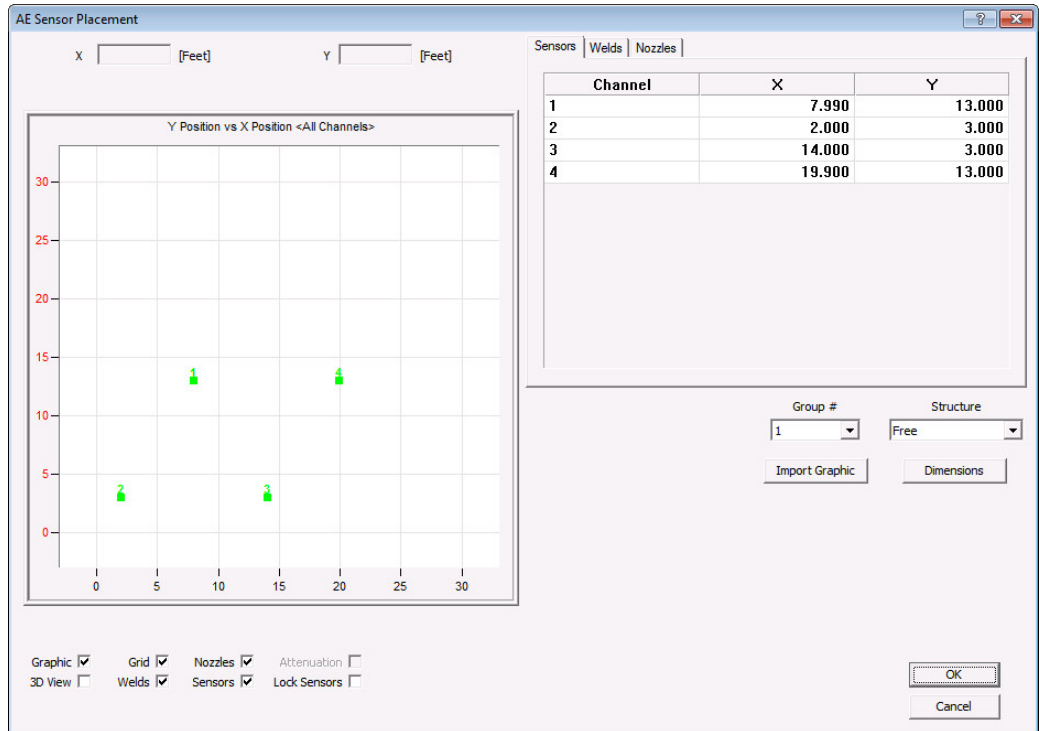


Figure 4-13. Sensor Placement Dialog

triangular grid made up of equilateral triangles, but the regression modes do not require exact equality of the triangle legs or even regular spacing between the sensors. Avoid creating triangles with one very long or very short side, however, as they do not work well for locating sources.

Pressing the Location View button in the Location Setup dialog opens the Sensor Placement dialog box shown above. The 2 main areas are the actual view of the sensors on the left-hand side of the dialog & a listing of their coordinates on the right-hand side. The grid area for the sensor coordinates also has tabs that will display the coordinates of any welds and nozzles created. You can change the color of the sensors using the Graphing menu to access Sensor Color on the Colors submenu.

When using the Sensor Placement dialog, there are 2 ways to add new sensors. When working in the grid area, pressing the INS key adds a new sensor. The F2 key allows you to edit its coordinates & move them to the correct location. Alternatively, you can Right-click the sensor view area to get a context menu like the one shown to the right. Place Sensor & Next Sensor allow you to put a new sensor at the current position. Next Sensor numbers it automatically using the next available sensor number. Place Sensor will ask for a sensor number. You can also use AutoPlace Sensors to create a grid of sensors with regularly varying coordinates. You work with welds & nozzles in a similar manner.

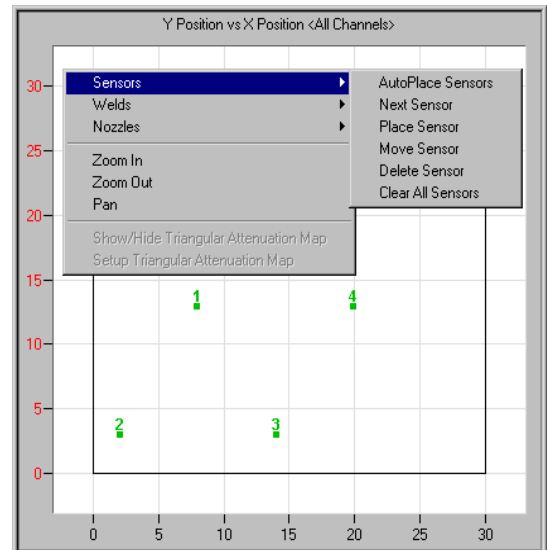


Figure 4-14. Sensor Menu

The exact menu you see may differ from that shown here as it depends on the Structure.

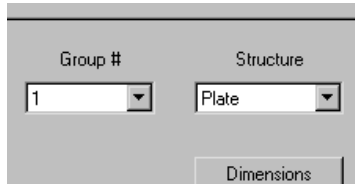


Figure 4-15. Structure List

The Structure list is just below the grid area. Support for welds, nozzles, manual placement of sensors & auto-placement of sensors varies with the Structure setting. When creating a sensor layout for a new location group, the first thing you should do in the Sensor Placement dialog is to set the Structure type. The Group # setting allows you to quickly switch to editing the sensors for another location group without having to return to the Location Setup dialog. If you made changes to the current settings, you are prompted to save them.

Below the Structure setting is the Dimension button. If you click it, one of the dimensions dialogs will appear. They are also specific to the type of Structure. The one shown to the right is for the Plate structure. When working with a new location group, the second thing you should do in Sensor Placement is to fill in the dimensions for the structure you are working with. The dimensions affect the scaling of the sensor view area. If you have incorrect values entered, some sensors may not be visible in the sensor view.

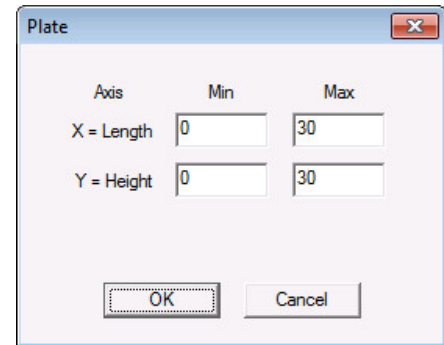


Figure 4-16. Plate



Figure 4-17. Sensor Label Checkboxes

Below the sensor view area, there are a number of checkboxes that control the sensor view. The Nozzles, Sensors & Welds checkboxes control whether or not those objects are displayed. Likewise, the Grid checkbox controls whether or not the sensor view shows a Cartesian grid to

facilitate the placing of sensors. The Graphic checkbox controls whether or not a representation of the structure is drawn on the graph. The Lock Sensors checkbox disables the moving of sensors with the mouse. The 3D View checkbox will toggle the display between 2D & 3D views of the sensor layout if that structure type supports both 2D & 3D views.

4.4 Notes on Individual Location Modes

As mentioned in the section describing the Sensor Placement Dialog, there is a point that all users should understand about the regression location modes & it is important enough to bear repeating. *You can place sensors anywhere, but you will get better results with the regression modes if your sensor layout is made up of an evenly spaced triangular array of sensors. The optimal arrangement would be a triangular grid made up of equilateral triangles, but the regression modes do not require exact equality of the triangle legs or even regular spacing between the sensors. Avoid creating triangles with one very long or very short side, however, as they do not work well for locating sources.*

In general, you can just read the section on the location mode you are interested in using. All the regression location modes are a variation on 2D Planar, however, so you might also want to read it as well. If you'd like more information on setting up a Zonal, Linear, or Tank Bottom location group, you can also see the Examples section.

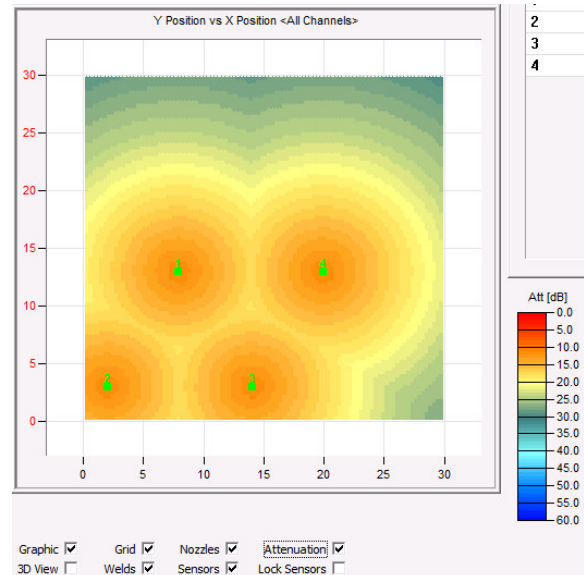


Figure 4-18. Attenuation

4.4.1 Zonal Location

Zonal location is the simplest location mode there is. It just returns the coordinates of the first sensor in the list of hits that make up the event. You can place sensors absolutely anywhere & you can get better resolution on source locations by just using more sensors. A better way to improve your source locations would be to switch to a more sophisticated location technique like Linear or 2D Planar. See section 4.5.1 for a detailed example of setting up a Zonal Location group.

4.4.2 Linear Location Modes

4.4.2.1 Linear Location

Linear location is a step up from Zonal location in that it actually attempts to locate the source from the timing data in the event rather than just restricting it to a region of space around the first hit sensor. Linear location uses 2 hits & the difference in their arrival times to calculate the source location using linear interpolation of the coordinates for the 2 sensors. Its drawback is that in a 2-dimensional area, it restricts the source to points on the line segments that connect the sensors when real sources do not face that restriction. Still, that can be good enough when looking for a source along with a long series of sensors on, for example, a pipeline.

Linear location requires the sensors to be laid out in a line, although the line does not have to be perfectly straight. You could lay them out in an arc. It is not possible, however, to have the line double back on itself, so laying out the sensors in a square or circular array would not work. Linear location relies on a list of the sensors sorted by the x coordinate when deciding which 2 hits to use in an event. You can also use a vertical line. If the x coordinate of the first & last sensor in a Linear location group is equal, the list of sensors gets sorted by the y coordinates. Linear location only forms events between adjacent sensors, so there are only 2 possible 2nd hit sensors for any 1st hit sensor. When there are hits on both, it decides which hit to use based on the ratio of the Δt value of the hit divided by the maximum Δt calculated from the sensor spacing & the user entered velocity. The hit with the lowest relative Δt value becomes the 2nd hit of the event used for calculating the source location. The logic is that the 2 sensors on either side of the source will have the shortest relative Δt value, while the Δt value between the 1st hit sensor & other sensors will always be higher. See section 4.5.1 for a detailed example of setting up a Linear Location group.

4.4.2.2 LinearXY Location

LinearXY location is related to Linear location in that the source location is still calculated by linear interpolation between the arrival times of the 2 hits in the events. The difference between LinearXY & Linear location is that LinearXY removes the assumption that the sensors will be laid out in a line & the sorting based on coordinates. With LinearXY location, you can put the sensors literally anywhere in 2-dimensional space, so it is perfectly acceptable to have your sensors laid out in a square, a circle, or a rectangular array. The advantage &, at the same time, the drawback to this method is that it allows events to form between any 2 sensors in the array. For a square, not only are events allowed on the sides of the square, but they can also occur on the diagonals. LinearXY does still look for the pair of sensors with the lowest relative Δt values.

4.4.2.3 Wrapped Linear Location

A common question from users of PAC's DOS software is how to turn on wrapping in a Linear location. The software doesn't have a checkbox to enable wrapping in Linear location like the old DOS software, but there are 2 methods you can use to set it up. The first method is an older long mentioned in the AEwin software manual & it works in any version of AEwin independent of when it was released or how the user's licensing options are configured. It requires the use of 2 linear location groups. The second method only works in the newest releases of AEwin licensed for Full Location. It makes use of a newer Wrapped Linear location mode.

4.4.2.3.1 Method #1 — Wrapped Linear Location Using 2 Location Groups

Below are a series of graphs from the sample layout, WrappedLinear.lay, showing how it is done with 2 Linear location groups.

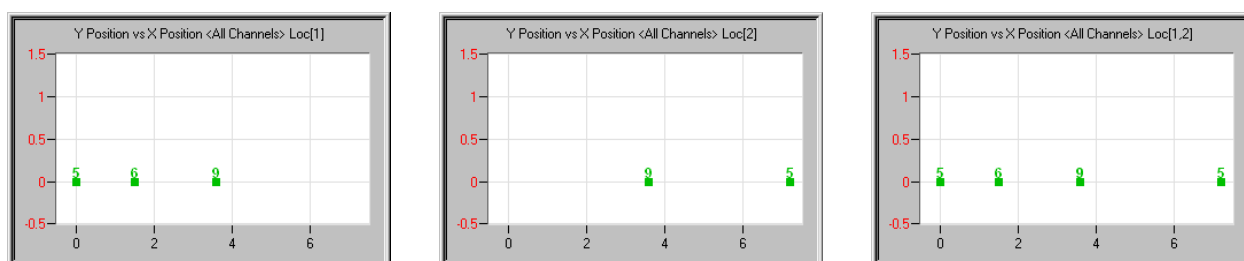


Figure 4-19. Wrapped Linear Location using 2 Groups

The graph on the top left shows location group 1, which consists of sensors 5, 6 & 9. The linear location allows events between sensors to pair 5 & 6 or 6 & 9 in this location group. Turning on wrapping requires allowing events between sensors 5 & 9, and it requires a second set of coordinates for the wrapped location of sensor 5. This is accomplished by creating location group 2 shown in the graph on the top right above. Sensor 9 is common to both location groups at the same coordinates. Sensor 5 in location group 2 is at its wrapped location, which is just the position of sensor 5 in group 1 offset by the structure circumference. The graph on the bottom just illustrates that you can superimpose both location groups 1 & 2 on a single graph using Graph Setup. You also need to turn on both groups in Location Setup. Do not enable Show All Points for both groups, though, if you want the software to decide whether events between sensors 6 & 9 for group 1 or events between sensors 5 & 9 for group 2 make the most sense based on relative Δt values. If you were to enable Show All Points, then you could end up with a single set of hits producing an event between sensors 6 & 9 as well as between 9 & the wrapped location of sensor 5.

4.4.2.3.2 Method #2 — Wrapped Linear Location Mode

Method #2 is illustrated in the WrappedLinear2.lay example layout. A Wrapped Linear location mode is currently available as a standard feature of the software. In older versions, it was part of the “Full Location” option.

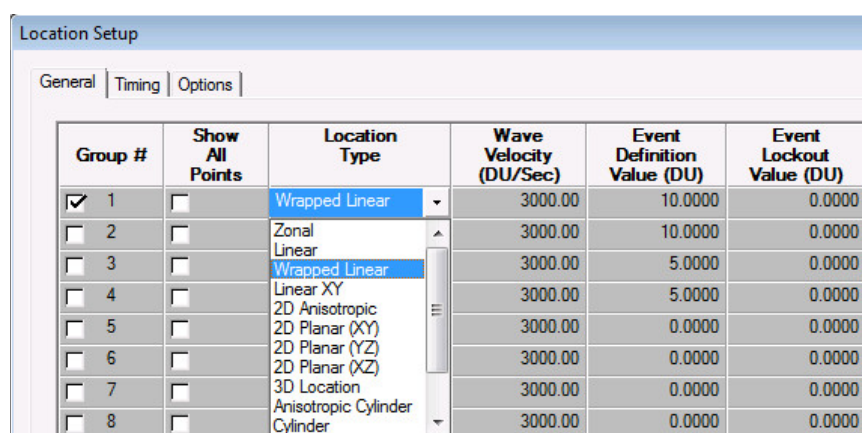


Figure 4-20. Wrapped Linear in Location Set up

If you can't see a Wrapped Linear location mode in Location Setup, you will have to either purchase the “Full Location” option, upgrade to the latest version of AEwin for your hardware, or use Method #1 for doing wrapped linear location. Contact MISTRAS Group Inc., Products & Systems Division, Customer Service (customer.service@mistrasgroup.com) for information on any needed upgrades or licensing options.

The Wrapped Linear location mode consolidates the 2 location group method into 1 location group. Here is the location graph from WrappedLinear2.lay, which demonstrates the newer Wrapped Linear location mode.

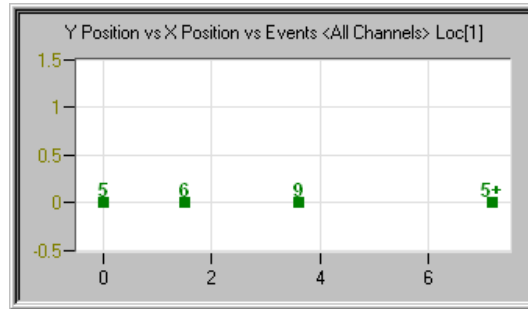


Figure 4-21. Graph Showing Wrapped Linear Location Group

Here the wrapped location for sensor 5 is marked on the graph as “5+”. Putting the same sensor in 2 locations is a bit of a complication for AEwin because the Sensor Placement dialog is only designed to have 1 set of coordinates for each sensor. The way around this is to invent a new sensor in Sensor Placement that is used for the wrapped sensor location. Here’s the Sensor Placement dialog showing the sensors for this location group:

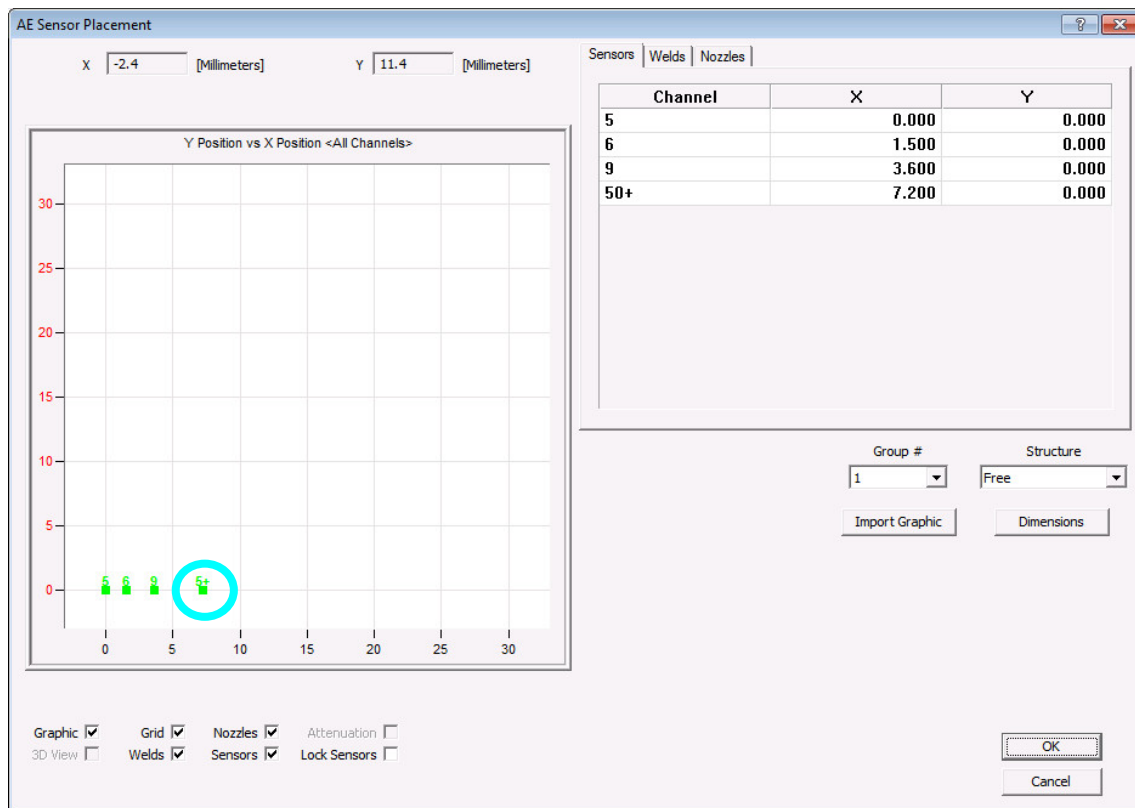


Figure 4-22. Wrapped Linear Location Group in Sensor Placement

Notice that the table of sensors lists sensor “50+”, but the sensor diagram shows sensor “5+” at the location of “50+”. These are the same wrapped sensor as indicated by the “+” signs. The only location mode that relabels a sensor on the graph is Wrapped Linear. If this location group is changed to a regular Linear location group, then the “+” signs will disappear in the grid & on the graph & the sensor “5+” on the graph would become sensor 50. There is nothing magical about sensor number 50 used in the table. It could have just as easily been labeled sensor 51, 52, or any other unused channel number.

Wrapped Linear location always replaces the label of the last sensor with the number of the first sensor & a plus sign to indicate that it is the wrapped sensor location. It picks the first & last sensors based on their coordinate values, not their sensor channel ID's. Internally it sorts the list of sensors based on the X coordinate to find the sensors with the lowest & highest X coordinates & then labels the highest X, the rightmost sensor on the graph with the ID of the lowest X, leftmost sensor & the "+" sign. It also adds the "+" sign in the table to let you know which sensor it picked as the last sensor. If you input a vertical array of sensors where all the sensors have the same X coordinate, Wrapped Linear will sort the sensors based on their Y coordinate to decide which sensors are the first & last sensors to identify the wrapped sensor.

4.4.3 2D Planar & 2D Planar (XY)

There is no fundamental difference between 2D Planar & 2D Planar (XY) locations except on the menus. Enabling 3D Location also enables 2D Planar (YZ) & 2D Planar (XZ) location. To distinguish the standard 2D Planar from those 2 modes, it is relabeled 2D Planar (XY) location whenever the 3D Location option is enabled. 2D Planar (YZ) & 2D Planar (XZ) locations are discussed in the section on 3D Location.

For 2D Planar location, a particularly difficult problem is locating the source of an acoustic emission event if all the hits are from sensors in a straight line (linear events). For linear events, even if a distance from the line can be determined, there is no way to distinguish positive & negative directions away from the line. Random error in the data can also make determining the distance from the line highly suspect. By default, the software filters out linear events and does not attempt 2D Planar location on them. You could always enable Allow Marginal Events (See discussion in section 4.2.4), but then you could end up flooding your graphs with lots of low-accuracy results. A better solution is to use a layout that minimizes the chances of linear events being observed. An example of a bad layout is shown below. It is bad because the horizontal sensor spacing is much smaller than the vertical spacing.

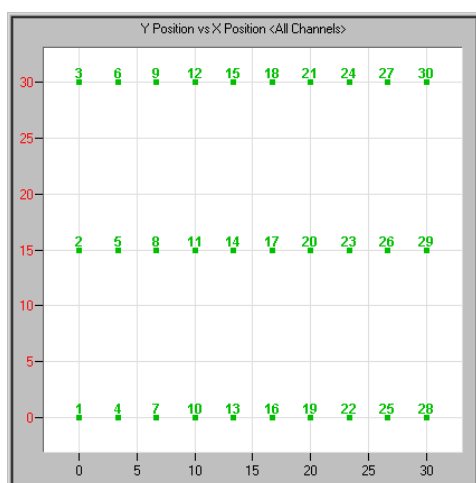


Figure 4-23. Bad 2D Planar Layout

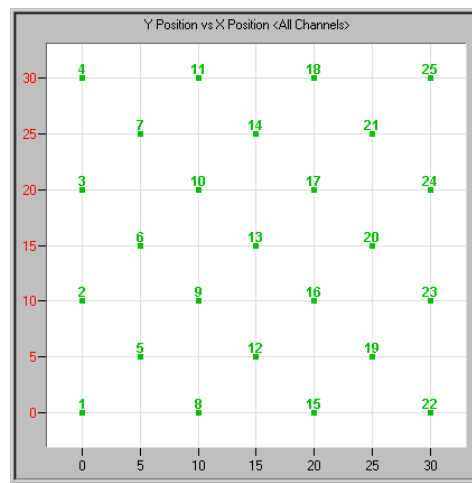


Figure 4-24. Good 2D Planar Layout

A better layout would have all distances in between the sensors approximately equal. Triangles with one very long side or one very short side can also cause problems as they approach becoming a straight line of sensors. The ideal 2D Planar layout would be made up of equilateral triangles, but exact equality of all sides is not required, as shown in the good layout above. As mentioned in the introduction, you should avoid creating triangles with one very long side or one very short side. Events based on triangles with one very long or very short side become linear events when one side goes to infinity or to 0. Short of those limits, though, they can still act similar to the linear events and won't accurately locate any sources outside of the triangle.

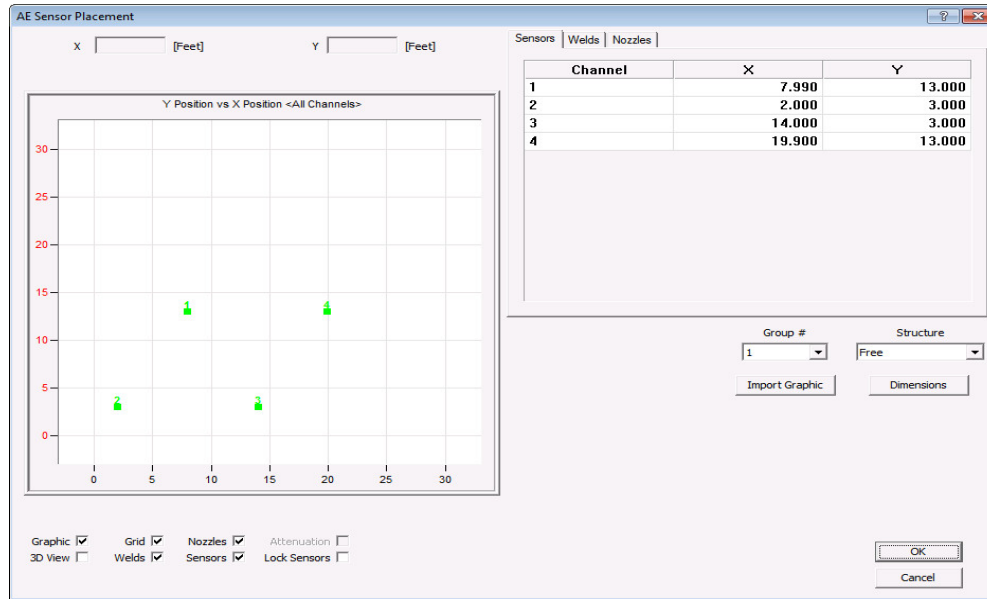


Figure 4-25. 2D Planar Sensor Placement

Here is Sensor Placement setup for a 2D Planar mode with the Plate structure. 2D Planar location is designed to work with either the Free or Plate structures. Although there isn't any difference in 2D Planar & 2D Planar (XY) location during analysis, there is one difference in Sensor Placement. A 2D Planar (XY) location group has the option of displaying the sensors in the 3D graph if the user wishes it. For a 2D Planar (XY) location group with the structure type set to Free, the 3D View checkbox enables. The user may use it to toggle between 2D & 3D views of the sensors. The checkbox disables the Plate structure, though.

The Dimensions dialog for both the Free & the Plate structure contains the max/min limit settings for both the X & Y axes. These settings control the scaling of the graph in the sensor view of Sensor Placement. For the Plate structure, they have added significance. When 2D Planar is analyzing events, it will restrict results location groups using the Plate structure to points on the plate. To allow locations close to the limits to report, the software uses a tolerance of 2% of the axis length when deciding whether it is an acceptable result. For example, if x has a minimum value of 0 and a maximum value of 100, then results with an x coordinate greater than 102 or less than -2 are thrown out as invalid. If you don't want to limit your results, you can either set the Structure type to Free or stick with Plate & enable the 'Allow Outside Events' checkbox in Location Setup. See section 4.2.3 for more details.

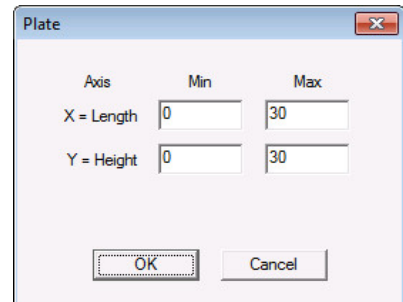


Figure 4-26. Plate

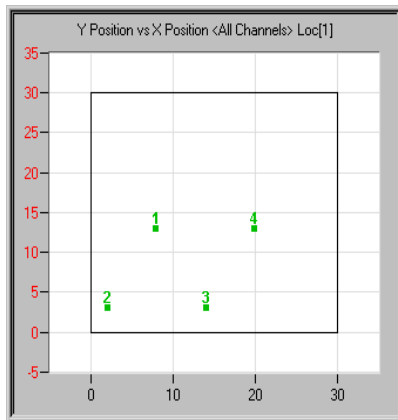


Figure 4-27. 2D Planar Graph

After setting up the sensors & a 2D Planar location group, you will want to create a location graph. Use the Graphing pulldown menu's New Graph option & then configure the graph using the Graph Setup dialog box (F3 shortcut key). To the right is a 2D X-Y location graph based on the 2D Binned Scatter graph type where the location group is using a Plate structure. If you create a 2D Binned Scatter, 2D Colored Scatter, or 2D Scatter graph that uses X & Y Position, the graph will automatically draw the sensors on the plot for any location groups selected on the Event Groups tab. In the Graph Setup dialog, make sure that the Input Data is set to Events and use the Event Groups tab to select which location group(s) to use for source information. Below is the corresponding Graph Setup dialog to illustrate the typical settings.

4.4.4 3D Location, 2D Planar (YZ) & 2D Planar (XZ)

Primarily this section focuses on 3D Location, but 2D Planar (YZ) & 2D Planar (XZ) are related. All 3 location modes are designed to use 3D coordinates. The 2D Planar (YZ) & 2D Planar (XZ) modes are similar to normal 2D Planar & 2D Planar (XY) locations, with the exception that you can enter 3D coordinates & have the software locate the source while ignoring either the X or Y coordinate. The reason these modes are useful relates to a problem with sensor placement & event handling for 3D Locations.

The equivalent of the linear event problem for 2D Planar location is a problem with planar events for 3D Location. A planar event is one in which all the hits are on sensors in the same plane. For planar events, it is hard to determine distance from the plane, not to mention distinguishing sources in the positive direction from those in the negative direction. By default, the 3D Location algorithm filters out planar events and does not attempt to locate them as the distance from the plane is often quite simply useless. You can override this behavior by enabling the Allow Marginal Events checkbox in Location Setup, but it exists primarily for compatibility with older versions of PAC's DOS software. See section 4.2.4 for more details.

Because of the planar problem, you will get better results if you set up your sensors to minimize the number of sensors that line up across the body of the test structure. Avoid using a rectangular grid layout that lines up sensors on opposing faces of your structure. Instead, try to stagger the sensors so that no 2 sensors are direct across from one another. The bad sample layout above uses a rectangular approach with 3 rows of 3 sensors directly opposite one another. The good layout uses a triangular approach by staggering each row on one side by half the sensor-sensor distance, and then it inverts the pattern on the opposite wall. The diagram may not show it well, but sensor 4 is below the midpoint of sensors 1&2, above the midpoint of 7&8, and across from the midpoint of 12&13. Both sides have 3 rows of sensors, but the left-hand side has a 3-2-3 pattern of sensors, while the right-hand side has a 2-3-2 pattern of sensors. This staggering minimizes the number of sensor planes through the body of the structure.

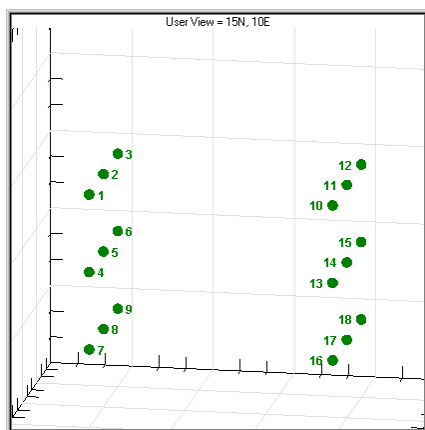


Figure 4-28. Bad 3D Location Layout

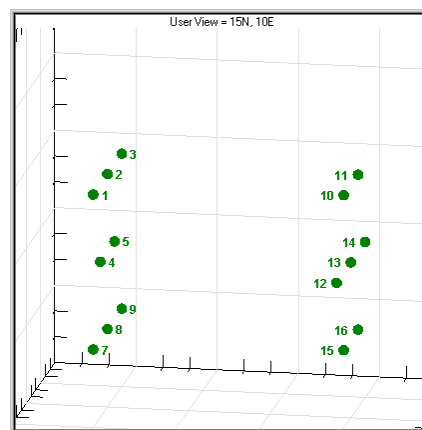


Figure 4-29. Good 3D Location Layout

One problem with 3D Location filtering out planar events is that pencil lead breaks will not locate with 3D Location unless the lead break is energetic enough to register with a sensor on one of the other faces. If you enable Allow Marginal Events, you may get results far outside the structure. A good alternative, in this case, are the 2D Planar (XZ) & (YZ) location modes. The distinction between them is what sensor coordinates they use when fitting the data. Both allow you to enter 3D coordinates but ignore one of the values when locating. 2D Planar (XZ) uses the X & Z sensor coordinates & ignores the Y values. 2D Planar (YZ) ignores the X coordinate. They are useful when doing pencil lead breaks by sensors confined to one face of your structure. With the 3D Location option enabled, 2D Planar (XY) will allow you to enter 3D coordinates & only locate using the x & y coordinates. Fundamentally that's no different than the standard 2D Planar location. Both are ignoring the z coordinate, but as a convenience, one allows you to enter a non-zero value for Z.

For best performance, any 2D Planar groups using 3D coordinates should be confined to just one face of the test structure. Parallel sensors on an opposing face could appear as duplicate sensors with identical coordinates when the 3rd unused coordinate is ignored & which would lead to errors in the event location calculations. You should also note that during acquisition, a result from a 2D location mode does not necessarily imply that the source was confined to the 2D location plane. Just as for planar events in 3D Location, distance from the plane is indeterminate. You can draw a perpendicular to the 2D plane at the best fit location, and the source event could be anywhere on the line. Of course, for pencil lead breaks, you will know in advance that the source is confined to a given face and be able to check your sensors before starting acquisition.

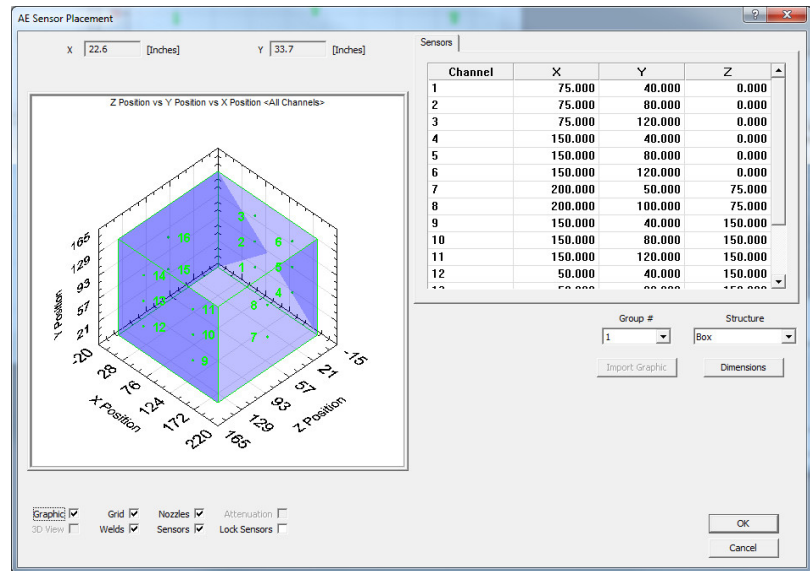


Figure 4-30. 3D Sensor Placement

For 3D Location, the Sensor Placement dialog shown here allows you to directly enter X, Y, Z coordinates for any groups that are considered a 3D Location group with the right structure type. 3D Location is designed to use either the Free & the Box structure types.

The Dimensions dialog for the Box & Free structures is similar to the one for the 2D Free or Plate structures with the addition of max/min settings for the Z-axis. Similar to the way the Plate structure limits 2D Planar results, the Box structure acts as search limits for 3D Location. If any location results are more than 2% above or below the limits, the result will be considered bad & is thrown out. For example, if Z Min is 50 & Z Max is 150, then any location results where z is above 152 or below 48 will be filtered out. As with 2D Planar, using the Free structure avoids imposing any limits on the results. Alternatively, you can enable Allow Outside Events in Location Setup. See section 4.2.3 for details.

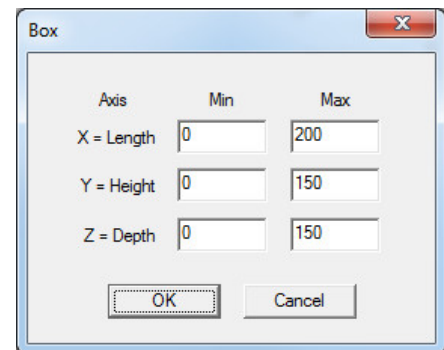


Figure 4-31. Box Structure Dimensions Dialog

You cannot graphically enter 3D sensor locations in Sensor Placement, but you can graphically see the sensors in the 3D view. If you do not see a grid column for Z or a 3D view of your sensors, go back to the Location Setup dialog & make sure the group type is set to 3D Location, 2D Planar (YZ), or (XZ). Also, make sure you have the structure type set to Free or Box.

Use the INS key & F2 to insert a new sensor & edit its coordinates in the grid. As an alternative to creating all the sensors manually, you can also use the AutoPlace Sensors option to create a grid of sensors with varying x,y coordinates & a fixed z coordinate of 0. You would then manually edit the z coordinates of each sensor. Note that you can use AutoPlace Sensors multiple times without erasing the old sensors. That allows you to build up a 3D grid one layer at a time. Right-click the graph in Sensor Placement to find AutoPlace Sensors.

To create a location graph for 3D Location, use the 3D Scatter graph & set each axis to X, Y & Z Position, which correspond to the x, y & z coordinates of a located source point. The graph to the right is a 3D Scatter graph setup as a location graph. As shown below, Graph Setup has the Input Data is set to Events & the Event Groups tab has the 3D Location group enabled.

If you create a graph that uses X, Y & Z Position, the graph will automatically draw the sensors and structures on the plot for enabled location groups.

For 3D Scatter plots, both the sensors & the data points are drawn as small cubes. You may use the Graphing Colors menu to change the colors. The data points use the color assigned to Plot 1 under Plot Colors and are drawn slightly smaller than the sensor points. If you would like larger points, Right-click your graph & turn on the Large Scatter Graph Points. Both the sensor points & the data points will get larger.

Graph Setup for the example graph above is shown here. It sets the X (Horizontal) Axis to X Position, Y (Vertical) Axis to Y Position & Z (Depth) Axis to Z Position. You may prefer to remap the axes if you are used to discussing z as height. For example, if you are used to z being vertical, y being horizontal, and x being out of the plane of the monitor(paper), set the X (Horizontal) Axis to Y Position, the Y (Vertical) Axis to Z position and the Z (Depth) Axis to X Position.

You may also want to change the positive direction of the Z (Depth) Axis. The Invert checkboxes will let you specify whether the positive direction is in or out of the monitor. Similarly, you can also invert the horizontal & vertical axes.

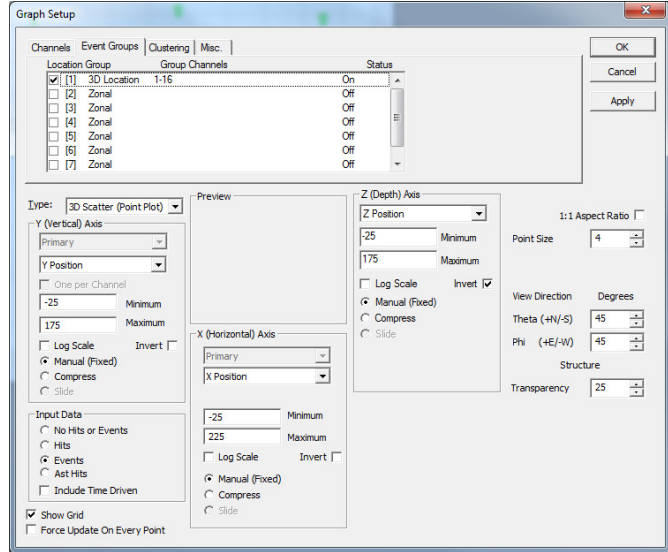


Figure 4-32. 3D Graph Set up

You can also create 2D location plots using X, Y & Z Position settings on a 2D Binned Scatter graph as done in the section on 2D Planar. If you do, the sensors will also draw on the 2D plots using 2 of the 3 sensor coordinates ignoring the unused dimension. For example, an X & Z Position graph would show the sensors at their x & z coordinates & ignore their y coordinate. This allows you to create front, top and side views of your structure, as shown below. These graphs also illustrate another benefit to staggering your sensor locations – legible sensor markers. If your sensors line up on opposing faces, you will end up with overlaid sensor markers on the 2D graphs. That can make them hard to read.

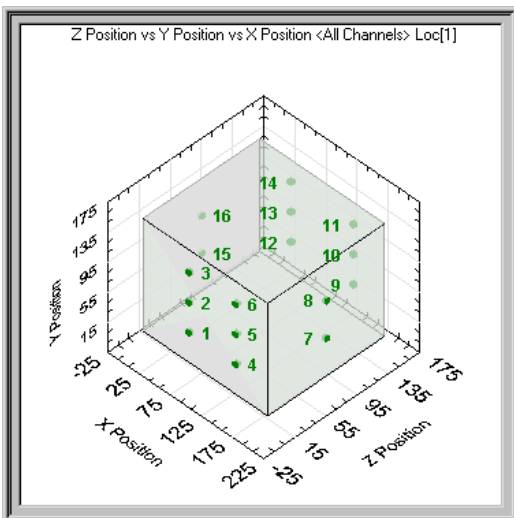


Figure 4-33. 3D Graph Example

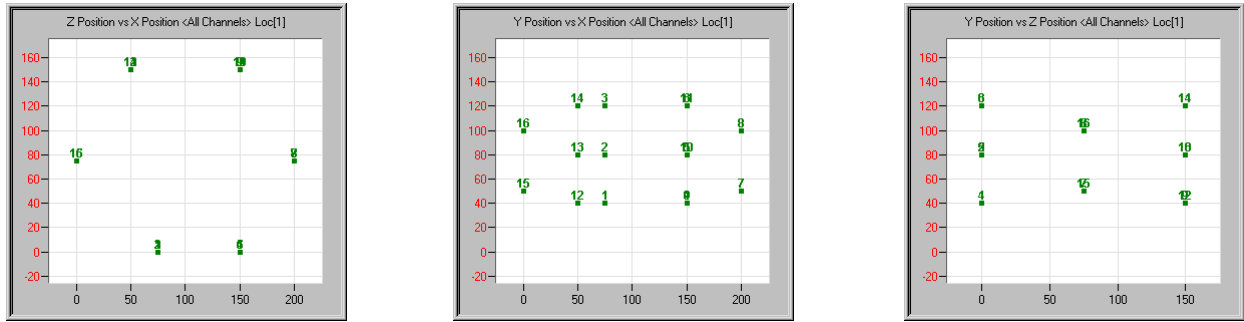


Figure 4-34. Screen 1 - Top View, Screen 2 - Front View, Screen 3 - Side View

4.4.5 3D Location on Solid Cylinders

3D Location has been enhanced to allow the use of a cylinder structure to determine the limits of acceptable 3D source locations. The 3dLocCylinderDemo.lay file is a sample layout file included with AEwin that demonstrates how to set up & use a cylinder structure with 3D Location. Location Setup is configured as for any other 3D Location group (see 3D Location section above). Sensor Placement controls the user's choice of structure. The user simply selects Horiz. or Vert. The vessel as the Structure setting:

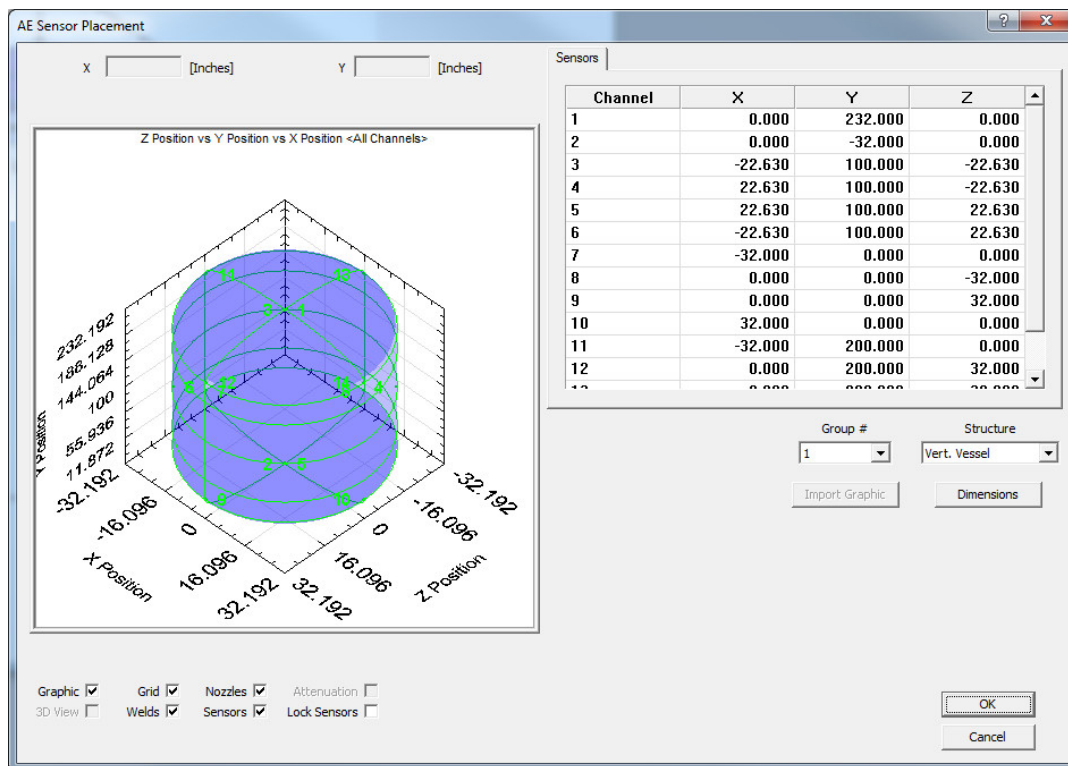


Figure 4-35. Sensor Placement for 3D Location on a Solid Cylinder

The user then uses the Dimensions dialog to configure the size of the cylinder:

Sensors are placed on the cylinder using the grid in the Sensor tab of the Sensor Placement dialog box, just as for the Box & Free structures. Simply click on the grid & press the Insert key to add a new sensor. Edit the coordinates to values needed. You can highlight a sensor in the grid & press the Delete key to remove an unwanted sensor.

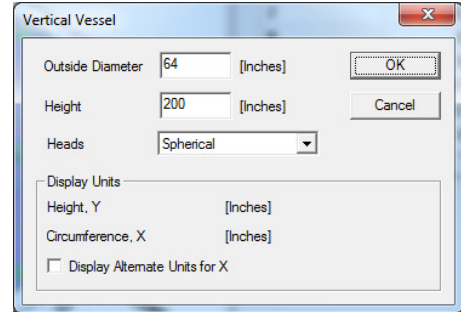


Figure 4-36. Vertical Vessel

The results of 3D Location on a solid cylinder can be displayed just like any other 3D Location result in a 3D Scatter graph that is setup to display the X, Y & Z Location results on each of the axes:

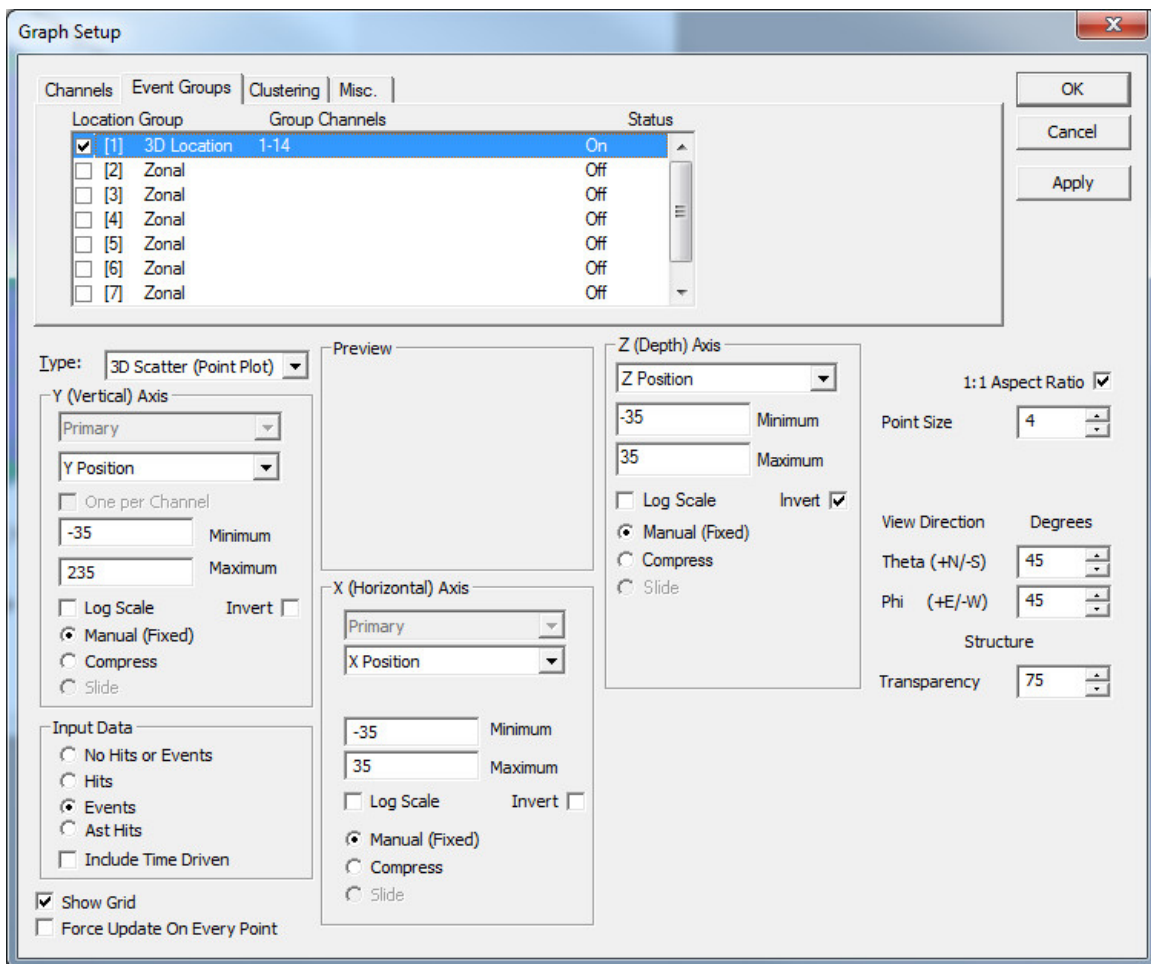


Figure 4-37. Graph Set up for 3D Location on a Solid Cylinder

Note that the user checks the desired location group on the Event Groups tab to determine which location group generates the location results. The user also can control the transparency level of the cylinder and the viewing angle in Graph Setup, as shown above. The results display on the 3D Scatter graph after a replay of the data:

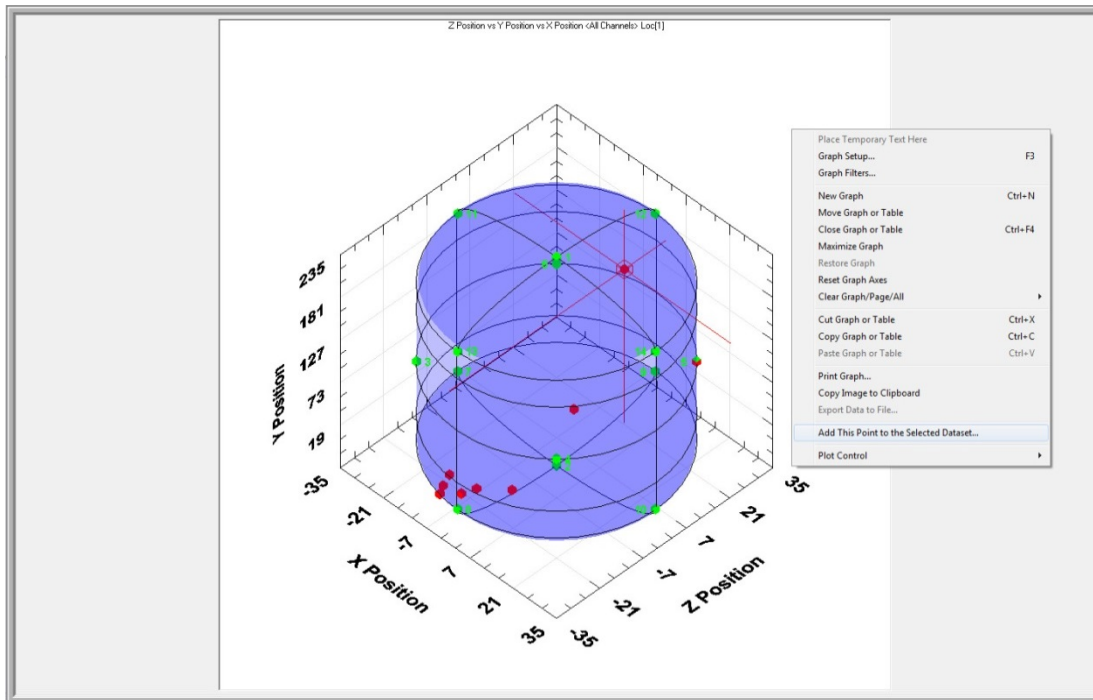


Figure 4-38. Data Selection with 3D Location on a Solid Cylinder

If desired, the user can use the Data Cursor to select a data point to add it to the Selected Dataset by selecting “Add this Point to the Selected Dataset” from the Right-click context menu of the graph. This launches the Selected Dataset Display dialog, which will then display the corresponding event results:

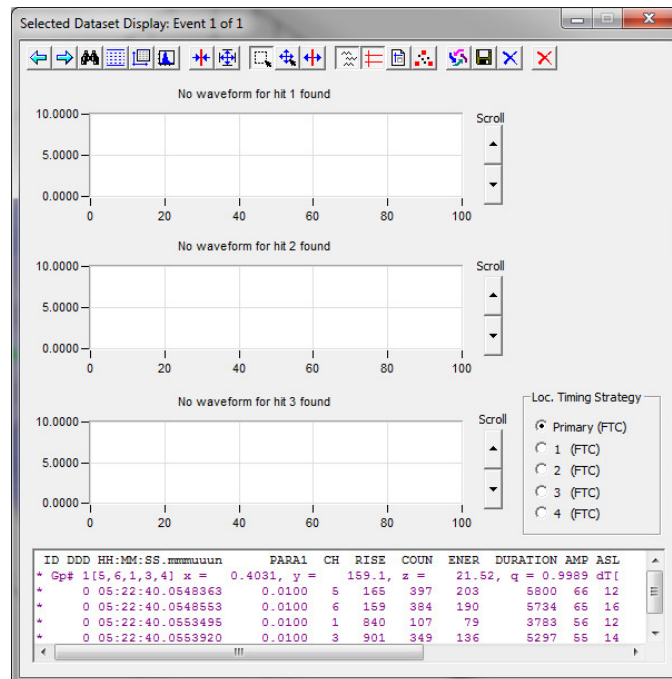


Figure 4-39. Selected Dataset Dialog for 3D Solid Cylinder Results

4.4.6 Spherical Location

Using Spherical location is similar to using the 2D Planar location mode, although there are differences in the coordinate system used & how a typical location plot is a setup. Mathematically, the Spherical location behaves like 2D Planar where both axes wrap back around on themselves. There aren't any new pitfalls in sensor placement, but the linear event problem that affects 2D Planar also affects Spherical. If you attempt location using just 4 sensors on the equator of a sphere, Spherical location would always fail as all events would be due to sensors in a straight line. You could override that behavior by enabling Allow Marginal Events in Location Setup at the risk of having many poorly located events. See section 4.2.4 for details. For better results, use at least 6 sensors (north & south poles & 4 evenly spaced on the equator). The maximum number of sensors is limited only by the number of available hardware channels. When using more than 6 sensors, it is recommended that you stagger alternate rows of the sensors to yield a triangular grid and produce more locatable events.

The Sensor Placement dialog has been enhanced to allow you to directly enter theta & phi coordinates anytime the structure type is set to Sphere, as shown here. Sensor coordinates are entered in degrees. Theta is defined as the north/south angle (latitude) and should be between +90 and -90. +90 is 90 degrees north (top of the sphere), while -90 is 90 degrees south (bottom of the sphere). Phi is the east/west angle (longitude) and ranges between +180 & -180. A phi of +90 is 90 east (right), while -90 is 90 west (left) relative to the origin. +180 (east) and -180 (west) are actually the same locations due to the wrapping of the sphere. A white + symbol marks the theta, phi origin of 0,0. You can quickly generate sensor layouts with the AutoPlace Sensors dialog. Just Right-click the graph in the Sensor Placement dialog.

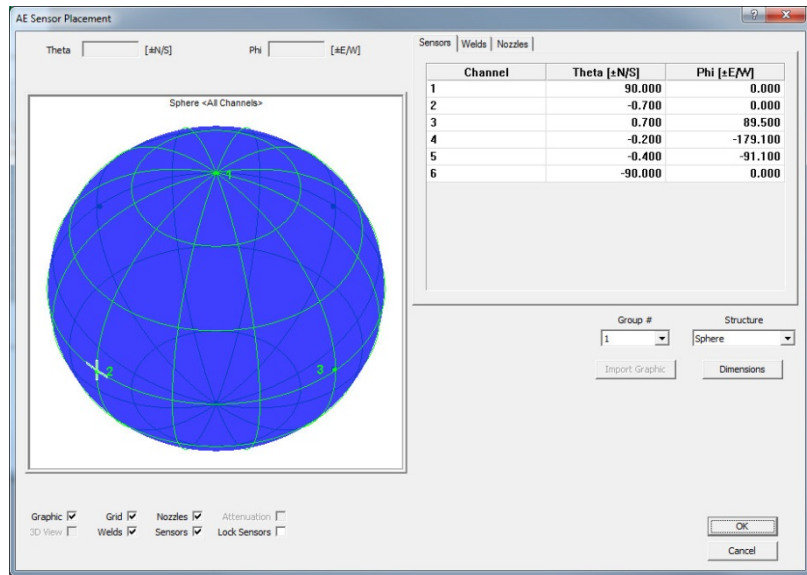


Figure 4-40. Spherical Sensor Placement

Alternatively, you can use the INS key to manually insert a new sensor & then edit the sensor coordinates on the sensor grid.

Spherical location is designed to work only with the Sphere structure type. You will receive a warning if you attempt acquisition or replay with a Spherical location group that attempts to use any other kind of structure. The Dimensions button allows you to set the diameter of the sphere. The drawing of the sphere in Sensor Placement does not change in size with the diameter, but the diameter is a key parameter required when making use of the sensor coordinates in the location calculations. If your diameter is set incorrectly, location will often fail all-spherical events.

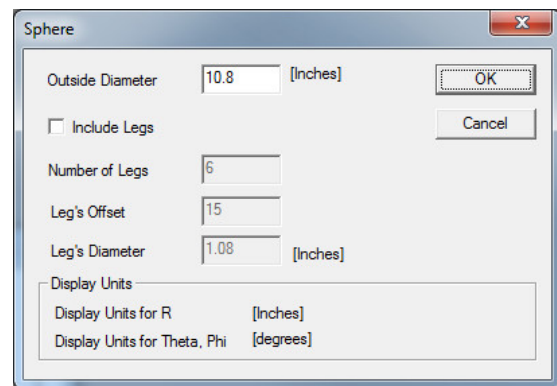


Figure 4-41. Sphere Dimension Dialog

There is an Include Legs option in the Dimensions dialog that allows the drawing of the sphere to show support columns if desired. Spherical location does not directly support location on the legs, but you could use a Cylinder location on the legs if the location on the legs is required. You can specify the number of legs, their diameter, and the offset in degrees of the first leg. The rest of the legs are drawn evenly spaced around the sphere.

After setting up the sensors & each of the location groups, you will want to create location graphs to display the results. It is possible to use any of the standard 2D graph types and display theta or phi on one of the axes. Just pick Theta Position or Phi Position from the list of available features. The easiest way, however, is to use a Spherical Location graph like the one shown on the right.

In use, a Spherical Location graph can contain a maximum of 4095 data points. During acquisition or replay, the newest data points replace the oldest ones on the graph once the 4095 point limit is reached.

When viewing the Spherical Location plot, you can use the mouse to rotate the sphere by clicking & dragging with the left mouse button. If you use the wheel on a wheel mouse or hold down the Alt key while clicking & dragging with the left mouse button, you can make the sphere larger or smaller. Holding down the Shift key while clicking & dragging pans the sphere up/down/left/right. The Reset Axes button on the toolbar will cause the sphere to revert to the view originally specified by the View Direction settings in Graph Setup.

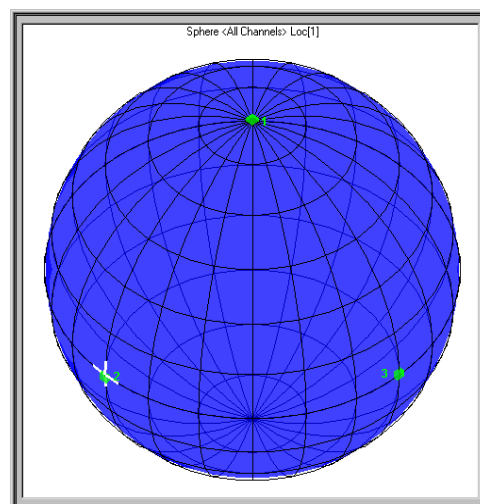


Figure 4-42. Spherical Location Graph

Here is the Graph Setup dialog configuration for the graph shown above. Unlike the normal 2D & 3D graphs, for the Spherical Location plot, you do not specify the axis quantities or the data type (always plots events), so the Input Data and X & Y-Axis settings are disabled. The graph will automatically draw sensors on the plot for any spherical location groups selected on the Event Groups tab. If you do not see a sphere on the graph after you exit Graph Setup, go back into Graph Setup & make sure at least one spherical location group is selected in Event Groups.

There is an Options section specific to the Spherical Location plot. It contains Transparency Level, Point Size, View Direction, Spherical Grid Spacing & Cartesian Grid Planes. Here is a brief description of what they do.

Transparency: The Transparency level can vary between 0 and 100%. When set to 0%, the sphere becomes opaque & points that plot on the back side of the sphere are not visible. If you use 25 or 50%, the points on the back side become visible but are a darker shade than those on the front side.

Point Size: Point Size directly affects the size of the data points & the sensor markers. The Spherical Location plot draws both the sensors & the data points as small diamonds. A bigger Point Size yields bigger data points & bigger sensor markers. The sensor markers are always slightly larger than the data points. You may also use the Graphing Colors menu to change the colors of the sensors, data points, and the sphere (structure color). The data points use the Plot 1 color in Plot Colors.

View Direction: The View Direction setting in Graph Setup controls what angle the sphere is viewed from using the same conventions as the sensor coordinates. This allows you to create a top, bottom, and side views of your sphere. For example, if you enter Theta = -90 and Phi = 0, the sphere will draw as if you are looking directly at the south pole. Below is a page of sample graphs from the SphereDemo.lay example file.

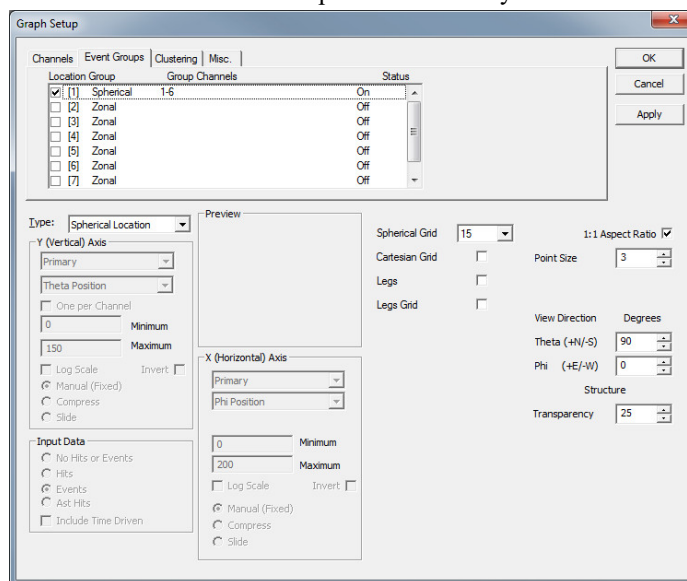


Figure 4-43. Spherical Graph Set up

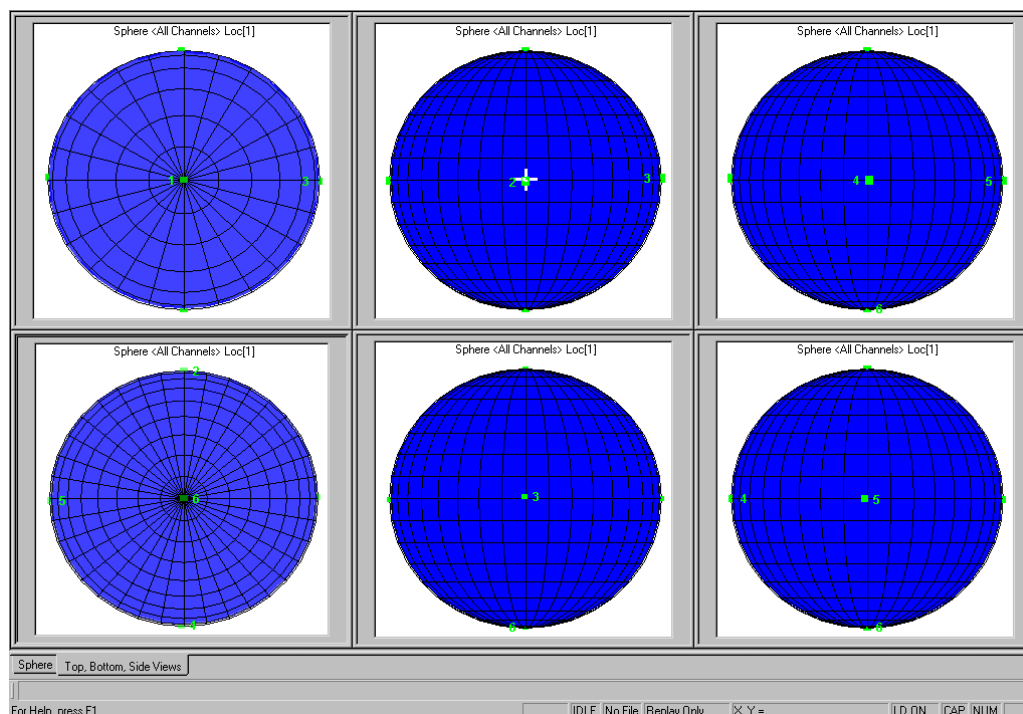


Figure 4-44. Spherical Location Graphs From Differing Viewpoints.

Spherical Grid: The Spherical Grid spacing parameter controls how a grid is overlaid on the sphere. Available choices are 10, 20, 30, 45, 90 degrees, and None.

Cartesian Grid: The Cartesian Grid planes checkbox toggles an X, Y, Z frame of reference on/off.

Legs: The Legs checkbox controls the display on the graph of any support columns defined for the structure.

Legs Grid: This option turns on an additional grid for the legs.

4.4.7 Cylinder Location

Cylinder Location is designed for locating source emissions on a hollow cylinder with either flat, elliptical, or hemispherical endcaps or no endcaps at all. (If you are interested in a location on solid cylinders, see section 4.4.5 3D Location on Solid Cylinders.) It is similar to 2D Planar Location with the x, y coordinates being the unwrapped cylinder distances, but one or both of the axes wraps around the body of the cylinder.

When setting up your sensors on your cylinder, there are several guidelines to keep in mind. As with 2D Planar, the software cannot successfully locate any source when all the hits are from sensors in a single straight line. You can force it to do so with the Allow Marginal Events checkbox in Location Setup, but the advice on good vs. bad sensor layouts for 2D Planar applies here too. Avoid rectangular sensor layouts where the distance between rows differs greatly from a distance between columns, as shown in the Bad 2D Planar Layout diagram. As stated before, the optimal arrangement would be a triangular grid made up of equilateral triangles. For Cylinder location, also be sure your sensor triangles on the heads are not significantly different in size than the triangles on the body, especially if the cylinder has flat heads. If not, you may get spurious event locations clustering along the head-body border.

Sensors that are 180 degrees apart can also cause mathematically invalid events too. Geometrically, it is similar to a problem with a straight line of sensors. If an event consists of a sensor on one side of the cylinder & 2 others exactly 180 degrees around the body, there is no way to distinguish source locations in the positive direction from the negative. As with linear events, you can force it to do so with the Allow Marginal Events checkbox in Location Setup. See section 4.2.4 for details. Generally, it is recommended that you use 3 sensors around the circumference of the cylinder in a given row. If you must use 2 sensors per ring around the cylinder, definitely stagger the alternate rows, or else you will not see any locatable events.

Use the Sensor Placement dialog shown below to set up the cylinder as well as place your sensors. Cylinder location is only designed to be used with the Horizontal or Vertical Vessel structure types. For Cylinder location, the Structure list in Sensor Placement *must be* set to either Horizontal or Vertical Vessel (cylinder) as shown above. A warning displays in the acquisition or replays with a Cylinder location group using any other kind of structure.

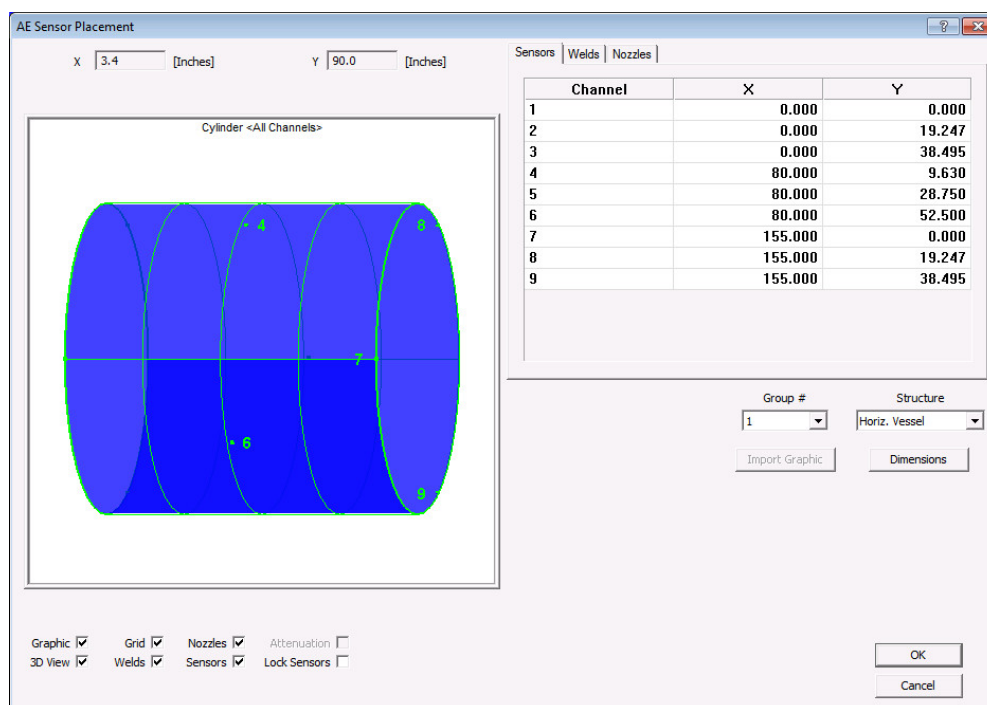


Figure 4-45. Cylinder Sensor Placement (3D View)

By default, Sensor Placement shows a 3D cylinder view. If you prefer a 2D view, simply uncheck the 3D View option at the bottom of the dialog & you will be presented with an unwrapped 2D view of the cylinder shown below.

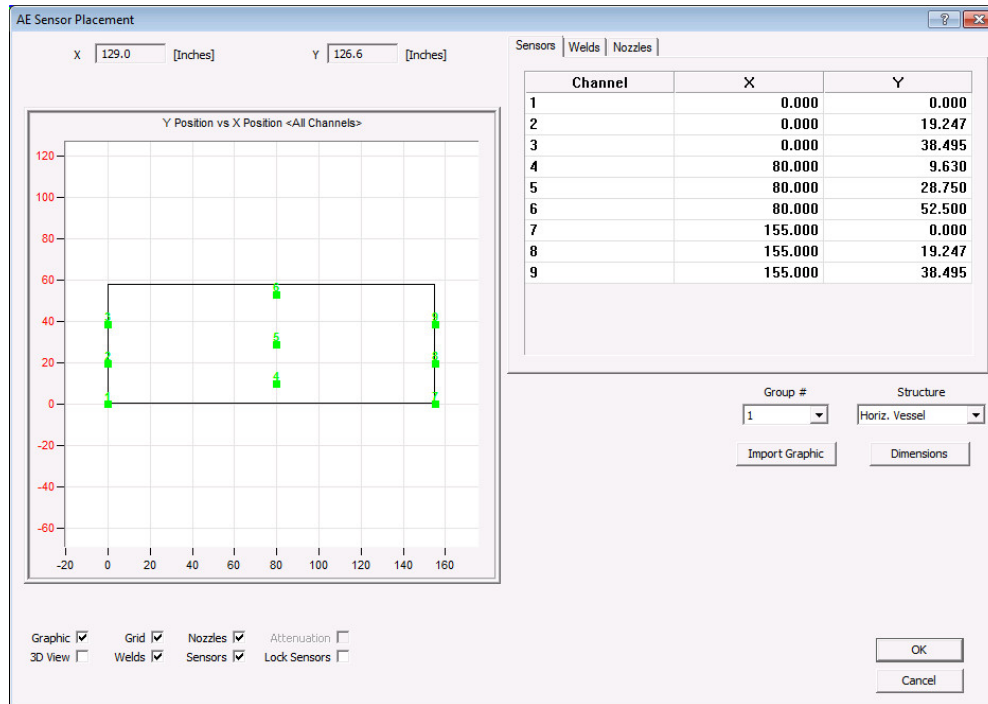


Figure 4-46. Cylinder Sensor Placement

The Sensor Placement dialog in 2D view & 2D location plots draws a rectangle to represent the body of the cylinder based on the user entered dimensions. You use the Dimensions dialog shown to the right to enter an Outside Diameter and the Height (vertical) or Length (horizontal). For a Horizontal Vessel, the rectangle extends along the X-Axis from 0 to Length and along the Y-Axis from 0 to the circumference calculated from the user entered Outside Diameter. For a Vertical Vessel, the rectangle extends along the X-Axis from 0 to the circumference and along the Y-Axis from 0 to Height.

Both dimensions are also vital to accurate source calculations. The Diameter is used to allow for the wrapping around the cylinder circumference. The Height or Length in combination with the Diameter is used in eliminating 2D locations off of the cylinder body caps itself.

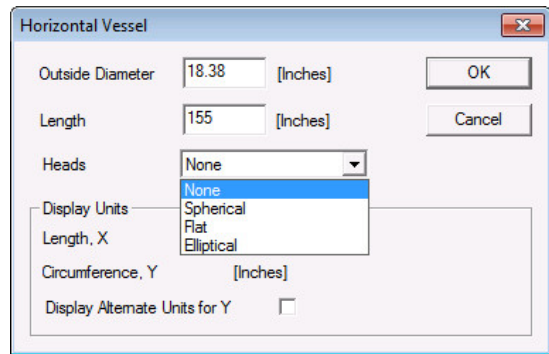


Figure 4-47. Vertical Vessel

There is also a dropdown list to select the type of Head on the cylinder. Available choices are Spherical, Flat, Elliptical & None. If a head type other than None is selected, the body rectangle drawn in Sensor Placement will have 2 additional rectangles on the top & bottom (Vertical Vessel) or left & right ends (Horizontal Vessel). The head rectangles extended the body rectangle by a distance determined by the head type. For Spherical or Flat, it is the length of the cap radius, which is also the body radius and half the user entered diameter. In the horizontal case, the rectangle will extend along the X-Axis from -radius to Length+radius, while in the vertical case, the Y-Axis will run from -radius to Height+radius. For Elliptical heads, a 2:1 ratio is assumed between the body radius & the perpendicular head radius, so the distance is ½ the body radius or ¼ the user entered diameter.

You can quickly generate rectangular & triangular sensor layouts with the AutoPlace Sensors dialog available by right-clicking on the sensor view. If you are using the 2D view of the cylinder, you can also manually place sensors on the cylinder by right-clicking. After using either option, you can edit the coordinates of the sensors if structural features (valves, nozzles, welds) require shifting sensors around. Just enter the x,y coordinates in the user's preferred distance units.

When you enter sensor coordinates & you are not including the caps, express your x, y coordinates with a lower limit of 0 and an upper limit of the circumference or the length/height. For example, when working with a vertical cylinder of height 150 and diameter 63.66 (circumference = 200), express x as a value from 0 to +200 and y as a value from 0 to +150. Basically, just make sure your sensor is visible in the body rectangle drawn in the Sensor Placement dialog.

If you are using Spherical or Flat heads, these limits are widened for the Y-Axis by the radius from -31.83 to +181.83. These extensions allow for the entry of sensors on the heads. For this vertical cylinder, a pole sensor on the bottom cap would have a y coordinate of -31.83. The x coordinate for a cap on a vertical cylinder corresponds to a rotation angle but is entered as a fraction of the body circumference. In this case, the limits are 0 to +200, and sensors placed 120 (=360/3) degrees apart are separated by 66.67 (=200/3). For a cap sensor placed directly at the pole of the hemisphere, the rotation angle is unimportant in the source location calculations, but a value of half the circumference will place the sensor halfway up the cap rectangle border. In the case of the vertical cylinder shown above, the pole sensor on the bottom cap is at -31.83, +100 (any x really), and +181.83, +100 for the top cap.

When you are finished setting up your sensors & ready to actually do data acquisition, you will need a location graph to display the results of the source location calculations. You can either use a standard 2D graph as described in the section on 2D Planar location, or you can use a Cylinder Location graph. Like the Spherical Location graph, a Cylinder Location graph is limited to 4095 data points, with newer points replacing older ones once the limit is reached. A Cylinder Location graph is shown at the right, and its corresponding Graph Setup dialog settings are below. It has an Options section that is very similar to the one for Spherical Location. See the section on Spherical Location for a description of the options common to both.

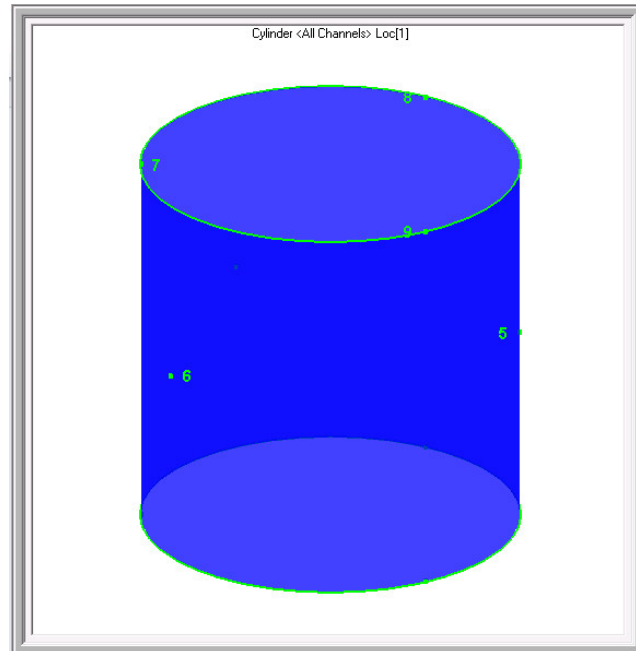


Figure 4-48. Cylinder Graph

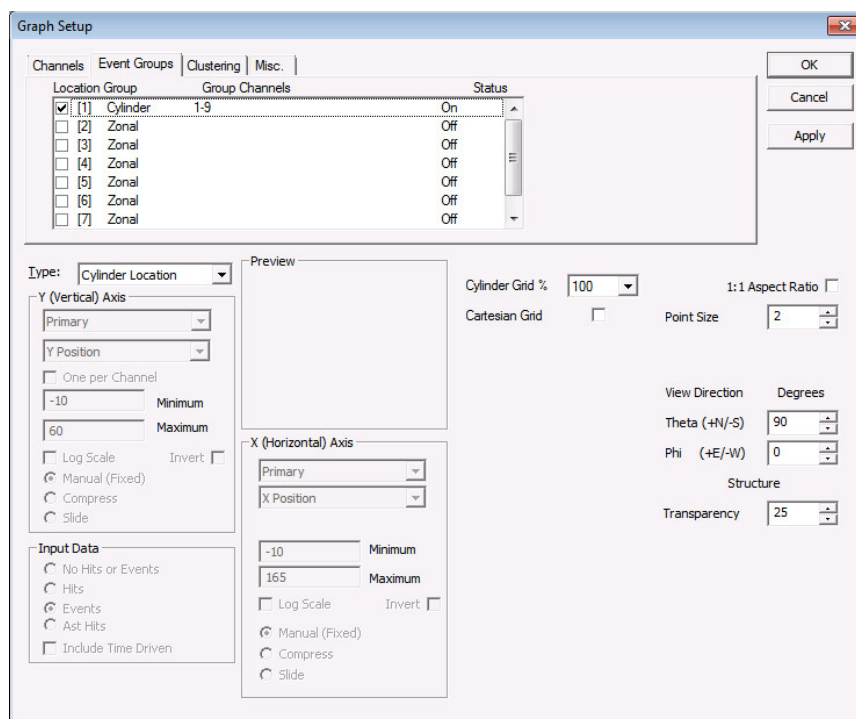


Figure 4-49. Cylinder Graph Set up

The Spherical Grid Spacing option is replaced by the Cylinder Grid Spacing option, although it behaves in a similar fashion. You may set the Cylinder Grid Spacing to None, 10%, 20%, 25%, 50%, or 100% to control the appearance of a grid overlaid onto the surface of the cylinder. The percentages are applied to the heads & the body separately as the dimensions for the 2 regions can vary greatly. If the user selects a grid spacing of 25%, the body will be divided up into 4 equal regions, and the head will also be divided up into 4 regions. If the user selects 100%, only the boundary between the heads & the body will be marked. Cylinder Location graphs always use Events as the Input Data, but as for the 2D graphs, make sure you set the Events Groups to the appropriate location group, or your graph will not show any data & will not draw the cylinder associated with the group.

4.4.8 Conical Location

Conical location is set up almost identically to Cylinder location mode. In Location Setup, the user selects Conical for the Location Type on the General tab. The other settings in the table are described in the section on the Location Setup dialog. In Sensor Placement, set the Structure list to 'Vert. Cone' or 'Horiz. Cone' as shown below.

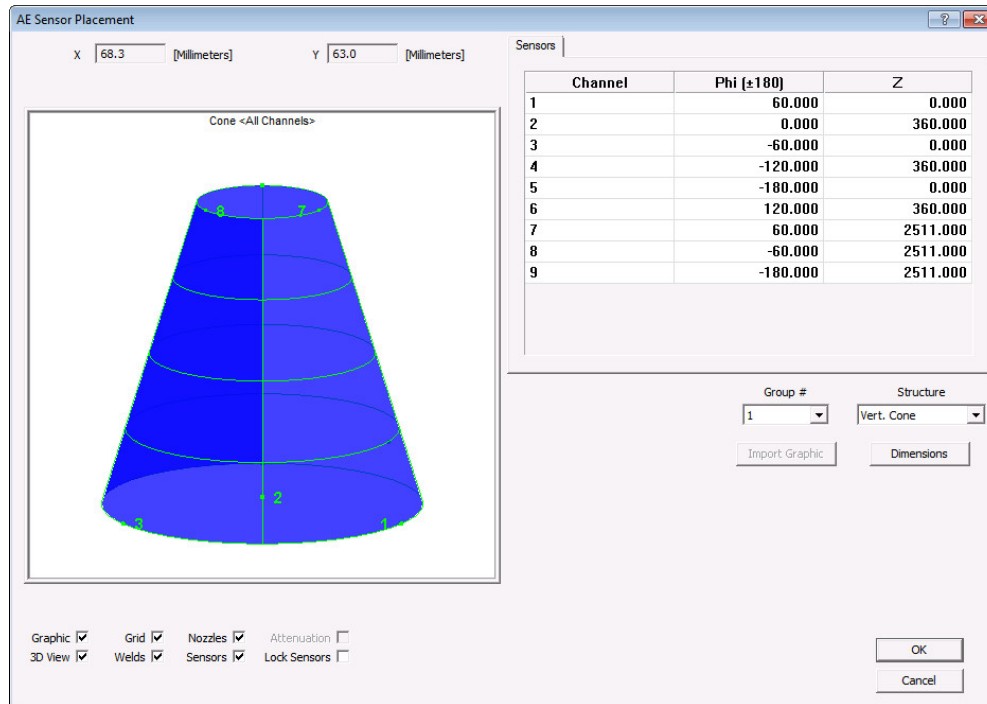


Figure 4-50. Conical Sensor Placement (3D View)

The Dimensions button calls up the Cone dimensions dialog allowing the user to enter the upper & lower diameters as well as the height (or the left & right diameters & length, if horizontal). The height or length has added significance when a conical location is analyzing data. Conical results are limited to source locations to within +/- 2% of the height. In other words, points located wildly off the cone body are thrown out as bad results.

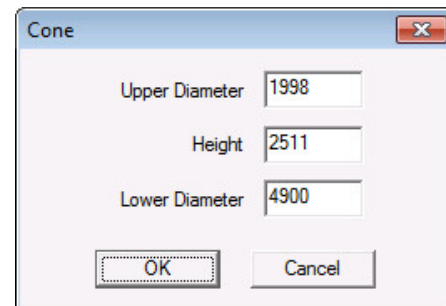


Figure 4-51. Cone

Sensor coordinates are entered as Z & Phi. The Z-axis is the cone axis independent of horizontal vs. vertical configuration. Z starts at 0 at the bottom of the cone & goes up to the height of the cone (or 0 at the left & up to the length). Phi should be entered between -180 to +180 degrees to keep the sensors within the 2D drawing of the unrolled cone, although the location will work with values from 0-360 as well.

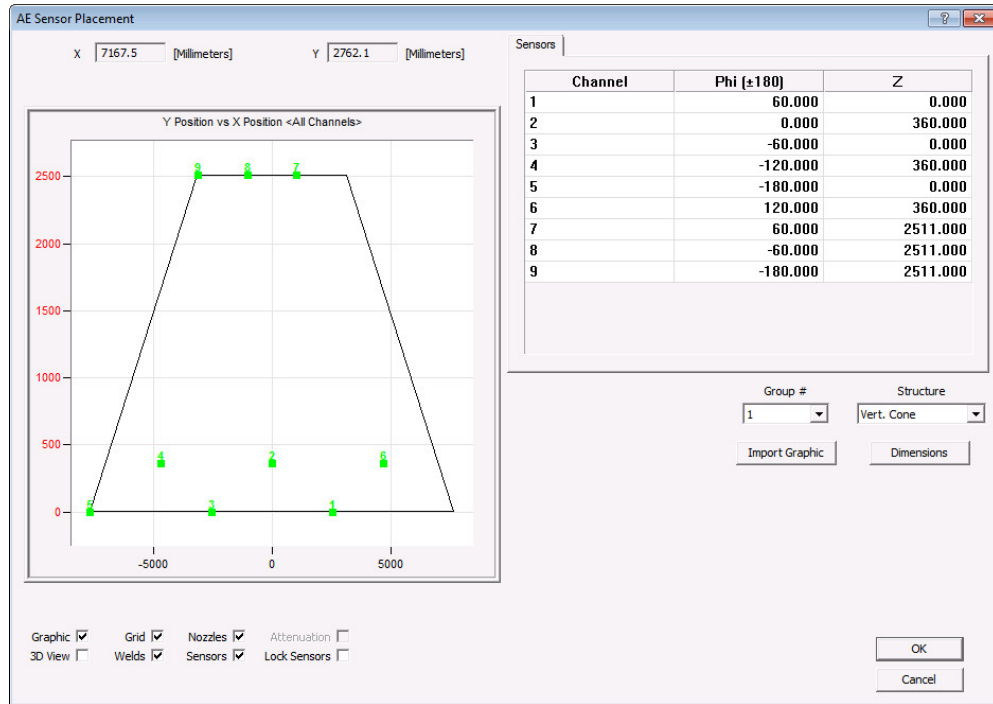


Figure 4-52. Conical Sensor Placement (2D View)

You can toggle between 2D & 3D graphs in Sensor Placement with the 3D View checkbox. The 2D view translates the phi value into a fraction of the circumference. The trapezoid is drawn from +circumference/2 to -circumference/2, where the diameter depends upon the z value in the cone.

There is also a 3D Conical Location graph similar to the Cylinder Location graph as another graph type in Graph Setup. It has options similar to those for the Cylinder Location graph & the Spherical Location graph. See their sections for a description of these options.

In use, the 3D Conical Location graph looks like the 3D graph in Sensor Placement. The sample layout, Conical.lay, gives an example on the 3D Cone page. A Conical Location graph can contain a maximum of 4095 data points. During acquisition or replay, the newest data points replace the oldest ones on the graph once the 4095 point limit is reached.

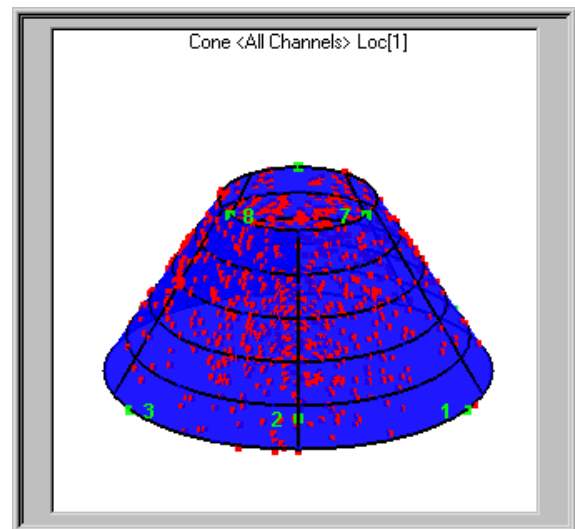


Figure 4-53. Conical Graph

4.4.9 Anisotropic Location Modes — 2D Anisotropic & Anisotropic Cylinder Location

These location modes allow for differing velocities in the x & the y directions. They are almost identical to their corresponding isotropic modes, 2D Planar & Cylinder Location. The velocity in the Wave Velocity column of the General tab corresponds to the velocity along the X-Axis for these modes. When one of these location modes is selected in Location Setup, another field will appear in Sensor Placement, as shown below, that allows the user to enter the Y-Axis wave velocity, V_y .

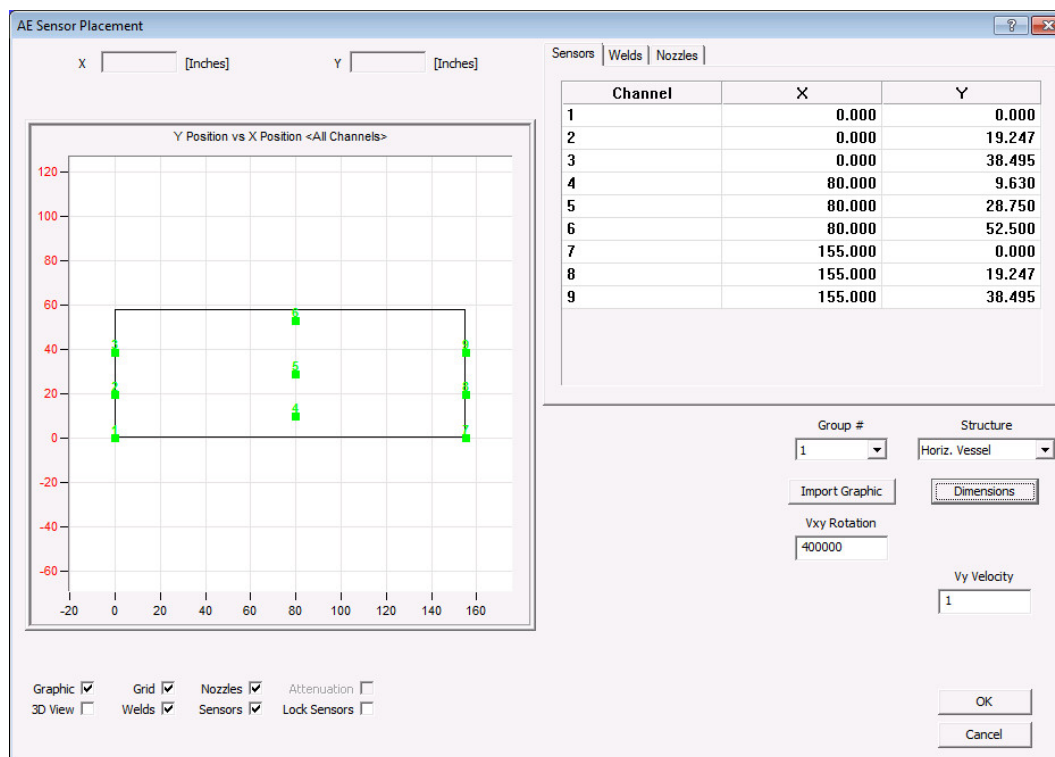


Figure 4-54. Sensor Placement for Anisotropic Location Modes

These location modes do not support the use of Timing Strategies other than Primary & the software will not let you select other strategies if you try. The reason is that the alternate strategies only have one velocity associated with them, but the anisotropic modes require 2. The anisotropic location modes rely on a coordinate transformation to enable the standard location modes, 2D Planar or Cylinder location, to analyze the data. That coordinate transformation is based on the ratio of the X & Y-Axis velocities being constant, but allowing alternate timing strategies with varying velocities would violate this assumption. Anisotropic cylinder location also does not support the location of the cylinder endcaps for much the same reason. One might reasonably expect that if the velocities along the X & Y-Axis differ substantially, then the velocities on the endcaps might also differ. For example, flat endcaps might use the velocity down the length of the cylinder, while spherical endcaps might need the velocity observed around the circumference of the body of the cylinder. The coordinate transformation cannot allow for those kinds of variations, so location on the endcaps is not supported. Except for those limitations, the anisotropic location modes, 2D Anisotropic & Anisotropic Cylinder, function just like their isotropic counterparts, 2D Planar & Cylinder.

4.4.10 Tank Bottom Location

Tank Bottom location is designed for use with evenly spaced 2-dimensional circular arrays of sensors. You must use it with the Tank Bottom structure. A direct consequence of requiring evenly spaced arrays of sensors is that the user is not allowed to manually place sensors on the Tank Bottom structure. Instead, one uses the Dimensions button in Sensor Placement to bring up the Tank Bottom dimensions dialog shown to the right. Unlike other dimensions dialog boxes, this one adds a section for automatically generating the evenly spaced circular array of sensors. By default, the sequence starts with sensor 1 placed at 0 degrees or 12:00 on the circle, but the user can start with other channel numbers and specify the rotational offset to the first sensor in degrees. You set the diameter of the tank bottom circle in this dialog box as well. After you click on OK, you will be asked if you want to delete the previously placed sensors. You must answer 'Yes' to have the changes take effect. If you answer no, the changes will be canceled. Once you have things configured correctly, Sensor Placement will reappear with the display updated based on your settings, as shown below.

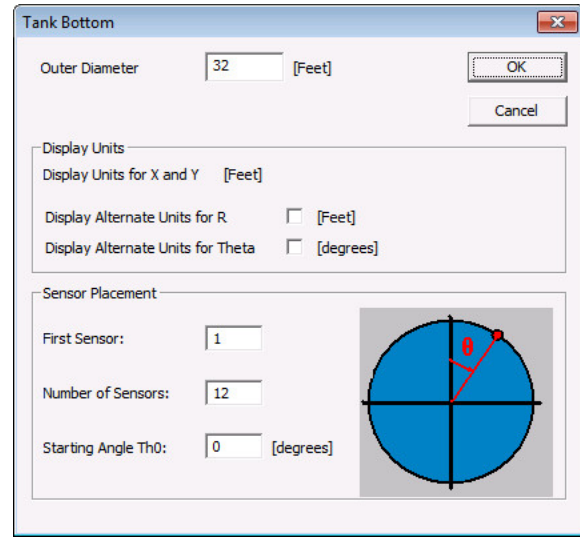


Figure 4-55. Tank Bottom Dimensions Dialog

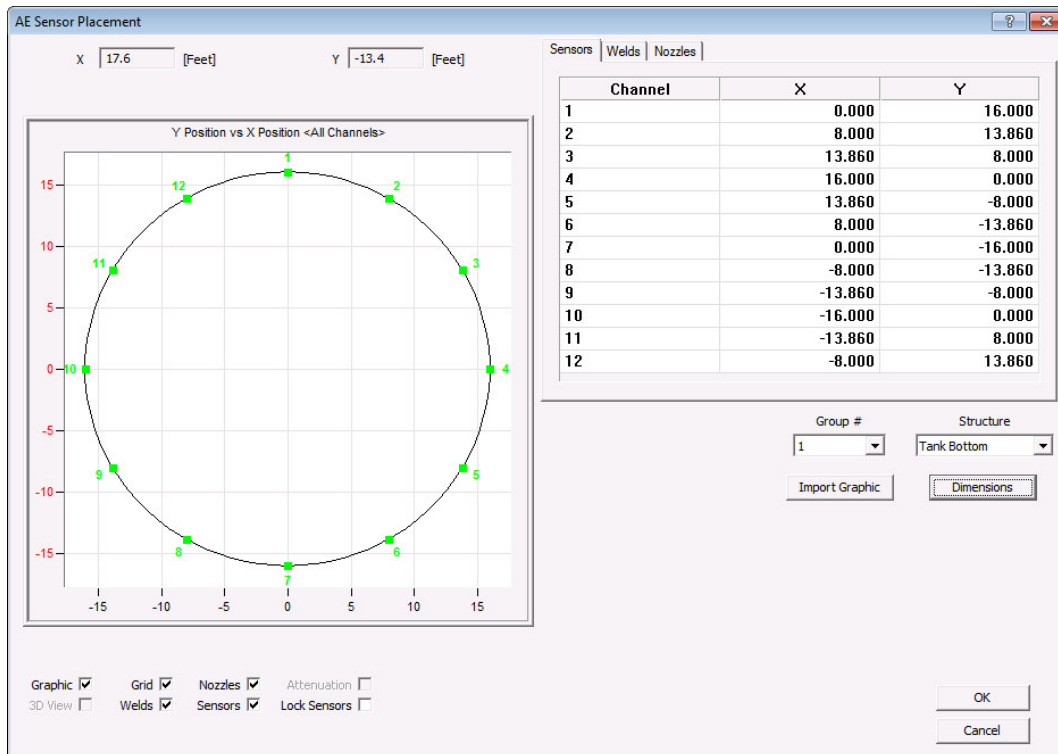


Figure 4-56. Sensor Placement for Tank Bottom Location.

Note that you cannot edit the X & Y coordinates of the sensors in the sensor grid array when using a Tank Bottom structure. You can edit the channel numbers, though, to allow non-sequential ordering of the sensors on the tank. In the example above, sensor 2 was renamed 12, even though the X & Y coordinates have been altered.

Tank Bottom location determines the source location by directly solving the time-distance equations generated from an event with 3 hits. It cannot directly make use of more than 3 hits when calculating a source location. When there are more than 3 hits available, however, it can vary which 3 hits are used in the calculation. The user can tell the software to do this by using the Tank Bottom Options dialog available thru the Tank Bottom Options button in Location Setup.

The top field in the dialog controls whether or not variations of the 3 hits are calculated & the results displayed on location plots. By default, the software will only show 1 location per event & variations are never shown in acquisition mode. When limited to 1 location per event & there are more than 3 hits, the algorithm uses the first 3 available hits on the assumption that those sensors were the closest to the source. This setting has no effect on events that only have 3 hits available. Below are 2 graphs showing how the results vary for a data file with about 600 events. The graph on the left is limited to 1 location per event, while the one on the right allows up to 10 locations per event. Note that the software counts these variations as extra events, so the total events for the graph on the right are reported to be about 2500 events.

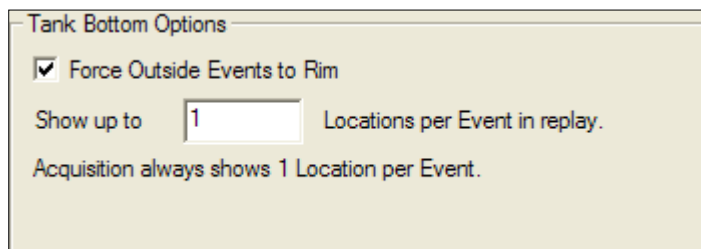


Figure 4-58. Tank Bottom Options

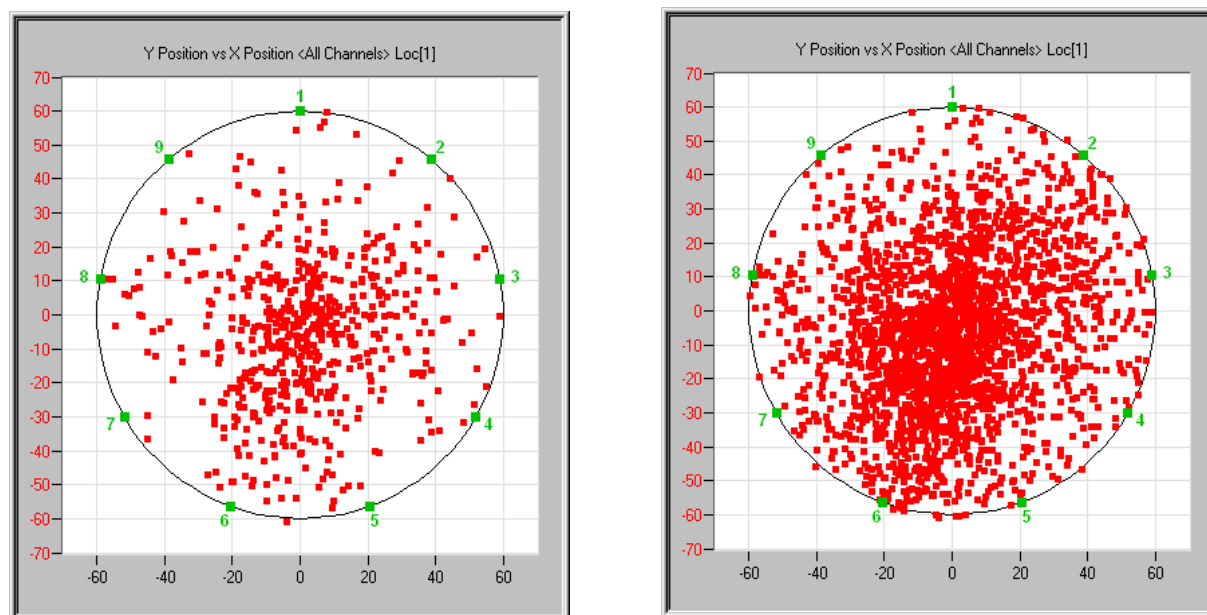


Figure 4-57. Tank Bottom Location Results with 1 (left) vs. 10 (right) Locations per Event.

The Force Outside Event to Rim checkbox at the bottom of the Tank Bottom Options dialog controls how events that locate outside the tank bottom area are displayed. It works in conjunction with the Allow Outside Events checkbox in Location Setup. Allow Outside Events applies to all the regression modes as well as Tank Bottom location, but Force Outside Events to Rim only affect Tank Bottom location. Allow Marginal Events does not affect Tank Bottom location. The effects of Allow Outside Events combined with Force Outside Events to Rim are shown in the 3 graphs below, which are restricted to showing just 1 location per event.

As discussed in the section on the Location Setup dialog box, “Allow Outside Events” controls whether or not the software reports events that locate outside the structure. If it is off, “outside events” are considered failing events and, in general, are not reported. Some events may occur very close to or just outside the tank rim & still be considered inside events because the algorithm uses a tolerance of $\pm 2\%$ beyond the radius when deciding whether an event is inside or outside the tank. That 2% tolerance allows for rounding errors that can push events at the rim just outside the tank bottom. The results of Tank Bottom location with Allow Outside Events disabled are shown below in the graph on the right. If Allow Outside Events is disabled, the Force Outside Events to the Rim setting has no effect on the results. If Allow Outside Events is enabled & Force Outside Events to Rim is disabled, all events plot at the

calculated source locations as shown below in the middle graph. If both Allow Outside Events & Force Outside Events to Rim are enabled, then events that would plot outside the tank bottom area are redirected to the rim of the tank, as shown below in the graph on the right. The software draws a line from the center of the tank to the calculated source location, finds the point this line segment crosses the tank rim & reports it as the source location. Although this location is not an accurate source location, it does allow the user to see that an event occurred. A drawback to enabling Force Outside Events to Rim is that it can obscure events that actually do locate at the edge of the tank. It is provided primarily for backward compatibility when comparing with results observed in PAC's older DOS applications.

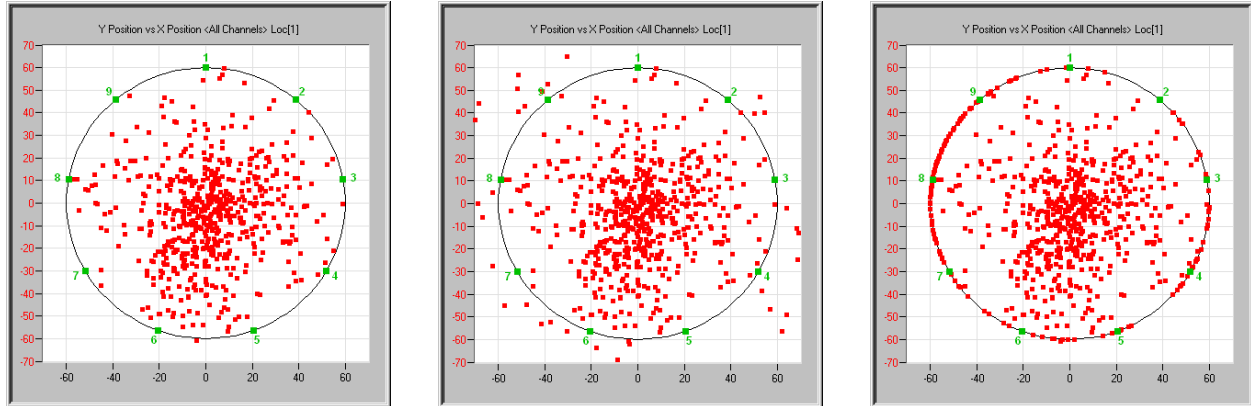


Figure 4-59. Tank Bottom Location Results

The left image shows Tank Bottom Location Results with 'Allow Outside Events' disabled.
The middle image shows them enabled with 'Force Outside Events to Rim' disabled.
The right image shows both settings enabled.

4.5 Location Examples

It is not feasible to provide detailed examples of all location modes in this manual. There are, however, example layout files & sample data files for each location mode. When you are starting out with a new location mode, you can simply recall the sample layout & inspect the location-related menus (Location Setup, Sensor Placement & Graph Setup dialogs). After you learn how the example is set up, you can make changes to better reflect your test setup & save the modified layout under a new name. Below is a table summarizing the example layouts & data files for the various location modes.

Location Mode	Layout File	Data File
Zonal	Zonal_Linear.lay	Zonal_Linear.dta
Linear	Zonal_Linear.lay	Zonal_Linear.dta
(Wrapped) Linear	WrappedLinear.lay	WrappedLinear.dta
Tank Bottom	TankBot.lay	TankBot.dta
2D Planar	2D-Planar.lay	HE-Link.dta
Cylinder	OpenCylinderDemo.lay	LongVessel.dta
Conical	ConicalDemo.lay	ConicalDemo.dta
Spherical	SphereDemo.lay	SphereDemo.dta
3D Location	3dLocDemo.lay	3dLocDemo.dta

In addition to the example files, there are detailed examples given here to illustrate how to use Zonal, Linear, 2D Planar & Tank Bottom locations.

The steps in setting up and running a location test are pretty similar, no matter what location mode you are using. Here is an overview of those steps:

1. Create a layout file with the hardware setup appropriately (e.g., the correct number of sensors turned on, detection threshold set properly).
2. Bring up the Location Setup dialog (=F8 key) and start by setting the distance units.
3. Next, fill in the location group information. Enable the group & select the Location Type. Enter the Wave Velocity, Event Definition Value & Overcal Value. If this is your first time working with location, leave the Event Lockout Value, Timing Strategies, Max Iterations, Min & Max Hits/Event at the default values. That applies to all the settings on the Timing page as well. These parameters are more fully described in the section on the Location Setup dialog under the headings of General Tab Settings & Timing Tab Settings.
4. Next, define the sensor locations on the structure using the Sensor Placement dialog. Use the Location View button in Location Setup to display it.
5. In Sensor Placement, select the Structure type, then click on Dimensions & enter the structure information.
6. Now create your sensors by right-clicking the sensor view area & or using INS & F2 in the sensor grid.
7. When finished, click OK in both the Sensor Placement dialog & in the Location Setup dialog.
8. Next, create a suitable location graph using the Graphing menu (New Graph) & configure it using Graph Setup (=F3 key). See the notes on individual location modes for information on the appropriate location graphs.
9. Now we are ready to run a trial AE test using pencil lead breaks to verify that everything is set up properly. If everything works, you're ready for acquisition.
10. If you end up staring at an empty graph, here is a list of things to double-check. Some are obvious, but personal experience shows them to be easy mistakes.
 - a. Is the Input Data set to Events in Graph Setup?
 - b. Is this location group checked on in the Event Groups tab of Graph Setup?
 - c. Is this location group checked on in the Group # column of Location Setup?
 - d. Are there any events from the group in the Line Display?
 - e. If there are events, are the axis limits in Graph Setup wide enough to see the positions?
 - f. If no events, are the Location Setup settings reasonable? Did you look up a velocity like 10,000 ft/sec & enter 10,000 when it should be 120,000 because the Distance Units are Inches? Verify the Event Definition & Lockout Value are reasonable given your sensor spacing & structure type. They all use the same units, so the values should be related to sensor-sensor spacing.

4.5.1 Setup and Testing of a Zonal or Linear Location Group

As an easy setup for the layout, we will be using two sensors located 30 inches apart. This sensor layout will be used to illustrate Zonal location in this section & Linear location in the next section. We will be using example layout and data files called Zonal_Linear.lay and Zonal_Linear.dta, which are installed in the Example Files folder under your MISTRAS Group folder, for example, "C:\MISTRAS Group\AEwin for DiSP\Example Files\." If you do not have these sample files, we will be showing the setup screens so that you can duplicate the setup and conduct your own acquisition test.

In this section, we will be setting up a Zonal Location Test following the steps identified above. For the example, we are assuming that we have two sensors in Channels 1 & 2 of the AE system.

Step 1. Start by using the File pulldown menu to access the Open Layout menu option or press Ctrl-O. Select the layout file named "Zonal_Linear.lay" & recall it.

Step 2. Enter the location setup menu by either pressing the F8 function key, clicking on the Toolbar icon that looks like a "bullseye" target called "Location Setup," or select Location on the Acquisition Setup pulldown menu.

The Location Setup dialog will be displayed as shown in this section. This is the first menu that needs to be set up in our Step-by-step procedure; by default, the General page displays. Notice the General tab near the top left of the dialog box. The General page of Location Setup allows you to control the setting of the individual location groups. Up to 8 location groups can be set up, although an additional licensing option provides for up to 32 groups.

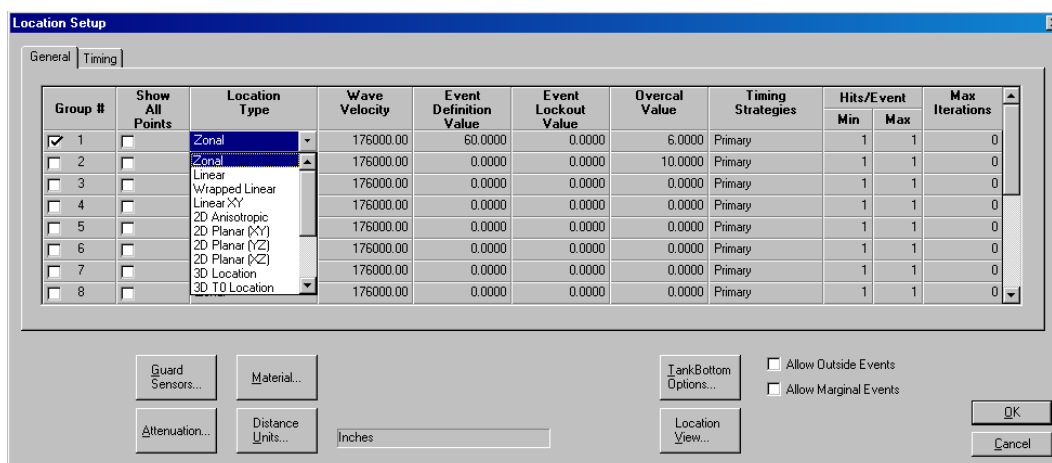


Figure 4-60. Setting up Location Group Information

In addition, near the bottom center of the dialog, the user can set up the location distance units. Various selections, including Metric, English, and time units, can be selected for distance units. Selecting a real distance value is most appropriate since the values read out can be directly measured when trying to locate a source on a structure. Use the Distance Units button to select inches, as in this example.

Step 3. In this example, we will select a Zonal location group as group #1. Follow the Group #1 line and enter the information as shown. First, select the zonal group. This is done by left-clicking the mouse on the down arrow on the right side of the Group #1, “Type” box. A dropdown list will appear to allow the user to select from the various alternatives (Zonal, Linear, etc.). Select Zonal.

For the Wave Velocity, the velocity entered will be related to your Distance Units setting. In this case, we've selected inches as our units, so the velocity is in inches/second. You can look up a velocity using the Material Properties dialog by clicking on the Material button. The database contains values for Longitudinal, Shear & Surface velocities. It is often better to directly measure the velocity using the AE system. Simply use the Line Display mode to determine the actual transit time (Δt) to traverse between two sensors. You can then divide the distance between the two sensors by the Δt to arrive at the estimated velocity. In order to determine this correctly, you really need to simulate the type of defect you are trying to detect. Enter the desired velocity in the edit box. In our example, we used an aluminum bar and a pencil lead break as a source. Although the Longitudinal velocity is 249,000 inches/second and the Shear velocity is 123,000 inches/second, we experimentally determined a test velocity of 176,000 inches/second.

For the Event Definition Value, we want to select a generous Event Definition using the same units as the Distance Units selected. This should be set to make sure that from any point in the location array, a minimum desired number of sensors will be hit, forming an event. In our example, two sensors 30 inches apart, the Event definition should be at least 30 inches to make sure that the two sensors get hit within a single event. We have entered 60 in the example above.

For Event lockout Value, we enter a value (in the selected distance units) to assure that there is enough time (or distance) for all additional hits and reflections related to the event to have had a chance to die down. The amount of time or distance selected will be the “dead” time as we wait for the event to die down. In this example, we will leave it at the default of 0 as reflections are not expected to be a problem for this setup.

The Overall value is also set in the selected distance units and is usually set at an amount of approximately 10% to assure that minor variations in wavepath velocity are not ignored by the location algorithm. When you are just learning to use location, it is best to leave the settings on the Timing page & the Timing Strategies column at the defaults. The Max Iterations, Min & Max Hits/Event cannot be changed for Zonal as they are not used by it. They only apply to the regression location modes. All of these parameters are more fully described in the section on the Location Setup dialog under the headings of General Tab Settings & Timing Tab Settings.

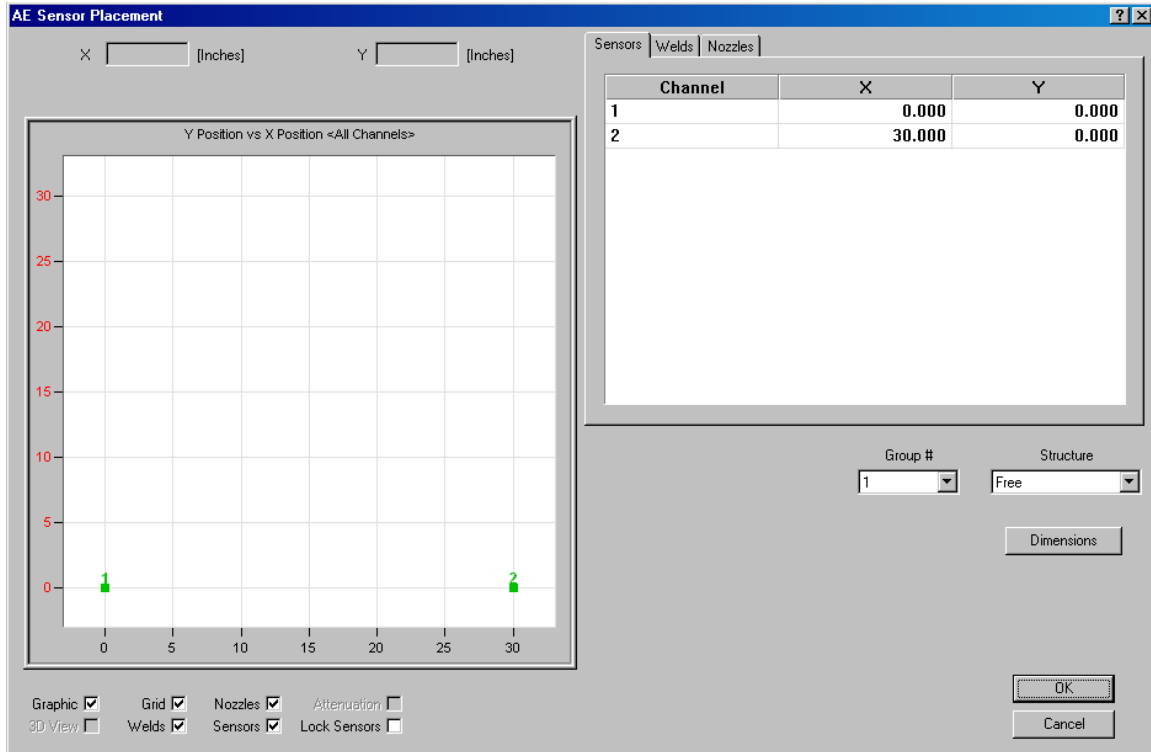


Figure 4-61. Sensor Placement Dialog

Step 4. Once the location group settings have been entered, we can define the sensor setup using the Sensor Placement dialog. Click on the Location View button on the bottom right side of Location Setup to display it. The figure above shows the Sensor Placement dialog box. There is 2 main area to the dialog box. The sensor layout view is on the left-hand side, and the sensor grid with the coordinates of individual sensors is on the right-hand side.

When we click Location View, the sensor layout for that particular location group is displayed. If we had group 1 highlighted but wanted to display the sensors for group 5, we could change the group displayed with the Group dropdown listbox just below the sensor grid. Select Group 1 if it is not already selected.

Step 5. Next, we need to select the “Structure.” The Structure setting is also just below the sensor grid & just to the right of the Group setting. This is currently set to Free. Free allows sensors to be placed anywhere without worrying about the specific structure geometry such as a plate, box, or sphere.

Next, we set the Structure dimensions. For the Free structure, this isn't required for location, but the dimensions control the axis limits of the graph in the sensor view area. To change them, we click the “Dimensions” button, and the Free dimensions dialog shown at the right appears. Here we've set the X and Y minimum and maximum values to 0 & 30.

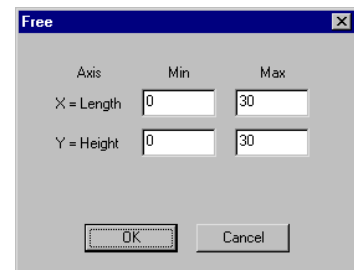


Figure 4-62. Free Dialog

Step 6. We are now ready to create our sensor layout. In the Sensor Placement example above, you can see that the sensors have already been placed at the bottom of the graph area, at locations of (0, 0) and (30, 0) using an (X, Y) arrangement. The software makes it easy to place and move sensors on the screen and to get them in an exact position.

To place a new sensor, we Right-click inside the graph region. The context menu to the right appears. The menu options we want to concentrate on are on the Sensors submenu allowing you to perform various functions related to sensor placement.

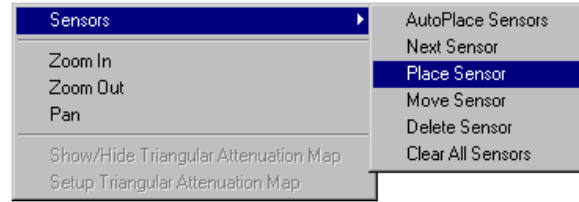


Figure 4-63. Place Sensor

Since we want to place a sensor, simply click on the “Place Sensor” selection. Upon doing so, a small menu box comes up asking you to verify the number of the sensor to be placed or to change it. When you're done, click OK, and the mouse cursor will change to a red sensor icon with a crosshair cursor. Each time you click, you will generate another sensor in your layout. Notice the x,y coordinates of the mouse cursor are shown above the sensor view area. When you are finished placing sensors, right-clicking will return the mouse to its normal arrow shape. Simply follow the Right-click with a Left-click off the menu if you don't need any of the context menu options. Whatever you do, don't press the ESC key. That is equivalent to pressing the Cancel button for the dialog & you'll lose all your changes to the sensor layout.

There are 2 ways to move sensors around to correct bad placement. First, the Right-click Sensors menu has a Move Sensor option. When you select it, the mouse cursor changes to a red sensor icon with a 4-headed crosshair cursor. To use it, click on the sensor you want to move. The sensor will turn red. Now click a second time at the spot where you want to move the sensor & it will shift to the new location. Right-clicking ends the Move Sensor mode.

If, after graphically placing or moving a sensor, its placement is not accurate enough, you can alter the position to an exact value by editing the X or Y value in the sensor grid. Left-click the mouse to highlight the desired box to be changed and enter the exact value, followed by the Enter key. Alternatively, press the F2 key to edit the current value.

Note the other sensor placement and edit capabilities provided by the submenu. You can automatically place the “Next” sensor by number by pressing the “Next Sensor” button. In this case, the sensor channel number selection box does not come up. It simply enters the Place Sensor mode using the next highest number sensor. Please note and experiment with these sensor placement management selections.

By now, you may have added several sensors to your setup that you want to delete. Do this by right-clicking over the sensor you want to delete and then select “Delete Sensor.” The mouse cursor will change to a red sensor with a crosshair labeled "del." When you click on a sensor, a "Delete this sensor?" message box asks for confirmation. Click “Yes” to delete the sensor from the graph and from the table. Again right-clicking puts the cursor back to normal. Return to the original sensor layout as shown above when you are finished exploring the Sensors menu options.

Step 7. In the previous steps, we have fully defined the location setup. The next step in the process is to set up graphs to see our location in acquisition or replay. We do this by first saving our sensor placement & location setup. Click the OK button in Sensor Placement & then in Location Setup. DO NOT CLICK Cancel in either one as you will lose any changes you made while in these dialog boxes. This is a very good time to save your layout on disk as well using the File pulldown Save Layout As menu option. When you don't need to rename the layout, you can just use Ctrl-S to update the current layout on disk.

Step 8. Next, we need to create a suitable location graph. If you have been following along with our example layout file called Zonal_Linear.lay, you will be seeing a screen like that shown below. On this screen, you can see two graphs. The first or top graph is already set up as a location graph. Since this is being set up for zonal location, we will only see the activity in a histogram or bar chart type of arrangement.

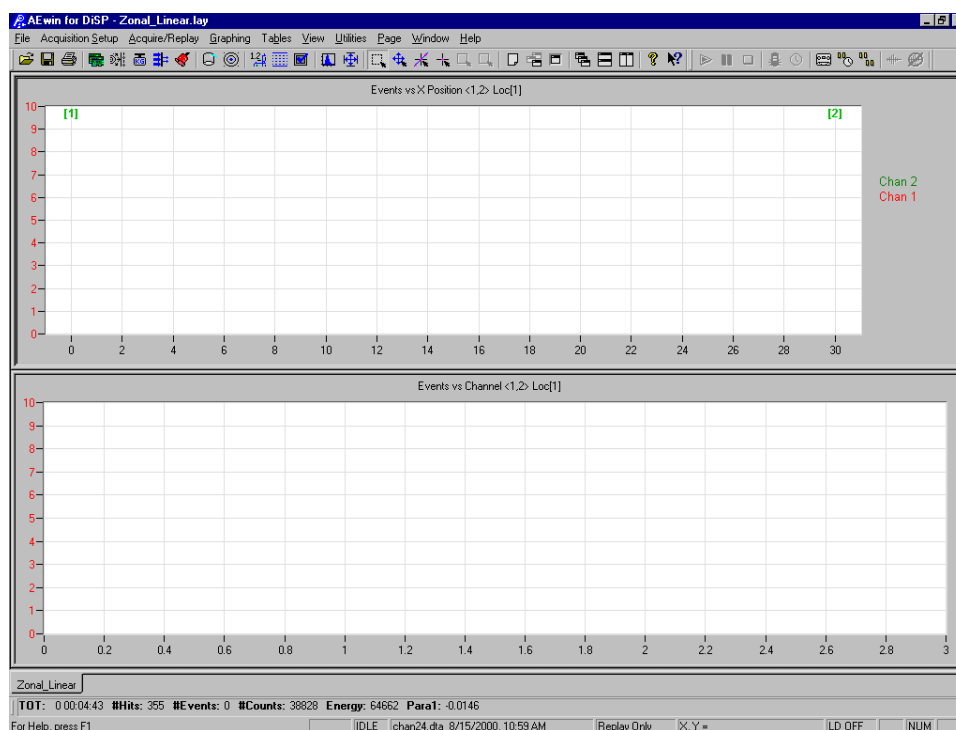


Figure 4-64. Zonal – Linear Location Graph

Assuming that this graph was not already set up in the desired location mode, let's go through the steps of setting up the graph for a Linear or Zonal location. First, we select the graph that we want to change to the location graph. We do this by right-clicking the mouse when it is inside the graph region of the graph that we want to work on. Once we do that, a menu appears. This menu gives us many choices, but we will select the top one called “Graph Setup.” Upon selecting this, a graph setup menu appears like that shown below.

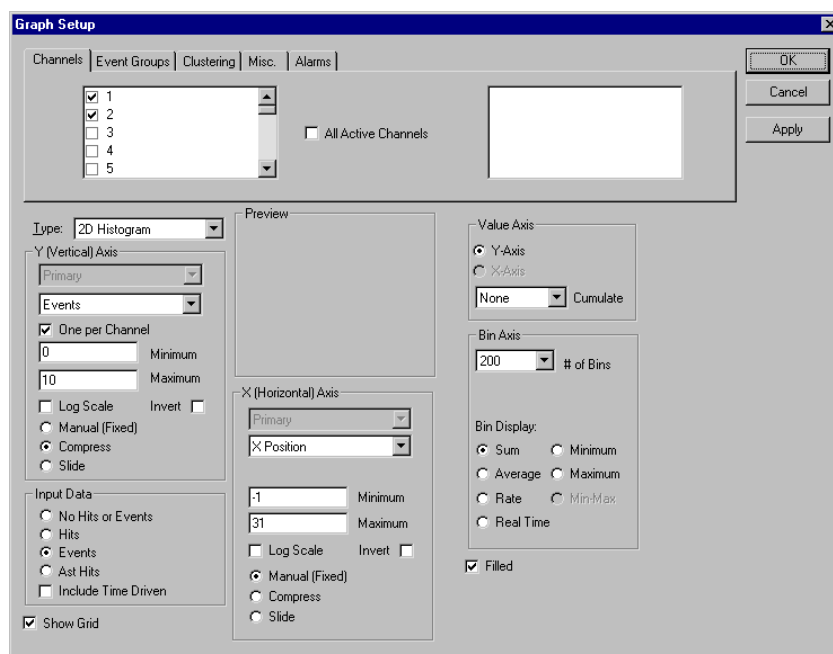


Figure 4-65. Graph Setup Dialog

This screen capture shows us how we have set up the Graph setup menu for a linear location. This gives us a lot of insight into how the graph setup menu works and how to set it up. Let's note first the layout of the menu. We see at the top of the menu a selection area for Channels, Event Groups, and Clustering. These settings allow us to select exactly those items. With the Channels Tab, we can select which channels will be displayed on the graph that we are setting up. In this case, we are only selecting channels 1 and 2 for viewing in the graph. You can see that we have set group #1 in Event Groups if you select that tab. Clustering allows you to set up parameters that link closely spaced hits/events together.

The next menu area is the Graph Type selection box. This is located on the left, just below the Channel, Event Group, and Clustering selection area. Various graph types can be selected, but for this application, we will use a 2D Histogram.

Below the Graph Type selection is the Y (Vertical) Axis setup menu area. Going down one item at a time in this menu is the plot # selection within the graph. This selection normally allows you to display more than one plot on a graph, but it is disabled for this 2D Histogram type because we have checked the One per Channel checkbox (see below). Basically, the selection (Primary, Secondary-1, 2, etc.) controls which plot the Y-Axis settings apply to.

Next is the Y-Axis parameter to be plotted on the graph. By clicking this, you can see many selections. However, we have selected Events. This will display the number of Events detected on the Y-Axis.

The checkbox One per Channel allows us to select a unique per channel plotting color that will give us multiple colors by channel plot. In our case, for events and 2 channels, we will get a different color for the channel that was hit first if this is checked. If not, we will only get one color plot.

The next two edit boxes allow us to set the starting Y-axis Minimum and Maximum settings.

The next checkbox, Log Scale, allows us to set a log or linear graph scale. Besides, it is an Invert checkbox that allows the user to reverse the positive to negative direction on the axis.

The next 3 radio buttons are related, allowing you to select one type of graph scaling. Manual(Fixed) will force the graph limits as set above to remain fixed. Compress will compress the graph so that the minimum graph value remains, but the maximum keeps changing as the graph limit is exceeded. And the Slide selection will allow both the minimum and maximum limits to change or slide as data is received that is beyond the current limits.

The next box allows us to select what type of Input Data we are plotting. In our case, since we are plotting a location graph, we want to choose the Event data for plotting.

Below is a Show Grid checkbox for displaying a grid on the graph background.

The next column to the right shows the X (Horizontal) Axis settings. The X-axis settings have a similar setup box as were described for the Y-Axis and will not be described here. Just note how they have been set up to display the X-Position for location and that we have set up the minimum and maximum graph values to include the sensor positions of 0 and 30 inches. Also, note that we have selected fixed scaling on the X-axis because we do not expect the X-Axis position to change. If this were a time axis, we would be using a "compression" or sliding graph scaling.

The last column on the right changes according to the graph type being set up. If a 3-D graph were being set up, this would be the column for the Z-axis setting. However, since we have selected a graph type of 2-D Histogram, settings for the Value Axis and the Bin Axis appear. Again, for now, just note the settings, and you can get familiar with what they do by making a change and seeing how it affects the graph later.

At the bottom of the third column is a checkbox that allows you to decide whether the histogram shows unfilled, outlined bars or filled solid bars.

Please note and try to make sense of all the selected menu settings. Although it might look complex at the start, if you have gone through these settings, one at a time, you will see that all is quite simple. Exit the graph menu by selecting the OK button at the top right of the Graph setup menu.

By looking at the top of the graph, we see its graph title of “Events vs. X Position <1,2> Loc(1)”. This summarizes what is being displayed. First is the graph type “Events vs. X-Position.” Next, we are displaying data from channels 1 and 2. Finally, we are displaying results from location group #1.

Now that we are finished, we see that we have set up a Linear location graph and are ready to test it. However, before we do, we have a second graph located below the first one. Let’s take a look at it. By the title, we see that it is an Events vs. Channel graph, plotting channel 1 and 2 data and also working with data from location group #1. We can look further to see how this graph is set up by entering the graph setup menu (right-clicking within the graph area and selecting the “graph menu” selection at the top of the submenu).

Take a look to see how this has been set up. The main difference is in the X-Axis graph parameter and the Bin Axis setting # of bins. This should help give a little insight into how you can set up different graphs.

Now we are ready to proceed with a test or a replay. As a reminder, as a precaution, always save your Layout file before trying it out. It is not fun to have all your work disappear because the power went down or something happened to lose the layout.

Step 9. The next step after setting up the graphs is to test the layout & verify that it functions properly during acquisition. We can do this by carrying out a trial acquisition and location test if we have set up our sensors, or we can do this by replaying our test file called **Zonal_Linear.DTA** . To do this, refer to the sections on going into acquisition or replay (section 3.4 and 3.5). In our example, we will replay the DTA file by going to the Replay function, selecting the file “Zonal_Linear.DTA and clicking Start. The data file eplays, and the results should look like the screen below.

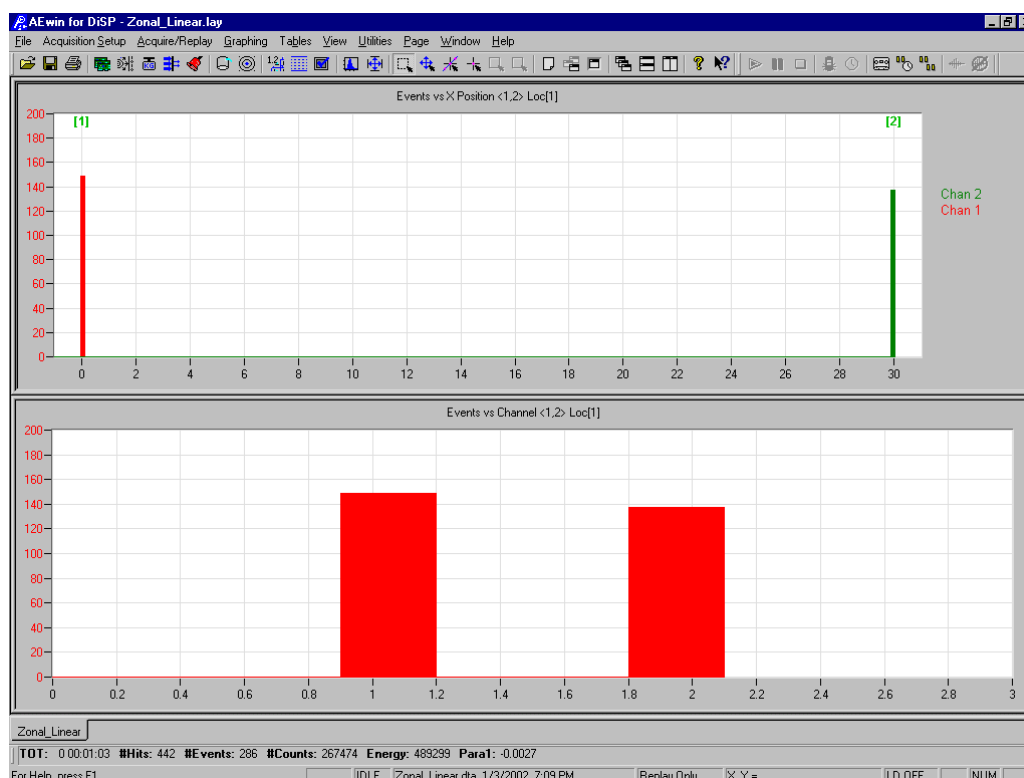


Figure 4-66. Zonal Location Graph

The top Events versus position graph shows that all the locations occurred at each of the sensor positions, even though, in reality, the events were distributed along the length between the two sensors. You can also see that there are fewer events on the right side or from sensor #2. This is confirmed in the Events versus channel graph below the location graph. The reason for the events showing up at the channel of a Zonal location group is that again the zonal group provides information on which sensor was hit first.

From the above display, it seems that our setup is good, and we could proceed on to test with a setup like this. However, you can modify some of the settings in the graph setup menu & repeat the replay of the data to see how the graphs are affected by changes and to get familiar with the graphing capabilities.

To illustrate the ease with which changes can be made, we'll now modify the location setup & transform the Zonal location group into a Linear location group. Simple use F8 to bring up the Location Setup dialog & change the Location Type of group #1 from Zonal to Linear, Next use F10 to replay Zonal_Linear.DTA again. Note how the results on the location graph below compare with the results for the zonal location shown above using the same Zonal_Linear.dta data file. There are now locations along the length of the top graph indicating that some events have been located in between the 2 sensors.

This example shows one of the very nice capabilities built into the software, the ability to change a setup and visualizing a different result upon replay. Remember that simply by changing the Zonal setup to a Linear location setup, we got two different graphical results. This is very useful when analyzing data.

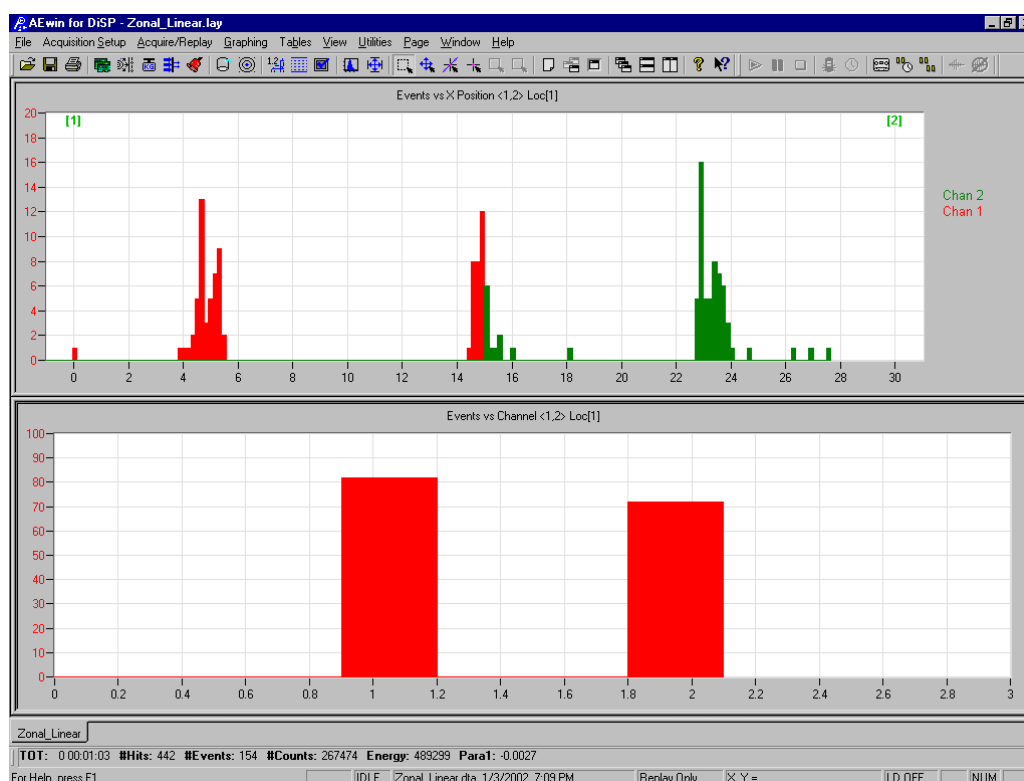


Figure 4-67. Linear Location Graph

4.5.2 Setup and Testing of a 2D Planar Location Group

As a detailed example, we will be setting up a 2D Planar location group with 4 sensors. We will be using the 2D-Planar.lay and chan24.dta example layout and the data file and which are in the product folder located in “C:\MISTRAS Group” by default. If you do not have these examples, we will be showing the setup screens so that you can duplicate the setup and conduct your own acquisition test.

As explained at the beginning of this chapter, 2D Planar location is part of the full location option. You can check for the full location option using the Options dialog available on the Help pulldown menu. Look for a checkmark next to Full Location. Alternatively, you can just look for 2D Planar in the list of available location modes under the Location Type column in Location Setup.

2D Planar location lets you place multiple sensors anywhere on your test structure without restricting you to a fixed grid arrangement. This is convenient as many structures have pipes or valves that may make a fixed triangular or rectangular arrays of sensors impossible to set up. The position of the source event is calculated from the differences in arrival times between the first 3 valid hits, $\geq 2 \Delta t$ values, & can use as many as 8 hits in the calculation. The positions of the sensors for the first 3 hits always determine a triangle. The ideal arrangement for 2D Planar would be a layout made up of equilateral triangles, but exact equality of the triangles & sides of the triangles is not an absolute requirement. You will get better results if you avoid creating sensor triangles where one side of the triangle is much larger or smaller than the other 2.

Setting up for 2D Planar location is very easy in the software and is similar to setting up Zonal or Linear location. The steps are identical to those outlined above; only some of the menu selections are changed for the 2D Planar location. Let's provide a quick run-through of setting up a 2D Planar location using the same steps outlined above for the other location setups. We will be using the example layout file, 2D-Planar.lay.

Step 1. Use the File pulldown menu's Open Layout option (Ctrl-O) to recall 2D-Planar.lay.

Step 2. Enter the Location Setup dialog (F8 key) & use the Distance Units button to select Inches.

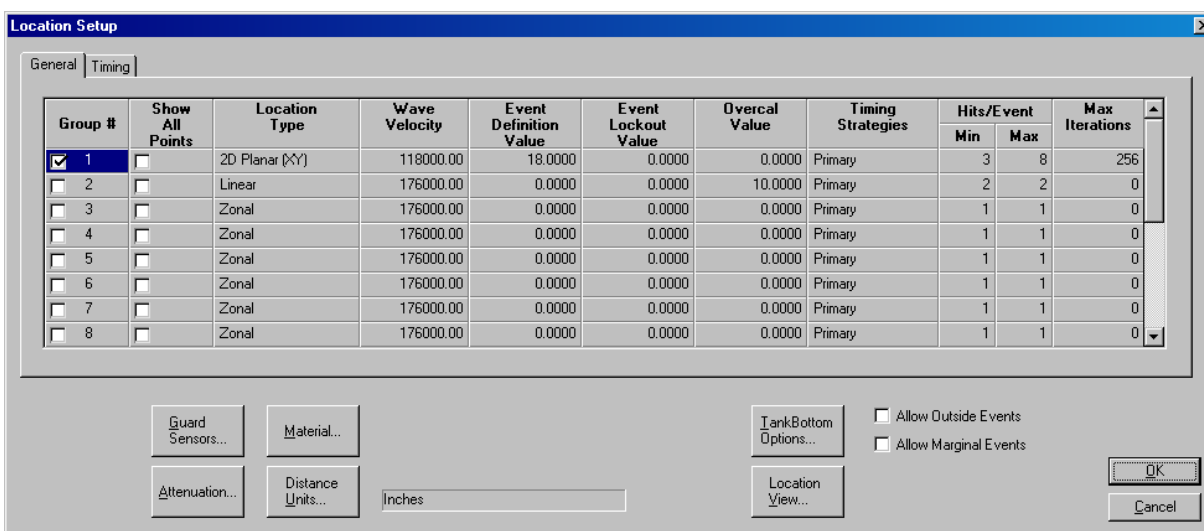


Figure 4-68. 2D Planar Location Set up

Step 3. Now configure the settings for location Group #1 as shown above. Enable the group by putting a check next to #1. Set the Location Type to 2D Planar. Enter 118,000 for Wave Velocity. Set the Event Definition Value to 18, Event Lockout to 0 & Overcal to 0. Leave the Timing Strategies, Min & Max Hits/Event, Max Iterations & all settings on the Timing page at the defaults.

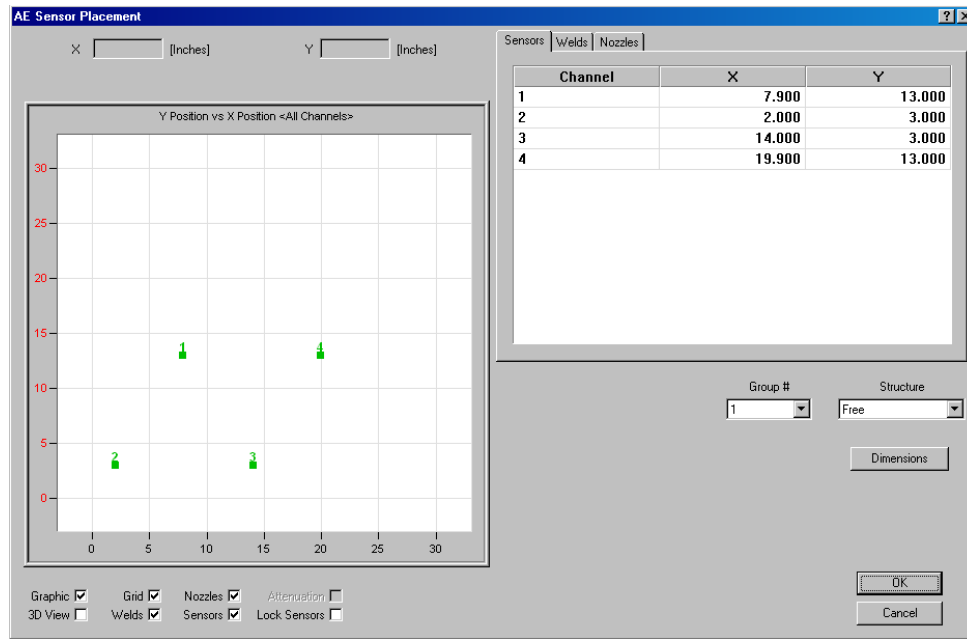


Figure 4-69. Sensor Placement for 2D-Planar.lay Location Setup

Steps 4-5. Click on Location View to bring up the Sensor Placement dialog. Set the Structure type is to Free & use Dimensions to set 0 & 30 for the Min & Max values for both x & y.

Steps 6-7. Setup the sensor locations on the screen using the same procedure described in the previous section for zonal location. The only difference now is that you will set up 4 sensors in a 2-dimensional layout. An example is shown above, along with the coordinates of the 4 sensors. Even though we are showing a triangular arrangement, 2D Planar is not restricted to a specific sensor pattern. When you are done placing sensors, click OK in Sensor Placement & Location Setup to return to the main menu.

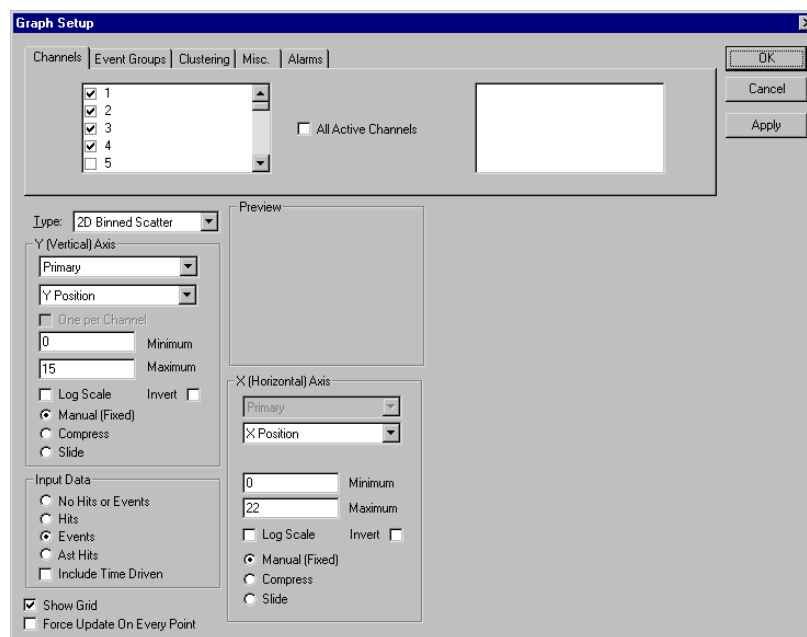


Figure 4-70. Graph Setup for 2D-Planar Location Plot

Step 8. Next, define a 2D X-Y location plot using the graphing menu following the procedure described in the section for zonal location. The screen capture above shows the graph setup menu for the location plot in 2D-Planar.lay, while the one below shows the graph (top graph) for that layout. You can compare the differences between the setup above and the graph setup for the zonal and linear location graph shown earlier. First, you can see that we have enabled all 4 sensors for display that have been set up in the location view menu. Next, you can see that we have selected a 2D Binned Scatter (point plot) for the graph type. Next, you can see that we have changed the Y-Axis to plot “Y-Position” instead of Events. You can also see that we have changed the axis coordinates in both the X and Y axes. Finally, you can see that there are no “bin or value axis settings” to the right of the X-Axis setup as there was for the 2D Histogram graph setup. 2D Binned Scatter graphs don't require any user choices about the binning strategy. All 2D Binned Scatter graphs use a fixed 200x200 binned grid when plotting the data. The point of using a 2D Binned Scatter graph rather than a 2D Scatter graph is speed. Its binning strategy decreases the number of data points actually plotted so that multiple points at one location graph as just one point. The 2D Colored Scatter graph is similar to the 2D Binned Scatter, but it will allow you to plot the points in different colors that reflect how many original data points are superimposed at any particular location.

The result of the graph setup is shown in the top graph of the screen below. In this graph you can see the sensor locations, although your graph will be empty at this point.

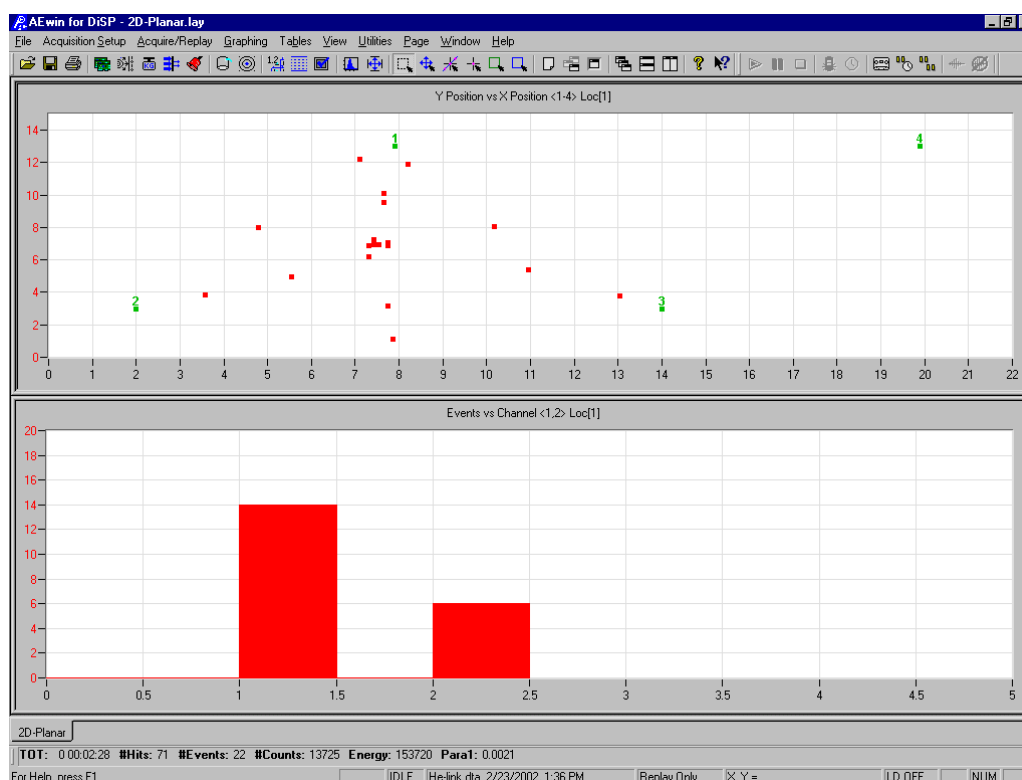


Figure 4-71. 2D-Planar Location Point Plot

Step 9. Now perform a test of the setup using the same procedure as described in the previous section for zonal location. Replay the data file HE-Link.dta. Results in a point plot will be plotted as dots on the screen where each dot represents the location of a detected event. Now your graphs should resemble those shown above. You can Right-click on the graph & look for the Large Scatter Graph Points for better visibility of the data points if you prefer.

4.5.3 Setup and Testing of a Tank Bottom Location Group

Tank Bottom location is a specialized AE source location mode developed specifically for monitoring the floors of large cylindrical vessels. The software assumes a tank with a circular bottom with sensors equally spaced around the circumference of the tank. Tank Bottom location should not be confused with TANKPAC. The difference between them is that Tank Bottom software performs a location calculation and projects the result of that location determination onto the tank floor graphic. In contrast, TANKPAC® is a knowledgeable & experienced-based procedure for assessing and grading the integrity of tank floors.

If you have purchased the Tank Bottom location software option, this section will help you to setup for your first Tank Bottom test. Please first verify that you have the Tank Bottom software option by selecting the Options dialog from the Help pulldown menu. If you have this option, it will be checked. If the option is not present and you desire it, please contact the Customer Service Department at customer.service@mistrasgroup.com.

Tank Bottom location follows along with the same location setup, menus, and setup procedure as do the other location modes. Setting up Tank-Bottom location is a simple 3 step process including:

1. Accessing the Location Setup menu (F8) from the “Acquisition Setup” main toolbar menu and setting up a Tank Bottom location group.
2. Accessing the Location View menu to set up the Tank bottom structure, dimensions, and sensor placement.
3. Setting up a Tank Bottom location graph in the Graph setup menu.
4. Acquiring and/or Replaying data.

Assuming one is familiar with the software, the following is a step-by-step procedure for setting up and running the Tank Bottom Location software.

4.5.3.1 Configuration of Location Setup

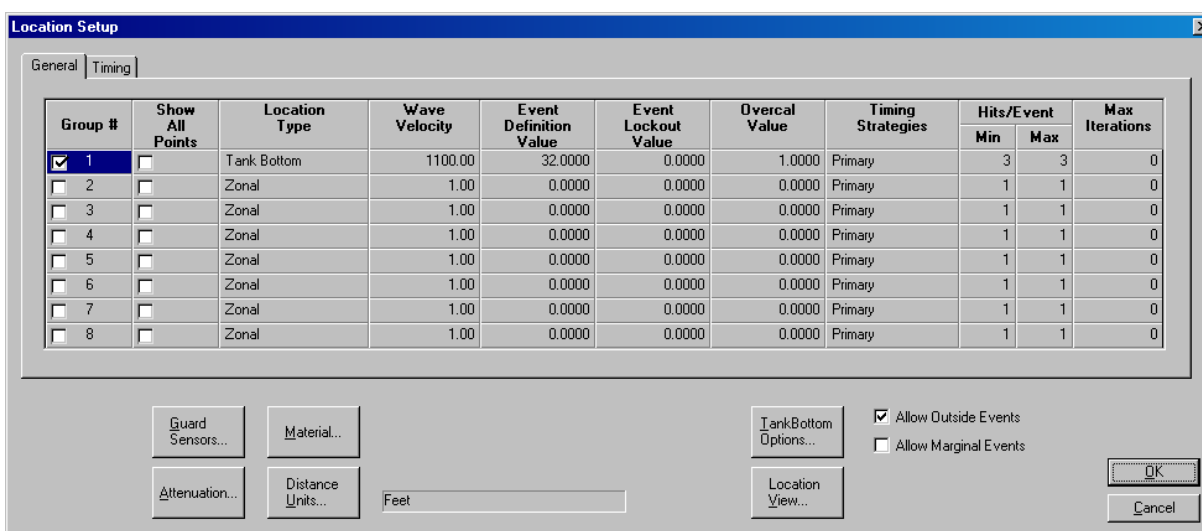


Figure 4-72. Location Setup Menu

The first step in setting up a Tank Bottom location application is to access the Location setup menu. This is done by selecting Location from the Acquisition Setup pull-down menu (or by pressing the F8 key). The Location Setup menu appears as shown above. As an example, we have shown a Tank Bottom location group as group number 1. We give an overview of setting up this location group below, but you may also wish to read section 4.2 on the Location Setup dialog box for more detailed information on the individual settings.

The first step in setting up a location group is to first select the distance units that you want to use for your work. You will know what distance units you are working in by viewing the textbox at the bottom center of the Location setup menu. In the example above, the current measurements are in feet. To change from the default distance units, left-click the “Distance Units” selection box at the bottom of the Location setup menu. Upon clicking this button, you will see the dialog shown to the right appear. Select whichever units you desire to work with and click “OK.” Upon clicking OK, you will see the units textbox at the bottom of the location setup menu show these new values.

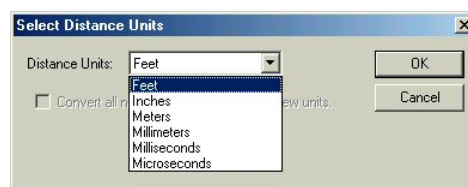


Figure 4-73. Distance Units Dialog

It is important to setup your distance units as early as possible into the location setup because many of the locations setup entries, including “Wave Velocity,” “Event Definition Value,” “Event Lockout value,” and “Overcal value,” all use the units selected or in the case of velocity it will be units/second. By dealing in direct units, the user can think in terms of distance values when setting up these entries.

Returning to setting up Group #1 (as shown by the first row of the Location setup menu), we will address each column entry below.

Group#: The first column shows that we have check-marked the Group #1 Location group to be an active location group. Putting a check in this box activates that location group and tells the software to perform a location calculation based on the setup data in this menu, the Location setup “Timing” menu (see the upper tabs), and the “Location View” and “Tank Bottom Options.”

Show All Points: This allows the user to show all source results for events that this group can successfully locate. Independent of which groups have Show All Points checked, the software always allows all enabled groups to analyze a given event to get the best results for each group and then compares the results from all successful groups to come up with the best result. For example, linear location is always presumed to give a better result than zonal location if there are enough hits(2) to produce a location. If no groups have Show All Points checked, the software will report one source location per event & it is the best result that is shown on graphs & in the line display. There are times when a side-by-side comparison of the results of 2 or more location groups is useful. In that case, you can mark the groups as Show All Points & then whatever source locations they can calculate will be reported. The software will appear to sometimes produce more than one source location per event & if you look at the line display, you will see extra event listings with the same first hit times. The extra location results will also show up as extra points on your graphs too, but they only count as 1 event in the Statistics bar. In all cases, you will see the software's best result as one of the points. There will be times when an event cannot be located by more than 1 group, so having Show All Points does not guarantee a result from a given location group. It is only a guarantee that you will see it if it can produce a useful result.

Location Type: In the next column, we have selected “Tank Bottom location” as the location type. By Left-clicking your mouse in the Location type area on this first row, a selection box will appear with all the available location option choices that are enabled within the software. “Tank Bottom” will be one of the choices; select this.

Wave Velocity: For Wave Velocity, we suggest that you use the velocity of the fluid that is in the tank as a starting point. It is recommended you perform some tests to try to empirically determine what that actual velocity is once you get on the tank. This can be done by creating an impact signal near one sensor that travels from one side of the tank to the other and monitoring the time it takes for the signal to travel the liquid path to the sensor on the opposite side. You can then measure the time it has taken to arrive and use that with the knowledge of the diameter to arrive at a velocity. However, you must be careful not to mistake the faster metal path arrival, traveling around the tank.

If you do not know the velocity of the fluid in the tank & cannot easily measure it, you can access a built-in “Material Properties” database within the software. To access the “Material Properties” database, simply click the “Material” button at the bottom of the Location Setup menu. The Material Properties dialog box will appear as shown below. Along the top of the menu, you can select what type of materials group that you want. In our case, the “liquid” database has been selected. By clicking on the down-arrow in the Material Name text entry box, a list of materials

(liquids) in our case will be shown. Simply scroll down and select the desired material (or liquid). In our case, we have selected Motor oil. You will note that the velocity will automatically read out in the distance units that you have selected in the Location setup menu. If the material selection you want is not in the database, you will be able to enter this into the material database by using the “Add,” “Delete,” and “Update” keys near the bottom of the Materials database menu. You will note that for the example of the Motor oil in the figure, the velocity is 5708. We can make use of this velocity by entering it into the “Wave Velocity” column of the location group setup row. The direct way is to simply left-click the cell in Location Setup, type the appropriate value into it, and hit the “Enter” key. Remember to enter the value in the same units as that listed at the bottom of the Location setup menu. As an alternative, you can Copy & Paste the value by highlighting it in the Material Properties dialog box & pressing Ctrl-C. In Location Setup, highlight the cell, press F2 to edit its contents & then press Shift-Ins

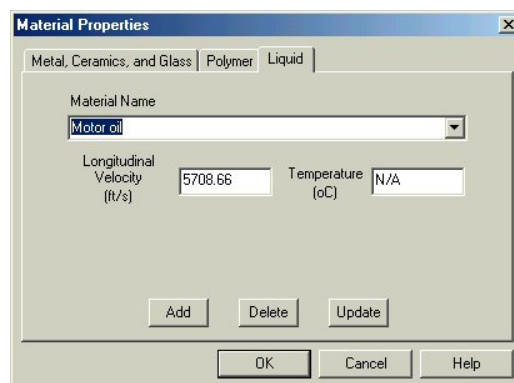


Figure 4-74. Material Properties Dialog

Event Definition Value: The Event Definition Value is the length of an event in the user's distance units. The Event Definition Value should be set to make sure that an AE source, from any point in the location array, will intersect a minimum desired number of sensors (3 is the minimum), forming an event. In our example of a 30-foot diameter tank, we have selected an Event Definition value of 32, which assures that all the sensors in the tank will have the capability of detecting an event regardless of where it originates.

Event Lockout Value: The Event Lockout Value is also entered in the user's distance units. The Event Lockout Value controls the interval between consecutive events. Enter an Event lockout Value starting from the beginning of the Event and long enough to assure that there is enough time (or distance) for all additional hits and reflections related to that event to have had a chance to die down. In order for the Event Lockout Value to be effective, it needs to be set longer than the Event Definition Value since they both start at the same time. There are no rules of thumb for setting Event Lockout Value since the event-related activity is dependent on the structure and sensitivity settings of the system. It is best to experiment to find the best selection.

Overcal Value: The Overcal value is also set in the selected distance units and is usually set equal to approximately 10% of the Event Definition Value. This is to assure that the location algorithm does not ignore minor variations in the wavepath velocity.

Timing Strategies: The user can select from some different timing strategies as set up in the “Timing” tab located at the top of the Location Setup menu. Please note that the Tank Bottom location algorithm does not support multiple timing strategies as the advanced regression location modes can.

Hits/Event (min/max): This entry is used with the regression location modes, which are able to use more than 3 hits when calculating the source location for an event. Tank Bottom location is not, however, a regression-based location mode. It only uses 3 arrivals to determine location, so these fields are fixed at 3 & are not adjustable for Tank Bottom location.

Max Iterations: Again, this setting only applies to regression location modes, not Tank Bottom location, so you cannot modify this setting for Tank Bottom.

Once you fill out all this information, you should proceed to the “Location View” menu, where you can setup up the Tank Bottom “Structure” and dimensions and set sensors around the tank. An important point with all these settings is that they do not affect the data that is stored in the DTA file during acquisition because the location results are not stored in the DTA file. They are recalculated each time you do a replay of the data, so you can change any of the above entries at any time & reanalyze the data to try to improve the location accuracy.

4.5.3.2 Configuration of Sensor Placement

The Sensor Placement dialog is entered from the Location Setup by clicking on the “Location View” button. Upon selecting this option, the dialog box shown below will appear. This menu is the base starting point for setting up the Tank bottom and other location-based structures. The Tank Bottom structure is one of the easiest location setups to create since the sensors can be placed and automatically. Please perform the following steps to set up all the aspects required in the Sensor Placement dialog shown below.

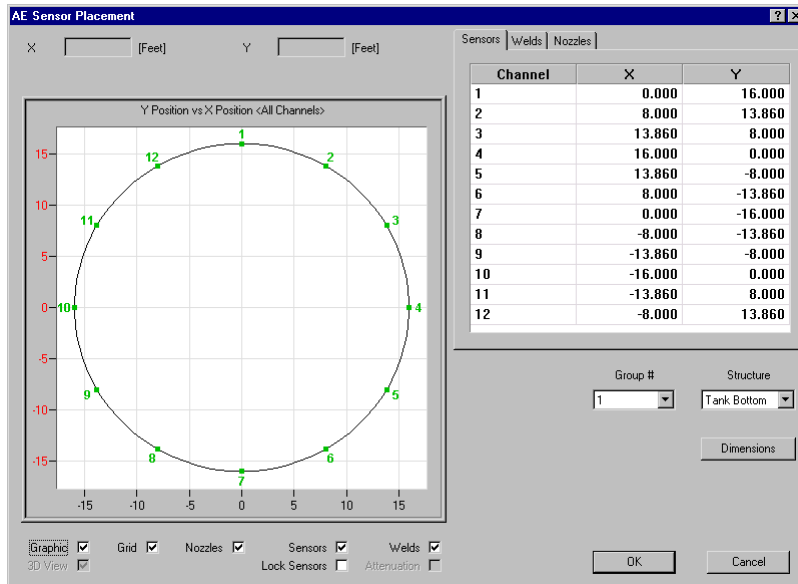


Figure 4-75. Tank Bottom Sensor Placement

1. First, make sure that you have selected the location group that you wish to work with. In our case, we had set up location group #1 in the Location Setup menu, so make sure Group #1 is selected in the Group selection text entry box.
2. Next, we go to the “Structure” menu and select the “Tank Bottom” structure as the desired structure. This gives us a chance to talk about our “structure” based on location capability. Since the user is setting up his application on a particular type of structure (such as a plate, a pipe, a vessel, a sphere, a tank bottom, etc., etc.), it is important for the location to think in terms of structures as well. This is what has been implemented in the software and is an extremely advanced concept that only PAC is doing.
3. Once you have selected the Tank Bottom as your Structure type, you should then click the “Dimensions” button. The dialog box shown to the right will appear. This menu is a customized user configuration menu for the type of structure you have selected and allows the customer to set up certain aspects of that structure. In the case of a Tank Bottom, the menu allows the user to enter the Tank diameter (which we have set to 30 feet in our example, allows you to set alternate units, and also allows you to enter information about the automatic sensor placement. First, it lets you select the starting sensor number as well as the total number of sensors that will be equally spaced around the tank and the starting phase, assuming that 0 degrees are that the top of the tank bottom (or North). In our example, we have set up for our sensors to start at sensor #1, starting at the top or 0 degrees, and the placement

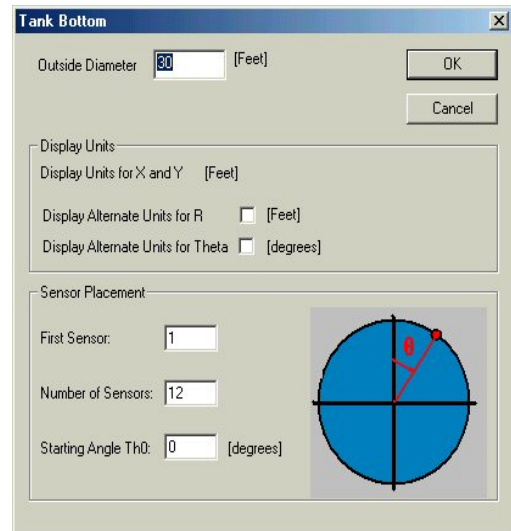


Figure 4-76. Tank Bottom Dimensions

of 12 sensors. When accepted (by clicking the OK button), the sensors will be automatically placed in a clockwise fashion at equal angular spacing on a circular tank bottom in the Location View area. This allows the user to view the spacing. Please note the table of sensor positions that are automatically generated. This table can be exported to the clipboard (by right-clicking on the grid and selecting 'copy to clipboard') for insertion into a report file.

4. After the sensor placement step, the user should then go and set up the sensors on the tank in the locations indicated by the Location view menu. Once done and the sensors have been connected to the appropriate AE channel number, you are ready to proceed to set up the Location graph and begin acquisition.

4.5.3.3 Setting Up a Tank Bottom Location Graph

The next step in the process of getting ready for carrying out a Tank Bottom location test is to set up the Tank Bottom Location screen. This is a straightforward process that is carried out in conjunction with the Graph Setup menu. The following are the steps in setting up a Tank Bottom Location Graph.

1. First, exit the Sensor Placement dialog and the Location Setup menu by clicking the OK button. Please note that if you click "Cancel," that you will lose all the setup changes that you have entered. Make sure you click the OK button before proceeding. Upon getting out of the Location setup menu, you will be back to the main screen showing some graph setups.
2. First, let's add a graph page by either right-clicking on one of the graph tabs at the bottom of the screen or by clicking on the "New Page" icon on the icon toolbar (the icon looks like a sheet of paper). This action will add a new tab to the bottom of the graph selection tabs and will create a blank screen. Next, let's name the new Graph Tab "Tank Bottom" by right-clicking on the tab and selecting the "Rename Page" selection.
3. Next, let's add a graph to the page by either right-clicking in the new "Tank Bottom" page or selecting the "New Graph" icon on the icon toolbar menu at the top of the screen (it's next to the "new page" icon). Once you do this, a graph window will pop up. Maximize the graph to be full screen by clicking on the "Tile Horizontal" or "Tile Vertically" icons on the top icon toolbar. You are now ready to create your Tank Bottom Location graph.
4. To setup for a Tank Bottom Location graph, first, you need to enter the Graph Setup menu by right-clicking in the Graph area of the graph you want to set up. This is the topmost selection in the context menu that comes up when you right-click. Please note the Graph setup is the dialog box to the right.
5. Next, select the "Event Groups" Tab at the top of the Graph setup menu and Check location group #1, which is identified as a Tank Bottom location group (if you have properly followed all previous steps).
6. Next, select the type of Graph. In the case of a location graph, you want to select a 2D Binned Scatter Plot. This text entry box is just above the Y-vertical axis settings.

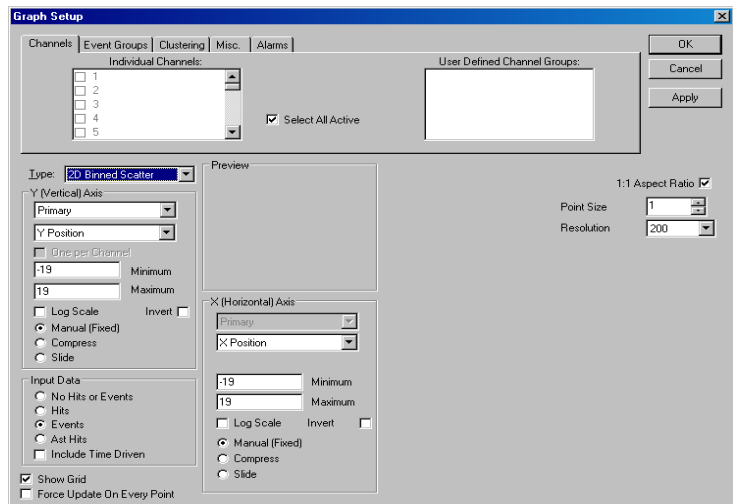


Figure 4-77. Tank Bottom Graph Set up

7. Next, set up the Y-(vertical axis) and X-(horizontal axis) for display. First, set the Y vertical axis to be a Primary graph (this means that you will only be plotting one graph on this screen plot. The next scrolling textbox allows you to select the feature or value that you want to plot. Since this is a location plot, you want to plot the Y position versus the X position, so select these for your Y and X settings.

8. Next, you need to select the graph scale. Tank Bottom location assumes that the center of the tank is the 0,0 position, so the axis limits should be a little larger than the tank radius. In our case, the radius is 15, so we'll use ± 19 as the limit for both X & Y. Since we want our Tank Bottom outline to look like a circle rather than an ellipse, we've checked the 1:1 Aspect Ratio. This checkbox controls whether the software is allowed to stretch the graph to fill the available space or whether it is required to keep the X & Y-Axis the same length.
9. The last important setting is to select Events as your Input data to the graph. Events will show the result of the location determination on the incoming AE data.
10. With all these settings done, we can exit the graph setup menu by clicking the OK button. If you did everything correctly, you would get a nice-looking tank bottom screen like that shown above. You can have other graphs on this same screen; you are not limited to just one graph if you want to just show a portion of the tank in different regions.
11. Note that the structure and the sensors were automatically plotted on the screen. This is because the software was instructed that the structure of the Location group #1 was a Tank Bottom location and was automatically able to draw this structure as soon as it saw that the graph which was being setup was an X Position versus Y Position location graph. If you do not see the Tank Bottom outline in the graph, go back to step 5 & make sure Event Groups has location group #1 checked & that it says Tank Bottom next to it.

You have now totally completed the setup of a Tank Bottom location test and are ready to acquire or replay data.

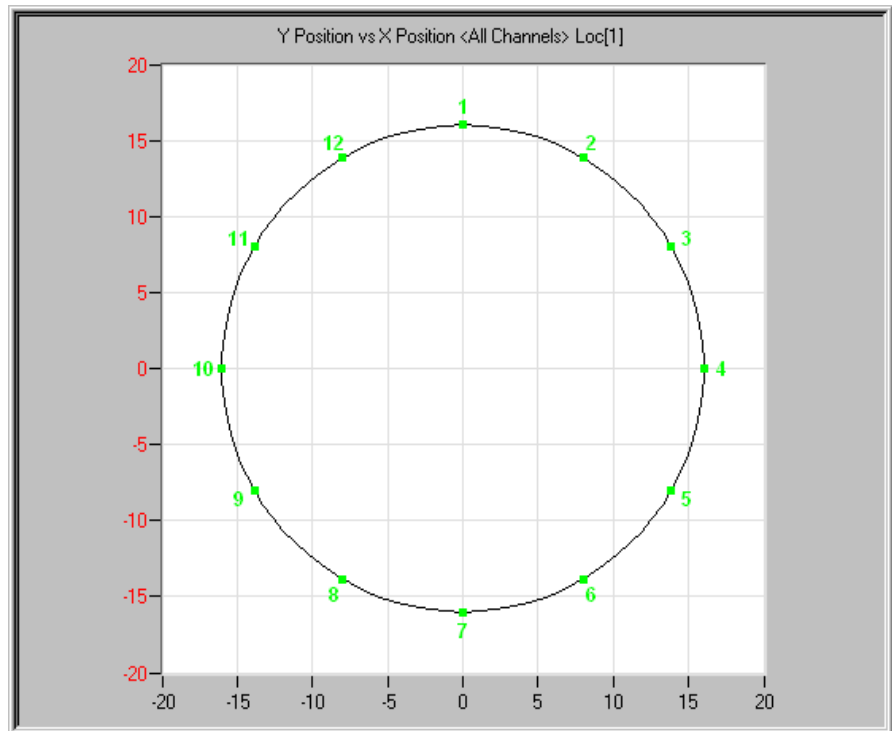


Figure 4-78. Tank Bottom Graph

4.5.3.4 Acquiring and Replaying and Displaying Data

Now that the entire setup is complete, you are ready to acquire or replay data from your Tank bottom test. To acquire data, simply enter data acquisition and begin recording data. You should always perform a test to determine if the location algorithm is operating properly before actually performing the true acquisition test. Verify that the location is performing properly and the AE hardware and software settings are all correct to minimize noise and maximize the types of signals you want to record. Once satisfied, start acquiring test data.

Once the acquisition is completed, you can start replaying and analyzing data. There is a Tank Bottom data file named TankBot.dta in the product folder located in “C:\MISTRAS Group” for you to experiment with. You can use it to test the Tank Bottom location. If you do a replay of TankBot.dta using the TankBot.lay layout file, you should see results like that shown here.

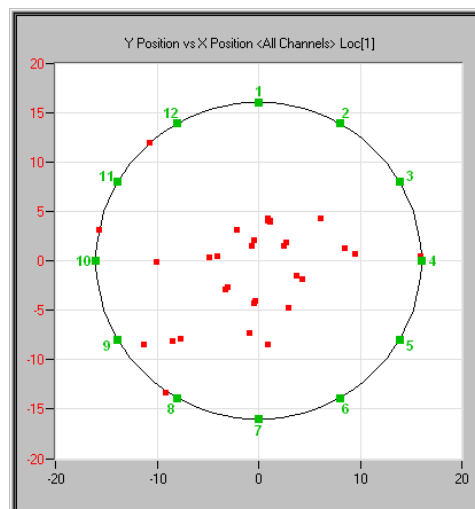


Figure 4-79. Tank Bottom Results with 1 Location per Event

All standard analysis and filtering tools can be used on the data to eliminate noise and assure relevant data.

There is another option that has been built into the Tank Bottom Software that allows multiple location calculations and plots to be displayed on the Tank Bottom location screen. To access this processing feature, look for the Tank Bottom Options button in Location Setup (F8 or select Location on the Acquisition Setup pulldown menu). This button is just above the “Location View” button, as shown in figure 139. The resulting Tank Bottom Options dialog box is shown here.

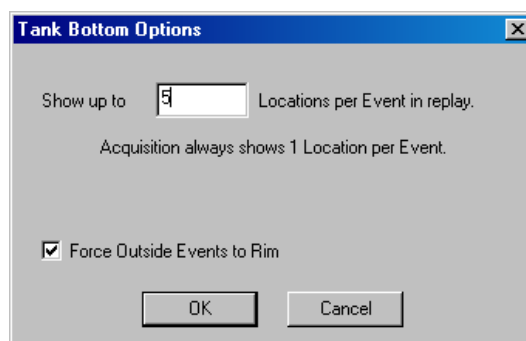


Figure 4-80. Tank Bottom Options Dialog

This option allows you to choose to display more than 1 location per event. Normally, the number is set to 1 as a default, but we have set it to 5 to allow up to 5 location calculations per event to process and display on the screen. This uses other hits in the event in order to calculate the location in case the first hits of the events have some problems. Note that this option only affects replay mode to keep acquisition at its highest speeds. Once this is selected, and the file is replayed, you can see the difference by comparing the graph on the right with the one above. As can be seen, there are more events displayed on the graph to the right, as there should be. What we learn from this is that the location is pretty good and matches pretty closely, telling us that the original location data in the previous graph is valid.

For a discussion of the Force Outside Events to Rim checkbox, its interaction with Allow Outside Events in Location Setup & how that can affect your results, see the Tank Bottom Location notes in section 4.4.9.

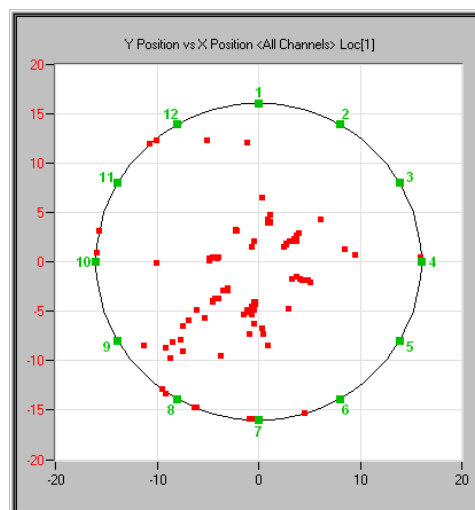


Figure 4-81. Tank Bottom Results with 5 Locations per Event

5. VIRTUAL PARAMETRICS

Virtual parametric data is data that the software collects from hardware usually manufactured by another company and treats it similar to the parametric data normally collected with the hardware manufactured by Physical Acoustics from the SAMOS, EXPRESS-8 boards, and the Sensor Highway Smart Monitor and Sensor Highway III systems. This section documents how virtual parametrics are set up and used in AEwin in general, along with specific details on using the individual instruments. The instruments supported include the Vaisala Weather Station, the Corr Instruments NanoCorr, Measurement Computing's PCI-DAS6033, and USB-2533 boards, the MISTRAS Group 9380-2114 Strain Gauge Module, and the MODBUS Module. MODBUS is a robust serial communications protocol in the industry that also supports TCP/IP communications. The MODBUS device connected to AEwin with serial lines or network cable is usually a small industrial computer (e.g., Programmable Logic Controller (PLC) and Distributed Control System (DCS) supplier) that controls one or more hardware devices.

5.1 Configure Virtual Parametrics Dialog Box

To make use of virtual parametrics in AEwin, it is necessary to tell the software what kind of virtual parametric hardware you will be using before acquisition starts. This is done with the Configure Virtual Parametric Modules dialog box shown below:

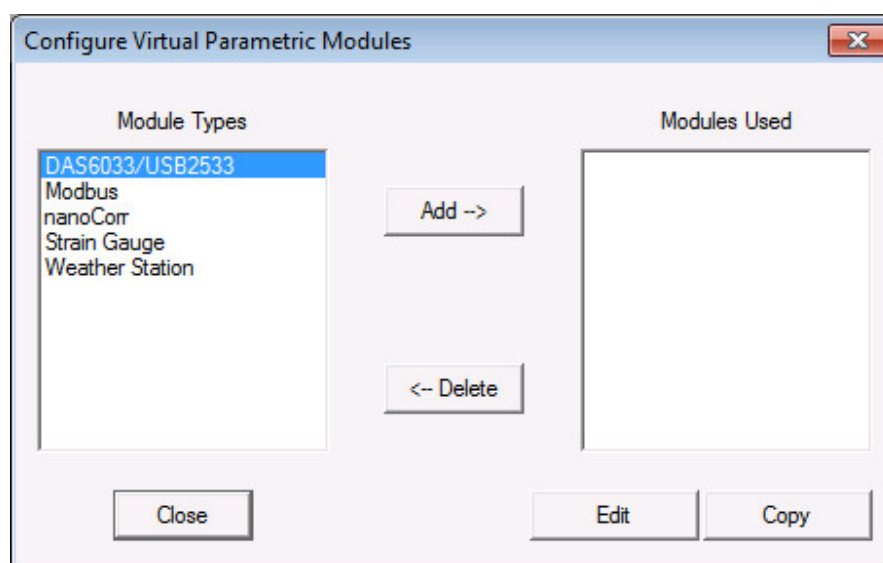


Figure 5-1. Configure Virtual Parametrics Dialog Box

This dialog box can be accessed in several ways. First, it is available directly on the Acquisition Setup dropdown menu. You can also use the hotkey combination Ctrl+F2 to bring it up when you are at the main menu. It can also be accessed thru a button on the Virtual Parametrics tab in the Hardware Setup dialog box. More on the Virtual Parametrics tab in Hardware Setup will be given in the section “Virtual Parametrics tab in Hardware Setup.”

The column on the left-hand side of this dialog box displays the Module Types list. This is just a list of the types of virtual parametric hardware that is currently supported by AEwin. Simply highlight the module you want to use and click on the “Add -->” button. Alternatively, you can double-click the module type to add it as well. Each module type will lead to a different Virtual Parametric Module Setup dialog box specific to that hardware. More on each dialog box is given below. Once the module has been configured, and the user clicks on OK, the module name will appear in the Modules Used column. If you no longer wish to use a module, you may highlight it in the Modules Used column and click on the “<-- Delete” button.

Modules listed in the Modules Used column can be double-clicked to edit their settings. Alternatively, you can highlight the module name and click Edit to launch the appropriate Virtual Parametric Module Setup dialog box. For some virtual parametric types, you may wish to work with more than one instrument. For example, you might want

to make use of two NanoCorr instruments for collecting data. If you wish, you may copy the setup for an existing module by highlighting it in the Modules Used column and clicking on the Copy button. A new setup dialog box will open with an exact duplicate of the settings.

Before you can successfully save the settings of the new module, you must change the Module Name, and for many modules, you need to change the Module ID. See the Virtual Parametric Module Setup dialog box for the DAS6033/USB2533 for an example. The Module ID is often used as an address or board identifier when communicating with the device. For example, Weather Station instruments can communicate via a shared USB port and come configured by default to address 0. If you wished to use more than one, the second Weather Station would have to be reconfigured to appear at another address like 1 or 2. The Module ID is how the user tells the software which instrument to access. It is not a problem if a NanoCorr module and a Weather Station module both use an ID of 0, but if you were using two Weather Stations, they cannot both have ID values of 0.

You can have up to four modules of a given type, like a NanoCorr instrument. There is also a maximum limit of 256 total channels of virtual parametric data that are allowed. Weather Station modules have 22 channels of data, NanoCorrs have 46, Measurement Computing DAS6033 & USB2533 boards have 64 channels, Strain Gauge modules have 4 channels, and the MODBUS modules have 48 channels. You may mix and match modules as needed, but you are limited to no more than four modules of a given type & no more than 256 channels total from all modules in use.

There are many settings that are common to all the Virtual Parametric Setup dialog boxes. Even if you are not using a Measurement Computing board to collect virtual parametric data, it is suggested that you read that section as well as the sections appropriate for the hardware you are using because the common settings are described in this first section.

5.2 Measurement Computing's PCI-DAS-6033 & USB-2533 Boards

The Measurement Computing boards are designed for 64 channels of analog input. The DAS-6033 uses a PCI slot in your computer, while the USB-2533 is an external board that connects to one of your computer's USB ports. They can monitor any instrument or sensor that outputs a voltage between +/- 10v, making them very versatile for virtual parametric input. Here is a description of the available options in the setup dialog box shown below.

5.2.1 Common to All Modules

Module Name – Module Name is just a string used to identify the source of virtual parametric data in the Line Display. Because of the potential for confusion, you cannot use the same name for two modules. It is also recommended that you keep the Module Name short as it does affect how much data can be displayed on one line in the Line Display.

Module ID – Module ID is an integer used to uniquely identify the module in the software. You may not have two modules of the same type with the same ID. In the case of the PCI-DAS6033 and the USB-2533 boards, this number corresponds to the board number used in Measurement Computing's InstaCal program. This defaults to 0 for the first board. If you have multiple boards, the second board number is usually 1. Enter the same number here that is used in InstaCal to access the hardware.

Channel	Name	Software Scaling			*Sampling Period (ms)
		Multiplier	Offset	Units	
☒ 1	Ch01	1.0000	0.0000	Un01	10000
☐ 2	Ch02	1.0000	0.0000	Un02	10000
☐ 3	Ch03	1.0000	0.0000	Un03	10000
☐ 4	Ch04	1.0000	0.0000	Un04	10000
☐ 5	Ch05	1.0000	0.0000	Un05	10000
☐ 6	Ch06	1.0000	0.0000	Un06	10000
☐ 7	Ch07	1.0000	0.0000	Un07	10000
☐ 8	Ch08	1.0000	0.0000	Un08	10000
☐ 9	Ch09	1.0000	0.0000	Un09	10000
☐ 10	Ch10	1.0000	0.0000	Un10	10000
☐ 11	Ch11	1.0000	0.0000	Un11	10000
☐ 12	Ch12	1.0000	0.0000	Un12	10000
☐ 13	Ch13	1.0000	0.0000	Un13	10000
☐ 14	Ch14	1.0000	0.0000	Un14	10000
☐ 15	Ch15	1.0000	0.0000	Un15	10000
☐ 16	Ch16	1.0000	0.0000	Un16	10000

*Sampling Period must be 100ms minimum. Higher values must be an integer multiple of 100ms.

Figure 5-2. Configure DAS6033/USB2533

Channel On/Off Checkbox – This is just a simple On/Off toggle that specifies whether or not to collect data from this channel. If you wish to turn all the channels on or off, simply check the box for the first channel and then press the F5 key. F5 sets all values in the column to the value of the selected cell.

Channel Name – This is a string that can help you identify what data a channel is collecting. For example, channel 9 could be a voltage reading that represents the current temperature reading. In that case, you might prefer to replace the default label of Ch09 with Temp. It allows a maximum number of 26 characters for each Channel Name.

Multiplier, Offset & Units – These 3 settings are identical to those for regular parametrics in AEwin. The Multiplier and Offset allow the user to rescale the reading from the hardware to units more to the user's liking. Multiplier & Offset are floating-point numbers. The raw reading is multiplied by the Multiplier, and then the Offset value is added to it. The Units setting holds a string to represent the new units. For example, if the original reading was in °C & the user wanted the reading in °F, then the Multiplier would be 1.8 (= 9/5) & the Offset would be set to +32. Units could then be set to F. For the DAS6033 and USB2533, all raw readings are actually expressed in volts, but the same principle applies if you are converting volts to loading pressure or a rotation speed. It allows a maximal number of 22 characters for each Unit Name.

Sampling Period – All modules have a Sampling Period setting, but some modules have only one setting for all channels. The DAS6033 & USB2533 boards can, however, vary the sampling rate for each channel independent of the others, so there is a column in their setup dialog box to allow a value for each channel. The values are entered in milliseconds, and the minimum value is 100ms. Higher values must be an integer multiple of 100ms. If you wish to set all channels to the same sampling rate, type the value desired into one cell in the column and then press F5.

Test Module – This is not an option per se, but this button is useful for testing whether or not AEwin can successfully communicate with your Measurement Computing board. It will either tell you “Could not detect the board” or “Board successfully detected.” If it was not detected, verify that the Module ID setting is set to the same value as the board number assigned in the Measurement Computing's InstaCal program. If InstaCal can't see the hardware, AEwin won't be able to either. If it is a USB-2533, you can double-check that the USB cable hasn't come loose. If it is a DAS6033, make sure the board is fully seated in its PCI socket.

5.2.2 Specific to PCI-DAS6033 & USB-2533 Boards

Samples to Average – The other modules do not have a “Samples to Average” setting. It is specific to the DAS6033 and USB2533 boards. You can enter a number from 1 to 256. This is the number of samples that the board acquires when making a reading. The samples are collected as fast as possible for the DAS6033 and USB2533 (about 640ms/reading) and then averaged to give the reading reported for that sampling period.

5.3 Vaisala Weather Station

The Vaisala Weather Station is designed to collect information on current weather conditions such as temperature, rainfall amounts & wind speed.

5.3.1 Common to All Modules

Module Name, Channel On/Off Toggle, Channel Name, Multiplier, and Offset – These are described in the section on the Measurement Computing boards. Please refer back to that section for detailed descriptions.

Module ID – Module ID is an integer used to uniquely identify the module in the software. You may not have two modules of the same type with the same ID. In the case of the Vaisala Weather Station, this number corresponds to the device address on the shared serial connection. This defaults to 0 for the Weather Station. If you were using multiple Weather Stations, the second one would have to be configured to use another address like 1. You can use the Vaisala Configuration Tool to change the device address using its Device Settings dialog box. Enter the same number here that is used in the Vaisala Configuration Tool to access the hardware.

Channel	Name	Software Scaling		
		Multiplier	Offset	Units
☒ 1	Wind Speed Min	1.0000	0.0000	m/s
☒ 2	Wind Speed Avg	1.0000	0.0000	m/s
☒ 3	Wind Speed Max	1.0000	0.0000	m/s
☒ 4	Wind Direction Min	1.0000	0.0000	deg
☒ 5	Wind Direction Avg	1.0000	0.0000	deg
☒ 6	Wind Direction Max	1.0000	0.0000	deg
☒ 7	Air Pressure	1.0000	0.0000	bar
☒ 8	Air Temperature	1.0000	0.0000	C
☒ 9	Internal Temperature	1.0000	0.0000	C
☒ 10	Humidity	1.0000	0.0000	%RH
☒ 11	Rain Accumulation	1.0000	0.0000	mm
☒ 12	Rain Duration	1.0000	0.0000	sec
☒ 13	Rain Intensity	1.0000	0.0000	mm/hr
☒ 14	Rain Peak Intensity	1.0000	0.0000	mm/hr
☒ 15	Hail Accumulation	1.0000	0.0000	hits/cm2
☒ 16	Hail Duration	1.0000	0.0000	sec

*Sampling Period must be 10000ms minimum. Higher values must be an integer multiple of 10000ms.

Figure 5-3. Configure Weather Station

Units – The Units string is common to all virtual parametric modules, but a discussion of the Weather Station is not complete without a few comments on the units. The default units listed in the Weather Station setup dialog are the units the data is transmitted in from the instrument over a serial port connection. AEwin specifies what units the Weather Station should use when it initializes the instrument at the start of data acquisition. The user can use the Vaisala Configuration Tool to modify the instrument's default units, BUT AEwin will override any changes made in the configuration program. Do not try to change the units this way. For example, if you desired wind speed expressed in km/hr instead of m/s and you enter km/hr here and try to use the Vaisala Configuration Tool to do switch to readings in km/hr, you will get a reading in m/s, and it would be misleadingly labeled with km/hr. Instead, set the Multiplier here to 3.6 (3600s/hr * 1km/1000m) and the Offset to 0 to scale the incoming m/s readings to km/hr. Do not forget to update the Units string to km/hr as well.

Sampling Period – For the Weather Station, the Sampling Period must be at least 10,000ms. You can set a higher value, but not a lower one. Higher values should be integer multiples of 10s (10000ms), so valid numbers are 10000, 20000, 30000, but not 15000, 25000.

5.3.2 Specific to Vaisala Weather Station

Serial Port – The Weather Station is designed to communicate over a serial connection. It can be configured to communicate over an RS-232, RS-422, RS-485, or via USB connection. AEwin expects the instrument to communicate at the defaults of 19200 baud, 8 data bits, no parity, and 1 stop bit. The only user-configurable parameter in AEwin is what serial port is connected to the Weather Station.

Test Module – You may use the Test Module button to verify that AEwin can properly communicate with the Weather Station. You may receive the cryptic Windows error message, “The system cannot find the file specified.” Windows treats serial ports like a file for reading and writing information which is why it says it cannot find the file. In this case, the file is the serial port and what Windows is saying is that it cannot open that port. Most likely, you have used a serial port that does not exist on your computer. If you receive a “Weather Station communication failed” error message, the port exists, but the Weather Station did not respond to the test message.

5.3.3 Troubleshooting Problems with an RS-485 Connection

If you happen to be using an RS-485 connection, there is one more instrument setting that may affect your ability to reliably communicate with the Weather Station. If you start up the Vaisala Configuration Tool and it cannot maintain a connection with your Weather Station, AEwin will not be able to maintain a connection with it either. In that case, you may need to adjust the RS-485 line delay parameter. It cannot be adjusted by AEwin, but it can be adjusted with the Vaisala Configuration Tool if the connection is somewhat reliable. Look for it in the “Device Settings” dialog box. If the Configuration Tool cannot maintain a connection, you will need to use Windows HyperTerminal to adjust the parameter.

Configure HyperTerminal to use the same serial port that the Vaisala Configuration Tool is using. In HyperTerminal, this is done using the File Properties menu option. On the “Connect To” tab, set the serial port to use 19200 baud, 8 data bits, no parity, and 1 stop bit. Next to the Settings tab, click the ASCII Setup button. Check “Send line ends with line feeds,” “Echo typed characters locally,” and “Append line feeds to incoming line ends” Click on OK to save your changes and then again to exit the Properties dialog box.

When you exit the properties dialog box, HyperTerminal will attempt to open the port. Assuming the port opened successfully, you should be able to type a command to the Weather Station instrument and have it respond. Simply type the device address, 0 by default, and press Enter. The instrument should respond with the same value. If you do not know the device address, type “?” and then press Enter. The response should be the device address. All commands to the Weather Station start with its device address, and commands are case sensitive. Assuming this instrument is using the default address of 0, you can check the line delay parameter by typing:

```
0XU<enter>
```

The instrument should respond with something like:

```
0XU, A=0, M=P, T=0, C=2, I=0, B=19200, D=8, P=N, S=1, L=0, N=WXT520, V=1.00<enter>
```

Look for the L=# entry. This is the line delay parameter in milliseconds. You can update the value to 100ms by typing:

```
0XU, L=100<enter>
```

which the instrument should respond to by echoing the command back to you:

```
0XU, L=100
```

You can double-check that the setting was stored by using the 0XU command again. Once you have adjusted the line delay, disconnect from the port in HyperTerminal and try the Vaisala Configuration Tool. If it still cannot maintain the connection with the instrument, close the Configuration Tool, return to HyperTerminal and readjust the line delay. Valid line delay values range from 0 to 10000ms. Start low and slowly make it larger by 50 or 100ms at a time. Otherwise, you may jump over the range of values that work. Most likely, you will only need a delay between 0 and 250 ms, but it does depend on the length of your RS-485 cable, so the exact value is hard to predict.

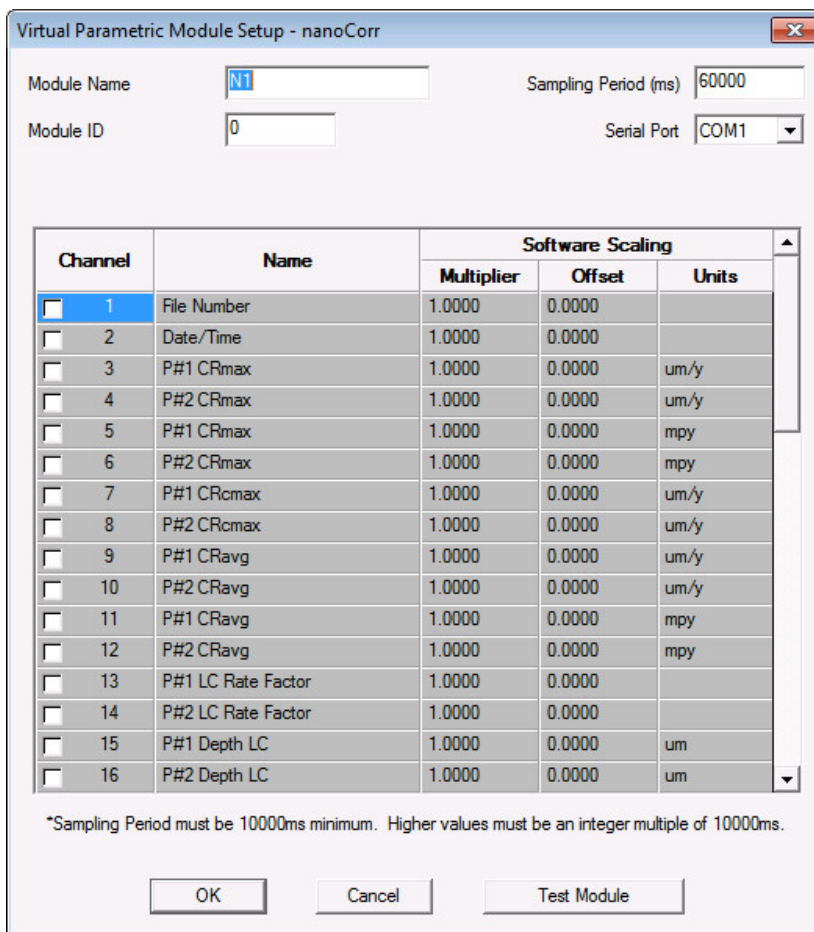
5.4 Corr Instruments' NanoCorr

Corr Instruments' NanoCorr is designed to measure corrosion rates of metal probes. It is unique in that it is the only instrument that the software cannot actively control. All the software can do is monitor its output on a serial port & import the data readings. The user must configure it with NanoCorr front panel & then make the AEwin software settings match the NanoCorr settings.

5.4.1 Common to All Modules

Module Name, Module ID, Channel On/Off Toggle, Channel Name, Multiplier, and Offset – These are described in the section on the Measurement Computing boards. Please refer back to that section for detailed descriptions.

Sampling Period – For NanoCorr, as for the Weather Station, the Sampling Period must be at least 10000ms. You can set a higher value, but not a lower one. Higher values should be integer multiples of 10s (10000ms), so valid entries are 10000, 20000, 30000, but not 15000, 25000.



Channel	Name	Software Scaling		
		Multiplier	Offset	Units
☐ 1	File Number	1.0000	0.0000	
☐ 2	Date/Time	1.0000	0.0000	
☐ 3	P#1 CRmax	1.0000	0.0000	um/y
☐ 4	P#2 CRmax	1.0000	0.0000	um/y
☐ 5	P#1 CRmax	1.0000	0.0000	mpy
☐ 6	P#2 CRmax	1.0000	0.0000	mpy
☐ 7	P#1 CRcmax	1.0000	0.0000	um/y
☐ 8	P#2 CRcmax	1.0000	0.0000	um/y
☐ 9	P#1 CRavg	1.0000	0.0000	um/y
☐ 10	P#2 CRavg	1.0000	0.0000	um/y
☐ 11	P#1 CRavg	1.0000	0.0000	mpy
☐ 12	P#2 CRavg	1.0000	0.0000	mpy
☐ 13	P#1 LC Rate Factor	1.0000	0.0000	
☐ 14	P#2 LC Rate Factor	1.0000	0.0000	
☐ 15	P#1 Depth LC	1.0000	0.0000	um
☐ 16	P#2 Depth LC	1.0000	0.0000	um

*Sampling Period must be 10000ms minimum. Higher values must be an integer multiple of 10000ms.

OK Cancel Test Module

Figure 5-4. Configure NanoCorr

5.4.2 Specific to NanoCorr

Serial Port – The NanoCorr is designed to communicate over an RS-232 or RS-485 serial connection. It can also be configured to communicate over an Ethernet connection masquerading as a serial connection. Select the port number of the serial port that the NanoCorr is hooked up to. If using an Ethernet connection, refer to Chapter 5 of the NanoCorr manual for details on how to set up the connection & determine the serial port number to use here. AEwin expects the instrument to communicate at the defaults of 115200 baud, 8 data bits, no parity, and 1 stop bit.

Test Module – You may use the Test Module button to verify that AEwin can properly communicate with the NanoCorr. As for the Weather Station, you may receive the cryptic Windows error message “The system cannot find the file specified.” Again Windows treats serial ports like a file for reading and writing information which is why it says it cannot find the file. In this case, the file is the serial port and what Windows is saying is that it cannot open that port. Most likely, you have used a serial port that does not exist on your computer. Assuming the port exists, you will receive the following message:

“COM Port opened. Must wait for nanoCorr to transmit a new reading.”

Click on OK when ready to check for a data string.

The NanoCorr test capability is very primitive due to the limitation that AEwin can only monitor the output of the NanoCorr over its serial port connection. It cannot send a command to the NanoCorr and have it respond. The “Test Module” button just checks for incoming data from the chosen serial port. At this point, you will have to wait until you feel the NanoCorr has transmitted data to the PC. If you used the front panel of the NanoCorr to tell it to collect new readings every 2 minutes, you would have to wait 2 minutes before pressing the OK button. If you press the button before data has been transmitted to the PC, you will see a “No data received from NanoCorr.” message. If data has been received, you will see a message that shows you the text received. It should be a series of numbers that looks something similar to this NanoCorr data record:

NanoCorr data string read =

```
"186 2008-07-23 15:22:54 591.0344848 23.269074 591.0344848 122.956886 4.840822
4.806843 0.0962117 0.0202236 4.757399 -0.554344 -109.181907 6.709134 0.0209924
-0.722047 -27 -89 -17531 602829 -220275 25 -73 49 -293030 -315945 -399937
-40374 -43860 -82 47"
```

If you do not end up seeing any data, you might want to try using HyperTerminal to monitor the serial port that the NanoCorr is connected to. Setup HyperTerminal as described in the “**Troubleshooting Problems with an RS-485 Connection**” section for the Weather Station, but use 115200 as the baud rate instead of 19200. The NanoCorr, by default, communicates at a 115200 baud rate, 8 data bits, no parity, and 1 stop bit. You still want to turn on “Append line feeds to incoming line ends.” Just leave HyperTerminal open till you see data strings come in.

If you leave the window open long enough to see two data records received, you can tell how long the time is between data readings. Note in the example data string above, the time of the data reading is the 3rd number in the string. Just compare the two readings to figure out the time interval. That is how long you need to leave the message box open when it is waiting on data from the NanoCorr. If HyperTerminal never receives any data, neither will AEwin. Either you are using the wrong serial port, or the NanoCorr is not actively collecting data.

5.5 MISTRAS Group 9380-2214 4-Channel Strain Gauge Module

As noted above, the Strain Gauge Module is only supported by the Sensor Highway Smart Monitor system. It is not available for other systems.

5.5.1 Common to All Modules

Module Name, Channel On/Off Toggle, Channel Name, Multiplier, and Offset – These are described in the section on the Measurement Computing boards. Please refer back to that section for detailed descriptions.

Sampling Period – For the Strain Gauge module, the Sampling Period must be at least 10ms. You set a higher value, but not a lower one. Higher values should be integer multiples of 10s (10000ms), so valid entries are 10000, 20000, 30000, but not 15000, 25000.

5.5.2 Specific to Strain Gauge Module

Test Module – You may use the Test Module button to verify that AЕwin can properly find the Strain Gauge hardware. If there is a problem, you will get a “Strain Gauge 0 could not be detected”. If all goes well, you will get a “Strain Gauge 0 successfully detected” message, and it will give you information on the firmware and PCB versions of the hardware.

Zero Selected Channels and Calibrate Selected Channels – These buttons work with the channels that are checked on in the grid. They just run thru the routines responsible for zeroing and recalibrating the channel readings.

5.6 MODBUS Module

In contrast to the setup of the former four virtual parametric models described above, the setup of the MODBUS module is much more complicated for three reasons. First, although the MODBUS protocol is a serial communication protocol, it also supports the network communication protocol TCP. The properties of both the serial port and TCP connection need to be configured in addition to the setup of MODBUS properties. Second, in the MODBUS protocol, data is stored and transferred by means of registers. There are four types of registers, each of which has 9999 units associated with unique addresses. Hence, in addition to the common VP columns such as Channel Name, Unit, Multiplier, Offset, etc., there are two more columns for MODBUS —Address and Data Type. Finally, it is possible to output AЕwin alarms over MODBUS, and in this case, the alarm outputs also have to be configured. Therefore, there are three configuration dialog boxes, namely, “MODBUS Channel Setup,” “MODBUS Property Setup,” and “MODBUS Alarm Setup,” each linked to a tab page. These three dialog pages are described below.

Channel	Name	Software Scaling		
		Multiplier	Offset	Units
☒ 1	SG01	1.0000	0.0000	ue
☐ 2	SG02	1.0000	0.0000	ue
☐ 3	SG03	1.0000	0.0000	ue
☐ 4	SG04	1.0000	0.0000	ue

Figure 5-5. Configure Strain Gauge

5.6.1 MODBUS Channel Setup

This is the first MODBUS dialog page. Double-clicking the first tab, “MODBUS Channel Setup,” will bring it up, as seen in the following figure.

VP MODBUS Module Setup

MODBUS Channel Setup | MODBUS Property Setup | MODBUS Alarm Setup

Module Name: B1 Module ID: 0

Channel	Name	Software Scaling			Endian	Address	Data Type	Sampling Period (S)
		Multiplier	Offset	Units				
☐ 1	unknown 1	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 2	unknown 2	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 3	unknown 3	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 4	unknown 4	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 5	unknown 5	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 6	unknown 6	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 7	unknown 7	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 8	unknown 8	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 9	unknown 9	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 10	unknown 10	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 11	unknown 11	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 12	unknown 12	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 13	unknown 13	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 14	unknown 14	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 15	unknown 15	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15
☐ 16	unknown 16	1.0000	0.0000	na	Little Endian	0001	16-bit unsigned	15

OK Cancel Apply Help

Figure 5-6. MODBUS Channel Set up

5.6.1.1 Common to All Modules

Module Name, Channel On/Off Toggle, Channel Name, Multiplier, and Offset – These are described in the section on the Measurement Computing boards. Please refer back to that section for detailed descriptions. By default, the MODBUS module name starts with a capital letter B following a number.

Module ID – Module ID is an integer used to uniquely identify the module in the software. You may not have two modules of the same type with the same ID. In the case of the MODBUS module, this number corresponds to the device address on the shared serial/TCP connection. This defaults to 0 for the MODBUS. If you were using multiple MODBUS modules, the second one would have to be configured to use another address like 1. Depending on a specific AEwin application, the number of MODBUS modules may vary from 1 to 8, corresponding to a Module ID of 0 to 7.

Channel, Name & Units – Each MODBUS module has 48 channels. If there are other VP modules already configured, the total number of enabled channels must not exceed 256; otherwise, a warning message box would pop up. There are no specific names and units for the channels. A general channel name “unknown” is used followed by an index from 1 to 48, and all units default to “na.” The names and units are suggested to be entered for all enabled channels. The maximum allowable length of the channel name and unit is 26 characters and 22 characters, respectively.

Sampling Period – unlike other VP modules, the MODBUS Sampling Period is in second (S). For the MODBUS module, the Sampling Period must be at least 1S. You can set a higher value, but not a lower one. The maximally allowable Sampling Period for the MODBUS module is 65535 (S).

5.6.1.2 Specific to MODBUS

Address – It is the address of the register used for an external operational parameter. Its range is from 1 to 9999. This column does not allow input of five or more digits. Only numbers are acceptable. Different **Enabled** MODBUS channels cannot share the same register(s). This means a registered address cannot be assigned to multiple single-register parameters. If two parameters share an identical address, the configuration will not be passed. This is called an address conflict. When an address conflict happens to the **Enabled** channels, clicking the “OK” button will pop up a warning message box saying "Overlapped register addresses! Please check yellow cells." and make all cells on this column with conflicting addresses become yellow. All the conflicts must be manually solved to pass the configuration in this column. However, address conflict with **disabled** channels does not count. In other words, multiple disabled channels may share the same address. If virtual operational parameters are multi-register long, address conflict means register overlapping among parameters. This type of conflict is addressed in the description of the next column: Data Type.

In addition to the overlapped address error, there is another address error: lower than the start address error. The address of an enabled channel cannot be lower than the start address, which is configured on the MODBUS Setup Page that is described in the section “MODBUS Property Setup.” Addresses lower than the start address cannot pass the setup validation when clicking the OK button. A warning message box will pop up, and the background of the corresponding cells will become cyan color, and the message is:

"Address must be no less than the Start Address!
Please check the cyan cells and Start Address on the MODBUS Property Setup page."

Data Type – As aforementioned, MODBUS supports four types of registers, or four data tables: Digital Read-only Contacts, Digital Read/Write Coils, Analog Read-only Registers, and Analog Read/Write Registers (also known as Holding Registers). A holding register can function what the other three types of data tables can, first because it is read/write, and second, it has 16 bits, each of which can be used as a digital read-only contact or a read/write coil. Thus, it is assumed in this section that only the holding registers are used. Under this assumption, the AEwin MODBUS module supports six data types: unsigned 16-bit integer, signed 16-bit integer, unsigned 32-bit integer, signed 32-bit integer, 32-bit float, and 64-bit double. These six data types are sufficient for most practical uses and are also supported by most PLC's and DCS's. Because a holding register is 16-bit, the size of these six data types is 1, 1, 2, 2, 2, and 4 registers, respectively. The properties of these six data types are summarized in the table below.

Type	Size in Bits	Size in Registers	Range
Unsigned Short	16	1	0 – 65535
Signed Short	16	1	-32768 – 32767
Unsigned Integer	32	2	0 – 4294967295
Signed Integer	32	2	-2147483648 to 2147483647
Float	32	2	3.4E +/- 38 (7 digits)
Double	64	4	1.7E +/- 308 (15 digits)

Use the drop list of the combo box to select the data type for a channel. The down arrow of the cells in this column is hidden; click the cell to make it visible. If the data type for all enabled channels is the same, it is unnecessary to make a choice for each channel. Select the correct data type for one channel and have it highlighted, then press the F5 key. F5 sets all values in the column to the value of the selected cell.

In order for any of these six data types to be used, the MODBUS device connected to AEwin must also support that type. Make certain that the data type of a virtual parameter configured in AEwin matches the data type for the same parameter on the MODBUS device (PLC or DCS); otherwise, the data received in AEwin will not be parsed correctly.

Now that one channel may use more than one register (if using one of the latter 4 data types) avoiding address conflicts among channels becomes tricky. For example, suppose one channel uses the Double type, which is 4-registers long, and its address is 108. Then registers 108 – 111 cannot be used by any other enabled channels. Address overlapping becomes less obvious because only the starting address of a channel is shown (in this example, it is 108), but the remaining registers are implied by the data type. This type of address conflict is caused by register overlapping. The good news is AEwin does the overlap checking for you: if overlapping conflict happens to **Enabled** parameters, the configuration cannot be accepted. Clicking the “OK” button will pop up an error message box and make all cells on this column with conflicting addresses become yellow.

Endian – Endianness is related to how the bytes of a register are organized and is typically a function of how the CPU stores multi-byte numbers in memory. In the MODBUS controller market, most PLC or DCS use little endian, but some MODBUS device vendors also use big endian. Because the little endian prevails, it is the default setting in the AEwin MODBUS module. The endianness setup dialog control is a dropdown box, which is visible if the cell is clicked. Unless AEwin is to be used with a big endian MODBUS controller, this column is needless to change—just use the default setting. To save the setup time, instead of working on every dropdown box, the hot key “F5” can be used to make all enabled channels have the same set of endianness as the highlighted cell when “F5” is pressed.

5.6.2 MODBUS Property Setup

This is the second MODBUS dialog page. Double-clicking the mid-tab “MODBUS Property Setup” will bring it up, as seen in the following figure. The MODBUS properties are grouped into different categories using group boxes according to their functions. There are 10 group boxes in total: Work Mode, ID and Data Table, Connection, Serial Port, Flow Control, Error/Exception, Response Timeout, Delays Between Polls, TCP/IP Server, and Transmission Mode. Among them, Work Mode, ID & Function Code, and Connection are standard MODBUS protocol setups; Serial Port, Flow Control, Error/Exception, Response Timeout, Delay Between Polls, and Transmission Mode are setups for serial communications; TCM/IP Server is MODBUS TCP setup. Each group and its setup are discussed below.

The screenshot shows the 'VP MODBUS Module Setup' window with the 'MODBUS Property Setup' tab selected. The window contains several configuration sections:

- Work Mode:** Radio buttons for 'As Slave' (selected) and 'As Master'.
- ID and Data Type:** 'Slave ID' text box with '1', and 'Data Type' dropdown menu with '03 Holding Registers'.
- Connection:** 'Start Address' text box with '1', and 'Modbus Type' dropdown menu with 'Serial Lines'.
- TCP/IP Server:** 'IP Address' text box with '. . .', 'Port' text box with '502', and 'Connect Timeout' text box with '0 [ms]'.
- Serial Port:** 'Port #' dropdown with 'COM1', 'Media Layer' dropdown with 'RS-232', 'Baud Rate' dropdown with '9600', 'Data Bits' dropdown with '8', 'Parity' dropdown with 'No', and 'Stop Bits' dropdown with '1'.
- Flow Control:** Checkboxes for 'DSR', 'CTS', 'DTR', and 'RTS'. Below is a text box with '100 [ms] RTS Disable Delay'.
- Error/Exception:** Checkboxes for 'Skip Response', 'Insert RC Error', and 'Busy'. Below is a text box with '400 [ms] Response Delay'.
- Response Timeout:** Text box with '0 [ms]'.
- Delay Between Polls:** Text box with '0 [ms]'.
- Transmission Mode:** Radio buttons for 'RTU' (selected) and 'ASCII'.

At the bottom of the window are four buttons: 'OK', 'Cancel', 'Apply', and 'Help'.

Figure 5-7. MODBUS Property Set up

5.6.2.1 Work Mode

Standard MODBUS communication uses a master-slave technique, in which only one device (the master) can initiate transactions (called “queries”). The other devices (the slaves) respond by supplying the requested data to the master or by taking the action requested in the query. In MODBUS TCP, the slave is the socket server, and the master is the socket client. The Work Mode group box consists of two radio buttons: As Slave and As Master, with the Slave radio button checked by default. The remaining MODBUS property groups on this dialog page are related to either the Slave mode or the Master mode, or both. In fact, there are two sets of MODBUS property configurations associated with the Slave mode and Master mode, respectively: switching from one to another will result in a different set of configurations to load up, with the temporary changes (if any) made to the previous set saved in a buffer to avoid being lost.

Although the slave and master share the most common setup items, there are slight differences in the setup dialog boxes between them. The Error/Exception group belongs to the Slave mode alone; thereby, it is disabled in the Master mode; on the other hand, the “Response Timeout” group and “Delays Between Polls” group belong to the Master mode alone; thereby, they are disabled in the Slave mode. All remaining property groups are shared by both modes.

5.6.2.2 ID and Data Table

Slave ID – In the MODBUS master-slave scheme, one master can communicate with up to 247 slaves, each identified by a unique ID, from 1 through 247. In the Slave mode, the slave ID is to identify itself for communicating with the master device; in the Master mode, the slave ID refers to the ID of slave device with which AEwin communicates. The slave ID is configured by a text box that only accepts integers between 1 and 247 inclusive. If entering a number out of this range, an error message box will pop up saying, “Please enter an integer between 1 and 247”.

Data Table – MODBUS provides four data tables (also known as point types) on the slave side for data transmission: storing received data from the master or put requested data for the master to fetch. The four data tables are divided into two categories: two tables store on/off discrete values (coils or contacts) and two store numerical values (registers). The coils/contacts and registers may have a read-only table or a read/write table. Each table has 9999 (270F in hexadecimal) values. Each coil or contact is 1 bit and assigned a data address between 0001 and 9999. Each register is 1 word (i.e., 16 bits or 2 bytes) and also has a data address between 0001 and 9999. The table below summarizes the four MODBUS data tables.

Table Name	Type	Segment Address	Offset Address
Discrete Output Coils	R/W	00000	1 – 9999
Discrete Input Contacts	R	10000	1 – 9999
Analog Input Registers	R	30000	1 – 9999
Analog Output Holding Registers	R/W	40000	1 – 9999

From this table, the segment address is uniquely determined by the data table type. Once the data table type is selected here, only the offset addresses of individual cells need to be configured, which is to be done in column “Address” on the first tab page “MODBUS Channel Setup” that is described above. The data table type is configured by the combo box right below the “Slave ID” text box.

5.6.2.3 Connection

Start Address – As aforementioned in the previous section, “ID and Data Table,” the address of all data tables ranges from 1 to 9999. Usually, an application only uses a small part of it, and the starting address is not necessarily 1. In addition, the used data block is defined on the Slave side, and this data block must be known to the Master side because any data block the Master reads and writes must be within this range. The starting address of the data block defined on the Slave side is configured by the “Start Address” text box.

Note that the starting address of the data block may not be the smallest address among all channels (in column “Address” of dialog page “MODBUS Channel Setup”) or all enabled channels because any part of the defined data block may not be used. However, all addresses in column “Address” must be no less than the start address. If this rule is violated, an error message box will pop up upon clicking the “OK” or “Apply” button, saying

“Address must be no less than the Start Address!
Please check the cyan cells and Start Address on MODBUS Property Setup page”.

Note also that the smallest possible starting address is 1 rather than zero, and the highest starting address must be no greater than 10000 minus the data block size. For example, if the data block size is 100, then the starting address must be smaller than or equal to 9900. Otherwise, an error message will pop up when trying to validate the setup.

MODBUS Type – The MODBUS module supports two standard MODBUS versions: MODBUS serial lines (including MODBUS RTU and MODBUS ASCII) and MODBUS TCP. If the serial type is used, AEwin is connected to the device with serial lines (RS-232, RS485; the serial type is set up in the next group); if MODBUS TCP is used, the two are connected with an Ethernet cable. The MODBUS type is configured by a combo box that has two options, and its value is Serial Lines by default.

Note: If MODBUS TCP/IP is selected, all serial communication-related setup groups on this dialog will be automatically disabled; likewise, if MODBUS serial type is selected, the “TCP/IP Server” setup group will be automatically disabled.

5.6.2.4 Serial Port

This group is enabled only when the MODBUS serial type is selected. It consists of six combo boxes: Port #, Media Layer, Baud Rate, Data Bits, Parity, and Stop Bits. These are all common serial port parameters, and they are summarized in the table below:

Property	# of Options	Options	Default
Port #	10	COM1 – COM10	COM1
Media Layer	2	RS – 232, RS – 485	RS – 232
Baud Rate	7	300, 600, 1200, 2400, 4800, 9600, 19200	9600
Data Bits	2	7, 8	8
Parity	3	No, Odd, Even	No
Stop Bits	3	1, 1.5, 2	1

Note 1) The Data Bits dialog box is not configurable because it is determined by the Transmission Mode configuration that is to be addressed next: 7-bit is associated with ASCII transmission mode, and 8-bit is associated with RTU transmission mode. If a switch happens from one transmission mode to another, the number displayed in this disabled textbox will automatically change.

Note 2) All AEwin settings of the serial port must match the same settings on the MODBUS device connected to AEwin. Otherwise, the serial connection cannot be established.

5.6.2.5 Transmission Mode

MODBUS defines two transmission modes, RTU (Remote Terminal Unit) and ASCII, for serial communications. In the RTU mode, one byte of information is transmitted as an 8-bit unit, while in the ASCII mode, the same information is transmitted as TWO 7-bit ASCII characters. For example, the number 254 is transmitted as 1111110B in RTU, but as FE in ASCII, which sends the character ‘F’ and then the character ‘E.’ In other words, the efficiency of RTU mode is about twice that of ASCII mode. For this reason, RTU transmission mode is the default mode in AEwin. The “Transmission Mode” setup dialog is a pair of radio buttons. A flip of transmission mode will trigger an instant change in the Data Bits box in the Serial Port group above.

5.6.2.6 Flow Control

AEwin MODBUS module supports four types of flow controls in serial communications:

- 1) DSR - Data Set Ready
- 2) CTS - Clear To Send
- 3) DTR - Data Terminal Ready
- 4) RTS - Request to Send

In general, the RTS and CTS are turned off and on from alternate ends to control data flow, for instance, when a buffer is almost full. DTR and DSR are usually on all the time and are used to signal from each end that the other equipment is actually present and powered up.

Four check boxes are used to set up the MODBUS serial flow control. They are all unchecked by default. The setup is straightforward and simple; just make sure that the setup must be the same as setting up on the MODBUS device.

RTS disable delay is configured by a text box with a default value of 100 (ms).

5.6.2.7 Error/Exception

Skip Response, Insert RC Error & Busy – “Skip Response” and “Busy” are reserved for future use. “Insert RC” determines the redundancy check to the transmission/response of MODBUS serial communication. To guarantee the correctness of information transmission, the redundancy check is available in the MODBUS communications. In the ASCII transmission mode, the longitudinal redundancy check (LRC) is used, and in the RTU transmission mode, the circular redundancy check (RCR) is used.

Response Delay – It is the maximal delay allowed before responding to a command from the MODBUS master device. Its default value is 400 (ms). The text box accepts up to four digits, and an upper limit of 1000 (ms) is enforced. Values above 1000 will be rejected with an error message box when the “OK” button of the MODBUS dialog window is clicked.

5.6.2.8 Response Timeout

When the MODBUS master sends out a command to a certain slave, it expects the slave to respond to the command within a specified amount of time; otherwise, response times out, so the master has to send the command again or generates an exception. This is called a response timeout. The text box accepts up to four digits, and the value must be no greater than 1000 (ms). The default is 200. Values above 1000 will be rejected with an error message box when the “OK” button of the MODBUS dialog window is clicked.

5.6.2.9 Delay Between Polls

Delay between polls is the minimal amount of time the master has to wait to issue the next poll after the previous poll is issued. It is a parameter to regulate the master polls. This is necessary because the slave device may become overwhelmed with data requests. Another reason is that some media (like radio modems or RS-232/485 converters) take time to turn around from TX to RX. So polling too fast might cause the first few bytes NOT to be transmitted. The text box accepts up to three digits, and an upper limit of 200 (ms) is enforced. The value is 30 by default. Values above 200 will be rejected with an error message box when the “OK” button of the MODBUS dialog window is clicked.

5.6.2.10 TCP/IP Server

In MODBUS, the slave is the server (serving data), and the master is the client; specifically, when the connection is TCP/IP, the slave is the server of the TCP socket, and the master is the socket client. In a typical TCP socket handshaking, the server opens the socket with a port number (502 by default) and waits for the client to connect to it. Keep in mind that the server (MODBUS TCP slave) must open the socket before trying to connect the Master to it.

The client needs the port number and the IP address of the server for identification. In other words, a slave (the server) does not need to set up the IP address; thus, the IP Address text box is disabled for the slave.

IP Address – In convention, the IP address consists of four integer numbers connected by three dots (.); each number is between 0 and 255. The default value of this text box is “0.0.0.0”. IP address input not matching this convention or any of the four numbers out of the range [0 255] is not acceptable: an error message box will pop up.

Port – The default port number is 502 for MODBUS. The text box accepts integer values between 0 and 65535. Out-of-range input is not accepted; otherwise, an error message box will pop up.

Connect Timeout – It is the maximal time allowed for the TCP client to take to successfully connect to the server. If the client tries to connect the server but cannot get the confirmation message in this amount of time, then the connection times out. An error message will be generated, and the client needs to connect again. It is 1000 (ms) by default, and only integers no greater than 1000 are accepted. Otherwise, an error message box will pop up.

5.6.3 MODBUS Alarm Setup

Clicking the third tab, “MODBUS Alarm Setup,” will bring up this dialog box, as shown in the figure below. The data grid view looks like the dialog box on the first tap page, “MODBUS Channel Setup,” but has fewer columns. By referring back to that section for detailed descriptions, the setup for this section will become a lot simpler.

There are 10 types of alarms in the current version of AЕwin supported here: Front End, Graph, LeakTEC, Online Crack, FFT, Multivariate, Cluster, Cluster Rate, System Status, TurnKey, Intensity, ASL, and RMS. There are three general alarm states, Off, Warn and Trip, shared by all alarm types except 1) Cluster Rate and Turnkey do not define “Warn”; 2) System (alarm) does not define “Warn” and “Trip.” Off, Warn, and Trip is represented by 0, 1, and 2, respectively. On the other hand, some alarms have other states. The details of all 10 alarms, including states and their encodings, are summarized in the table below. Although 4 digits (bits) are sufficient to encode the states of all alarms, one holding register, which has 16 bits, is assigned for each alarm, reserving remaining bits for possible future use (e.g., more states are defined for alarms). The encodings are needed to parse the alarm states on the MODBUS device, which receives the alarm data from AЕwin.

Here is a description of the available options in the setup dialog box shown below.

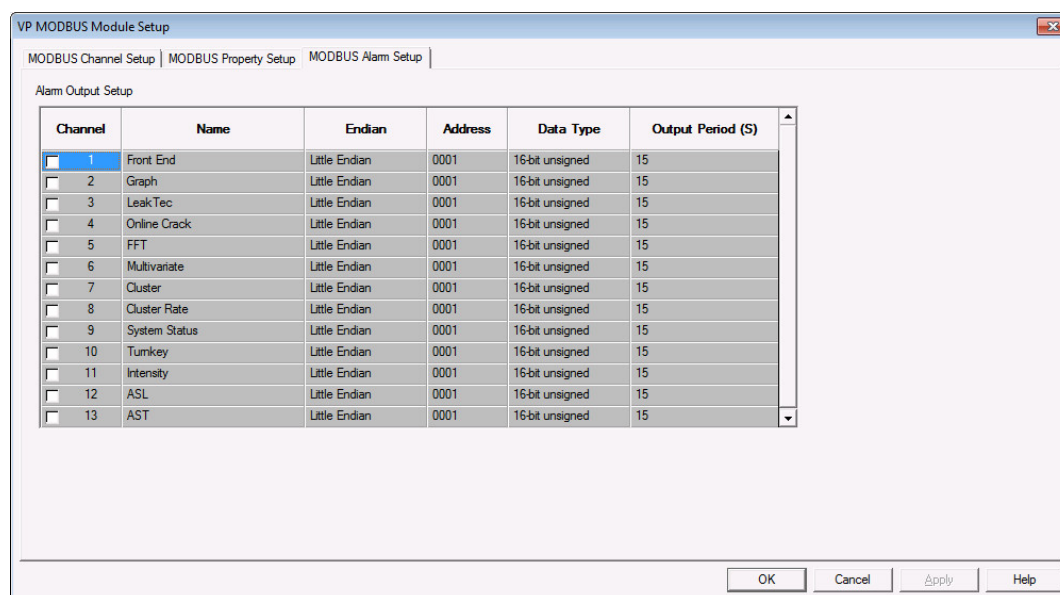


Figure 5-8. MODBUS Alarm Set up

Channel On/Off Toggle – While there are only 10 alarms defined in AEdwin, the data grid view dialog box has 16 rows: the latter six rows are reserved for future use, thereby unchecked and with a general name “unknown” followed by a number. These reserved rows can be used to output other values if necessary. The configuration for outputting values will be treated separately at the end of this section. Uncheck any channels corresponding to the alarms that are not interested. For some applications, it is not necessary to output alarms over MODBUS; in this case, uncheck all channels.

Channel Name – It is not suggested to configure the names of the 10 alarms, but they are configurable if necessary, e.g., if wanting to reorder the alarms or to output values using the reserved channels. Renaming is necessary if any alarms are not needed, but instead, you want to output other parameters' values. The length of the alarm name is limited to 26 characters.

Address – As stated, the holding registers are used for alarm output, so only the offset addresses are to be entered in this column. The address range is 1 – 9999.

Alarm Type	Register Address	Bits Used	Alarm States
Front End	xxxx	0, 1	0000-Off; 0001-Warn; 0010-Trip
Graph	xxxx	0, 1	0000-Off; 0001-Warn; 0010-Trip
LeakTEC	xxxx	0, 1, 2, 3	0000-Off; 0001-Low; 0010-High; 0100-High High; 1000-Sputtering
Online Crack	xxxx	0, 1, 2, 3	0000-Off; 0001-Hit/Event Warn; 0010-Hit/Event Trip; 0100-Energy Trip; 0110-Both Trip; 1000-Any Trip
FFT	xxxx	0, 1, 2, 3	0000-Off; 001-Warn; 0010-Trip; 0100-Low; 1000-Max Number Box
Multivariate	xxxx	0, 1	0000-Off; 0001-Warn; 0010-Trip
Cluster	xxxx	0, 1	0000-Off; 0001-Warn; 0010-Trip
Cluster Rate	xxxx	0, 1	0000-Off; 0001-Warn; 0010-Trip
Intensity	xxxx	0, 1	0000-Off; 0001-Warn; 0010-Trip
*ASL	xxxx to xxxx +31	Depends on channel #, 2 bits/channel	0000-Off; 0001-Low; 0010-High
*AST	xxxx to xxxx +31	Depends on channel #, 2 bits/channel	0000-Ok or not tested 0001-Failed
System Status	xxxx	0, 1, 2, 3	0000-Off; 0001-Heart Beat; 0010-Media Low; 0100-Not in Acquisition; OR on the latter three
Turnkey	xxxx	0, 1	0000-Off; 0001-Warn; 0010-Trip

* ASL and RMS define a block of 32 registers each. Each register holds the result for 8 channels (2 bits/channel). To find the register for a channel, take the channel number, subtract 1, and divide by 8 (channels 1-8 are the lowest numbered register, followed by 9-16, ...). To find the 2 bits in the register for the channel, take the channel number, subtract 1 and take the modulo 8 and multiply that by 2 (channel 2 would be bits 2 and 3, channel 9 would be bits 0 and 1).

Channel	Register	Bits
1	Start+0	XXXX XXXX XXXX XX00
2	Start+0	XXXX XXXX XXXX 00XX
9	Start+1	XXXX XXXX XXXX XX00
10	Start+1	XXXX XXXX XXXX 00XX

Data Type – Because only the four lower bits of a register are used for an alarm, the data type is fixed to unsigned short for the first 10 channels. That is to say, it is unnecessary to configure the data type for the alarms, or the data type configuration has no effect even though a type other than “16-bit unsigned” is selected. The configuration is available because these channels may be used for outputting values other than alarms. Configuring the combo box is described in the section on the first dialog page, “MODBUS Channel Setup.” Please refer back to that section for detailed descriptions.

Output Period – Output period defines how often a channel outputs the alarms or other parameter values over MODBUS. It is in second (S). The configuration is the same as configuring the “Update Period” described in the section on the first dialog page, “MODBUS Channel Setup.” Please refer back to that section for detailed descriptions.

Output Values – The latter six reserved channels can be used to output values of parameters. Any of the first 10 channels can also be used for outputting values if the corresponding alarms are not needed. To do this, check the channels and configure the names, addresses, data types, and output periods the same way as described above.

5.7 Virtual Parametrics Page in Hardware Setup

The Configure Virtual Parametric Modules dialog box can also be accessed thru the Virtual Parametrics tab in Hardware Setup. This tab in Hardware Setup (shown below) will let you adjust the setup of all the virtual parametric modules without having to jump in and out of the individual setup dialog boxes for each module. You cannot adjust all the settings for all modules, but you can control the main ones. You can turn channels on and off, and revise the channel Name, Multiplier, Offset, and Units string. The Module Name is displayed, but it is a read-only column. To change the Module Name, you will have to display the “Configure Virtual Parametric Modules” dialog box and use it to bring up the setup dialog box for the module you want to rename. The Sampling Period also requires accessing the individual setup dialog boxes as some modules have one setting for the whole module, and others allow individual settings for each channel.

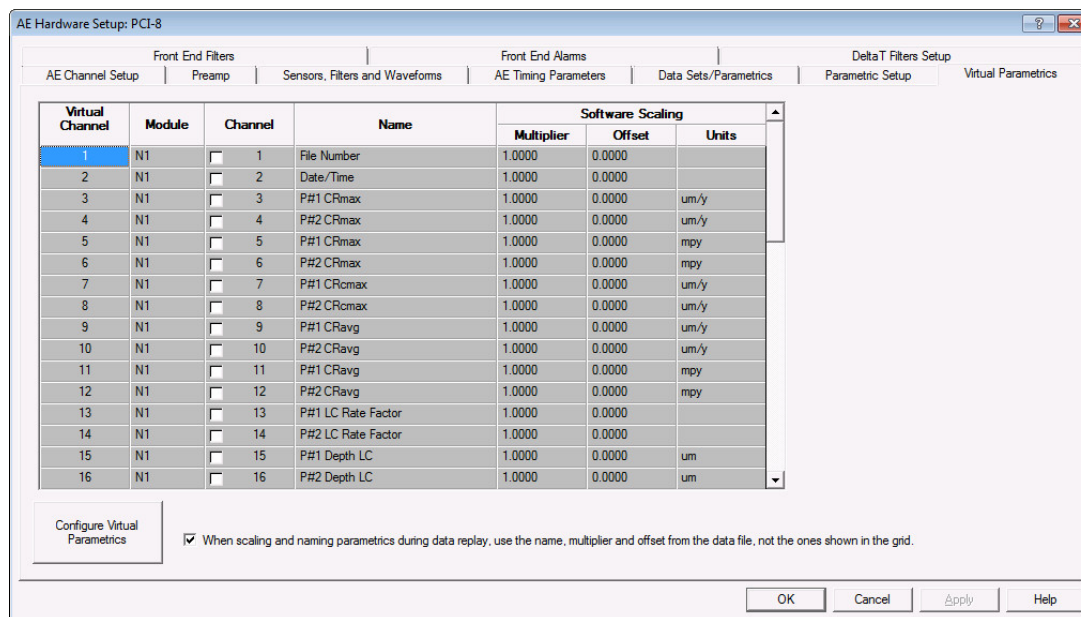


Figure 5-9. Virtual Parametrics Page of Hardware Set up

Note: For compatibility reasons, only the first tab page of the MODBUS Module is displayed in the Hardware Setup window, not including the “Address” and “Data Size” columns because all other virtual parametric modules do not have these two columns. For information on the MODBUS Property Setup and MODBUS Alarm Setup, one needs to click the “Configure Virtual Parametrics” button to bring up the setup dialog box for the MODBUS Module.

There is one setting on this tab that you cannot adjust anywhere else. At the bottom of the tab is a checkbox labeled:

“When scaling parametric values during data replay, use the multiplier and offset from the data file, not the ones shown in the grid.”

During data acquisition, the virtual parametric settings that are in use are stored in the DTA file. This checkbox controls whether you use the settings in your layout file or from the DTA file when replaying the data. This allows you flexibility in the units that are displayed during a replay. If you use the settings from the DTA file, you will see the data in the same format as it was collected. If, however, you want to change some units, you would want to use the settings from your layout file so that you can rescale the data using the Multiplier and Offset values and display the correct units string. If you do not need to change any units, you should normally stick with using the settings in the DTA file so that your data displays properly.

If you use the settings from your layout, you should start with the layout that you used to collect the data because it will have the correct modules already defined. When you create a layout and tell AEwin which modules you are using, it keeps track of the data using a virtual channel number. These are shown in the leftmost column in the grid. They range from one to the total number of virtual parametric channels available for the modules in use. If you are

only using one module, the virtual channel numbers are identical to the module channel numbers. If you are using two or more modules, though, the virtual channel numbers for the 2nd module start where the 1st module channels end.

For example, the Measurement Computing DAS6033 and USB2533 have 64 channels of data available, while the Weather Station has 22. When they are both being used, the virtual channel numbers for the Weather Station start at the end of the 64 from the Measurement Computing module, so channel 1 for the Weather Station corresponds to virtual channel number 65. These are important because you need them to graph virtual parametric data. In this case, if you want to graph channel 1 for the Weather Station, you need to look for Virtual Parametric 65 in Graph Setup. These numbers can also show up in the Line Display. More on this is given in the next section.

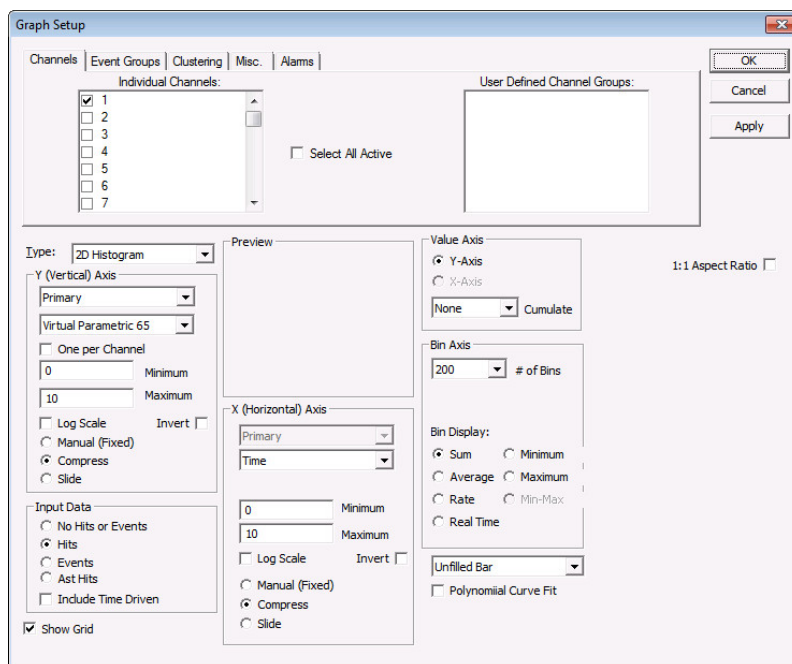


Figure 5-10. Accessing Virtual Parametric in Graph Set up

5.8 Replay of ACMS EOPS DTA Files

ACMS External Operational Parameters (EOPS) are similar to the Virtual Parametrics in AEwin but are referred to as EOPS in ACMS, and they are also acquired by MODBUS. Although the effort was taken to make ACMS DTA files compatible in format with AEwin DTA files, there are still some differences between the two: there are no such terms as Module Name or Module ID in ACMS. For this reason, the ACMS DTA is always called “ACMS” as a module name while being replayed. The following tabulated data gives an example in ACMS DTA file replay in the Line Display window. Except for this difference in line display, there is no difference in operating the replay of ACMS DTA: Press “F10” and choose an ACMS DTA in the dialog window and click “OK.”

Note: Make sure the checkbox with the following label on “VParametrcs” tab of the Hardware Setup” dialog box is checked before the replay:

“When scaling parametric values during data replay, use the multiplier and offset from the data file, not the ones shown in the grid.”

9 0 00:00:14.9999000	ACMS:GEN RPM 1	= 37568.0000	RPM	ACMS:GEN RPM 2	= 16813.0508	RPM
	ACMS:Wind Speed	= 2516.0000	C	ACMS:Outside Temp	= 9589.0000	C
9 0 00:00:29.9999000	ACMS:GEN RPM 1	= 10940.0000	RPM	ACMS:GEN RPM 2	= 24223.4512	RPM
	ACMS:Wind Speed	= 2898.0000	C	ACMS:Outside Temp	= 8888.0000	C
9 0 00:00:44.9999000	ACMS:GEN RPM 1	= 28544.0000	RPM	ACMS:GEN RPM 2	=149079.2813	RPM
	ACMS:Wind Speed	= 3430.0000	C	ACMS:Outside Temp	= 20500.0000	C
9 0 00:00:59.9999000	ACMS:GEN RPM 1	= 6544.0000	RPM	ACMS:GEN RPM 2	= 15896.1709	RPM
	ACMS:Wind Speed	= 40050.0000	C	ACMS:Outside Temp	= 13452.0000	C
9 0 00:01:14.9999000	ACMS:GEN RPM 1	= 8060.0000	RPM	ACMS:GEN RPM 2	= 17428.4902	RPM
	ACMS:Wind Speed	= 57340.0000	C	ACMS:Outside Temp	= 48090.0000	C
9 0 00:01:29.9999000	ACMS:GEN RPM 1	= 8980.0000	RPM	ACMS:GEN RPM 2	= 6529.5503	RPM
	ACMS:Wind Speed	= 1890.0000	C	ACMS:Outside Temp	= 59487.0000	C
9 0 00:01:44.9999000	ACMS:GEN RPM 1	= 111788.0000	RPM	ACMS:GEN RPM 2	= 62556.5703	RPM
	ACMS:Wind Speed	= 19200.0000	C	ACMS:Outside Temp	= 11524.0000	C

5.9 Line Display & Virtual Parametric Data

To get virtual parametric data to show up in the Line Display, you will need to turn on both the Line Display and make sure virtual parametric is being output. Use the “Line Display Setup” dialog box shown below. Check “Enable Line Display” in the upper left-hand corner to turn on the line display.

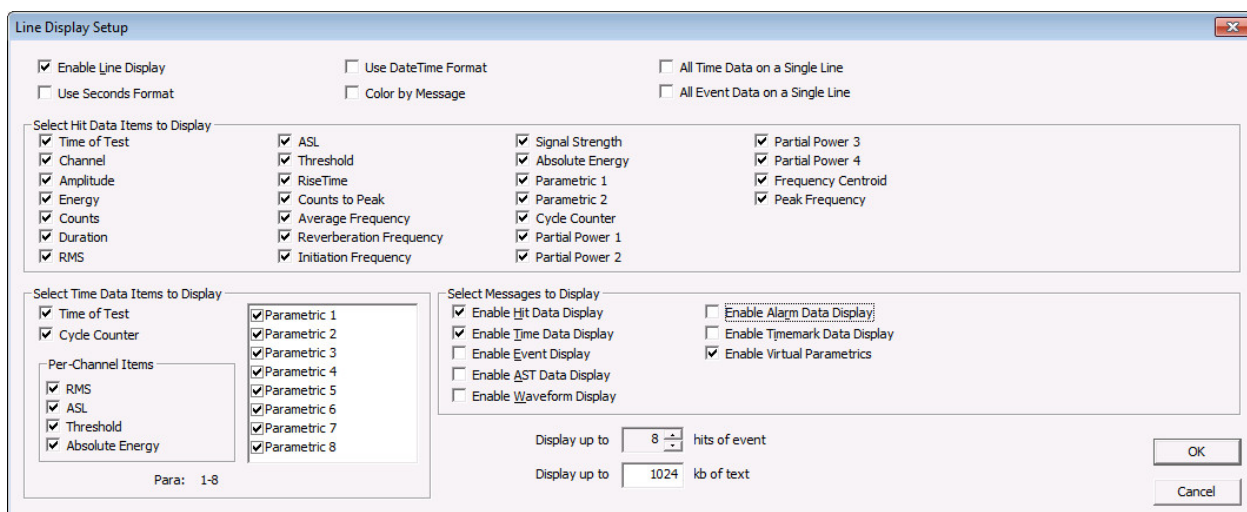


Figure 5-11. Enabling Virtual Parametrics in Line Display Set up

Check “Enable Virtual Parametrics” in the “Select Messages to Display” section in the lower right-hand side of the dialog box. Note there is a difference between the line display being on and being visible. You can have it on but not have it visible. You can use the F7 key to show or hide the Line Display window itself. When you replay a DTA file with virtual parametric data in it, it will show up in the Line Display as message ID 9. Here is a sample for data collected from a NanoCorr instrument:

```

9 0 00:00:18.2061840
N1:P#1 CRmax = 0.3219 um/y N1:P#2 CRmax = 0.1372 um/y
N1:P#1 CRmax = 0.0127 mpy N1:P#2 CRmax = 0.0054 mpy
N1:P#1 CRCmax = 0.3219 um/y N1:P#2 CRCmax = 0.1372 um/y
N1:P#1 CRAvg = 0.0522 um/y N1:P#2 CRAvg = 0.0207 um/y
N1:P#1 CRAvg = 0.0021 mpy N1:P#2 CRAvg = 0.0008 mpy
N1:P#1 LC Rate Factor = 6.1660 N1:P#2 LC Rate Factor = 6.6439
N1:P#1 Depth LC = 0.0000 um N1:P#2 Depth LC = 0.0000 um
N1:P#1 Depth Avg = 0.0000 um N1:P#2 Depth Avg = 0.0000 um
N1:P#1 LC Depth Factor = 1.8404 N1:P#2 LC Depth Factor = 1.2792
N1:P#1E = 0.0013 V N1:P#1E = 0.0013 V
N1:Temperature = 24.8901 oC N1:pH = 7.0001
N1:ORP = 0.0072 V N1:Remarks = 0.0000
N1:P#1I Offset = 18.0000 pA N1:P#2I Offset = -170.0000 pA
N1:P#1E#1 = -75.0000 pA N1:P#1E#2 = -217.0000 pA
N1:P#1E#3 = -10.0000 pA N1:P#1E#4 = 61.0000 pA
N1:P#1E#5 = 16.0000 pA N1:P#1E#6 = -13.0000 pA
N1:P#1E#7 = 130.0000 pA N1:P#1E#8 = 83.0000 pA
N1:P#1E#9 = 26.0000 pA N1:P#2E#1 = 30.0000 pA
N1:P#2E#2 = -99.0000 pA N1:P#2E#3 = 30.0000 pA
N1:P#2E#4 = -7.0000 pA N1:P#2E#5 = -27.0000 pA
N1:P#2E#6 = 10.0000 pA N1:P#2E#7 = 2.0000 pA
N1:P#2E#8 = 50.0000 pA N1:P#2E#9 = 10.0000 pA

```

Figure 5-12. Sample Line Display Output for NanoCorr

If you replay a DTA file with virtual parametric data in it and choose to scale the data using the settings in the layout file but are not using the same virtual parametric modules as being used to collect the data, the software may not be able to correctly display the virtual parametric data. Here is the same NanoCorr data as above replayed with a layout that did not have any virtual parametric modules defined to illustrate how that looks:

```

9  0 00:00:18.2061840
   ??:vch1 = 202.0000    ??:vch2 = 0.0000
   ??:vch3 = 0.3219     ??:vch4 = 0.1372
   ??:vch5 = 0.0127     ??:vch6 = 0.0054
   ??:vch7 = 0.3219     ??:vch8 = 0.1372
   ??:vch9 = 0.0522     ??:vch10 = 0.0207
   ??:vch11 = 0.0021    ??:vch12 = 0.0008
   ??:vch13 = 6.1660    ??:vch14 = 6.6439
   ??:vch15 = 0.0000    ??:vch16 = 0.0000
   ??:vch17 = 0.0000    ??:vch18 = 0.0000
   ??:vch19 = 1.8404    ??:vch20 = 1.2792
   ??:vch21 = 0.0013    ??:vch22 = 0.0013
   ??:vch23 = 24.8901   ??:vch24 = 7.0001
   ??:vch25 = 0.0072    ??:vch26 = 0.0000
   ??:vch27 = 18.0000   ??:vch28 = -170.0000
   ??:vch29 = -75.0000  ??:vch30 = -217.0000
   ??:vch31 = -10.0000  ??:vch32 = 61.0000
   ??:vch33 = 16.0000   ??:vch34 = -13.0000
   ??:vch35 = 130.0000  ??:vch36 = 83.0000
   ??:vch37 = 26.0000   ??:vch38 = 30.0000
   ??:vch39 = -99.0000  ??:vch40 = 30.0000
   ??:vch41 = -7.0000   ??:vch42 = -27.0000
   ??:vch43 = 10.0000   ??:vch44 = 2.0000
   ??:vch45 = 50.0000   ??:vch46 = 10.0000

```

Figure 5-13. Line Display Output Without Any Virtual Parametrics Modules Defined.

Notice that the Module Name is displayed as “??” and the Channel Name is VCh#n, where #n is the virtual channel number. The readings are the raw unscaled values from the DTA file. You can also see a display like this if you rely on the settings in the DTA file, and it is an older DTA file that does not have the virtual parametric settings stored in it. In this case, you just need to define your virtual parametric modules and change to scaling the data using the settings in the layout.

The number of channels displayed will allow you to guess at the virtual parametric modules in use when the data was collected because the total number of channels differs for each type of virtual parametric data. The Measurement Computing boards have 64 channels of data, the NanoCorr has 46, the Weather Station 22, the MODBUS module 48, and the Strain Gauge Module has only 4 channels. If you were to mistakenly use a Weather Station module and replay NanoCorr data, you would get a display like the following one where some channels are defined, and some are not:

```

9  0 00:00:18.2061840
   w1:Wind Speed Min      = 202.0000 m/s      w1:Wind Speed Avg      = 0.0000 m/s
   w1:Wind Speed Max      = 0.3219 m/s       w1:Wind Direction Min  = 0.1372 deg
   w1:Wind Direction Avg  = 0.0127 deg       w1:Wind Direction Max  = 0.0054 deg
   w1:Air Pressure        = 0.3219 bar        w1:Air Temperature    = 0.1372 C
   w1:Internal Temperature = 0.0522 C        w1:Humidity           = 0.0207 %RH
   w1:Rain Accumulation   = 0.0021 mm        w1:Rain Duration      = 0.0008 sec
   w1:Rain Intensity      = 6.1660 mm/hr     w1:Rain Peak Intensity = 6.6439 mm/hr
   w1:Hail Accumulation   = 0.0000 hits/cm2   w1:Hail Duration      = 0.0000 sec
   w1:Hail Intensity      = 0.0000 hits/cm2h  w1:Hail Peak Intensity = 0.0000 hits/cm2h
   w1:Heating Temperature = 1.8404 C        w1:Heating Voltage    = 1.2792 V
   w1:Supply Voltage      = 0.0013 V        w1:Reference voltage  = 0.0013 V
   ??:vch23               = 24.8901         ??:vch24              = 7.0001
   ??:vch25               = 0.0072         ??:vch26              = 0.0000
   ??:vch27               = 18.0000         ??:vch28              = -170.0000
   ??:vch29               = -75.0000        ??:vch30              = -217.0000
   ??:vch31               = -10.0000        ??:vch32              = 61.0000
   ??:vch33               = 16.0000         ??:vch34              = -13.0000
   ??:vch35               = 130.0000        ??:vch36              = 83.0000
   ??:vch37               = 26.0000         ??:vch38              = 30.0000
   ??:vch39               = -99.0000        ??:vch40              = 30.0000
   ??:vch41               = -7.0000         ??:vch42              = -27.0000
   ??:vch43               = 10.0000         ??:vch44              = 2.0000
   ??:vch45               = 50.0000         ??:vch46              = 10.0000

```

Figure 5-14. Line Display Output Without All Virtual Parametrics Modules Defined.

Anytime you see virtual parametric data in the Line Display showing “??” for the Module Name and virtual channel numbers, you know that your virtual parametric setup is incorrect. Just modify the modules in use in your layout and replay the data again till you get the number of channels defined in the layout to match the data in the DTA file. Most often, the simplest solution to these problems is to display the data using the settings stored in the DTA file. As stated earlier, though, this does not work for early version files that do not contain the virtual parametric settings used to collect the data.

6. OPTIONS FOR AEwin

6.1 Supervisor Option

The 'Supervisor Option' allows the user to restrict access to the software by the use of a password. With this option installed, the software will start up in **Operator Mode** by default.

In **Operator Mode**, the user has limited access to the system. He can:

- Load a layout file.
- Print/Export graphs or copy them to the clipboard.
- Change the Test Title.
- Edit the Logistics Information.
- Setup the Audio Monitor (if present).
- Perform a test or AST.
- Replay an existing data file.
- Toggle the Line Listing Display on/off.

The user can switch to supervisor mode by clicking 'Switch to Supervisor Mode' from the File menu and entering the password.

In **Supervisor Mode**, the user has full access to the system. He can perform all the functions possible in operator mode, plus he can:

- Change the password.
- Create and save layout files.
- Access the various Test Setup menus.
- Create, modify and delete graphs.
- Modify the arrangement of the graphs, create and delete pages, etc.

How to set the password the first time: The first time the user tries to switch to supervisor mode, the software will ask for an initial password. Enter a password of your choice, which must be 4-8 alphanumeric characters in length. It is not case-sensitive. From this point on, the user must enter this password to switch to supervisor mode.

Note: If your system came with AEwin pre-installed and the program doesn't request an initial password, this means that the initial password was set at the factory. The password, in this case, is 'AEwin.'

How to change the password: The user can change the password by switching to supervisor mode and clicking 'Change Password' from the file menu.

What to do if you forget the password: If you forget the password, please contact Customer Service for assistance.

6.2 LeakTEC Alarms

LeakTEC Alarms (part of the 'LeakTEC/On-Line-Crack Alarm' Option) are used to monitor ASL or RMS levels (using Time Data) during acquisition and alert the user if they stray beyond certain pre-set lower and upper limits. If ASL or RMS levels stray beyond these limits for a certain period of time, an Alarm is generated. Additionally, 'Sputtering Alarms' are available to monitor hit-based noise.

6.2.1 Setup

If you have the 'LeakTEC/On-Line-Crack Alarm' Option installed, then you can enable and customize these alarms by using the LeakTEC Alarm Setup dialog box located on the Acquisition Setup menu. Its controls are described below.

The LeakTEC Alarm Setup dialog box contains the following elements:

Channel	(Warning) Low Alarm ≤ Level	(Warning) High Alarm ≥ Level	(Alarm Trip) High High ≥ Alarm Level	Sputtering Hits Amplitude Threshold	(Alarm Trip) # Sputtering Hits ≥
☒ 1	0.0000	0.0000	0.0000	70	40
☐ 2	0.0000	0.0000	0.0000	70	40
☐ 3	0.0000	0.0000	0.0000	70	40
☐ 4	0.0000	0.0000	0.0000	70	40
☐ 5	0.0000	0.0000	0.0000	70	40
☐ 6	0.0000	0.0000	0.0000	70	40
☐ 7	0.0000	0.0000	0.0000	70	40
☐ 8	0.0000	0.0000	0.0000	70	40
☐ 9	0.0000	0.0000	0.0000	70	40
☐ 10	0.0000	0.0000	0.0000	70	40
☐ 11	0.0000	0.0000	0.0000	70	40
☐ 12	0.0000	0.0000	0.0000	70	40
☐ 13	0.0000	0.0000	0.0000	70	40
☐ 14	0.0000	0.0000	0.0000	70	40
☐ 15	0.0000	0.0000	0.0000	70	40
☐ 16	0.0000	0.0000	0.0000	70	40

Below the table, there are settings for:

- Low Alarm Hold-off: 10 [msec/sec] (radio buttons)
- High Alarm Hold-off: 10 [msec/sec] (radio buttons)
- High High Alarm Hold-off: 10 [msec/sec] (radio buttons)
- Rain Channel (For High/High-High Alarms): ☐ Enable, 1 [dropdown]
- Sputtering Alarm Monitor Interval: 10 [msec/sec] (radio buttons)
- ☐ Disable Conditional Auto-Reset of High-High Alarms
- ☐ Disable Automatic Popup of Alarm Display Dialog
- ☐ Suppress Quick Retriggering of Manually Cleared Alarms (When enabled, LeakTEC alarms that are cleared by the user won't trigger again until the condition that triggered them subsides.)
- Alarm Reset Setup button
- Alarm Type: ASL [dropdown]
- OK and Cancel buttons

Figure 6-1. LeakTEC Alarm Set up

Settings Common to Both ASL/RMS Alarms and to Sputtering Alarms:

- **Channel** – Alarm monitoring for each channel can be enabled and customized independently. Click a channel's checkbox to enable alarm monitoring for that channel.
- **Disable Automatic Popup of Alarm Display Dialog** – Check this box to prevent the Alarm Display popup box from being displayed automatically when a LeakTEC alarm is triggered. The dialog can still be opened manually.
- **Suppress Quick Retriggering of Manually Cleared Alarms** – When enabled, LeakTEC alarms that are cleared (reset) by the user won't trigger again until the condition that triggered them subsides.
- **Alarm Reset Setup** – Click this button to enter the 'Alarm Preferences' dialog box, where the 'Auto Alarm-Reset' settings can be customized. These can be used to set the system to automatically clear the alarms periodically or when the condition(s) that triggered the alarm(s) subside.

Settings Specific to ASL/RMS Alarms:

- Low Alarm – Set the ‘Low Alarm’ Level for a channel in this cell. An alarm ‘warning’ is triggered if ASL/RMS levels drop to this level or below for a length of time equal to the ‘Low Alarm Hold-off’ period.
- High Alarm – Set the ‘High Alarm’ Level for a channel in this cell. An alarm ‘trip’ is triggered if ASL/RMS levels equal or exceed this level for a length of time equal to the ‘High Alarm Hold-off’ period.
- High-High Alarm – Set the ‘High-High Alarm’ Level for a channel in this cell. An alarm ‘trip’ is triggered if ASL/RMS levels equal or exceed this level for a length of time equal to the ‘High-High Alarm Hold-off’ period.
- Low Alarm Hold-off – This is the period of time that a channel’s ASL/RMS must remain at or below the Low Alarm Level before a Low alarm ‘warning’ is triggered.
- High Alarm Hold-off – This is the period of time that a channel’s ASL/RMS must remain at or above the High Alarm Level before a High alarm ‘trip’ is triggered.
- High-High Alarm Hold-off – This is the period of time that a channel’s ASL/RMS must remain at or above the High-High Alarm Level before a High-High alarm ‘trip’ is triggered.
- Rain Channel – This allows you to enable and select a ‘Rain Channel.’ The purpose of this is to help the system ignore ASL/RMS noise caused by environmental sources. See the section below for more information on how this works.
- Disable Conditional Auto-Reset of High-High Alarms – Check this box to prevent High-High alarms from being automatically reset by the Conditional Auto-Alarm Reset feature.
- ASL/RMS – Select which Time Dataset feature to monitor using this control.

Settings Specific to Sputtering Alarms: (See the section below for more info on Sputtering Alarms)

- Sputtering Hits Amplitude Threshold – Set the amplitude threshold for hits processed by Sputtering Alarm functionality.
- # Sputtering Hits – Set the number of hits that must meet the Amplitude Threshold within the Interval to trigger a Sputtering Alarm.
- Sputtering Alarm Monitoring Interval – This is the period of time in which hits (passing the threshold) are accumulated towards a Sputtering alarm.

6.2.2 Low, High and High-High Alarm Operation

If the ASL/RMS for a particular channel violates an alarming level (low, high, or high-high) and remains continuously in violation of that level for a duration of time equal to the appropriate ‘Hold-Off,’ then an alarm is triggered.

6.2.3 Sputtering Alarm Operation

Sputtering alarms work differently than Low, High, and High-High alarms. These are more similar to On-Line Crack alarms (described in a section below) in that they measure a hit rate. If a certain number (or more) hits that meet/exceed a specified threshold are recorded within a specified interval of time, a “Sputtering” alarm is triggered.

6.2.4 How the ‘Rain Channel’ Works

The use of a ‘Rain Channel’ is intended to reduce the number of High and High-High alarms erroneously triggered by environmental factors such as rain. Typically the sensor used as the rain channel is placed in an exposed position near to the other sensors. If it triggers an alarm, then alarms of the same type from other channels are prevented.

- A ‘High’ alarm on the Rain Channel suspends all other channels from triggering High alarms until the High alarm on the Rain Channel is cleared.
- A ‘High-High’ alarm on the Rain Channel likewise suspends all other channels from triggering High-High alarms until the High-High alarm on the Rain Channel is cleared.

Note that Sputtering Alarms are not affected by Rain Channel functionality.

6.2.5 What happens when an alarm is triggered

- 1) An alarm message (16) is generated. If the auto dump is on, then this message is saved to the datafile. If the line display is active, then it will be displayed there with the time it was triggered and the conditions that triggered it.
- 2) The Alarm Display dialog box is opened (or updated if already open). The details of the alarm and the conditions that triggered it are shown. The alarm can be cleared by clicking the 'Clear Alarms' button.
- 3) If the alarm is an alarm 'warning' then the Alarm Warning output line is set high. If the alarm is an alarm 'trip,' then both the Alarm Warning and Alarm Trip lines are set high.
- 4) An audible alarm is triggered if you have an audio board installed and enabled.

6.3 On-Line Crack Alarms

On-Line Crack Alarms (part of the 'LeakTEC/On-Line-Crack Alarm' Option) are used to monitor hits or events during acquisition and alert the user if the hit/event or energy rates exceed certain pre-set levels. If these rates exceed these limits within a certain period of time, an Alarm is generated.

6.3.1 Setup

If you have the 'LeakTEC/On-Line Crack Alarm' Option installed, then you can enable and customize these alarms by using the On-Line Crack Alarm Setup dialog box located on the Acquisition Setup menu. Its controls are described below.

Channel	Hits/Events Amplitude Threshold	(Warning) Hits/Events	(Trip) Hits/Events	(Trip) Energy	(Trip) Logic Condition
☒ 1	70	15	40	60000	AND
☐ 2	70	15	40	60000	AND
☐ 3	70	15	40	60000	AND
☐ 4	70	15	40	60000	AND
☐ 5	70	15	40	60000	AND
☐ 6	70	15	40	60000	AND
☐ 7	70	15	40	60000	AND
☐ 8	70	15	40	60000	AND
☐ 9	70	15	40	60000	AND
☐ 10	70	15	40	60000	AND
☐ 11	70	15	40	60000	AND
☐ 12	70	15	40	60000	AND
☐ 13	70	15	40	60000	AND
☐ 14	70	15	40	60000	AND
☐ 15	70	15	40	60000	AND
☐ 16	70	15	40	60000	AND
☐ 17	70	15	40	60000	AND
☐ 18	70	15	40	60000	AND

Alarm Parameter: ☒ Hits ☐ Events

Warning Alarm Monitor Interval: msec ☒ sec

Trip Alarm Monitor Interval: msec ☒ sec

OK Cancel

Figure 6-2. On-Line Crack Alarm Set up

'Hits/Events' alarm will be triggered if at least this many hits/events that meet the amplitude threshold are acquired within a period of time not exceeding the Trip Alarm Monitor Interval. If the logic condition is 'ENERGY,' then this alarm type is disabled and will not trigger. If the logic condition is 'AND,' then the Trip Energy condition must be met as well before an alarm will be triggered.

[Trip] Energy – A 'Trip Energy' alarm will be triggered if the total energy of all hits/events (with any amplitude) acquired within a period of time not exceeding the Trip Alarm Monitor Interval meets or exceeds this value. If the logic condition is 'Hits/Events,' then this alarm type is disabled and will not trigger. If the logic condition is 'AND,' then the 'Trip Hits/Events' condition must be met as well before an alarm will be triggered.

[Trip] Logic Condition – The logic condition controls how the Trip alarms are calculated. If it is set to 'Hits/Events,' then the 'Trip Energy' condition is ignored, and no alarms will be triggered using it. If it is set to 'Energy,' then the 'Trip Hits/Events' condition is ignored, and no alarms will be triggered using it. If it is set to 'AND,' then both the 'Trip Hits/Events' and 'Trip Energy' conditions must be met before an alarm is triggered. If it is set to 'OR,' then meeting either condition will trigger a Trip alarm.

Warning Alarm Monitor Interval – This is the period of time in which hits/events (passing the threshold) are accumulated towards a Warning alarm.

Trip Alarm Monitor Interval – This is the period of time in which hits/events (passing the threshold) and energy are accumulated towards a Trip alarm.

Alarm Parameter – Select whether to use hits or the first hit of events for alarm determination.

Channel – Alarm monitoring for each channel can be enabled and customized independently. Click a channel's checkbox to enable alarm monitoring.

Hits/Events Amplitude Threshold – This is the amplitude that a hit (or first hit of an event) must equal or exceed in order to be counted towards a 'Warning Hit/Events' or 'Trip Hit/Events' alarm. This does not affect energy alarms.

[Warning] Hits/Events – A 'Hits/Events Warning' alarm will be triggered if at least this many hits/events that meet the Amplitude Threshold are acquired within a period of time not exceeding the Warning Alarm Monitor Interval.

[Trip] Hits/Events – A 'Trip

Alarm Reset Setup – Click this button to enter the ‘Alarm Preferences’ dialog box, where the ‘Auto Alarm-Reset’ settings can be customized. These can be used to set the system to automatically clear the alarms periodically or when the condition(s) that triggered the alarm(s) subside.

6.3.2 Operation

Four kinds of alarms can be generated by this option.

- Warning Hits/Event – A number of hits/events that met the Amplitude Threshold were acquired within a period of time not exceeding the Warning Alarm Monitor Interval. The number of hits/events met or exceeded the [Warning] Hits/Events value.
- Trip Hits/Event – A number of hits/events that met the Amplitude Threshold were acquired within a period of time not exceeding the Trip Alarm Monitor Interval. The number of hits/events met or exceeded the [Trip] Hits/Events value.
- Trip Energy – The total energy of all hits/events (with any amplitude) that were acquired within a period of time not exceeding the Trip Alarm Monitor Interval met or exceeded the [Trip] Energy value.
- Trip Hits/Event and Energy – Both the ‘Trip Hits/Event’ and ‘Trip Energy’ conditions were met.

6.3.3 What happens when an alarm is triggered

- 1) An alarm message (16) is generated. If the auto dump is on, then this message is saved to the datafile. If the line display is active, then it will be displayed there with the time it was triggered and the conditions that triggered it.
- 2) The Alarm Display dialog box is opened (or updated if already open). The details of the alarm and the conditions that triggered it are shown. The alarm can be cleared by clicking the ‘Clear Alarms’ button.
- 3) If the alarm is an alarm ‘warning’ then the Alarm Warning output line is set high. If the alarm is of the ‘trip’ variety, then both the Alarm Warning and Alarm Trip lines are set high.
- 4) An audible alarm is triggered if you have an audio board installed and enabled.

6.4 Intensity Option

The readme file discusses the Intensity Analysis option of the AEwin software. If you are unsure if you have this license option, look for the Options dialog box under the Help menu. Intensity Analysis is option #39. If it's not checked, you won't be able to access the features described in this file. You can contact customer service (customerservice@mistrasgroup.com) if you decide you need this option.

With the Intensity Analysis option, you can create intensity charts, graphs of historic index or severity vs. time (or any other parameter), and customize the calculations & the charts for your preferred applications. You can create as many intensity charts & intensity-related calculations as you want. There are sample layouts shipped with the software. Look for Intensity.lay in the example layout directory. Configuring intensity charts or simple plots like Severity vs. Time are actually very similar. All of the settings common to the calculations are on the Intensity tab at the top of Graph Setup. Graph Setup is shown below with the Intensity Chart as the selected graph type.

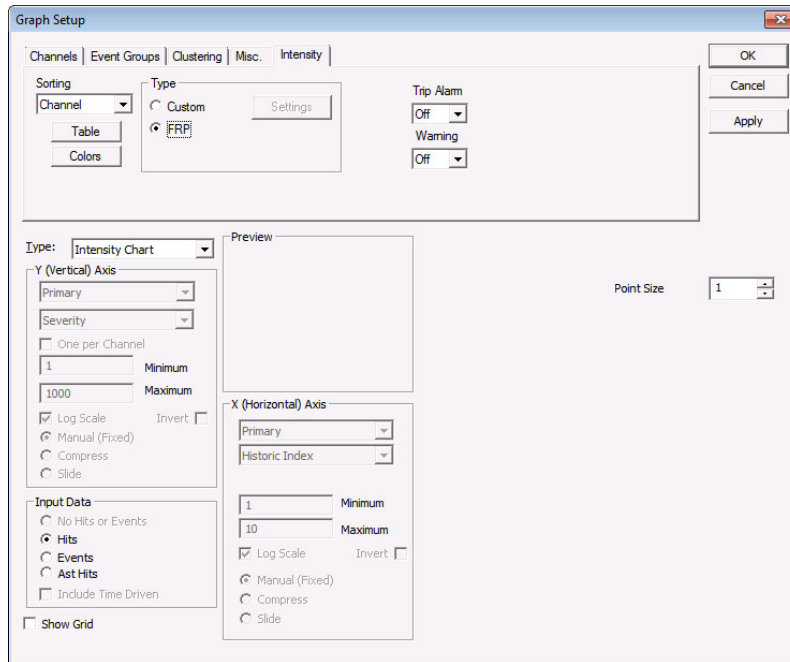


Figure 6-3. Intensity Set up

You may choose between a built-in chart for testing fiber-reinforced plastic (FRP) storage tanks & a custom chart of your own design. More details on custom charts are available in the Custom Intensity Chart section. Sample charts are shown below.

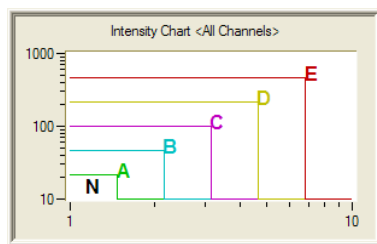


Figure 6-4. Custom

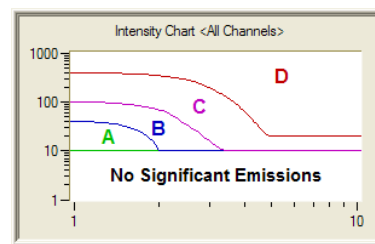


Figure 6-5. FRP

You can change the color associated with each grade level on the Intensity Chart using the FRP Intensity Colors or Custom Intensity Colors options on the Colors submenu of the Graphing dropdown menu shown to the right. You can also use the Colors button on the Intensity tab of the Graph Setup dialog box shown above.

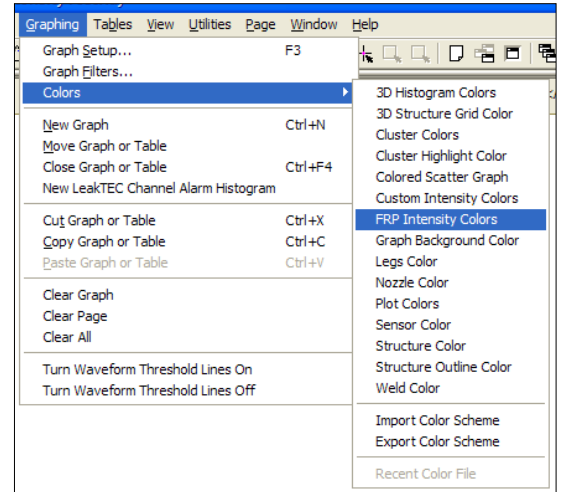


Figure 6-6. Color Menu

Both the Colors button & the Colors menu options lead you to the small dialog shown to the right. Simply double-click one of the colors in the list to change it. A standard Windows color dialog will appear to allow you to select a new color. Note that there are separate sets of colors for the FRP charts & the custom charts, but the colors are common to all Intensity charts of the same type.

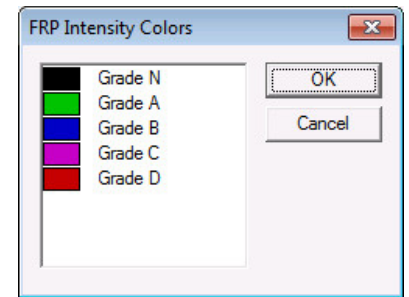


Figure 6-7. FRP Intensity Colors

The Table button on the Intensity tab leads to the same Table dialog used for Cluster calculations as shown to the right. If you use it to turn on the Intensity table, you will see a listing of the data in the Intensity Chart similar to the one shown below. In addition to using the Table dialog, you can also use the F6 key to toggle the Table on/off & shift-F6 shifts its position among the top, bottom, right & left settings.

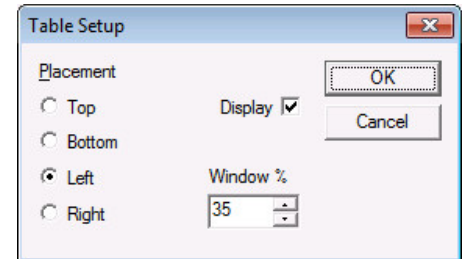


Figure 6-8. Table Set up

Alarm functionality can be enabled via the 'Trip' and 'Warning' controls. Set each to the desired grade. An alarm will be triggered when the grade reaches the specified level. Each channel is evaluated on an independent basis, so if 5 channels go above the threshold, you will get 5 alarms. Setting an alarming level to 'OFF' disables that level. As with other alarm types, alarm information is shown on the line display and in the Alarm Display dialog box.

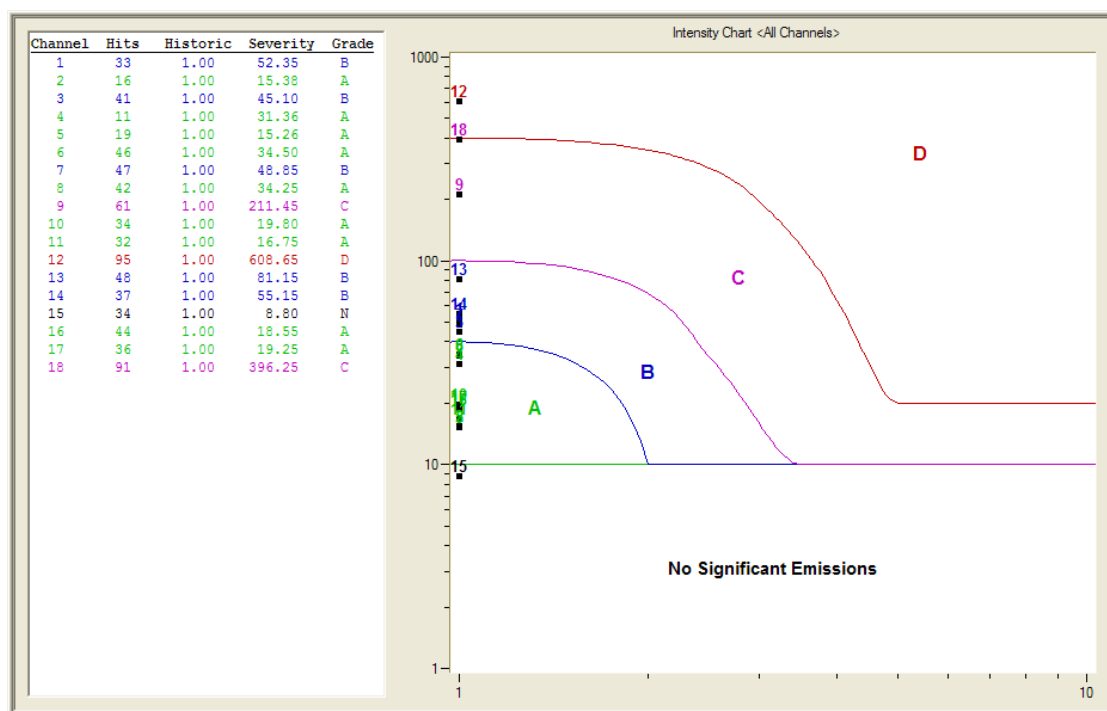


Figure 6-9. Intensity Table

The Sorting drop-down list on the Intensity tab will allow you to change the sort order of the Intensity Table. The table can be sorted by channel, grade level, severity, or historic index. Outside of Graph Setup, you can also use Ctrl-F6 to control the sort order of both Intensity & Cluster Tables. Each time you press Ctrl-F6, the sort order will be changed to the next available order cycling through channel, grade, severity & historic index for the Intensity table.

The other available Intensity Chart settings are similar to those for standard graphs. Since an Intensity Chart plots Severity vs. Historic Index with fixed limits, the axis settings are disabled for this type of graph. You can choose between Hits or Events as the Input Data, although traditionally, Intensity calculations use all available hits. If you select Events, use the Event Groups tab to specify which Location Group the graph should monitor. For both Hits & Events, the Channel tab will allow you to limit the graph to select specific channels rather than just analyzing all channels.

The Show Grid option allows you to display gridlines for all the major (1,10,100 etc.) & minor tick marks (2,3,4...9, 20-90, etc.) as shown here. The graphs can be a bit busy with gridlines & data points. As an alternative, you may prefer to leave the grid turned off & just use a data cursor to display the coordinates of individual data points.

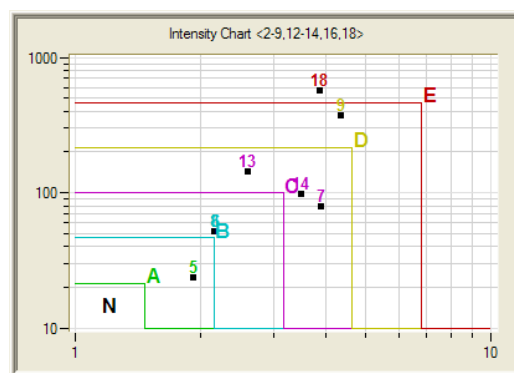


Figure 6-10. Intensity Chart

If you wish, you may use Historic Index & Severity as parameters on any other graph type. Simply look for them in the list of parameters available for plotting. When an axis parameter is set to Historic Index or Severity, the Intensity tab will automatically appear to allow you to specify the type of intensity analysis to perform. Here's an example of using Graph Setup to create a graph of Historic Index vs. Time graph.

You should note that a graph of Historic Index vs. Time shows you the actual ongoing Historic Index values, while an Intensity Chart only uses the maximum Historic Index observed up to any point in time. This can lead to apparent discrepancies. If you look at the current value on an Intensity Chart, it should be the highest maximum currently on the time plot assuming the axis limits aren't cutting it off.

The screenshot shows the 'Graph Setup' dialog box for a '2D Binned Scatter' plot. The 'Y (Vertical) Axis' is set to 'Primary' and the 'X (Horizontal) Axis' is set to 'Time'. The 'Y-axis' scale is 'Linear' with a 'Minimum' of 1 and a 'Maximum' of 10. The 'X-axis' scale is 'Linear' with a 'Minimum' of 0 and a 'Maximum' of 1. The 'Log Scale' checkbox is checked for both axes. The 'Input Data' section shows 'Hits' selected. The 'Show Grid' checkbox is checked, and the 'Force Update On Every Point' checkbox is unchecked.

Figure 6-11. Historic Index vs. Time

Custom Intensity Analysis

If the user wishes, it is possible to set up your own intensity chart using the Custom type. When intensity is set to Custom, the Settings button enables allowing the user to bring up the Custom Intensity Settings dialog box.

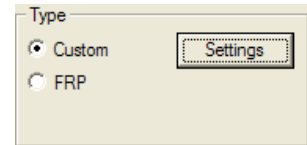


Figure 6-12. Type

The Custom Intensity Settings dialog box has 3 main areas. In the upper left corner of the dialog box, there are settings that control the low-level Historic Index & Severity calculations. The bottom half of the dialog box displays a series of tabs that correspond with the borders between the various grade levels. There is also a tab for the label location for each of the grades. The upper right-hand corner shows a preview of the grade borders & label locations which updates as the user edits the data points in the bottom section.

Each of the border tabs has a grid that displays the data points that make up that particular grade boundary. These borders form the boundary between 2 grades. When grades are assigned to a data point on the intensity, the point that is above or falls on the border is considered to be the higher grade.

You can add new data points using the INS key or F5 key. The INS key inserts another data point at the current row & copies the values from the previous data point. The F5 key appends another data point after the current row & copies the values from the current row. The DEL key deletes the data point in the current row.

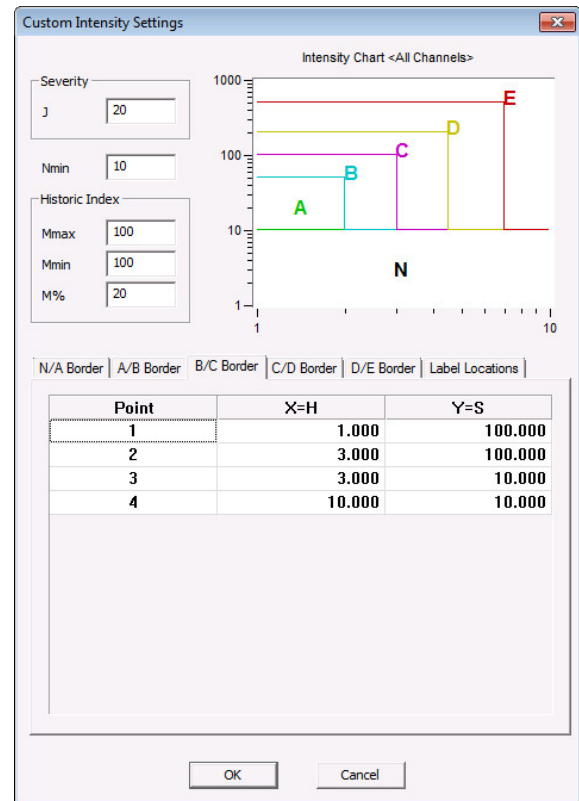


Figure 6-13. Custom Intensity Settings

You can copy all the data points in the grid by right-clicking & picking the “Copy Grid to Clipboard” option. You can also paste information back into the grid by right-clicking & using “Paste from Clipboard.” Copy/Paste is useful if you wish to use a program like Microsoft Excel to generate the data points in a border based on a mathematical formula. Copy the grid, paste it into Excel & then modify it as much as you wish. The borders can contain up to 256 data points. If “Paste from Clipboard” is disabled when you go to paste the border back into AEwin, either there is no data on the clipboard, or you haven’t included the column headers along with the data. The software checks for the column headers to avoid trying to paste incompatible data into the grid. Otherwise, you might accidentally try to paste in pictures or other binary data that should not be interpreted as a series of border points.

Border		A/B Border	B/C Border	C/D Border	D/E Border
Point	X=H				
1	1.000				
2	4.641				
3	4.641				
4	10.000				

Copy Grid to Clipboard
Paste from Clipboard
Clear All

Figure 6-14. Intensity Grid

When you are editing your border, keep in mind that the grading algorithm interpolates between the data points in the border to find the coordinates of the border at a given value of the historic index. That means that for straight-line segments, you only need to store the 2 endpoints. This makes it easy to create borders with fairly complicated shapes using straight line segments. You should also store your border points in ascending order by the Historic Index value (=X). The software does not force you to do this while editing the curve, but it will sort the points when you close the dialog box. That does rule out borders that fold back on themselves like a circle, but the arc of a semi-circle is feasible.

You do not have to define all the borders if you do not wish to use all the grades from N (Not Significant) thru grade E. You can delete all the points in a border using Clear All in the Right-click Menu. The software will only use the grades up to the missing border. For example, if you delete all the points in the C/D Border, both the C/D & D/E borders disappear from the intensity chart & grade C will be the highest grade assigned by the analysis. When a border is not being used, the software also removes the corresponding grade label from the chart. Alternatively, if you wish to hide a label, you can just give it a 0 coordinate or coordinates outside of the axis limits, but this is not required if the border is missing.

Severity & Historic Index Low-Level Calculation Settings

The low-level Historic Index & Severity calculations can be controlled using the 5 settings in the upper left-hand corner of the Custom Intensity Settings dialog box. To understand them, you first need to know more about how Severity and Historic Index are defined:

Severity, S = Average of J *Highest* Energies

Historic Index, H = $\frac{\text{(Average Energy of } M \text{ Most Recent Hits)}}{\text{(Average Energy of All } N \text{ Hits)}}$

Some useful definitions:

N or N_{hits} = Total number of hits on a given channel.

J = Number of highest energy hits used to calculate Severity.

M or M_{hits} = Subset of N_{hits} on a given channel used to calculate Historic Index.
Varies in a complex manner based on the value of N_{hits} .

M_{max} = Maximum number of M_{hits} saved for high values of N_{hits} .

M_{min} = Minimum number of M_{hits} saved for low values of N_{hits} . For $N_{\text{hits}} < M_{\text{min}}$, $H = 1$.

$M_{\%}$ = Percentage of hits saved for intermediate values of N_{hits} , ie. $M_{\text{hits}} = N_{\text{hits}} * M_{\%}$

N_{min} = Minimum number of hits before Severity & Historic Index start being evaluated.
For $N_{\text{hits}} < N_{\text{min}}$, $S = 0$ & $H = 0$.

The image shows a screenshot of a software settings dialog box. It has a title bar and a light-colored background. There are two main sections: 'Severity' and 'Historic Index'. Under 'Severity', there is a label 'J' and a text input field containing the number '20'. Under 'Historic Index', there are three labels: 'Mmax', 'Mmin', and 'M%'. Each has a corresponding text input field. 'Mmax' and 'Mmin' both contain the number '100', and 'M%' contains the number '20'. There is also a label 'Nmin' with a text input field containing '10'.

Figure 6-15. Settings

In simplest terms, Severity is based on the J highest energy hits for a given channel & Historic Index is based on the M most recent hits where the value of M varies as the total number of hits, N , changes. There are several subtle considerations here, though. For Severity, there is the question of what Severity evaluates to when N_{hits} is less than J . The value could be defined as 0, but the intensity chart is a log plot. The intensity chart will not display data for any channels where S or H is 0. In some cases, it might be better to define Severity as a simple average of all the hits that are available up to that point so that the data can appear on the intensity chart. For example, assume J is set to a high number like 100. A value of 0 for channels with less than 100 hits would not show any data for those channels till 100 hits had been observed on that sensor. The N_{min} user setting allows the user to decide what value of N_{hits} is required before Severity is calculated. If N_{min} is 40 & J is 100, then the software will average all hits from 40 up to 100 hits as the data accumulate. To be more specific, when there are 42 hits on the channel, Severity will be the average of those 42 hit energies. When there are 43, it is based on 43 hits. Once there are 101 hits, then the Severity calculation will drop the lowest energy value & S will be based on the 100 highest energy hits.

For Historic Index, the discussion is complicated by the fact that the number of hits used, M_{hits} , varies depending on the total number of hits.

Value of N_{hits}	implies	Value of M_{hits}
$N_{\text{hits}} \leq M_{\text{min}}$	$\rightarrow$	$M_{\text{hits}} = M_{\text{min}}$
$M_{\text{min}} < N_{\text{hits}} < M_{\text{max}}$	$\rightarrow$	$M_{\text{hits}} = M_{\%} * N_{\text{hits}}$
$N_{\text{hits}} \geq M_{\text{max}}$	$\rightarrow$	$M_{\text{hits}} = M_{\text{max}}$

Again as for Severity, there is the question of what value to assign to Historic Index when M_{hits} is less than M_{min} . Again the simple answer could be that it is 0, but then the data will not appear on the intensity chart. Sometimes that may be what the user wants & sometimes, it may not be. In this case, the software uses the following 2 rules. If $M_{hits} < M_{min}$, the Historic Index calculation will define $H = 1$. If $M_{hits} < N_{min}$, $H = 0$. What if M_{hits} is less than both N_{min} & M_{min} , what happens? Remember N_{min} is the number of hits where the software will start evaluating S & H, so it is the more important number in this case as H can't be set to 1, if H is not being evaluated. Some concrete numbers will help clarify. Assume for example $M_{min} = 20$ & $N_{min} = 10$. For $N_{hits} < 10$, H will be defined as 0. For $10 \leq N_{hits} \leq 20$ hits, H will be defined as 1. Above 20 hits, M will vary according to the other user settings & H will vary as it is calculated. Now reverse the values & assume $M_{min} = 10$ & $N_{min} = 20$. In this case, for $N_{hits} < 20$, H is 0 & it doesn't matter whether or not $N_{hits} > M_{min}$. Once $N_{hits} \geq N_{min}$, then $N_{hits} > M_{min}$ already & M_{hits} will be a percentage ($M\%$) of N_{hits} . This gives the user the flexibility to control when H is 0, when it is defined as 1 & when H is actually calculated.

You may wish to duplicate the low-level settings that correspond to the FRP calculation if you are not sure how you want to default these settings. The software uses the following values for the built-in FRP calculation:

FRP Equivalent Custom User Settings

J	= 20	
M_{max}	= 100	Applies to $N_{hits} > 500$ because 100 is 20% of 500.
M_{min}	= 100	Applies to $N_{hits} < 500$. $H = 1$ for 10-100 hits.
$M\%$	= 20%	Doesn't apply due to high M_{min} .
N_{min}	= 10	$H = S = 0$ for $N_{hits} < 10$ & data doesn't appear on chart.

6.5 C-Factor Option

The European standards NEN-EN12817 and NEN-EN12819 describe testing of LPG tanks using acoustic emission (appendix C for both 12817 and 12819). Both describe a user's definable evaluation factor C to help make the pass/fail decision for the LPG tank. (See section C.5.1e in 12817 and C.6.1.3e in 12819). C-Factor is an enhanced version of clustering that allows the user to define his own formula for this evaluation factor. The formula uses standard mathematical operations and can access statistical information about the cluster being evaluated. There is a licensing option related to C-Factor. You can check your licensing options by selecting Options under the Help pulldown menu. The Options dialog shown below will appear:

If you don't see the C-Factor option (#45) in the list in the Options dialog box, your AEwin license does not include it. You will need to contact [Physical Acoustics \(customerservice@mistrasgroup.com\)](mailto:customerservice@mistrasgroup.com) to obtain the proper license.

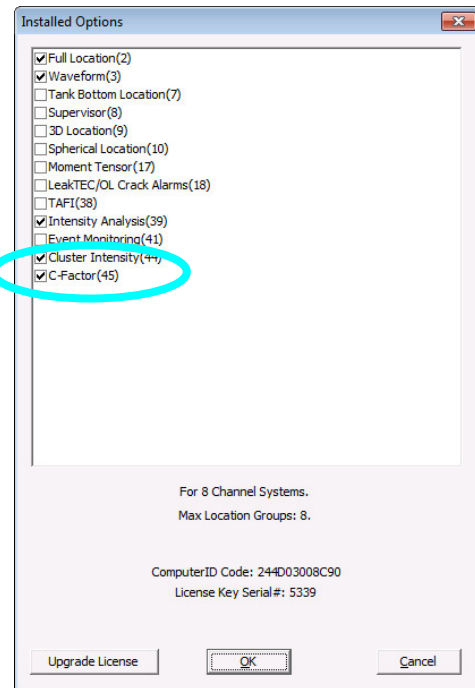


Figure 6-16. Options

C-Factor Settings

The Clustering tab of Graph Setup has several fields that are being used with C-Factor. The Clustering tab is shown below:

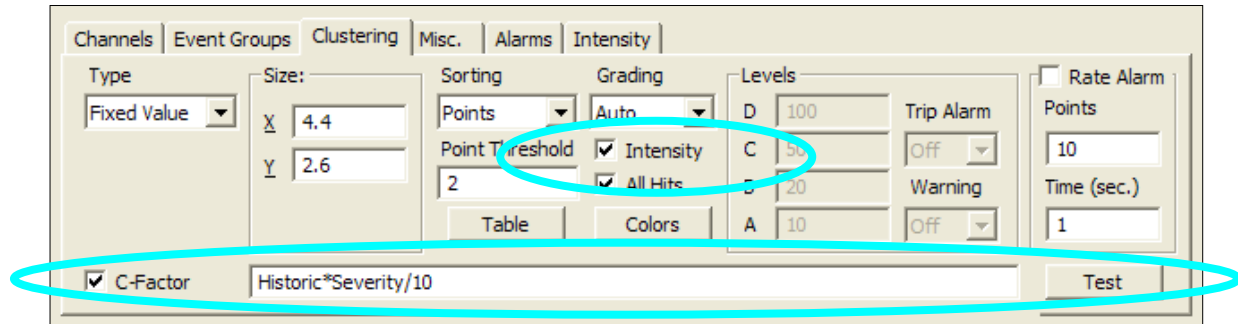


Figure 6-17. C-Factor Settings

The Intensity checkbox is not new or specific to C-Factor but is available when the AEwin software has been licensed for C-Factor. It allows the user to calculate cluster intensities and use the Severity and Historic Index of the cluster in the C-Factor formula.

The “All Hits” checkbox is not specific to the C-Factor checkbox either but can influence the results. It is of use only when the Input Data for the graph is set to Events. If the Input Data is set to Hits, there is 1 hit for each data point on the graph. If the Input Data is set to Events, the event can contain up to 12 hits. For events, “All Hits” controls which hits in the event are used in calculating the totals for Energy, Counts, Amplitude (= Source Amplitude, if available), Severity, and Historic Index. If “All Hits” is not checked, those quantities are just the totals of the 1st hit of all the events that fall within the cluster. If “All Hits” is checked and the Input Data is events, the totals will be higher. The cluster table will also list the number of Points in the cluster as well as the number of Hits in the cluster. If Input Data is set to Hits, then the number of data points on the graph and in the cluster is the same as the number of hits. When Input Data is set to Events, the Points column will have a smaller number than in the Hits column.

Cluster	Points	Hits	Energy	Counts	Amplitude	Historic	Severity	Grade	C-Factor	X	avg.	Y
K1	2	7	11	105	0	1.27	1.57	D	0.20	1417.00		-2.50

The “C-Factor” checkbox is a simple on/off toggle controlling whether or not the software evaluates the user’s formula listed in the text field next to the checkbox when data is collected or replayed. The user’s formula can refer to many of the cluster numbers listed in the table, and they may be combined using the usual mathematical operators like + and -, * and /. Quantities available for use in the formula include Points, Hits, Energy, Counts, Amplitude, Historic (Index), and Severity. These quantities are totals for the cluster at the point in time that the C-Factor formula is evaluated. For example, to make the C-Factor formula calculate the average energy of all the hits in the cluster, simply enter Energy/Hits. The Test button will allow the software to check the formula for errors like typos in the names of the quantities. You should test your formula before using it to evaluate data during acquisition or replay. If you don’t and there is an error, C-Factor will not be calculated. Clicking on the Test button will yield either an error message or the value of the formula:

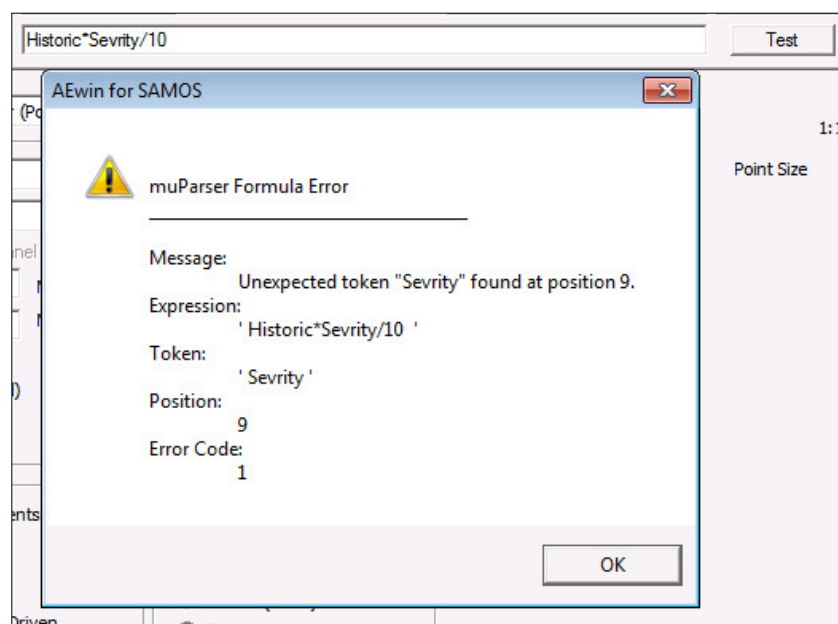


Figure 6-18. C-Factor Formula Error

If you have correctly entered the formula, you should get a message box containing sample values of the cluster quantities used in calculating the value of the C-Factor formula:

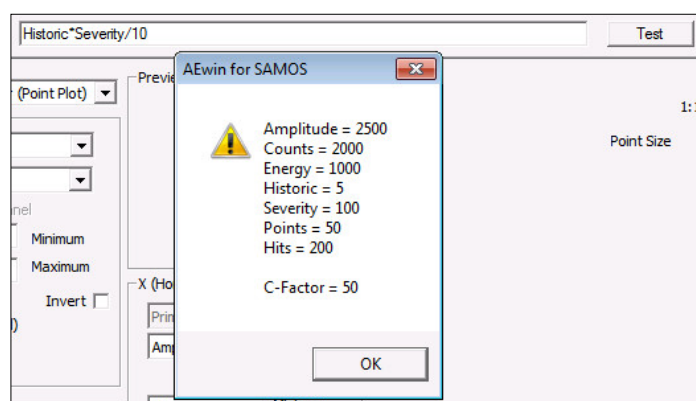


Figure 6-19. C-Factor Formula Test

The Parser reference section at the end of this document has a table of other math functions that are allowed in the formula like log(), exp(), or sqrt(). The C-Factor value is not stored with the user's DTA data file, so you can revise the formula in your layout file and replay the data at any time to see the values from an updated formula.

Graphing/Drawing Options

C-Factor Cluster Histograms

There are several other data display options that are not on the Clustering tab of the Graph Setup dialog box. First, you should be aware of the ability to create cluster histogram graphs by right-clicking the cluster graph itself. If you do, you will see the following options on the context menu that appears:

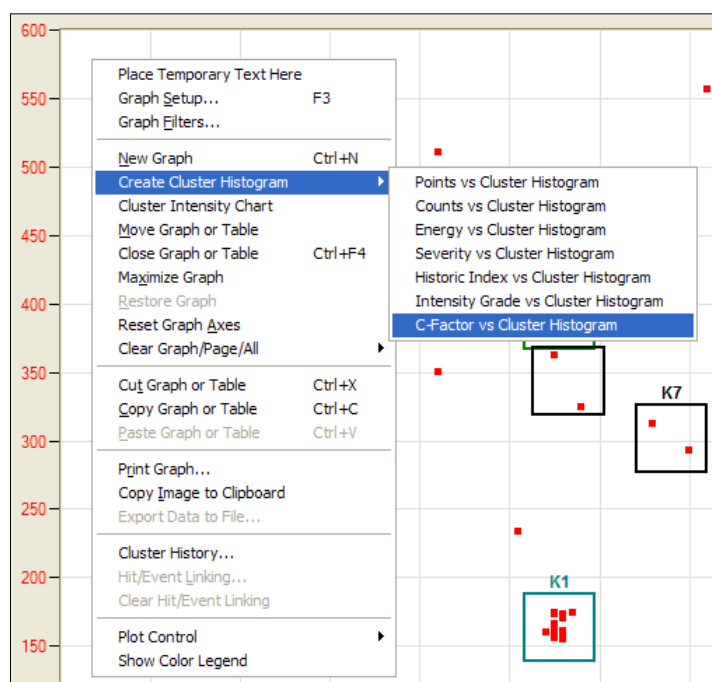


Figure 6-20. Cluster Histogram Menu

The new addition to this menu is the C-Factor vs. Cluster Histogram graph. If you pick it, it will just create a histogram allowing you to quickly compare the C-Factor values for all the clusters detected:

You can move and resize cluster histograms, and they can be saved as part of your layout file too. They are auto-scaled graphs, so you cannot access a Graph Setup dialog box for the cluster histogram, but it is possible to use the mouse to zoom in on the graph. You can also use the Maximize Graph option in the right-click context menu to make them larger as well.

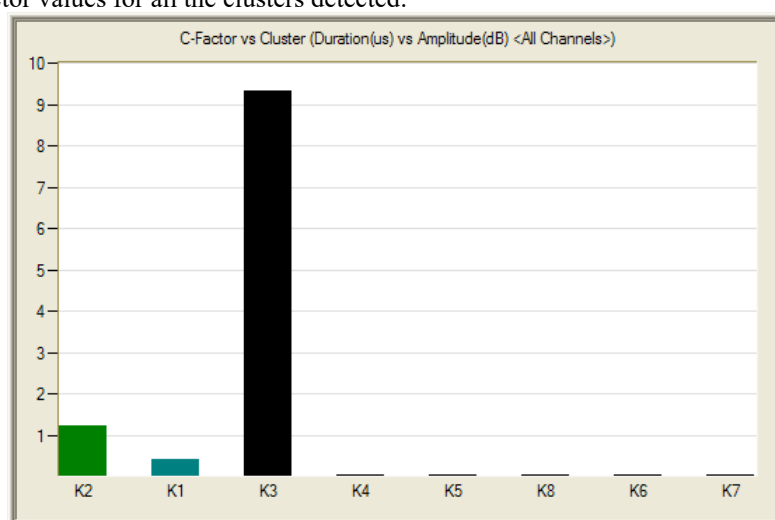


Figure 6-21. Cluster Histogram

Cluster Drawing Methods in Preferences

The Preferences dialog box on the Acquisition Setup dropdown menu has 2 menu options related to clustering, which control how the clustering results are presented:

The “*Label Clusters with K1, K2, K3... instead of z to a*” checkbox controls how the clusters are labeled on the graph and in the cluster table. In the past, clusters have been shown with the labels ‘z’ through ‘a’ to avoid confusing the cluster labels with grade labels (‘A’ to ‘E’) shown on some graphs like the Cluster Intensity chart. The Clustering tab of the Graph Setup dialog box has a field that controls how the clusters are sorted and displayed in the cluster table. The ‘z’ cluster is the cluster that is highest in the sort order, and the ‘a’ cluster is lowest in the sort order. This can be confusing when the table sort order is changed because it changes which cluster is the ‘z’ cluster. There is also a Cluster-ID shown in the table in that case. The cluster with ID 1 formed; first, ID 2 was second, etc. Now instead of ‘z’ to ‘a’ labels and a separate cluster-ID, the user can have the clusters labeled with K and the cluster-ID#, so that the cluster label does not change when the user changes the sort order in the cluster table. Those that prefer the traditional ‘z’ to ‘a’ labeling can simply leave this option unchecked.

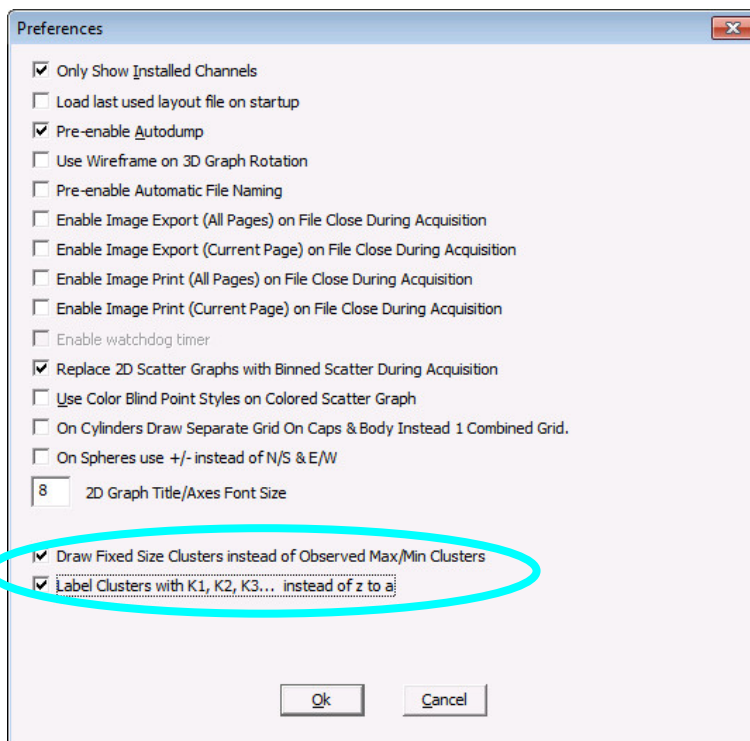


Figure 6-22. Preferences for Clustering

The other option available is the “*Draw Fixed Size Clusters instead of Observed Max/Min Clusters*” checkbox. Clusters have traditionally been drawn on the graph using the max/min values from data points that go into the cluster. This option allows the user to draw the max/min limits of the clusters based on the user’s cluster size settings. Here are 2 graphs. The first one has the clusters labeled ‘z’ to ‘a’ and uses observed cluster max/min limits, while the second uses K1 to K26 labels along with the fixed cluster max/min limits based on the user’s cluster size settings.

Cluster Drawing with Observed Max/Min Limits and z to a Labels

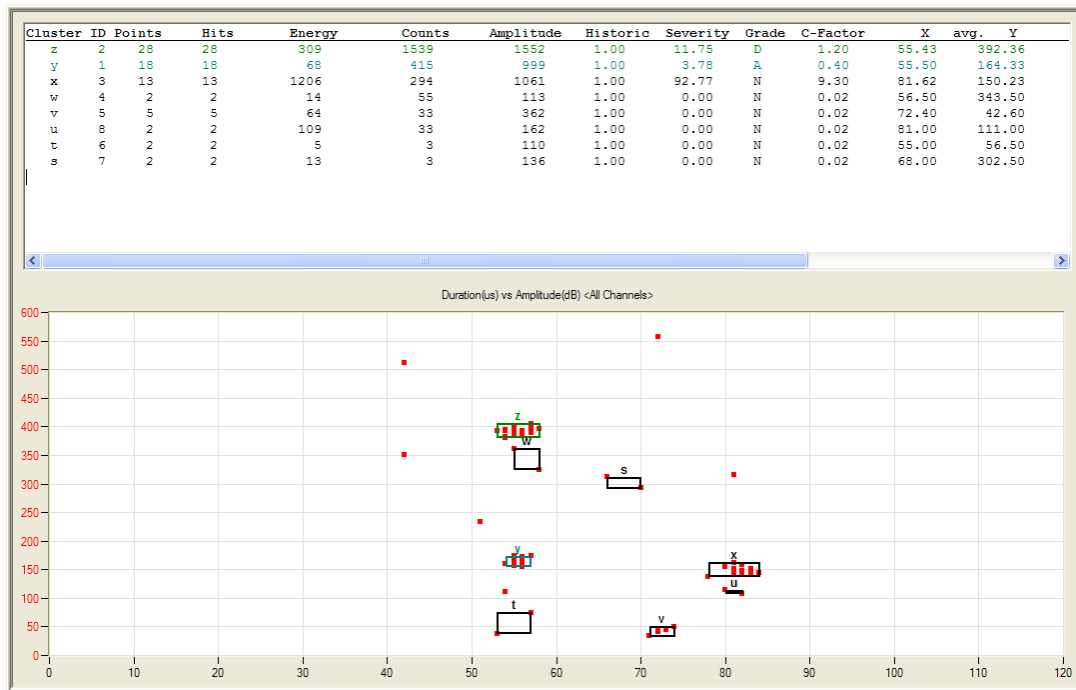


Figure 6-23. Observed Limits and z-to-a Labels

Cluster Drawing with Fixed Size Max/Min Limits and K1 to K26 Labels

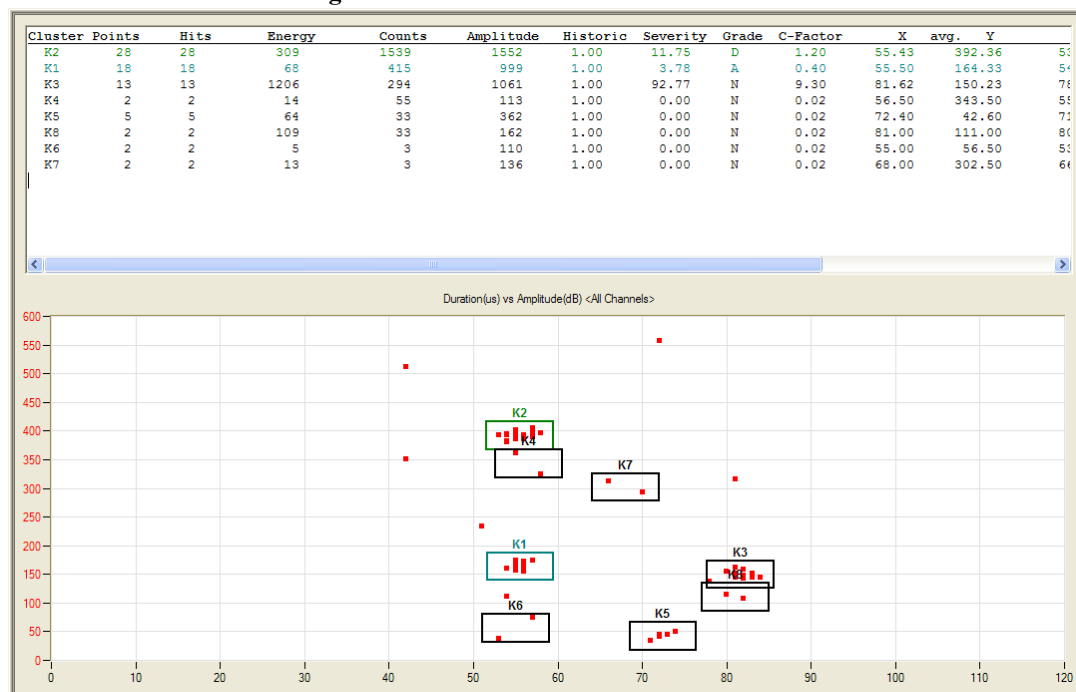


Figure 6-24. Fixed Size Limits and K1-to-K26 Labels

2D Binned and 2D Colored Scatter Fixed Clustering

Drawing fixed-size clusters have special behavior when dealing with the 2D Binned and 2D Colored Scatter graphs. Many users like to use 2D Colored Scatter graphs as a form of pseudo-clustering by setting the color to be tied to the number of data points in the bin for the graph. By checking the “*Draw Fixed Size Clusters instead of Observed Max/Min Clusters*” checkbox in the Preferences dialog box, clustering will use the bins as a real cluster so that now you can see the Energy, Counts, and number of Points in the bin. When it is off, the 2D Binned and 2D Colored Scatter graphs will behave like the 2D Scatter graph results shown above. When “*Draw Fixed Size Clusters instead of Observed Max/Min Clusters*” is on, the Clustering tab will hide the cluster size settings for the 2D Binned and 2D Colored Scatter graphs:

Figure 6-25. 2D Binned/Colored Scatter Fixed Clustering

If you want to change the size of the clusters, you have to change the bin resolution of the graph, so the size is always tied to the length of the axis divided by the number of bins. Here's a 2D Binned Scatter graph showing how the clusters are drawn when using fixed-size clusters:

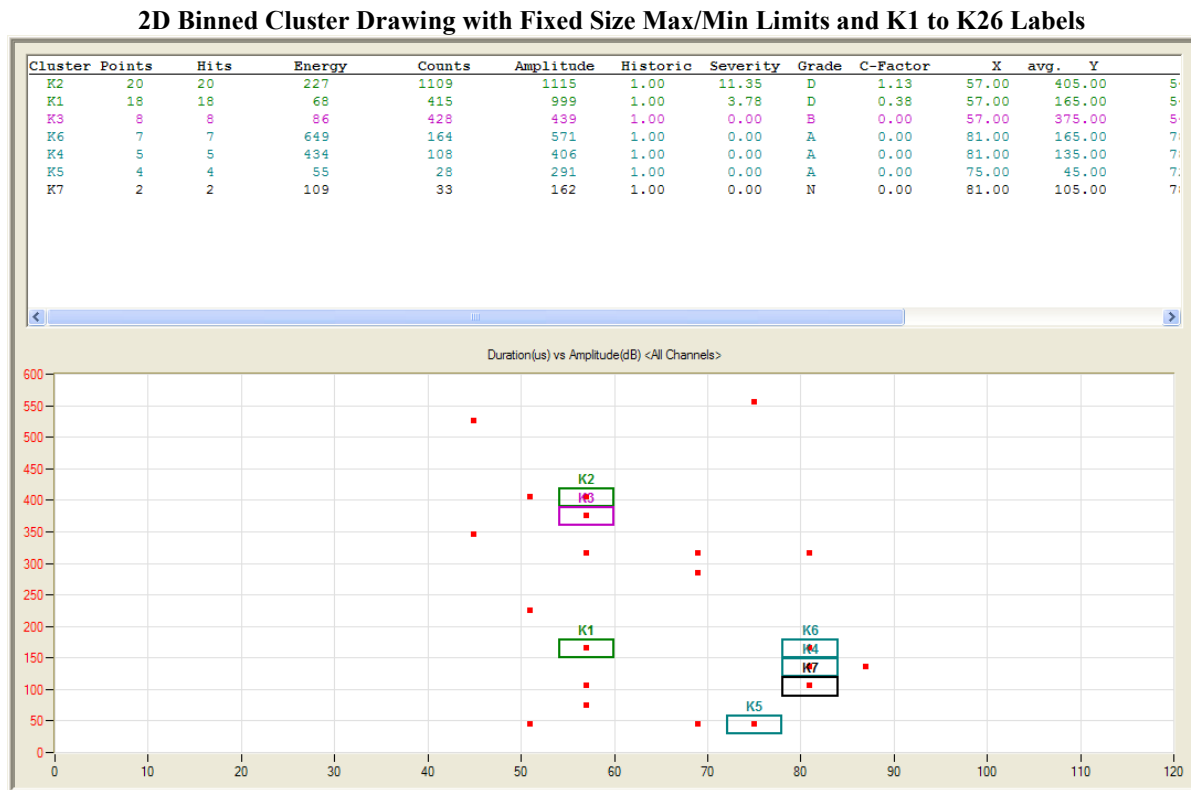


Figure 6-26. 2D Binned Cluster Example

Parser Formula Reference

The following reference material on what you can and can't do in your C-Factor formula is taken from the Parser documentation. AEwin uses muParser, a free code library available through SourceForge.net, copyrighted 2005-2013 by Ingo Berg.

C-Factor Alarms and Notifications

Selecting C-Factor in the Sorting box, Manual in the Grading box, setting level values in the Levels group, and setting Trip and/or Warning levels will set the graph to alarm at definable C-Factor levels. The alarms will output a digital high level on the Warning and Trip digital output pins as well as show a message on the line display (3.3.2.19 of AEwin manual) and optionally output a notification to one or more email recipients (enable Cluster Alarms notification).

Built-in Functions

The following table gives an overview of the functions supported by the default implementation. It lists the function names, the number of arguments, and a brief description.

Name	Argc.	Explanation
sin	1	sine function
cos	1	cosine function
tan	1	tangent function
asin	1	arc sine function
acos	1	arc cosine function
atan	1	arc tangent function
sinh	1	hyperbolic sine function
cosh	1	hyperbolic cosine
tanh	1	hyperbolic tangent function
asinh	1	hyperbolic arc sine function
acosh	1	hyperbolic arc cosine function
atanh	1	hyperbolic arc tangent function
log2	1	logarithm to the base 2
log10	1	logarithm to the base 10
log	1	logarithm to the base 10
ln	1	logarithm to base e (2.71828...)
exp	1	E raised to the power of x
sqrt	1	square root of a value
sign	1	sign function -1 if x<0; 1 if x>0
rint	1	round to nearest integer
abs	1	absolute value
min	var.	min of all arguments
max	var.	max of all arguments
sum	var.	sum of all arguments
avg	var.	mean value of all arguments

Built-in Binary Operators

The following table lists the default binary operators supported by the parser.

Operator	Meaning	Priority
=	assignment	-1
&&	logical and	1
	logical or	2
<=	Less or equal	4
>=	greater or equal	4
!=	not equal	4
==	equal	4
>	greater than	4
<	less than	4
+	addition	5
-	subtraction	5
*	multiplication	6
/	division	6
^	Raise x to the power of y	7

*The assignment operator is special since it changes one of its arguments and can only be applied to variables.

Other Operators

The Parser has built-in support for the 'if-then else' operator. It uses lazy evaluation in order to make sure only the necessary branch of the expression is evaluated.

Operator	Meaning	Remarks
? :	if then else operator	C++ style syntax

You can use this operator to avoid dividing by 0 in a formula. For example, if you're C-Factor formula is Historic/Severity, Severity can be 0 at the start. To avoid a divide by 0, you can use this more complicated formula:

Historic/(Severity>0 ? Severity : 1)

The Severity>0 section reads like an 'if...then...else' statement in deciding whether to use Severity or 1. If Severity>0, then use Severity; else, use 1.

6.6 TAFI Option

This option is designed to aid in the detection and analysis of Partial Discharge (PD) in power transformers.

Its full name is 'Time Arrival Feature Index.' It is an AE feature automatically extracted from the AE hits to synchronize the AE signal with respect to partial electric discharge. It is developed based on PAC's proprietary technology and is used to reveal partial discharge patterns, which can be displayed graphically.

TAFI is a testing time-related feature. A 'TAFI versus Time' graph, which is a scatter plot, can reveal a special pattern indicating partial discharge. When PD or other transformer defect is occurring, the dots are clustered around integer values 1, 2, 3, 4, 5 and 6, and appear as horizontal straight lines. If no PD happens, either the dots are scattered randomly along the test time, or the scattered dots are not lined up around integer values. The "AC Frequency" field in the AE Hardware setup dialog must contain the frequency of the transformer (typically 60 or 50Hz).

To purchase this option, or obtain more information on it, please contact our Sales department.

6.7 Material Property Calculation Option

The following paper gives an overview of the “Material Properties Calculation” option. This is a standard part of AEwinRockTest and can also be purchased as an option for AEwin. It is designed to allow the user to understand the purpose of the Material Properties Calculation dialog box, what inputs are required to be entered into it, how to acquire the time difference inputs via new menu options in hit-based ‘data selection’ (section 3.9), and how the user can format the output.

The Material Properties Calculation dialog box is designed to calculate important properties of the material based on its length, density, and time propagation of AST hits that are pulsed through the material. The dialog box is shown to the right.

The purpose of this is to test the properties of the material while it's undergoing a test. Input variables into the dialog box are the following: length of material (in meters), density (kg/m^3), Longitudinal time difference (microseconds), shear wave 1-time difference (microseconds), shear wave 2-time difference (microseconds), time correction for longitudinal time difference (microseconds), and time correction for shear time difference (microseconds). The “time differences” measures the time it takes for an AST pulse to go from a pulsing sensor to a receiving sensor. There are three pairs of these sensors: Longitudinal, shear wave 1, and shear wave 2. A longitudinal sensor pair is where the sensors are placed on opposite ends lengthwise of the material. The pulse would be transmitted by one sensor and received by the other at the end of the material. A shear wave sensor pair is where two sensors are placed on opposite ends widthwise of the material. A pulse would be transmitted lengthwise, creating a “shear wave” that travels along the width of the material; to be received by the other sensor. “Shear Wave 1” and “Shear Wave 2” are shear waves that have polarization directions perpendicular to each other. They are both used as a reference to the longitudinal wave, where the shear velocities are to be roughly half of that of the longitudinal. Finally, the “time correction” parameters are pre-determined values that the user enters to correct the time differences on account of the patterns above and below the material that will be present in the test.

Figure 6-27. Material Properties Calculation

The output variables are the following: velocity of AST pulse (shear wave 1, shear wave 2 and longitudinal: calculated in m/s), Young's modulus (MPa), Modulus of Rigidity (MPa), Poisson's Ratio, Lamé's Constant (MPa), and Bulk Modulus (MPa). All values except the velocities are calculated with both shear wave and longitudinal variables. Since we have two shear waves, there will be two values of each corresponding output relative to Shear Wave 1 and Shear Wave 2. The formulas for these outputs are below:

Formulas for output variables:

P = density

L = length

Tp = longitudinal time difference

Ts = shear wave time difference

$V_p = L / T_p$; $V_s = L / T_s$; (Longitudinal and shear wave velocities)

$E = [P \cdot V_p^2 \cdot (3 \cdot V_p^2 - 4 \cdot V_s^2)] / (V_p^2 - V_s^2)$ (Young's Modulus)

$G = P \cdot V_s^2$ (Modulus of rigidity)

$\mu = (V_p^2 - 2 \cdot V_s^2) / [2 \cdot (V_p^2 - V_s^2)]$ (Poisson's Ratio)

$\lambda = P \cdot (V_p^2 - 2 \cdot V_s^2)$ (Lamé's Constant)

$K = P \cdot (3 \cdot V_p^2 - 4 \cdot V_s^2) / 3$ (Bulk Modulus)

In the Material Properties dialog box, the user enters a name in “Material” to indicate the type of material they are working with. In all other text fields, the user is required to enter numerical values. If the user enters an invalid entry or leaves an entry blank, an error message will pop up indicating this and asking the user to fix the entry. This is shown to the right.

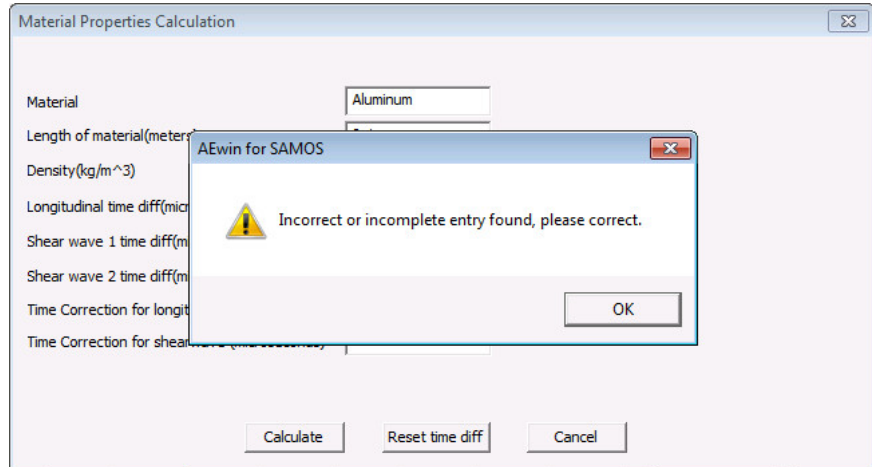


Figure 6-28. Incorrect/Incomplete Entry Found

The user cannot, however, manually enter values into the following fields: “Longitudinal time diff,” “Shear Wave 1 time diff”, and “Shear Wave 2 time diff”. This is because these values need to be calculated directly from the AST hits themselves. This can be done via hit-based data selection. To access this, load up a layout file that has graphs preset for AST hits with the sensor pairs. Replay a file containing these AST hits. Using the “Area Data Selection Mode” (the menu item containing a green box in AEwin), highlight a sensor pair on one of the graphs. Data selection will start containing two graphs and display the graph from the pulsing sensor. If you right-click on the graph, you will see three new options: “Select as Pulsing Time,” “Select as Receiving Time,” and “Sensor Pair Selection,” where “Sensor Pair Selection” has three submenus: “Longitudinal,” “Shear Wave 1”, and “Shear Wave 2”. This is demonstrated at right.

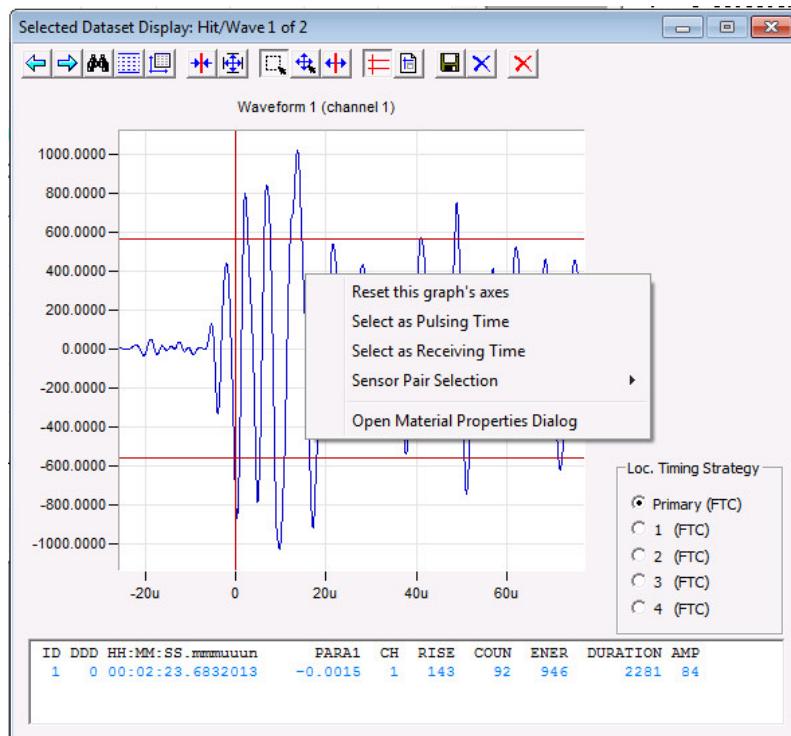


Figure 6-29. Starting Menu

These menu options are designed for calculating the time difference in an AST pulse and sending it to the Material Properties dialog box. “Select Pulsing Time” and “Select Receiving Time” are designed to select which graphs were respectively pulsing and receiving. “Sensor Pair selection” is designed to indicate which of the sensor pairs it belongs to. Selecting all three options will automatically compute the time difference and send it to the corresponding sensor pair. To illustrate this, Right-click on the first graph in Hit/Event Linking and select “Select Pulsing Time.” Now, Right-click again. You will see the menu as seen at right.

Here, the pulsing time option is selected, and receiving time is grayed out. This is to prevent the user from making more than one selection on the first graph.

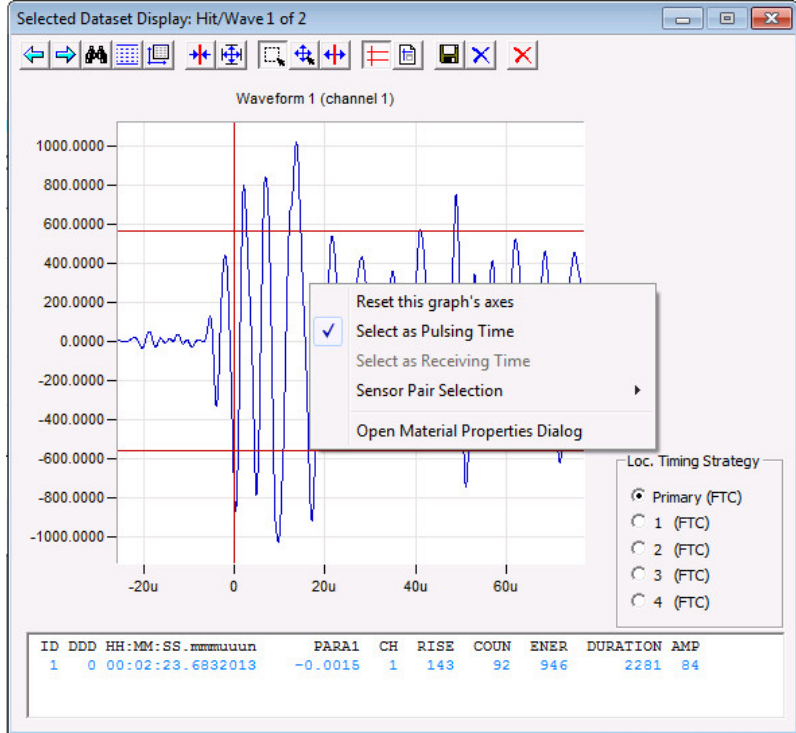


Figure 6-30. Select as Pulsing Time

Now go to the next graph in the selected dataset and Right-click here. The menu will be like that shown on the right.

Here, the pulsing option is grayed out while the receiving time option is enabled. This allows the user to only select the receiving option. Doing so will make the menu option look as shown below.



Figure 6-31. Pulsing Time Entry Disabled

Here, the pulsing time option is still disabled while the receiving option is selected. We have now selected which graph was the pulsing and which was the receiving. As a note, the user may wish to select a different pulsing or receiving time than the one given by default in the graphs. This can be done via the cursor mode in the Selected Dataset Display dialog box. (Found on the toolbar). The user slides over the cursor to the time they feel is appropriate. Once done, you can select the pulsing or receiving option to indicate your time selection. The cursor info time will be added to the starting time on your graph. To finally compute our value, we need to make a sensor pair selection. Select Longitudinal under “Sensor Pair Selection.” You will get the following, as shown below.

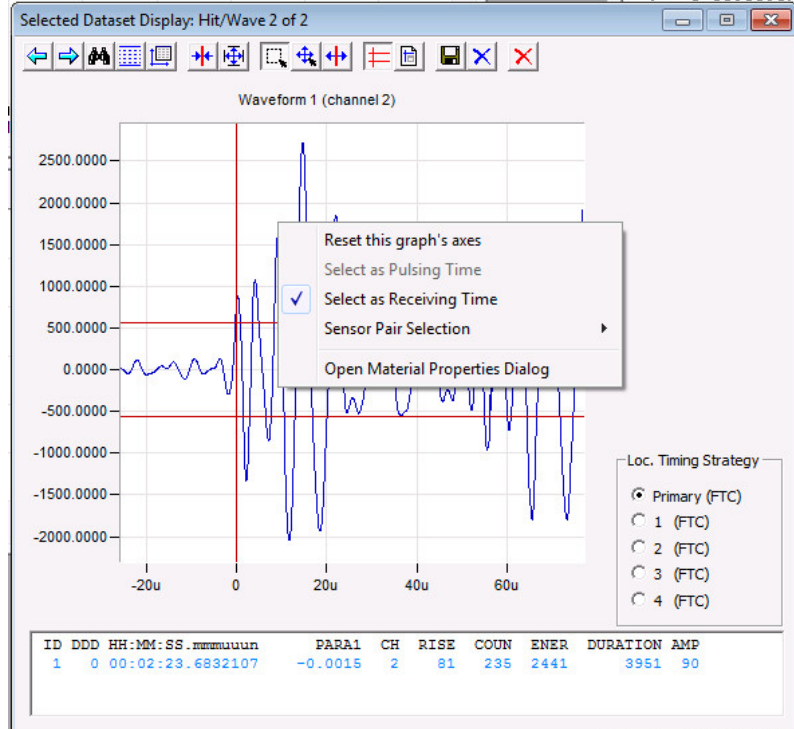


Figure 6-32. Select as Receiving Time

Here, longitudinal has a checkmark next to it while the other sensor pairs are grayed out. This will be the same when you Right-click on all the other graphs. Now that we have all three required menu selections, we will have a time difference calculated. Open the Material Dialog box by either going to Utilities->Material Properties or right-clicking on a graph in the Selected Dataset Display dialog and selecting “Open Material Properties Dialog.” You should now see the result as shown below.

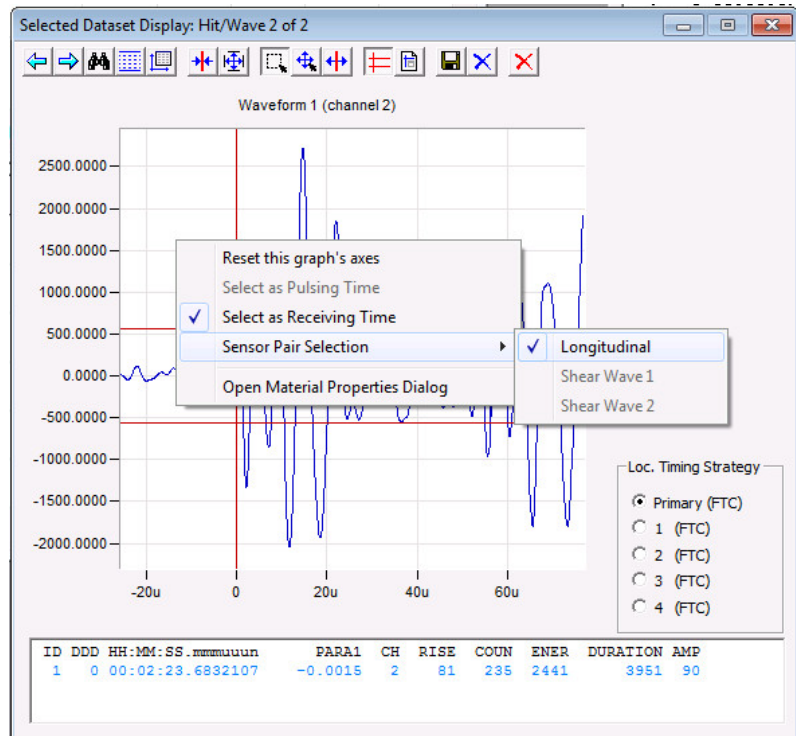


Figure 6-33. Select Longitudinal

Here, the time difference is shown along with the channel pair from which it originated. Since the AST graph was done using sensors on channels 1 and 2, we see “Channel 1,2” next to the calculated time difference.

As an additional note, it doesn't matter the order you make your three selections. You can select a sensor pair or receiving option first rather than pulsing and select any other of the two types. If, however, you incorrectly select a pulsing graph as receiving and vice-versa, a negative time difference will be generated. This will generate an error message.

Material Properties Calculation

Material		
Length of material(meters)		
Density(kg/m ³)		
Longitudinal time diff(microseconds)	22.411346	Channel 1,2
Shear wave 1 time diff(microseconds)		
Shear wave 2 time diff(microseconds)		
Time Correction for longitudinal (microseconds)		
Time Correction for shearwave (microseconds)		

Buttons: Calculate, Reset time diff, Cancel

Figure 6-34. Longitudinal Calculated

After clicking OK, all check-marked options will be off, allowing for the user to reselect the proper receiving and pulsing graphs. After calculating our shear wave, 1, and shear wave 2 values, the dialog box should now look like this.

Material Properties Calculation

Material		
Length of material(meters)		
Density(kg/m ³)		
Longitudinal time diff(microseconds)	22.411346	Channel 1,2
Shear wave 1 time diff(microseconds)	24.080276	Channel 3,4
Shear wave 2 time diff(microseconds)	22.888184	Channel 1,2
Time Correction for longitudinal (microseconds)		
Time Correction for shearwave (microseconds)		

Buttons: Calculate, Reset time diff, Cancel

Figure 6-35. All Calculated

For our calculations, we need now to populate all editable text fields. For this example, we will enter in “Aluminum” for the material, 0.1 for length, 2800 for density, and 0 for both time corrections. Note, if we enter a text into the “Time correction” values, we'll get an error reporting this. The populated text field should now look like this.

Material Properties Calculation

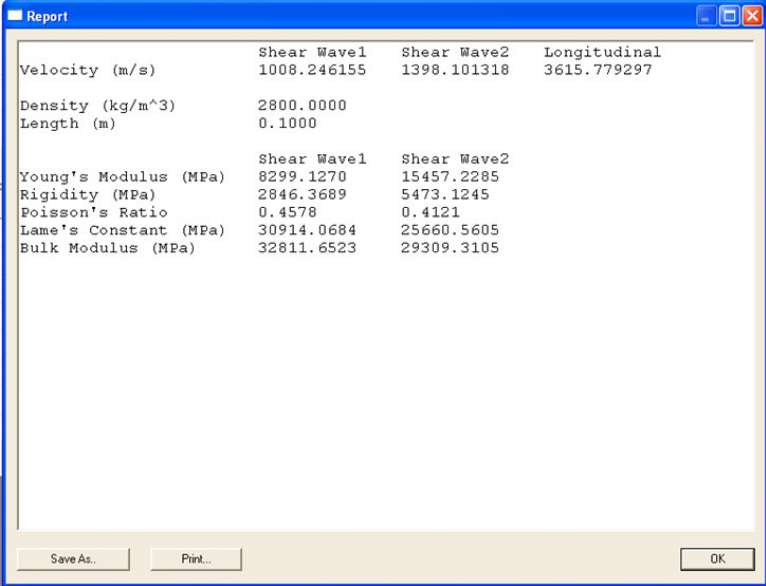
Material	Aluminum	
Length of material(meters)	0.1	
Density(kg/m ³)	2800	
Longitudinal time diff(microseconds)	22.411346	Channel 1,2
Shear wave 1 time diff(microseconds)	24.080276	Channel 3,4
Shear wave 2 time diff(microseconds)	22.888184	Channel 1,2
Time Correction for longitudinal (microseconds)	0	
Time Correction for shearwave (microseconds)	0	

Buttons: Calculate, Reset time diff, Cancel

Figure 6-36. Completed

To get our output values, click on “Calculate.” The report window will arise as shown here.

Here, the velocity is shown relative to Shear Wave 1, Shear Wave 2, and Longitudinal. For reference purposes, the entered density and length are also shown. Finally, the calculated values are shown with respect to Shear Wave 1 and Shear Wave 2. If the user wishes to save this output to a text file, they can click on “Save As...”.



	Shear Wave1	Shear Wave2	Longitudinal
Velocity (m/s)	1008.246155	1398.101318	3615.779297
Density (kg/m ³)	2800.0000		
Length (m)	0.1000		
	Shear Wave1	Shear Wave2	
Young's Modulus (MPa)	8299.1270	15457.2285	
Rigidity (MPa)	2846.3689	5473.1245	
Poisson's Ratio	0.4578	0.4121	
Lame's Constant (MPa)	30914.0684	25660.5605	
Bulk Modulus (MPa)	32811.6523	29309.3105	

Figure 6-37. Report

6.8 Event Classification Option

Event Classification (previously called “Event Monitoring”) is an enhanced version of location designed to decide whether or not events are important enough to report via email or STA files. It assesses whether located events are potentially important by applying a series of tests to the hits in the event. It can work with any location mode. There is a licensing option related to this. You can check your licensing options by selecting Options under the Help pulldown menu. The Options dialog shown below will appear:

You may have to scroll the list of options to the bottom to see the option, Event Classification (41). If there isn't a check next to it, your AEwin license does not include it. You will need to contact Physical Acoustics (customerservice@mistrasgroup.com) to obtain the proper license.

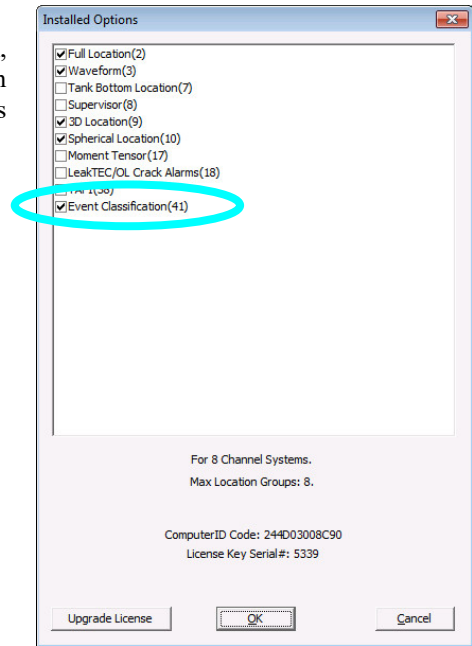


Figure 6-38. Event Classification Option

The Event Classification option controls whether or not the user can turn on Event Classification on the Options tab of the Location Setup dialog box & whether or not its Options button appears as well. Event Classification can be used with all location modes.



Figure 6-39. Enabling Event Classification

Event Classification Options Dialog

The Event Classification Options dialog box allows adjustments to the event classification criteria limits as well as controlling which tests are applied to an event. The event classification criteria involve a series of up to 6 tests on the hits included in the event. The first 3 tests are applied to the hits at the start of the event, beginning with the first hit. Subsequent tests apply to hits after the first hit. The parameter limits for these tests are adjustable using this dialog box. The default values are shown in the dialog below.

Event Classification Options

☒ Email Reporting Level: 5

☒ STA Reporting Level: 5

☐ STA Event Binning: 100 Bins

Tests for Hits in Event starting with First Hit

☒ 1. Frequency Centroid: 10 Min, 22 Max

☒ 2. Amplitude Threshold: 90

☒ 3. Energy Threshold: 1000

Hits Tested: 2

Tests for Hits in Event after First Hit

☒ 4. Amplitude Decay Rate: 0 dB

☒ 5. Velocity Range: 16 %

☒ 6. Total Hits Required: 3

Event Labelling

Level	Source Type
0.	Source Type 0
1.	Source Type 1
2.	Source Type 2
3.	Source Type 3
4.	Source Type 4
5.	Source Type 5
6.	Source Type 6

OK Cancel

Figure 6-40. Options Dialog

Tests for Hits in Event starting with First Hit

These tests are designed to ensure that the calculated source location is based on high-quality hits. Since Event Classification is available for all location modes, the minimum number of hits in the event used to calculate the source location can vary from 2-4. These 3 tests simply check the values of the frequency centroid, amplitude & energy of those hits to make sure they are in the appropriate ranges. You can use the Hits Tested field to include more hits for testing or to limit the testing to the first hit. If you don't want to perform one of these 3 tests, simply uncheck the test to turn it off.

Tests for Hits in Event starting with First Hit

☒ 1. Frequency Centroid	10	Min
	22	Max
☒ 2. Amplitude Threshold	90	
☒ 3. Energy Threshold	1000	
Hits Tested	2	

Figure 6-41. Tests 1-3

1. Frequency Centroid $\geq$ Min & $\leq$ Max? Limits default to 10 & 22.
2. Amplitude $\geq$ Amplitude Threshold? Defaults to 90.
3. Energy $\geq$ Energy Threshold? Defaults to 1000 dB.

Tests for Hits in Event after First Hit

These tests are not applied to the first hit for the simple reason that they cannot be. They are consistency checks on the other hits in the event.

The amplitude decay test looks for hits that are too high in amplitude compared to the amplitude of the first hit. High amplitudes would indicate that the hit came from an event other than the one that generated the first hit.

Tests for Hits in Event after First Hit

☒ 4. Amplitude Decay Rate	0	dB
☒ 5. Velocity Range	16	%
☒ 6. Total Hits Required	3	

Figure 6-42. Tests 4-6

The velocity test looks for hits that traveled significantly faster or slower than expected. It's not possible to calculate the velocity of the first hit because the exact time the event started is an unknown amount of time before the first hit. It is possible to calculate velocity for the later hits using their delta T values relative to the first hit & the coordinates of the sensors & the calculated source location.

4. Amplitude $\leq$ 1st Hit Amplitude – Amplitude Decay Rate * (Distance to Source – Distance to Source from 1st Hit.)? Amplitude Decay Rate has units of dB per user distance unit. Defaults to 0.
5. User Velocity(=100%) + Velocity Range > Calculated Hit Velocity > User Velocity - Velocity Range? Range defaults to 16% of user-entered velocity.
6. Hits in Event $\geq$ Total Hits Required? Presumes the event will be intense enough to reach a minimum of 2, 3, or 4 sensors.

Email Reporting Level, STA Reporting Level & STA Event Binning

The STA & Email Reporting Levels control which events go out via STA files & email notifications. The Internet Monitoring dialog still controls whether or not STA files are generated & whether or not any email notifications are sent out as there are other things recorded in the STA files & other kinds of email notifications possible too. The STA & Email Reporting Levels just control which events are required to be reported. The level can be set to 0-6.

Figure 6-43. Reporting Levels and Binning

The levels directly correspond to the 6 tests applied to the event. If you wish to only know about events that pass at least the first 6 tests, that is, Source Type 6 events (see discussion below), then set the corresponding reporting level to 6. If you want to see all events reported via the STA file, set the STA Reporting Level to 0. If you don't want any STA or Email reporting of events, just uncheck the setting to turn off the reporting. Note that while you can turn off event reporting here via the STA files & via email, these checkboxes do not turn off the creation of STA files & other notifications going out via email. Again that is controlled by the Internet Monitoring dialog.

The STA Event Binning checkbox controls how events are reported in the STA files. If it is unchecked, then each event that is reported via the STA file occurs as a single line of text with the coordinates & time of the test. If STA Event Binning is checked, then the location structure is divided up into bins & each bin occupies a single line in the STA file. Binning makes for smaller STA files but does not include the details of each event. Instead, it just tracks where the events happened & reports the total number of events occurring during an STA interval in that bin. Note if you are going to turn on STA Event Binning, make sure you have set the structure limits for your location group. For Linear, Zonal & 2D Planar, use either the Free or Plate structure to set the limits. For the other modes, just set up the structure associated with the mode (Sphere for Spherical Location, Horizontal or Vertical Vessel for Cylinder Location, etc.). The number of bins is user-selectable, with values ranging from 10 to 1000. All axes use the same number of bins.

Event Classification Results & Source Type

After the software calculates the source location based on the delta T values, it applies these 6 tests to the event, in order, one by one. The event is assigned a Source Type equal to the highest level that passes before any failures. Once a particular level fails, no higher levels are tested. If the lowest enabled level fails, then the Source Type is set to 0.

Figure 6-44. Event Labelling

For example, if an event passes the first and second tests, but fails the third one, then the Source Type is set to 2, and the later tests are omitted.

The higher the Source Type value the more likely it is that the event is an important one. The user can supply his own labels for each test level in the Event Labeling section. The default labels are just Source Type 0-6. The Source Type label is sent out in emails generated by location & is shown in the event listing in the Line Display:

```
* Gp# 1[12,15,9] x = 1550, y = 0, Source Type 6 dT[ 460, 1221] Src Amplitude = 99.0
* 0 05:40:11.8431613 12 42 6630 3771 99 19 189.715E+06 81 19 19 10
* 0 05:40:11.8436210 15 55 11539 5628 99 17 301.201E+06 94 6 18 8
* 0 05:40:11.8443817 9 15 1029 2248 87 18 5.244E+06 95 5 12 8
```

Graphs can also display Source Type as an integer on an axis similar to the way Events or Channel is displayed, so you could, for example, create a histogram of Source Type vs. Time as shown below to see when important events occur.

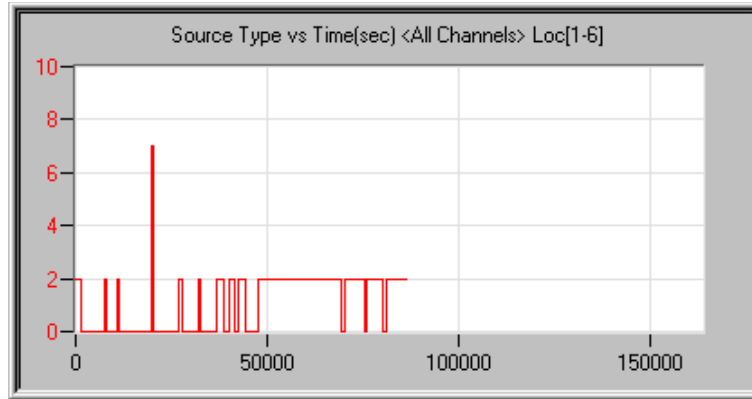


Figure 6-45. Source Type vs. Time

The availability of Source Type as a graphable quantity allows for its use in the Graph Filter dialog box to restrict input to graphs based on the Source Type. Here are 2 location graphs. The one on the left shows all events that occur & the one on the right shows only events that have a Source Type of 4 or higher.

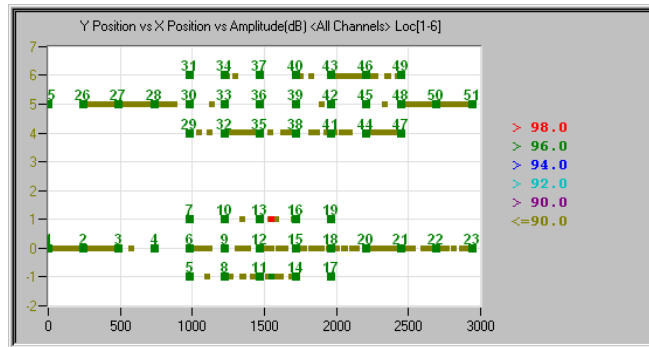


Figure 6-47. Y vs. X Position

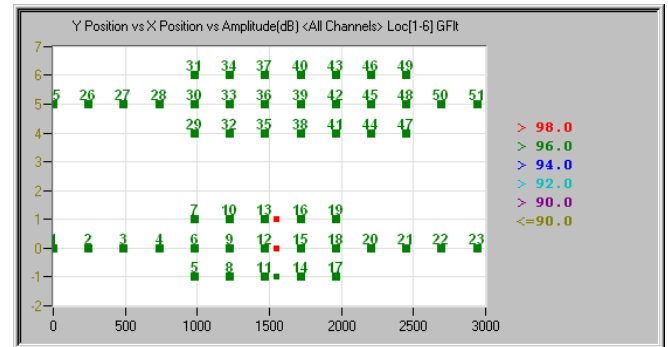


Figure 6-46. Y vs. X Position with GFLT

Here are the Graph Filter settings that restrict the input of the second location graph:

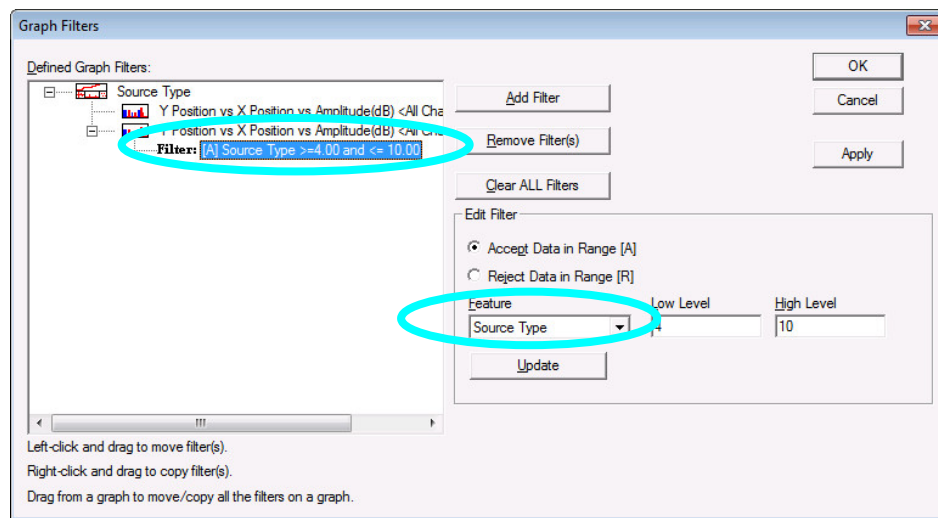


Figure 6-48. Graph Filters

7. COMMAND-LINE OPTIONS

7.1 Auto Start Acquisition

7.1.1 Setting AEwin for Replay Only Mode

Sometimes it is desirable to have AEwin start in replay only mode. In this mode, AEwin will not try to initialize the data acquisition system. For example, if you are running AEwin to do data analysis on a computer that does not have any data acquisition boards, AEwin will emit warning messages about the lack of boards. To avoid these messages, put AEwin in replay mode.

By specifying /r as a command-line argument in the AEwin Windows® shortcut, AEwin will start in the replay the only mode.

Follow the instructions below in the section “How to Create a Custom Shortcut for AEwin” using /r in place of /X in the procedure.

7.1.2 Setting AEwin to Load a Specific Layout File upon Program Launch

Sometimes it is desirable to have AEwin start with a specific layout file when it first loads.

By specifying /o and the layout file path and name as a command-line argument in the AEwin Windows® shortcut, AEwin will load the specified layout.

For example, if you want to load C:\My Layouts\planar.lay specify /o “C:\My Layouts\planar.lay “.

Note that because there is a space in ‘My Layouts,’ the file specification is enclosed in quotes.

Follow the instructions below in the section “How to Create a Custom Shortcut for AEwin” using /o “file path and name” in place of /X in the procedure.

7.1.3 Setting AEwin to Go Directly into Acquisition Upon Program Launch

Sometimes it is desirable to have AEwin startup and go directly into an acquisition without any user interaction.

By specifying /s as a command-line argument in the AEwin Windows® shortcut, AEwin will go directly into acquisition when it is launched. It takes user interaction to come out of the acquisition.

Follow the instructions below in the section “How to Create a Custom Shortcut for AEwin” using /s in place of /X in the procedure.

7.1.4 Setting AEwin to Go into Acquisition Based on the State of the Test Start Input

Sometimes it is desirable to control the acquisition state of AEwin from an external digital input. There are two available command-line switches for this feature.

By specifying /sm as a command-line argument in the AEwin Windows® shortcut, AEwin will monitor the test start input and go into and out of acquisition whenever the state of the input transitions between a low and a high state. AEwin ignores the level; it is the transition between states that controls acquisition.

By specifying /ssm as a command-line argument in the AEwin Windows® shortcut, AEwin will monitor the test start input and go into acquisition whenever the state of the input is high and go out of acquisition whenever it is low.

For most products, digital input 1 on the board that contains AE channel 1 is the test start input. Consult the hardware section of the system manual for more information about digital inputs and their assignments.

Follow the instructions below in the section “How to Create a Custom Shortcut for AEwin” using /sm or /ssm in place of /X in the procedure.

7.1.5 Assigning Specific Data Acquisition Boards to AEwin

Separate instances of the AEwin program can be configured to use 1 or more data acquisition boards that are in the system. For example, if the system contains two PCI-2 boards, one instance of the program can be configured to use the first board, and another instance can use the second board. In this case, each program will have access to 2 AE channels, numbered from 1 to 2.

There are several items that need to be addressed before this feature can work in a system:

- 1) The DIP switch on each board must be unique. This will be done at the factory if a system is purchased with multiple boards installed. If additional boards are purchased to increase the number of channels in a system, it is the responsibility of the user to correctly set the DIP switches. Refer to the system manual for address switch settings.
- 2) If more than one board is to be used by a given instance of the program, there must be a Real-Time Control Cable connected to each board that is to be used. The cable must NOT be connected among boards that are controlled by different instances of AEwin. For example, if there are 4 boards, the first controller by one instance, the second two by another instance, there need to be 2 RTC cables. One is connecting the first two boards and another connecting the last two boards.
- 3) The AEwin program must be told which boards are assigned to it. AEwin uses the DIP switch setting as the board id. The AEwin command-line must be modified to set the board ids for the program. Here is the procedure:
 - a. Follow the instructions below in the section “How to Create a Custom Shortcut for AEwin” using the instruction in the next step for specifying the modifications to the Target field.
 - b. Right-click the newly created icon and select ‘Properties.’ In the ‘Target’ field, append the following (excluding the quotes): ‘ /B X-Y’ where X and Y are the board id(s) (0 is board 1, 1 is board 2...). ‘-Y’ is not used for single board configuration. If the board ids are not sequential, then they can be listed using a comma as a separator. Note: there must be a space between the letter B and X.
 - c. Click Ok to dismiss the properties window.

Notes:

- If an instance of AEwin is running and it uses the /B command-line switch to assign board usage, DO NOT run an instance of AEwin without any /B switch or one that uses the same board assignments. Doing so causes two programs to try to access the same board, which must be avoided.

7.1.6 How to Create a Custom Shortcut for AEwin

AEwin supports several command-line arguments that modify the operation of the program. For instance, /s (go directly into acquisition) or /sm (use the test start input to control the acquisition), among others. This procedure explains how to make a new shortcut where the command-line switches can be added. The procedure will use /X as a placeholder for whatever command-line argument is to be inserted.

Here is the recommended procedure:

1. From the Windows Start menu
 - a. Go to All Programs-> MISTRAS Group and find your AEwin shortcut. Right-click on it and select ‘Send to’ then Desktop (create shortcut).’ Note that depending on the version of your AEwin, it may be under either MISTRAS Group or Physical Acoustics.
 - b. Locate the shortcut that was just created on the desktop. Right-click it and select Rename. Delete the old name and replace it with ‘AEwin Autostart’, or whatever text describes the command-line argument you will be adding.
 - c. On the same shortcut, Right-click, and select Properties.
 - d. In the ‘Shortcut’ tab of the Properties window:
 - i. In the ‘Target’ field, go to the end of the text (to the right of the closing quote) and add a space and then the additional command-line argument ‘/X’ (no quotes around /X).
 - ii. Select Ok to save the changes.
2. If AEwin is to start up each time the system is booted, drag the shortcut from the desktop into the Startup group of the Start->All programs menu. Typically (unless you are using the /b command-line), only one instance of AEwin should be in the startup group.

7.2 OPC Interface for AE

7.2.1 How to install under Windows 7:

1. Copy the folder AEOPC from the CD to the computer's hard disk at the C:\ location
2. Browse to the AEOPC folder on the hard disk, click once on the REGISTER file name
3. Right-click on the REGISTER file name and select 'Run as Administrator' from the pop-up menu
4. Click 'Yes' to the question "Do you want to allow the program to make changes to your computer?"
5. On the Desktop, click the Start icon at the lower-left, then 'All Programs,' 'Accessories,' and 'Run.'
6. Enter DCOMCNFG and click 'OK.' The 'Component Services' window appears
7. Double-click on 'Component Services' to make 'Computers' appear underneath
8. Double-click on 'Computers' to make 'My Computer' appear underneath
9. Double-click on 'My Computer' to make 4 items appear underneath
10. Double-click on 'DCOM Config' to bring up the list of icons on the right
11. Locate the 'Mistras AE OPC Server' icon among the many icons on the right
 - Right-click on it and select 'Properties'
 - Click on the 'Authentication Level' list, select 'None.'
 - Click the 'Identity' tab at the top
 - Check 'The Interactive User' then click OK
12. Close the 'Component Services' window. Installation is complete.

7.2.2 How to install under Windows XP:

1. Copy the folder AEOPC from the CD to the computer's hard disk at the C:\ location
2. Browse to the AEOPC folder on the hard disk and double-click the REGISTER.BAT file
3. On the Desktop, click 'Start' then 'Run.'
4. Enter DCOMCNFG and click 'OK.' The 'Component Services' window appears
5. Double-click on 'Component Services' to make 'Computers' appear underneath
6. Double-click on 'Computers' to make 'My Computer' appear underneath
7. Double-click on 'My Computer' to make 4 items appear underneath
8. Double-click on 'DCOM Config' to bring up the list of icons on the right
 - A few 'DCOM Configuration Warning' messages may appear; click 'No' on each one
9. Locate the 'Mistras AE OPC Server' icon among the many icons on the right
 - Right-click on it and select 'Properties'
 - Click on the 'Authentication Level' list, select 'None.'
 - Click the 'Identity' tab at the top
 - Check 'The Interactive User' then click OK
10. Close the 'Component Services' window. Installation is complete.

7.2.3 How to enable the AEwin interface to the OPC Server:

1. Browse to the AEwin program folder:
 - Windows XP and Windows 7 32-bit - C:\Program Files\Physical Acoustics\AEwin for ...
 - Windows 7 64-bit - C:\Program Files (x86)\Physical Acoustics\AEwin for ...
2. Right-click the AEwin.exe file icon, select 'Send To,' then 'Desktop (create shortcut).'
3. On the desktop, right-click on the AEwin shortcut icon and select 'Properties'
4. Click the 'Shortcut' tab at the top, locate the 'Target' field
 - 5a. Windows 7:
 - At the right end of the 'Target' string AFTER the " add a space followed by:
 - s -memfile
 - 5b. Windows XP:
 - At the right end of the 'Target' string BETWEEN AEwin.exe and the " add a space followed by:
 - s -memfile
6. Click 'OK.'
7. On the desktop, right-click on the AEwin shortcut icon and select 'Copy.'
8. On the desktop, click the Start icon at the lower-left, then 'All Programs'

9. Right-click on 'Startup' and select 'Open.'
10. Right-click inside the 'Startup' folder and select 'Paste.'
11. AEwin configuration for autostart with OPC is complete.

7.2.4 OPC items available:

1. OPC Server name - Mistras.AE.OPC
2. OPC tags
 - Heatbeat- alternates between 0 and 1
 - ActiveTest - 0 if idle, 1 for active
 - AlarmsActive - 0 if no alarms, 1 if some alarms below
 - ACTMSwarnings- number of ACTMS warning alarms
 - ACTMSalarms - number of ACTMS alarms
 - RateAlarms - number of rate alarms

8. PRODUCT SPECIFIC FUNCTIONS

8.1 AEwin Wireless Software

8.1.1 Power up procedure

The following procedure is recommended for starting up a system based on 1282 wireless AE nodes:

1. Connect the Wireless AE Basestation to a USB port on the computer
2. Follow instructions on the CD to install the Windows drivers (if using that USB port for the first time)
3. Turn on the power on the first 1282 Wireless AE Node
4. Observe the side of the basestation opposite the antenna. A red light should be on, and 3 green lights should light up within about 5 seconds and then turn off. This indicates that the 1282 node was detected and made a wireless connection to the basestation.
5. If the green lights did not light up, turn off the 1282 node, wait 10 seconds and turn on again. Then, follow step 4 above. If again there are no green lights in about 5 seconds, there probably is no wireless connection to that node. The node's battery may be discharged, or the distance to the basestation exceeds the wireless range of the system.
6. Repeat steps 3 and 4 for each 1282 node in the system.
7. Start AEwin for Wireless, then start data acquisition and review the Line Display for time-driven data records. Verify that parametrics 5, 6,..., N+4 (where N is the number of nodes) show numbers greater than zero. These are the battery readings for each node.

8.1.2 Wireless system configuration

A 1282 system is configured with matching wireless parameters for the basestation and associated nodes. It is not possible for the user to reconfigure the system and add nodes taken from another 1282 system.

Two different 1282 systems can operate in the same area. Each basestation will recognize its matching nodes and communicate only with those nodes. However, depending on configuration, the two systems may have to share the same wireless bandwidth and operate at a reduced performance level.

8.1.3 Acquisition Setup Menu

8.1.3.1 Hardware

This is the dialog box where all hardware settings are selected before the test. There are a number of property pages that each deal with a different part of the hardware setup. Each is described below.

8.1.3.2 AE Channel Setup

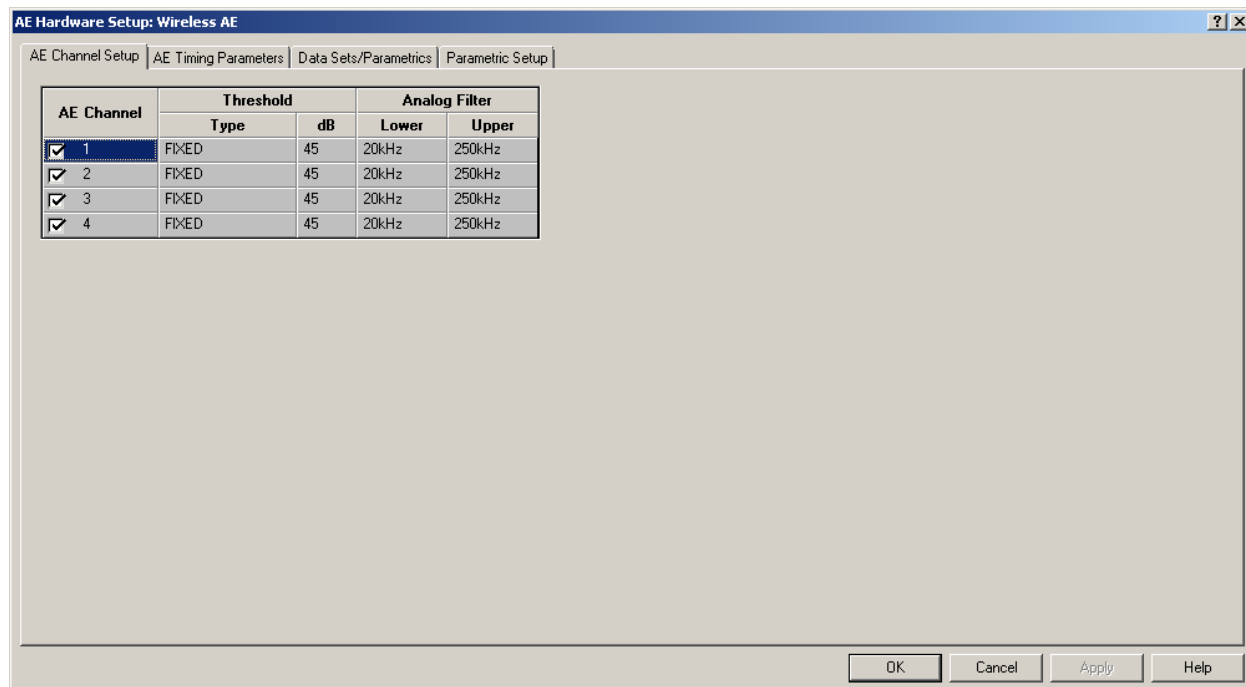


Figure 8-1. AE Channel Setup

On the 'AE Channel Setup' page, you can enable or disable each channel and edit their common settings.

AE Channel – You can enable a channel by clicking its checkbox.

Threshold – 'Fixed' threshold is used when you want the threshold to remain unchanged at the 'dB' level you set (in the dB column). 'Floating' threshold is not currently implemented. The threshold is the prime variable that controls channel sensitivity.

Analog Filter – The Lower and Upper drop-down list boxes allow you to select from the available analog filter values for low pass and high pass filters on each active channel. This is done by simply selecting the desired value from the drop-down list boxes.

Note: it may be difficult to adjust certain fields using a mouse when only one node is installed. An alternative way to change a field is to click on the field contents, press the Enter key, use the Up or Down arrow keys to change the selection, and then press the Enter key to lock in the new selection.

8.1.3.3 AE Timing Parameters Setup

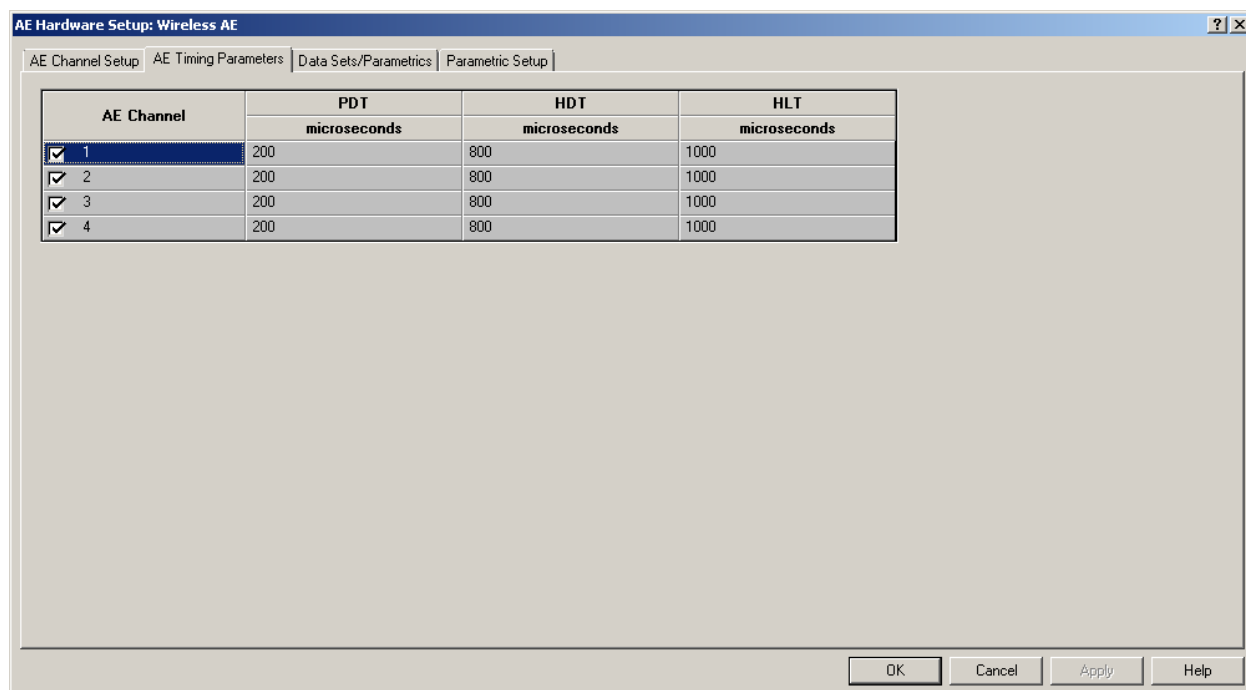


Figure 8-2. AE Timing Parameters Setup

On the 'AE Timing Parameters' page, you can enable or disable each channel and edit Peak Definition Time (PDT), Hit Definition Time (HDT), and Hit Lockout Time (HLT).

PDT, HDT, and HLT are timing parameters of the signal measurement process. Their function is discussed in full in Appendix 1. In brief, a proper setting of the PDT ensures correct identification of the signal peak for risetime and peak amplitude measurements. Proper setting of the HDT ensures that each AE signal from the structure is reported as one and only one hit. With the proper setting of the HLT, spurious measurements during the signal decay are avoided. Given the wireless bandwidth limitations, HLT should be set to higher values than other AE systems so as to prevent saturating the system with AE data activity.

8.1.3.4 Data Sets/Parametrics

AEwin features user-selectable data sets. This means that if you do not plan to use all the measurable AE parameters, you can leave some out and enjoy the compensating advantages of higher data throughput speed and more economical use of storage space. Selecting Data Sets/Parametrics from the Hardware Setup dialog box displays the Data Sets/Parametrics page that follows.

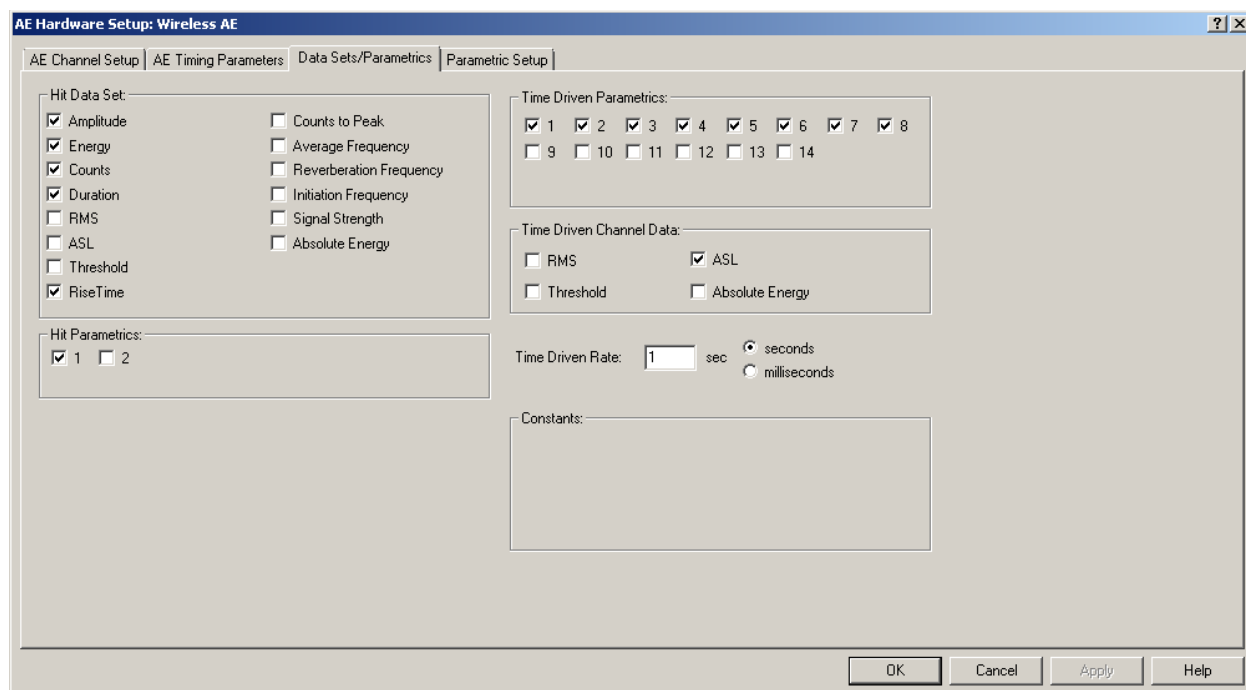


Figure 8-3. Data Sets/Parametrics

Hit Data Set

This box allows you to select which measured parameters are to be included in the description of each AE hit. The parameters to be included are so marked with a check in the checkbox next to the parameter. Parameters can be included/excluded from the data set by selecting or deselecting the checkbox next to the parameter. "Time of test" is not shown because it is always included in the data set.

Hit Parametrics

This box allows you to select which parametric inputs are to be included in the description of each hit. A maximum of two parametric inputs (e.g., external load signals) can be included in the Hit data set. These items, however, must also be included in the Time-driven Data Set; otherwise, an error message will appear.

Time-driven Parametrics and Time-driven Channel Data

Additional to the hit descriptions, AEwin allows the recording of RMS, ASL, Threshold, Absolute Energy, and Parametric inputs at regular time intervals even in the absence of AE activity. These are called Time-driven messages. This box allows you to select a set of features for them.

Time-driven Rate –Time-driven data is useful for leak monitoring and for recording changes in load and background noise levels. Also, time-driven data can be used for updating hit data graphs involving parametric values or time, even when no-hits are coming in. Each 1282 node reports time-driven data once every second. This control is reserved for future versions that may implement a different interval.

RMS/ASL Time Constants

RMS/ASL Time Constant – The time constant is fixed at 500ms.

8.1.3.5 Parametric Setup

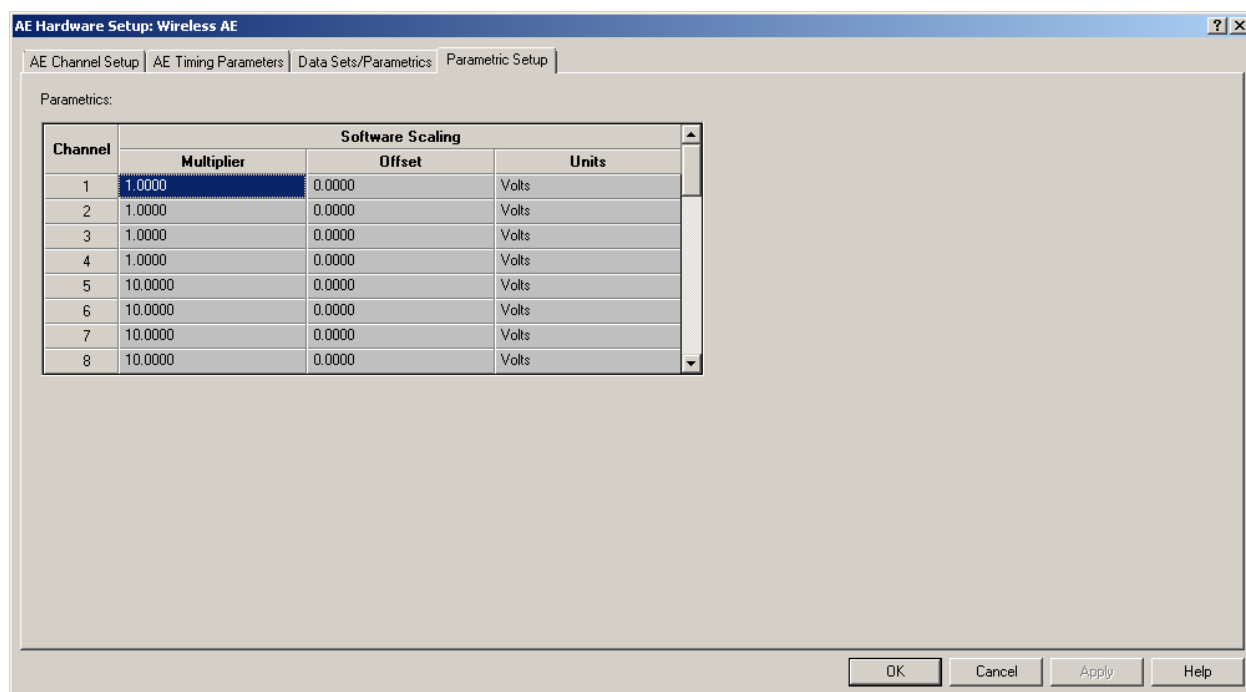


Figure 8-4. Parametric Setup

The Parametric Setup page allows you to scale the voltages measured at the parametric input such that the system displays the corresponding load or pressure values. This page also allows you to tell the system what units (kN, psi) you want to be displayed on your graph axis and printouts. The Hardware Gain and Filter are inactive (reserved for future versions).

The Parametric Scaling process used by AEwin is: $\text{Displayed Value} = (\text{Measured Voltage} \times \text{Multiplier}) + \text{Offset}$

The default values set up in the delivered Autoload file are 1.00, 0.00, and "Volts" so that the system will display the raw voltage delivered to the Parametric Input socket on the rear panel.

To scale the output of a linear load-measuring transducer, the easiest way to find the correct Multiplier and Offset is to go into a simulated test and turn on the Line Listing Display (F7). From the display, read the parametric voltages for two widely separated loads. Suppose these readings are V1 at load L1 and V2 at load L2. The correct values for Multiplier and Offset are then:

$$\text{Multiplier} = (L1 - L2) / (V1 - V2)$$

$$\text{Offset} = (V1 * L2 - V2 * L1) / (V1 - V2) = L1 - \text{Multiplier} * V1$$

Once you click OK to exit this dialog, the scaled values of the parametric input will now appear on all graph displays and data listings. Minimum and maximum values for graph axes should be specified in the scaled values, and the entered unit will appear on the graph (e.g., 0–1000 psi). The raw data on the disk will contain the original voltage readings, and the scaling will be done again each time the file is replayed.

Parametrics 5, 6,..., N+4 (where N is the number of nodes) show the battery readings for each node. Parametric 5 is the battery level of node 1, parametric 6 corresponds to the battery for node 2, and so on until node N. A multiplier of 10.0 should be used to scale the value reported from the node to the actual battery voltage.

8.2 AEwin for wireless-4 Software

8.2.1 Menus

8.2.1.1 Acquisition Setup Menu

Location...

The Location Setup dialog box is used to define and enable event groups (location groups). The wireless AE node provides support for one location group (Group #1). Enable this location group as follows:

1. Click the checkbox next to 'Group # 1' to enable the event. Only the first group is available for use with 1284.
2. Click the Distance Units button and change to a time-based unit (milliseconds or microseconds).
3. Change the Wave Velocity, Event Definition Value, and Event Lockout Value as needed.

NOTE: Events are supported, but the full location is not, so the fields just described are the only ones in use. The only Location type supported by the 4-channel wireless AE node is Zonal.

Group #	Show All Points	Location Type	Wave Velocity	Event Definition Value	Event Lockout Value	Overall Value	Timing Strategies	Hits/Event		Max Iterations
								Min	Max	
☒ 1	☐	Zonal	1000.00	250.0000	500.0000	0.0000	Primary	1	1	0
☐ 2	☐	Zonal	1000.00	0.0000	0.0000	0.0000	Primary	1	1	0
☐ 3	☐	Zonal	1000.00	0.0000	0.0000	0.0000	Primary	1	1	0
☐ 4	☐	Zonal	1000.00	0.0000	0.0000	0.0000	Primary	1	1	0
☐ 5	☐	Zonal	1000.00	0.0000	0.0000	0.0000	Primary	1	1	0
☐ 6	☐	Zonal	1000.00	0.0000	0.0000	0.0000	Primary	1	1	0
☐ 7	☐	Zonal	1000.00	0.0000	0.0000	0.0000	Primary	1	1	0
☐ 8	☐	Zonal	1000.00	0.0000	0.0000	0.0000	Primary	1	1	0

Buttons: Guard Sensors..., Material..., Attenuation..., Distance Units..., Event Hit Filter..., Location View..., OK, Cancel

Distance Units: Milliseconds

Figure 8-5. Location Dialog Box

Guard Sensors...

Clicking this button will open the Guard Sensors dialog box. Guard sensors can be used to lockout events in the location group whenever a hit is first detected on a guard channel. It is designed for situations in which one wishes to ignore events that originate outside of the location group. **NOTE:** Only the first guard sensor setup is used in the 4-chan wireless AE node.

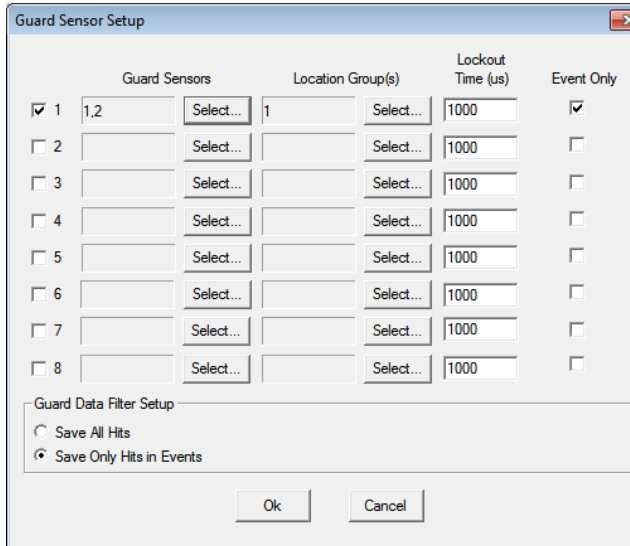


Figure 8-6. Guard Sensors Dialog Box

Event Hit Filter...

Clicking this button will open the Event Hit Filter dialog box shown below. The Hit Filter can be used for Local Area Monitoring. It is used in situations where you wish to ignore events that don't register a minimum number of hits.

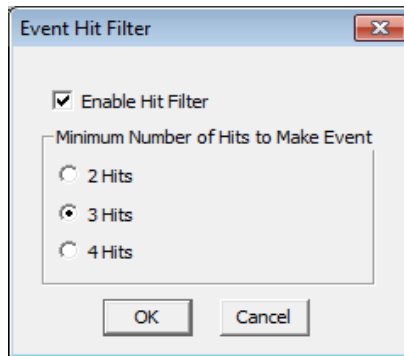


Figure 8-7. Event Hit Filter set up

For example, if there is a certain area on a structure that needs to be monitored, the sensors on the four channels of the node can be arranged around it. Click the checkbox next to 'Enable Hit Filter' and select '4 Hits'. Now, if all four sensors detect AE activity in the area being monitored, the event will be recorded. If three or fewer sensors detect activity, the event will be ignored.

8.2.1.2 Wireless Menu

This menu handles wireless communication between the AEwin and the wireless node, except for starting, pausing, and stopping/aborting acquisition, which is explained in section 4.4.4. The status of the wireless communication prompted by these menu items will show up in a Wireless Status window shown below.

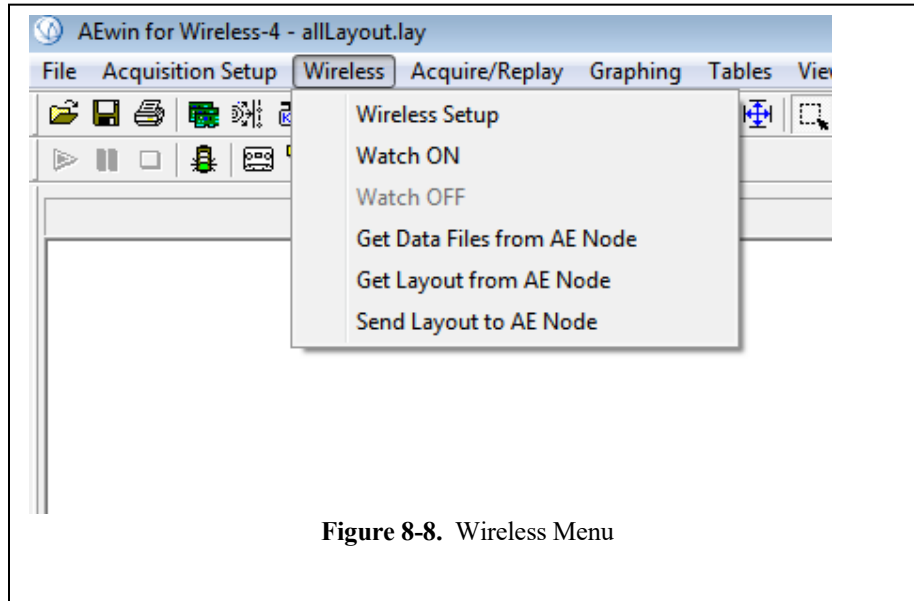


Figure 8-8. Wireless Menu

Wireless Setup

This selection opens a dialog box shown below where you can choose the frequency of the wireless modules. Different countries have varying regulations on the frequency bands they allow for wireless transmissions. We will ship you the appropriate modules for your region. 900MHz modules have a far greater range than the 2.4GHz and are more capable of penetrating through physical obstructions. Unfortunately, they are not allowed in some countries, whereas 2.4GHz modules are allowed throughout most of the world.

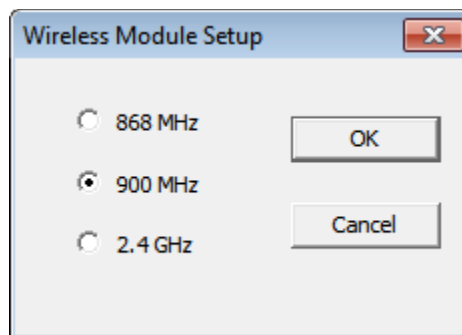


Figure 8-9. Wireless Setup Dialog Box

Select the appropriate option in the dialog box and press 'OK.' You will need to restart AEwin for the changes to take effect.

NOTE: North America, South America, Australia, New Zealand, Singapore, and Israel allow the 900MHz modules while most of Europe allows the equivalent 868MHz modules. Japan is restricted to low-power 2.4GHz modules.

Watch ON

This selection enables wireless transmission of real-time AE hit and Time-Driven Data messages when the node is already actively acquiring. In order to save battery power, it's best that the node does not constantly transmit wireless data, opting instead to only record data onto the SD card, if present. However, this is a nice feature that allows you to 'peek' at the current activity at times without disrupting acquisition in progress.

When you select 'Watch ON,' AEwin will begin establishing a connection to the node. If the acquisition is currently running, the Wireless Status Box will report that the node is ACTIVE. The node will then transmit the hardware setup so AEwin can synch its settings with that of the node. The node will then start transmitting real-time acquisition data.

If the node is not in acquisition when 'Watch ON' is selected, the Wireless Status box will report that the node is IDLE and exit.

NOTE: Due to their size, waveforms will never be wirelessly transmitted. When enabled, they will only be stored in the DTA file on the SD card and can be replayed later.

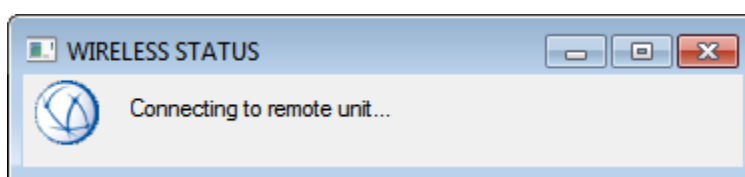


Figure 8-10. Wireless Status Window

Watch OFF

This selection ends the 'Watch' session and puts the wireless module back to sleep. The node will cease wireless transmissions without interrupting acquisition. Data will continue to be written to the SD card if present.

Get Data Files from AE Node

This selection will command the wireless node to begin transmitting all the data files stored on its SD card to AEwin, where they can then be replayed. After establishing a connection with the node, AEwin will open a dialog box where you can select which folder to save the DTA files as well as a filename prefix. The files will be sent from oldest to newest. After a successful transmission, the file will be deleted from the SD card. If a transmission error occurs, the process is aborted, and the file will remain on the SD card. You are free to attempt the transmission again at this point. This feature should only be selected when the node is not actively acquiring.

An alternative to using this method would be to physically remove the SD card from the node and insert it into your PC through an SD card reader or USB adapter. If you can reach the node, this method may be preferable as the wireless transfer can take some time to complete for large files.

Get Layout from AE Node

This selection will command the node to wirelessly transmit the layout file it has on its SD card to AEwin. A 'Save As' dialog box will appear, allowing you to choose a destination folder and filename. It's important to keep the layout file synched between AEwin and the node, so acquisition configuration and message formats are identical on both sides of the lines of communication. This item should only be selected when the node is not actively acquiring.

The alternative to getting the layout file wirelessly would be to physically remove the SD card from the node and insert it into your PC and manually copy the 'setup.lay' file to your disk drive.

Send Layout to AE Node

This selection will allow the user to send a layout file saved on the PC to the wireless node. An 'Open' dialog box will appear, allowing the user to select a saved layout file. This is a powerful feature that enables a user to change acquisition settings in AEwin and transfer them to the node without having physical access to the node itself. The layout file, once successfully transferred, will be saved on the SD card as 'setup.lay'. If a layout file of this name already exists on the SD card, it will be overwritten. This item should only be selected when the node is not actively acquiring.

The alternative to sending the layout file wirelessly would be to physically remove the SD card from the node and insert it into your PC and copy the layout file manually to the card. The file **MUST** be named 'setup.lay' on the SD card for the node to recognize it.

8.2.2 Loading a Layout File

Once a layout file is loaded in AEwin, it must also be transferred to the AE node.

To transfer wirelessly, ensure the node is powered on and click 'Wireless' on the Menu bar. Select 'Send Layout to AE Node' and choose the same layout file that is loaded in AEwin. A wireless connection will be established, and the layout file will begin sending.

8.2.3 Entering Acquisition

NOTE: Make sure you have saved and transferred the layout file to the SD card on the node using one of the methods described in 'Loading a Layout File.'

To enter acquisition wirelessly: Make sure the node is powered on. Click Acquire... in the Acquire/Replay menu. Alternately, you can click the 'Acquire' button in the Acquisition Controls toolbar or simply press F9 on the keyboard. If you wish to save acquisition data in AEwin, select 'Save to DTA file (Auto-Dump)' and enter a filename. Click the Start button. The wireless connection will be made, and the acquisition will begin with an implied 'Watch ON' session. The node will begin transmitting test data to AEwin.

8.2.4 Viewing Acquisition Data

If you started acquisition using the wireless method, the node would automatically begin transmitting data wirelessly since this start method comes with an implied 'Watch ON' command. You should see your graphs updating, and you can press F7 to enable the line display listing to view detailed AE data line by line.

If you started acquisition using the Start button on the node, click on 'Wireless' in the menu bar and select 'Watch ON.' A wireless connection will be made, and the node will begin sending data, viewable on the Line Display or any graphs you have set up.

NOTE: No waveforms are ever wirelessly transmitted during acquisition due to their large size.

If you plan to run acquisition for an extended period of time and don't need a real-time display, click on 'Wireless' in the menu bar and select 'Watch OFF.' This will disable the wireless transmissions without disturbing acquisition. Data will continue to be written to the SD card and can be replayed later.

8.2.5 Exiting Acquisition

NOTE: The red ALM LED on the node will illuminate when the node receives the command to stop acquisition if there are buffers of AE data still being written to the SD card. Wait until the LED goes out before removing the SD card.

8.2.6 Replaying AE Data

To retrieve DTA files wirelessly: Click on 'Wireless' in the menu bar and select 'Get Data Files from AE Node.' Enter the filename prefix and file location to save the DTA files and click 'Save.' A wireless connection will be established, and the node will begin sending DTA files from oldest to newest.

NOTE: If you expect there to be large DTA files on the node (i.e., waveforms were enabled), and the node is accessible, it is preferable to manually copy the files by ejecting the SD card from the node and plugging it into an SD card reader on your PC. Wireless transfer can take quite some time for large files.

AE GENERAL PRACTICES

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Acoustic Emission Inspection

Dr. Adrian A. Pollock
Physical Acoustics Corporation

TECHNICAL REPORT**TR-103-96-12/89**

Acoustic Emission Inspection

Adrian A. Pollock, Physical Acoustics Corporation

ACOUSTIC EMISSIONS are stress waves produced by sudden movement in stressed materials. The classic sources of acoustic emissions are defect-related deformation processes such as crack growth and plastic deformation. The process of generation and detection is illustrated in Figure 1. Sudden movement at the source produces a stress wave, which radiates out into the structure and excites a sensitive piezoelectric transducer. As the stress in the material is raised, many of these emissions are generated. The signals from one or more sensors are amplified and measured to produce data for display and interpretation.

The source of the acoustic emission energy is the elastic stress field in the material. Without stress, there is no emission. Therefore, an acoustic emission (AE) inspection is usually carried out during a controlled loading of the structure. This can be a proof load before service, a controlled variation of load while the structure is in service, a fatigue test, a creep test, or a complex loading program. Often, a structure is going to be loaded anyway, and AE inspection is used because it gives valuable additional information about the performance of the structure under load. Other times, AE inspection is selected for reasons of economy or safety, and a special loading procedure is arranged to meet the needs of the AE test.

Relationship to Other Test Methods

Acoustic emission differs from most other nondestructive testing (NDT) methods in two key respects. First, the signal has its origin in the material itself, not in an external source. Second, acoustic emission detects movement, while most other methods detect existing geometrical discontinuities. The consequences of these fundamental differences are summarized in Table 1.

Often in NDT, there is no one method that can provide the whole solution; for cost effectiveness, technical adequacy, or both, it is best to use a combination of methods. Because acoustic emission has features that distinguish it so sharply from other methods, it is particularly useful when used in combination with them.

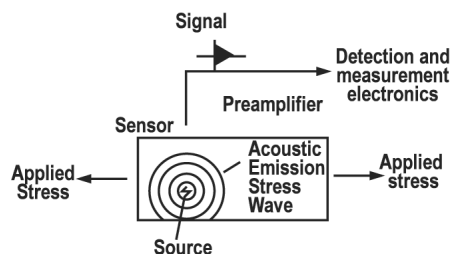


Figure 1: Basic principle of the acoustic emission method.

A major benefit of AE inspection is that it allows the whole volume of the structure to be inspected nonintrusively in a single loading operation. It is not necessary to scan the structure looking for local defects; it is only necessary to connect a suitable number of fixed sensors, which are typically placed 1 to 6 m (4 to 20 ft.) apart. This leads to major savings in testing large structures, for which other methods require removal of insulation, decontamination for entry to vessel interiors, or scanning of very large areas.

Typically, the global AE inspection is used to identify areas with structural problems, and other NDT methods are then used to identify more precisely the nature of the emitting defects. Depending on the case, acceptance or rejection can be based on AE inspection alone, other methods alone, or both together.

Acoustic Emission	Other Methods
Detects movement of defects	Detect geometric form of defects
Requires stress	Do not require stress
Each loading is unique	Inspection is directly repeatable
More material-sensitive	Less material-sensitive
Less geometry-sensitive	More geometry-sensitive
Less intrusive on plant/process	More intrusive on plant/process
Requires access only at sensors	Requires access to whole area of inspection
Tests whole structure at once	Scan local regions in sequence
Main problems: noise related	Main problems: geometry related

Table 1: Characteristics of acoustic emission inspection compared with other inspection methods

Range of Applicability

Acoustic emission is a natural phenomenon occurring in the widest range of materials, structures and processes. The largest-scale acoustic emissions are seismic events, while the smallest-scale processes that have been observed with AE inspection are the movements of small numbers of dislocations in stressed metals. In between, there is a wide range of laboratory studies and industrial testing.

In the laboratory, AE inspection is a powerful aid to materials testing and the study of deformation and fracture. It gives an immediate indication of the response and behavior of a material under stress, intimately connected with strength, damage and failure. Because the AE response of a material depends on its microstructure and deformation mode, materials differ widely in their AE response. Brittleness and heterogeneity are two major factors conducive

to high emissivity. Ductile deformation mechanisms, such as microvoid coalescence in soft steels, are associated with low emissivity.

In production testing, AE inspection is used for checking and controlling welds (Ref 1), brazed joints (Ref 2), thermocompression bonding (Ref 3), and forming operations such as shaft straightening (Ref 4) and punch press operations. In general, AE inspection can be considered whenever the process stresses the material and produces permanent deformation.

In structural testing, AE inspection is used on pressure vessels (Ref 5), storage tanks (Ref 5) pipelines and piping (Ref 6), aircraft and space vehicles (Ref 7), electric utility plants (Ref 7), bridges (Ref 7), railroad tank cars (Ref 8), bucket trucks (Ref 7), and a range of other equipment items. Acoustic emission tests are performed on both new and in-service equipment. Typical uses include the detection of cracks, corrosion, weld defects and material embrittlement.

Procedures for AE structural testing have been published by The American Society of Mechanical Engineers (ASME), the American Society for Testing and Materials (ASTM) and other organizations. Successful structural testing comes about when the capabilities and benefits of AE inspection are correctly identified in the context of overall inspection needs and when the correct techniques and instruments are used in developing and performing the test procedure (Ref 9).

Acoustic emission equipment is highly sensitive to any kind of movement in its operating frequency range (typically 20 to 1200 kHz). The equipment can detect not only crack growth and material deformation, but also such processes as solidification, friction, impact, flow and phase transformations. Therefore, AE techniques are also valuable for:

- In-process weld monitoring (Ref 10)
- Detecting tool touch and tool wear during automatic machining (Ref 10)
- Detecting wear and loss of lubrication in rotating equipment (Ref 10) and tribological studies (Ref 11)
- Detecting loose parts and loose particles (Ref 12)
- Detecting and monitoring leaks, cavitation and flow (Ref 12, 13)
- Monitoring chemical reactions, including corrosion processes (Ref 14), liquid-solid transformations and phase transformations (Ref 14)

When these same processes of impact, friction, flow, and so on, occur during a typical AE inspection for cracks or corrosion, they constitute a source of unwanted noise. Many techniques have been developed for eliminating or

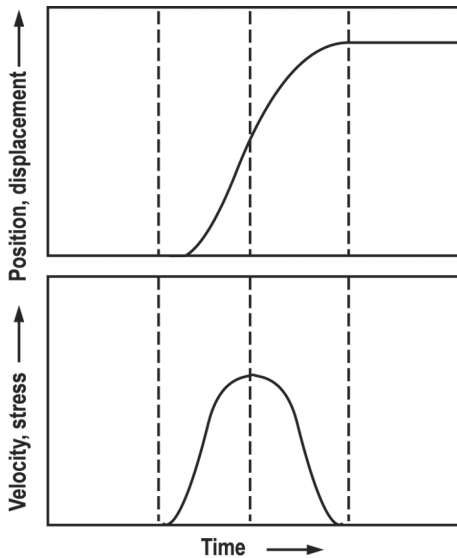


Figure 2: Primitive AE wave released at a source. The primitive wave is essentially a stress pulse corresponding to a permanent displacement of the material.

discriminating against these and other noise sources. Noise has always been a potential barrier to AE applicability. This barrier is constantly being explored and pushed outward, bringing previously impractical projects into the realm of feasibility.

Acoustic Emission Waves and Propagation

The primitive wave released at the AE source is illustrated in Figure 2. The displacement waveform is basically a steplike function corresponding to the permanent change associated with the source process. The corresponding velocity and stress waveforms are basically pulselike. The width and height of the primitive pulse depend on the dynamics of the source process. Source processes, such as microscopic crack jumps and precipitate fractures, are often completed in a few microseconds or fractions of a microsecond, so the primitive pulse has a correspondingly short duration. The amplitude and energy of the primitive pulse vary over an enormous range from submicroscopic dislocation movements to gross crack jumps. The primitive wave radiates from the source in all directions, often having a strong directionality depending on the nature of the source process, as shown in Figure 3. Rapid movement is necessary if a significant amount of the elastic energy liberated during deformation is to appear as an acoustic emission.

The form of the primitive wave is profoundly changed during propagation through the medium, and the signal emerging from the sensor has little resemblance to the original pulse. This transformation of the AE waveform is important, both to the researcher interested in source function analysis and to the practical NDT inspector interested in testing structures. The researcher who wants to determine the original source waveform uses broadband sensors and performs a detailed analysis of the early part of

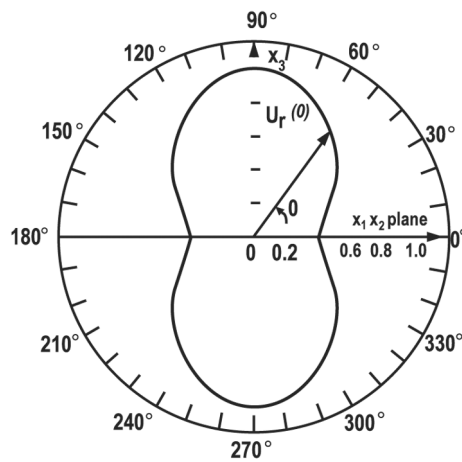


Figure 3: Angular dependence of acoustic emission radiated from a growing microcrack. Most of the energy is directed in the 90 and 270° directions, perpendicular to the crack surfaces.

the received signal. This is an important but very demanding line of inquiry. It may take an hour of computer time to process a single waveform. Most materials-oriented researchers, along with NDT inspectors, are interested in broader statistical features of the AE activity and do not need to know the precise details of each source event. They use narrowband sensors and electronic equipment that measures only a few features of the received waveform but is able to process hundreds of signals per second. The salient wave propagation factors are different for these two lines of work, as discussed below.

Factors in Source Function Analysis. The relationship between the source pulse and the resulting movement at the point of detection has been intensively studied during the last 10 to 15 years. Research groups at the NDT Centre at Harwell, UK (Ref 15), the National Bureau of Standards (Ref 16), Cornell University (Ref 17), and the University of Tokyo (Ref 18) have led the attack on this surprisingly difficult problem. A long-term goal of these studies has been to learn how to calculate a description of the source event from observation of the output of a distant sensor.

The difficulty of the problem is indicated in Figure 4, which shows the vertical component of the surface movement at point B, resulting from the abrupt application of a vertical force at point A on a semi-infinite body. Even with this simple geometry and source function, the resulting waveform is quite complicated. In the case of a plate, it is more complicated yet, because the second surface also plays its role in the elastodynamics of the wave propagation process. In plates, the motion at the point of detection depends strongly on the ratio of source distance to plate thickness.

In addition, the source function is not instantaneous, it will in fact be a force dipole and/or double couple rather than a point force, its orientation is in general unknown, and horizontal as well as vertical components of motion need to be considered. With these complications, it has understandably taken many years of effort to de-

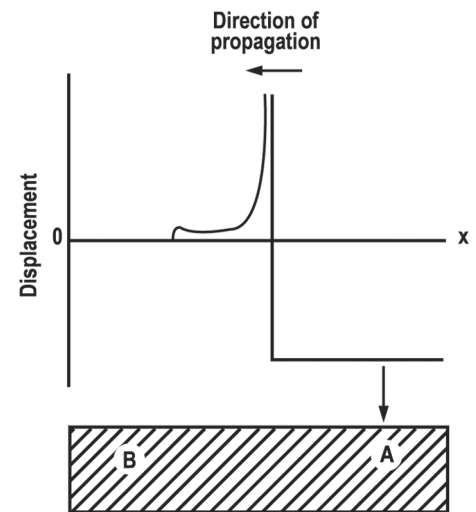


Figure 4: Displacement waveform produced by an abrupt application of a downward force at point A. The waveform can be viewed as either a "snapshot" of surface displacement (in which case x = position) or as a graph of the displacement at point B as a function of time (x = time).

velop mathematical theory, computational tools, and experimental techniques equal to the task of calculating the source function.

In recent years, the leading laboratories have attained the ability to quantify crack growth increments, orientations and time characteristics in some of the simpler specimen geometries ordinarily encountered (Ref 19). High-fidelity sensors must be used, and the analysis involves only the first part of the AE waveform, which is recorded in full detail with a high-performance transient recorder. This effort in the characterization of the source pulse has been one of the leading areas of AE research, and it can be expected eventually to yield returns in applied NDT.

Factors in Source Location and Typical AE Measurements. Whereas source function analysis utilizes only the first part of the AE waveform, mainstream AE technology accepts the waveform in its entirety. The later part of this waveform is made up of many components reaching the sensor by a variety of paths. Figure 5 illustrates this principle, but shows only a few of the indefinitely large number of possible paths by which the wave can reach the sensor. Typically, the highest peak in the waveform is produced, not by the first component, but by the constructive interference of several of the later components. The AE wave bounces around the testpiece, repeatedly exciting the sensor until it finally decays away. This decay process may

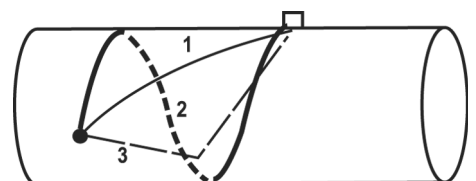


Figure 5: Three possible paths from source to sensor in a water-filled pipe. 1, direct path; 2, spiral path; 3, waterborne path

take 100 m s in a highly damped, nonmetallic material or tens of milliseconds in a lightly damped, metallic material—much longer than the source event, which is usually finished in a few microseconds or less.

It is important to understand that the shape of the received waveform is fundamentally the result of these wave propagation processes. Other important aspects of wave propagation in typical AE testing are attenuation and wave velocity. Attenuation is the loss of signal amplitude due to geometric factors and material damping as the wave travels through the material (Ref 20). Attenuation governs detectability at a distance and is therefore an important factor in choosing sensor positions and spacing. Acoustic emission procedures typically call for attenuation measurements to be made before a test and specify permissible sensor spacing based on these measurements.

Wave velocity is an additional factor to be considered when AE technology is used for source location. Source location is an important technique that is widely used both in laboratory studies and in structural testing. It is particularly significant in testing large structures, for which AE inspection is used to identify active regions for conclusive follow-up inspection with other NDT methods. Large cost savings have been realized through this combination of global AE inspection and focused inspection by other methods.

There are several strategies for source location. Zone location places the source within a broad area. Point location places the source precisely, by calculating from the relative arrival times of the AE wave at several sensors. Wave velocity is involved in these calculations. The attainable accuracy is governed by wave propagation processes and depends on such factors as geometry, plate thickness, and contained fluids. In effect, these factors render the wave velocity uncertain and this leads to errors in source location. In favorable cases, the attainable accuracy is better than 1% of the sensor spacing; in unfavorable cases, worse than 10%. The wave propagation effects underlying these variations are reviewed in Ref 20.

Acoustic Emission Sensors and Preamplifiers*

The key element in an AE resonant sensor is a piezoelectric crystal (transducer) that converts movement into an electrical voltage. The crystal is housed in a suitable enclosure with a wear plate and a connector, as shown in Fig. 6. The sensor is excited by the stress waves impinging on its face, and it delivers an electrical signal to a nearby preamplifier and then to the main signal-processing equipment. The preamplifier can be miniaturized and housed inside the sensor enclosure, facilitating setup and reducing vulnerability to electromagnetic noise.

Sensor Response. One of the most sought-after properties in an AE sensor is high sensitiv-

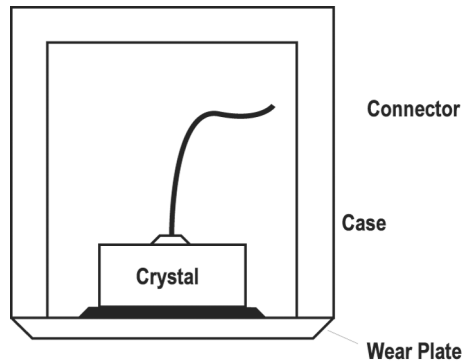


Figure 6: Typical construction of an AE resonant sensor

ity. Although high-fidelity, flat frequency response sensors are available, most practical AE testing employs resonant-type sensors that are more sensitive, as well as less costly, than the flat frequency response type. These sensors have one or more preferred frequencies of oscillation, governed by crystal size and shape. These preferred frequencies actually dominate the waveform and spectrum of the observed signal in typical AE testing.

The sensitivity calibration of AE sensors was the subject of a substantial developmental program at the National Bureau of Standards (NBS) through the late 1970s. This program has led to the routine availability of NBS-traceable plots showing the absolute sensitivity of AE sensors in volts per unit velocity as a function of frequency (Ref 21).

Acoustic Emission Waveform Transformation. In addition to the wave propagation factors discussed earlier, the transformation of a single AE waveform is further compounded by the sensor response. When a resonant sensor is excited by a broadband transient pulse, it rings like a bell at its own natural frequencies of oscillation. Therefore, the electrical signal at the sensor output is the product of this ringing, thus compounding the effects of multiple paths and multiple wave modes by which the wave travels from source to sensor. A typical AE signal from a piezoelectric sensor is shown in Fig. 7; the radical difference between this observed signal and the simple waveform at the source (Fig. 2) cannot be overemphasized.

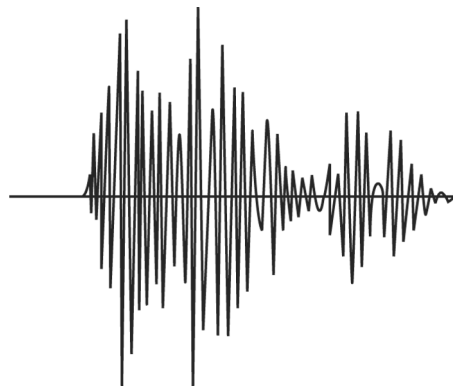


Figure 7: Typical AE burst-type waveform recorded from a flat-bottom storage tank with a transient recorder

Frequency Response. By selecting a resonant sensor from the wide range available, one can effectively choose the monitoring frequency. This is a useful feature that allows the inspector to make a suitable trade-off between the desired detection range and the prevailing noise environment. In practice, the vast majority of AE testing is well performed with sensors that are resonant at about 150 kHz.

Preamplifier Response. The signals generated by the sensor are amplified to provide a higher, more usable voltage. This is accomplished with a preamplifier, which is placed close to (or even inside) the sensor so as to minimize pickup of electromagnetic interference. The preamplifier has a wide dynamic range and can drive the signal over a long length of cable so that the main instrumentation can be placed hundreds of meters from the testpiece if necessary.

The preamplifier typically provides a gain of 100 (40 dB) and includes a high-pass or bandpass filter to eliminate the mechanical and acoustical background noise that prevails at low frequencies. The most common bandpass is 100 to 300 kHz, encompassing the 150 kHz resonant frequency of the most commonly used sensor. Other operating frequencies can be used, but there are limitations. At lower frequencies, there are increasing problems with mechanical background noise. At higher frequencies, the wave attenuates (damps out) more rapidly, and the detection range of the sensor will be smaller. Choice of operating frequency is therefore a trade-off between noise and detection range. Lower frequencies are used on pipelines, where detection range is at a premium, and in geological work because rocks and soils are highly attenuating. Higher frequencies are used to test steam lines in electricity generating stations, where background noise is unusually high.

Attainable Sensitivity. Preamplifiers inevitably generate electronic noise, and it is this noise that sets the ultimate limit to the smallest movement detectable with AE equipment. The smallest signal that can be detected is about 10 mV at the transducer output, corresponding to a surface displacement of about 25 pm (1×10^{-6} in.) for a typical high-sensitivity sensor. This sensitivity is more than enough for most practical NDT applications.

Installation. Typically, the sensor is coupled to the testpiece with a fluid couplant and is secured with tape, an adhesive bond, or a magnetic hold-down device. In some applications, however, the AE sensor may be mounted on a waveguide, as in Example 1.

After the sensor is installed and connected to the monitoring equipment, system performance is checked by “lead break” before monitoring begins. This involves the breaking of a lead pencil near the sensor to verify the response from an acoustic signal. Properly performed, the lead break delivers a remarkably reproducible signal that closely matches the “point impulse loading” source discussed above (see the section “Fac-

tors in Source Function Analysis” in this article).

Example 1: Acoustic Waveguide Sensors Used in Monitoring the Cooling of Molten Vitrified Nuclear Waste. Acoustic emission monitoring was used to help correlate cracking in vitrified high-level waste with cooling procedures. There was a need for a method capable of performing in an environment consisting of approximately 900° C (1650° F) temperatures and 500 Gy/h (50 000 rad/h) gamma radiation for the continuous monitoring of vitrified waste in canisters during cooling to detect glass cracking. Waveguide sensors about 4.6 m (15 ft) long were used; one end was submerged in the glass, and a sensing crystal and preamplifier were positioned on the other end. The signal from the sensor was passed through coaxial cables to the outside of the hot cell, where it was received by an AE monitor system for analysis. At the end of the testing, the AE sensors had been in the environment for 120 days, and the accumulated dose of gamma radiation had reached 14×10^5 Gy (14×10^7 rad). The sensors were still functioning properly.

Instrumentation Principles

During an AE test, the sensors on the testpiece produce any number of transient signals. A signal from a single, discrete deformation event is known as a burst-type signal. This type of signal has a fast rise time and a slower decay, as illustrated in Fig. 7. Burst-type signals vary widely in shape, size, and rate of occurrence, depending on the structure and the test conditions. If there is a high rate of occurrence, the individual burst-type signals combine to form a continuous emission. In some cases, AE inspection relies on the detection of continuous emission (see the sections “Mechanisms of AE Sources” and “Leak Testing” in this article).

The instrumentation of an AE inspection provides the necessary detection of continuous emissions or detectable burst-type emissions. Typically, AE instrumentation must fulfill several other requirements:

- The instrumentation must provide some measure of the total quantity of detected emission for correlation with time and/or load and for assessment of the condition of the testpiece
- The system usually needs to provide some statistical information on the detected AE signals for more detailed diagnosis of source mechanisms or for assessing the significance of the detected signals
- Many systems can locate the source of detectable burst-type emissions by comparing the arrival times of the wave at different sensors. This is an important capability of great value in testing both large and small structures
- The systems should provide a means for discriminating between signals of interest and noise signals from background noise sources such as friction, impact, and electromagnetic interference

Instruments vary widely in form, function, and price. Some are designed to function automati-

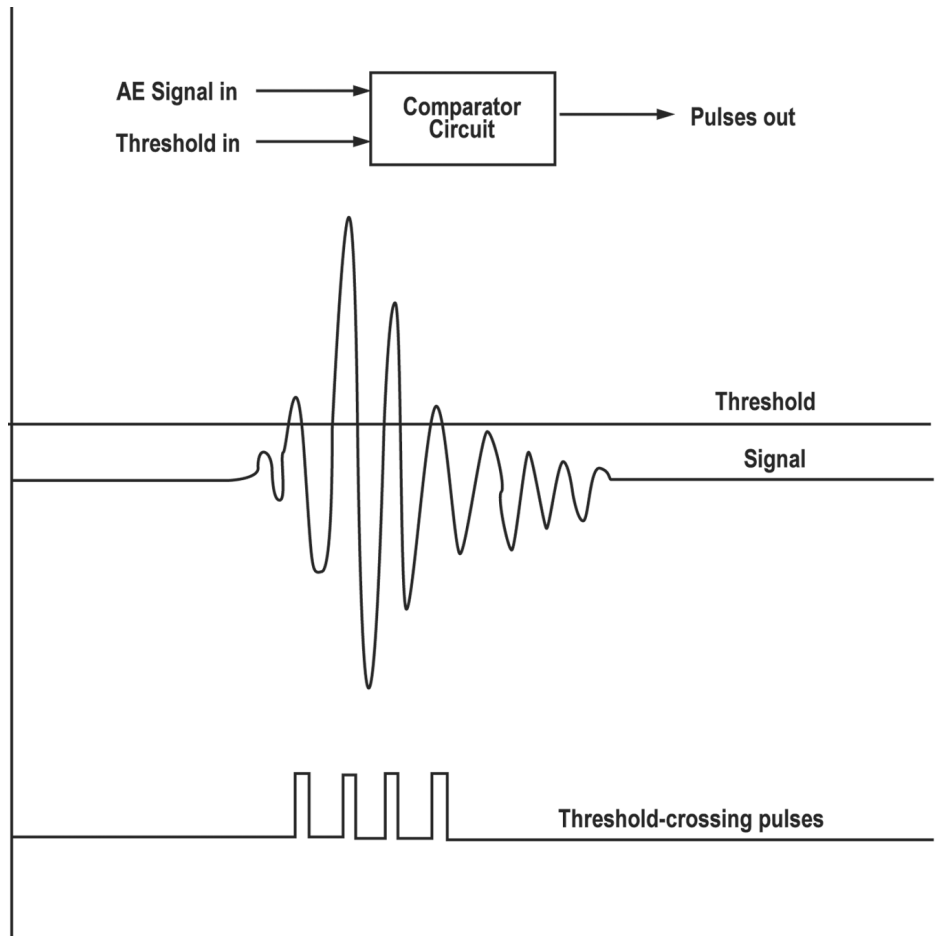


Figure 8: Principle of AE signal detection and threshold-crossing counts

cally in automated production environments. Others are designed to perform comprehensive data acquisition and extensive analysis at the hands of skilled researchers. Still others are designed for use by technicians and NDT inspectors performing routine tests defined by ASME codes or ASTM standards.

Signal Detection and Emission Counts. After sensing and preamplification, the signal is transmitted to the main instrument, where it is further amplified and filtered. Next is the critical step of detecting the signal. This is accomplished with a comparator circuit, which generates a digital output pulse whenever the AE signal exceeds a fixed threshold voltage. The relationship between signal, threshold, and threshold-crossing pulses is shown in Fig. 8. The threshold level is usually set by the operator; this is a key variable that determines test sensitivity. Depending on instrument design, sensitivity may also be controlled by adjusting the amplifier gain.

One of the oldest and simplest ways to quantify AE activity is to count the threshold crossing pulses generated by the comparator (Fig. 8). These acoustic emission counts are plotted as a function of time or load, either as an accumulating total or in the form of a count rate histogram. The all-hardware AE systems of the early 1970s could draw these count and count rate displays on x-y recorders as the test proceeded, and much of the early AE literature presents results in this

form. Figure 9, a typical plot of this type, shows cumulative counts as a function of applied load during a rising-load test on a precracked specimen of high-strength steel. The vertical scale is 10 000 counts full-scale. The vertical steps on the first parts of the plot are individual AE events. The larger events score several hundred counts each. By 35 kN (8000 lbf), 10 000 counts have been accumulated. The pen resets to the bottom of the graph, and resumes plotting. As

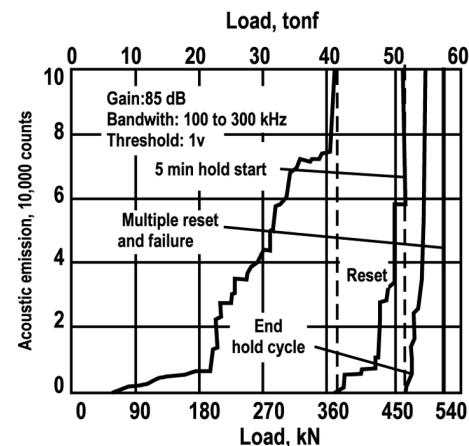


Figure 9: Acoustic emission from a welded three-point bend specimen of 12% Ni maraging steel. Steps in the curve are discrete, burst-type emissions caused by plastic zone growth and later, crack front movement.

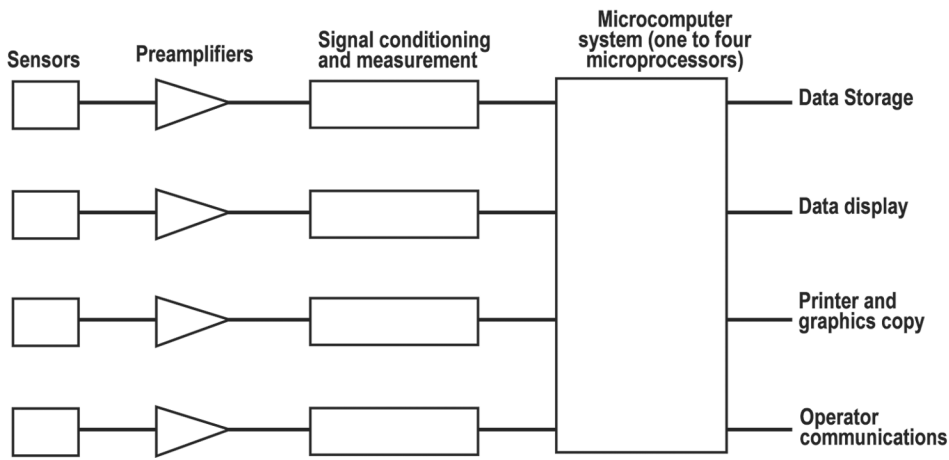


Figure 10: Generic block diagram of a four-channel acoustic emission system

the load rises, the AE rate increases, and the individual events are no longer discernible on the plot. As the specimen approaches failure, there are multiple resets of the pen corresponding to the generation of hundreds of thousands of AE counts.

Hit-Driven AE Systems. All-hardware systems reached an apex of development in the late 1970s, but they were eventually superseded by computer-based systems. The development of AE technology coincided with the development of computers, and computers were probably used earlier for AE inspection than for any other NDT method. Computers were first used for AE multichannel source location systems around 1970. Although source location was the first task (and a very advanced one), computers soon came into use for the more general purposes of AE data storage, analysis, and display. At the same time, personnel involved in AE inspection became interested in other signal features of burst-type emissions beyond the threshold-crossing counts (see the section "Signal Measurement Parameters" in this article).

These trends led to a new principle of AE instrumentation that has dominated the technology ever since. This principle involves the measurement of key parameters of each hit, that is, each AE signal that crosses the threshold. A digital description of each hit is generated by the front-end hardware and is passed in sequence with other hit descriptions through a computer system, which provides data storage, a variety of graphical displays, and replay for posttest analysis.

A generic block diagram is shown in Fig. 10, and a typical modern system is shown in Fig. 11. The larger, multichannel systems divide the data-processing tasks among many microprocessors. In the system shown in Fig. 11, for example, a separate microprocessor serves each pair of signal measurement channels. The highest priority for this microprocessor is to read the results of each signal measurement as soon as the measurement process is completed, so that the measurement circuitry can be reset for the next event. The front-end microprocessor can rapidly store

several hundred hit descriptions in its buffer, pending further processing. With this parallel processing architecture, added channels will automatically bring added data processing power. With the front-end buffers supplemented by other, even larger buffers in the later stages of the microcomputer network, the system has the versatility to absorb sudden surges of AE activity and to handle widely varying data rates in an optimum manner (Ref 22).

Signal Measurement Parameters. The five most widely used signal measurement parameters are counts (Fig. 8), amplitude, duration, rise time, and the measured area under the rectified signal envelope (MARSE) (Fig. 12). Some tests make do with fewer parameters, and some tests use others, such as true energy, counts-to-peak, average frequency, or spectral moment.

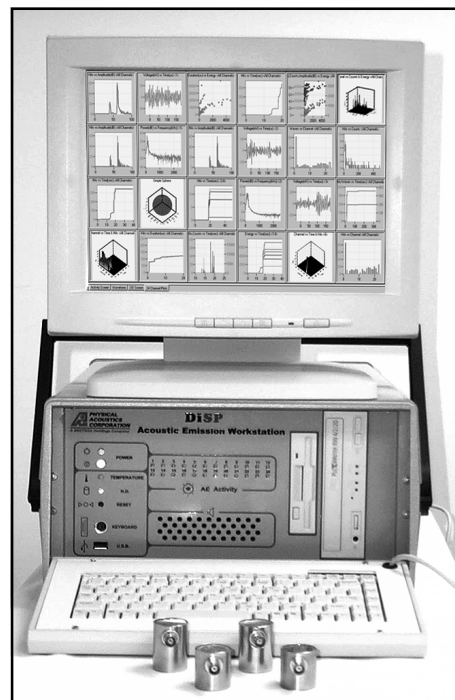


Figure 11: Typical general purpose AE instrument - a 24-channel data acquisition system with computer for data display, storage, and analysis. Courtesy of Physical Acoustics Corporation

However, the five principal parameters have become well standardized and accepted through the market processes of the last 10 years.

Along with these signal parameters, the hit description passed to the computer typically includes important external variables, such as the time of detection, the current value of the applied load, the cycle count (if it is a cyclic fatigue test), and the current level of continuous background noise. The length of the total hit description is usually between 20 and 40 bytes.

Amplitude, A , is the highest peak voltage attained by an AE waveform. This is a very important parameter because it directly determines the detectability of the AE event. Acoustic emission amplitudes are directly related to the magnitude of the source event, and they vary over an extremely wide range from microvolts to volts. Of all the conventionally measured parameters, amplitude is the one best suited to developing statistical information in the form of distribution functions (Ref 23). The amplitudes of acoustic emissions are customarily expressed on a decibel (logarithmic) scale, in which 1 mV at the transducer is defined as 0dBae, 10 mV is 20dBae, 100 mV is 40dBae, and so on.

Counts, N , are the threshold-crossing pulses (sometimes called ringdown counts) discussed above. This is one of the oldest and easiest ways of quantifying the AE signal. Counts depend on the magnitude of the source event, but they also depend strongly on the acoustic properties and reverberant nature of the specimen and the sensor.

MARSE, sometimes known as energy counts, E , is the measured area under the rectified signal envelope. As a measure of the AE signal magnitude, this quantity has gained acceptance and is replacing counts for many purposes, even though the required circuitry is relatively complex. MARSE is preferred over counts because it is sensitive to amplitude as well as duration, and it is less dependent on threshold setting and operating frequency. Total AE activity must often be measured by summing the magnitudes of all the detected events; of all the measured parameters, MARSE is the one best suited to this purpose.

Duration, D , is the elapsed time from the first threshold crossing to the last. Directly measured in microseconds, this parameter depends on source magnitude, structural acoustics, and reverberation in much the same way as counts. It is valuable for recognizing certain long-duration source processes such as delamination in composite materials (Ref 24), and it can be useful for noise filtering and other types of signal qualification.

Rise time, R , is the elapsed time from the first threshold crossing to the signal peak. Governed by wave propagation processes between source and sensor, this parameter can be used for several types of signal qualification and noise rejection.

Multichannel Considerations. Measure-

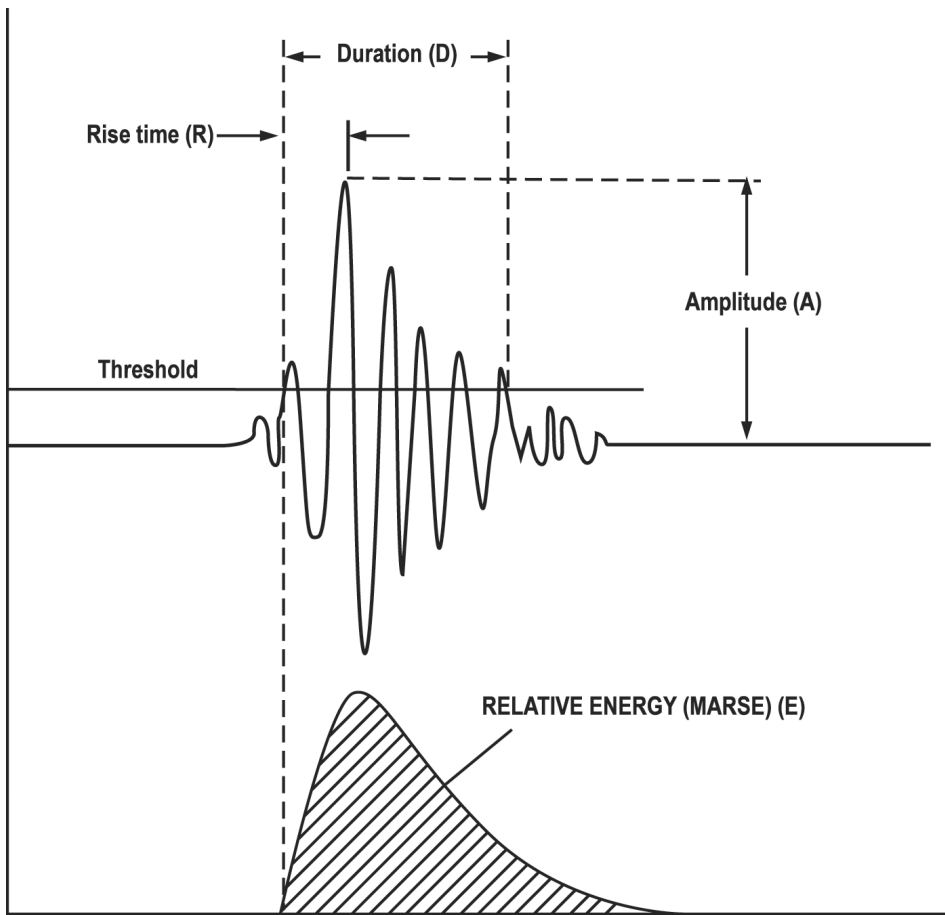


Figure 12: Commonly measured parameters of a burst-type acoustic emission signal

ment of the signal proceeds simultaneously on every channel that detects (is hit by) the AE wave. Acoustic emission systems are available in sizes from 1 channel to over 100 channels, depending on the size and complexity of the structure to be tested. Typical laboratory systems have 2 to 6 channels, while most structural tests are accomplished with 12 to 32 channels.

An individual AE event may hit just one channel or it may hit many channels, depending on the strength of the event, the wave attenuation in the structure, and the sensor spacing. Therefore, an early task for the multichannel system is to determine whether a group of closely spaced hits on different channels is from the same source event. Depending on the system design, this can be accomplished either in hardware or in software. The second, third, and later hits from a source event can be either retained for the purposes of source location or discarded to keep the data clean and simple. After this task of event/hit identification has been performed, the system can deal in event descriptions as well as hit descriptions. The event description usually includes channel identification and relative timing information for all the channels involved, along with the signal characteristics of the first hit and perhaps the other hits as well.

The stream of hit (or event) descriptions is passed through a central processor that coordinates the tasks of data storage, display, and operator communications. In larger systems, these

tasks can be divided among several processors. In many systems, the entire stream of hit descriptions is stored to disk; this provides unlimited posttest analysis capability. Full data storage is a vital aspect of applied AE technology. It reduces dependence on the on-site operator for ultimate test results, allowing him to concentrate on the vital task of correct data collection (Ref 11).

Data Displays. A software-based, hit-driven AE system can produce many types of graphic displays. The operator is not limited to what can be observed during the test, because the results can be refined, filtered, and redisplayed in any manner during the posttest analysis.

Broadly, AE data displays can be classed as:

- History plots that show the course of the test from start to finish
- Distribution functions that show statistical properties of the emission
- Channel plots showing the distribution of detected emissions by channel
- Location displays that show the position of the AE source
- Point plots showing the correlation between different AE parameters
- Diagnostic plots showing the severity of AE indications from different parts of the structure

Some of these generic display types are illustrated in Fig. 13.

Figures 13(a) and (b) show history plots of AE data versus time in cumulative and rate form, respectively. A cumulative plot is the more convenient format for reading off a total emission quantity, while a rate plot highlights the changes in activity that occur during the test.

Figure 13(c) is a history plot of AE data versus load. This is the most fundamental plot because it directly relates cause to effect. This type of plot is especially useful for separating good parts from bad; bad parts characteristically begin to emit at lower loads and give more emission than good parts at all load levels. This basic plot of AE data versus load is also the best way to display the Kaiser and Felicity effects, as shown in Fig. 14.

Figures 13(d) and (e) show the cumulative and differential forms, respectively, of the amplitude distribution function. The x-axis shows amplitude, and the y-axis shows how many hits had that amplitude (differential form) or exceeded it (cumulative form). The differential amplitude distribution (Fig. 13e) is useful for distinguishing between deformation mechanisms and for observing changes in AE intensity as the test proceeds. The cumulative form (Fig. 13d) is more useful for quantitative modeling and for assessing how the detectability of AE will be affected by changes in test sensitivity. The amplitude distribution is a standard AE display, and the underlying theory is well developed (Ref 23). Distribution functions using the other signal measurement parameters are also employed for special purposes.

Figure 13(f) is a planar source location display. This display is basically a map of the structure, with the computed location of each emission event shown as a single point in the appropriate position. Sensor locations are shown as large dots, providing a reference frame. The eye is drawn to clusters of located events, which correspond to the most active sources, typically structurally significant defects.

Figure 13(g) is a point plot of counts (or duration) versus amplitude. Each hit is shown as one point on the display, and its position shows information about the size and shape of the waveform. This type of display is used for data quality evaluation, specifically for identifying some commonly encountered types of unwanted noise (Ref 25). Acoustic emission signals from impulsive sources typically form a diagonal band running across this display. Noise signals from electromagnetic interference fall below the main band (circled area, lower right, Fig. 13g) because they are not prolonged by acoustic reverberation. Noise signals from friction and leaks fall above the main band (circled area, upper left, Fig. 13g) because the source process is extended in time, not a short impulse. This is only one of the many point plots that have proved useful in practical AE testing.

The typical software-based system can generate many displays simultaneously in memory while the test is running, presenting them to the

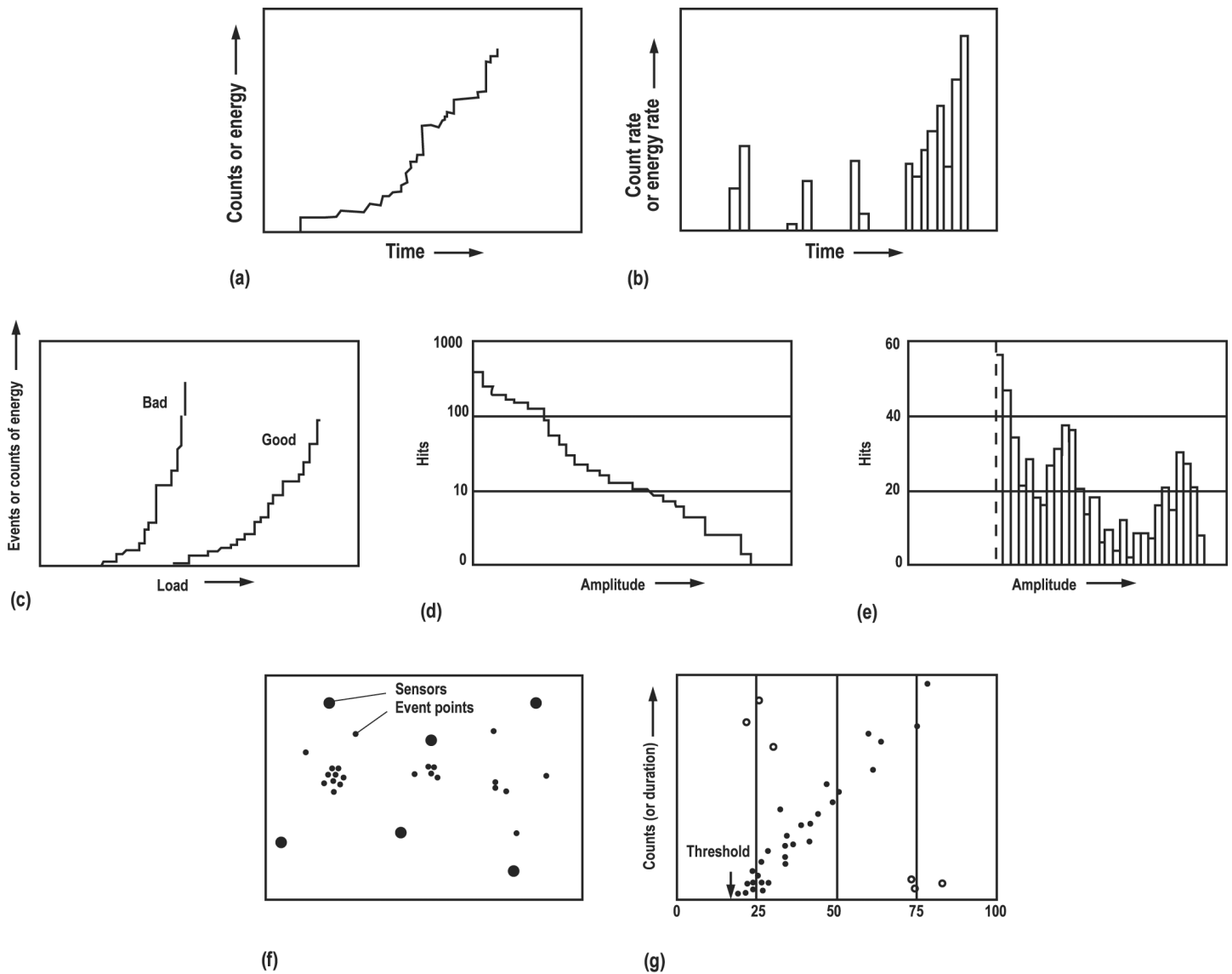


Figure 13: Typical AE data displays. (a) History plot of the cumulative count or energy. (b) History plot of the count rate or energy rate. (c) History plot of AE data versus load. (d) Cumulative amplitude distribution showing the number of hits that exceeded an amplitude. (e) Differential amplitude distribution showing the number of hits of a particular amplitude. (f) Planar source location display. (g) Point plot of counts (or duration) versus amplitude

operator upon demand. In addition to these graphic displays, the system may present tabulated data and/or listings of the individual event or hit descriptions.

Special-Purpose AE Systems. The software-based, hit-driven AE system has the architecture of choice for application development and general-purpose laboratory and structural testing, but not all AE systems require this kind of computational power and versatility of display. Once the needs of the test have been defined, simpler equipment is often appropriate for routine application.

Production testing can often be done with a basic, all-hardware instrument that simply measures counts or energy and trips an alarm when the emission exceeds a predetermined quantity. Automatic self-checking for good sensor contact can be incorporated into the function of such an instrument.

Resistance weld monitoring and feedback control is accomplished with all-hardware systems

that have special gates, timers, and interfaces to synchronize the AE monitoring with the operation of the weld controller. Other types of weld monitoring instruments incorporate pattern-recognition algorithms for automatically recognizing and classifying specific kinds of weld defects.

Leak testing is a major and relatively simple application of AE instrumentation (see the section "Structural Test Applications" in this article). Leak testing can be performed with instruments that measure only the root mean square (rms) voltage of the continuous emission from a leak. Sometimes, detect ability is enhanced by the occurrence of burst-type signals from particle impact or structural degradation of the local material. Small size is a major advantage when the instrument has to be carried around an industrial complex or power generating plant.

Specific Applications. Instrument manufacturers have also developed special instruments for specific, well-established applications, such as bucket truck testing and tank car testing. These

instruments are based on the applicable codes or standard procedures for performing the test. Simplification of hardware and software leads to a lower-cost instrument. Customized software provides more positive guidance and fewer operator choices, so that a lower level of skill can be used on-site and the test can be performed reliably and economically.

Noise

Precautions against interfering noise are an integral part of AE technology. Enormous progress has been made since the early days when students worked at night, using specially constructed loading machines in underground laboratories to avoid disruption of their experiments by street traffic and people moving nearby. With current technology, many tests can be performed without special measures, and a wide range of techniques have been developed to make AE inspection applicable in extremely noisy environments.

A basic starting point is the selection of an appropriate frequency range for AE monitoring. The acoustic noise background is highest at low frequencies. The 100 to 300 kHz range has proved suitable for perhaps 90% of all AE testing. In noisy environments (an electric power plant, for example), higher frequencies, such as 500 kHz, have been necessary to reduce the noise detected from fluid flow. Because higher frequencies bring reduced detection range, there is an inherent trade-off between detection range and noise elimination.

Acoustic noise sources include fluid flow in pumps and valves, friction processes such as the movement of structures on their supports, and impact processes such as rain and windblown cables striking the structure. Electrical and electromagnetic noise sources include ground loops, power switching circuits, radio and navigation transmitters, and electrical storms.

Noise problems can be addressed in many ways. First, it may be possible to stop the noise at the source. Second, it may be possible to eliminate an acoustic source by applying impedance-mismatch barriers or damping materials at strategic points on the structure. Electrical noise problems, which are often the result of poor grounding and shielding practices, can be eliminated by proper technique or by using differential sensors or sensors with built-in preamplifiers. If these measures are inadequate, the problem must be dealt with by hardware or software in the AE instrument itself.

Sensitivity adjustments, including floating-threshold techniques, can be very effective as long as they do not also cause the loss of essential AE data. Methods for selective acceptance and recording of data based on time, load, or spatial origin are well developed. Beyond this, because noise sources often give characteristically different waveforms, they can often be separated from true acoustic emissions by computer inspection of the measured signal characteristics (Ref 25). This can be accomplished immediately after measurement (front-end filtering), during the display process (graphical filtering), or after the test by playing the data through a posttest filtering program or advanced waveform analysis package.

Through the development and application of these techniques, AE inspection has been brought into service in increasingly demanding environments, and this trend is expected to continue. Examples of difficult applications in which noise elimination was key to the successful use of AE inspection include the on-line monitoring of welding (Ref 1, 26) and the detection of fatigue crack growth in flying aircraft (Ref 7).

Load Control and Repeated Loadings

Because acoustic emission is produced by stress-induced deformation of the material, it is highly dependent on the stress history of the

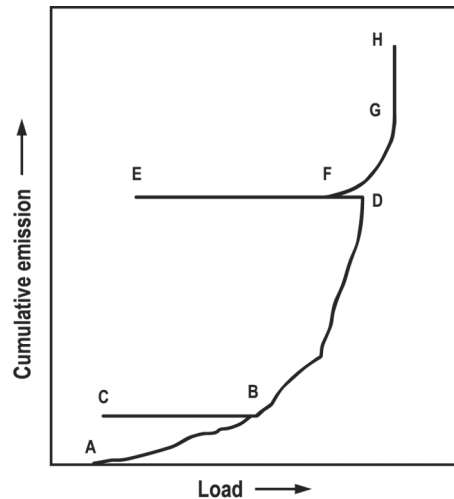


Figure 14: Basic AE history plot showing Kaiser effect (BCB), Felicity effect (DEF), and emission during hold (GH)

structure. Emission/stress/ time relationships also depend on the material and on the type of deformation producing the emission. Some materials respond almost instantly to applied stress, emitting and then quickly stabilizing. Other materials take some time to settle down after a load is applied; this is readily observed in materials that show viscoelastic properties, such as resin-matrix composites. In other cases, a constant load may produce ongoing damage, and the structure may never stabilize. An example of this is hydrogen-induced cracking, which may proceed under constant load to failure, with continual emission.

Acoustic emission testing is often carried out under conditions of rising load. The first load application will typically produce much more emission than subsequent loadings. In fact, for instantaneously plastic materials, subsequent loadings should produce no emissions at all until the previous maximum load is exceeded. This behavior was first reported by Kaiser in 1950 (Ref 27) and has been a leading influence in the development of AE test methodology. Dunegan (Ref 28) showed that for materials that obey the Kaiser effect, emission on a repeat loading will indicate that structural damage occurred between the first loading and the repeat. This became the conceptual basis of much of the AE testing of the 1970s, when the first AE field test organizations undertook periodic inspection of pressure vessels and other structures.

Recent test strategies pay much attention to emission that occurs at loads below the previous maximum and to emission that continues when the load is held at a constant level. The evidence is that structurally significant defects will tend to exhibit these behaviors, while emission related to stabilization of the structure, such as the relief of residual stress, will tend not to recur when the structure is loaded again.

Figure 14 is a generic illustration of these contrasting behaviors. Emission is observed upon

initial loading from A to B, but not upon unloading (B to C). Upon reapplying the load, there is no emission (line is horizontal) until B is reached again; this is the Kaiser effect. The load is increased to D, with more emission, and another unload-reload cycle is applied. This time, because of the higher stress levels, significant defects begin to emit at point F, below the previous maximum load. This behavior is known as the Felicity effect. It can be quantified with the Felicity Ratio (FR):

Technically, the Kaiser effect can be construed as a Felicity Ratio of 1.0 or greater. Systematic decreases in the Felicity Ratio as material approaches failure have been well documented for fiber-reinforced plastics (Ref 29) and a Felicity

$$FR = \frac{\text{Load at which emission begins again}}{\text{Previous maximum load}}$$

Ratio less than 0.95 is cause for rejection of an FRP tank or pressure vessel tested by AE inspection according to ASME Article II (Ref 30). Under ASME Article 12 (Ref 31) for the AE testing of metal pressure vessels, it is in some cases admissible to ignore AE data from the first loading of a vessel and to consider only AE data from a second loading. The basis for this is that much emission on the first loading comes from local yielding (structurally insignificant), while only the significant defects will emit on the second loading (Felicity Ratio < 1).

Figure 14 also illustrates the graphical appearance of emission continuing during a load-hold period (G to H). The Felicity effect and the occurrence of emission during load holds may share a common underlying explanation; both are associated with the unstable nature of structurally significant defects. Emission during load holds

has been known since the early years of AE inspection (Ref 28) and was incorporated in FRP evaluation criteria in the mid 1970s. In the late 1980s, emission during hold has been made the entire basis of Monsanto's successful procedure for the AE testing of railroad tank cars (Ref 8). In this interesting development, data analysis is greatly simplified because the background noise sources present during rising load are much less obtrusive during the load-hold periods.

Careful attention must be paid to the loading schedule if AE testing is to be successful. Procedures for an AE test typically specify the loads that must be applied (relative to the working load or design load) and the upper and lower limits on the loading rate. Fiber-reinforced plastic tanks and vessels must be conditioned by a period at reduced load before the AE test is conducted (Ref 30). An AE test can be invalidated if the structure is inadvertently loaded beforehand to the AE test pressure. For success in dealing with these points, there must be good communication and coordination between the personnel loading the structure and those collecting the AE data.

Acoustic Emission in Materials Studies*

Acoustic emission is a remarkable tool for studying material deformation because the information it provides is both detailed and immediate. With its sensitivity to microstructure and its intimate connection with failure processes, AE inspection can give unique insights into the response of material to applied stress. Acoustic emission analysis is most useful when used in conjunction with other diagnostic techniques, such as stress-strain measurements, microscopy, crack-opening-displacement measurements and potential drop (for crack growth), or ultrasonic damping measurements (for dislocation studies). Acoustic emission complements these techniques and offers additional information on the dynamics of the underlying deformation processes, their interplay, and the transitions from one type of deformation to another.

Many materials studies involve the development of a test approach for eventual field application. Such work can be valuable, but it is subject to the difficulty of simulating defect emissivities and other field conditions in the laboratory. Laboratory tests are often done with simple uniaxial stresses applied parallel to the rolling direction, while materials in industrial service are often subjected to complex biaxial or triaxial stress fields. In such cases, the acoustic emissions from the laboratory tests will not be a good model of the acoustic emissions from materials in industrial service.

Mechanisms of AE Sources. Needless to say, acoustic emissions are not generated by the reversible, homogeneous alteration of interatomic spacings that constitutes elastic deformation. Acoustic emissions are only generated when some abrupt and permanent change takes place somewhere in the material. Mechanisms that produce acoustic emissions in metals include the movement and multiplication of dislocations; slip; twinning; fracture and debonding of precipitates, inclusions, and surface layers; some corrosion processes; microcrack formation and growth; small and large crack jumps; and frictional processes during crack closure and opening.

The amount of AE energy released depends primarily on the size and speed of the local deformation process. The formation and movement of a single dislocation does produce an AE stress wave, but it is not a large enough process to be detected in isolation. However, when millions of dislocations are forming and moving at the same time during yielding of a tensile specimen, the individual stress waves overlap and superimpose to give a detectable result. The result is a continuous excitation of the specimen and sensor that is detectable as soon as the voltage it produces becomes comparable with the background noise. The higher the strain rate and the larger the specimen, the larger this signal becomes. This so-called continuous emission is different from burst-type emission in that the individual source events are not discernible. Continuous emission

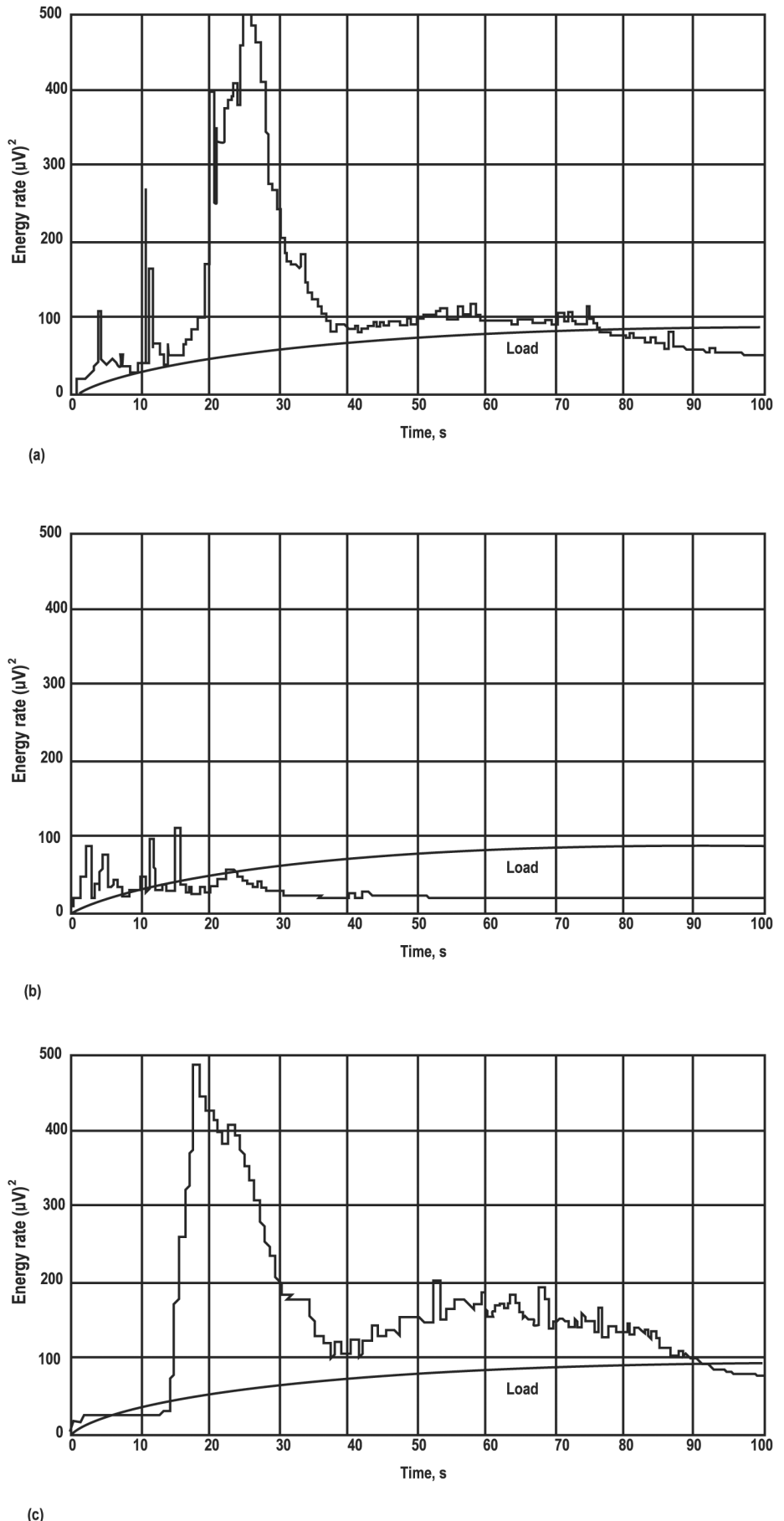


Figure 15: Acoustic emission energy rate (expressed as the mean square of sensor voltage) and load versus time. Optimum formability corresponds to the lowest emissivity in a deep-drawing steel subjected to spheroidization heat treatment. (a) Underannealed: 80% pearlite, 20% spheroids. (b) Optimally annealed: 100% spheroids. (c) Overannealed: 30% elongated spheroids and doglegs.

is best measured with rms or energy rate measuring circuitry.

Continuous emission from the plastic deformation of steels, aluminum alloys, and many other metals has been extensively studied, and there have been many detailed findings relating acoustic emissions to dislocation activity and precipitates, microstructure, and materials properties (Ref 32). Such studies can yield valuable insights for alloy and material development. Most studies have focused primarily on continuous emission during and after yield; burst-type emissions sometimes observed in the nominally elastic region are less well explained.

The following example illustrates the dependence of continuous emission on microstructure. The fracture of small-scale precipitates (in this case, pearlite lamellae) generates continuous emission, which can be related to the microstructure that results from heat treating.

Example 2: Relation of Acoustic Emissions With the Optimum Heat Treating of Ferritic/Pearlitic Steel. Figure 15 illustrates the dependence of continuous acoustic emissions on the microstructure of a deep-drawing ferritic/pearlitic steel subjected to a spheroidizing heat treatment to improve its formability. Data are shown from representative underannealed, optimally annealed, and overannealed conditions. Figure 15 shows AE energy rate as a function of time from dog-bone tensile specimens pulled to failure in a screw-driven test machine. All the graphs display peaks around the yield region, a common feature in the high-sensitivity tensile testing of unflawed specimens. Figure 15 also shows a second, shallow peak at higher strain levels.

The interesting result is that the optimally annealed specimen shows a much smaller peak (gives much less emission) than the other two specimens. The explanation is found by carefully relating AE behavior to microstructural deformation processes. It is known that dislocations can pile up against pearlite lamellae during plastic deformation, eventually causing the lamellae to fracture. This fracture of pearlite lamellae is believed to be the cause of the first peaks in Fig. 15.

With the test material in the underannealed condition, microscopy reveals the presence of many untransformed pearlite lamellae that can intercept the moving dislocations, so the peak is high. With the test material in the optimally annealed condition, microscopy shows that virtually all the lamellae have been transformed to spheroids. These have a smaller cross-sectional area and present less of an obstacle to the moving dislocations, so deformation can proceed without breaking pearlite. Ductility is enhanced, and there is very little emission from this optimally annealed material.

With the test material in the overannealed condition, microscopy shows that additional carbon has come out of solution, growing the spheroids and forming doglegs at the grain boundaries. These

larger particles interfere more strongly with dislocation motion and produce larger emissions when fractured, so the emission peak is strong again. It is an interesting result that the optimum material condition is the condition of lowest emissivity, suggesting that AE inspection could be used for inspection and quality control of this material as well as for research.

Acoustic emission from crack growth is of the greatest interest for practical NDT applications of the AE phenomenon. By virtue of the stress concentrations in their vicinity, cracks and other defects will emit during rising load, while unflawed material elsewhere is still silent. Acoustic emissions from crack initiation and growth have been extensively reported in the literature. Many of these reports deal with specialized forms of crack growth, such as fatigue, stress-corrosion cracking, and hydrogen embrittlement (Ref 33).

It is useful to distinguish between AE signals from the plastic zone at the crack tip and AE signals from movement of the crack front itself. Growth of the plastic zone typically produces many emissions of rather low amplitude. These emissions are typically ascribed to the fracture of precipitates and inclusions (for example, manganese sulfide stringers in steels), and the triaxial nature of the stress field is implicated in the emissivity of these sources.

Acoustic emissions from crack front movement depend critically on the nature of the crack growth process. Microscopically rapid mechanisms such as brittle intergranular fracture and transgranular cleavage are readily detectable, even when the crack front is only advancing one grain at a time at subcritical stress levels. Slow, continuous crack growth mechanisms such as microvoid coalescence (ductile tearing) and active path corrosion are not detectable in themselves, but if general yield has not occurred, they may be detectable through associated plastic zone growth. Quantitative theory, which explains why some processes are detectable and others are not, was developed by Wodley and Scruby (Ref 33). The possibility of silent crack growth in ductile materials caused much consternation when it was first recognized in laboratory conditions, but it has not been a deterrent in real-life NDT, in which emission from defects is characteristically enhanced by environmental embrittlement, emissive corrosion products, crack face friction, or emissive nonmetallic materials entrained in the defect during the fabrication process.

Many fracture mechanics models have been developed to relate acoustic emissions to crack growth parameters. An important early approach was to relate acoustic emissions to the plastic zone size with the hope of estimating directly the stress intensity factor at defects found in the field (Ref 34, 35). Other models relate acoustic emissions to crack tip movement in situations of cyclic fatigue (Ref 33) or stress-corrosion cracking (Ref 36) for various materials. These models are commonly framed as power-law rela-

tionships, with the acoustic emission described by conventional parameters such as threshold-crossing counts, N . In the more recent but difficult technique of source function analysis discussed in the section "Wave Propagation Effects" in this article, individual crack growth increments can be quantified in absolute terms by computer-intensive analysis of the early portion of the AE waveform.

Nonmetallic layers on metal surfaces also exhibit acoustic emissions for potential NDT applications. Examples of acoustic emissions from nonmetallic layers include:

- The acoustic emission from high-temperature oxidation (Ref 37)
- The extensive study of acoustic emissions from room-temperature corrosion processes (Ref 38, 39)
- The use of acoustic emissions to optimize the performance of ceramic coatings used in high-temperature components (Ref 40)

Metal-Matrix Composites. The following example illustrates one application involving the testing of a metal-matrix composite.

Example 3: Acoustic Emissions From the Microcracking of the Brittle Reaction Zones of Two Metal-Matrix Composites.

During the tensile testing of two metal-matrix composites, microcracking of the brittle interphase between the fibers and the matrix produced distinguishable peaks in AE count rates well before ultimate failure from ductile failure of the matrix. This may suggest a potential application of AE inspection for the real-time monitoring of structures made of these or similar metal-matrix composites to provide indications of structural problems before critical damage occurs.

The materials tested were metal-matrix composites that consisted of a titanium (Ti-6Al-4V) matrix reinforced with continuous, large-diameter silicon carbide (SiC, ≈ 0.142 mm, or 0.0056 in., in diameter) or boron carbide coated boron (B₄C, ≈ 0.145 mm, or 0.0057 in., in diameter) fibers (with fiber volume fractions of 0.205 and 0.224, respectively). Standard straightedge tensile test coupons were used. Specimens were cut with the fibers either parallel or perpendicular to the load axis (longitudinal or transverse tension specimens, respectively). Steel end tabs were used, and all surfaces were sanded and cleaned.

Specimens were tested to failure in a servohydraulic testing machine operated at constant crosshead displacement. For each test, a single AE transducer was coupled to the midpoint of the specimen (within the gage section) with vacuum grease, and the acoustic count rate was measured as a function of the longitudinal displacement (strain). After each test, the fracture surface was examined with optical and scanning electron microscopes to determine the fracture processes that occurred.

The values given in Table 2 for rupture or fail-

Metal-matrix composite	Brittle compound	Failure strain, %
B(B ₄ C)/Ti-4V	Titanium diboride	0.25
	Boron carbide	0.57
	Boron	0.80
SiC/Ti-6Al-4V	Titanium carbide	0.28
	Titanium silicide	0.66
	Silicon carbide	0.91

Table 2: Brittle phase mechanical properties

ure strains of the fibers and the brittle reaction compounds formed at the fiber/matrix interface during the hot pressing process were used to find the correlations between fracture processes and AE count rates, with differences in the AE count rates between the two materials related to the differences in their brittle components. As shown in Fig. 16(a), for longitudinal tension specimens of B(B₄C)/Ti-6Al-4V, there was a distinguishable rise in the AE count rate near the rupture strain of titanium diboride and a peak near the rupture strain for boron carbide. The final peak resulted from ultimate fiber failure. For transverse tension specimens, Fig. 16(b) and (c) show large peaks in AE count rate near the rupture strains of the major brittle components (titanium diboride in B(B₄C)/Ti-6Al-4V and titanium carbide in SiC/Ti-6Al-4V). There were also minor peaks near the rupture strains of the other brittle components. The larger brittle reaction zone in B(B₄C)/Ti-6Al-4V relative to SiC/Ti-6Al-4V results in the larger area for the AE count rate plot.

The ultimate failure in the transverse specimens consisted largely of ductile matrix failure with lower AE count rates (relative to the microcracking).

Use of AE Inspection in Production Quality Control

In a small but important class of applications, AE inspection is applied during a manufacturing process to check the quality of the product or one of its components before final assembly and/or delivery. Of the production testing applications discussed in the section "Range of Applicability" in this article, common application of AE inspection in production quality control is the monitoring of welding and shaft straightening processes. Other efforts have been

directed toward the inspection of integrated circuits. In the early 1970s, for example, an entire satellite launch mission failed because of a loose particle inside the cavity of a single integrated circuit. As a result, integrated circuits for critical applications are now routinely tested by particle impact noise detection technology, an inexpensive derivative of AE testing (Ref 12). During the manufacturing process, other types of flaws in integrated circuits can also be effectively controlled with AE inspection. Acoustic emissions from bonding processes and from ceramic substrate cracking were investigated by Western Electric researchers during the 1970s and were used as accept/reject criteria for parts on automated assembly lines (Ref 3).

The AE monitoring of welding processes has been part of the technology since its early days. Slag-free, more-automated weld techniques such as resistance welding, laser and electron beam welding, and gas tungsten arc and gas metal arc welding are the easiest to monitor. In the case of resistance welding, AE monitoring is carefully synchronized to the weld cycle, and the various phases of the weld process are treated separately. Emission during solidification and cooling is correlated with nugget size and therefore with weld strength, while high-amplitude signals from expulsions can be used to switch off the weld current at the optimum time to avoid overwelding and to save power and electrode life. In the case of gas tungsten arc and gas metal arc welding, real-time computer algorithms have been developed to recognize the characteristic AE signatures of particular types of defects and to report these defects while the weld is being made. These procedures are effective even in the presence of substantial background noise. Gas tungsten arc welded injector tubes for the space shuttle are among the welded components routinely monitored by AE inspection in the production environment.

Shaft straightening is another production process that lends itself to quality control by AE monitoring. Forged shafts are routinely straightened in special machinery that detects any imperfections in alignment and applies suitable bending forces to correct them. The quality of the product is threatened by microcracking of the hardened surface of the shaft during the bend-

ing process. Acoustic emission inspection detects this very effectively and is incorporated into the machinery to warn personnel and to halt the process when potentially damaging microcracking occurs (Ref 4).

In welding and shaft straightening, the stresses that activate acoustic emissions are already present in the normal production processes (in welding, they are thermal stresses). In other cases, the stress is applied for the express purpose of AE testing. This is akin to the loadings routinely applied for the AE inspection of new and in-service pressure vessels and other large structures. Examples include the production testing of brazed joints (Ref 2), and the proof testing of welds in steel ammunition-belt links described in the following example.

Example 4: Acoustic Emission Inspection of Projection Welds in 1050 Steel Ammunition-Belt Links. The ammunition-belt link shown in Fig. 17(a) was made of 0.81 mm (0.032 in.) thick 1050 steel strip. The steel was preformed into link halves that were joined by two projection welds on each side where the sections overlapped. Although the welding schedules were carefully controlled to produce good resistance welds, there was a significant potential for producing some faulty welds in the mass-produced links. In a good weld, a weld nugget is formed that is usually stronger than the base metal in tension; that is, the base metal will tear before the weld will break. In a poor weld, the joint interface is literally just stuck together, and a moderate force, particularly one imposed by impact, will cause the joint to fracture at the interface. A preliminary feasibility investigation showed that poor welds produced more acoustic emission under load than good welds, even though the load was insufficient to break a poor weld.

Proof-Testing Equipment. A mechanical link tester (Fig. 17b) was designed to apply both a shear load and a bending load to the ammunition-belt link at the welded joints. This simulated the service load that would be imposed on the link.

Initially, piezoelectric sensors were attached to each link before testing to monitor acoustic emissions. This was the simplest and most direct method of confirming feasibility. Because attaching sensors directly to the link was not feasible for production testing, piezoelectric sen-

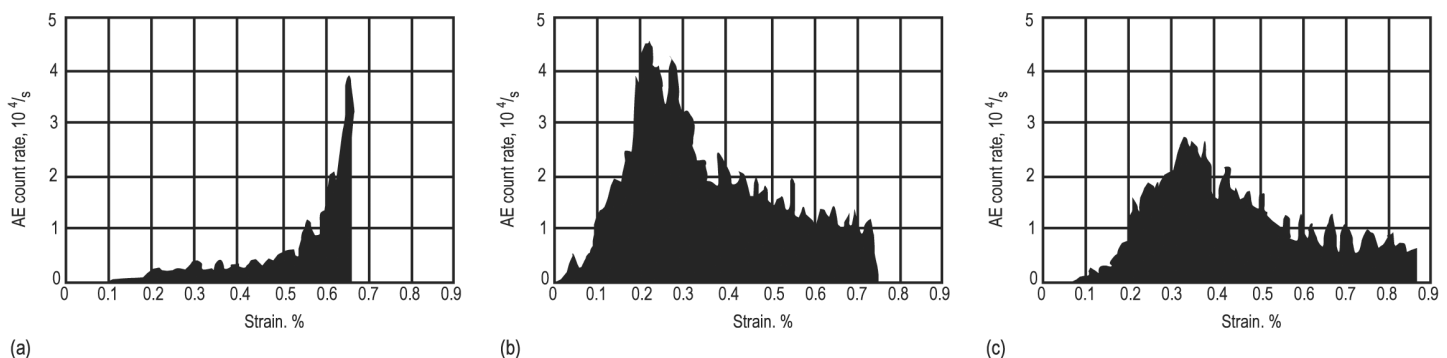


Figure 16: Area plot of AE count rate versus strain. (a) Longitudinal tension test of a B(B₄C)/Ti-6Al-4V specimen. (b) Transverse tension test of a B(B₄C)/Ti-6Al-4V specimen. (c) Transverse tension test of a SiC/Ti-6Al-4V specimen

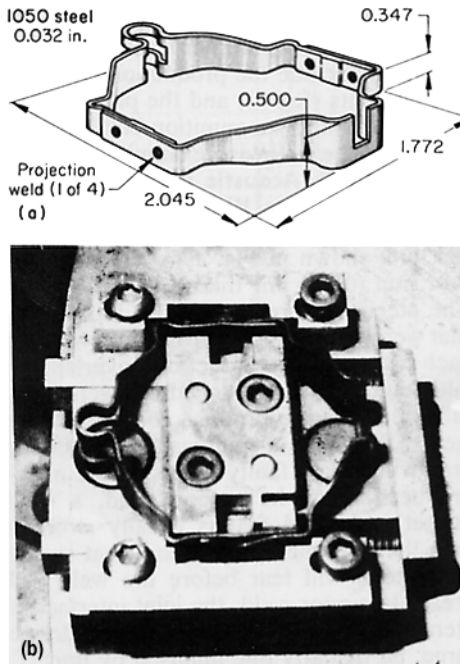


Figure 17: Ammunition-belt link, of 1050 steel, joined by four projection welds that were inspected by AE monitoring during proof testing in the fixture shown. Dimensions given in inches. Source: Ref 41

sors were embedded in the spreader arms of the link-test fixtures in an area adjacent to the welded joints in the link.

A spreader force of 270 N (60 lbf) on the link provided a link-spreader-arm interface pressure of about 35 MPa (5 ksi), which provided good coupling of acoustic information across the interface. The sliding action of the spreader mechanism produced a wide-frequency noise range that could not be electronically filtered without also filtering the acoustic emission. This problem was overcome by gating out the noise from moving parts of the link-stressing mechanism and monitoring for acoustic emission during static stressing of the link after the spreader arms had reached full displacement. A microswitch was installed in the fixture to turn on the AE monitoring system in proper relation to operation of the spreader arm.

Acceptance Levels. The form of signal energy analysis that produced the best results consisted of electronically integrating for the area under the half wave rectified envelope of the emission signal in terms of volts amplitude and time duration. The analyzer used for the production application produced a dc voltage proportional to the total AE energy measured. The system sensitivity was adjusted so that an energy analog output voltage of 10 V represented the division point between a good and a bad projection weld in a link. If the welded joint generated enough acoustic emission to produce a 10-V energy output, the link was rejected. If the value was less than 10 V, the link was accepted. The selective ejection function of the mechanical tester was designed to eject the links into the accept or reject container based on an electronic switching function that was controlled by the output volt-

age of the emission analyzer.

The monitoring system was calibrated by introducing an artificial signal into sensors in the spreader arms, where it was detected and processed by the monitoring system. A 10-V, 10- μ s pulse was fed into these sensors from a pulse generator. The resulting signal was reproducible, was a reasonable simulation of the real data, and was simple to generate. Monitoring by acoustic emission was the only available nondestructive method that could perform the necessary 100% inspection of these projection resistance welds.

Structural Test Applications

Acoustic emission inspection has been successfully applied in the structural testing of aircraft, spacecraft, bridges, bucket trucks, buildings, dams, military vehicles, mines, piping systems, pipelines, pressure vessels, railroad tank cars, rotating machinery, storage tanks, and other structures. The typical goal of an AE structural test is to find defects and to assess or ensure structural integrity. Acoustic emission inspection has been described as condition monitoring of static plant (Ref 42), which is parallel to the vibration monitoring techniques that are effectively used to monitor the condition of rotating plant. Both of these methods are useful for predicting failures (Ref 43) and reducing maintenance costs.

Key to structural testing with AE inspection are the stress concentrations that cause defects and other areas of weakness to emit while the rest of the structure is silent. Acoustic emission inspection thus highlights the regions that threaten the integrity of the structure. As a whole-structure test using fixed sensors, AE inspection is normally complemented by other NDT methods that are used to follow up the AE findings and to assist in determining the type, severity, and acceptability of the AE sources.

A major advantage of AE inspection is that it does not require access to the whole examination area. Removal of external insulation or internal process fluids, typically a major expense associated with other NDT methods, is not required for the AE test. In fact, this procedure can be avoided altogether if the AE test indicates that the structure is in good condition.

For AE inspection to function reliably as a whole-structure test, the structure must be loaded in such a way as to stimulate emission from all structurally significant defects. Continuous monitoring in service is a possible test approach that has been applied, for example, to aircraft (Ref 7) and nuclear reactors. This approach guarantees appropriate stressing, but it is difficult because a small amount of emission from defect growth must be separated from a large amount of noise over a long time period. More commonly, the AE test is conducted over the course of a few minutes or hours, during which the structure is stimulated by applying a controlled stress (Ref 44). In most cases, this is satisfactorily accomplished by going somewhat above the normally apparent service loads (for example, 110% of

the working pressure for an in-service pressure vessel, 200% of the rated load for an aerial manlift device). However, there are cases in which this approach will not work. For example, if defects are being induced in service by thermal stresses rather than mechanical stresses, an applied mechanical loading may not give a good match to the stress field that is causing the defects to grow in service. The defects, effectively unstressed, may not emit. To overcome this problem, inspectors testing steam lines in electric power plants have conducted AE monitoring during thermal overloads and cool-downs and have reported better success from this type of stressing.

In performing a successful AE test, careful attention must be paid to the type, magnitude, and rate of the applied stimulation (loading). Previously applied stresses will have a very strong influence on the emission that will be observed, as discussed in the section "Load Control and Repeated Loadings" in this article. Precautions must be taken to avoid inadvertent loadings of the structure. Many tests have been spoiled when site personnel, eager to ensure that the pressure system is leaktight, have taken the vessel up to the test pressure before the arrival of the AE inspectors. Accurate load measurement and the ability to hold load at a constant value are other requirements that may demand special attention from site personnel.

Load history is less important for leak testing because leak testing relies primarily on the detection of turbulent flow through the leak orifice. Major structural test applications of acoustic leak testing include flat-bottom storage tanks and nuclear reactor components. In the case of nuclear reactor components, millions of dollars have been saved through the selected use of AE instrumented inspection technology as an alternative to hydrotesting (Ref 45).

Data evaluation procedures depend on the context and content of the test. In one-of-a-kind and developmental testing, the skill and experience of the investigator are of prime importance. This fact inhibited the widespread use of AE inspection until standard test procedures started to become available in the late 1970s. The development of standard test procedures made it possible for AE tests to be efficiently conducted by regular NDT inspectors (given the proper training), while the more innovative investigators moved on to the development of new application areas. Some of the well developed and standardized structural test applications of AE inspection will be briefly summarized below.

Bucket Trucks. The AE inspection of aerial manlift devices (bucket trucks) was pioneered by the author for Georgia Power Company in 1976 and was carried forward into routine practice by independent testing laboratories and several electric utilities in the years that followed. The ASTM F-18 Committee on Electrical Protective Equipment for Workers published the Standard Practice on the subject in 1985 (Ref 46).

First intended for use on the fiberglass boom sections of insulated bucket trucks, the method was soon extended to cover the pedestal, pins, and other metal components. An estimated 70 000 to 100 000 AE tests have been conducted up to 1988. Bucket trucks can develop problems through accidents, overloads, and fatigue in service. A thorough, regular inspection and test program can identify potential problems before they cause injuries or downtime (Ref 47).

Acoustic emission inspection is a major part of the structural integrity evaluation that complements functional tests of the bucket truck. Of all inspection methods, it is the most effective for detecting problems in fiberglass components, while for metal parts and 100% structural coverage, it serves as a cost-saving screening test that directs the inspector's attention to problem areas. The AE test is preceded by a visual inspection, and any AE indications are normally followed up with magnetic particle, dye penetrant, or ultrasonic inspection.

The AE test typically requires 12 to 16 sensors. System performance is checked by lead break before monitoring begins (see the section "Installation" in this article). Monitoring begins with a noise-check period, followed by two loadings to a predetermined proof load. Emissions during rising load, load hold and load release are separately recorded.

Data evaluation procedures are difficult to spell out because of the wide variety of possible noise and AE sources and the wide range of bucket truck constructions. The experienced inspector uses his knowledge of the truck as he evaluates high- and low-amplitude emission on the different channels during the different stages of the test. Aware of possible noise sources, he looks for indications that may lead to confirmation of damaged fiberglass, cracked metal components, or maintenance problems such as lack of lubrication in visually inaccessible areas.

Using an AE instrument specially designed for economical bucket truck testing, an experienced test crew can perform five to ten AE tests in a single day. When the required visual, operational, ultrasonic, magnetic particle, and dye penetrant testing operations, are also performed, typically two to three trucks can be inspected per day.

Jumbo Tube Trailers. Acoustic emission inspection for testing jumbo tube trailers was developed by Blackburn and legitimized in lieu of hydrotest by a Department of Transportation exemption granted in 1983 (Ref 48). These tube trailers carry large volumes of industrial gases on the public highways, typically at a pressure of 18 200 kPa (2640 psi). Fatigue cracks can grow in service, but the hydrotest will not detect these cracks unless they actually cause rupture of the tube. The AE test will detect the cracks while they are still subcritical, during a 10% overpressure applied during the normal filling operation. The AE test is therefore more meaningful than the hydrotest. The AE test is also less expensive and avoids disassembly of the trailer and con-

tamination of the internals with water.

The trailer typically holds 12 tubes, which are all tested at the same time. The AE test requires just two sensors on each 10 m (34 ft) long tube; wave propagation and attenuation characteristics are very favorable in this structure, and AE sources can be accurately located. If ten or more valid events are located within a 200 mm (8 in.) axial distance on the cylindrical portion of the tube, ultrasonic inspection is carried out and the tube is accepted or rejected based on the ultrasonic evaluation of flaw depth. The accept/reject criteria are based on a conservative fracture mechanics analysis of in-service fatigue crack growth. Between March 1983 and March 1988 about 1700 jumbo tubes were AE tested, and the method has been further extended to other shippers of compressed gas and other tube types.

Fiberglass Tanks, Pressure Vessels, and Piping. In the 1970s, the chemical industry was experiencing worldwide failures of fiberglass storage tanks and pressure vessels. Causes included inadequate design and fabrication, mishandling during transportation, and misuse of this relatively unfamiliar material. The situation was aggravated by the lack of a viable inspection method.

An AE test methodology was pioneered by Monsanto, resulting in elimination of the tank failure problem (Fig. 18). The method came into widespread acceptance through the formation and activities of the Committee for Acoustic Emission in Reinforced Plastics (CARP), which is currently affiliated with the American Society for Nondestructive Testing. The written procedure developed by CARP and first published by the Society for the Plastics Industry in 1982 was the basis for the introduction of AE inspection into the ASME Boiler and Pressure Vessel Code in 1983 (Ref 30, 49). An estimated 5000 tests have been carried out using this procedure as of 1988. The work of CARP was also extended to cover methodology for testing fiberglass piping (Ref 50).

The AE test typically requires 8 to 30 sensors, depending on the size of the vessel or tank. High-frequency (typically 150 kHz) channels are used to cover regions of known high stress, such as knuckles, nozzles, and manways; low-frequency (typically 30 kHz) channels having a larger detection range are used to complete the coverage of less critical regions. In the case of a storage vessel, the test is typically conducted during filling with process fluid after an appropriate conditioning period with the contents at a reduced level. In the case of a pressure vessel, appropriate overpressure is applied. The loading is conducted in several stages, with load-hold emission, Felicity Ratio, and other accept/reject criteria evaluated at each stage. System performance checks and background noise checks are part of the test procedure.

Metal Pressure Vessels and Storage Tanks. In the 1970s, many research and engineering organizations and test companies were active in

the AE testing of metal pressure vessels. A 1979 survey estimated that about 600 pressure vessels had been tested by AE inspection on a production basis up to that time, mostly in the petroleum, chemical, and nuclear industries (Ref 51). (Although tests on pipes, heat exchanger tubing, and miscellaneous components were much more numerous, pressure vessels have always been a focal point for AE testing.) Much of this testing was done with *ad hoc*, undocumented procedures that relied heavily on the individual experience of the teams performing the work. The main emphasis was usually placed on source location, technically a most attractive feature of AE testing. Located sources would be graded according to their AE activity and/or intensity, and the vessel owner would be advised regarding which areas to inspect with other NDT methods. Many structural defects were successfully identified with these methods.

A significant maturing of the technology took place when Fowler at Monsanto engaged on a systematic program of methods development, using the results of follow-up inspection to refine and improve the AE data analysis procedures. Starting in 1979, this program included destructive tests on decommissioned vessels, field tests on many hundreds of in-service vessels and tanks, and development of analytical procedures for recognizing and eliminating extraneous noise (Ref 25). The program de-emphasized the calculation of source location (which requires several sensors to be hit) because in practice many AE events hit only one sensor. To use all the AE hits, zone location was employed instead of point location. The outcome of this program was a comprehensive test procedure backed by detailed case histories and available to licensees under the trademark MONPAC. This procedure has been applied to approximately 2000 metal vessels and storage tanks as of 1988 (Ref 8).

A typical MONPAC test result is shown in Fig. 19. Here a 30-year old ethylene storage bullet was AE tested on-line, raising the pressure by turning off a compressor. Results are presented in the form of an unfolded map of the vessel, with the evaluation of appropriate zones on the map being displayed in color codes (Fig. 19 only shows the color as a dark region). The zone evaluations indicated "no significant emissions," so a hazardous and possibly deleterious internal inspection was avoided (Ref 42).

Other MONPAC tests have found external and internal corrosion, stress-corrosion cracking, weld cracking, lack of fusion, lack of penetration, and embrittled material. Plant shutdowns have been shortened through early diagnosis of major problems, or have been eliminated through positive demonstration of structural integrity. Savings achieved through this method run to tens of millions of dollars.

An AE methodology for metal vessel testing has also been introduced into the ASME Boiler and Pressure Vessel Code as of the December 1988 Addendum (Ref 31). The article states re-

quirements for written test procedure, personnel qualification, equipment, system calibration, pre-examination measurements, background noise check, and vessel pressurization. An illustrative loading schedule and sensor layouts are included. Evaluation criteria are to be supplied, class by class, by the referencing code section; they will be based on emission during load hold, count rate, number of hits, large amplitude hits, MARSE, and activity trends. This entry into the ASME Code represents a major milestone in the establishment and maturity of AE technology.

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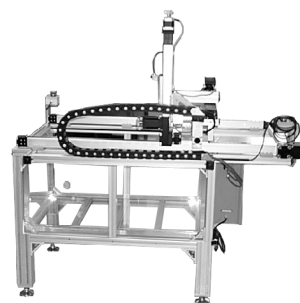


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APPENDIX

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1. UTILITIES MENU

The utilities Submenu shows the list of utility programs to help the user perform post-test analysis.

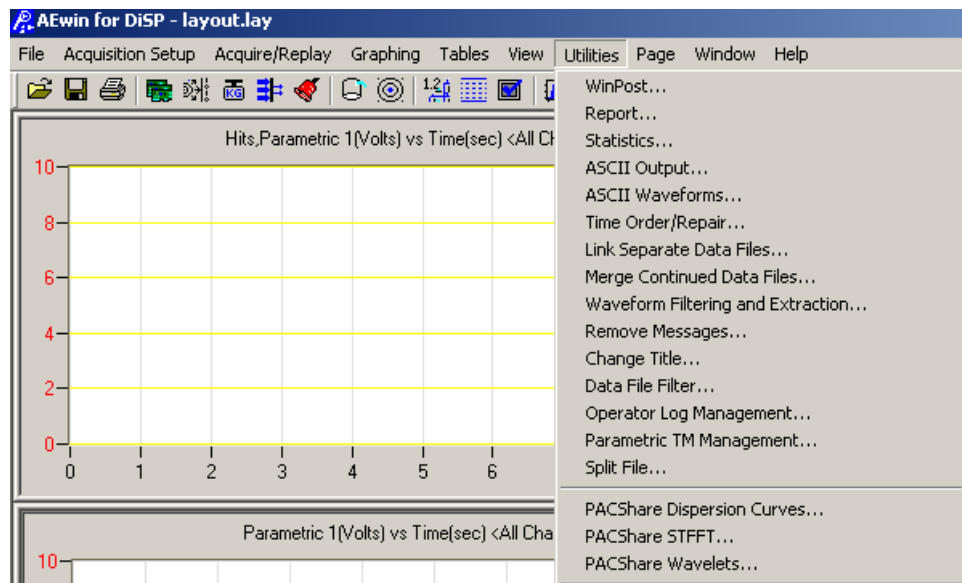


Figure 1. AEwin Utilities Menu

The utility programs and their functions are:

- **WinPOST:** AEwinPOST is an optional graphical data visualization post-acquisition tool that provides useful insight into your AE data. In addition, there are some fantastic data manipulation and filtering tools available in this optional utility that are not available in AEwin. Please note that this utility is an option. A brochure for AEwinPOST can be found in your AEwin directory in a file called AEwinPOST Brochure. PDF. Also, a separate manual exists in PDF form in your AEwin directory called AEwinPOST_v1_Manual.pdf.
- **Report:** Reads a Data file and creates a report containing: System Information, Test Information, Hardware Setup as well as Alarm and Filter setup information. This is useful when making reports of AE tests to provide full Test Setup documentation.
- **Statistics:** Provides minimum, maximum, average, and standard deviation of all features in a DTA file.
- **ASCII Output:** Extracts AE Feature data from a DTA file and writes it into a new ASCII format text file.
- **Waveform to ASCII:** Converts each waveform found in a DTA or TDA file into an ASCII text file.
- **Time Order/Repair:** Performs a time ordering of all messages in a DTA file and checks the file for proper file close messages.
- **Link Separate Data Files:** Puts together at least two data files with the header information of the first one.
- **Merge Continued Data Files:** Puts together at least two continued data files.
- **Waveform Filtering and Extraction:** Filters waveforms and extracts hits based on a user-specified dataset.
- **Remove Messages:** Removes the selected types of messages from a DTA file

- [**Change Title:**](#) Lets the user change the title from a given DTA file.
- [**Data File Filter:**](#) Filters a DTA files based on based on a user-defined filter.
- [**Operators Log Management:**](#) Allows the user to view, modify, add and manipulate comment user time marks in a DTA file.
- [**Time Mark and Parametric:**](#) Allows the user to view, modify, add and manipulate parametric time marks in a DTA file.
- [**Split File:**](#) Lets the user to split a given DTA file into smaller DTA files.

Each of these utilities is described more comprehensively in the following sections.

2. REPORT

REPORT is utility program that reads PAC, AE Multi-channel system Data (DTA) files, extracts the file "header" contents, and creates a report containing; System Information, Test Information, Hardware setup, as well as Alarm and Filter setup information. This file can be displayed on the screen, saved as a text (ASCII) file, copied, or imported into a word processor file.

The Data (DTA) file includes hardware setup information and also includes Test Information as shown below:

- The filename of the file being processed
- Software Product Name and software Version Number (which was used to generate the file)
- Test Title
- Test Duration
- Date & Time of Test
- Complete logistics table
- Version Number of the software where the file was created.
- Status of the waveforms (ON/OFF)
- Number of channels
- List of channels (Channel number, gain, HDT, HLT, PDT, THR, and THRTYPE)
- Smart Threshold (Channel, Low Hits, High Hits, Internal, Step, Low Limit, and Duration Limit)
- Front End Filters (Filter ID, Channel, Characteristic, Type, Lower Level, High Level)
- Front End Alarms (Alarm ID, Channel, Characteristic, Type, Warning, Trigger)
- Waveform Front End Filters (Waveform Filter ID, Channel, Characteristic, Type, Lower Level, High Level)

This utility program is very easy to use and is a very convenient way of documenting and archiving a test setup as part of a test report or as a standalone file for later access.

Don't expect to see all the above information because not all the files have this information. In other words, the "report" can read all information, but the user only will see what has been saved in the file under examination.

When the user runs this utility, the program will ask him to select the input file:

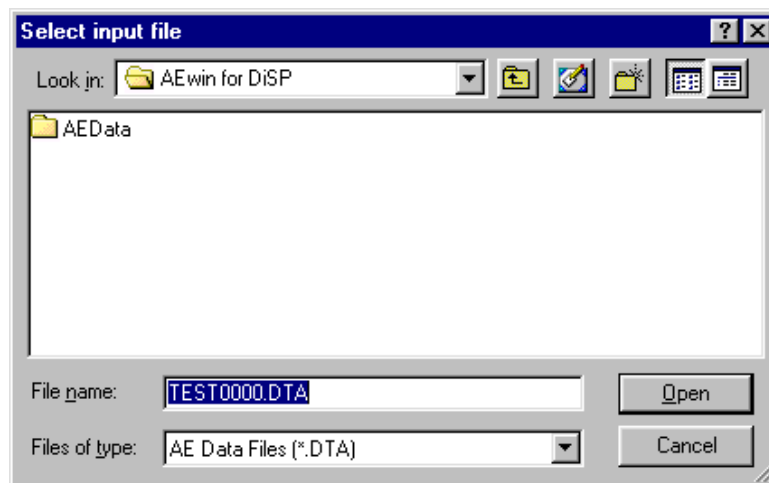


Figure 2. Select Input File

If the user selects "Open," the report program immediately will begin to process the selected file; meanwhile, the user can see the progress bar at the right bottom in the main window. When the program finishes, the user will see the information about the file displayed in the report results window as shown in the following image.

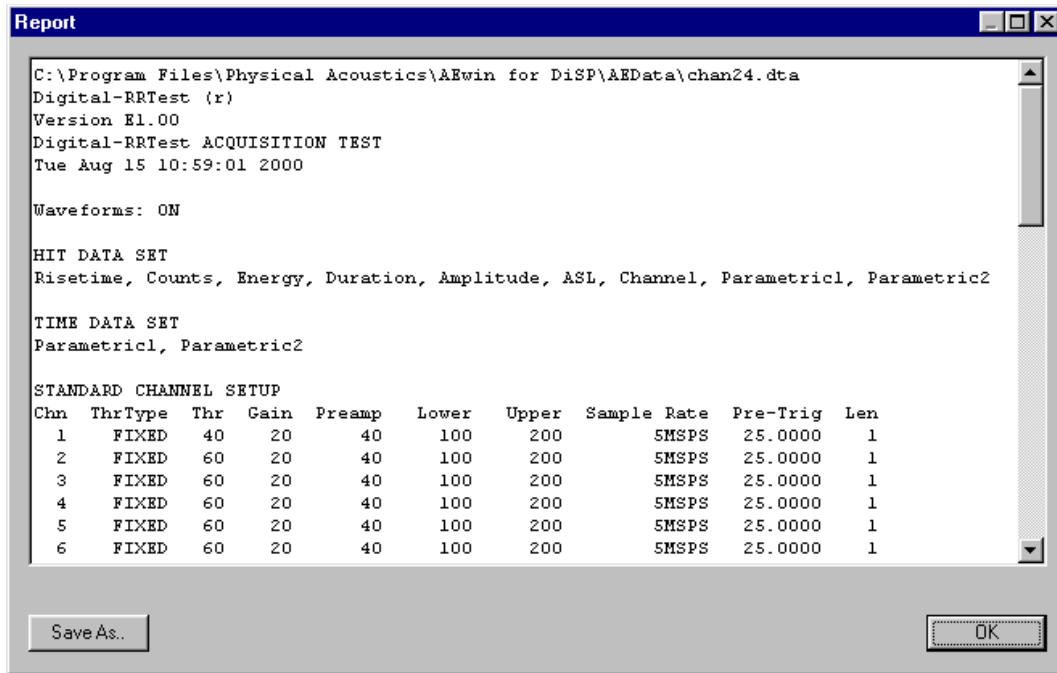


Figure 3. Report Utility, Result Window

The user has the option to copy this information to the clipboard for inserting in other documents, directly save the information in an ASCII format text file by clicking on the "Save As." button. Or simply view the information on the screen.

If the user selects more than one data file and clicks "Open," the program will enter batch mode, whereby it will process all selected files and output text files for each selected file. Clicking "Open" on multiple.DTA files will produce the following window:

This window will allow you to select which folder you want to store your results in. Once you select the folder and click "OK," the user can see the progress bar. Once finished, the user will find text files under the same name as the data files. Note, no window will be displayed showing the information on the data file, as opposed to the case when one file is selected.

If a text file with the same name already exists in the folder, the following dialog box will appear:

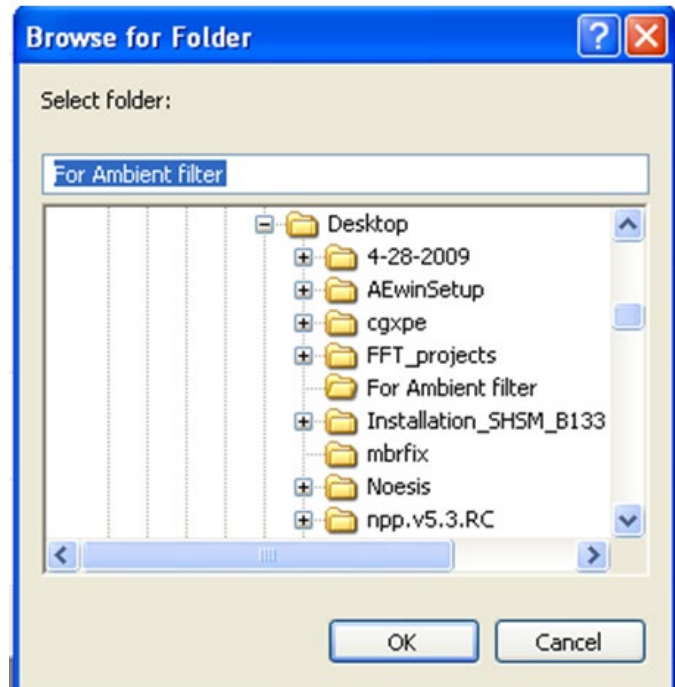


Figure 4. Browse Folder Dialog

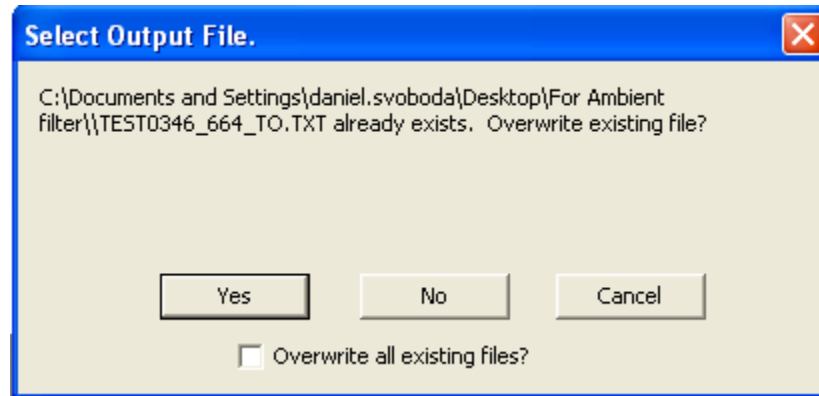


Figure 5. Overwrite File Dialog

If the user clicks "Yes," then the output file produced by the computer will overwrite the one already existing. Selecting "No" will cause the program not to process the data file at hand and keep the original text output. "Cancel" will exit out of batch mode, and no further files will be processed (apart from the ones that were already done).

More text files may exist on which the program will attempt to overwrite. It would be tedious for the user to select "Yes" or "No" for each text file, specifically if there are dozens of files. The checkbox "Overwrite all existing files?" helps the user to skip through all this. By selecting this checkbox and clicking "Yes," the program will automatically overwrite all conflicting files without further prompting the user.

3. STATISTICS

STATISTICS is a utility program that reads an AE multi-channel Data (DTA) file and processes the file to provide a summary report of the AE data inside the data file in terms of the number of hits for each channel, total hits, total waveforms, and time-based records as well as the maximum, minimum average and standard deviation per feature per channel.

When the user selects the option in the menu, the program asks to select the input name for the data file.

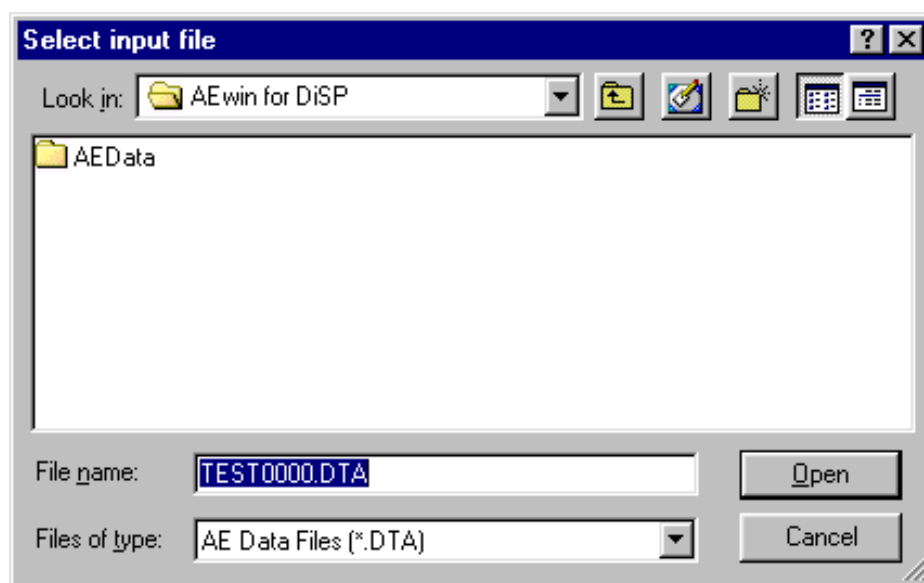


Figure 6. Select Input File

The Statistics process begins when the user clicks on the "Open" button over a file (if the user clicks Open over a folder, the Open button expands the folder).

When the process finishes, the user can see the statistics result in the window as shown below:

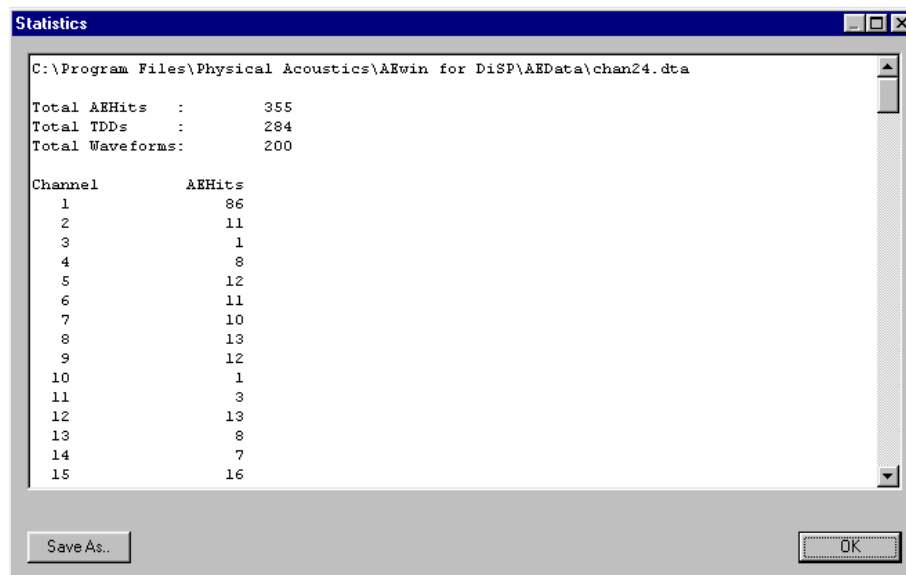


Figure 7. Statistics Utility, Result Window

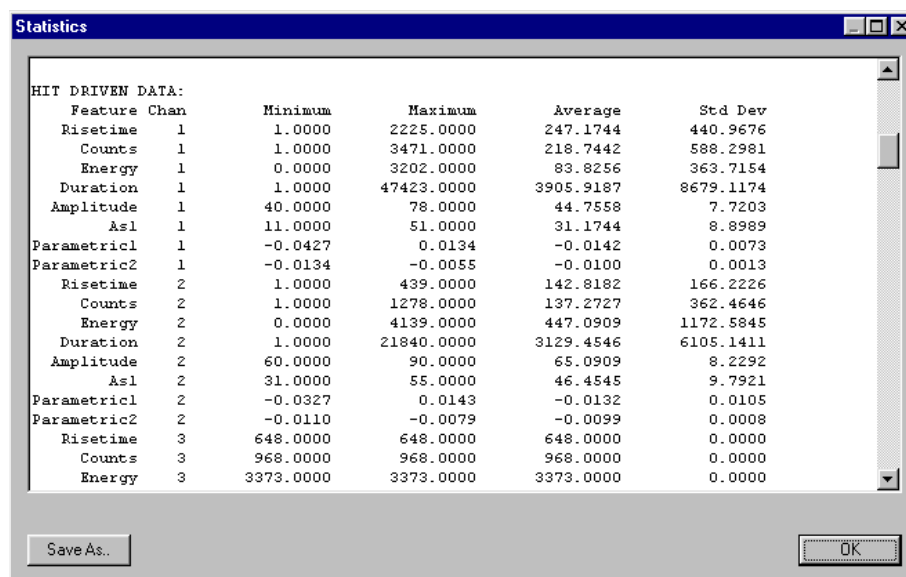


Figure 8. Statistics Utility Result Window

The user has the option to copy this information to the clipboard for inserting in other documents, directly save the information in an ASCII format text file by clicking on the "Save As.." button. Or simply view the information on the screen.

This option also allows for batch mode processing, whereby if the user selects more than one data file, the Browse Folder Dialog will be displayed, asking the user which folder to save the processed batch files. When the location is selected, and the "OK" button pressed, the program will store the information as text files in the given location, under the same name as the data files. Support for overwriting text files, as shown by the Overwrite File Dialog, is also provided.

4. ASCII OUTPUT

The ASCII Output utility will convert an AE data file (.DTA) into an ASCII format text file that can be read in word processors and spreadsheets. The program requires the user to input the file name to be processed, the output file name, and the selection of messages and data content the user is interested in saving to the ASCII output file.

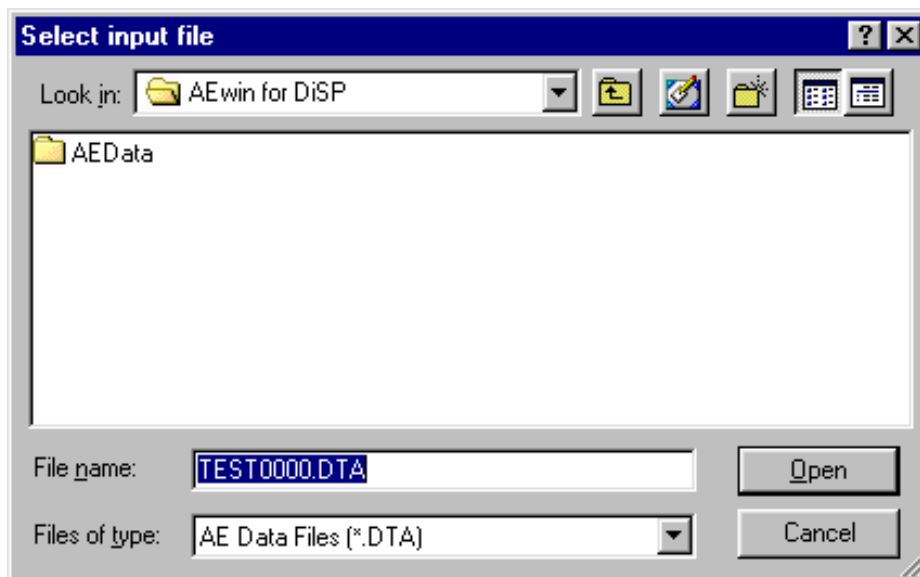


Figure 9. Select Input File

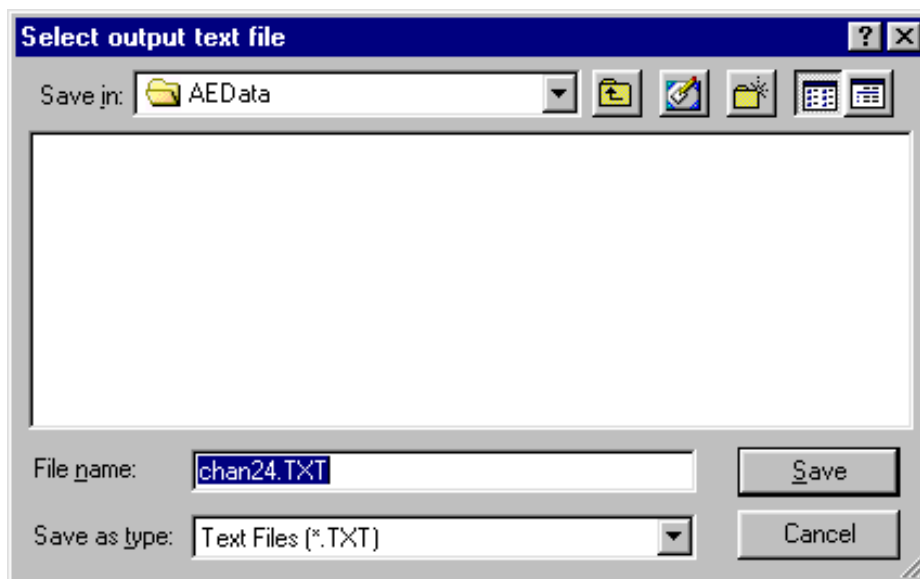


Figure 10. Select Output Text File

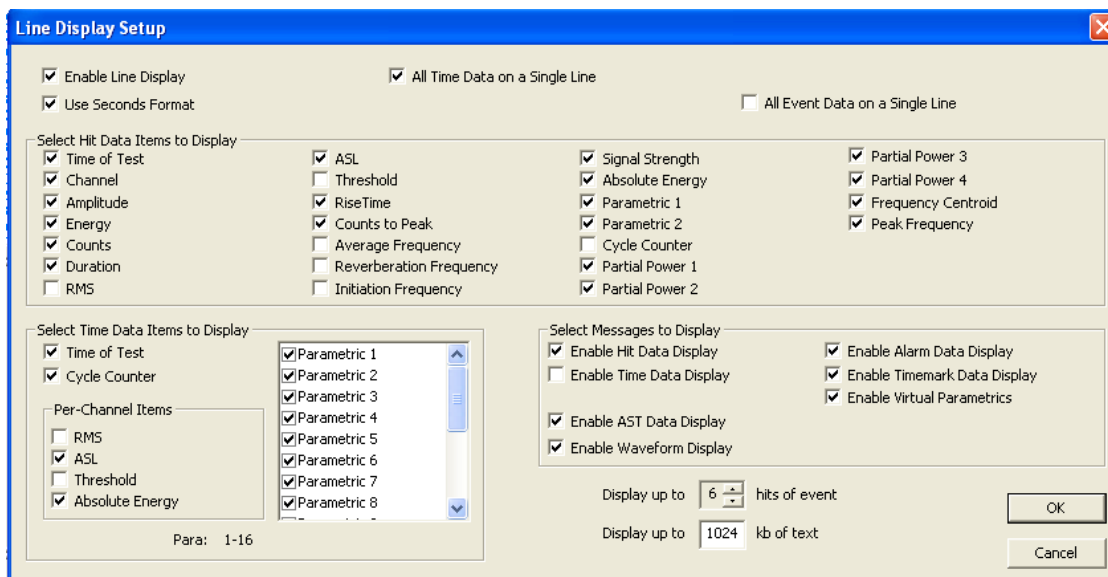


Figure 11. Options to Include in the Text File Output

This Dialog box (shown above) lets the user select which should be recorded to the resultant text file (if existing on the file)

The user will know that the program has finished when the following message appears:

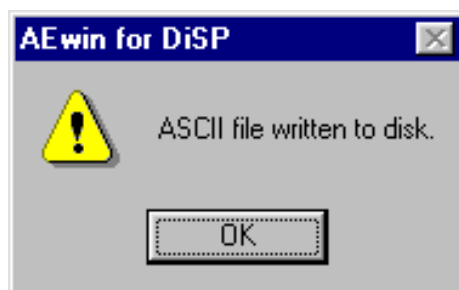


Figure 12. ASCII Output Utility, Final Message

Note – If Virtual Parametrics is checked, then the output text file will have all Virtual Parametrics output on a single line instead of each being on a separate line. These values will also appear without their units of measurement and will be tabbed evenly. This is to allow for easy importing into Excel or any other type of spreadsheet program.

As with Report and Statistics, ASCII Output supports Batch mode. Selecting multiple files allows you to select which folder to save all processed files along with the dialog box to handle any potential conflicting files.

5. CII WAVEFORMS

The waveform to ASCII utility processes a DTA file or a TDA file containing waveforms takes each waveform found in the file and saves each into a separate ASCII text file (Excel compatible). The program offers the user two types of conversion: simple and complete. Simple is when the output files only contain one column of data without any information header, meanwhile complete contains the information header and the data column.

When the user selects to run this utility, the program will ask him the name for the input file.

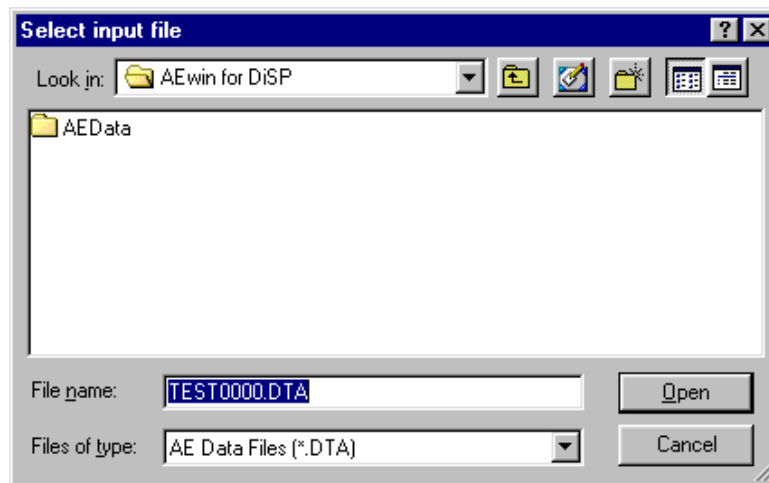


Figure 13. Select Input File

After selecting the input file, a dialog is displayed for selecting the output folder name. Notice that this is not the common window to select a file because, in this case, you need to provide a folder for the multiple waveform files to reside, so you are going to see the window "Browse for Folder." As a default, the same folder as the input file is displayed, but the user can change it.

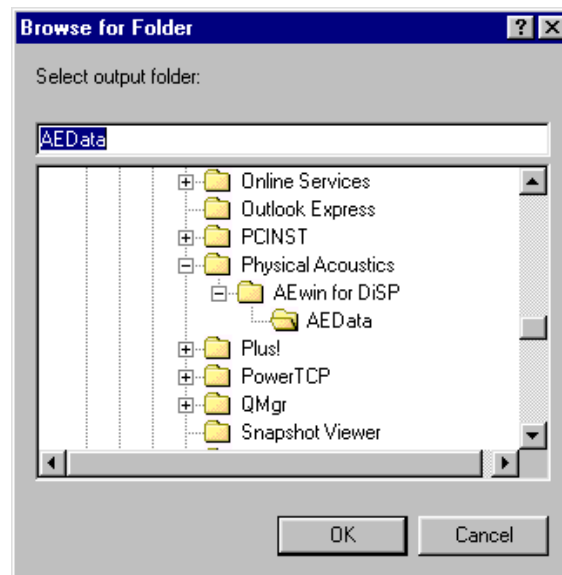


Figure 14. Browse for Output Folder

Now the program will ask the user to select some options for the output file.

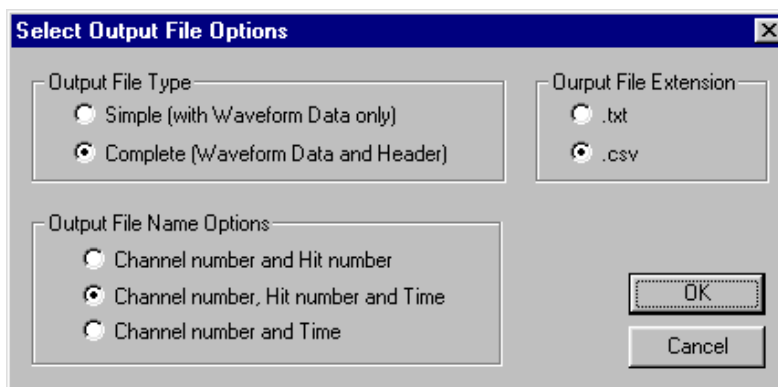


Figure 15. Select Output File Options

The user can choose the Type, the file extension, and the name for the output files. Type has two choices: simple, with waveform data only, and complete with waveform data and header. Complete is the default option. The information header lists the following information:

- Input file name
- Date
- Time
- Sample Interval
- Units
- Channel number
- Hit number
- Time of Test

Another option for the user is the output file extension. If he wants to analyze the data using EXCEL he should select the "csv" extension to open directly in EXCEL without being asked to convert the file in EXCEL. The "csv" option is the default. The "txt" output should be selected for non-EXCEL applications and is more generic.

The last question to answer is the file name options, there are three of them: (1) channel number and hit number, (2) channel number, hit number and time, and (3) channel number and time. The second one is the default option. These allow an automatic file naming convention for recognition and identification of the waveform file after conversion. The user should choose the option that he feels is best.

When the program is processing the input file, the user will see the progress bar located at the right bottom in the main window. As soon as the program finishes, the summary window will show the input file name, the output folder with the base name, and a list with the number of processed waveforms per channel.

The user can go to the output folder he selected to find the created files, one per waveform.

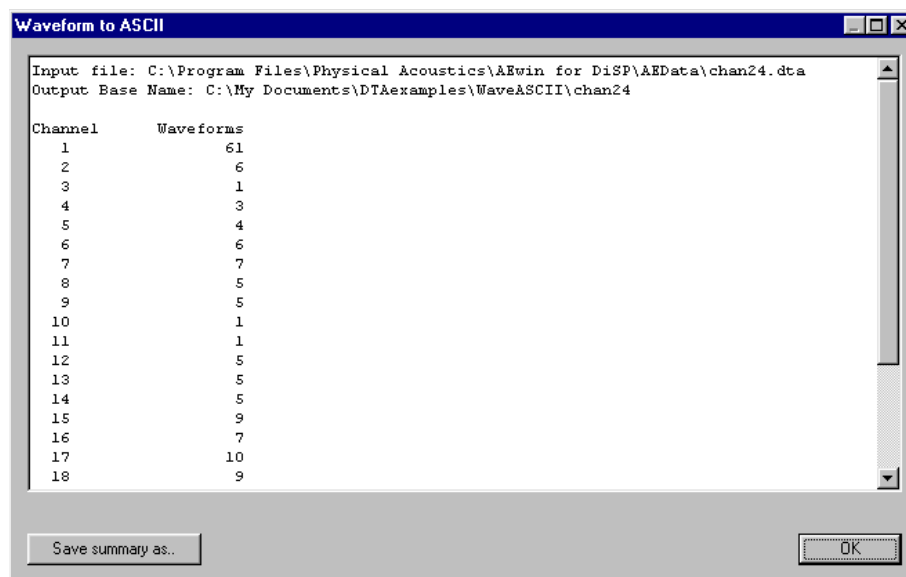


Figure 16. ASCII Waveform Utility Summary Window

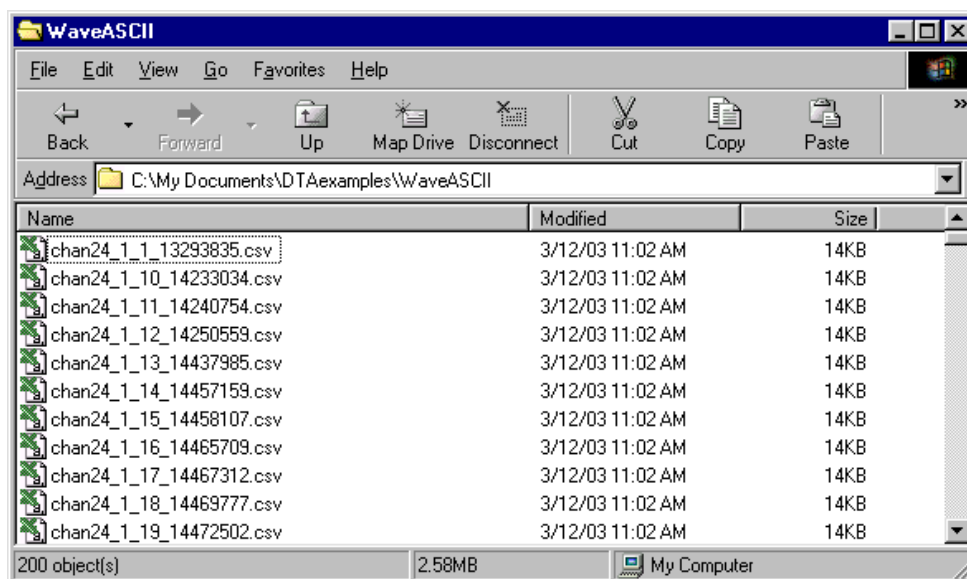


Figure 17. Output Folder Showing the Text Files Created by ASCII Waveform Utility

6. TIME ORDER/REPAIR

It should be noted that in AEwin, data is already saved in a Time Ordered sequence, so time ordering is not a necessity for data collected by AEwin. However, all DTA data files generated outside of AEwin (all DOS-based AE programs) should be processed by this utility before being replayed into AEwin.

Time Order, as its name suggests, is a utility to time order the messages in AE Data files. The data file "header" is transferred to the output file unchanged. The remaining messages, which constitute the actual data, are transferred to the output file in time order. That is, the data is sorted into order of increasing time.

Extensive error checking of the messages in the data file is provided in the process, and, when possible, corrections are made.

To run Time order over a file, the user will need to provide the input file name and the output file name.

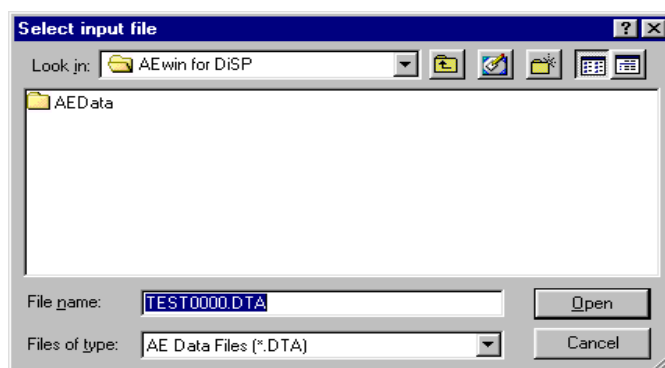


Figure 18. Select Input File

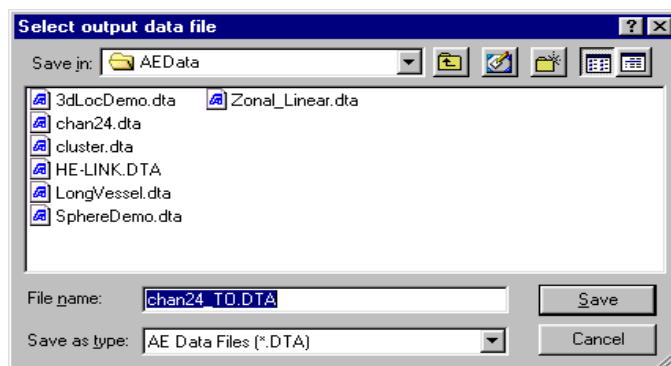


Figure 19. Select Output DTA File

Once the output file name is provided, then the process for time ordering begins. When it finishes, the user will see the following message:

Batch mode is also supported for this option. By clicking on multiple files, the user has the option to save multiple files onto one folder. The user also can overwrite certain files or all of them based on the Overwrite File Dialog.

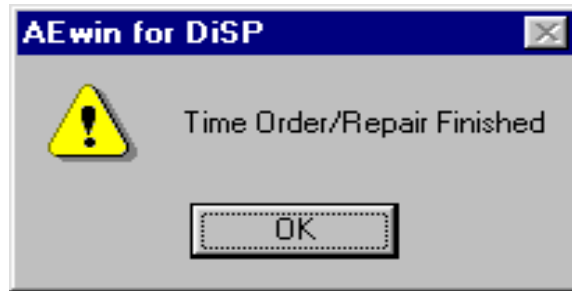


Figure 20. Time Order Final Message

7. LINK SEPARATE DATA FILES

Link is a utility that allows the user to combine files from different tests into one file so that for all other purposes, the data will appear as if they came from a single test. A typical case where this may be desirable might be: if for some reason a test had to be aborted and was resumed at a later time, the user will then have two separate files, but for analysis purposes, the user will need to have just one file.

When the user selects to run Link Utility from the Utilities Menu in AEwin, the first window that will appear asks him about the time between files. The default value is 1 second. This time becomes the time gap between the ending of the first data file and the beginning time of the second when the files get linked into one.

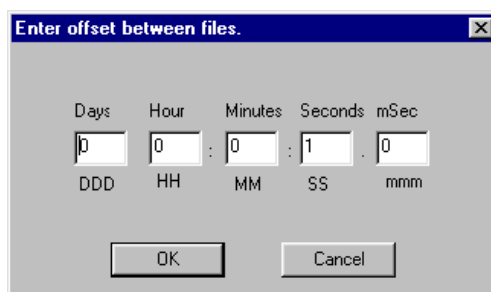


Figure 21. Dialog for Entering Offset Time Between Files

There is one restriction in the way to use this utility. If the user needs to link more than two files with different time gaps between them, then the user should run the program twice, the first time with file1 and file2 with a time gap of time1 with an output file to file12, and the second with file12 and file3 with a time gap of time2 with an output of file123. Notice that this is not necessary for linking more than two files with the same time gaps in between them.

After the user selects the offset time between files, then he should select and provide an order for the files to combine. This allows the linking of two or more files into one.

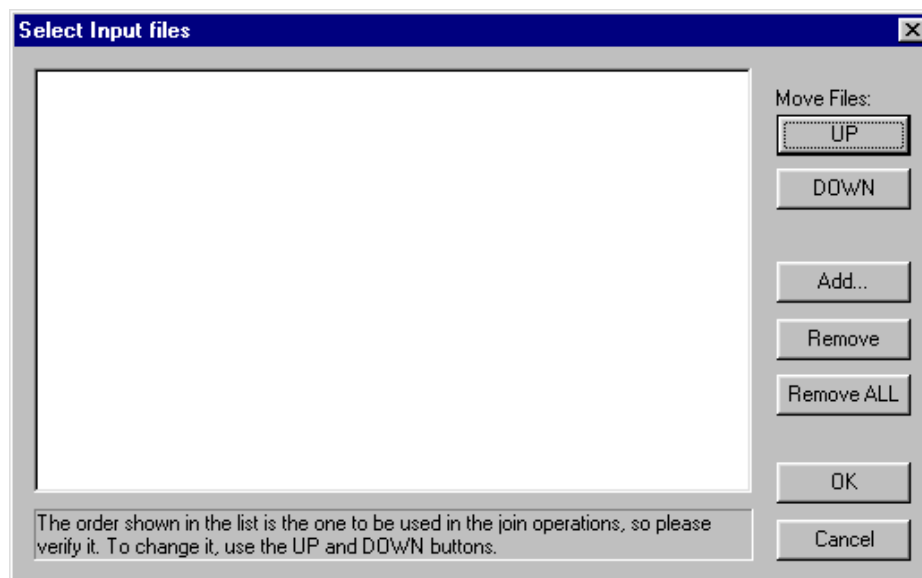


Figure 22. Select and Order Multiple Input Files

Button Actions:

- **Add:** Use this button to select the files to be linked.
- **Remove:** Select a file and then press this button to remove the selected file.
- **Remove ALL:** With this one, the user can remove in one step all the files. Doesn't have to select them.
- **Up:** Select the file you want to move one position before and press this button.
- **Down:** Select the file you want to move one position after and click on this button.
- **OK:** To proceed with the process
- **CANCEL:** To stop the selected process.

After selecting the files to link and establishing the order, the user must provide the program with an Output file name. This file will be approximately, actually, less than the sum of the sizes of the files selected as input files. So when selecting the output file, take this fact into account.

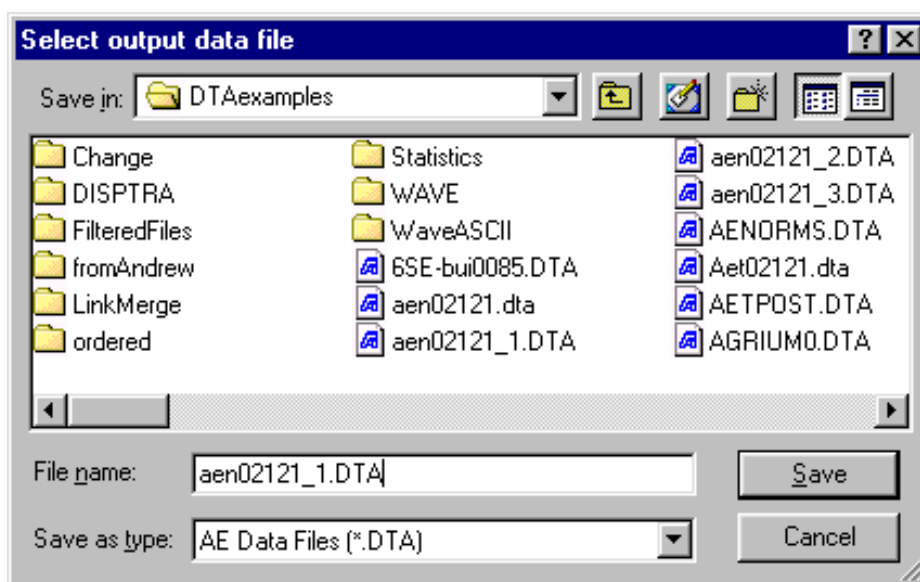


Figure 23. Select Output Data File

When the program finishes the linking, it will show the next dialog message to the user:

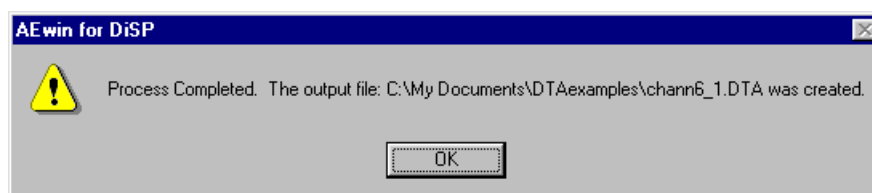


Figure 24. Link Utility, Final Message

8. MERGE CONTINUED DATA FILES

Continued files are an option in the AEwin software, which allows the user to close a DTA file based on an elapsed time, the number of hits, or file size, and immediately re-open a new "continued" file with links in the filename and data header to the previous file.

Merge is the utility that allows the user to merge multiple continued files into one large data file. The user has to provide the input files names and order for merging them in the following window:

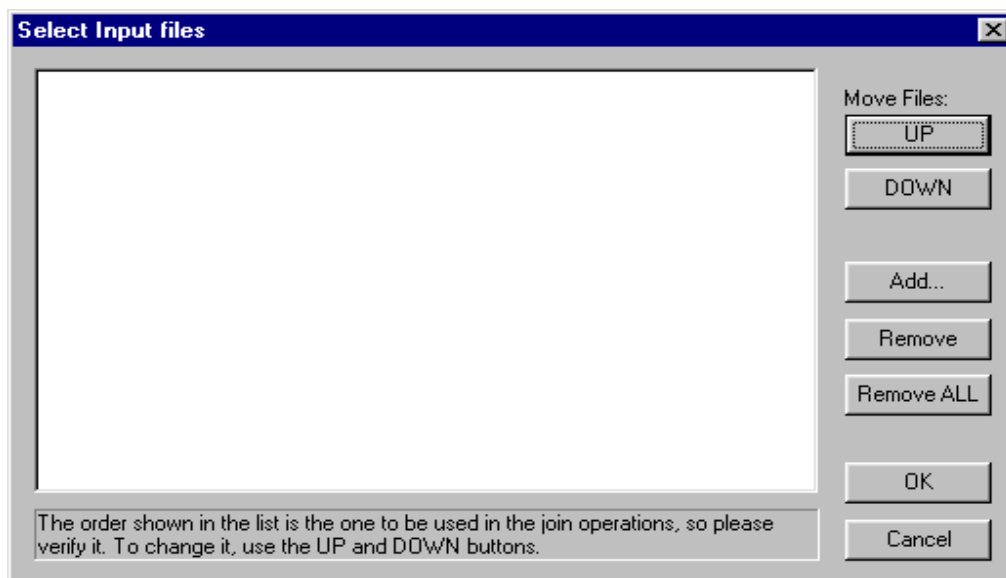


Figure 25. Select and Order Multiple Input Files

Then the user also has to provide the name for the output file:

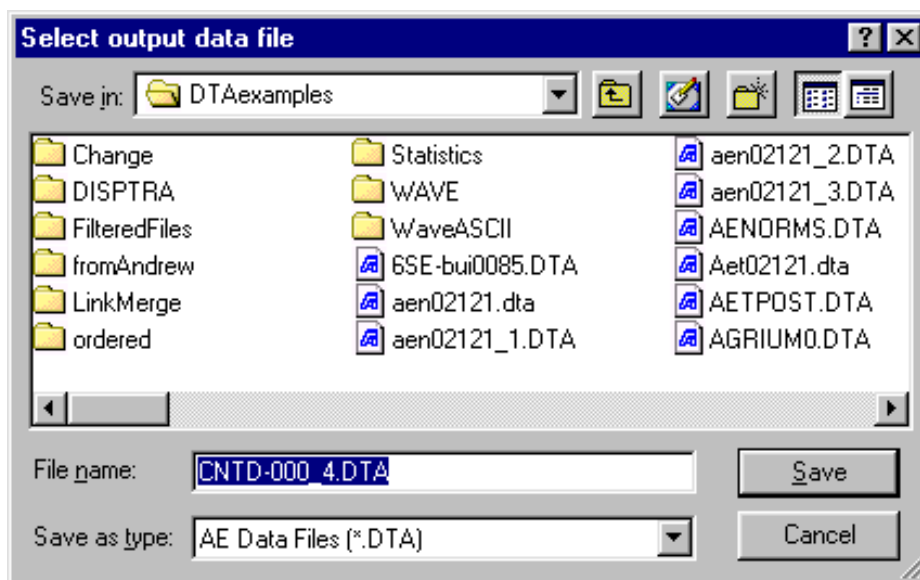


Figure 26. Select Output Data File

When the process finishes, will show the next dialog message to the user:

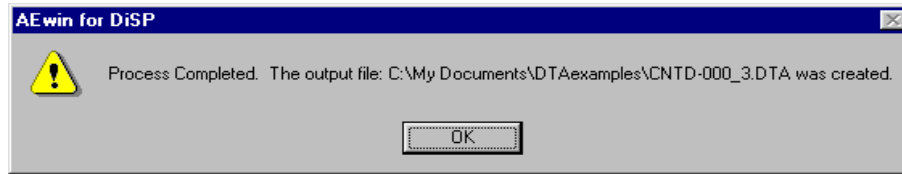


Figure 27. Merge Utility, Final Message

9. WAVEFORM FILTERING AND FEATURE EXTRACTION

9.1. Overview

"Waveform Filter and Extraction" is a 'built-in' utility within AEwin used in post-test to filter waveforms and extract hit-based data on a user-specified data file. This utility can be run by clicking on "Waveform Filter and Extraction" on the 'Utility' menu.

9.2. Setup

On startup, the user is prompted to enter an input data file. Be sure that this data file was created by the same data acquisition board type (AEDSP, PCI-DSP4, etc.) used by your version of AEwin. The input data file is then read, and the Setup screen is displayed. The information displayed reflects the setup information of the input data file.

Waveform & Hit Output:

Choose which output mode to use. The utility can save either:

- Original Waveforms and Original Hits.
- Filtered Waveforms and Extracted Hits.

In both cases, the waveforms/hits are only saved if the filtered waveform (or original if no filter defined) passes the new threshold. Note that Feature Extraction is disabled if "Save Original" is selected.

General Setup:

Here the user can set a new threshold and pre-amp gain if desired.

Waveform Filtering: Here, the user can specify any High, Low, or Band Pass waveform filter.

Feature Extraction:

This is enabled only if 'Saving Filtered Waveform' is selected. It replaces every hit with a new one based on its associated waveform.

The user can select or remove hit data set features and change HDT and PDT at this time. Certain hit dataset features (RMS, ASL, Parametrics, Cycle Counter) cannot be 'extracted' from the waveform and are copied directly from the original hit. Therefore, these features can't be enabled if they were not present in the original hit dataset.

Figure 28. Waveform filter and extraction setup dialog

9.3. Program Process

Once the 'Start' button is clicked and an output filename is entered, the main process begins. Each waveform and its associated hit are read from the input data file and processed as follows:

- 1) The filter (if enabled) is applied to the waveform. This may alter its arrival time and that of any hit extracted from it.
- 2) If the filtered waveform (or original if no filter is enabled) does not reach or exceed the threshold, then it and its associated hit are NOT written to the output file, and the process begins again with the next waveform and associated hit.
- 3) If the filtered waveform (or original if no filter is enabled) does reach or exceed the threshold, then:
 - If 'Saving Original Waveform...' is selected, then the original waveform and associated hit are written to the output data file. The process begins again with the next waveform and associated hit.
 - If 'Saving Filtered Waveform...' is selected, then the filtered waveform is saved to the output data file, and the process continues to step 4.
- 4) A new hit is extracted from the filtered waveform and is saved to the output data file. If no hit is associated with the waveform, then a new hit is NOT extracted and written.

9.4. Tips and Considerations

- 1) The input data file must have been collected using the same data acquisition board type (AEDSP, PCI-DSP4, etc.) as the version of AEwin this utility is running on. I.E. If the user is running AEwin for DiSP, then this utility can use data files created by AEwin for DiSP, DiSP, and DiSP-LOC. If the user tries to filter/extract a data file created by a different data acquisition board type, the setup may be incompatible. If it is, then an error message "Error Reading Hardware Setup, Aborted" will be shown, and the utility will abort.
- 2) Unassociated hits (hits not associated with a waveform) are NOT copied to the output data file.
- 3) The arrival times of filtered waveforms and their associated extracted hits may change during the filter process.
- 4) For the extracted features to be the same as the original features, the waveforms must be equal to a length defined as Pre-Trigger + Duration + HDT. If the waveform length is shorter than this, then the calculation of the features may terminate prematurely.

10. REMOVE MESSAGES

"Remove Messages" is a utility that lets the user remove some messages from a given DTA file and saves the result in a new DTA file. The user has to provide the input and output names for the files. The default name for the output file is the input file + the suffix QF (for quick filter).

The following figure shows the options for removing.

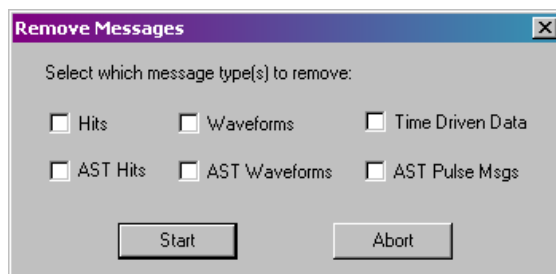


Figure 29. Remove Messages Dialog

11. CHANGE TITLE

"Change Title" is a utility that lets the user change the title from a given DTA file and saves the result in a new DTA file. The user has to provide the input and output file names. The default name of the output file is the name of the input file + the suffix "title."

The only restriction for the new title is the length. Currently, it is 40 characters.

The following figure shows the change title dialog.

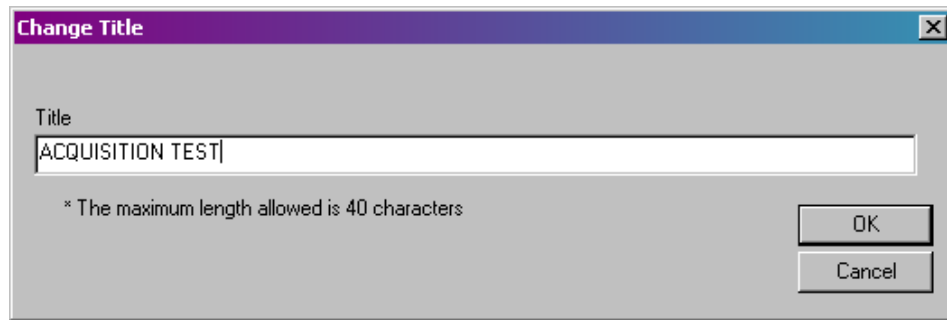


Figure 30. Change Title Dialog

12. DATA FILE FILTER

"Data File Filter" is a very helpful utility in AEwin that lets the user filter data files based on a user-defined filter. Clicking on the "Data File Filter" submenu at the 'Utility' menu can run this utility.

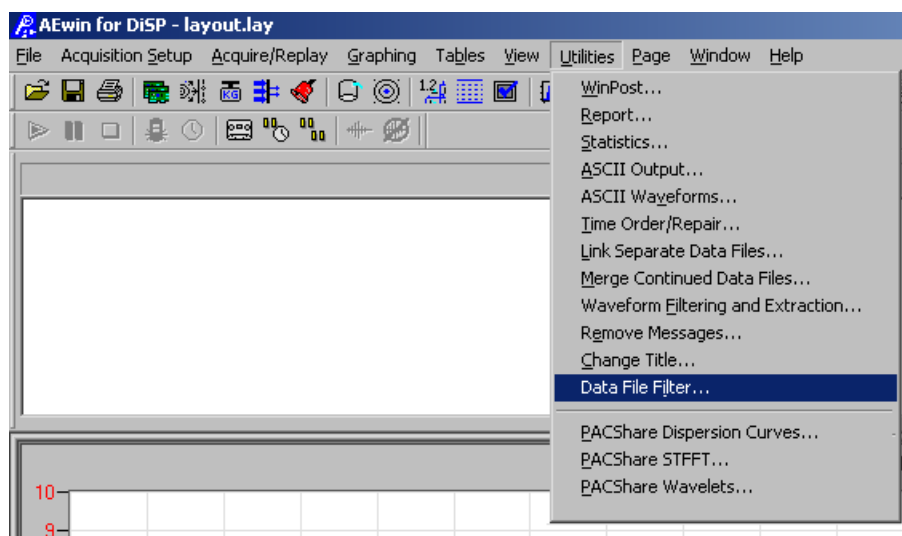


Figure 31. Utilities Menu

The user will be prompted to select an input data file for filtering. This utility program assumes the file is in time order. This is a valid assumption for any "DTA" data file collected under AEwin acquisition software.

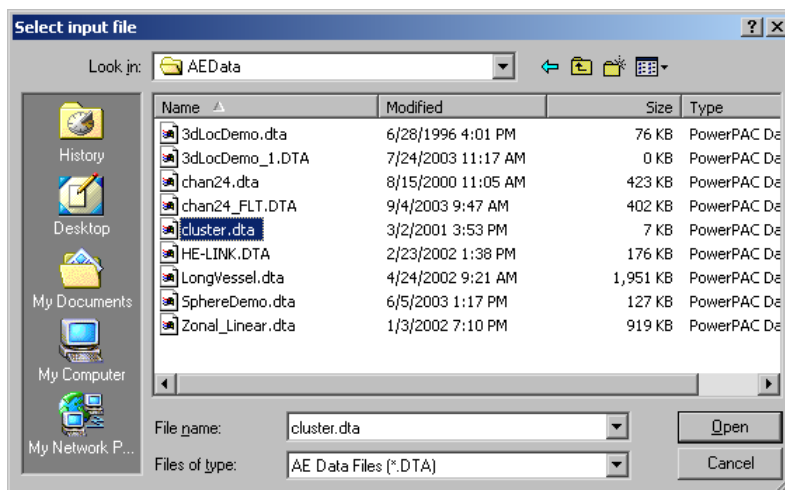


Figure 32. Select Input File

When the user opens the input file, the "Filter Setup Dialog" is shown. (Figure 31)

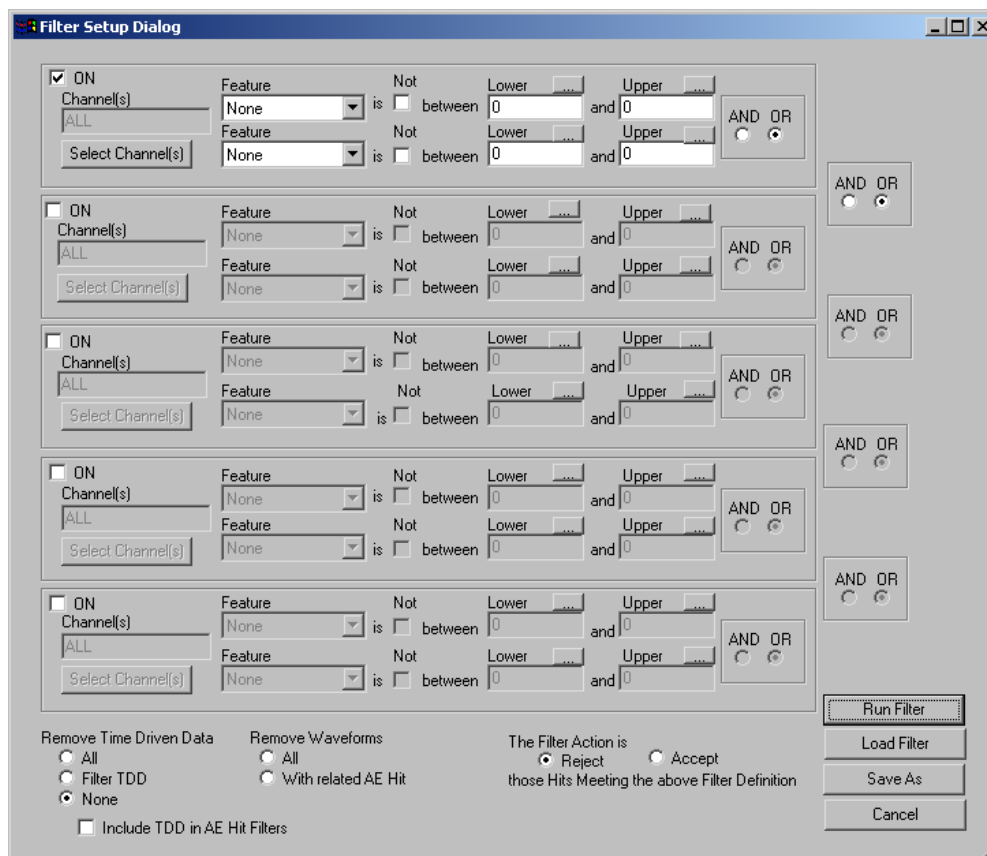


Figure 33. Filter Setup Dialog

The filter definition menu consists of five (5) groups, each of them linked by a logical condition (AND/OR). Each group operates on a pair of conditions, also linked by a logical condition and a set of channels to which those conditions apply. The user can set the group ON or OFF.

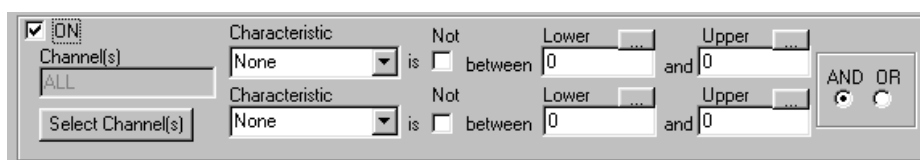


Figure 34. Group

To set a condition, the user selects a characteristic to look for from the drop-down list, then specifies a range by typing a lower and an upper value and also can check the "NOT" box if he wants to select the values out of the range.



Figure 35. Condition

When the characteristic selected is "Time," the lower and upper values should be given in seconds, but the software provides a time converter by clicking the little button at the right top of the edit box.

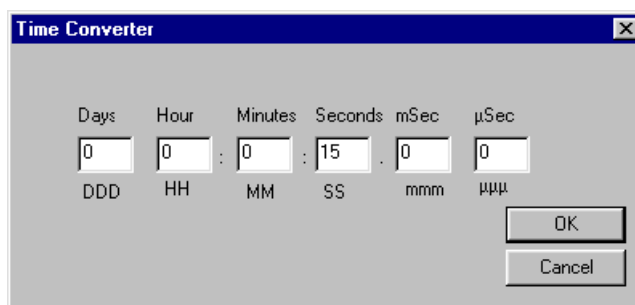


Figure 36. Time Converter User Interface

To specify the channels to filter or the condition based on certain channels, the user needs to click on the "Select Channel(s)" button of the specified condition. The interface that he will see is shown below.

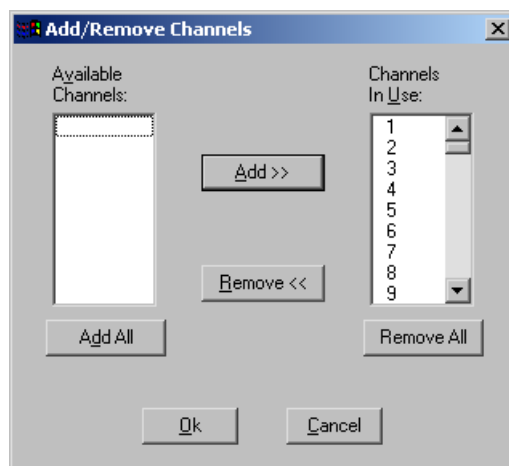


Figure 37. Select Channels User Interface

At the bottom part of the filter definition menu window, the user has other options applicable to the Waveforms and the Time-Driven Data messages. Also in this section is the definition of the Filter Action. By selecting "Reject," all the values that fulfill the condition will be rejected while selecting "Accept," all the values that fulfill the condition are going to be saved to the output file.

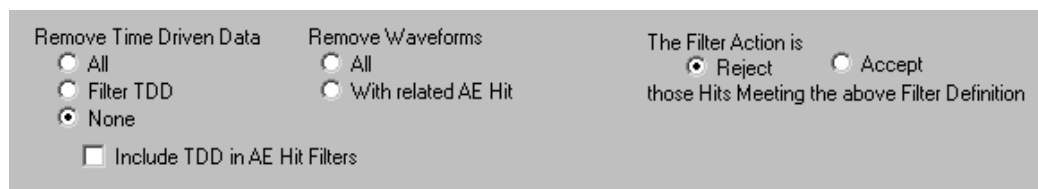


Figure 38. Other Options in Filter Definition Interface

Options to "Remove Time-Driven Data."

- **All.** Removes all TDD messages without checking anything.
- **None.** Keep all TDD messages without checking anything.
- **Filter TDD.** Checks the value of the TDD and filters if the characteristic selected is Time or any parametric.

If the check box for "Include TDD in AE Hit Filters" is checked, this means that the user wants to look at TDD values and filtering AE hits. To enable this, you have to choose the Filter TDD option.

Options to "Remove Waveforms."

- **All.** Removes all Waveforms
- **With related AE Hit.** Keep the waveform if the related AE Hit is saved or remove it when the AE Hit is removed

Filter Definition Filter User Interface the user has four option buttons at the right bottom:

- **Run Filter.** This option prompts the user for an output file name for the filtered file. Once the user selects the file, then the program runs the filter definition file to filter the input data file. While this action is being carried out, the progress bar will indicate how much of the file has been processed; the name of the file in the process is shown below the progress bar. This is helpful when working with big files. A message box will appear when the process is finished.
- **Load Filter.** This option prompts the user to recall a saved filter definition file (*.flt) to load and use in the filter for use directly or to use it as a base to create a new filter definition.
- **Save as.** This option prompts the user to name the filter definition file (*.flt) for future use/archiving.
- **Cancel.** This option lets the user exit the utility without saving any changes to the filter since the last time it was saved.

Note: The Report Utility can be used to view the history of filters applied to a data file.

As an extra note, batch mode is supported for the selection of multiple data files. Choosing more than one data file in Figure 32: "Select Input File, "0 will bring up the Folder Browse dialog box that will allow you to choose what folder to save all the processed data files. The Overwrite File Dialog will help to handle any conflicting data files.

13. OPERATORS LOG MANAGEMENT

The Operators Log Management is a utility tool in AEwin that allows users to view, modify and manipulate time mark data in a file during post-test analysis. It is accessible by selecting 'Operators Log Management...' from the Utilities menu.

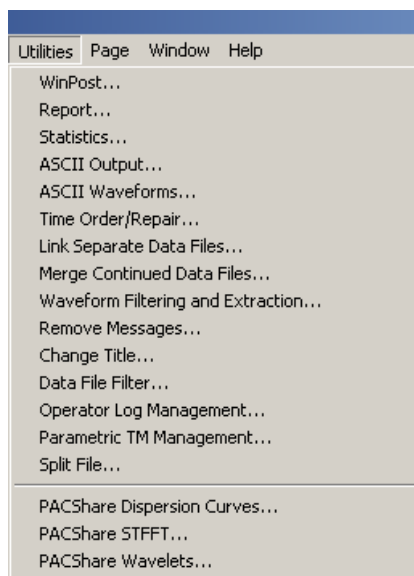


Figure 39. Utilities Menu

You will be prompted to select a data file for analysis. This utility program assumes that the file is in time order. Perform time ordering only if the data file was generated by an acquisition program other than AEwin.

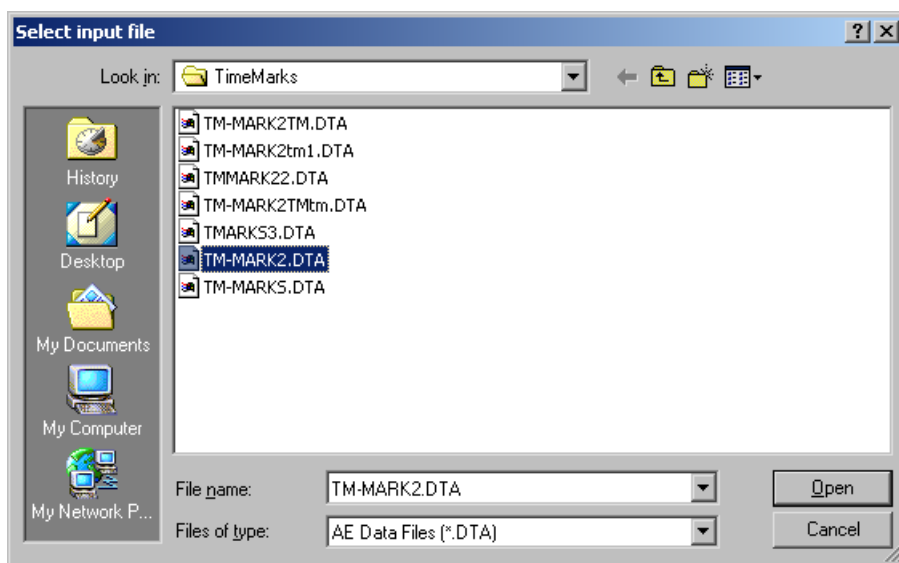


Figure 40. Select Input File

Once loaded, a list of existing comment user time marks will appear on the Operators Log Management screen (shown below).

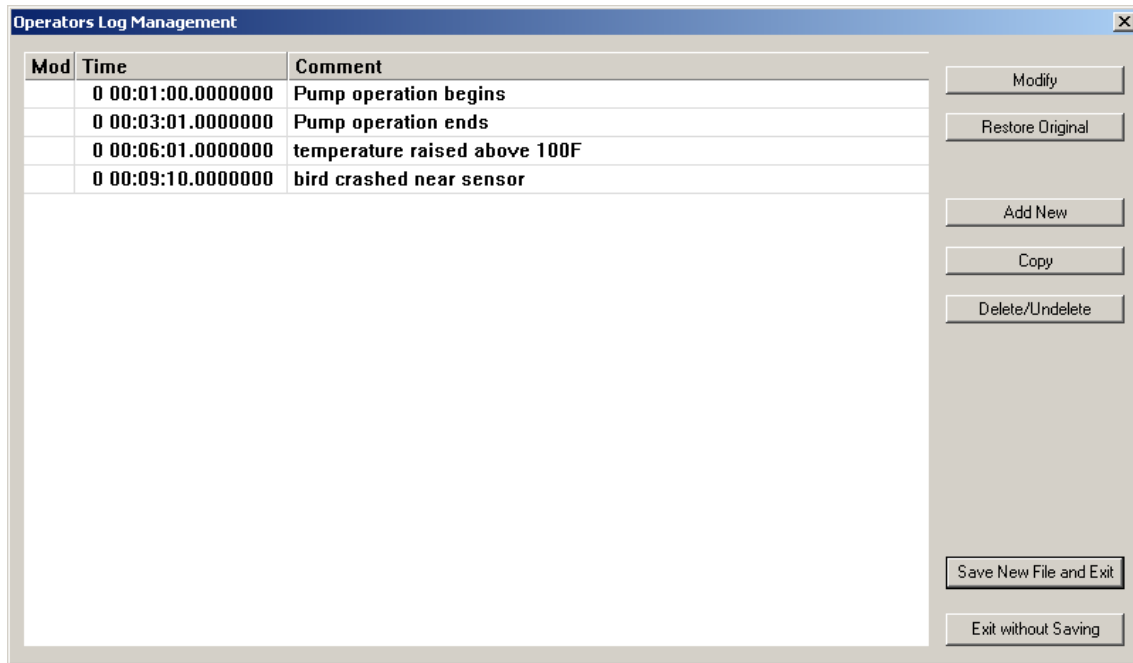


Figure 41. Operators Log Management

Each row in the grid pertains to a single comment time mark. There are three columns.

'**Mod**' shows an indication of any changes the user has made to the time-mark.

- 'N' means that the comment time mark was just created.
- 'M' means that the comment time mark is an existing one that was just modified.
- 'D' means that the comment time mark is an existing one to be deleted.
- 'DM' means that the comment time mark is an existing one, modified and to be deleted.

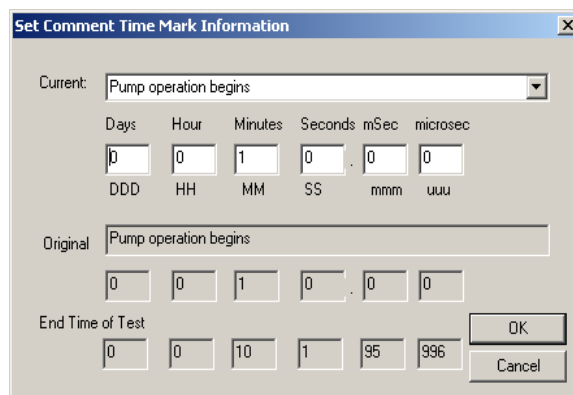
'**Time**' is the time of test of the comment time mark.

'**Comment**' is the label/comment (if any) of the time mark. The maximum length of the comment is 40 characters.

13.1. Modify

To change the time of test or comment of a comment time mark, select it by clicking on the row and pressing the "Modify" button. This will bring up the dialog box shown:

The "Set Comment Time Mark Information" dialog box displays the original comment (if any) and time of the test. A new comment can be entered by selecting one of the standard comments from its drop-down box or by typing a new one. You can enter a new time in the 'Time of Test' fields. You can see the end time of the test at the bottom of the dialog; you should be aware of this time because the time of test for any time mark should be less than the end time of the test. When your changes are complete, press 'OK' to keep your changes or 'Cancel' to discard them. A modified (but non-new) time-mark will have an 'M' in its 'Mod' column.



The dialog box titled "Set Comment Time Mark Information" contains the following fields:

- Current:** A dropdown menu showing "Pump operation begins".
- Time of Test:** A row of six input boxes for Days, Hour, Minutes, Seconds, mSec, and microsec. Below them are labels: DDD, HH, MM, SS, mmm, and uuu. The values shown are 0, 0, 1, 0, 0, 0.
- Original:** A dropdown menu showing "Pump operation begins".
- End Time of Test:** A row of six input boxes for Days, Hour, Minutes, Seconds, mSec, and microsec. Below them are labels: DDD, HH, MM, SS, mmm, and uuu. The values shown are 0, 0, 10, 1, 95, 996.
- Buttons:** "OK" and "Cancel" buttons at the bottom right.

Figure 42. Modify

13.2. Restore Original

To undo a modification of a comment time mark, select the time mark row from the grid and press "Restore Original." This will remove the 'M' from its 'Mod' column and prevent it from being modified in a future file update.

13.3. Add New

To add a new comment time mark to the grid, press "Add New." This will open the "Set Comment Time Mark Information" dialog box (shown previously). Once you have set the comment and time, press 'OK' to keep your changes or 'Cancel' to discard them. The new comment time mark will be inserted in time order to the grid, marked by an 'N' in the 'Mod' column. It will be added to the new file during a file update.

13.4. Copy

A quick way to create a comment time mark is to copy an existing one and modify it as desired. To do this, select a time mark row from the grid and press "Copy." This copies the selected comment time mark and opens the "Set Comment Time Mark Information" dialog box (shown previously). Once you have completed your changes, select 'OK' to keep the new time mark or 'Cancel' to discard it. The new comment time mark will be inserted in time order to the grid, marked by an 'N' in the 'Mod' column.

13.5. Delete/Undelete

To delete a comment time mark, select the row from the grid and press "Delete/Undelete." If the time mark is a new one (marked by 'N'), it is immediately and permanently removed from the grid. If not, then a 'D' will appear in the 'Mod' column, except when the time mark is a modified one (with an 'M' in the 'Mod' column); for this case, a 'DM' will appear in the 'Mod' column. This action can be undone at any time before a file update. During a file update, any comment time marks designated for deletion will be removed from the new file.

To undelete a comment time mark, select it from the grid and press "Delete/Undelete." This will remove the 'D' or 'DM' mark from the 'Mod' column and prevent the comment time mark from being deleted during a future file update.

13.6. Save New File and Exit

To create a new modified file from your old one, press "Save New File and Exit." You will be prompted to enter a new output file name (see image below); by default, you will have the input file name and the suffix 'tm'.

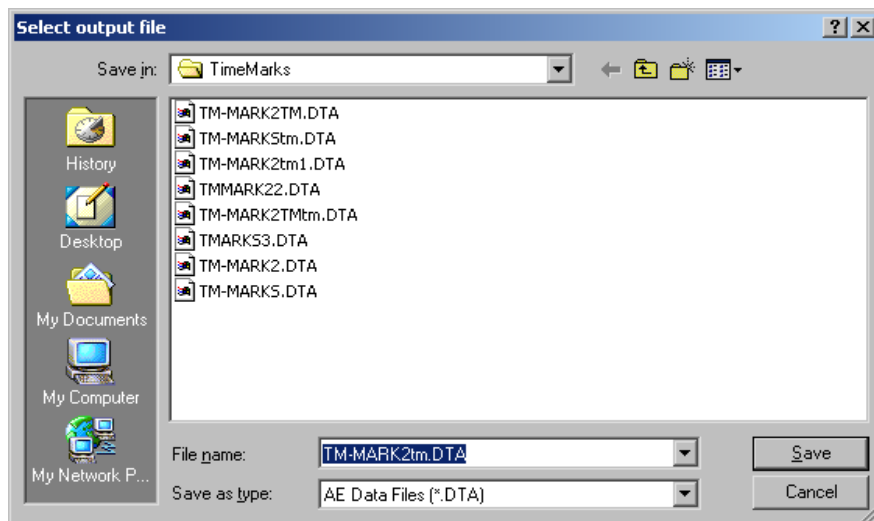


Figure 43. Select Input File

If you select the "Save" button, it will then copy your original file, modifying or deleting existing time marks and adding new ones as needed based on the changes you made. The original file is left unmodified.

13.7. Exit without Saving

To exit without saving changes to a file, press the "Exit without Saving" button.

14. PARAMETRIC TM MANAGEMENT

Parametric TM Management is a utility tool in AEwin that allows users to view, modify, add and manipulate parametric time marks in a file during post-test analysis. Many times a user is not able to connect to a parametric source for input into the AE system. This feature allows the user to enter and record parametric information (as might be read and recorded from a meter or CRT screen) and add it to the existing AE data file in Post Test. This is accomplished through the use of 'parametric' time marks. These are created by the user and hold a (user-specified) value for parametric 1 and comment. Once setup is complete, these are written to the file, and the parametric 1 value for all hit and time-based messages are changed to the value of the most recent parametric time mark.

It is accessible by selecting 'Parametric TM Management...' from the Utilities menu.

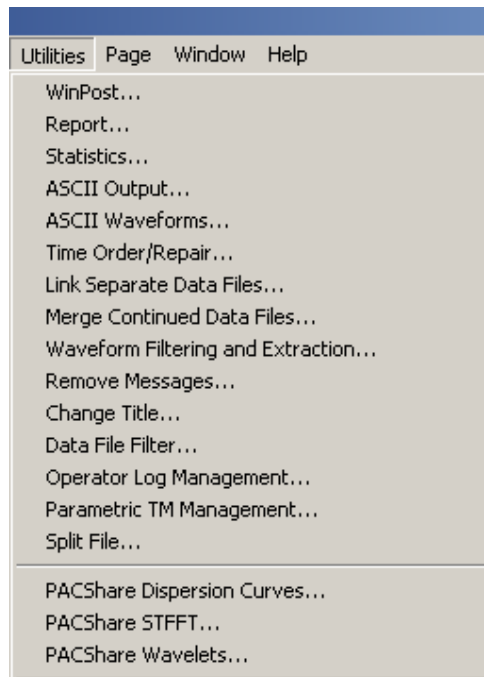


Figure 44. Utilities Menu

You will be prompted to select a data file for analysis. This utility program assumes that the file is in time order. Perform time ordering only if the data file was generated by an acquisition program other than AEwin.

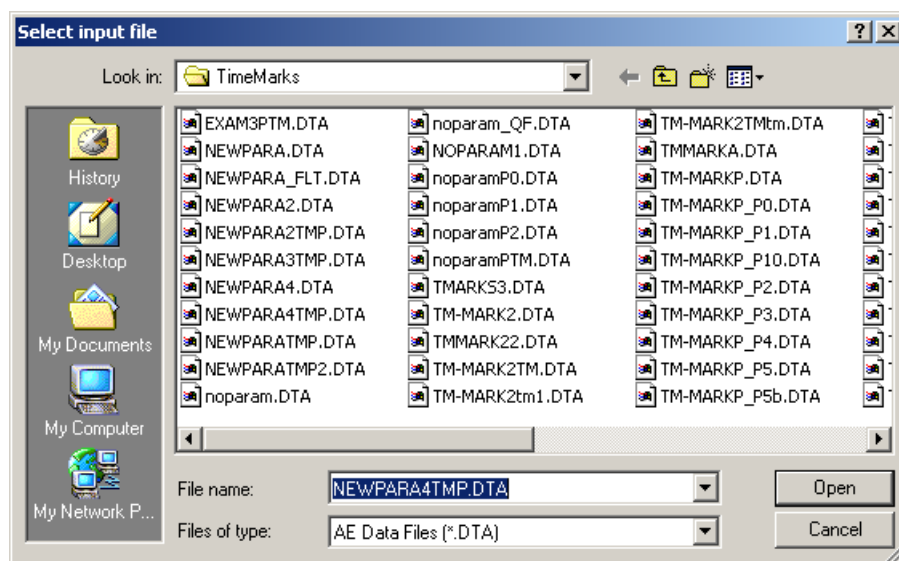


Figure 45. Select Input File

Once loaded, a list of existing time marks will appear on the Parametric TM Management screen (shown below).

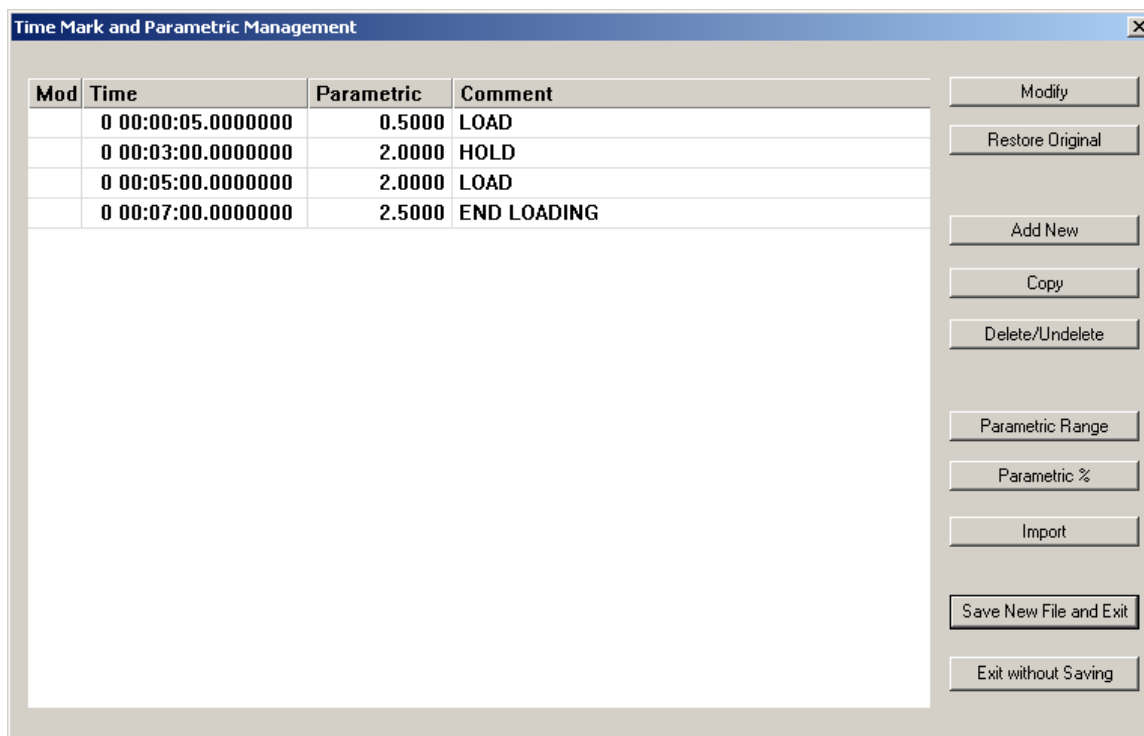


Figure 46. Parametric FM Management

Each row in the grid pertains to a single time mark. There are four columns.

1. **'Mod'** shows an indication of any changes the user has made to the time-mark.
 - **'N'** means that the time mark was just created.
 - **'M'** means that the time mark is an existing one that was just modified.
 - **'D'** means that the time mark is an existing one to be deleted.
 - **'DM'** means that the time mark is an existing one, modified and to be deleted.
2. **'Time'** is the time of test of the time mark.
3. **'Parametric'** displays the value of parametric 1 archived with the time mark. This number is scaled in the same units the user defines in the 'Scaling' dialog box in hardware setup (F2). **Or** it may display this same value as a percentage of the user-defined range. See the section on 'Changing the Parametric Range' for more on this.
4. **'Comment'** is the label/comment (if any) of the time mark. The maximum length of a time mark message is 40 characters.

14.1. Modify

To change the time of test, comment, or parametric value of a time mark, select it by clicking on the row and pressing the "Modify" button. This will bring up the dialog box shown below:

The dialog box titled "Set Time Mark and Parametric Information" contains the following fields and controls:

- Current:** A drop-down menu.
- Parametric Value:** A text box containing "1" and a "Volts" label.
- Time of Test:** A set of six spin boxes for Days (0), Hour (0), Minutes (1), Seconds (0), mSec (0), and microsec (0). Below these are labels DDD, HH, MM, SS, mmm, and uuu.
- Original:** A text box.
- Original Parametric Value:** A text box containing "1" and a "Volts" label.
- Original Time of Test:** A set of six spin boxes for Days (0), Hour (0), Minutes (1), Seconds (0), mSec (0), and microsec (0).
- End Time of Test:** A set of six spin boxes for Days (0), Hour (0), Minutes (10), Seconds (1), mSec (95), and microsec (996).
- Buttons:** "OK" and "Cancel" buttons at the bottom right.

Figure 47. Set Information

The "Set Time Mark and Parametric Information" dialog box displays the original comment (if any), the parametric value, and the time of the test. A new comment can be entered by selecting one of the standard comments from its drop-down box or by typing a new one. You can enter a new parametric value and a new time in the 'Time of Test' fields. You can see the end time of the test at the bottom of the dialog; you should be aware of this time because the time of test for any time mark should be less than the end time of the test. When your changes are complete, press 'OK' to keep your changes or 'Cancel' to discard them. A modified (but non-new) time-mark will have an **'M'** in its **'Mod'** column.

14.2. Restore Original

To undo a modification of a time mark, select the time mark row from the grid and press "Restore Original." This will remove the **'M'** from its **'Mod'** column and prevent it from being modified in a future file update.

14.3. Add New

To add a new time mark to the grid, press "Add New." This will open the "Set Time Mark and Parametric Information" dialog box (shown previously). Once you have set the comment and time, press 'OK' to keep your changes or 'Cancel' to discard them. The new time mark will be inserted in time order to the grid, marked by an 'N' in the '**Mod**' column. It will be added to the new file during a file update.

14.4. Copy

A quick way to create a time mark is to copy an existing one and modify it as desired. To do this, select a time mark row from the grid and press "Copy." This copies the selected time mark and opens the "Set Time Mark and Parametric Information" dialog box (shown previously). Once you have completed your changes, select 'OK' to keep the new time mark or 'Cancel' to discard it. The new time mark will be inserted in time order to the grid, marked by an 'N' in the '**Mod**' column.

14.5. Delete/Undelete

To delete a time mark, select the row from the grid and press "Delete/Undelete." If the time mark is a new one (marked by 'N'), it is immediately and permanently removed from the grid. If not, then a 'D' will appear in the 'Mod' column, except when the time mark is a modified one (with an 'M' in the 'Mod' column); for this case, a 'DM' will appear in the 'Mod' column. This action can be undone at any time before a file update. During a file update, any time marks designated for deletion will be removed from the new file.

To undelete a time mark, select it from the grid and press "Delete/Undelete." This will remove the '**D**' or '**DM**' mark from the '**Mod**' column and prevent the time mark from being deleted during a future file update.

14.6. Parametric Range

To display parametric 1 in percent form, you need to set a valid parametric range. Pressing "Parametric Range" will open the 'Set Parametric Range' dialog box (shown below).

Figure 48. Parametric Range

Here you can set the values for 0% and 100% of the range. There are also shown the minimum and maximum recorded parametric 1 values. When you are done, press 'OK' to apply your changes or 'Cancel' to discard them.

14.7. Displaying Parametric % or Value

To toggle the parametric display between Value and % (percent), press the button "Parametric % / Value" on the Time Mark and Parametric Management Dialog. This is labeled as either "Parametric %" or "Parametric value," depending on what is currently being displayed.

14.8. Import

This option lets the user write a txt file with the information instead of modifying or adding one by one the parametric time marks to the grid. The file needs to have two columns; the first one is the time in seconds and the second one is the parametric value.

14.9. Save New File and Exit

To create a new modified file from your old one, press "Save New File and Exit." You will be prompted to enter a new output file name (see image below); by default, you will have the input file name and the suffix ' tmp.'

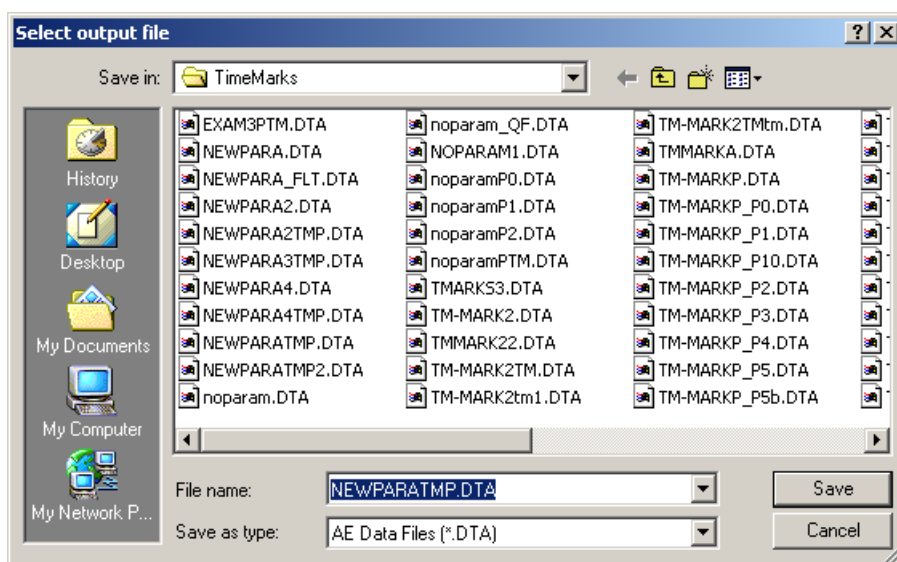


Figure 49. Select Output File

If you select the "Save" button, it will then copy your original file, modifying or deleting existing time marks and adding new ones as needed based on the changes you made. The original file is left unmodified.

14.10. Exit without Saving

To exit without saving changes to a file, press the "Exit without Saving" button.

15. SPLIT FILE

Split File Utility is a tool in AEwin that allows the user to split a given file into smaller files. It is accessible by selecting 'Split File...' from the Utilities menu.

The user will be prompted to select an input data file. This utility program assumes the file is in time order. This is a valid assumption for any "DTA" data file collected under AEwin acquisition software.

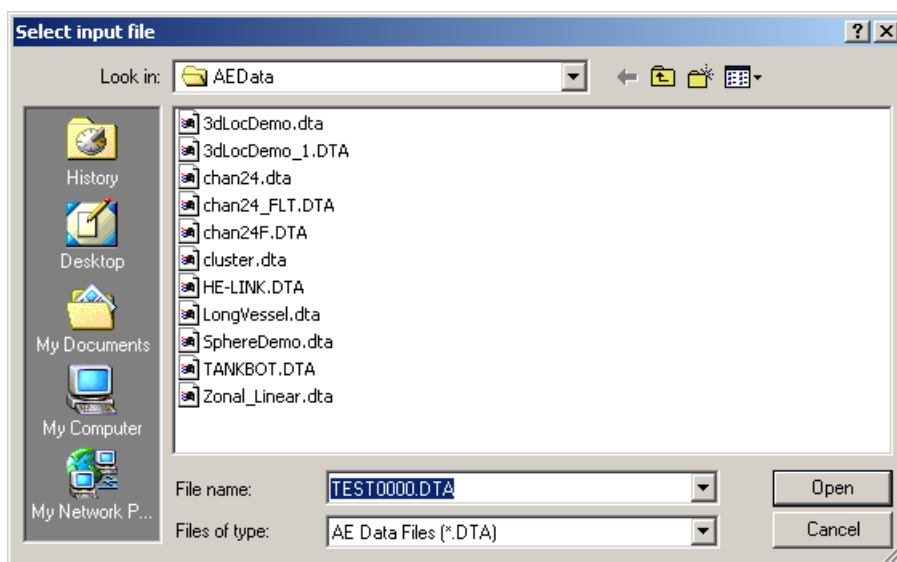


Figure 50. Select Input File

Once the user selects the name of the input file, then the utility will prompt for the base name for the output files. The program will take the base name the user selected and will add to it a consecutive number as a suffix.

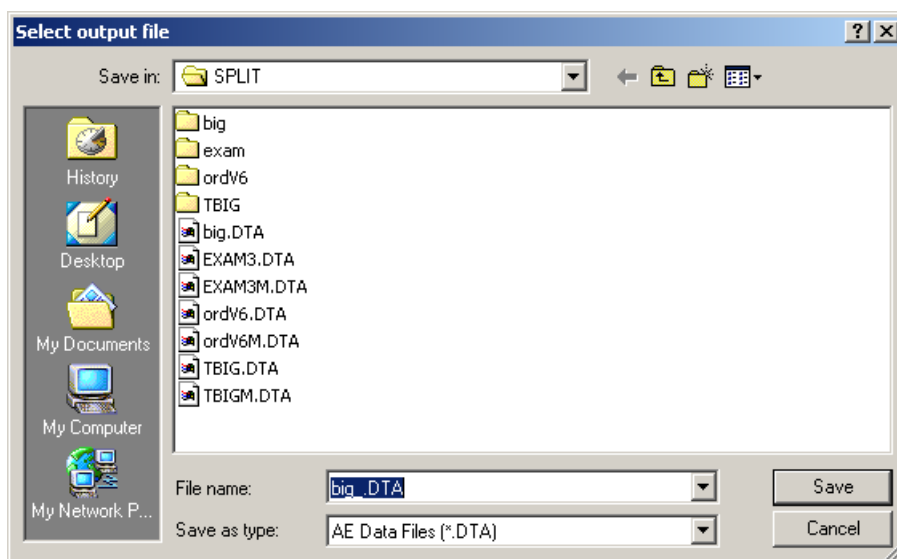


Figure 51. Output Type

Then the utility will ask the user to type the desired size for the resultant files, which has to be specified in Kb (and for the program is taking 1 Kb equal to 1000 bytes).

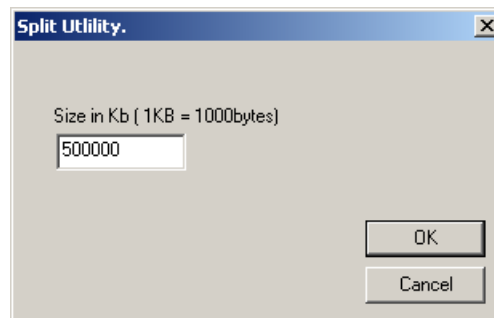


Figure 52. Split Utility

APPENDIX I

The AE Timing Parameters: HDT, HLT, PDT, and MAX Duration

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The AE Timing Parameters: HDT, HLT, PDT, and MAX Duration

This appendix describes the purpose and functioning of the three-timing parameters used in AE waveform measurements:

HIT DEFINITION TIME (HDT) — Formerly known as Single Channel Event Time-Out (SCETO)

HIT LOCKOUT TIME (HLT) — Formerly known as Re-arm Time Out (RTO)

PEAK DEFINITION TIME (PDT) — Formerly known as Rise Time - Time Out (RTTO)

This appendix also offers methods for selecting values of these parameters for practical AE testing.

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1. INTRODUCTION

The four AE Timing Parameters are found under the second tab of the AE Hardware Setup menu. They have an important influence on how each channel's data processor (FPGA) forms hits and/or calculates signal features after the sensor output has been conditioned and digitized. Therefore, the operator sets the four timing parameters before the test.

Briefly stated, the four timing parameters have the following functions and typical settings:

- **Hit Definition Time (HDT):** this is a time constant used to terminate the measurement of a signal. It functions like a re-triggerable one-shot whose expiration brings the hit to an end. This is the most important of the four timing parameters. *Traditional, material-oriented settings are 100-200 μ s in attenuating non-metals, 400-2000 μ s in metals. When waveforms are being recorded in modern systems, "flatlines" are conveniently avoided by using longer HDT (800 μ s).*
- **Hit Lockout Time (HLT):** this is used to create a "dead time" after the end of a hit. Any threshold crossings occurring during this time will be ignored. *It is a common practice to set the HLT equal to the HDT.* Another common practice is to set the HLT to the minimum permissible value, such as 2 μ s). Other special uses are discussed below.
- **Peak Definition Time (PDT):** for signals whose envelopes have several rising peaks, this timing parameter gives some control over which peaks are used for reporting risetime and amplitude. *The commonest practice is to set PDT to the maximum allowed, i.e., one-half of the HDT.*
- **Max Duration:** This sets a limit to the maximum duration allowed for a hit. Unlike the other three parameters, which are entered in microseconds, this one is entered in milliseconds. The Max Duration function is important when the signal stays above the threshold more-or-less continuously due to noise or extremely high AE activity. If the channel is allowed to "lock-up" and process this activity as a single ultra-long hit, there is a real danger that the operator or data analyst will not realize that the channel is in saturation, thinking instead that it is merely inactive. Proper use of the Max Duration parameter mitigates this risk. *A preferred value is 99ms, for reasons explained later.*

Usually, the four timing parameters are set to the same values for each channel. However, if different kinds of channels are being used on the same test - high frequency and low-frequency channels, for example - the timing parameters might be set differently for the two kinds.

It is possible, after the test, to re-process the captured signal waveforms using different values for the timing parameters. However, this kind of analysis has some limitations, and it is not performed very often.

Many years ago, in previous generations of AE system HDT, PDT and HLT were known by the acronyms Single Channel Event Time Out (SCETO), Rise Time Time Out (RTTO), and Inhibit Time Out, respectively.

The four timing parameters will now be discussed individually in detail.

2. HIT DEFINITION TIME (HDT)

A signal from a discrete AE event is detected when its voltage first exceeds the detection threshold, whether this is in the positive-going or negative-going direction. This is the beginning of the "hit." The measurement of signal features begins at this point. Almost as important is the end of the hit, the moment when all measurement of signal features stops. The hit automatically ends when an amount of time equal to the HDT has elapsed without any threshold crossings. The tacit assumption is that the signal has subsided below the threshold, and after such a long time, it is not going to rise above it again; all is finished, and it is time to close out this hit and get ready for the next one.

The selection of HDT can substantially affect how many hits are recorded when burst-type AE is taking place. Consider, for example, the AE activity shown in Figure 1 coming from a plastic billet undergoing stress corrosion cracking. At this point in the test, two events took place less than one millisecond apart. Each had a fine structure giving rise to a rather complex waveform. We will see how the value selected for HDT affects the formation of hits in this situation. The horizontal lines indicate a 26dB_{AE} threshold.

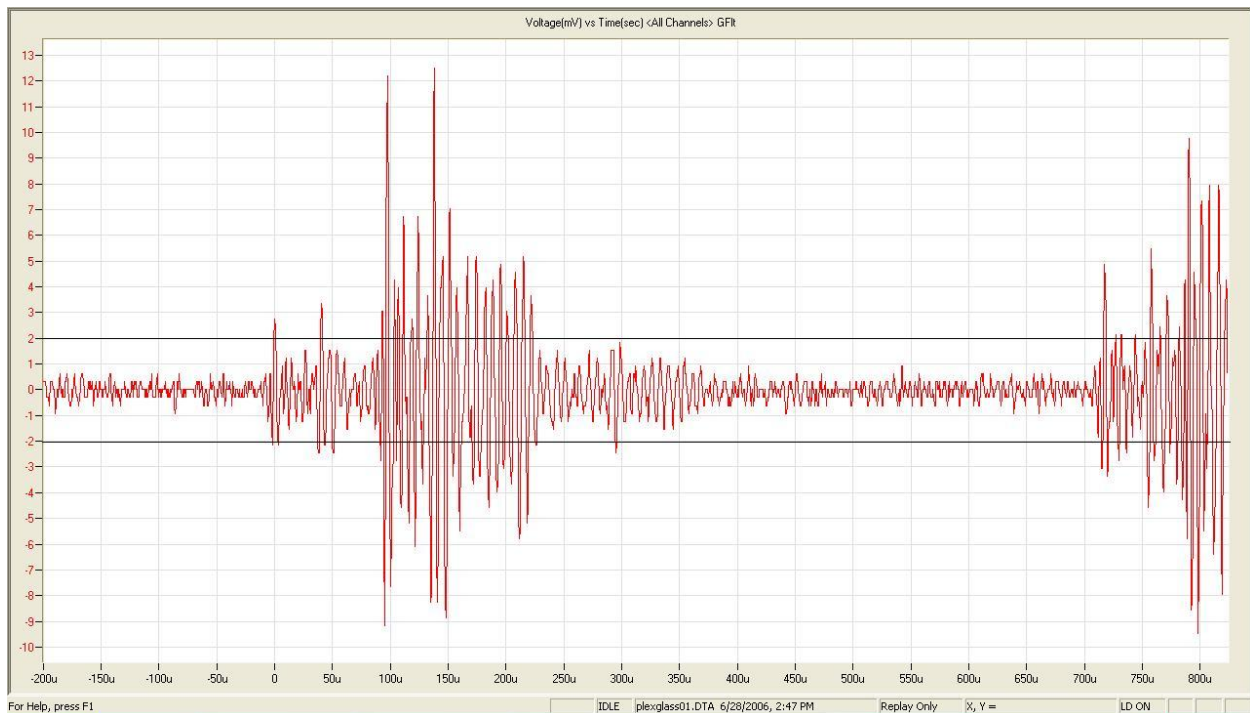


Figure 1. Recorded AE Activity (Waveforms from Two Events) Showing Threshold

The hit begins at time zero (this is how AEwin defines its timescale for waveforms). Within the first $5\mu\text{s}$, there are three threshold crossings, one positive and two negatives. There is then a gap without further threshold crossings until time $40\mu\text{s}$ when there is a negative crossing, and so on.

- Suppose the HDT is made extremely short, $25\mu\text{s}$. The hit will end at $30\mu\text{s}$. Ignoring the effect of HLT, which will be discussed later, we can see that a second hit will be started when the threshold is crossed again at $40\mu\text{s}$. This hit will last until $75\mu\text{s}$ ($25\mu\text{s}$ after the negative-going threshold crossing at $50\mu\text{s}$). A little later, the large-amplitude part of the signal crosses the threshold at $90\mu\text{s}$, starting a third hit that will last until $250\mu\text{s}$. Finally, a fourth hit starts at $295\mu\text{s}$, and eventually a final hit from the second event at $715\mu\text{s}$. Four hits from a single event is not a good way to record the AE activity. Logically one would want to see one hit from each event. Clearly, we need an HDT greater than $25\mu\text{s}$ to achieve this.

- Suppose the HDT is set to 50 μ s. This will work better; from 0 μ s to 220 μ s, there are no gaps longer than this without threshold crossings. But the hit will end at 270 μ s, and there will still be a second hit from the small-signal excursion 295 μ s. So the HDT needs to be even longer.
- The longest gap without threshold crossings, in the signal resulting from the first event, is the gap of 70 μ s from 225 μ s to 295 μ s. Any value of HDT longer than 70 μ s will produce the desired result, spanning the gaps in the first event's threshold crossings to produce a single hit. It is common and important for hit formation that the longest gaps between threshold crossings are found at the *end* of the signal. This is especially noticeable for large signals in very reverberant test articles.
- However, if the hit definition time is made too long, a different problem arises. Specifically, if it is made longer than 420 μ s, it will also span the gap from the 295 μ s tail end of the first-event response to the 715 μ s beginning of the second. The system will then count the two responses as a single hit. With the two events scrambled together in this way, many of the resulting features will be compromised or even meaningless; for example, the amplitude will be the amplitude of the larger event, and the time interval between them will determine the duration.

Summarizing the results of this analysis is clear that:

- The HDT should be set longer than any anticipated "gaps" between consecutive threshold crossings but shorter than any anticipated minimum time intervals between consecutive events.
- The limitations are that if the setting is too short, single events will be split into multiple hits, whereas if the setting is too long, adjacent events will be combined into single hits.

Long ago, the practical experience led to the use of values in the following ranges for a large majority of tests:

- Non-metals (composite materials, concrete, etc.): 100 μ s -200 μ s. Shorter values would apply to smaller specimens, higher frequencies (150kHz and up, rather than 60kHz and below), and materials with more damping (fiberglass is notably more damping than graphite-epoxy). Longer values would apply to larger structures, lower frequencies, and materials with less damping. If the activity levels are low, higher values can be selected without concern; if it is required to make valid measurements of signal characteristics at high data rates, the choice becomes important, and one tries to keep the HDT as low as possible.
- Metals: 400 μ s -1500 μ s. 400 μ s is the commonest setting, used with 150kHz sensors in MONPAC testing of metal tanks and pressure vessels, in railroad tank car testing, etc. In opting for this low-end value, it is considered that the most serious error would be to "lose" events by having two or more of them forming a single hit. The alternative viewpoint emphasizes the importance of not having single events reported as multiple hits. This viewpoint leads to the selection of larger HDT values and will find more favor on highly reverberant, undamped structures where the signal includes sustained echoes from distant parts of the structure. In these circumstances, HDT's of 1000 μ s or even 2000 μ s have proved effective. The use of longer HDT's will typically accompany the use of lower frequencies.

There are no codified measurement procedures for determining an appropriate value for HDT; the appropriate value for any given application is usually worked out by the developers and written into the test procedure. But here is one practical approach that makes use of pencil lead breaks (PLB's). PLB's are performed, starting with a low HDT such as 200 μ s (likely producing multiple hits) and then doubling it until only a single hit from each PLB. Given that the PLB is an unusually large source and will thus give an unusually large quota of late-arriving but still detectable echoes, one would then run the test with the value *before* the one that first produced single hits.

With the introduction of waveform recording systems such as the PCI series, a new factor was added to the business of selecting the HDT. The way these systems operate, the recording of the waveform stops *either* on reaching the end of the selected waveform length *or* at the end of the HDT, whichever comes first. In the latter case, the displayed waveform ends with a section of "flatline," as shown in Figure 2. (In this example, the pretrigger was 400µs, the HDT was 400µs, and the record length was 1024µs). If one wants to be sure that all recorded waveforms extend over the whole record timeframe without any flatlines, one should make sure that:

$$\text{HDT} > (1024 \times L)/S - P$$

Where L is the Length in k, S is the Sample Rate in MSPS (millions of samples per second), and P is the pretrigger (in microseconds). This was neatly accomplished in the default settings of software shipped from ~2007 onwards: Sample Rate 1MSPS, Pretrigger 256µs, Length 1k, and HDT 800µs. With these settings, the 1024 µs waveform record will always end before the expiry of the HDT (256 + 800 = 1056µs), so there will never be any flatline.

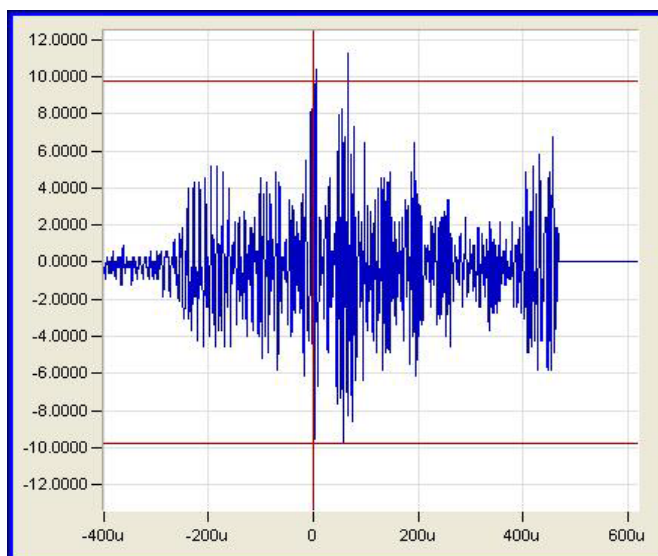


Figure 2. "Flatline" in the recorded waveform when the HDT expires before the end of the waveform "length."

3. HIT LOCKOUT TIME (HLT)

This setting determines the Length of a "dead time" that begins at the end of the hit, i.e., at the expiry of the hit definition time. During this "dead time," any new threshold crossings, which could otherwise start a new hit, are ignored.

The original design concept of this dead time was to get rid of the insignificant, low-amplitude echoes at the end of a signal and allow the system to get on with storing the measurements made on the more significant, early part of the signal. This made an important contribution to system throughput in the 1980s when it took as long as 300 μ s for a channel to transfer the measured signal features into the buffer memory. As systems became faster, the HLT became less significant. As a reasonable rule of thumb, it became customary to set it equal to the HDT. With that setting, getting rid of tail-end echoes in larger signals would potentially do a useful job without disproportionately enlarging the total dead time (HDT + HLT). This setting, HLT = HDT, is spelled out in many well-established test procedures that are still in use today.

When the AE system revolution of the 1990s replaced hardware measurement circuits with A-D converters and FPGAs, the HLT could be made arbitrarily short. In AEwin, entries down to 2 μ s are accepted by the software. There is a distinct benefit in working with this minimum value. If one uses a large HLT, a signal from a new event may begin during the HLT dead time, but the measurement process will only begin when the HLT expires. The measured values will therefore be false. This is often evidenced when signals show up with unreasonably short risetimes, etc. While much more could be written about the problems of signal measurement at high event rates in general, this particular problem is practically eliminated by setting the HLT to the minimum permitted value.

The above values (HLT = HDT and HLT $\approx$ 0) are most often used for regular data acquisition. However, there are some other strategies that can be useful in particular circumstances. For example, during pencil lead break (PLB) operations, it is desirable to eliminate the small hits that are sometimes caused by accidental movements of the pencil, broken lead impacting the test article, etc. Setting the HLT to its maximum permitted value (65535 μ s) will disable the system for 65ms after each lead break, greatly reducing the unwanted hits from secondary effects. The record can be made cleaner still by setting the HDT to the same large maximum value.

4. PEAK DEFINITION TIME (PDT)

Of the four timing parameters, this one has been of the least interest to users. It was introduced with the notion that the user might want to measure a signal's early wave arrivals rather than on its ultimate peak. The underlying assumption is that the signal is separated into distinct arrivals with space in between. Figure 3 shows such a signal; it was produced by pencil lead break on a ¼ inch aluminum plate, at a distance of about 18 inches from an R15I sensor. It illustrates the well-known pattern of an extensional-wave arrival (starting at 0µs) followed by a flexural-wave arrival (starting at 90µs).

Suppose the threshold is set to 40dB_{AE} (10mV on the diagram) to detect the extensional wave. PDT controls the measurement of risetime and amplitude in the following manner. As the signal waveform is processed in the FPGA, the latter keeps track of the largest excursion (peak) in the signal so far. If no higher peak occurs within the PDT, that excursion becomes the peak of the record, and its amplitude becomes the amplitude of the record. In the example of Figure 3, the detection threshold is 40dB_{AE}, crossed by the first negative-going excursion at time zero. The next peak is a little larger, 15.6mV (44dB) positive going at 3µs. After that, there are no higher peaks until 70µs. So if the PDT is set below 67µs, the peak amplitude will be measured as 44dB and the risetime as 3µs. But if the PDT is set above 67µs, the peak amplitude will be measured as 200mV (66dB_{AE}) and the risetime as 100µs.

This is how in concept, the PDT might be used to select the wavemode to be measured. But in practice, it is difficult to achieve a setup that will do this reliably because there is so much variation in the source-sensor distances and other factors affecting the signal. So this use of PDT is seldom attempted. Instead, normally the user takes the simplest approach and aims to measure the ultimate highest peak. To this end, the PDT is set to its highest permissible value, which is one-half of the PDT.

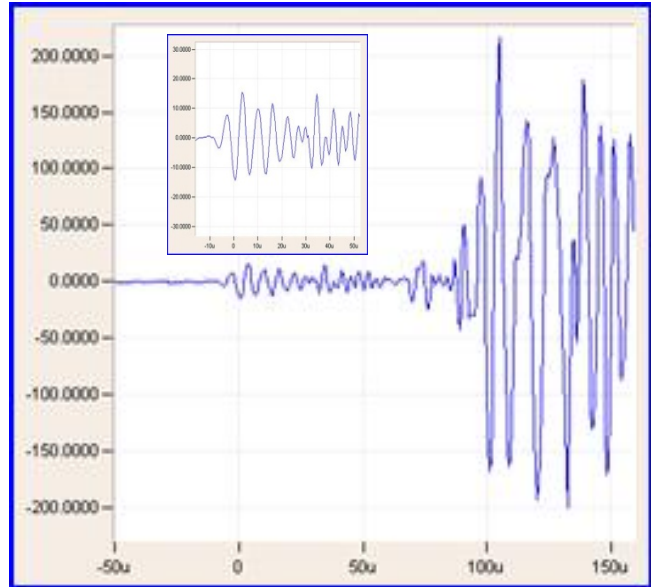


Figure 3. Waveform Showing Extensional and Flexural Waves

5. MAX DURATION

The duration of normal AE burst-type signals ranges from $\sim 1\mu\text{s}$ to $\sim 10\text{ms}$. Recall that duration is the time from first threshold crossing to last and that the expiration of the HDT determines the last threshold crossing. Thus, under abnormal conditions, the crossings of the threshold may persist much longer than $\sim 10\text{ms}$ without the HDT expiring. Such conditions include final failure processes such as fast tearing of cracks in emissive metals, approach to rupture in fiber-reinforced composites and the collapse of concrete and rocks; continuous emission from leaks or yielding when the signal amplitude is above the threshold; and certain kinds of noise.

There is a limit to the duration that the AE system can record. In the early days, it was set by the capacity of the hardware counters that could record durations up to $65535\mu\text{s}$. Later, this was increased up to several seconds. But this brought on a real risk that the ultra-long hits would escape notice. They would be very few in number, and the recorded values of counts, duration, and energy could be extraordinarily large, falling outside the scale of incautiously prepared graphs. The operator or naïve analyst could easily miss this data and form incorrect conclusions. To mitigate this risk, the maximum recordable duration was put under the control of the operator. When the duration of a hit reaches the entered value, the hit is automatically terminated without waiting for the expiry of the HDT. The usual HLT follows the termination of the hit, and then the channel is ready to detect the next hit.

Since Max Duration comes into play only under abnormal conditions, the selected value has little or no effect under normal conditions. A value of 99ms is convenient because it lends itself to a graph with a duration on one axis, logarithmically scaled with a maximum of $100,000\mu\text{s}$. For example, this scaling is widely used in the scatter plot of duration versus amplitude, which is one of the most important graphs for data interpretation. Also, with a Max Duration of 99ms , when AE is continuously above the threshold, the hit rate will be high enough to be readily noticed, low enough not to fill the record with useless data, and suitable for recording changes in signal amplitude with time.

There is one other known creative use of Max Duration. This was an unusual application where the need was to record very rare transient signals of very large amplitude and very long ($\sim 1\text{s}$) reverberation time. The normal arrangement would be to use a long Max Duration and let the signal produce a single hit - but then only the first part of the signal waveform would be captured, and important information might go unrecorded. In this case, it was desired to record the signal waveforms in as much detail as possible, including the reverberating "tail" of the signal. The normal waveform recording lengths and sampling rates were inadequate to capture such a reverberant signal fully. So an innovative approach was taken. To maximize the signal acquisition, first, the waveform length was made as long as possible, and then the Max Duration was set slightly greater than the timeframe of the waveform recording. The reverberant signal would be broken up into segments with these settings, but each segment would have its waveform fully captured for analysis. Later, using programs other than AEwin, the full signal could be reconstructed. Again, this would be rare, and engineering time would be available for analysis, which was deemed preferable to the normal arrangement.

6. SUMMARY OF RECOMMENDATIONS

The following settings are recommended:

- HDT (Hit Definition Time): (a) Materials-oriented strategy: 400-2000 μ s for metals (400 μ s most commonly but can be set higher in the reverberant situation to reduce false hits); 100-200 μ s for composite materials and other non-metals. (b) Strategy for good waveform capture: 800 μ s, in conjunction with 1MSPS Sample Rate, 256 μ s Pretrigger, and 1k Length in the Waveform Setup.
- HLT (Hit Lockout Time): Usually set equal to the HDT or the minimum available (2 μ s); other settings can be useful for special purposes.
- PDT (Peak Definition Time): $\frac{1}{2}$ of the HDT.
- Max Duration: 99ms.

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APPENDIX II

Data File Definition for AEwin™

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DATA FILE DEFINITION FOR AEwin™

This document gives the format of the AEwin data files. These files are created by the AEwin when acquiring data in AUTODUMP mode. When data is dumped to disk, a file specified by the user, with the extension.DTA (e.g., TEST0000.DTA) is created.

The data file consists of three basic types of variable-length messages. These messages are (I) Products Definition Messages, (II) Test Setup Messages, and (III) Messages from the test. Each message has two bytes for length, 1-byte for ID (2 bytes for ID followed by 2 bytes for the format version number in messages with ID41 to 44 only), and a variable-length body. Messages 41-44 are special messages in that they are also included in the INI files.

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1. INTRODUCTION

This document describes the messages used to communicate between various processes within the data acquisition and replay environment of PAC products, including SPARTAN, MISTRAS, DiSP, PCI-2, PCI-8(SAMOS), and Digital AIMS ADTI. These messages may be commands and/or data depending on where they are generated within the system and where they are sent. The same types of messages are saved in the data files.

The message structure was developed for the first SPARTAN implementation to allow communication between the SPARTAN main chassis (GCC) and High End (HE), but it has been extended as our products have matured and have grown technically. Other general principles have been applied when assigning message ID numbers: the IDs from 200 to 255 were reserved for applications, ID = 0 is illegal, once an ID assigned meaning and implemented for some product, it can only be used by a different product for the same function. By the time the second application program was designed, it seemed that the 55 message ID values would quickly be used up, and therefore later application messages have added structure by using a message "subfunction" byte (e.t., 172, 42 for MI-TRA).

Once assigned, we try to avoid changing the meaning of a message ID according to where it is at a given time or which direction it is going. Thus, AE data messages always have a message ID of 1, and the Resume message has an ID of 128. Commands and messages may be built up using other messages by "sequence" or by using a new ID to combine a series of messages into a "single" block. The "Start" command is actually 2 commands: a Clear Time of Test Clock command and a "Resume" command. The "Resume" command and the "Resume" message share the same message ID; user commands are sent to the GCC as 1-byte versions of the message, and the returned (data) message with the Time of Test confirms that the GCC has completed the requested action. So the "command" message is "hidden" from the rest of the world. Any subsequent use of the message may or may not use the time of test attached to it, depending on the application. The "Resume," which is a start, normally has its time of test IGNORED (the program treats it as 0 since it follows a Clear Clock), but subsequent "Resume" messages have a new time of test which the programs can use.

This structure has been extended for the TRA products and the AEDSP software.

2. GENERAL MESSAGE INFORMATION

Classes of Messages

(I) HARDWARE STATUS AND CONTROL Messages [NOT IN THE DATA FILES]

(II) PRODUCT-SPECIFIC Messages

These are messages only found in the data files, which have a general form, but their details will vary with the products.

Product Identification Message: Identifies the Product and its Version.

Product Special Message: Test Product-specific information PRIMARILY for post-test report generation but may contain configuration information.

Product Post Test Filter Description Message: While the form of this message should be the same for all the products, the details of the filter process are Product-specific. Similar to the setup message, THIS OUGHT TO CONTAIN ALL THE INFORMATION FOR THE FILTER AND ALSO SERVE AS THE BASE FOR ANY CODE TO SAVE OR RESTORE FILTERS FOR FUTURE USE.

(III) TEST SETUP Messages

These messages are used to set up the system hardware and the test parameters. The Test Setup messages include Test Label, Date and Time, Location Setup, and the Hardware Setup Messages.

Test Label: Title or Name of the test.

Date and Time: Date and Time of Day (ASCII string) of the start of the test.

Location Setup: Defined in Location version data file only.

Hardware Setup: The Hardware setup message is a group of setup messages.

(IV) Messages from the TEST

(A) DATA Messages: DATA messages include AE Hit Data, Time-Driven Data, Sample Data, Alarm Data, and User Comment Messages.

(B) CONTROL Messages: PAUSE, RESUME, STOP, and ABORT.

(V) Synchronization and Control Messages for "external" tasks

These include commands sent to the mGCC or AEDSP processors.

(VI) Data File Messages

Typical raw data (.DTA) files contain the following messages:

Type	Id	Description of Message Body
II	41	Product Definition
III	7	Test Label
III	99	Time of Day of start of test
III	44	Location setup message
III	42	Hardware setup message
V	128	Start/Resume test
IV	1	AE hit data
IV	2	Time Driven data
IV	3	Sample data
IV	7	User Comment
IV	130	Test Pause
IV	129	Test Stop
IV	15	Test Abort message

Message Format

Each message has two bytes for length and a body of that length of bytes. Thus the total length of the "transmission" is Length+2 bytes. The body of the message starts with a message-id byte. Unfortunately, it was felt that the software would be easier if the messages which "define" the file (Presently, they have ID values from 40 to 49) if the ID space was an integer quantity. In fact, because the files contain messages with 1-byte id values, all the message IDs should be identified by the byte value. The legal range for an ID is 1 to 255.

Special setup files would contain only (primary) messages with 2 byte ID values, for example, the ".INI" files for setup of LOCAN 420, MISTRAS, or SPARTAN 2000 data acquisition and replay. I would expect that we will expand this concept to at least the post-test filtering programs when they have to be coded to save and restore user setup information.

The INI file contains the messages:

41 Product Definition
 42 Hardware Setup
 43 Graph Setup
 44 Location Group Setup (even in non-location versions)
 45 Acquisition Control
 46 Auto Run
 7 Test Title

and possibly:

49 Product-specific Information

In order to help the user to understand the structure of various types of messages, a set of them from the data file header will be briefly described. This should help to explain the message concept and give a good idea of the structure of a DTA or INI file.

ID 41 — ASCII Product Definition
(With example values)

LEN	b1, b2	2 bytes
ID	b1= 4 (29H) b2= 00	Product Definition Message ID high byte of ID (2 bytes)
PVERN	b1= 200 (C8H) b2= 00	Product version number (200) (2 bytes)
Prod string	"LOCAN-AT," CR, LF ,"VERSION 2.00",CR,LF number, CR, LF	ASCII Product name, Carriage Return, Line Feed, Version number, CR, LF

ID 7 — User Comments/Test Label

LEN	b1, b2	2 bytes
ID	b1=7 (07H)	User Comments or Test Label
	b1 bn	Text in ASCII

ID 99 — Time and Date of Test Start

EN	b1, b2	2 bytes
ID	b1= 99 (63H)	Time and date of test start ID
"Sun Jul 03, "	(For Example)	ASCII date string,
"08:49:55 1988"		ASCII time and year string.

ID 44 — Location Definition
(Dummy message in Non-Location version)

LEN	b1, b2	2 bytes
ID	b1= 44 (2CH) b2= 00	Location Definition ID high byte of ID (2 bytes)

The MISTRAS DTA file also includes two TRA setup messages when the waveforms are being recorded.

ID 173,42 — Hardware Setup
ID 172,29 — Digital Filter Setup

ID 42 — Hardware Setup

LEN	b1, b2	2 bytes
ID	b1= 42 (2AH) b2= 00	Hardware Setup ID high byte of ID (2 bytes)
MVERN b2= 00	b1= 200 (C8H) (0 filled)	Message version number
LSUB 1 SUBID 1 SUBMSG 1 		length of sub-message 1 (2 bytes) submessage 1 ID (1-bytes), submessage 1 body.
LSUB I SUBID I SUBMSG I 		length of sub-message I (2 bytes), submessage I ID (1-bytes), submessage I body.
LSUB n SUBID n SUBMSG n		length of sub-message n (2 bytes), submessage n ID (1-bytes), submessage n body

Note: Submessages are:

Start of Test Setup	ID = 100 (64H)
Set Sampling Interval	ID = 27 (1BH)
Set Demand Sampling Rate	ID = 102 (66H)
Pulser Rate	ID = 133 (85H)
Event Data Set Definition	ID = 5 (05H)
Demand Data Set Definition	ID = 6 (06H)
Group Definition	ID = 106 (6AH)
Alarm Definition	ID = 28 (1CH)
AE Filter Definition	ID = 29 (1DH)
Group Parametric Assignment	ID = 110 (6EH)
Group Settings	ID = 111 (6FH)
End of Group Settings	ID = 124 (7CH)
End of Setup	ID = 101 (65H)

3. DETAILED MESSAGE DESCRIPTIONS

The following is a list in numeric order of the messages. Each message has the two bytes for length explicitly shown, 1-byte for ID. (Remember there are a few messages with 2 bytes for the ID field EVEN THOUGH the ID is 1-byte long. They are shown with the O BYTE in the ID field.

Notation:

AM	-	Acquisition Module
#	-	'the number (of)'
PID	-	the Parametric Channel ID
CID	-	an AE channel ID
CHID	-	the AE characteristic ID number (List after Message #5) b1, b2, b3, b4, ...
	-	bytes (least significant first)
V	-	value
RTOT	-	Time of test relative to the start of test
PVERN	-	Product version number
MVERN	-	Message version number
LEN	-	length of a message
Internal	-	Defined but never appear in the data file
Defined	-	Assigned for some product(s) may appear in a data file at some later date
Not Used	-	Reserved by PAC for future use

Setup Message Units

Threshold and Gain Units:	dB
HDT, Rearm Time:	2 μ sec
PDT	1 μ sec
Sampling Interval	1 msec
[Also MID 102 is in msec steps]	
msec	== milliseconds
μ sec	== microseconds

Front End Filter Types (ICC filtering)

SCSH Single Channel, Single Hit

GCC or Parametric Voltage Filtering

Alarms

SCSH	Single Channel, Single Hit
SCTC	Single Channel, Test Cumulative
GTC	All Channels (Group), Test Cumulative
Parametric	

ID 1 — AE Hit or Event Data

LEN	b1, b2	2 bytes
ID	b1=1 (01H)	AE Hit ID
RTOT	b1 b2	Relative time of test
	b6	
CID	b1	AE Channel Number
V1	b1	Value of First AE Characteristic
V1	bn	
		as many as defined
Vm	b1	Value of Last AE Characteristic
Vm	bx	
PID1	b1	Hit Parametric Channel ID
V1	b2	Parametric value (low byte)
V2	b3	Parametric value (high byte)
V3	(b4)	Cycle Counter MSB byte
PIDj	b1	Last Hit Parametric Channel ID
V1	b2	
V2	b3	
V3	(b4)	

ID 2 — "Time-Driven" Sample Data

LEN	b1, b2	2 bytes
ID	b1 = 2 (02H)	Time Driven ID
RTOT	b1 b2	Relative time of test
	b6	
PID1	b1	First Parametric Channel ID
V1	b2	Parametric value (low byte)
V2	b3	Parametric value (high byte)
(V3)	(b4)	(Cycle Counter MSB byte)
PIDj	b1	Last Parametric Channel ID
V1	b2	
V2	b3	
V3	(b4)	
CID	b1	The ID of the first channel in use
V1	b1	Value of first AE Characteristic
.	.	
V1	bn	
Vm	b1	Value of Last AE Characteristic
.	.	
Vm	bx	end of the first channel
CID	b1	The ID of the last channel in use
V1	b1	Value of First AE Characteristic
V1	bn	
Vm	B1	Value of Last AE Characteristic
Vm	bx	end of the last channel
		end of Time data set

ID 3 — "User Forced" Sample Data — Now also a GCC Command

The message body is the same as the "Time-Driven" Sample Data except that the ID changes from 2 to 3. The GCC forces one after all PAUSE commands or when the user invokes the keypad function.

ID 4 - GCC Module Detection of Error

Defined for mGCC Ver. 1.53 and later.
(V1.8+ planned to have control from H.E.)

LEN	b1, b2	2 bytes
ID	b1 = 4 (04H)	GCC detection of error
Error Code	b1	Error identification
Error Bytes	eb1 ebX	Error information. May be text or garbled message from ICC.

Error Codes

b1 = 1 This is ICC-GCC transmission error. Bad CID, bad MID, or bad message length. MID is part of the fifo information (1,2...). Also, the fifo presents a total byte count = 2 + message length. The CID can only be the ICC's low or high channel.

eb1...ebX The message that was not received correctly

b1 = 2 Internal GCC error detection: Error message displayed on LCD. {May be only in debug version.} and saved in error bytes field.

eb1...ebX The text of the front panel error message.

b1 = 100 and
200-203 Used by the Combustion Engineering special software.

ID 5 — Event Data Set Definition

LEN	b1, b2	2 bytes
ID	b1=5 (05H)	Event Data set definition ID
# CHID	b1	Number of AE characteristics
V1	b1	Value of CHID 1
		Value of CHID m (last one)
Vm	b1	Value of CHID m (last one)
# PID	b1	Maximum number of Hit parametric

<u>CHID Value</u>	<u>Size</u>	<u>Item</u>
1	2	Rise Time of AE signal
2	2	AE Counts to Rise Time (or "Counts to peak")
3	2	Total AE counts
4	2	Energy counts (ASTM has another name for this)
5	4	Duration
6	1	Amplitude
7	1	RMS (volts) - After both were defined, the H/W
8	1	ASL (dB) - design allowed only 1 at a time.
9	1	Gain
10	1	Threshold
11	1	Pre-Amp Current
12	4	Lost Hits (per channel) WAS Overlap flag
13	2	Average Frequency (kHz) (1000* [3]/[5])

Tentative Assignments

14	2	Reserved (2-byte duration)
15	6	Reserved (6-byte TOT)
16	(4)	Reserved (Vari. length TOT with Message 12) (Example: Msg12 used in TI for 4-byte TOT.)

This is clearly a system with expansion capability, but we will need to try to do the expansion in an intelligent manner.

ID 6 — Time-Driven/Demand Data Set Definition

LEN	b1, b2	2 bytes
	b2	High byte value
ID	b1=6 (06H)	Demand Data Set Definition ID
# CHID	b1	The number of AE char. in the demand data set
V1	b1	First CHID
Vn	b1	Last CHID
# PID	b1	Number of Parametric in user
V1	b1	First PID
Vn	b1	Last PID

ID 7 — User Comments/Test Label

LEN	b1, b2	2 bytes
ID	b1=7 (07H)	User Comments or Test Label
	b1	Text in ASCII
	bn	

ID 8 — Message for Continued File TYPE A

This one is at the END of the file and indicates that there was more data when space ran out on the disk.

LEN	b1, b2	2 bytes
ID	b1=8 (08H)	File continuation
	b1	Time & Date of continuation in DOS format so that 'gaps' in time in the continuation can be detected.
		{CX,DX of DOS Time, CX,DX of DOS Date INT 21 DOS calls.}
	b8	

ID 8 — Message for Continued File TYPE B

Appears only at the beginning of a file and indicates that the file is NOT the start of a test.

LEN	b1, b2	2 bytes
ID	b1=8 (08H)	File continuation
	b1	Time of continuation in DOS format
		exactly the same as written at the
		End of the preceding file.
	b8	
{complete setup record from the first file of the test}		

This is all the messages up to the "Start Test Command," which were written at the beginning of the first file. If that file were to be unavailable, then this allows this file to be analyzed.

ID 9/ID 10 — Not Used**ID 11 — Reset Absolute Time Clock**

LEN	b1, b2	2 bytes
ID	b1=11 (0BH)	Reset Absolute Time Clock

ID 12 — Define the Size of the Clock Base ("Tick") Reserved**ID 13/ID 14 — Not Defined****ID 15 — Abort Data Acquisition**

LEN	b1, b2	2 bytes
ID	b1=15 (0FH)	Abort data acquisition/transfer ID
RTOT	b1	Relative time of test
	b2	
	b6y	

ID 16 — Alarm Data

LEN	b1, b2	2 bytes
ID	b1=16 (10H)	Alarm message ID
RTOT	b1	Relative time of test
	b2	
	b6	
Level	b1	0 = warning, 1 = trip [C.E. 16,17]
AID	b1	Alarm ID
CID	b1	Channel number that generated the alarm
		0 if time alarm (C.E.)]
V	b1	Value of the alarm (LSB)
	b2	
	b3	
	b4	

ID 17 — Reserved**ID 18 — Reserved****ID 19 — Reserved****ID 20 — Reserved for Recording Pre-amp Gain**

TRA 212 uses this as a sub-function code.

ID 21 — Reserved**ID 22 — Set Threshold**

LEN	b1, b2	2 bytes
ID	b1= 22 (16H)	Set threshold value (dB) ID
ID		
CID	b1	Which channel
V	b1	the byte value for the threshold (MSBit = 1 float, = 0 fix)

ID 23 — Set Gain

LEN	b1, b2	2 bytes
ID	b1= 23 (17H)	Set gain value (dB) ID
ID		
CID	b1	Which channel
V	b1	1-byte value for gain

ID 24 — Set Hit Definition Time

LEN	b1, b2	2 bytes
ID	b1= 24 (18H)	Set Hit Definition Time (mS) ID
ID		
CID	b1	Which channel
V	b	low byte value
V	b2	high byte value in steps of 2 mS

ID 25 — Set Hit Lockout Time

LEN	b1, b2	2 bytes
ID	b1= 25 (19H)	Set Hit Lockout Time (mS) ID
ID		
CID	b1	Which channel
V	b1	low byte value
V	b2	high byte value in steps of 2 mS

ID 26 — Set Peak Definition Time

LEN	b1, b2	2 bytes
ID	b1= 26 (1AH)	Set Peak Definition Time (mS) ID
ID		
CID	b1	Which channel
V	b1	low byte value
V	b2	high byte value in steps of 1 mS

ID 27 — Set the Sampling Interval

LEN	b1, b2	2 bytes
ID	b1= 27 (1BH)	Set the sampling interval (mS) ID
V	b1	low byte value
V	b2	high byte value in steps of 1 mS

ID 28 — Alarm Definition

LEN	b1, b2	2 bytes
ID	b1= 28 (1CH)	Alarm definition ID
j	b1	Number of alarm definitions sent
Alarm #	b1	Which one is being defined
CID	b1	Which channel
Type code	b1	Type of alarm
CHID #	b1	AE characteristic used
Level 1	b1	Warning level
	b2	
	b3	
	b4	
Level 2	b1	Trip level
	b2	
	b3	
	b4	j times

ID 29 — AE Filter Definition

LEN	b1, b2	2 bytes
ID	b1= 29 (1DH)	AE filter definition ID
k	b1	Number of filter definitions sent
Filter #	b1	Which one is being defined
CID	b1	Which channel
Type code	b1	Type of filter (scsh only)
CHID #	b1	AE characteristic used
Level 1	b1	Low level
	b2	
	b3	
	b4	
Level 2	b1	High level
	b2	
	b3	
	b4	k times

ID 30 — Delta-T AE Filter Definition
 (INI and DTA, AEDSP board)

LEN	b1, b2	2 bytes
ID	b1 = 30	Location definition ID
FS		1-byte # of filter definitions to follow
FID		1-byte Delta-T filter ID number
BDID		1-byte Board number (2 channels per board) 0 = all boards 1 = first board (channels 1,2) 2 = second board (channels 3,4),etc
TYPE		1-byte type of delta-T operation = accept events (hit pairs) with delta-T between LOW and HIGH reject other events Only one type is presently defined
INDHITS		1-byte independent hit control 0 = reject independent hits delta-T over CAL) 1 = accept independent hits
LOW		4 bytes Low level in usec (signed long integer)
HIGH		4 bytes High level in usec (signed long integer)
EDT		4 bytes Calibration time in usec (unsigned long)
		FS times

ID 31 — Reserved
ID 32 — Reserved
ID 33 — Reserved
ID 34 — Reserved
ID 35 — Reserved
ID 36 — Reserved
ID 37 - INI Write Protect Password
 (INI file)

LEN	b1, b2	2 bytes
ID	b1 = 37	Location definition ID
MVERN		1-byte = Message version*100 e.g., V1.00=100
TOT		6 bytes (least significant first) Time of hit
SZPW		n bytes Zero terminated string

ID 38 - Test Information (INI and DTA files)
--

LEN	1, b2	2 bytes
ID	1 = 38	Location definition ID
MVERN		1-byte = Message version*100 e.g., V1.00=100
ENEDIT		1-byte Enable edit of title fields
EDISPF9		1-byte Enable display of test info.at the start of the test
ZTITLE		n bytes Zero terminated string title
SZFIELD		n bytes Zero terminated string field

These two fields repeat for the duration of the message. If there is no entry, the field will be 1 '0' byte as a placeholder.

ID 39 — Reserved (Not Implemented)

ID 40 — Reserved (Not Implemented)

ID 41 — Product Definition Message

LEN	b1, b2	2 bytes
ID	b1= 41 (29H) b2= 00	ASCII Product Definition ID high byte of ID (2 bytes)
PVERN	b1= pv# b2= 00	Product version number (2 bytes) MESSAGE FORMAT=100*Product ID ONLY FOR THIS ONE
TEXT	a1 a2 a3 an	SCII Product name, Carriage return, Line feed, Version number, Carriage return, Line feed

ID 42 — Hardware Setup

LEN	b1, b2	2 bytes
ID	b1= 42 (2AH) b2= 00	Hardware Setup ID high byte of ID (2 bytes)
MVERN	b1= fv#	Message version number (2 bytes)
	b2= 00	(200. at present)
LSUB 1		length of submessage 1 (2 bytes),
SUBID 1		submessage 1 ID (1-bytes),
SUBMSG 1		submessage 1 body
LSUB I		length of sub-message I (2 bytes),
SUBID I		submessage I ID (1-bytes),
SUBMSG I		submessage I body
LSUB n		length of sub-message n (2 bytes),
SUBID n		submessage n ID (1-bytes),

submessage n body.

Start of Test	ID = 100 (64H)
Set Sampling Interval	ID = 27 (1BH)
Set Demand Sampling Rate	ID = 102 (66H)
Pulser Rate	ID = 133 (85H)
Event Data Set Definition	ID = 5 (05H)
Demand Data Set Definition	ID = 6 (06H)
Group Definition	ID = 106 (6AH)
Alarm Definition	ID = 28 (1CH)
AE Filter Definition	ID = 29 (1DH)
Group Parametric Assignment	ID = 110 (6EH)
Group Settings	ID = 111 (6FH)
End of Group Settings	ID = 124 (7CH)
End of Setup	ID = 101 (65H)

LEN	b1, b2	2 bytes
ID	b1= 43 (2BH) b2= 00	Graph Definition ID high byte of ID (2 bytes)
MVERN	b1= fv# b2= 00	Message version number (2 bytes)
Product-specific Information		

(if found in a non-location data file, length is 1 or 2).

ID 45 — Acquisition Control Information information (in the Multichannel AT products, "F8" information)	Product-specific
---	------------------

ID 46 — Autorun Message — "Command" for Acquisition Program — Product-specific information

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ID 47 — Reserved**ID 48 — Filtered File Information**

LEN	b1, b2	2 bytes
ID	b1= 48 (30H) b2= 00	Location Definition ID high byte of ID (2 bytes)
MVERN	b1= fv# b2= 00	Message version number (2 bytes)
Text follows		Product-specific subkeys & maybe some product specific keys (see Appendix A, Part 1)

ID 49 — Special Product-specific Information

LEN	b1, b2	2 bytes
ID	b1= 49 (31H) b2= 00	Location Definition ID high byte of ID (2 bytes)
MVERN	b1= fv# b2= 00	Message version number (2 bytes)
Product-specific Information		

Note: Messages from 50 to 53 are defined in some versions of LOCAN AT software. These messages will not be supported in LOCAN AT version 2.00 and higher.

ID 50 — Reserved**ID 51 — Reserved****ID 52 — Reserved****ID 53 — Reserved****ID 54 to ID 98 — Not Used**

See 59 (V1.8) and C.E. specific MIDs

ID 59 — Turn Off Alarm Command from HE to mGCC

Version 1.8 supports this.

LEN	b1, b2	2 bytes
ID	b1= 59 (3BH)	Turn off alarm

ID 99 — Time and Date of Test Start

LEN	b1, b2	2 bytes
ID	b1= 99 (63H)	Time and date of test start ID
"Sun Jul 03, " (For Example)		ASCII date string,
"08:49:55 1988"		ASCII time and year string.

ID 100 — Begin Setup

LEN	b1, b2	2 bytes
ID	b1= 100 (64H)	Begin Setup ID

ID 101 — End of Setup

LEN	b1, b2	2 bytes
ID	b1= 101 (65H)	End of Setup ID

ID 102 — Set Demand Sampling Rate

LEN	, b2	2 bytes
ID	= 102 (66H)	Set Demand Sampling Rate ID
V		Multiple of Parametric Sampling Rate)
V		

ID 103 to ID 105 (Not Used)**ID 106 — Define a Group**

LEN	b1, b2	2 bytes
ID	b1= 106 (6AH)	Define a group ID
Grp #	b1	Which group
# CHNs	b1	How many channels IDs in the list
CHID1	b1	First channel id
CHID2	b1	
CHIDn	b1	Last channel

ID 107 — Reserved**ID 108 and 109 Not Used****(110) Group Parametrics Assignment**

LEN	b1, b2	2 bytes
ID	b1= 110 (6EH)	Group parametrics assignment ID
# PIDs		Number of PIDs to follow
PID1	b1	First PID
PIDm	b1	Last PID

ID 111 — Group Settings

LEN	b1, b2	2 bytes
ID	b1= 111	Group settings ID
j		Number of group messages
[body 1]		Body of first message (ID # = 22-26)
[body 2]		Body of the second message
[body j]		Last body

ID 112 to ID 123 — Not Used**ID 124 — End of Group Setting**

LEN	b1, b2	2 bytes
ID	b1= 124 (7CH)	End of group message ID

ID 125 — Internal**ID 126 — Internal****ID 127 — Internal****ID 128 — Resume Test or Start of Test**

Test Start only when the Clear Clock was sent after Legal Setup, and no intervening Resume Stop nor Pause was sent. Normally the HE sends the messages sequentially.

LEN	b1, b2	2 bytes
ID	b1= 128 (80H)	Resume/start a test ID
(End of command)		
RTOT	b1	Relative time of test
	b2	Added when the message is built from
		The command.
		ALSO FOR STOP AND PAUSE messages
	b6	

ID 129 — Stop the Test

len	b1, b2	2 bytes
ID	b1= 129 (81H)	Stop the test ID
{End of Command}		
RTOT	b1	Relative time of test
	b2	
	b6	

ID 130 — Pause the Test

LEN	b1, b2	2 bytes
ID	b1= 130 (82H)	Pause the test ID
{End of Command}		
RTOT	b1	Relative time of test
	b2	
	b6	

ID 131 — Configuration Status/Report (Response to Command Mess. 132) NEW definition for PAC Bus 488 Inquire/Report (BOEING)
--

Len	b1 b2	Length of body
Message-ID	131 (H)	MID byte
Prod ID or Version.	a1 a2 a3 a4	ASCII Prod ID code (byte 1) byte 2 byte 3 byte 4
Prod Ver	b1 b2	Product Version Number (16 bits)

Product-specific information follows**SPARTAN mGCC Response in version 1.8 is shown below**

LEN	b1, b2	2 bytes
ID	b1=131	Status report message ID
	b1 b2 b3 b4	GCC version string (e.g., V1.8)
	b1 b2	GCC version number (e.g., 180)
	b1 b2 b3 b4	ICC version string (e.g., 2.11 or MIX)
STAT	b1	Test state code
AECH	b1	Number of active AE channels(=number of bytes to follow)
	b1 bn	number of first active channel number of last active channel
PCH	b1	Number of active parametric channels (= 4 for mGCC, 8 for PACbus)

The GCC valid states are 0 through 7.

0 Power Up	1 In Setup	2 Setup Done	3 T.O.T cleared
4 Active(Acq.)	5 Waiting	6 Test Paused	7 Stopped

ID 132 — Status Report (Command HE ® GCC) {C.E. only}
--

ID 133 — Pulser Rate

LEN	b1, b2	2 bytes
ID	b1= 133 (85H)	Pulser rate in milliseconds ID
v	b1 b2	low byte of rate high byte of rate

ID 134 — Reserved

ID 135 — Reserved**ID 136 — Analog Filter Definition**

(INI and DTA files only, AEDSP software)

LEN	b1, b2	2 bytes
ID	b1= 136	
AFID	b1	First analog multi-filter ID number
HFS	b1	k = # of highpass filters to follow
HPF1	b2	First highpass cutoff frequency, in kHz
HPF2	b2	
HPFk	b2	Last highpass cutoff frequency, in kHz
LFS	b1	n = # of lowpass filters to follow
LPF1	b2	First lowpass cutoff frequency, in kHz
LPF2	b2	
LPFn	b2	Last lowpass cutoff frequency, in kHz

Repeat above for as many different multi-filters as need to be defined

NOTE: one multi-filter is produced by the combination of available filters on a plug-in filter module with the fixed filters on the underlying board (e.g., AEDSP-32/16) if selections exist to bypass the plug-infilter.

Presently, only one combination is available. In the future, different options for plug-in and onboard filters may become available and will be described, per channel, in a modified configuration report message. At that point, the software will process the configuration report, determine all possible combinations, and generate this message.

EXAMPLE: Use this for the first version of the code	
LEN	20
ID	136
AFID	1
HFS	4
HPF1	10 kHz
HPF2	20 kHz
HPF3	100 kHz
HPF4	200 kHz
LFS	4
LPF1	1200 kHz
LPF2	100 kHz
LPF3	200 kHz
LPF4	400 kHz

ID 137 - Analog Filter Selection (INI and DTA files, AEDSP board)

LEN	b1, b2	2 bytes	
ID	b1 = 137		
CID		1-byte	Channel number, if 0 then all channels
AFID		1-byte	Multi-filter ID type number must be one defined with message 136
+			
Highpass code		1-byte	0-15, in the order that is defined for the plug-infilter
Lowpass code		1-byte	0-15, in the order that is defined for the plug-infilter

NOTE: the AFID number is ignored by the AEDSP board, which will just take the highpass and lowpass selection codes and set the plug-in filter control lines to these codes.

ID 138 — Analog Parametric Setup (INI and DTA files, AEDSP board)

LEN	b1, b2	2 bytes	
CID		1-byte	Channel number, if 0 then all channels
Gain code		1-byte	
	0 = x 1		
	1 = x 10		
	2 = x 100		
	3 = x 1000		
Filter code		1-byte	
	0 = no filter		
	1 = filter ON		

ID 139 — Cycle Counter Analog Setup (INI and DTA files, AEDSP board)
--

LEN	b1, b2	2 bytes	
CID		1-byte	Channel number for future expansion
Threshold		2 bytes	Threshold in millivolts
Range: -5080 to 5120			
Source		1-byte	
0 = parametric 1			
1 = filtered parametric 1			
2 = parametric 2			
3 = filtered parametric 2			

ID 140–ID 170 (Not defined)

ID 171 — TRA Messages (TRA2.5)

ID 171 — TRA Messages 1(TRA212)
--

ID 172,29 — Digital AE Data Filter Definition (INI and DTA, AEDSP board)
--

LEN	b1, b2	2 bytes
ID	b1 = 173	Message ID
Sub-ID	b1 = 29	Submessage ID

Exact copy of the body of ID 29 message (AE filter definition):

FS	1-byte	# of filter definitions to follow
FID	1-byte	Waveform filter ID number
CID	1-byte	Channel number
TYPE	1-byte	Type of filter operation 1 = SCSH (single channel, single hit)
		Only one type is presently defined
CHID	1-byte	AE characteristic 0 = none (reject all waveforms) 1 = risetime, 2 = counts to peak, 3 = counts, 4 = ICC energy 5 = duration, 6 = amplitude 13 = average frequency
LOW	4 bytes	Low level in msec (signed long integer)
HIGH	4 bytes	High level in msec (signed long integer)
		FS times

Will determine which hits are eligible to be recorded as waveforms

To block all waveforms, the AE characteristic (CHID) is set to 0, and the LOW and HIGH limits are also set to 0.

ID 172, 42 Hardware Setup

LEN	b1, b2	2 bytes
ID	b1 = 172 (ACH)	Message ID
Sub-ID	b1 = 42 (2AH)	Message sub-ID
MVERN	b1 = 100 b2 = 0	Message version number, 100 is V 1.00
ADT	b1	A/D converter data type 2 = 16 bit signed (the only type currently defined for MI-TRA)
SETS	b1 = n b2	Number of TRA channels (setups)
SLEN	b1 b2	Size of hardware setup, in bytes
TRA setup 1		
CHID	b1	Channel ID. If 0, setup for all channels
HLK	b1 b2	Hit length, in K samples (K = 1024)
HITS	b1 b2	Not used for MI-TRA.
SRATE	b1 b2	Sampling rate, in kHz (8000, 4000, 2000, 1000, 500, 200, 100)
TMODE	b1 b2	Trigger mode 0 = individual
TSRC	b1 b2	Trigger source 1 = digital
TDLY	b1 b2	Trigger delay in samples (negative is pretrigger)
MXIN	b1 b2	Maximum input voltage 10 = 10 Volts, only one for AEDSP
THRD	b1 b2	Trigger threshold in dBae
TRA setup 2		
TRA setup n		

ID 172,134 - Reserved

ID 173,1 — Digital AE Waveform Data
 (DTA file, AEDSP board)

LEN	b1, b2	2 bytes
ID	b1 = 173	Message ID
Sub-ID	b1 = 1	Submessage ID
TOT		6 bytes (least significant first) time of hit
CID		1-byte Channel number
ALB		1-byte Alignment byte (dummy)
N		2 bytes Number of 16-bit samples following
s1		2 bytes First sample of the waveform
s2		2 bytes Second sample
sN		2 bytes Last sample
AEF		Many bytes Copy of part of message 1 after CID (AE features and TD data)

ID 173,3 — Digital AE Power Spectrum Data (AEDSP board)

Reserved for external use

ID 174 to ID 196 — Not defined
ID 197 IDIBVALSCRN — Screen Template for Display of Improved b-Value data

LEN	b1, b2	2 bytes
ID	b1	Message ID byte
VID	b1	Version of the current message
b1		TRUE/FALSE flag Controls which event plot is shown
b2		not used
		struct graph def array defines the special graphs
bn		

ID 198 IBVALCALC — Improved b-Value Calculation Definition

LEN	b1, b2	2 bytes
ID	b1	Message ID byte
VID	b1	Version of the current message
b1		First byte of the information which contains the user setting to control the b-value Calculations.
bn		

ID 199 — Internal
ID 200–ID 254 — Reserved for Application Programs
 (See MID = 211)

ID 211 — Time Mark [Command] @ mGCC

(Lets the user make a time mark in the data set to indicate when some special action or external event occurs.)

ID 211 — (Extended) Time Mark Message(s) (Time Stamp)

LEN	b1, b2	2 bytes
ID	b1= 211 (D3H)	Time mark ID
RTOT	b1	Relative time of test
	b2	
	bn	(n = 6 if MONPAC, SPARTAN, ... (n = 4 if TRANSPORTATION) *

IF LEN = (5 or) 7, then the 211 message is a plain time mark message.

IF LEN >7, then it is a SPARTAN-AT style Extended time mark. Version 1.54 and higher of the mGCC will send to the high-end gain and threshold changes with the Time of Test stamp.

Extended message: Len = 7 + total length of the appended message.
 Body of 211 messages (7 bytes)
 Len of appended message Body length (2 bytes)
 Body of appended message {Total is 2+Body}

The "unnecessary" length of the appended message is, so that appended message can be processed as though it didn't have a time stamp (i.e. skip the first 9 bytes and pass the rest onto the message processing for appropriate message type).

***NOTE: The Transportation Instrument does not have extended messages.**

ID 220 & ID 221 — Reserved for Boeing Programs

NOTE: Short Hand W == Word, DW == Double Word, TW == Triple Word

W1 Length of Message

B Message ID = 206

W SubFunction Code [0-8]

X subfunction body

	SubFunction 0:	Velocity conversion Factor
	DW	Value
	SubFunction 1:	Display Limits
	DW	Value {Horiz. Low Lim}
	DW	Value {Horiz. High Lim}
	DW	Value {Vert. Low Lim}
	DW	Value {Vert. High Lim}
Opt. [	DW	Value {Depth Low Lim}
	DW	Value {Depth High Lim}
]		
	SubFunction 2:	Sensor Position
	DW	Horiz. Value
	DW	Vert. Value
Opt. [	DW	Depth Value
]		
	SubFunction 3:	Event Data
	W	+1 Start -1 Stop
	SubFunction 4:	Channel Timing
	W	CID (Channel ID)
	TW	6 byte Time
	W	Time to Peak
	SubFunction 5:	Turns ON/OFF S100 Data Display
	W	1 == ON , else OFF
	SubFunction 6:	Special 3-D algorithm control values
	SubFunction 7:	Special 3-D setup Command to Draw
		Either (1) All Sensors or (2) Last one
	SubFunction 8:	Use Rise Time in 3-D location calculate
	W	1==Yes, 0==No

ID 255 — Reserved**ID 255 — Reserved**

4. SUMMARY

Condensed System Messages & Commands

ID	Description
1	AE hit/Event Data
2	Time Demand Data
3	Sample/Last Time Demand Data
4	GCC detected an error condition.
5	Hit Data Set Definition
6	Time driven/Demand Data Set Definition
7	User Comments/Test Label
8	Continued File Mark
9–10	Not Used
11	Reset Real Time Clock
12	Reserved
13–14	Not Used
15	Abort acquisition/transfer
16	Alarm Data
17	Reserved
18	ICC self test (Internal)
19	Single ICC Reset (Internal)
20	{Reserved for pre-amp gain}
21	(AST messages)
22	Set Threshold
23	Set Gain
24	Set Hit Definition Time
25	Set Hit Lockout Time
26	Set Peak Definition Time
27	Set Parametric Sampling Time
28	Alarm Definition
29	Filter Definition
30	AEDSP Delta-T AE Filter Definition
31	Reserved
32	High end Alarm Detected (Internal)
33–36	Reserved
37	INI Write-Protect Password
38	Test Info
39–40	Reserved
41	ASCII Product Definition
42	Hardware Setup
43	Graph setup (*.ini file Only)
44	Location Setup
45	Acquisition Control Information
46	Auto Run Message
47	Reserved
48	Post Filter Definition and Information
49	Product-specific Setup & Configuration Information
50	Reserved
51	Reserved

ID	Description
52	Reserved
53	Reserved
54–58	Not Used
59	Ignore Alarm" - HE command to mGCC
60–98	Not Used
99	Time and Date of Test Start
100	Begin setup
101	End Setup
102	Set Demand Data Sampling Rate
103–105	Not Used
106	Begin a Group Setup
107	Reserved
108–109	Not Used
110	Define Group Parametric Channels
111	Group Parametric Settings (msgs 22-26)
112–123	Not Used
124	End of Current Group Setup
125	Internal
126	Internal
127	Internal
128	Begin Test
129	Stop Test
130	Pause Test
131	Configuration report [PRODUCT-SPECIFIC]
132	Status report [PRODUCT-SPECIFIC]
133	Pulser Rate
134	Reserved
135	Reserved
136	Analog Filter Definition (INI and DTA Files only, AEDSP software)
137	Analog Filter Selection (INI and DTA Files only, AEDSP board)
138	Analog Parametric Setup (INI and DTA Files only, AEDSP board)
139	Cycle Counter Analog Setup (INI and DTA Files only, AEDSP board)
140–170	Not Used
171	TRA2.5
172	TRA212
173	AEDSP waveform recording
174–196	Not Used
197	Screen template for display of improved b-value data
198	Improved b-value calculation definition
199	Internal
200–203	Not Used
204–210	Reserved
211	Time Mark (also in the MONPAC and Transportion Instrument)
220-221	Reserved {Boeing}
222-254	Not defined
255	Reserved

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APPENDIX III

MISTRAS IMPORTANT DOCUMENTS

TABLE OF CONTENTS

1. Mistras System Defect Report And Product Enhancement Request Form
2. Mistras Product Warranty
3. Mistras Software End User License Agreement
4. Mistras Important Notice

MISTRAS SYSTEM DEFECT REPORT AND PRODUCT ENHANCEMENT REQUEST FORM

This report is designed for you to communicate to MISTRAS, formerly Physical Acoustics Corp., any defects that you may encounter in your system so that we can take corrective action. Therefore, please send the information to MISTRAS on a copy of this form if you encounter any defects. Thank you.

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Group Headquarters:
195 Clarksville Road
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Phone: (609) 716-4044 Fax: (609) 716-0706

PRODUCTS & SYSTEMS DIVISION

SYSTEM DEFECT REPORT

AND PRODUCT ENHANCEMENT REQUEST

To provide continuous improvement of our products and to solicit your idea's regarding product enhancements, we provide this "System Defect and Product Enhancement Request Report" for you. We appreciate your comments, suggestions, and concerns. This form can be used whenever you feel the need. Additionally, we have included our email and web page addresses for correspondence and information over the internet.

Hardware _____ Software _____ (Check One)

Name: _____ Date: _____

Company: _____

Address: _____

Phone: _____ Fax: _____

System Name/Model #: _____ System S/N #: _____

Software Title: _____ Version #: _____

Software Release Date: _____

Enhancement or Defect Description: _____

Please send attention to: **Customer Service Department**

MISTRAS Group Inc. Or: E-mail report to: customer.service@mistrasgroup.com
195 Clarksville Road Fax #: (609)716-0706
Princeton Junction, NJ 08550-5303 Web pages: www.mistrasgroup.com

This area to be filled in by MISTRAS Group Inc.

Verified by: _____ Date: _____

Additional MISTRAS Comments:

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PRODUCTS & SYSTEMS DIVISION

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OEM parts used in MISTRAS systems or sold as accessories will be warranted for **six (6) months**. OEM items include monitors, printers, modems and non-MISTRAS manufactured computer items and **ninety (90) days** in the case of DVD/CD Drives, hard disk drives and battery packs.

NO OTHER WARRANTIES ARE EXPRESSED OR IMPLIED INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. MISTRAS IS NOT LIABLE FOR CONSEQUENTIAL DAMAGES.

Please read the Instruction Manual thoroughly as to installation, operation, and maintenance. Should you require additional assistance in servicing this equipment, please contact your nearest MISTRAS Regional Office or the factory directly. Returned products will not be accepted at the factory without a **Return Authorization (RMA)** number. This is obtainable from our Princeton Repair Department (609) 716-4044 or by email at: customer.service@mistrasgroup.com.

Management Approval

Document ID: Product Warranty

Revision No: 3

Date: January 3, 2017

A handwritten signature in black ink, appearing to read "David Kaskie", is written over a horizontal line.

David Kaskie, VP Products & Systems

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PRODUCTS & SYSTEMS DIVISION

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PRODUCTS & SYSTEMS DIVISION

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